

# Collected Frontier Reports for the Fabius–Rvachev System

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## Contents

|                                                                                                              |           |
|--------------------------------------------------------------------------------------------------------------|-----------|
| <b>I Tail Quadrature, Exact Dyadic Error Laws, and q-Binomial Acceleration for the Fabius–Rvachev System</b> | <b>18</b> |
| <b>1 Corpus audit, novelty boundary, and theorem status</b>                                                  | <b>19</b> |
| 1.1 What was audited . . . . .                                                                               | 19        |
| 1.2 The two explicit gaps used as starting points . . . . .                                                  | 19        |
| 1.3 Status discipline . . . . .                                                                              | 20        |
| 1.4 Executive list of new results . . . . .                                                                  | 22        |
| <b>2 The master dictionary: probability, sinc products, and Thue–Morse splines</b>                           | <b>22</b> |
| 2.1 The up-law and its finite prefixes . . . . .                                                             | 22        |
| 2.2 The finite Thue–Morse formula . . . . .                                                                  | 23        |
| 2.3 A saturated derivative plateau . . . . .                                                                 | 24        |
| <b>3 Rational moments from the infinite sinc product</b>                                                     | <b>25</b> |
| 3.1 Moment generating function and cumulants . . . . .                                                       | 25        |
| 3.2 Hankel determinants and exact quadrature constants . . . . .                                             | 25        |
| 3.3 Low-degree orthogonal data . . . . .                                                                     | 26        |
| <b>4 The exact ordinary-prefix error law</b>                                                                 | <b>26</b> |
| 4.1 A convex-order potential for the omitted tail . . . . .                                                  | 27        |
| 4.2 Sharp $C^m$ constants . . . . .                                                                          | 27        |
| 4.3 Why the equality is finite-stage exact . . . . .                                                         | 28        |
| <b>5 Gaussian tail quadrature: a positive <math>4^{-rn}</math> hierarchy</b>                                 | <b>29</b> |
| 5.1 The Gaussian rule for the up-law . . . . .                                                               | 29        |
| 5.2 The Gaussian-corrected finite spline . . . . .                                                           | 30        |
| 5.3 Explicit Gaussian rules . . . . .                                                                        | 31        |
| <b>6 Gauss–Lobatto tails: positivity and exact support</b>                                                   | <b>31</b> |
| 6.1 Construction and positivity . . . . .                                                                    | 32        |
| 6.2 The Lobatto Peano kernel . . . . .                                                                       | 32        |
| 6.3 Explicit support-preserving rules . . . . .                                                              | 33        |

|           |                                                                                           |           |
|-----------|-------------------------------------------------------------------------------------------|-----------|
| <b>7</b>  | <b>Higher-convex order, sharp weak errors, and scheme comparison</b>                      | <b>34</b> |
| 7.1       | A two-sided quadrature sandwich . . . . .                                                 | 34        |
| 7.2       | Four representative finite-tail models . . . . .                                          | 35        |
| 7.3       | A design principle . . . . .                                                              | 35        |
| <b>8</b>  | <b>An all-orders expansion for the ordinary prefix</b>                                    | <b>35</b> |
| 8.1       | Reciprocal-sinc coefficients . . . . .                                                    | 35        |
| 8.2       | The expansion theorem . . . . .                                                           | 36        |
| 8.3       | Comparison with the repeated-integration series . . . . .                                 | 37        |
| <b>9</b>  | <b><math>q = 1/4</math> Richardson acceleration and Gaussian <math>q</math>-binomials</b> | <b>37</b> |
| 9.1       | Geometric Lagrange weights . . . . .                                                      | 38        |
| 9.2       | Accelerated combinations . . . . .                                                        | 39        |
| 9.3       | Low-order filters . . . . .                                                               | 40        |
| <b>10</b> | <b>Exact and high-precision verification</b>                                              | <b>40</b> |
| 10.1      | Exact rational checks . . . . .                                                           | 40        |
| 10.2      | Third-order checks . . . . .                                                              | 41        |
| 10.3      | Minimal exact-arithmetic pseudocode . . . . .                                             | 41        |
| 10.4      | Verification philosophy . . . . .                                                         | 42        |
| <b>11</b> | <b>Consequences and additional deductions</b>                                             | <b>42</b> |
| 11.1      | Exact dyadic values reappear as extremal certificates . . . . .                           | 42        |
| 11.2      | A rational hierarchy despite algebraic nodes . . . . .                                    | 42        |
| 11.3      | A compact-support hierarchy of increasingly smooth splines . . . . .                      | 43        |
| 11.4      | Uniform approximation of integrals and observables . . . . .                              | 43        |
| 11.5      | Asymptotic bracketing of smooth observables . . . . .                                     | 43        |
| 11.6      | Why $q = 1/4$ is unavoidable . . . . .                                                    | 44        |
| <b>12</b> | <b>Lean formalization roadmap</b>                                                         | <b>44</b> |
| 12.1      | Phase I: probability measures and scaling . . . . .                                       | 44        |
| 12.2      | Phase II: the ordinary-prefix potential . . . . .                                         | 44        |
| 12.3      | Phase III: abstract positive quadrature . . . . .                                         | 45        |
| 12.4      | Phase IV: exact low-order certificates . . . . .                                          | 45        |
| 12.5      | Phase V: reciprocal sinc and $q$ -binomial acceleration . . . . .                         | 46        |
| <b>13</b> | <b>Open problems and next frontier</b>                                                    | <b>46</b> |
| 13.1      | Arithmetic of the Hankel constants . . . . .                                              | 46        |
| 13.2      | Asymptotics as the quadrature order grows . . . . .                                       | 46        |
| 13.3      | Nested positive tail rules . . . . .                                                      | 47        |
| 13.4      | Positive acceleration versus signed Richardson filters . . . . .                          | 47        |
| 13.5      | Fourier-decay interaction . . . . .                                                       | 47        |
| 13.6      | Optimal support-exact positive rules . . . . .                                            | 47        |
| 13.7      | Beyond the dyadic scale . . . . .                                                         | 48        |
| <b>14</b> | <b>Conclusion</b>                                                                         | <b>48</b> |
| <b>A</b>  | <b>Coefficient and constant tables</b>                                                    | <b>48</b> |
| A.1       | Moments and cumulants . . . . .                                                           | 48        |
| A.2       | Reciprocal-sinc coefficients . . . . .                                                    | 49        |
| A.3       | Quadrature constants . . . . .                                                            | 49        |

|           |                                                                        |           |
|-----------|------------------------------------------------------------------------|-----------|
| <b>B</b>  | <b>Derivation of the low-order quadrature rules</b>                    | <b>49</b> |
| B.1       | Gaussian order two . . . . .                                           | 49        |
| B.2       | Gaussian order three . . . . .                                         | 49        |
| B.3       | Lobatto order two . . . . .                                            | 50        |
| B.4       | Lobatto order three . . . . .                                          | 50        |
| <b>C</b>  | <b>A reusable abstract sharpness lemma</b>                             | <b>50</b> |
| <b>D</b>  | <b>Corpus coverage map</b>                                             | <b>51</b> |
| <b>E</b>  | <b>Notation index</b>                                                  | <b>52</b> |
| <b>II</b> | <b>Arithmetic Spectra of the Rvachev Sinc Product</b>                  | <b>54</b> |
| <b>1</b>  | <b>Audit boundary and research method</b>                              | <b>55</b> |
| 1.1       | Why the snapshot is pinned . . . . .                                   | 55        |
| 1.2       | Collision policy . . . . .                                             | 55        |
| <b>2</b>  | <b>Normalization and established inputs</b>                            | <b>56</b> |
| 2.1       | The centered random series . . . . .                                   | 56        |
| 2.2       | Canonical product and exact orders . . . . .                           | 56        |
| <b>3</b>  | <b>The arithmetic zero spectrum</b>                                    | <b>57</b> |
| 3.1       | Equivalent descriptions . . . . .                                      | 57        |
| 3.2       | Ordinary generating function and natural boundary . . . . .            | 57        |
| 3.3       | Spectral traces and the Taylor law . . . . .                           | 58        |
| <b>4</b>  | <b>Thue–Morse signs from zero counting</b>                             | <b>58</b> |
| 4.1       | The neighboring-bit identity . . . . .                                 | 58        |
| 4.2       | Exact lobe signs . . . . .                                             | 58        |
| 4.3       | A Thue–Morse lobe decomposition of the bump . . . . .                  | 59        |
| <b>5</b>  | <b>Laguerre–Polya geometry and real critical points</b>                | <b>59</b> |
| <b>6</b>  | <b>Exact 2-adic Strang–Fix hierarchy</b>                               | <b>60</b> |
| 6.1       | Weighted periodizations . . . . .                                      | 60        |
| 6.2       | Exact polynomial reproduction . . . . .                                | 61        |
| 6.3       | A formalization-friendly version . . . . .                             | 62        |
| <b>7</b>  | <b>Spectral zeta function and complex dimensions</b>                   | <b>62</b> |
| <b>8</b>  | <b>Mellin transform on the imaginary axis</b>                          | <b>63</b> |
| 8.1       | Exact transform . . . . .                                              | 63        |
| 8.2       | Complete large- $y$ expansion . . . . .                                | 63        |
| 8.3       | Endpoint duality . . . . .                                             | 64        |
| <b>9</b>  | <b>Finite products, <math>q</math>-integers, and pole condensation</b> | <b>64</b> |
| 9.1       | Truncated multiplicities . . . . .                                     | 64        |
| 9.2       | Gaussian $q$ -binomial layer algebra . . . . .                         | 65        |
| 9.3       | The common-zero zeta tower . . . . .                                   | 65        |
| <b>10</b> | <b>The zero heat trace</b>                                             | <b>66</b> |

|                                                                                                                    |               |
|--------------------------------------------------------------------------------------------------------------------|---------------|
| <b>11 A base-<math>b</math> theory</b>                                                                             | <b>66</b>     |
| <b>12 Numerical and symbolic verification</b>                                                                      | <b>67</b>     |
| 12.1 Imaginary-axis residual . . . . .                                                                             | 68            |
| 12.2 Exact digital checks . . . . .                                                                                | 68            |
| 12.3 A reproducible verification recipe . . . . .                                                                  | 68            |
| <b>13 What is genuinely added to the pinned corpus</b>                                                             | <b>68</b>     |
| <b>14 Lean formalization roadmap</b>                                                                               | <b>69</b>     |
| 14.1 Layer I: elementary arithmetic . . . . .                                                                      | 69            |
| 14.2 Layer II: finite algebra and Dirichlet series . . . . .                                                       | 70            |
| 14.3 Layer III: exact lattice reproduction . . . . .                                                               | 70            |
| 14.4 Layer IV: complex analysis . . . . .                                                                          | 70            |
| <b>15 Open frontier</b>                                                                                            | <b>70</b>     |
| <b>16 Conclusion</b>                                                                                               | <b>71</b>     |
| <b>A Residue calculation at the origin</b>                                                                         | <b>71</b>     |
| <b>B First coefficients and moments</b>                                                                            | <b>72</b>     |
| <b>C Corpus concordance</b>                                                                                        | <b>72</b>     |
| <br><b>III Zero-Divisor-Preserving <math>q</math>-Richardson Extrapolation for the Fabius–Rvachev Sinc Product</b> | <br><b>74</b> |
| <b>1 Corpus boundary, audit method, and collision control</b>                                                      | <b>76</b>     |
| 1.1 Pinned recursive corpus . . . . .                                                                              | 76            |
| 1.2 The mature baseline . . . . .                                                                                  | 76            |
| 1.3 Candidates deliberately discarded . . . . .                                                                    | 77            |
| 1.4 External literature boundary . . . . .                                                                         | 77            |
| <b>2 Established logarithmic <math>q</math>-Richardson structure</b>                                               | <b>77</b>     |
| 2.1 The lacunary sinc product . . . . .                                                                            | 77            |
| 2.2 Bernoulli–zeta coefficients . . . . .                                                                          | 78            |
| 2.3 Geometric Lagrange weights . . . . .                                                                           | 78            |
| <b>3 Factorization through the complete zero divisor</b>                                                           | <b>79</b>     |
| 3.1 Definition and holomorphic continuation . . . . .                                                              | 79            |
| 3.2 Exact error on the whole disk . . . . .                                                                        | 80            |
| 3.3 Uniform complex bounds . . . . .                                                                               | 80            |
| 3.4 Local jets at every included zero . . . . .                                                                    | 81            |
| 3.5 Bernoulli and Bell transfer at zeros . . . . .                                                                 | 82            |
| <b>4 Exact classification of same-parity nesting</b>                                                               | <b>82</b>     |
| 4.1 Positive error magnitudes . . . . .                                                                            | 82            |
| 4.2 Boundary asymptotics . . . . .                                                                                 | 83            |
| 4.3 Necessary and sufficient condition . . . . .                                                                   | 83            |
| 4.4 Golden-ratio threshold and eventual nesting . . . . .                                                          | 84            |
| 4.5 Threshold curve and its asymptotics . . . . .                                                                  | 84            |

|           |                                                                                         |            |
|-----------|-----------------------------------------------------------------------------------------|------------|
| <b>5</b>  | <b>The dyadic Fabius–Rvachev product</b>                                                | <b>85</b>  |
| 5.1       | Normalization and integer zeros . . . . .                                               | 85         |
| 5.2       | Two monotone zero-preserving chains . . . . .                                           | 85         |
| 5.3       | Exact preservation of Thue–Morse lobe signs . . . . .                                   | 86         |
| 5.4       | Low-order dyadic weights . . . . .                                                      | 86         |
| 5.5       | Adaptive global evaluation . . . . .                                                    | 86         |
| <b>6</b>  | <b>Structural connections and extensions</b>                                            | <b>87</b>  |
| 6.1       | $q$ -binomial interpolation is the common finite engine . . . . .                       | 87         |
| 6.2       | Canonical products and spectral determinants . . . . .                                  | 87         |
| 6.3       | Connection to inverse-Fabius and Lambert phases . . . . .                               | 87         |
| 6.4       | What the theorem does not say . . . . .                                                 | 88         |
| <b>7</b>  | <b>Numerical experiments</b>                                                            | <b>88</b>  |
| 7.1       | Reproducibility . . . . .                                                               | 88         |
| 7.2       | Dyadic nesting across the full tail disk . . . . .                                      | 88         |
| 7.3       | Unique reversal beyond the threshold . . . . .                                          | 89         |
| 7.4       | A multiplicity-five zero . . . . .                                                      | 89         |
| <b>8</b>  | <b>Conjectures and frontier directions</b>                                              | <b>90</b>  |
| <b>9</b>  | <b>Lean formalization blueprint</b>                                                     | <b>91</b>  |
| <b>10</b> | <b>Conclusion</b>                                                                       | <b>92</b>  |
| <b>A</b>  | <b>Complete recursive TeX manifest</b>                                                  | <b>92</b>  |
| <b>B</b>  | <b>Reproduction files</b>                                                               | <b>95</b>  |
| <b>IV</b> | <b>Midpoint Transmutation, Dyadic Cardinal Reproduction, and Holonomic Obstructions</b> | <b>97</b>  |
| <b>1</b>  | <b>Corpus audit and non-duplication protocol</b>                                        | <b>98</b>  |
| 1.1       | What was audited . . . . .                                                              | 98         |
| 1.2       | The overlap map . . . . .                                                               | 98         |
| 1.3       | Status labels used below . . . . .                                                      | 99         |
| <b>2</b>  | <b>Established framework and notation</b>                                               | <b>99</b>  |
| 2.1       | The three functions . . . . .                                                           | 99         |
| 2.2       | Probability, Fourier, and moments . . . . .                                             | 99         |
| 2.3       | The imported endpoint coordinate . . . . .                                              | 100        |
| <b>3</b>  | <b>An exact midpoint–endpoint transmutation</b>                                         | <b>100</b> |
| 3.1       | The complete midpoint jet . . . . .                                                     | 101        |
| 3.2       | Central integral identities . . . . .                                                   | 101        |
| <b>4</b>  | <b>The inverse midpoint defect and its Lambert–W transfer</b>                           | <b>102</b> |
| 4.1       | An exact fixed-point equation . . . . .                                                 | 102        |
| 4.2       | Certified monotone inversion . . . . .                                                  | 102        |
| 4.3       | Sharp leading comparison . . . . .                                                      | 103        |
| 4.4       | A rigorous Lagrange–Bell jet . . . . .                                                  | 103        |

|           |                                                                    |            |
|-----------|--------------------------------------------------------------------|------------|
| <b>5</b>  | <b>Thue–Morse blocks and the dyadic zero architecture</b>          | <b>104</b> |
| 5.1       | Box convolutions . . . . .                                         | 104        |
| 5.2       | Exact integer multiplicities . . . . .                             | 105        |
| <b>6</b>  | <b>A dyadic cardinal reproduction ladder</b>                       | <b>106</b> |
| 6.1       | Strang–Fix vanishing at every scale . . . . .                      | 106        |
| 6.2       | The sampling operator . . . . .                                    | 106        |
| 6.3       | Reciprocal-moment Appell correctors . . . . .                      | 107        |
| <b>7</b>  | <b>Bernoulli cumulants and Bell-polynomial correctors</b>          | <b>108</b> |
| 7.1       | Closed cumulants . . . . .                                         | 108        |
| 7.2       | Complete Bell form . . . . .                                       | 108        |
| 7.3       | Bernoulli-polynomial dyadic transport . . . . .                    | 109        |
| <b>8</b>  | <b>q-Lagrange interpolation and exact Richardson closure</b>       | <b>109</b> |
| 8.1       | Interpolation in the scale variable . . . . .                      | 109        |
| 8.2       | Exact q-binomial annihilation . . . . .                            | 110        |
| <b>9</b>  | <b>Dense nonanalyticity and holonomic obstructions</b>             | <b>111</b> |
| 9.1       | D-finite and P-recursive objects . . . . .                         | 111        |
| 9.2       | Backward propagation through the tent map . . . . .                | 111        |
| 9.3       | The entire Fourier transform . . . . .                             | 112        |
| 9.4       | Moments and Appell coefficients . . . . .                          | 112        |
| <b>10</b> | <b>Numerical and symbolic verification</b>                         | <b>112</b> |
| 10.1      | Reproducible implementation . . . . .                              | 112        |
| 10.2      | Midpoint identity and inverse defect . . . . .                     | 113        |
| 10.3      | Exact dyadic arithmetic . . . . .                                  | 114        |
| 10.4      | Cardinal and q-scale checks . . . . .                              | 114        |
| <b>11</b> | <b>A formalization roadmap</b>                                     | <b>115</b> |
| 11.1      | Low-cost Lean targets . . . . .                                    | 115        |
| 11.2      | Finite combinatorial and algebraic targets . . . . .               | 115        |
| 11.3      | Fourier and periodization targets . . . . .                        | 115        |
| 11.4      | Holonomic infrastructure . . . . .                                 | 116        |
| <b>12</b> | <b>Further frontier problems</b>                                   | <b>116</b> |
| <b>13</b> | <b>Conclusion</b>                                                  | <b>117</b> |
| <b>A</b>  | <b>Additional proof details</b>                                    | <b>117</b> |
| A.1       | Barycentric denominator calculation . . . . .                      | 117        |
| A.2       | Why the inverse correction is beyond every Lambert order . . . . . | 118        |
| A.3       | Ordinary-point analyticity . . . . .                               | 118        |
| <b>B</b>  | <b>Explicit Appell polynomials through degree eight</b>            | <b>118</b> |
| <b>C</b>  | <b>Reproducibility files</b>                                       | <b>118</b> |
| <b>D</b>  | <b>Complete recursive TeX manifest</b>                             | <b>119</b> |

|           |                                                                                                    |            |
|-----------|----------------------------------------------------------------------------------------------------|------------|
| <b>V</b>  | <b>Gamma Duality, Total Positivity, and Lambert-Localized Moments in the Fabius–Rvachev System</b> | <b>122</b> |
| <b>1</b>  | <b>Corpus audit, baseline, and novelty discipline</b>                                              | <b>123</b> |
| 1.1       | Scope of the audit . . . . .                                                                       | 123        |
| 1.2       | What was already present . . . . .                                                                 | 123        |
| 1.3       | Status ledger . . . . .                                                                            | 123        |
| <b>2</b>  | <b>Normalization and the master determinant</b>                                                    | <b>124</b> |
| 2.1       | The random series . . . . .                                                                        | 124        |
| 2.2       | Fourier and Wick-rotated products . . . . .                                                        | 124        |
| 2.3       | The spectral power sums . . . . .                                                                  | 125        |
| <b>3</b>  | <b>A <math>q</math>-binomial and multiple-zeta formula for every even moment</b>                   | <b>125</b> |
| 3.1       | Finite depth and one-site occupancy . . . . .                                                      | 125        |
| 3.2       | Distinct-index zeta sums . . . . .                                                                 | 126        |
| 3.3       | Reduction to ordinary zeta values . . . . .                                                        | 127        |
| 3.4       | First examples . . . . .                                                                           | 127        |
| <b>4</b>  | <b>Laguerre–Pólya structure and an infinite hierarchy of moment inequalities</b>                   | <b>127</b> |
| 4.1       | Classification of the Wick-rotated determinant . . . . .                                           | 127        |
| 4.2       | Pólya frequency and Jensen hyperbolicity . . . . .                                                 | 128        |
| <b>5</b>  | <b>Optimal sub-Gaussian concentration without infinite divisibility</b>                            | <b>129</b> |
| <b>6</b>  | <b>The generalized gamma-convolution dual</b>                                                      | <b>130</b> |
| 6.1       | Construction and Thorin measure . . . . .                                                          | 130        |
| 6.2       | An exact $q = 1/4$ perpetuity . . . . .                                                            | 131        |
| 6.3       | Thue–Morse parity as Thorin-mass parity . . . . .                                                  | 131        |
| 6.4       | Complete Bernstein and Stieltjes consequences . . . . .                                            | 131        |
| <b>7</b>  | <b>The Lévy heat kernel and its exact theta completion</b>                                         | <b>132</b> |
| 7.1       | Lévy density and Mellin transform . . . . .                                                        | 132        |
| 7.2       | Exact completion . . . . .                                                                         | 132        |
| <b>8</b>  | <b>Endpoint tilting: truncated exponentials, mod-Gamma limits, and Lambert <math>W</math></b>      | <b>133</b> |
| 8.1       | Exact negative-Laplace decomposition . . . . .                                                     | 133        |
| 8.2       | The tilted binary coordinates . . . . .                                                            | 134        |
| 8.3       | Tilted cumulants and exact phase dependence . . . . .                                              | 134        |
| 8.4       | Central and local concentration under the tilt . . . . .                                           | 135        |
| 8.5       | Lambert- $W_{-1}$ localization . . . . .                                                           | 136        |
| <b>9</b>  | <b>A universal endpoint-transfer theorem</b>                                                       | <b>137</b> |
| 9.1       | Universal correction polynomials . . . . .                                                         | 137        |
| 9.2       | The first two corrections . . . . .                                                                | 138        |
| 9.3       | Ordinary Fabius moments . . . . .                                                                  | 138        |
| 9.4       | Centered moments and the $q$ /MZV coefficient bridge . . . . .                                     | 139        |
| 9.5       | Numerical validation . . . . .                                                                     | 139        |
| <b>10</b> | <b>Synthesis: one determinant, two probability laws, three asymptotic regimes</b>                  | <b>140</b> |

|                                                                                                         |                |
|---------------------------------------------------------------------------------------------------------|----------------|
| <b>11 Next frontier targets</b>                                                                         | <b>141</b>     |
| 11.1 Phase-resolved Edgeworth and local limits . . . . .                                                | 141            |
| 11.2 All-orders moment transfer . . . . .                                                               | 141            |
| 11.3 Inverse recovery of the periodic endpoint function . . . . .                                       | 141            |
| 11.4 A $q$ -deformed gamma dual . . . . .                                                               | 141            |
| <b>12 Lean formalization roadmap</b>                                                                    | <b>142</b>     |
| 12.1 Low-risk algebraic modules . . . . .                                                               | 142            |
| 12.2 Analytic modules . . . . .                                                                         | 142            |
| 12.3 Library-heavy modules . . . . .                                                                    | 143            |
| <b>A Exact coefficients and normalized Turán ratios</b>                                                 | <b>143</b>     |
| <b>B Proof-dependency map</b>                                                                           | <b>143</b>     |
| <b>C Corpus map used in the audit</b>                                                                   | <b>144</b>     |
| <b>D Bibliographic notes</b>                                                                            | <b>144</b>     |
| <br><b>VI Confluent Digital Extrapolation and Lambert-Phase Tomography in the Fabius–Rvachev System</b> | <br><b>146</b> |
| <b>1 Corpus boundary and nonduplication audit</b>                                                       | <b>148</b>     |
| 1.1 Pinned recursive inventory . . . . .                                                                | 148            |
| 1.2 Strong negative controls . . . . .                                                                  | 148            |
| 1.3 Result ledger . . . . .                                                                             | 148            |
| <b>2 Established framework and normalization</b>                                                        | <b>149</b>     |
| <b>3 Confluent shift-polynomial extrapolation</b>                                                       | <b>150</b>     |
| 3.1 The master identity . . . . .                                                                       | 150            |
| 3.2 Exact conditioning for positive roots . . . . .                                                     | 151            |
| 3.3 The Gaussian $q$ -Richardson row as the simple-root case . . . . .                                  | 151            |
| <b>4 Prouhet–Richardson factorizations and bounded digital filters</b>                                  | <b>151</b>     |
| 4.1 A general factorized family . . . . .                                                               | 152            |
| 4.2 The twisted Thue–Morse filter . . . . .                                                             | 152            |
| 4.3 Why the powers of two are canonical . . . . .                                                       | 153            |
| 4.4 Mixed digital filters . . . . .                                                                     | 154            |
| <b>5 Rational multiplicative Lambert phase locks</b>                                                    | <b>154</b>     |
| 5.1 Exact phase geometry . . . . .                                                                      | 154            |
| 5.2 Confluent endpoint extraction . . . . .                                                             | 154            |
| 5.3 Conditioning versus the additive lock . . . . .                                                     | 155            |
| <b>6 The maximal Gamma–zeta strip</b>                                                                   | <b>156</b>     |
| 6.1 Exact exponential type of the Fourier coefficients . . . . .                                        | 156            |
| 6.2 Maximal holomorphy . . . . .                                                                        | 156            |
| <b>7 Weighted Wiener algebras and Bell-polynomial stability</b>                                         | <b>157</b>     |



|            |                                                                   |            |
|------------|-------------------------------------------------------------------|------------|
| <b>8</b>   | <b>Radial–angular Lambert phase tomography</b>                    | <b>158</b> |
| 8.1        | Radial interpolation of the coefficient jet . . . . .             | 158        |
| 8.2        | Angular DFT and exact aliasing . . . . .                          | 159        |
| 8.3        | The combined tomography theorem . . . . .                         | 159        |
| <b>9</b>   | <b>Numerical and exact-arithmetic experiments</b>                 | <b>160</b> |
| 9.1        | Exact filter verification . . . . .                               | 160        |
| 9.2        | Gamma–zeta modes . . . . .                                        | 161        |
| 9.3        | Tomography experiment . . . . .                                   | 161        |
| <b>10</b>  | <b>Consequences for the Fabius–Rvachev research program</b>       | <b>162</b> |
| 10.1       | A single annihilator language . . . . .                           | 162        |
| 10.2       | Endpoint coefficient spectroscopy . . . . .                       | 162        |
| 10.3       | Bell and Bernoulli closure . . . . .                              | 162        |
| 10.4       | Generalized Pascal–Rvachev hierarchies . . . . .                  | 163        |
| <b>11</b>  | <b>Conjectures and promising research directions</b>              | <b>163</b> |
| 11.1       | A concrete Lean formalization order . . . . .                     | 164        |
| <b>12</b>  | <b>Conclusion</b>                                                 | <b>164</b> |
| <b>A</b>   | <b>Auxiliary identities</b>                                       | <b>164</b> |
| A.1        | Complete homogeneous form of transformed moments . . . . .        | 164        |
| A.2        | Additive phase weights . . . . .                                  | 165        |
| A.3        | Digital condition limit . . . . .                                 | 165        |
| <b>B</b>   | <b>Reproduction guide</b>                                         | <b>165</b> |
| <b>C</b>   | <b>Complete pinned LaTeX inventory</b>                            | <b>165</b> |
| <b>VII</b> | <b>Dyadic Spectral Determinants for the Fabius–Rvachev System</b> | <b>169</b> |
| <b>1</b>   | <b>Scope, corpus audit, and novelty protocol</b>                  | <b>170</b> |
| 1.1        | What was inspected . . . . .                                      | 170        |
| 1.2        | Three status classes . . . . .                                    | 170        |
| 1.3        | Result map . . . . .                                              | 171        |
| <b>2</b>   | <b>Established baseline and notation</b>                          | <b>171</b> |
| 2.1        | The twin functions . . . . .                                      | 171        |
| 2.2        | Thue–Morse and binary notation . . . . .                          | 171        |
| 2.3        | $q$ -notation . . . . .                                           | 171        |
| 2.4        | The established zero divisor . . . . .                            | 172        |
| <b>3</b>   | <b>The dyadic spectral operator</b>                               | <b>172</b> |
| 3.1        | Definition and self-similarity . . . . .                          | 172        |
| 3.2        | The sinc product as a Fredholm determinant . . . . .              | 172        |
| 3.3        | Multiplicity as spectral degeneracy . . . . .                     | 173        |

|           |                                                                           |            |
|-----------|---------------------------------------------------------------------------|------------|
| <b>4</b>  | <b>Exact zero arithmetic and spectral parity</b>                          | <b>173</b> |
| 4.1       | The positive counting function . . . . .                                  | 173        |
| 4.2       | Thue–Morse as spectral-flow parity . . . . .                              | 174        |
| 4.3       | Dyadic-block statistics . . . . .                                         | 174        |
| 4.4       | A certified canonical-product truncation . . . . .                        | 174        |
| <b>5</b>  | <b>Spectral zeta, determinant coefficients, and cumulants</b>             | <b>175</b> |
| 5.1       | The exact spectral zeta function . . . . .                                | 175        |
| 5.2       | Trace logarithm and a coefficient recurrence . . . . .                    | 175        |
| 5.3       | A spectral interpretation of the known cumulants . . . . .                | 176        |
| <b>6</b>  | <b>Heat trace and dyadic complex dimensions</b>                           | <b>176</b> |
| 6.1       | Mellin representation . . . . .                                           | 176        |
| 6.2       | Complete heat expansion . . . . .                                         | 176        |
| 6.3       | Size of the oscillation . . . . .                                         | 177        |
| <b>7</b>  | <b>Exact local jets at all integer zeros</b>                              | <b>177</b> |
| 7.1       | A normalized factorization . . . . .                                      | 177        |
| 7.2       | The next coefficient and dyadic-ray recursion . . . . .                   | 178        |
| 7.3       | Thue–Morse signs of the leading jets . . . . .                            | 179        |
| 7.4       | Numerical checks . . . . .                                                | 179        |
| <b>8</b>  | <b>The dyadic <math>q</math>-Euler determinant</b>                        | <b>179</b> |
| 8.1       | Definition and $q$ -Pochhammer tower . . . . .                            | 179        |
| 8.2       | Lambert series and plethystic logarithm . . . . .                         | 180        |
| <b>9</b>  | <b>Finite multiscale Gaussian-binomial formulae</b>                       | <b>180</b> |
| 9.1       | A finite determinant . . . . .                                            | 180        |
| 9.2       | Dyadic endpoints and a multiscale factorial . . . . .                     | 181        |
| <b>10</b> | <b>Mellin asymptotics of the <math>q</math>-Pochhammer tower</b>          | <b>181</b> |
| 10.1      | The subcritical regime $ u  < 1$ . . . . .                                | 181        |
| 10.2      | The critical product $u = 1$ . . . . .                                    | 182        |
| <b>11</b> | <b>A symmetric <math>q</math>-integer deformation of the sinc product</b> | <b>183</b> |
| 11.1      | Symmetric $q$ -integers . . . . .                                         | 183        |
| 11.2      | Exact $q$ -scaling . . . . .                                              | 183        |
| 11.3      | Exact $q$ -Pochhammer factorization . . . . .                             | 184        |
| 11.4      | Convergence to the Rvachev Fourier transform . . . . .                    | 184        |
| 11.5      | Exact $q$ -zero arithmetic and a Thue–Morse sign law . . . . .            | 185        |
| 11.6      | The $q$ -spectral zeta and coefficient recurrence . . . . .               | 185        |
| <b>12</b> | <b>A base-<math>b</math> extension</b>                                    | <b>186</b> |
| <b>13</b> | <b>Computational verification and reproducible checks</b>                 | <b>186</b> |
| 13.1      | What was checked . . . . .                                                | 186        |
| 13.2      | $q \uparrow 1$ sample values . . . . .                                    | 187        |
| 13.3      | A practical evaluation hierarchy . . . . .                                | 187        |

|             |                                                                                                                |            |
|-------------|----------------------------------------------------------------------------------------------------------------|------------|
| <b>14</b>   | <b>Connections back to the Fabius and Thue–Morse theories</b>                                                  | <b>187</b> |
| 14.1        | Two independent appearances of $q$ -binomial structure . . . . .                                               | 187        |
| 14.2        | Thue–Morse as a determinant sign . . . . .                                                                     | 187        |
| 14.3        | Moments and zeros from one spectrum . . . . .                                                                  | 187        |
| 14.4        | Heat smoothing versus Fourier decay . . . . .                                                                  | 188        |
| <b>15</b>   | <b>Lean formalization roadmap</b>                                                                              | <b>188</b> |
| 15.1        | Layer 1: arithmetic and finite products . . . . .                                                              | 188        |
| 15.2        | Layer 2: locally finite zero factorization and cofactor jets . . . . .                                         | 188        |
| 15.3        | Layer 3: infinite products and traces . . . . .                                                                | 188        |
| 15.4        | Layer 4: Mellin asymptotics . . . . .                                                                          | 189        |
| <b>16</b>   | <b>Frontier questions</b>                                                                                      | <b>189</b> |
| <b>17</b>   | <b>Conclusion</b>                                                                                              | <b>189</b> |
| <b>A</b>    | <b>Recursive TeX corpus ledger</b>                                                                             | <b>191</b> |
| A.1         | Living and canonical documents: 18 . . . . .                                                                   | 191        |
| A.2         | Frozen historical archive: 32 . . . . .                                                                        | 191        |
| A.3         | Imported paper sources: 7 . . . . .                                                                            | 192        |
| <b>B</b>    | <b>Residue calculations for the critical <math>q</math>-Euler product</b>                                      | <b>193</b> |
| <b>C</b>    | <b>Selected source map</b>                                                                                     | <b>193</b> |
| <b>VIII</b> | <b>Logarithmic <math>q</math>-Richardson Acceleration and Lambert Phase Locking in Fabius–Rvachev Analysis</b> | <b>194</b> |
| <b>1</b>    | <b>Scope, corpus audit, and nonduplication boundary</b>                                                        | <b>196</b> |
| 1.1         | The audited corpus . . . . .                                                                                   | 196        |
| 1.2         | What is new relative to the repository . . . . .                                                               | 196        |
| 1.3         | Proof-status convention . . . . .                                                                              | 197        |
| <b>2</b>    | <b>Notation and structural background</b>                                                                      | <b>197</b> |
| 2.1         | Sinc products and the Rvachev transform . . . . .                                                              | 197        |
| 2.2         | $q$ -notation . . . . .                                                                                        | 198        |
| 2.3         | Bernoulli and Bell polynomials . . . . .                                                                       | 198        |
| 2.4         | The lower-Lambert coordinate . . . . .                                                                         | 198        |
| <b>3</b>    | <b>Exact logarithmic tails and multiplicative <math>q</math>-Richardson acceleration</b>                       | <b>198</b> |
| <b>4</b>    | <b>The exact Bernoulli–zeta tail of a lacunary sinc product</b>                                                | <b>198</b> |
| 4.1         | The logarithmic sinc series . . . . .                                                                          | 198        |
| 4.2         | Exact omitted-tail formula . . . . .                                                                           | 199        |
| <b>5</b>    | <b>Geometric Lagrange weights and their <math>q</math>-binomial form</b>                                       | <b>200</b> |
| 5.1         | Interpolation at $1, q, \dots, q^r$ . . . . .                                                                  | 200        |
| 5.2         | An exact condition number . . . . .                                                                            | 200        |

|           |                                                                              |            |
|-----------|------------------------------------------------------------------------------|------------|
| <b>6</b>  | <b>Multiplicative <math>q</math>-Richardson extrapolation</b>                | <b>201</b> |
| 6.1       | Definition and exact transformed error . . . . .                             | 201        |
| 6.2       | Certified alternating brackets . . . . .                                     | 201        |
| 6.3       | Leading term and explicit remainder bound . . . . .                          | 202        |
| 6.4       | Richardson table and multiplicative recurrence . . . . .                     | 202        |
| 6.5       | Explicit dyadic levels . . . . .                                             | 202        |
| 6.6       | Hybrid Bernoulli–Richardson correction . . . . .                             | 203        |
| <b>7</b>  | <b>Bell-polynomial error calculus and derivatives</b>                        | <b>203</b> |
| 7.1       | From logarithmic error to relative error . . . . .                           | 203        |
| 7.2       | Logarithmic derivatives . . . . .                                            | 203        |
| 7.3       | Application to interzero extrema . . . . .                                   | 204        |
| <b>8</b>  | <b>Transfer to finite signed Thue–Morse sums</b>                             | <b>204</b> |
| <b>9</b>  | <b>Numerical verification and observed orders</b>                            | <b>205</b> |
| <b>10</b> | <b>Lambert phase-locked extraction of the endpoint oscillation</b>           | <b>206</b> |
| <b>11</b> | <b>The all-orders endpoint template</b>                                      | <b>206</b> |
| <b>12</b> | <b>Phase-locked Lambert nodes</b>                                            | <b>206</b> |
| 12.1      | Exact phase locking . . . . .                                                | 206        |
| 12.2      | Lagrange weights in reciprocal scale . . . . .                               | 207        |
| 12.3      | All residual moments . . . . .                                               | 207        |
| 12.4      | Extraction of the periodic endpoint factor . . . . .                         | 209        |
| 12.5      | Coefficient extraction beyond $\Psi$ . . . . .                               | 210        |
| <b>13</b> | <b>Bernoulli-polynomial geometry and conditioning of the Lambert lattice</b> | <b>210</b> |
| 13.1      | A stable evaluation strategy . . . . .                                       | 210        |
| <b>14</b> | <b>Unified interpretation, formalization path, and research program</b>      | <b>211</b> |
| <b>15</b> | <b>Two extrapolation geometries</b>                                          | <b>211</b> |
| <b>16</b> | <b>Consequences for Fabius and Rvachev computations</b>                      | <b>211</b> |
| 16.1      | Certified Fourier evaluation . . . . .                                       | 211        |
| 16.2      | Peak localization and envelope testing . . . . .                             | 211        |
| 16.3      | Independent extraction of the endpoint oscillation . . . . .                 | 211        |
| 16.4      | A bridge from exact dyadic values to asymptotics . . . . .                   | 212        |
| 16.5      | Base- $b$ generalized up-functions . . . . .                                 | 212        |
| <b>17</b> | <b>Lean formalization blueprint</b>                                          | <b>212</b> |
| 17.1      | Suggested definitions and compiled Lambert crosswalk . . . . .               | 212        |
| 17.2      | Algebraic theorem chain . . . . .                                            | 212        |
| 17.3      | Remaining sinc-side analytic theorem chain . . . . .                         | 215        |
| <b>18</b> | <b>Open problems and next frontier steps</b>                                 | <b>215</b> |
| 18.1      | Optimal order under finite precision . . . . .                               | 215        |
| 18.2      | Monotonicity within each bracket parity . . . . .                            | 215        |
| 18.3      | Global continuation across zeros . . . . .                                   | 216        |
| 18.4      | Direct formulas for the phase coefficients . . . . .                         | 216        |
| 18.5      | A sign theory for Lambert corrections . . . . .                              | 216        |

|           |                                                                              |            |
|-----------|------------------------------------------------------------------------------|------------|
| 18.6      | Thue–Morse arithmetic of transformed products . . . . .                      | 216        |
| <b>19</b> | <b>Conclusion</b>                                                            | <b>216</b> |
| <b>A</b>  | <b>Original audited TeX inventory</b>                                        | <b>217</b> |
| A.1       | Living top-level and canonical sources (6) . . . . .                         | 217        |
| A.2       | Living Rvachev Fourier-decay drafts (12) . . . . .                           | 217        |
| A.3       | Bundled source-paper TeX (7) . . . . .                                       | 218        |
| A.4       | Frozen archive: early notes and standalone studies (5) . . . . .             | 218        |
| A.5       | Frozen archive: primary expositions and supplements (8) . . . . .            | 218        |
| A.6       | Frozen archive: $q$ -calculus, walkthrough, and glossary (6) . . . . .       | 218        |
| A.7       | Frozen archive: research frontiers (13) . . . . .                            | 219        |
| <b>B</b>  | <b>Additional algebraic details</b>                                          | <b>219</b> |
| B.1       | Derivation of the $q$ -moment formula . . . . .                              | 219        |
| B.2       | Why all transformed sinc-tail terms have one sign . . . . .                  | 220        |
| B.3       | Bell form of complete homogeneous functions . . . . .                        | 220        |
| <b>C</b>  | <b>Reproducibility notes</b>                                                 | <b>220</b> |
| <b>IX</b> | <b>Binary Spectral-Endpoint Bridges for the Fabius and Rvachev Functions</b> | <b>222</b> |
| <b>1</b>  | <b>Scope, audit method, and proof-status discipline</b>                      | <b>223</b> |
| 1.1       | What was inspected . . . . .                                                 | 223        |
| 1.2       | Status labels . . . . .                                                      | 223        |
| 1.3       | The structural gap that motivates the report . . . . .                       | 223        |
| <b>2</b>  | <b>Established framework and normalization</b>                               | <b>224</b> |
| 2.1       | The three functions . . . . .                                                | 224        |
| 2.2       | Probability model . . . . .                                                  | 224        |
| 2.3       | Sinc product . . . . .                                                       | 224        |
| 2.4       | Thue–Morse product . . . . .                                                 | 225        |
| 2.5       | Negative Laplace transform and periodic correction . . . . .                 | 225        |
| 2.6       | The Fourier-decay spectrum is not one number . . . . .                       | 225        |
| <b>3</b>  | <b>A geometric <math>q</math>-Rvachev family</b>                             | <b>226</b> |
| 3.1       | Definition and probabilistic realization . . . . .                           | 226        |
| 3.2       | $q$ -Pochhammer factorization . . . . .                                      | 226        |
| 3.3       | Cumulants in the $q$ -family . . . . .                                       | 227        |
| <b>4</b>  | <b>The binary specialization: a 2-adic zero divisor</b>                      | <b>227</b> |
| 4.1       | Finite triple factorization: sinc, $q$ -Pochhammer, and Thue–Morse . . . . . | 228        |
| <b>5</b>  | <b>The endpoint correction is a Thue–Morse logarithm</b>                     | <b>229</b> |
| 5.1       | Exact identification . . . . .                                               | 229        |
| 5.2       | The 2-adic Lambert series . . . . .                                          | 229        |
| 5.3       | The spectral–endpoint dictionary . . . . .                                   | 230        |
| <b>6</b>  | <b>Mellin transform and the exact periodic phase</b>                         | <b>231</b> |
| 6.1       | Mellin transform . . . . .                                                   | 231        |
| 6.2       | Residues and the Fourier coefficients of $P$ . . . . .                       | 231        |
| 6.3       | The periodic correction to the saddle mean . . . . .                         | 232        |

|           |                                                                          |            |
|-----------|--------------------------------------------------------------------------|------------|
| <b>7</b>  | <b>Tilted cumulants and exact periodically perturbed Lambert saddles</b> | <b>232</b> |
| 7.1       | Exact tilted mean . . . . .                                              | 232        |
| 7.2       | All tilted cumulants . . . . .                                           | 233        |
| 7.3       | Exact Lambert fixed point and branch threshold . . . . .                 | 233        |
| 7.4       | Numerical check of the exact fixed point . . . . .                       | 234        |
| <b>8</b>  | <b>All-orders sinc tails and centered Rvachev moments</b>                | <b>234</b> |
| 8.1       | Exact base-1/4 tail law . . . . .                                        | 234        |
| 8.2       | Centered cumulants and a Bernoulli–Mersenne recurrence . . . . .         | 235        |
| 8.3       | Certified Thue–Morse acceleration . . . . .                              | 236        |
| <b>9</b>  | <b>How the new identities reorganize the existing theory</b>             | <b>236</b> |
| 9.1       | A commutative diagram of transforms . . . . .                            | 236        |
| 9.2       | Two compatible $q$ -geometries . . . . .                                 | 236        |
| 9.3       | Endpoint asymptotics from spectral data . . . . .                        | 237        |
| 9.4       | Relation to the Fourier-decay audits . . . . .                           | 237        |
| <b>10</b> | <b>Further frontier problems</b>                                         | <b>237</b> |
| 10.1      | High-priority formalization targets . . . . .                            | 237        |
| 10.2      | A general- $q$ decay spectrum . . . . .                                  | 238        |
| 10.3      | Bridging the bases $1/2$ and $1/4$ . . . . .                             | 238        |
| 10.4      | A spectral derivation of the full endpoint expansion . . . . .           | 238        |
| 10.5      | Arithmetic nature of the periodic mean . . . . .                         | 238        |
| 10.6      | Weighted zero statistics . . . . .                                       | 238        |
| <b>11</b> | <b>Conclusions</b>                                                       | <b>238</b> |
| <b>A</b>  | <b>Supplementary derivations</b>                                         | <b>239</b> |
| A.1       | The coefficient formula by partitions . . . . .                          | 239        |
| A.2       | A direct proof of the first-harmonic scale . . . . .                     | 239        |
| A.3       | The unperturbed and perturbed Lambert equations . . . . .                | 239        |
| <b>B</b>  | <b>Recursive source inventory</b>                                        | <b>240</b> |
| B.1       | Living documents: five files . . . . .                                   | 240        |
| B.2       | Active frontier drafts: thirteen files . . . . .                         | 240        |
| B.3       | Frozen archive: thirty-two files . . . . .                               | 240        |
| B.4       | Source-paper transcriptions: seven files . . . . .                       | 241        |
| <b>C</b>  | <b>Notation crosswalk</b>                                                | <b>241</b> |
| <b>X</b>  | <b>Beyond the Dyadic Fabius Web</b>                                      | <b>243</b> |
| <b>1</b>  | <b>Corpus audit and the novelty boundary</b>                             | <b>247</b> |
| 1.1       | What was inspected . . . . .                                             | 247        |
| 1.2       | What the repository already does very well . . . . .                     | 247        |
| 1.3       | Deliberate exclusions . . . . .                                          | 247        |
| 1.4       | The transverse direction . . . . .                                       | 248        |
| <b>2</b>  | <b>The geometric <math>q</math>-Fabius–Rvachev family</b>                | <b>248</b> |
| 2.1       | Recovery of the classical functions . . . . .                            | 251        |
| 2.2       | Distributional refinement . . . . .                                      | 251        |

|           |                                                                                      |            |
|-----------|--------------------------------------------------------------------------------------|------------|
| <b>3</b>  | <b>Infinite sinc products and smooth compact support</b>                             | <b>252</b> |
| <b>4</b>  | <b>Bernoulli cumulants, Bell moments, and <math>q</math>-binomial scale geometry</b> | <b>253</b> |
| 4.1       | Exact cumulants . . . . .                                                            | 253        |
| 4.2       | Bell-polynomial moments and a triangular recurrence . . . . .                        | 254        |
| 4.3       | A Bernoulli–Appell inverse to geometric averaging . . . . .                          | 254        |
| 4.4       | Finite-stage Gaussian-binomial geometry . . . . .                                    | 255        |
| <b>5</b>  | <b>Parameter geometry: majorization, shape, and concentration</b>                    | <b>256</b> |
| 5.1       | A convex-order flow . . . . .                                                        | 256        |
| 5.2       | A uniform sub-Gaussian bound . . . . .                                               | 256        |
| 5.3       | Log-concavity and an exact plateau transition . . . . .                              | 257        |
| <b>6</b>  | <b>Gaussian degeneration and an Edgeworth frontier</b>                               | <b>257</b> |
| <b>7</b>  | <b>A rigorous endpoint law and Lambert-<math>W</math> inversion</b>                  | <b>259</b> |
| 7.1       | Finite-simplex upper bound . . . . .                                                 | 260        |
| 7.2       | A matching product lower bound . . . . .                                             | 260        |
| 7.3       | Inverse endpoint law . . . . .                                                       | 261        |
| 7.4       | The expected periodic completion . . . . .                                           | 261        |
| <b>8</b>  | <b>Fourier-zero arithmetic and the special role of base two</b>                      | <b>262</b> |
| 8.1       | Why the contiguous Thue–Morse block is uniquely dyadic . . . . .                     | 263        |
| <b>9</b>  | <b>Phase-aware Lagrange and Richardson extrapolation</b>                             | <b>263</b> |
| 9.1       | A phase-preserving interpolation theorem . . . . .                                   | 263        |
| 9.2       | Fourier phase averaging . . . . .                                                    | 264        |
| 9.3       | Exact Lambert- $W$ sample nodes . . . . .                                            | 265        |
| 9.4       | A practical composite algorithm . . . . .                                            | 265        |
| <b>10</b> | <b>Deterministic numerical experiments</b>                                           | <b>265</b> |
| 10.1      | Fixed-point density solver . . . . .                                                 | 266        |
| 10.2      | Cumulant checks . . . . .                                                            | 266        |
| 10.3      | Endpoint bounds . . . . .                                                            | 267        |
| 10.4      | Phase-aware extrapolation . . . . .                                                  | 267        |
| 10.5      | Zero multiplicities . . . . .                                                        | 267        |
| 10.6      | Reproducibility . . . . .                                                            | 267        |
| <b>11</b> | <b>Conjectures and promising research directions</b>                                 | <b>267</b> |
| 11.1      | Analytic shape . . . . .                                                             | 267        |
| 11.2      | Uniform Gaussian asymptotics . . . . .                                               | 268        |
| 11.3      | Endpoint phase and inverse transfer . . . . .                                        | 268        |
| 11.4      | A two-parameter large-deviation/Gaussian phase diagram . . . . .                     | 268        |
| 11.5      | Arithmetic zero collisions . . . . .                                                 | 268        |
| 11.6      | Beyond binary Thue–Morse . . . . .                                                   | 268        |
| 11.7      | Numerical analysis and conditioning . . . . .                                        | 269        |
| <b>12</b> | <b>A Lean formalization roadmap</b>                                                  | <b>269</b> |
| 12.1      | Probability layer . . . . .                                                          | 269        |
| 12.2      | Analysis layer . . . . .                                                             | 269        |
| 12.3      | Algebraic layer . . . . .                                                            | 270        |
| 12.4      | Order and limit layer . . . . .                                                      | 270        |

|           |                                                     |            |
|-----------|-----------------------------------------------------|------------|
| 12.5      | Endpoint and numerical layer . . . . .              | 270        |
| <b>13</b> | <b>Conclusion</b>                                   | <b>270</b> |
| <b>A</b>  | <b>Complete TeX corpus inventory</b>                | <b>271</b> |
| <b>B</b>  | <b>Auxiliary proof details</b>                      | <b>272</b> |
| B.1       | Schur-convexity of weighted i.i.d. sums . . . . .   | 272        |
| B.2       | A direct expansion behind the inverse law . . . . . | 272        |
| B.3       | Endpoint upper and lower constants . . . . .        | 272        |
| B.4       | Phase-aware weights as finite differences . . . . . | 273        |
| <b>C</b>  | <b>Reproducibility manifest</b>                     | <b>273</b> |



## Provenance

This volume consolidates ten source reports. Their theorem and proof bodies are preserved modulo removal of standalone document, title-page, and table-of-contents wrappers and mechanical prefixing of labels, citation keys, macro names, and asset paths. Later edits add status-only Lean crosswalk annotations throughout, including exact-support, absolute-continuity, null-singleton, contractive affine fixed-point, analytic-series-filter, and fractional-Volterra results proved after the reports’ pinned audits. The hashes below identify the unmodified source files at their former paths. The absorbed standalone drafts are deleted from the working tree; git history is the archive.

- Fabius\_Rvachev\_Frontier\_Report/Fabius\_Rvachev\_Frontier\_Report.tex  
SHA-256 7b342573c12b29316cac302c5f52bd67521e0e40335319f7e384af20a7374c5d
- Fabius\_Rvachev\_Frontier\_Report-2/Fabius\_Rvachev\_Frontier\_Report.tex  
SHA-256 607256e734ce082af18e1f982b40b8540adef17de329c7747d58450e4803a2a1
- Fabius\_Rvachev\_Frontier\_Report-3/fabius\_frontier\_report.tex  
SHA-256 70451d19a60a9dac4969a4329d3de0938971856ac762ce1f043107565ee3a463
- Fabius\_Rvachev\_Frontier\_Report\_2026-08-27/fabius\_frontier\_report.tex  
SHA-256 90614c605720093d557cbf7731af0e9889f26fe890243dd2d2f97d6a37152019
- Fabius\_Rvachev\_New\_Frontiers/Fabius\_Rvachev\_New\_Frontiers.tex  
SHA-256 f727940c971cd8fbbf90d1637ea9745672c4736ebf8714a8dbf739d9ee0bd44e
- fabius\_frontier\_report\_bundle/fabius\_frontier\_report.tex  
SHA-256 a9622f122f4bb2f0ba4a8642bb3766622150e5dc44cba51d7dc7f3e925a2585f
- fabius\_frontier\_results\_bundle/fabius\_frontier\_results.tex  
SHA-256 c76ca9aff35145e96e2c8a0ad944521216e3841920e01e7ea6a97d07aba74640
- fabius\_frontier\_new\_results/fabius\_frontier\_new\_results.tex  
SHA-256 2d4e2c3b3a705488e68ec678de449f5c617e730d23361e097cbe2a27d2162e1d
- fabius\_frontier\_spectral\_endpoint\_report\_bundle/fabius\_frontier\_spectral\_endpoint\_report.tex  
SHA-256 4d1c6cab49ab033a40ecd5ada834d39538184da43004d56412dc7dca0b4ffa9e
- beyond\_dyadic\_fabius\_web\_report/fabius\_frontier\_report.tex  
SHA-256 9cba234d935b3a48db7203cc64c5df2db73fcdc9b97b3b55b2b923ec86b0ee40

## Part I

# Tail Quadrature, Exact Dyadic Error Laws, and $q$ -Binomial Acceleration for the Fabius–Rvachev System

## Abstract

The Fabius function and Rvachev’s compactly supported up-function sit at an unusually fertile intersection of probability, dyadic self-similarity, Thue–Morse cancellation, exact rational arithmetic,  $q$ -binomial interpolation, and lacunary Fourier analysis. The accompanying ProveIt corpus already contains a broad human-readable exposition, a Lean walkthrough, a mathematical glossary, exact dyadic evaluators, a bridge from coefficient formulas to endpoint asymptotics, sharp repeated-integration laws, extensive Thue–Morse formula atlases, and a detailed analysis of the infinite sinc product. The purpose of the present report is therefore not to repeat that material, but to isolate gaps explicitly left by the corpus and to push them to theorem-level conclusions.

The first new result is an exact uniform and  $C^m$  error law for the *ordinary finite convolution prefix*

$$p_n = \beta_{2^{-1}} * \beta_{2^{-2}} * \cdots * \beta_{2^{-n}},$$

a case that the existing repeated-integration report deliberately left open. For every  $m \geq 0$  and  $n \geq m + 3$ ,

$$\left\| p_n^{(m)} - \text{up}^{(m)} \right\|_{\infty} = \frac{2^{\binom{m+3}{2}}}{18} 4^{-n},$$

and equality occurs at the explicit dyadic point  $-1 + 2^{-(m+2)}$ . In particular,  $\|p_n - \text{up}\|_{\infty} = \frac{4}{9} 4^{-n}$  for  $n \geq 3$ .

The second result resolves a more ambitious direction suggested in the corpus: replace the exact self-similar tail by a *positive compactly supported law matching additional moments*. Gaussian quadrature for the up-law gives an  $r$ -node positive tail model with exact rate  $4^{-rn}$ , while a Gauss–Lobatto variant includes the endpoints and therefore preserves the support  $[-1, 1]$  exactly. Both families admit sharp  $C^m$  constants, explicit extremal points, nonnegative Peano kernels, and rational constants expressed by Hankel determinants of the up moments. The first support-preserving variance-matched scheme is

$$\frac{1}{18} p_n(x + 2^{-n}) + \frac{8}{9} p_n(x) + \frac{1}{18} p_n(x - 2^{-n}),$$

whose exact uniform error is  $\frac{7168}{2025} 16^{-n}$  for  $n \geq 5$ .

Finally, the reciprocal infinite sinc product yields an all-orders expansion of the ordinary prefix, and the geometric error scale  $q = 1/4$  produces a new Richardson family whose weights are Gaussian  $q$ -binomial coefficients. These results supply a direct four-way bridge:

$$\begin{aligned} \text{Thue–Morse plateaux} &\longleftrightarrow \text{positive quadrature kernels} \\ &\longleftrightarrow \text{reciprocal sinc coefficients} \longleftrightarrow q\text{-binomial acceleration.} \end{aligned}$$

All proofs are given in ordinary analysis, with exact rational and high-precision checks. The results are new relative to the audited corpus. The finite geometric  $q$ -binomial/Lagrange bridge identified below now has matching Lean declarations; the analytic approximation, error, remainder, conditioning, and convergence theorems are not thereby claimed formalized or externally peer reviewed.

# 1 Corpus audit, novelty boundary, and theorem status

## 1.1 What was audited

The source tree under Analysis/FabiusFunction/docs has several layers rather than a single linear manuscript. The audit used the current document families as follows.

| Document family                                        | Mathematical role in the audit                                                                                                                                                          |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fabius_Function_and_Rvachev_Up                         | Canonical definitions, normalizations, probability law, infinite convolution and sinc product, exact dyadic arithmetic, $q$ -binomial/Thue–Morse formulas, and endpoint asymptotics.    |
| FabiusFunction_Mathematical_Glossary                   | Terminology and coverage map across analysis, probability, Fourier methods, $q$ -calculus, computability, and formalization.                                                            |
| fabius_lean_walkthrough                                | Structure and dependency map of the Lean development; used to separate established formal results from exploratory claims.                                                              |
| Non_Elementarity_of_the_Fabius_Function                | Differential-algebraic and non-elementarity direction; audited to avoid misclassifying an approximation theorem as a transcendence result.                                              |
| Exact-value and asymptotic papers                      | Dyadic evaluators, alternative representations, repeated integration, Thue–Morse convergence, and the coefficient-to-saddle bridge.                                                     |
| Canonical non-formalized frontier volume               | Consolidated research claims, open directions, and explicit formalization boundary. Its absorbed notebooks were treated as one mathematical corpus rather than as independent evidence. |
| Active drafts on Thue–Morse formulas and Fourier decay | Recent formula atlases and sharp analysis of the infinite sinc product; used to avoid duplicating the newest work.                                                                      |
| Archive and audit material                             | Provenance, superseded variants, and contradiction checks. Byte-identical or explicitly frozen historical copies were not counted as separate mathematical contributions.               |

The principal baseline established by this corpus includes:

1. the random-series and infinite-convolution construction of up;
2. the exact Fourier product  $\widehat{\text{up}}(\xi) = \prod_{k \geq 1} \text{sinc}(2^{-k}\xi)$ ;
3. exact rational values at dyadic arguments and executable recurrences;
4. finite Thue–Morse and Gaussian  $q$ -binomial formulas;
5. sharp endpoint asymptotics with periodic corrections;
6. a sharp law for the repeated-integration approximation, whose last dyadic box is duplicated;
7. extensive typical-ray and exceptional-ray results for the Fourier product.

## 1.2 The two explicit gaps used as starting points

The repeated-integration report compares its support-corrected spline  $f_n$  with the ordinary prefix  $p_n$ . It proves

$$\|f_n - \text{up}\|_\infty = \frac{8}{9 \cdot 4^n}$$

from the second stage onward, but labels the corresponding exact law for  $p_n$  as outside the scope of that paper. It then observes that a positive compactly supported tail model matching the true variance

would cancel the leading second-derivative term and should produce a  $16^{-n}$  scheme, while leaving positivity and support as nontrivial constraints.

Those two sentences are the main launch point of this report. The first becomes [Theorem 4.2](#); the second becomes the Gaussian and Gauss–Lobatto hierarchies in [Sections 5](#) and [6](#). The Gauss–Lobatto construction is especially decisive: it is positive, matches the variance, and includes the two endpoints, so the final spline has support exactly  $[-1, 1]$ .

### 1.3 Status discipline

**Mathematical status.** Every displayed frontier theorem below is accompanied by a proof or by an explicit reduction to standard Gaussian/Lobatto quadrature facts. The low-order constants were independently checked by exact rational arithmetic or high-precision evaluation. The results are presented as new relative to the audited ProveIt corpus. A broad literature search did not locate these Fabius-specific formulas, but this report does not assert an exhaustive priority claim. The finite algebraic bridge cross-referenced to `GeometricQBinomialLagrange.lean` is Lean-formalized, and `AnalyticSeriesFilter.lean` proves the abstract exact Gaussian tail under unconditional summability, with no unweighted convergence condition at a zero-weight node. The later `RvachevQBinomialFilter.lean` specializes that mechanism to the actual infinite Rvachev sinc product for complex scale  $c$ , with Gaussian base  $q = c^2$  and only finite node injectivity; its  $c = 1/2$ ,  $q = 1/4$  form is assumption-free. This infinite-product theorem is not an identification of the report’s finite-prefix quotient or its reciprocal-sinc/Bell coefficients, and it proves none of the report’s analytic signs, bounds, uniform convergence, or asymptotics. Separately, `FractionalVolterraCalculus.lean` proves positive affine covariance on ordered real intervals for arbitrary real order, and Gamma-normalized order raising for real  $\alpha > 0$  from a continuous Banach-valued primitive with an interval-integrable right derivative; for  $\alpha \leq 0$ , affine covariance is an identity for the totalized Lean interval-integral definition, and a classical Riemann–Liouville or integrability interpretation is asserted only for positive order. `FabiusFractionalVolterra.lean` defines the total causal Rvachev fractional primitive, proves its support cutoff, positive-natural bridge, and positive-order semigroup on  $x \geq -1$ , and specializes order raising to the signed extension for  $x \geq 0$ , bounded Fabius for  $0 \leq x \leq 1$ , and the up bridge from  $-1$  for  $x \geq -1$ . The fractional APIs were checked at source checkpoint `149332f9d`; the infinite-product filter is a current-tree result. Complex, weighted-monomial, iterated, negative-branch, shifted-lattice, endpoint-moment, transform/tail, piecewise/refinement, and inverse/quantile fractional formulas, and the report-specific analytic theorem families, remain frontier.

**Post-snapshot finite moment, Gram–Stieltjes, and Jacobi status.** The generic layer begins with `FiniteMomentGram.lean`, which defines the generic polynomial moment `Functional`, symmetric moment `Pairing`, moment `HankelMatrix`, moment `HankelDet`, and `finiteMomentPairing`. The declarations `momentFunctional_of_linearMap`, `momentFunctional_map`, `momentHankelMatrix_succ_submatrix`, `map_momentHankelDet`, `finiteMomentPairing_toMatrix`, and `finiteMomentPairing_nondegenerate_iff` give completeness, base change, leading-principal truncation, the exact monomial-basis Gram matrix, and—over an integral domain—fixed-order nondegeneracy exactly when the corresponding Hankel determinant is nonzero. The downstream `GramStieltjes.lean` constructs the fraction-free `gramStieltjesNumerator` and field-normalized `gramStieltjesPolynomial`. Its public endpoint declarations `momentPairing_gramStieltjesNumerator_eq_coeff_mul_det`, `momentPairing_gramStieltjesNumerator_self`, `gramStieltjesPolynomial_isMonicOfDegree`, `momentPairing_gramStieltjesPolynomial_eq_zero`, `eq_gramStieltjesPolynomial_of_isMonicOfDegree_of_orthogonal`, and `momentPairing_gramStieltjesPolynomial_self` prove finite Gram–Stieltjes orthogonality, uniqueness under a nonzero leading minor, and the quotient of consecutive Hankel determinants for the self-pairing. `FiniteMomentJacobi.lean` continues without a measure: it defines `gramStieltjesNorm`, `gramStieltjesJacobiDiagonal`, and `gramStieltjesJacobiSubdiagonal`; proves the determinant cross-ratio for the subdiagonal; proves the degree-zero equation through `gramStieltjesPolynomial_three_term_zero`; and proves every higher finite three-term recurrence through `gramStieltjesPolynomial_three_term` from the three consecutive nonzero minors. These three generic modules themselves assume no measure and prove no positivity. The complementary up-measure pipeline in `MomentHankelMatrix.lean`, `MomentHankelValues.lean`, and the `OrthogonalPolynomial*.lean` modules proves strict positivity, constructs the normalized polynomials, proves parity and the symmetric three-term recurrence, and evaluates the first exact Jacobi data. `OrthogonalPolynomialGramBridge.lean` now identifies the generic up-moment functional and pairing with integration, both Hankel objects, both determinant polynomials, and both monic normalizations; its Jacobi wrappers identify the norm, zero diagonal, and subdiagonal ratio with the up-specific data. Root location or simplicity, Gaussian or Lobatto nodes, weights, exactness, or error signs, finite or infinite Jacobi/Stieltjes-fraction identification, and convergence remain outside Lean. This post-snapshot proof-status update does not alter the absorbed reports’ provenance hashes or their pinned novelty comparison.

## 1.4 Executive list of new results

### New results proved in this report.

1. An exact ordinary-prefix law in every fixed  $C^m$  norm, including exact extremal dyadic points.
2. A positive  $r$ -node Gaussian tail hierarchy with sharp rate  $4^{-rn}$  and exact constants  $h_r/(2r)!$ .
3. A positive support-exact Gauss–Lobatto hierarchy with the same rate and constants  $\ell_r/(2r)!$ .
4. A two-sided order for all  $2r$ -convex observables, placing the true tail between its Gaussian and Lobatto surrogates.
5. Rational Hankel-determinant formulas for all sharp constants, tied to the rational moment sequence of the infinite sinc product.
6. A complete reciprocal-sinc expansion of the ordinary prefix in powers of  $4^{-n}$ .
7. A  $q = 1/4$  Richardson family whose weights are explicit Gaussian  $q$ -binomial coefficients.

## 2 The master dictionary: probability, sinc products, and Thue–Morse splines

### 2.1 The up-law and its finite prefixes

For  $a > 0$ , write

$$\beta_a(x) := \frac{1}{2a} \mathbf{1}_{[-a, a]}(x). \quad (2.1)$$

Let  $U_1, U_2, \dots$  be independent random variables uniformly distributed on  $[-1, 1]$ , and define

$$Y := \sum_{k=1}^{\infty} 2^{-k} U_k, \quad S_n := \sum_{k=1}^n 2^{-k} U_k. \quad (2.2)$$

The series converges absolutely,  $Y \in [-1, 1]$ , and its density is Rvachev’s even up-function. We write

$$\mu := \mathcal{L}(Y), \quad p_n := \beta_{2^{-1}} * \beta_{2^{-2}} * \dots * \beta_{2^{-n}}. \quad (2.3)$$

Thus  $p_n$  is the density of  $S_n$  and

$$\text{supp}(p_n) = [-1 + 2^{-n}, 1 - 2^{-n}]. \quad (2.4)$$

The omitted tail is an exact scaled copy of the original law:

$$Y - S_n \stackrel{d}{=} 2^{-n} Y. \quad (2.5)$$

If  $\mu_a$  denotes the pushforward of  $\mu$  under  $x \mapsto ax$ , then, with  $a = 2^{-n}$ ,

$$\boxed{\text{up} = p_n * \mu_a}. \quad (2.6)$$

This elementary identity is the source of every approximation theorem below.

We use the Fourier convention

$$\widehat{g}(\xi) = \int_{\mathbb{R}} g(x) e^{-ix\xi} dx, \quad \text{sinc } z = \frac{\sin z}{z}, \quad \text{sinc } 0 = 1. \quad (2.7)$$

Then

$$P_n(\xi) := \widehat{p}_n(\xi) = \prod_{k=1}^n \operatorname{sinc}(2^{-k}\xi), \quad (2.8)$$

$$\Phi(\xi) := \widehat{\operatorname{up}}(\xi) = \prod_{k=1}^{\infty} \operatorname{sinc}(2^{-k}\xi), \quad (2.9)$$

$$\Phi(\xi) = P_n(\xi)\Phi(2^{-n}\xi). \quad (2.10)$$

On the unit interval, the bounded Fabius function is the shifted left half of the bump:

$$F(x) = \operatorname{up}(x-1), \quad 0 \leq x \leq 1. \quad (2.11)$$

Consequently every approximation to  $\operatorname{up}$  immediately gives one for  $F$  by translating the argument by one.

## 2.2 The finite Thue–Morse formula

Let

$$\varepsilon_j := (-1)^{s_2(j)}, \quad (2.12)$$

where  $s_2(j)$  is the binary digit sum. The generalized Irwin–Hall formula for unequal dyadic boxes becomes a Thue–Morse spline.

**Proposition 2.1** (Finite Thue–Morse box-spline formula). *For  $n \geq 1$ ,*

$$p_n(x) = \frac{2^{\binom{n}{2}}}{(n-1)!} \sum_{j=0}^{2^n-1} \varepsilon_j (x+1-2^{-n}-j2^{1-n})_+^{n-1}. \quad (2.13)$$

For  $0 \leq r \leq n-1$ , differentiation gives

$$p_n^{(r)}(x) = \frac{2^{\binom{n}{2}}}{(n-1-r)!} \sum_{j=0}^{2^n-1} \varepsilon_j (x+1-2^{-n}-j2^{1-n})_+^{n-1-r} \quad (2.14)$$

away from the knots, and distributionally everywhere.

*Proof.* Write  $2^{-k}U_k = V_k - 2^{-k}$  with  $V_k$  uniform on  $[0, 2^{1-k}]$ . The density of  $\sum V_k$  is the inclusion–exclusion spline

$$\frac{1}{(n-1)! \prod_{k=1}^n 2^{1-k}} \sum_{A \subseteq \{1, \dots, n\}} (-1)^{|A|} \left( t - \sum_{k \in A} 2^{1-k} \right)_+^{n-1}.$$

Now  $\prod_{k=1}^n 2^{1-k} = 2^{-\binom{n}{2}}$ , the centering shift is  $\sum_{k=1}^n 2^{-k} = 1 - 2^{-n}$ , and the map

$$A \mapsto j(A) := \sum_{k \in A} 2^{n-k}$$

is a bijection from subsets to  $\{0, \dots, 2^n - 1\}$ . It satisfies  $\sum_{k \in A} 2^{1-k} = j(A)2^{1-n}$  and  $(-1)^{|A|} = (-1)^{s_2(j(A))}$ , proving (2.13). Differentiation gives (2.14).  $\square$

### 2.3 A saturated derivative plateau

The sharp constants below come from a simple but powerful fact about Thue–Morse prefix sums:

$$\sum_{j=0}^{2q-1} \varepsilon_j = 0, \quad \sum_{j=0}^{2q} \varepsilon_j = \varepsilon_q. \quad (2.15)$$

Thus every partial sum is 0 or  $\pm 1$ .

**Lemma 2.2** (Thue–Morse derivative plateau). *For  $r \geq 0$  put*

$$C_r := 2^{\binom{r+1}{2}}, \quad x_r := -1 + 2^{-r}. \quad (2.16)$$

*If  $n \geq r + 1$ , then*

$$\left\| p_n^{(r)} \right\|_{\infty} = C_r. \quad (2.17)$$

*Moreover,*

$$p_n^{(r)}(x) = C_r \quad \text{for almost every } x \in [x_r - 2^{-n}, x_r + 2^{-n}]. \quad (2.18)$$

*Proof.* First take  $n = r + 1$ . In (2.14), the  $r$ th derivative is

$$p_{r+1}^{(r)}(x) = C_r \sum_{\substack{0 \leq j < 2^{r+1} \\ j2^{-r} < x+1-2^{-(r+1)}}} \varepsilon_j.$$

By (2.15), its absolute value never exceeds  $C_r$ . On the first knot interval

$$[-1 + 2^{-(r+1)}, -1 + 3 \cdot 2^{-(r+1)}],$$

only the  $j = 0$  term is active, so the value is exactly  $C_r$ . This interval has center  $x_r$  and half-width  $2^{-(r+1)}$ .

For  $n > r + 1$ , factor

$$p_n = p_{r+1} * \beta_{2^{-(r+2)}} * \cdots * \beta_{2^{-n}}.$$

The remaining convolution factor is a probability density supported on an interval of radius

$$\sum_{k=r+2}^n 2^{-k} = 2^{-(r+1)} - 2^{-n}.$$

Convolution with a probability density cannot increase the  $L^\infty$  norm. It preserves the constant value  $C_r$  wherever the translated support remains inside the first plateau, leaving precisely the interval in (2.18). Hence the upper bound is attained.  $\square$

*Remark 2.3.* The same binary fact controls three phenomena already prominent in the corpus: the finite spline formula, the exact derivative plateau, and the terminating derivative tower of  $F$  at dyadic arguments. In the present report it acquires a fourth role: it turns every positive Peano-kernel upper bound into an exact equality.



### 3 Rational moments from the infinite sinc product

#### 3.1 Moment generating function and cumulants

The moment generating function of  $Y$  is entire and factors as

$$M(t) := \mathbb{E}e^{tY} = \prod_{k=1}^{\infty} \frac{\sinh(t/2^k)}{t/2^k}. \quad (3.1)$$

The classical expansion

$$\log \frac{\sinh z}{z} = \sum_{m=1}^{\infty} \frac{2^{2m-1} B_{2m}}{m(2m)!} z^{2m} \quad (3.2)$$

therefore yields

$$\log M(t) = \sum_{m=1}^{\infty} \frac{2^{2m-1} B_{2m}}{m(2m)!(4^m - 1)} t^{2m}. \quad (3.3)$$

If  $\kappa_j$  denotes the  $j$ th cumulant, then

$$\kappa_{2m} = \frac{2^{2m-1} B_{2m}}{m(4^m - 1)}, \quad \kappa_{2m+1} = 0. \quad (3.4)$$

The factors  $4^m - 1$  are the moment-side trace of the dyadic sinc scales.

Let  $\mu_j := \mathbb{E}Y^j$ . The standard moment–cumulant recurrence gives

$$\mu_0 = 1, \quad \mu_n = \sum_{k=1}^n \binom{n-1}{k-1} \kappa_k \mu_{n-k}. \quad (3.5)$$

In particular, every moment is rational. The first values needed below are

$$\begin{aligned} \mu_0 &= 1, & \mu_2 &= \frac{1}{9}, & \mu_4 &= \frac{19}{675}, & \mu_6 &= \frac{583}{59535}, \\ \mu_8 &= \frac{132809}{32531625}, & \mu_{10} &= \frac{46840699}{24405225075}, & \mu_{12} &= \frac{4068990560161}{4133856862760625}. \end{aligned} \quad (3.6)$$

All odd moments vanish.

#### 3.2 Hankel determinants and exact quadrature constants

Define the Hankel determinants

$$\Delta_r := \det(\mu_{i+j})_{0 \leq i, j \leq r-1}, \quad \Delta_0 := 1. \quad (3.7)$$

Because  $\mu$  has a positive smooth density on  $(-1, 1)$ , every  $\Delta_r$  is positive. Let  $\pi_r$  be the monic orthogonal polynomial of degree  $r$  for  $\mu$  and put

$$h_r := \int_{-1}^1 \pi_r(x)^2 d\mu(x). \quad (3.8)$$

Then the Gram-determinant identity gives

$$h_r = \frac{\Delta_{r+1}}{\Delta_r}. \quad (3.9)$$

Thus every  $h_r$  is a positive rational number. Equivalently,

$$h_r = \min \left\{ \int P(x)^2 d\mu(x) : P \text{ monic of degree } r \right\}. \quad (3.10)$$

For the Lobatto construction, introduce the weighted moment functional

$$\int q(x) d\nu(x) := \int (1 - x^2)q(x) d\mu(x) \quad (3.11)$$

and its Hankel determinants

$$\Lambda_s := \det(\mu_{i+j} - \mu_{i+j+2})_{0 \leq i,j \leq s-1}, \quad \Lambda_0 := 1. \quad (3.12)$$

If  $\chi_{r-1}$  is monic of degree  $r - 1$  and orthogonal for  $\nu$ , define

$$\ell_r := \int (1 - x^2)\chi_{r-1}(x)^2 d\mu(x). \quad (3.13)$$

Then

$$\ell_r = \frac{\Lambda_r}{\Lambda_{r-1}} \quad (3.14)$$

and

$$\ell_r = \min \left\{ \int (1 - x^2)P(x)^2 d\mu(x) : P \text{ monic of degree } r - 1 \right\}. \quad (3.15)$$

These rational norms will be the exact sharp constants of the two quadrature hierarchies.

### 3.3 Low-degree orthogonal data

Symmetry forces the expected parity. From (3.6),

$$\pi_1(x) = x, \quad h_1 = \frac{1}{9}, \quad (3.16)$$

$$\pi_2(x) = x^2 - \frac{1}{9}, \quad h_2 = \frac{32}{2025}, \quad (3.17)$$

$$\pi_3(x) = x^3 - \frac{19}{75}x, \quad h_3 = \frac{19808}{7441875}. \quad (3.18)$$

For the weighted Lobatto measure,

$$\chi_1(x) = x, \quad \ell_2 = \frac{56}{675}, \quad (3.19)$$

$$\chi_2(x) = x^2 - \frac{7}{75}, \quad \ell_3 = \frac{78976}{7441875}. \quad (3.20)$$

The rational coefficients coexist with algebraic quadrature nodes; this arithmetic split is one of the useful structural features of the construction.

## 4 The exact ordinary-prefix error law

The ordinary prefix simply omits the tail. At first sight this seems less structured than the support-corrected repeated-integration scheme. In fact the omitted tail has a canonical positive second-order potential, and the same Thue–Morse plateau used in the repeated-integration analysis makes the resulting bound exact.

#### 4.1 A convex-order potential for the omitted tail

Define

$$J(t) := \frac{1}{2} (\mathbb{E} |t - Y| - |t|). \quad (4.1)$$

**Lemma 4.1** (Positive tail potential). *The function  $J$  is even, nonnegative, continuous, and supported on  $[-1, 1]$ . In the sense of distributions,*

$$J'' = \mu - \delta_0. \quad (4.2)$$

Moreover,

$$\int_{\mathbb{R}} J(t) dt = \frac{\mu_2}{2} = \frac{1}{18}. \quad (4.3)$$

*Proof.* The law of  $Y$  is centered. Jensen's inequality for the convex map  $y \mapsto |t - y|$  gives

$$\mathbb{E} |t - Y| \geq |t - \mathbb{E} Y| = |t|,$$

so  $J \geq 0$ . If  $t \geq 1$ , then  $t - Y \geq 0$  and  $\mathbb{E} |t - Y| = t - \mathbb{E} Y = t$ ; the case  $t \leq -1$  is symmetric. Hence  $J$  vanishes outside  $[-1, 1]$ .

Since  $(|t - y|/2)'' = \delta_y$ , averaging in  $y$  and subtracting the case  $y = 0$  yields (4.2). Finally,  $J$  and its first derivative vanish at the boundary of its support, so two integrations by parts give

$$2 \int J(t) dt = \int t^2 J''(t) dt = \int t^2 d\mu(t) - 0 = \mu_2 = \frac{1}{9}.$$

□

For  $a > 0$ , set

$$J_a(t) := aJ(t/a). \quad (4.4)$$

Then

$$J_a'' = \mu_a - \delta_0, \quad \text{supp}(J_a) \subseteq [-a, a], \quad \int J_a = \frac{a^2}{18}. \quad (4.5)$$

Combining this with (2.6) gives the exact factorization

$$\boxed{\text{up} - p_n = p_n * (\mu_a - \delta_0) = p_n'' * J_a, \quad a = 2^{-n}.} \quad (4.6)$$

#### 4.2 Sharp $C^m$ constants

**Theorem 4.2** (Exact ordinary-prefix law). *Let  $m \geq 0$  and  $n \geq m + 3$ . Then*

$$\boxed{\|p_n^{(m)} - \text{up}^{(m)}\|_{\infty} = \frac{2^{\binom{m+3}{2}}}{18} 4^{-n}.} \quad (4.7)$$

At the dyadic point

$$x_{m+2} = -1 + 2^{-(m+2)}, \quad (4.8)$$

we have the signed identity

$$\boxed{(p_n^{(m)} - \text{up}^{(m)})(x_{m+2}) = -\frac{2^{\binom{m+3}{2}}}{18} 4^{-n}.} \quad (4.9)$$

*Proof.* Differentiate (4.6)  $m$  times:

$$\text{up}^{(m)} - p_n^{(m)} = p_n^{(m+2)} * J_a. \quad (4.10)$$

The kernel is nonnegative and has mass  $a^2/18$ . By Lemma 2.2,

$$\|p_n^{(m+2)}\|_\infty = C_{m+2} = 2^{\binom{m+3}{2}}.$$

Therefore

$$\|\text{up}^{(m)} - p_n^{(m)}\|_\infty \leq C_{m+2} \frac{a^2}{18}.$$

At  $x = x_{m+2}$ , the whole integration interval  $[x - a, x + a]$  lies inside the plateau from (2.18), because  $n \geq m + 3$ . Hence

$$(p_n^{(m+2)} * J_a)(x_{m+2}) = C_{m+2} \int J_a = C_{m+2} \frac{a^2}{18}.$$

Thus the upper bound is attained, and the sign follows from (4.10).  $\square$

**Corollary 4.3** (Uniform law). *For every  $n \geq 3$ ,*

$$\|p_n - \text{up}\|_\infty = \frac{4}{9} 4^{-n}. \quad (4.11)$$

*The negative extremum occurs at  $x = -3/4$ :*

$$p_n(-3/4) - \text{up}(-3/4) = -\frac{4}{9} 4^{-n}. \quad (4.12)$$

**Corollary 4.4** (Bounded Fabius version). *Define  $P_n(x) := p_n(x - 1)$  on  $[0, 1]$ . Then, for  $m \geq 0$  and  $n \geq m + 3$ ,*

$$\|P_n^{(m)} - F^{(m)}\|_{L^\infty[0,1]} = \frac{2^{\binom{m+3}{2}}}{18} 4^{-n}, \quad (4.13)$$

*with negative equality at  $x = 2^{-(m+2)}$ .*

**Remark 4.5** (Repeated integration). The support-corrected repeated-integration density is  $f_n = p_n * \beta_{2^{-n}}$ . The corpus proves

$$\|f_n^{(m)} - \text{up}^{(m)}\|_\infty = \frac{2^{\binom{m+3}{2}}}{9} 4^{-n}$$

for the stable range, exactly twice the ordinary-prefix constant and with the opposite local sign at the corresponding saturated plateau. The two laws are governed by positive second-order kernels of masses  $1/18$  and  $1/9$  respectively.

### 4.3 Why the equality is finite-stage exact

A formal Taylor expansion would predict

$$p_n = \text{up} - \frac{1}{18} 4^{-n} \text{up}'' + O(16^{-n}).$$

At  $x = -3/4$ , however, the theorem is not merely asymptotic. The kernel  $J_a$  is supported entirely inside a region where  $p_n''$  is exactly the constant 8, so all higher corrections collapse into the same finite convolution identity. This is the same mechanism that made the repeated-integration error exact, but with the omitted-tail potential replacing the uniform-tail potential.

## 5 Gaussian tail quadrature: a positive $4^{-rn}$ hierarchy

The ordinary prefix corresponds to replacing the true tail by the single atom  $\delta_0$ . Gaussian quadrature replaces it by  $r$  positive atoms while matching the first  $2r - 1$  moments. Because the tail scale is  $a = 2^{-n}$ , matching through degree  $2r - 1$  forces the first possible error to have size  $a^{2r} = 4^{-rn}$ .

### 5.1 The Gaussian rule for the up-law

Let  $\pi_r$  be the monic orthogonal polynomial from (3.8). Its roots

$$-1 < \xi_{r,1} < \cdots < \xi_{r,r} < 1 \quad (5.1)$$

are simple. There are unique positive weights  $w_{r,j}$  such that

$$Q_r := \sum_{j=1}^r w_{r,j} \delta_{\xi_{r,j}} \quad (5.2)$$

is exact on polynomials of degree at most  $2r - 1$ :

$$\int q \, dQ_r = \int q \, d\mu, \quad \deg q \leq 2r - 1. \quad (5.3)$$

Because  $\mu$  is even, the nodes and weights occur in symmetric pairs, with a central node at 0 when  $r$  is odd.

**Proposition 5.1** (Positive Gaussian Peano kernel). *The error functional*

$$\mathcal{E}_r^G(\phi) := \int \phi \, d\mu - \int \phi \, dQ_r \quad (5.4)$$

is nonnegative whenever  $\phi \in C^{2r}[-1, 1]$  and  $\phi^{(2r)} \geq 0$ . It has a nonnegative Peano kernel  $K_r^G$ , supported on  $[-1, 1]$ , such that

$$\mathcal{E}_r^G(\phi) = \int_{-1}^1 \phi^{(2r)}(t) K_r^G(t) \, dt, \quad (5.5)$$

$$(K_r^G)^{(2r)} = \mu - Q_r, \quad \int K_r^G = \frac{h_r}{(2r)!}. \quad (5.6)$$

*Proof.* Let  $H$  be the Hermite interpolant of degree at most  $2r - 1$  satisfying

$$H(\xi_{r,j}) = \phi(\xi_{r,j}), \quad H'(\xi_{r,j}) = \phi'(\xi_{r,j}) \quad (1 \leq j \leq r).$$

The remainder formula is

$$\phi(x) - H(x) = \frac{\phi^{(2r)}(\eta_x)}{(2r)!} \pi_r(x)^2 \quad (5.7)$$

for some  $\eta_x \in (-1, 1)$ . Hence  $\phi - H \geq 0$  when  $\phi^{(2r)} \geq 0$ . Gaussian exactness and interpolation give

$$\mathcal{E}_r^G(\phi) = \int (\phi - H) \, d\mu \geq 0.$$

The Peano-kernel theorem now gives a nonnegative kernel. For  $\phi(x) = x^{2r}$ , the difference  $\phi - H$  is the monic degree- $2r$  polynomial with double zeros at all Gaussian nodes, hence equals  $\pi_r^2$ . Therefore

$$\mathcal{E}_r^G(x^{2r}) = \int \pi_r^2 \, d\mu = h_r.$$

Since  $(x^{2r})^{(2r)} = (2r)!$ , the kernel mass is  $h_r/(2r)!$ . The distributional derivative identity follows directly from (5.5).  $\square$

For  $a > 0$ , let  $Q_{r,a}$  be the pushforward of  $Q_r$  under  $x \mapsto ax$ , and set

$$K_{r,a}^G(t) := a^{2r-1} K_r^G(t/a). \quad (5.8)$$

Then

$$(K_{r,a}^G)^{(2r)} = \mu_a - Q_{r,a}, \quad \int K_{r,a}^G = a^{2r} \frac{h_r}{(2r)!}. \quad (5.9)$$

## 5.2 The Gaussian-corrected finite spline

Define

$$G_{n,r} := p_n * Q_{r,2^{-n}} = \sum_{j=1}^r w_{r,j} p_n(\cdot - 2^{-n} \xi_{r,j}). \quad (5.10)$$

This is a positive probability density and a finite mixture of shifted Thue–Morse splines. Its support is strictly contained in  $[-1, 1]$  because all Gaussian nodes lie in  $(-1, 1)$ .

**Theorem 5.2** (Sharp Gaussian hierarchy). *Let  $r \geq 1$ ,  $m \geq 0$ , and  $n \geq m + 2r + 1$ . Then*

$$\left\| G_{n,r}^{(m)} - \text{up}^{(m)} \right\|_{\infty} = \frac{2^{\binom{m+2r+1}{2}} h_r}{(2r)!} 2^{-2rn}. \quad (5.11)$$

At

$$x_{m+2r} = -1 + 2^{-(m+2r)}, \quad (5.12)$$

we have

$$(G_{n,r}^{(m)} - \text{up}^{(m)})(x_{m+2r}) = -\frac{2^{\binom{m+2r+1}{2}} h_r}{(2r)!} 2^{-2rn}. \quad (5.13)$$

*Proof.* Using (2.6) and (5.9), with  $a = 2^{-n}$ ,

$$\begin{aligned} \text{up} - G_{n,r} &= p_n * (\mu_a - Q_{r,a}) \\ &= p_n^{(2r)} * K_{r,a}^G. \end{aligned} \quad (5.14)$$

After  $m$  derivatives,

$$\text{up}^{(m)} - G_{n,r}^{(m)} = p_n^{(m+2r)} * K_{r,a}^G.$$

The nonnegative kernel has mass  $a^{2r} h_r / (2r)!$ . The derivative plateau lemma gives the sharp upper bound

$$C_{m+2r} a^{2r} \frac{h_r}{(2r)!}.$$

At  $x_{m+2r}$  the full kernel support  $[-a, a]$  lies inside the constant positive plateau, so equality holds. The sign follows from the displayed factorization.  $\square$

**Corollary 5.3** (Rate ladder). *For fixed  $r$  and  $m$ , the exact error ratio between successive levels is*

$$\frac{\left\| G_{n+1,r}^{(m)} - \text{up}^{(m)} \right\|_{\infty}}{\left\| G_{n,r}^{(m)} - \text{up}^{(m)} \right\|_{\infty}} = 4^{-r} \quad (5.15)$$

throughout the stable range. Thus  $r = 1, 2, 3, \dots$  gives the geometric rates

$$4^{-n}, 16^{-n}, 64^{-n}, \dots$$

### 5.3 Explicit Gaussian rules

**Example 5.4** (One node: ordinary truncation). For  $r = 1$ ,  $\pi_1(x) = x$ , so  $Q_1 = \delta_0$  and  $h_1 = 1/9$ . Hence  $G_{n,1} = p_n$ , and [Theorem 5.2](#) is exactly [Theorem 4.2](#).

**Example 5.5** (Two nodes: a variance-matched  $16^{-n}$  scheme). For  $r = 2$ ,

$$\pi_2(x) = x^2 - \frac{1}{9}, \quad Q_2 = \frac{1}{2}\delta_{-1/3} + \frac{1}{2}\delta_{1/3}, \quad h_2 = \frac{32}{2025}.$$

Therefore

$$G_{n,2}(x) = \frac{1}{2}p_n\left(x - \frac{2^{-n}}{3}\right) + \frac{1}{2}p_n\left(x + \frac{2^{-n}}{3}\right). \quad (5.16)$$

For  $n \geq m + 5$ ,

$$\left\|G_{n,2}^{(m)} - \text{up}^{(m)}\right\|_{\infty} = \frac{4}{6075} 2^{\binom{m+5}{2}} 16^{-n}. \quad (5.17)$$

In particular,

$$\|G_{n,2} - \text{up}\|_{\infty} = \frac{4096}{6075} 16^{-n}, \quad n \geq 5, \quad (5.18)$$

with negative equality at  $x = -15/16$ .

**Example 5.6** (Three nodes: a  $64^{-n}$  scheme). For  $r = 3$ ,

$$\pi_3(x) = x^3 - \frac{19}{75}x.$$

The Gaussian nodes and weights are

$$0, \pm\sqrt{\frac{19}{75}}, \quad \frac{32}{57}, \frac{25}{114}, \frac{25}{114}, \quad (5.19)$$

respectively, and

$$h_3 = \frac{19808}{7441875}, \quad \frac{h_3}{6!} = \frac{1238}{334884375}.$$

Thus, for  $n \geq m + 7$ ,

$$\left\|G_{n,3}^{(m)} - \text{up}^{(m)}\right\|_{\infty} = \frac{1238}{334884375} 2^{\binom{m+7}{2}} 64^{-n}. \quad (5.20)$$

The extremal point is  $-1 + 2^{-(m+6)}$ .

*Remark 5.7* (Arithmetic form). The nodes are algebraic roots of rational orthogonal polynomials and the weights lie in the corresponding splitting fields, but the exact norm constants are rational. This is useful for formalization: one may prove existence and positivity abstractly, while the sharp constant reduces to rational moment and determinant arithmetic.

## 6 Gauss–Lobatto tails: positivity and exact support

Gaussian nodes lie strictly inside the support, so  $G_{n,r}$  has slightly shorter support than  $\text{up}$ . To preserve  $[-1, 1]$  exactly, force the tail rule to include  $-1$  and  $1$ . The correct interior nodes are given by orthogonality for the weighted measure  $(1 - x^2) d\mu$ .

## 6.1 Construction and positivity

Let  $\chi_{r-1}$  be the monic orthogonal polynomial of degree  $r - 1$  for the measure  $\nu$  from (3.11). Its roots

$$-1 < \eta_{r,1} < \cdots < \eta_{r,r-1} < 1 \quad (6.1)$$

are simple. There is a unique interpolatory rule on the  $r + 1$  nodes

$$-1, \eta_{r,1}, \dots, \eta_{r,r-1}, 1 \quad (6.2)$$

that integrates polynomials of degree at most  $r$ . Orthogonality improves its degree of exactness to  $2r - 1$ . Denote it by

$$L_r = \lambda_{r,-}\delta_{-1} + \sum_{j=1}^{r-1} \lambda_{r,j}\delta_{\eta_{r,j}} + \lambda_{r,+}\delta_1. \quad (6.3)$$

**Proposition 6.1** (Exactness and positive weights). *The rule  $L_r$  is exact on polynomials of degree at most  $2r - 1$ , and every weight in (6.3) is positive.*

*Proof.* Let  $p$  have degree at most  $2r - 1$ . Divide it as

$$p(x) = q(x)(x^2 - 1)\chi_{r-1}(x) + s(x),$$

where  $\deg q \leq r - 2$  and  $\deg s \leq r$ . The first term vanishes at every quadrature node, while

$$\int q(x)(x^2 - 1)\chi_{r-1}(x) d\mu(x) = - \int q(x)\chi_{r-1}(x) d\nu(x) = 0.$$

Thus both the integral and the rule reduce to  $s$ , proving degree  $2r - 1$  exactness.

For an interior node  $\eta_{r,j}$ , the polynomial

$$\frac{(1 - x^2)\chi_{r-1}(x)^2}{(x - \eta_{r,j})^2}$$

is nonnegative, has degree  $2r - 2$ , and vanishes at every quadrature node except  $\eta_{r,j}$ . Exactness therefore forces the corresponding weight to be positive. At 1, use  $(1 + x)\chi_{r-1}(x)^2$ ; at  $-1$ , use  $(1 - x)\chi_{r-1}(x)^2$ . These have degree  $2r - 1$ , are nonnegative, and isolate the endpoint weights.  $\square$

## 6.2 The Lobatto Peano kernel

**Proposition 6.2** (Positive Lobatto Peano kernel). *For*

$$\mathcal{E}_r^L(\phi) := \int \phi dL_r - \int \phi d\mu, \quad (6.4)$$

*we have  $\mathcal{E}_r^L(\phi) \geq 0$  whenever  $\phi^{(2r)} \geq 0$ . There is a nonnegative kernel  $K_r^L$  supported on  $[-1, 1]$  such that*

$$\mathcal{E}_r^L(\phi) = \int_{-1}^1 \phi^{(2r)}(t) K_r^L(t) dt, \quad (6.5)$$

$$(K_r^L)^{(2r)} = L_r - \mu, \quad \int K_r^L = \frac{\ell_r}{(2r)!}. \quad (6.6)$$

*Proof.* Let  $H$  interpolate  $\phi$  at the two endpoints and interpolate both  $\phi$  and  $\phi'$  at the  $r - 1$  interior nodes. Then  $\deg H \leq 2r - 1$  and

$$\phi(x) - H(x) = \frac{\phi^{(2r)}(\zeta_x)}{(2r)!} (x^2 - 1)\chi_{r-1}(x)^2. \quad (6.7)$$



The nodal polynomial is nonpositive on  $[-1, 1]$ . Hence  $\phi - H \leq 0$  for  $\phi^{(2r)} \geq 0$ . Since  $L_r$  is exact on  $H$  and uses the same values as  $\phi$  at all nodes,

$$\mathcal{E}_r^L(\phi) = \int (H - \phi) d\mu \geq 0.$$

For  $\phi(x) = x^{2r}$ , the difference  $x^{2r} - H(x)$  is exactly  $(x^2 - 1)\chi_{r-1}(x)^2$ . Therefore

$$\mathcal{E}_r^L(x^{2r}) = \int (1 - x^2)\chi_{r-1}(x)^2 d\mu(x) = \ell_r.$$

The Peano-kernel theorem and the same normalization argument as before give (6.6).  $\square$

Scale  $L_r$  and  $K_r^L$  by  $a = 2^{-n}$  as in the Gaussian case, and define

$$H_{n,r} := p_n * L_{r,2^{-n}}. \quad (6.8)$$

We use the letter  $H$  to avoid confusing the Lobatto approximant with its kernel. Because  $L_r$  has positive endpoint masses, the support is exactly

$$\text{supp}(H_{n,r}) = [-1, 1]. \quad (6.9)$$

**Theorem 6.3** (Sharp support-preserving hierarchy). *Let  $r \geq 1$ ,  $m \geq 0$ , and  $n \geq m + 2r + 1$ . Then*

$$\left\| H_{n,r}^{(m)} - \text{up}^{(m)} \right\|_{\infty} = \frac{2^{\binom{m+2r+1}{2}} \ell_r}{(2r)!} 2^{-2rn}. \quad (6.10)$$

At  $x_{m+2r} = -1 + 2^{-(m+2r)}$ ,

$$(H_{n,r}^{(m)} - \text{up}^{(m)})(x_{m+2r}) = + \frac{2^{\binom{m+2r+1}{2}} \ell_r}{(2r)!} 2^{-2rn}. \quad (6.11)$$

*Proof.* The scaled Lobatto kernel satisfies

$$(K_{r,a}^L)^{(2r)} = L_{r,a} - \mu_a, \quad \int K_{r,a}^L = a^{2r} \frac{\ell_r}{(2r)!}.$$

Therefore

$$H_{n,r} - \text{up} = p_n^{(2r)} * K_{r,a}^L. \quad (6.12)$$

Differentiate  $m$  times and apply the positive-kernel argument and Lemma 2.2. Equality and sign follow at the center of the saturated plateau.  $\square$

### 6.3 Explicit support-preserving rules

**Example 6.4** (Three-point variance-matched rule). For  $r = 2$ ,  $\chi_1(x) = x$ . The nodes are  $-1, 0, 1$  and exactness gives

$$L_2 = \frac{1}{18}\delta_{-1} + \frac{8}{9}\delta_0 + \frac{1}{18}\delta_1. \quad (6.13)$$

This is the unique symmetric positive law on  $\{-1, 0, 1\}$  with variance  $1/9$ . Since

$$\ell_2 = \mu_2 - \mu_4 = \frac{56}{675}, \quad \frac{\ell_2}{4!} = \frac{7}{2025},$$

the support-exact finite spline is

$$H_{n,2}(x) = \frac{1}{18}p_n(x + 2^{-n}) + \frac{8}{9}p_n(x) + \frac{1}{18}p_n(x - 2^{-n}). \quad (6.14)$$

For  $n \geq m + 5$ ,

$$\left\| H_{n,2}^{(m)} - \text{up}^{(m)} \right\|_{\infty} = \frac{7}{2025} 2^{\binom{m+5}{2}} 16^{-n}. \quad (6.15)$$

In particular,

$$\left\| H_{n,2} - \text{up} \right\|_{\infty} = \frac{7168}{2025} 16^{-n}, \quad n \geq 5, \quad (6.16)$$

with positive equality at  $x = -15/16$ .

**Example 6.5** (Four-point  $64^{-n}$  support-exact rule). For  $r = 3$ ,

$$\chi_2(x) = x^2 - \frac{7}{75}.$$

The nodes are

$$\pm 1, \quad \pm \sqrt{\frac{7}{75}}, \quad (6.17)$$

and their weights are

$$\frac{1}{102}, \frac{1}{102} \quad \text{at } \pm 1, \quad \frac{25}{51}, \frac{25}{51} \quad \text{at } \pm \sqrt{\frac{7}{75}}. \quad (6.18)$$

Furthermore,

$$\ell_3 = \frac{78976}{7441875}, \quad \frac{\ell_3}{6!} = \frac{4936}{334884375}.$$

Hence, for  $n \geq m + 7$ ,

$$\left\| H_{n,3}^{(m)} - \text{up}^{(m)} \right\|_{\infty} = \frac{4936}{334884375} 2^{\binom{m+7}{2}} 64^{-n}. \quad (6.19)$$

*Remark 6.6* (Resolution of the variance-matching problem). The rule (6.13) is positive, supported on the exact normalized tail interval  $[-1, 1]$ , and matches mass, mean, variance, and—by symmetry—the third moment. After scaling and convolution with  $p_n$ , it gives the  $16^{-n}$  support-exact scheme anticipated but not constructed in the earlier corpus.

## 7 Higher-convex order, sharp weak errors, and scheme comparison

### 7.1 A two-sided quadrature sandwich

The Gaussian and Lobatto remainder signs imply a clean order relation for observables.

**Theorem 7.1** ( $2r$ -convex sandwich). *Let  $\phi \in C^{2r}[-1, 1]$  with  $\phi^{(2r)} \geq 0$ . Then*

$$\int \phi \, dQ_r \leq \int \phi \, d\mu \leq \int \phi \, dL_r. \quad (7.1)$$

More quantitatively,

$$0 \leq \int \phi \, d\mu - \int \phi \, dQ_r \leq \frac{h_r}{(2r)!} \left\| \phi^{(2r)} \right\|_{\infty}, \quad (7.2)$$

$$0 \leq \int \phi \, dL_r - \int \phi \, d\mu \leq \frac{\ell_r}{(2r)!} \left\| \phi^{(2r)} \right\|_{\infty}. \quad (7.3)$$

Both constants are sharp, with equality for  $\phi(x) = x^{2r}$ .

*Proof.* The signs follow from [Propositions 5.1](#) and [6.2](#). Bounding the Peano representations by the kernel masses proves the inequalities. For  $x^{2r}$ , the derivative is the constant  $(2r)!$  and the errors are  $h_r$  and  $\ell_r$ .  $\square$

Let  $Z_G \sim Q_r$  and  $Z_L \sim L_r$ , independent of  $S_n$ . Scaling [Theorem 7.1](#) gives, for every  $2r$ -convex  $\phi$ ,

$$\mathbb{E}\phi(S_n + 2^{-n}Z_G) \leq \mathbb{E}\phi(Y) \leq \mathbb{E}\phi(S_n + 2^{-n}Z_L). \quad (7.4)$$

This is a weak-order statement, not a global pointwise ordering of the three densities. Pointwise signs depend on the sign of the appropriate derivative of  $p_n$ ; the exact extrema in [Theorems 5.2](#) and [6.3](#) occur where that derivative is saturated and positive.

## 7.2 Four representative finite-tail models

| Scheme                     | Normalized tail                         | Matched degrees  | Support after convolution | Exact rate and constant     | uniform                 |
|----------------------------|-----------------------------------------|------------------|---------------------------|-----------------------------|-------------------------|
| Ordinary $p_n$             | $\delta_0$                              | 0, 1             | radius $1 - 2^{-n}$       | $\frac{4}{9} 4^{-n}$        |                         |
| Repeated integration $f_n$ | uniform on $[-1, 1]$                    | 0, 1             | $[-1, 1]$ exactly         | $\frac{8}{9} 4^{-n}$        | (established in corpus) |
| Gaussian $G_{n,2}$         | atoms at $\pm 1/3$                      | through degree 3 | strictly inside $[-1, 1]$ | $\frac{4096}{6075} 16^{-n}$ |                         |
| Lobatto $H_{n,2}$          | weights $1/18, 8/9, 1/18$ at $-1, 0, 1$ | through degree 3 | $[-1, 1]$ exactly         | $\frac{7168}{2025} 16^{-n}$ |                         |

Gaussian quadrature has the smaller high-order constant and uses fewer atoms. Lobatto quadrature pays a constant-factor price for exact endpoint support. Both improve the geometric ratio from  $1/4$  to  $1/16$ .

## 7.3 A design principle

The constructions expose a general approximation design triangle:

1. *positivity* is obtained by quadrature weights;
2. *order* is obtained by moment exactness;
3. *support exactness* is obtained by including the endpoints.

For the self-similar Fabius tail, these choices are unusually transparent because the tail at every stage is a scaled copy of one fixed compactly supported measure and because the prefix derivatives have exact Thue–Morse plateaux.

# 8 An all-orders expansion for the ordinary prefix

The exact law in [Theorem 4.2](#) comes from a positive physical-space kernel. The full correction series is more naturally read from the reciprocal of the tail sinc product.

## 8.1 Reciprocal-sinc coefficients

Near the origin, define

$$\frac{1}{\Phi(z)} = \sum_{j=0}^{\infty} b_j z^{2j}, \quad b_0 = 1. \quad (8.1)$$

Euler's product for  $\sin z$  gives

$$\begin{aligned} \log \frac{1}{\Phi(z)} &= - \sum_{m=1}^{\infty} \log \operatorname{sinc}(2^{-m}z) \\ &= \sum_{k=1}^{\infty} \gamma_k z^{2k}, \end{aligned} \quad (8.2)$$

where

$$\gamma_k = \frac{\zeta(2k)}{k\pi^{2k}(4^k - 1)}. \quad (8.3)$$

The ratios  $\zeta(2k)/\pi^{2k}$  are rational, so every  $\gamma_k$  and every  $b_j$  is rational. Logarithmic differentiation yields the effective recurrence

$$b_j = \frac{1}{j} \sum_{k=1}^j k \gamma_k b_{j-k}, \quad j \geq 1. \quad (8.4)$$

The first coefficients are

$$b_0 = 1, \quad b_1 = \frac{1}{18}, \quad b_2 = \frac{31}{16200}, \quad (8.5)$$

$$b_3 = \frac{2347}{42865200}, \quad b_4 = \frac{1904369}{1311675120000}. \quad (8.6)$$

## 8.2 The expansion theorem

**Theorem 8.1** (Reciprocal-sinc expansion). *Fix integers  $m \geq 0$  and  $J \geq 1$ . As  $n \rightarrow \infty$ ,*

$$\left\| p_n^{(m)} - \sum_{j=0}^{J-1} (-1)^j b_j 4^{-jn} \operatorname{up}^{(m+2j)} \right\|_{\infty} = O_{m,J}(4^{-Jn}). \quad (8.7)$$

*The implied constant is independent of  $n$ .*

*Proof.* Let

$$B_{J-1}(z) := \sum_{j=0}^{J-1} b_j z^{2j}.$$

By the defining Taylor expansion,

$$1 - \Phi(z)B_{J-1}(z) = O(z^{2J}) \quad (z \rightarrow 0). \quad (8.8)$$

Since  $|\Phi(z)| \leq 1$  on the real axis and  $B_{J-1}$  is a fixed polynomial, the local bound extends to the global estimate

$$|1 - \Phi(z)B_{J-1}(z)| \leq C_J |z|^{2J}, \quad z \in \mathbb{R}. \quad (8.9)$$

Indeed, use (8.8) for  $|z| \leq 1$  and the trivial polynomial bound for  $|z| \geq 1$ .

Write  $a = 2^{-n}$ . From (2.10),

$$P_n(\xi) - \Phi(\xi)B_{J-1}(a\xi) = P_n(\xi) [1 - \Phi(a\xi)B_{J-1}(a\xi)]. \quad (8.10)$$

After multiplying by  $|\xi|^m$ , (8.9) gives

$$|\xi|^m |P_n(\xi) - \Phi(\xi)B_{J-1}(a\xi)| \leq C_J a^{2J} |\xi|^{m+2J} |P_n(\xi)|. \quad (8.11)$$

Choose a fixed  $N > m + 2J + 1$ . For  $n \geq N$ ,

$$|P_n(\xi)| \leq \prod_{k=1}^N \left| \text{sinc}(2^{-k}\xi) \right|,$$

and the right-hand side decays as  $O(|\xi|^{-N})$ . Thus  $|\xi|^{m+2J} P_n(\xi)$  has an  $L^1$  majorant independent of  $n$ . Fourier inversion yields an  $O(a^{2J})$  uniform error. Finally,

$$\mathcal{F}[\text{up}^{(m+2j)}](\xi) = (i\xi)^{m+2j} \Phi(\xi) = (i\xi)^m (-1)^j \xi^{2j} \Phi(\xi),$$

which gives the signs in (8.7). □

The first four terms are therefore

$$\boxed{p_n^{(m)} = \text{up}^{(m)} - \frac{\text{up}^{(m+2)}}{18 \cdot 4^n} + \frac{31 \text{ up}^{(m+4)}}{16200 \cdot 16^n} - \frac{2347 \text{ up}^{(m+6)}}{42865200 \cdot 64^n} + O_m(256^{-n}).} \quad (8.12)$$

The exact law of [Theorem 4.2](#) is compatible with the first term but is stronger: at the saturated dyadic point the physical-space kernel gives a finite-stage equality rather than merely an asymptotic statement.

### 8.3 Comparison with the repeated-integration series

The repeated-integration approximation has a different tail multiplier and begins

$$f_n = \text{up} + \frac{1}{9} 4^{-n} \text{up}'' + \frac{2}{2025} 16^{-n} \text{up}^{(4)} + \dots$$

Thus ordinary truncation and repeated integration approach up from opposite second-order directions. The two-node Gaussian and three-node Lobatto tails cancel that second-order term entirely, leaving fourth-order coefficients

$$G_{n,2} = \text{up} - \frac{4}{6075} 16^{-n} \text{up}^{(4)} + O(64^{-n}), \quad (8.13)$$

$$H_{n,2} = \text{up} + \frac{7}{2025} 16^{-n} \text{up}^{(4)} + O(64^{-n}). \quad (8.14)$$

The signs agree with the Gaussian/Lobatto Peano kernels and with the exact extrema.

## 9 $q = 1/4$ Richardson acceleration and Gaussian $q$ -binomials

The corpus already contains Gaussian  $q$ -binomial formulas at base  $q = 1/2$ , arising from geometric interpolation in the dyadic spatial scale. The ordinary-prefix expansion exposes a different geometric variable: its error powers are

$$4^{-n}, 16^{-n}, 64^{-n}, \dots$$

Hence the natural extrapolation base is

$$q := \frac{1}{4}. \quad (9.1)$$

This produces a second, conceptually distinct appearance of Gaussian  $q$ -binomial coefficients.

### 9.1 Geometric Lagrange weights

For  $s \geq 1$  and  $0 \leq j \leq s-1$ , let  $\lambda_{s,j}$  be the Lagrange cardinal weight for evaluation at 0 on the nodes  $1, q, \dots, q^{s-1}$ :

$$\lambda_{s,j} := \prod_{\substack{0 \leq \ell \leq s-1 \\ \ell \neq j}} \frac{-q^\ell}{q^j - q^\ell}. \quad (9.2)$$

A direct factorization gives

$$\lambda_{s,j} = \frac{(-1)^{s-1-j} q^{\binom{s-j}{2}}}{(q; q)_j (q; q)_{s-1-j}} = \frac{(-1)^{s-1-j} q^{\binom{s-j}{2}}}{(q; q)_{s-1}} \begin{bmatrix} s-1 \\ j \end{bmatrix}_q. \quad (9.3)$$

Here  $(q; q)_k = \prod_{h=1}^k (1 - q^h)$ .

Because these are interpolation weights,

$$\sum_{j=0}^{s-1} \lambda_{s,j} q^{kj} = \begin{cases} 1, & k = 0, \\ 0, & 1 \leq k < s. \end{cases} \quad (9.4)$$

For the first un-cancelled power,

$$\sum_{j=0}^{s-1} \lambda_{s,j} q^{sj} = (-1)^{s-1} q^{\binom{s}{2}}. \quad (9.5)$$

*Proof of (9.5).* Interpolate  $f(x) = x^s$  at the  $s$  nodes  $q^j$ . Since  $f(0) = 0$ , the interpolation remainder at 0 gives

$$-\sum_j \lambda_{s,j} q^{sj} = \prod_{j=0}^{s-1} (0 - q^j) = (-1)^s q^{0+1+\dots+(s-1)}.$$

Rearranging proves the identity.  $\square$

**Formal crosswalk.** Set  $r = s-1$  and  $k = j$ . The compiled module `GeometricQBinomialLagrange.lean` defines a numerator that is zero outside  $k \leq r$  and, by definition inside that range, has the reversed form  $(-1)^{r-k} q^{\binom{r-k+1}{2}} \text{gaussianBinomial } q \ r \ (r-k)$ . The theorem `gaussianBinomial_symm` in `FiniteQBinomialCore.lean` is division-free over every commutative semiring. Together with zero extension it yields the global forward formula

$$\text{geometricQBinomialWeightNumerator } q \ r \ k = (-1)^{r-k} q^{\binom{r-k+1}{2}} \text{gaussianBinomial } q \ r \ k$$

as `geometricQBinomialWeightNumerator_eq_forward`. Thus the Gaussian coefficient in (9.3) now agrees literally with the proved forward-index form. The same core supplies `map_gaussianBinomial`, both  $q$ -Pascal recurrences, finite  $q$ -Pochhammer shift and concatenation laws, the full denominator-free product–Gaussian factorial identity, and the finite  $q$ -binomial theorem. Its `reversed_finite_qBinomial_theorem` proves the denominator-free identity with product  $\prod_{h < r} (z - q^{h+1})$ . If  $k \mapsto q^k$  is injective on  $\{0, \dots, r\}$ , `geometricLagrangeWeight_eq_gaussianBinomial_div` identifies the normalized numerator with the Lagrange weight, and `sum_geometricLagrangeWeight_mul_pow_eq_gaussianBinomial` proves for every  $d > 0$

$$\sum_{k=0}^r \lambda_{s,k}(q) (q^k)^d = (-1)^r q^{r(r+1)/2} \text{gaussianBinomial } q \ (d-1) \ r.$$

This discharges the finite weight and all positive residual-moment obligations generically, hence also at  $q = 1/4$  whenever the node injectivity side condition is supplied. At  $q = 1/2$ , the existing `discreteLimitWeight_eq_geometricLagrangeWeight` supplies the half-base Toeplitz specialization.

The finite operator itself is also formalized coefficientwise. `FinitePowerSeriesFilter.lean` proves over every commutative semiring that

$$[X^m] \sum_{k \in S} w_k f(a_k X) = \left( \sum_{k \in S} w_k a_k^m \right) [X^m] f,$$

and proves compatibility with a further rescaling and with coefficient-ring homomorphisms. Its specialization `GeometricPowerSeriesFilter.lean` packages the mass-one degree-zero boundary and every positive Gaussian residual multiplier; in particular `geometricLagrangeSeriesFilter_eq_residual_mk` is an all-index identity of formal power series. The separate module `AnalyticSeriesFilter.lean` now proves the direct series-evaluation identity under Mathlib’s unconditional `Summable` hypothesis: arbitrary finite filters act diagonally and split into exact heads and tails, and the geometric-Lagrange specialization has the full Gaussian tail, including shifted scales. Summability is required only at nonzero-weight samples; a zero-weight node is automatic. The later module `RvachevQBinomialFilter.lean` identifies the named even-moment series with the actual infinite Rvachev product and gives its exact Gaussian filter; at quarter base it discharges node injectivity internally. It does not identify the reciprocal-sinc coefficients of the finite-prefix quotient used here. Conditionally-only boundary summation, that finite-prefix/Bell bridge, and the report’s signs, bounds, asymptotics, and uniform convergence remain open. Consequently the concrete expansion and convergence rate in [Theorem 9.1](#) remain frontier.

## 9.2 Accelerated combinations

Define the signed combination

$$R_{n,s} := \sum_{j=0}^{s-1} \lambda_{s,j} p_{n+j}. \quad (9.6)$$

Although  $R_{n,s}$  is generally not positive, it retains exact mass one because the weights sum to one.

**Theorem 9.1** (*q*-binomial depth acceleration). *Fix  $m \geq 0$  and  $s \geq 1$ . Then*

$$\boxed{R_{n,s}^{(m)} = \text{up}^{(m)} - b_s q^{\binom{s}{2}} q^{sn} \text{up}^{(m+2s)} + O_{m,s}(q^{(s+1)n}), \quad q = \frac{1}{4}.} \quad (9.7)$$

*In particular,*

$$\left\| R_{n,s}^{(m)} - \text{up}^{(m)} \right\|_{\infty} = O_{m,s}(4^{-sn}). \quad (9.8)$$

*Proof.* Apply [Theorem 8.1](#) to  $p_{n+j}^{(m)}$  through order  $s$ . The coefficient of  $\text{up}^{(m+2k)}$  is

$$(-1)^k b_k q^{kn} \sum_{j=0}^{s-1} \lambda_{s,j} q^{kj}.$$

The moment identities (9.4) cancel  $1 \leq k < s$ . At  $k = s$ , use (9.5); the two signs multiply to  $-1$ . The uniform remainder remains of order  $q^{(s+1)n}$  because  $s$  is fixed.  $\square$

### 9.3 Low-order filters

For  $s = 2$ ,

$$(\lambda_{2,0}, \lambda_{2,1}) = \left(-\frac{1}{3}, \frac{4}{3}\right), \quad (9.9)$$

so

$$R_{n,2} = \frac{4p_{n+1} - p_n}{3} = \text{up} - \frac{31}{64800} 16^{-n} \text{up}^{(4)} + O(64^{-n}). \quad (9.10)$$

For  $s = 3$ ,

$$(\lambda_{3,0}, \lambda_{3,1}, \lambda_{3,2}) = \left(\frac{1}{45}, -\frac{4}{9}, \frac{64}{45}\right), \quad (9.11)$$

so

$$R_{n,3} = \frac{p_n - 20p_{n+1} + 64p_{n+2}}{45} = \text{up} - \frac{2347}{2743372800} 64^{-n} \text{up}^{(6)} + O(256^{-n}). \quad (9.12)$$

For  $s = 4$ , the weights are

$$\left(-\frac{1}{2835}, \frac{4}{135}, -\frac{64}{135}, \frac{4096}{2835}\right). \quad (9.13)$$

*Remark 9.2* (Two bases, two mechanisms). The base  $1/2$  in the corpus's exact dyadic formulas is the spatial refinement ratio. The base  $1/4$  here is the leading *error* ratio, forced by symmetry and the first nonzero tail moment. Both yield Gaussian binomial rows, but they act on different variables and solve different problems.

## 10 Exact and high-precision verification

The proofs above are analytic. Exact computation nevertheless provides useful independent checks, especially for signs, powers of two, and index thresholds.

### 10.1 Exact rational checks

At  $x = -3/4$ ,  $\text{up}(x) = F(1/4) = 5/72$ . Direct evaluation of the Thue–Morse spline (2.13) gives:

| $n$ | $p_n(-3/4)$ | $p_n(-3/4) - \text{up}(-3/4)$ | predicted value |
|-----|-------------|-------------------------------|-----------------|
| 3   | 1/16        | -1/144                        | $-(4/9)4^{-3}$  |
| 4   | 13/192      | -1/576                        | $-(4/9)4^{-4}$  |
| 5   | 53/768      | -1/2304                       | $-(4/9)4^{-5}$  |
| 6   | 71/1024     | -1/9216                       | $-(4/9)4^{-6}$  |
| 7   | 853/12288   | -1/36864                      | $-(4/9)4^{-7}$  |
| 8   | 3413/49152  | -1/147456                     | $-(4/9)4^{-8}$  |

For the two-node Gaussian rule, equality is attained at  $x = -15/16$ . Exact rational arithmetic gives:

| $n$ | $G_{n,2}(-15/16) - \text{up}(-15/16)$ | exact prediction      |
|-----|---------------------------------------|-----------------------|
| 5   | -1/1555200                            | $-(4096/6075)16^{-5}$ |
| 6   | -1/24883200                           | $-(4096/6075)16^{-6}$ |
| 7   | -1/398131200                          | $-(4096/6075)16^{-7}$ |
| 8   | -1/6370099200                         | $-(4096/6075)16^{-8}$ |
| 9   | -1/101921587200                       | $-(4096/6075)16^{-9}$ |



The support-preserving rule has the opposite sign at the same point:

$$H_{5,2}(-15/16) - \text{up}(-15/16) = \frac{7}{2073600}, \quad (10.1)$$

followed by an exact factor  $1/16$  at every subsequent level.

## 10.2 Third-order checks

For  $r = 3$ , the nodes involve square roots. Eighty-digit evaluation of the algebraic-node mixtures agrees with the exact constants. For example,

| $n$ | $ G_{n,3}(-63/64) - \text{up}(-63/64) $ | exact value               |
|-----|-----------------------------------------|---------------------------|
| 7   | $1.7627708126288924 \times 10^{-12}$    | $619/351151718400000$     |
| 8   | $2.7543293947326444 \times 10^{-14}$    | $619/22473709977600000$   |
| 9   | $4.3036396792697570 \times 10^{-16}$    | $619/1438317438566400000$ |

The ratio is exactly  $1/64$ . The Lobatto errors at the same point are four times larger to first displayed precision and have positive sign, in agreement with (6.19).

## 10.3 Minimal exact-arithmetic pseudocode

The following code fragment computes the moment sequence, the Thue–Morse prefix, and the low-order constants using rational arithmetic.

Listing 1: Exact arithmetic for moments and dyadic prefix values.

```

from fractions import Fraction
from math import comb, factorial

def tm_sign(j):
    return -1 if j.bit_count() & 1 else 1

def positive_power(x, k):
    return x**k if x > 0 else Fraction(0)

def prefix_density(n, x):
    x = Fraction(x)
    A = Fraction(1) - Fraction(1, 2**n)
    s = Fraction(0)
    for j in range(2**n):
        knot = Fraction(j, 2**(n-1))
        s += tm_sign(j) * positive_power(x + A - knot, n-1)
    return Fraction(2**(n*(n-1)//2), factorial(n-1)) * s

# Given exact cumulants kappa[0..N], moments obey:
def moments_from_cumulants(kappa, N):
    mu = [Fraction(0)] * (N + 1)
    mu[0] = Fraction(1)
    for n in range(1, N + 1):
        mu[n] = sum(Fraction(comb(n-1, k-1))
                    * kappa[k] * mu[n-k]
                    for k in range(1, n + 1))
    return mu

mu2 = Fraction(1, 9)
mu4 = Fraction(19, 675)

```

```

mu6 = Fraction(583, 59535)
h2 = mu4 - mu2**2
h3 = mu6 - mu4**2 / mu2
ell2 = mu2 - mu4
assert h2 == Fraction(32, 2025)
assert h3 == Fraction(19808, 7441875)
assert ell2 == Fraction(56, 675)

```

## 10.4 Verification philosophy

The numerical checks are not used to infer the theorems. Their role is narrower:

1. detect normalization mismatches between  $\sin z/z$  and  $\sin(\pi z)/(\pi z)$ ;
2. confirm the signs of the Gaussian and Lobatto kernels;
3. verify the exact dyadic extremal points;
4. catch missing factorials in Peano-kernel masses;
5. test the geometric ratios  $4^{-r}$  before formalization.

The actual certificates are the positive-kernel factorizations and the Thue–Morse plateau.

## 11 Consequences and additional deductions

### 11.1 Exact dyadic values reappear as extremal certificates

The sharp points are not arbitrary. The order- $d$  derivative plateau is centered at

$$x_d = -1 + 2^{-d},$$

which corresponds under  $F(x) = \text{up}(x - 1)$  to the dyadic argument  $2^{-d}$ . Thus the exact dyadic arithmetic of  $F$  provides the values at precisely the points where the finite convolution errors saturate. For example,

$$\text{up}(-3/4) = F(1/4) = \frac{5}{72}, \quad \text{up}(-15/16) = F(1/16), \quad \text{up}(-63/64) = F(1/64).$$

The approximation theory and the dyadic-value theory therefore meet at a natural sequence of certificate points.

More generally, the derivative relation for the global Fabius extension implies that higher derivatives of  $\text{up}$  at these points reduce to scaled dyadic values. The exact-error constants are consequently compatible with the finite derivative tower already developed in the corpus.

### 11.2 A rational hierarchy despite algebraic nodes

The construction separates into three arithmetic layers:

1. The moments  $\mu_{2k}$  are rational because their cumulants contain only Bernoulli numbers and  $4^k - 1$ .
2. The orthogonal polynomials  $\pi_r$  and  $\chi_{r-1}$  have rational coefficients because their defining Gram systems have rational entries.
3. The nodes and weights are algebraic, but the sharp norm constants  $h_r/(2r)!$  and  $\ell_r/(2r)!$  are rational determinant ratios.

Thus the approximation itself may use algebraic shifts while its certified worst-case error remains a rational multiple of a power of two.

**Corollary 11.1** (Rational sharp constants). *For every  $r \geq 1$  and  $m \geq 0$ , the normalized quantities*

$$2^{2rn - \binom{m+2r+1}{2}} \left\| G_{n,r}^{(m)} - \text{up}^{(m)} \right\|_{\infty} = \frac{h_r}{(2r)!}, \quad (11.1)$$

$$2^{2rn - \binom{m+2r+1}{2}} \left\| H_{n,r}^{(m)} - \text{up}^{(m)} \right\|_{\infty} = \frac{\ell_r}{(2r)!} \quad (11.2)$$

are positive rational numbers throughout the stable range.

### 11.3 A compact-support hierarchy of increasingly smooth splines

For fixed  $r$ , both  $G_{n,r}$  and  $H_{n,r}$  are finite mixtures of copies of  $p_n$ . Hence they are piecewise polynomials of degree  $n - 1$  and belong to  $C^{n-2}(\mathbb{R})$ . The number of mixture components is fixed while  $n$  grows, so the cost of the tail correction is independent of the spline degree. Using (2.13), every component has an explicit finite Thue–Morse representation. For example,

$$\begin{aligned} H_{n,2}(x) = \frac{2^{\binom{n}{2}}}{(n-1)!} \sum_{j=0}^{2^n-1} \varepsilon_j \left[ \frac{1}{18} (x + 2^{-n} + 1 - 2^{-n} - j2^{1-n})_+^{n-1} \right. \\ \left. + \frac{8}{9} (x + 1 - 2^{-n} - j2^{1-n})_+^{n-1} \right. \\ \left. + \frac{1}{18} (x - 2^{-n} + 1 - 2^{-n} - j2^{1-n})_+^{n-1} \right]. \end{aligned} \quad (11.3)$$

Thus the new positive support-exact approximant is still an explicitly computable Thue–Morse spline.

### 11.4 Uniform approximation of integrals and observables

Let  $X_n^G = S_n + 2^{-n}Z_G$  and  $X_n^L = S_n + 2^{-n}Z_L$ . For an arbitrary  $\phi \in C^{2r}(\mathbb{R})$ , the Peano kernels yield

$$|\mathbb{E}\phi(Y) - \mathbb{E}\phi(X_n^G)| \leq 2^{-2rn} \frac{h_r}{(2r)!} \left\| \phi^{(2r)} \right\|_{\infty}, \quad (11.4)$$

$$|\mathbb{E}\phi(X_n^L) - \mathbb{E}\phi(Y)| \leq 2^{-2rn} \frac{\ell_r}{(2r)!} \left\| \phi^{(2r)} \right\|_{\infty}. \quad (11.5)$$

These bounds require no derivative plateau and therefore hold at every  $n$ . The density norm theorems are stronger and more specialized: they exploit the exact shape of  $p_n^{(m+2r)}$  to prove that the generic Peano constant is attained.

### 11.5 Asymptotic bracketing of smooth observables

For a  $2r$ -convex observable, (7.4) gives a certified bracket at every level. In the first nontrivial case  $r = 2$ ,

$$\begin{aligned} \frac{1}{2} \mathbb{E} \left[ \phi \left( S_n - \frac{2^{-n}}{3} \right) + \phi \left( S_n + \frac{2^{-n}}{3} \right) \right] &\leq \mathbb{E}\phi(Y) \\ &\leq \frac{1}{18} \mathbb{E}\phi(S_n - 2^{-n}) + \frac{8}{9} \mathbb{E}\phi(S_n) + \frac{1}{18} \mathbb{E}\phi(S_n + 2^{-n}) \end{aligned} \quad (11.6)$$

whenever  $\phi^{(4)} \geq 0$ . The bracket width is at most

$$\left( \frac{4}{6075} + \frac{7}{2025} \right) 16^{-n} \|\phi^{(4)}\|_{\infty} = \frac{5}{1215} 16^{-n} \|\phi^{(4)}\|_{\infty}. \quad (11.7)$$

This gives a practical positive-probability certificate without evaluating the full infinite convolution.

### 11.6 Why $q = 1/4$ is unavoidable

The law  $\mu$  is centered, so every symmetric tail surrogate automatically matches degree one. The first possible defect is the variance and scales as  $a^2 = 4^{-n}$ . Once the variance is matched, the next possible defect is the fourth moment and scales as  $a^4 = 16^{-n}$ . Thus  $q = 1/4$  is not merely a convenient extrapolation parameter: it is the square of the dyadic spatial scale and the fundamental ratio of the even-moment hierarchy. The Richardson weights in (9.3) are therefore the natural Gaussian-binomial coefficients for symmetric tail errors.

## 12 Lean formalization roadmap

The proofs were chosen to decompose into reusable lemmas that fit the existing formal development. A plausible formalization order is as follows.

### 12.1 Phase I: probability measures and scaling

1. Define the up-law  $\mu$  as the distribution of the established random series and define its scaled pushforward  $\mu_a$ .
2. Prove the exact tail split

$$\mu = \mathcal{L}(S_n) * \mu_{2^{-n}}.$$

3. Connect the density of  $S_n$  to the existing finite box-spline object  $p_n$ .

Most of this layer should be available from the current infinite-convolution and probability modules.

### 12.2 Phase II: the ordinary-prefix potential

1. Define

$$J(t) = \frac{1}{2} \int (|t - y| - |t|) d\mu(y).$$

2. Prove  $J \geq 0$  by Jensen, compact support by the support of  $\mu$ , and evenness by symmetry.
3. Formalize  $J'' = \mu - \delta_0$  either in distributions or through a twice-integrated identity against compactly supported test functions.
4. Prove the mass identity  $\int J = \mu_2/2$ .
5. Combine the existing derivative plateau with a nonnegative-kernel convolution lemma to obtain [Theorem 4.2](#).

The only delicate issue is the a.e. nature of the finite spline derivative at knots. One clean route is to state the plateau for the  $L^\infty$  representative and then use continuity of the final convolution.

### 12.3 Phase III: abstract positive quadrature

A reusable finite-dimensional module can avoid committing immediately to explicit roots.

1. Develop monic orthogonal polynomials for a compactly supported probability measure whose moment functional is strictly positive on the square of every nonzero polynomial; for a fixed finite algebraic tranche, one may instead assume the required leading Hankel minors are nonzero.
2. Prove the Gram determinant formula  $h_r = \Delta_{r+1}/\Delta_r$ .
3. Import or prove Gaussian quadrature existence, positivity, and exactness.
4. Formalize Hermite interpolation and the sign of the Gaussian remainder.
5. Define the Peano kernel by the truncated-power formula and prove nonnegativity and mass.
6. Repeat with the weighted measure  $(1 - x^2)\mu$  for Lobatto quadrature, including the positive-weight proof from isolating nonnegative polynomials.

The generic finite-algebraic portions of the first two items are now supplied by `FiniteMomentGram.lean` and `GramStieltjes.lean`: the latter gives the unique normalized monic polynomial under a nonzero leading Hankel determinant and proves  $h_r = \Delta_{r+1}/\Delta_r$  as a self-pairing identity. Those generic modules do not connect the hypotheses to a measure or prove positivity. `FiniteMomentJacobi.lean` also proves the generic finite three-term recurrence, including its degree-zero base equation, and the determinant formula for its subdiagonal coefficient. Separately, the up-law modules now prove strict positivity, construct the normalized orthogonal polynomials, and establish their symmetric three-term recurrence, while `OrthogonalPolynomialGramBridge.lean` proves that their objects are exactly the generic construction specialized to upMoment. The positive Gaussian/Lobatto construction, root and weight theory, Hermite-remainder sign, and Peano-kernel tasks in this phase remain open.

The density error theorem then becomes a generic lemma:

If a signed compactly supported measure  $\sigma$  has a nonnegative  $2r$ th-order potential  $K$  of mass  $M$ , and if a density  $p$  has a derivative plateau of height  $C$  covering  $\text{supp } K$ , then

$$\|p * \sigma\|_{\infty} = CM.$$

This abstraction would also subsume the existing repeated-integration theorem and the ordinary-prefix theorem.

### 12.4 Phase IV: exact low-order certificates

The cases  $r = 2, 3$  require only rational moment arithmetic plus elementary algebra:

1. verify the moments in (3.6);
2. prove the displayed orthogonal polynomials;
3. verify nodes and weights by moment exactness;
4. normalize the rational constants;
5. instantiate the generic plateau theorem.

These cases can be formalized before a fully general quadrature library is complete.

## 12.5 Phase V: reciprocal sinc and $q$ -binomial acceleration

1. Define the formal reciprocal coefficients  $b_j$  from the power series of  $1/\Phi$ .
2. Prove the logarithmic coefficient formula (8.3) from Euler’s sine product.
3. Establish the fixed-order Fourier majorant used in [Theorem 8.1](#).
4. Use the compiled `GeometricQBinomialLagrange.lean` bridge for the forward Gaussian weight formula and all positive residual moments, and `GeometricPowerSeriesFilter.lean` for their complete coefficientwise action on a formal series.
5. Use `AnalyticSeriesFilter.lean` for the exact abstract unconditionally summable Gaussian tail and `RvachevQBinomialFilter.lean` for its actual infinite-product specialization; then separately identify the finite-prefix quotient’s reciprocal-sinc coefficients and prove the controlled uniform remainder in [Theorem 9.1](#).

The finite algebraic and formal-series steps are now available at arbitrary ratio under node injectivity, and the actual infinite Rvachev product has the corresponding entire filter, assumption-free at  $q = 1/4$ . The remaining steps are the finite-prefix reciprocal-sinc coefficient bridge and its combination with controlled uniform remainders. The  $q = 1/2$  specialization is the existing half-base Toeplitz bridge.

## 13 Open problems and next frontier

The present results close two gaps but open a larger approximation theory for self-similar smooth laws.

### 13.1 Arithmetic of the Hankel constants

The sequences

$$h_r = \frac{\Delta_{r+1}}{\Delta_r}, \quad \ell_r = \frac{\Lambda_r}{\Lambda_{r-1}} \quad (13.1)$$

are positive rationals built from Bernoulli numbers and Mersenne factors. Their reduced denominators, odd-prime valuations, and possible normalized integrality properties appear unexplored. Natural questions include:

1. Is there a useful factorization of  $\Delta_r$  or  $\Lambda_r$  into predictable Mersenne components?
2. Do suitable powers of two and products  $\prod_{j \leq r} (4^j - 1)$  clear all denominators sharply?
3. Are the numerators related to known moment or continued-fraction sequences?

A Jacobi continued fraction for  $M(t)$  or for the Cauchy transform of  $\mu$  could encode these determinants more efficiently.

### 13.2 Asymptotics as the quadrature order grows

The fixed- $r$ , large- $n$  theory is exact. A different regime lets  $r \rightarrow \infty$ . The up density is positive in the interior but flat at the endpoints, so the zero distribution of  $\pi_r$  should be governed at first order by logarithmic potential theory on  $[-1, 1]$ , while endpoint spacing and Christoffel functions may feel the extreme flatness of the weight. Quantitative questions include:

1. asymptotics of  $h_r$  and  $\ell_r$ ;
2. the smallest distance of Gaussian nodes from  $\pm 1$ ;
3. comparison of Gaussian and Lobatto constants;
4. uniform algorithms when  $r = r(n)$  grows with the convolution depth.

### 13.3 Nested positive tail rules

Gaussian rules are not generally nested. Lobatto rules share endpoints but not necessarily interior nodes. A Fabius-specific Gauss–Kronrod or Patterson-type construction could provide nested positive tail laws, allowing one to reuse shifted prefix evaluations. The relevant optimization problem is:

Construct nested symmetric positive measures  $Q^{(1)} \subset Q^{(2)} \subset \dots$  on  $[-1, 1]$  that maximize moment exactness for the up-law, while keeping exact rational error certificates whenever possible.

The strong dyadic arithmetic of the moments may make this more tractable than for a generic weight.

### 13.4 Positive acceleration versus signed Richardson filters

The  $q$ -binomial Richardson rule reaches order  $4^{-sn}$  with a signed linear combination of different depths. Gaussian/Lobatto rules reach the same order using positive mixtures at one depth. It would be valuable to compare:

1. total variation of the signed Richardson coefficients;
2. number and location of positive quadrature atoms;
3. exact support;
4. numerical stability near flat endpoints;
5. compatibility with exact rational evaluation.

A hybrid method could use a low-order positive tail correction followed by a short  $q = 1/4$  extrapolation, reducing both the atom count and the signed coefficient growth.

### 13.5 Fourier-decay interaction

The recent Fourier-decay reports analyze

$$\Phi(\xi) = \prod_{k \geq 1} \text{sinc}(2^{-k}\xi)$$

along typical and exceptional rays. The corrected approximants have transforms

$$\widehat{G}_{n,r}(\xi) = P_n(\xi) \widehat{Q}_r(2^{-n}\xi), \quad (13.2)$$

$$\widehat{H}_{n,r}(\xi) = P_n(\xi) \widehat{L}_r(2^{-n}\xi). \quad (13.3)$$

Their low-frequency errors are controlled by moment matching, while their high-frequency behavior is inherited from the finite product  $P_n$ . A two-parameter analysis in  $n$  and  $\xi$  could determine the largest frequency window on which the relative error remains small. This is especially relevant because  $\Phi$  has many exact zeros, so global relative error requires zero-aware normalization.

### 13.6 Optimal support-exact positive rules

The Lobatto rule is canonical among interpolatory rules with both endpoints fixed and degree  $2r - 1$ . Other positive support-exact tail measures with more atoms may reduce the sharp  $C^m$  constant while keeping the same order. One can pose a finite moment problem:

$$\min_Q |\mu_{2r} - Q[x^{2r}]| \quad (13.4)$$

subject to positivity, support in  $[-1, 1]$ , inclusion of the endpoints, and matching moments through degree  $2r - 1$ . Understanding the extreme points of this feasible set may yield better support-exact constants than Lobatto at the cost of additional atoms.

### 13.7 Beyond the dyadic scale

The method is not intrinsically restricted to base two. For a self-similar random series

$$Y_b = \sum_{k \geq 1} b^{-k} U_k, \quad b > 1,$$

when the support geometry and finite-prefix derivative plateaux are controlled, moment-matched quadrature predicts rates  $b^{-2rn}$ . What is special about  $b = 2$  is the exact unique-subset-sum structure, which turns the finite spline signs into Thue–Morse signs and gives the saturated derivative constants in closed form. A classification of bases or weight sequences admitting an equally sharp plateau theorem would generalize the entire framework.

## 14 Conclusion

The finite dyadic prefix, the infinite sinc product, and the Thue–Morse sequence are three descriptions of one object. The new approximation theory developed here makes that unity explicit.

The omitted self-similar tail has a nonnegative second-order potential of mass  $1/18$ . Convolving this potential against a saturated Thue–Morse derivative plateau gives the exact ordinary-prefix law

$$\left\| p_n^{(m)} - \text{up}^{(m)} \right\|_{\infty} = \frac{2^{\binom{m+3}{2}}}{18} 4^{-n}.$$

Replacing the omitted tail by Gaussian quadrature raises the order from  $4^{-n}$  to  $4^{-rn}$  while preserving positivity. Replacing it by Gauss–Lobatto quadrature preserves both positivity and the exact support. The sharp constants are rational orthogonal-polynomial norms and are attained at explicit dyadic points because the finite prefix has exact derivative plateaux.

On the Fourier side, the reciprocal sinc product generates the complete correction series. Its powers are geometric in  $q = 1/4$ , and geometric Lagrange interpolation produces Gaussian  $q$ -binomial Richardson weights. Hence the four central themes of the corpus meet in one coherent mechanism:

$$\begin{array}{lcl} \text{dyadic random tail} & \xrightarrow{\text{moment quadrature}} & \text{positive high-order spline,} \\ \text{infinite sinc product} & \xrightarrow{\text{reciprocal series}} & \text{even-power error expansion,} \\ \text{Thue–Morse prefix sums} & \xrightarrow{\text{plateau saturation}} & \text{exact } C^m \text{ constants,} \\ q = \frac{1}{4} \text{ geometric interpolation} & \xrightarrow{\text{Gaussian binomials}} & \text{depth acceleration.} \end{array} \quad (14.1)$$

The most immediate next step is formalization of the ordinary-prefix potential and the explicit  $r = 2$  Gaussian and Lobatto schemes. These cases require little new infrastructure, resolve a documented frontier gap, and provide compact theorem statements with exact rational certificates.

## A Coefficient and constant tables

### A.1 Moments and cumulants

| $m$ | $\kappa_{2m}$     | $\mu_{2m}$          | comment                              |
|-----|-------------------|---------------------|--------------------------------------|
| 1   | $\frac{1}{9}$     | $\frac{1}{9}$       | variance                             |
| 2   | $-\frac{2}{225}$  | $\frac{19}{675}$    | first non-Gaussian correction        |
| 3   | $\frac{16}{3969}$ | $\frac{583}{59535}$ | used by the three-node Gaussian rule |



| $m$ | $\kappa_{2m}$              | $\mu_{2m}$                               | comment                              |
|-----|----------------------------|------------------------------------------|--------------------------------------|
| 4   | $-\frac{16}{3825}$         | $\frac{132809}{32531625}$                | determines degree-eight data         |
| 5   | $\frac{256}{33759}$        | $\frac{46840699}{24405225075}$           | rational Mersenne structure          |
| 6   | $-\frac{353792}{16769025}$ | $\frac{4068990560161}{4133856862760625}$ | contains the Bernoulli numerator 691 |

The cumulant entries follow directly from (3.4); the displayed higher fractions are included mainly as exact-arithmetic checkpoints.

## A.2 Reciprocal-sinc coefficients

| $j$ | $\gamma_j$ in $\log(1/\Phi)$ | $b_j$ in $1/\Phi$            |
|-----|------------------------------|------------------------------|
| 0   | —                            | 1                            |
| 1   | 1/18                         | 1/18                         |
| 2   | 1/2700                       | 31/16200                     |
| 3   | 1/178605                     | 2347/42865200                |
| 4   | 1/9639000                    | 1904369/1311675120000        |
| 5   | 1/478533825                  | 3309760579/88561680752160000 |

## A.3 Quadrature constants

| $r$ | Gaussian polynomial $\pi_r$ | Lobatto interior polynomial $\chi_{r-1}$ | $h_r/(2r)!$    | $\ell_r/(2r)!$ |
|-----|-----------------------------|------------------------------------------|----------------|----------------|
| 1   | $x$                         | 1                                        | 1/18           | 4/9            |
| 2   | $x^2 - 1/9$                 | $x$                                      | 4/6075         | 7/2025         |
| 3   | $x^3 - (19/75)x$            | $x^2 - 7/75$                             | 1238/334884375 | 4936/334884375 |

For  $r = 1$ , the Lobatto convention has the two endpoint rule, exact through degree one. It is not the ordinary prefix; it corresponds to a first-order support-exact correction. The higher-order statements in the body begin with the practically important  $r = 2$  case.

# B Derivation of the low-order quadrature rules

## B.1 Gaussian order two

Symmetry gives  $\pi_2 = x^2 - c$ . Orthogonality to 1 gives  $c = \mu_2 = 1/9$ . The roots are  $\pm 1/3$ , and symmetry plus total mass forces weights 1/2. The norm is

$$h_2 = \mu_4 - \mu_2^2 = \frac{19}{675} - \frac{1}{81} = \frac{32}{2025}.$$

## B.2 Gaussian order three

Write  $\pi_3 = x^3 - cx$ . Orthogonality to  $x$  gives

$$c = \frac{\mu_4}{\mu_2} = \frac{19}{75}.$$

If the two outer weights are  $w$  and the central weight is  $1 - 2w$ , moment matching gives

$$2w\frac{19}{75} = \frac{1}{9}, \quad w = \frac{25}{114}, \quad 1 - 2w = \frac{32}{57}.$$

Finally,

$$h_3 = \mu_6 - \frac{\mu_4^2}{\mu_2} = \frac{19808}{7441875}.$$

### B.3 Lobatto order two

The interior polynomial is  $\chi_1 = x$ . Symmetric weights  $(a, b, a)$  at  $(-1, 0, 1)$  obey

$$2a + b = 1, \quad 2a = \mu_2 = \frac{1}{9}.$$

Thus  $a = 1/18$ ,  $b = 8/9$ , and

$$\ell_2 = \int (1 - x^2)x^2 d\mu = \mu_2 - \mu_4 = \frac{56}{675}.$$

### B.4 Lobatto order three

Write  $\chi_2 = x^2 - c$ . Weighted orthogonality to 1 gives

$$0 = \int (1 - x^2)(x^2 - c) d\mu = \mu_2 - \mu_4 - c(1 - \mu_2),$$

so  $c = 7/75$ . Let  $a$  be each endpoint weight and  $b$  each interior weight. Then

$$a + b = \frac{1}{2}, \quad 2a + 2b\frac{7}{75} = \frac{1}{9},$$

which gives  $a = 1/102$  and  $b = 25/51$ . Direct substitution yields

$$\ell_3 = \int (1 - x^2) \left(x^2 - \frac{7}{75}\right)^2 d\mu = \frac{78976}{7441875}.$$

## C A reusable abstract sharpness lemma

The ordinary, Gaussian, Lobatto, and repeated-integration laws all fit one template.

**Theorem C.1** (Positive-kernel plateau principle). *Let  $K \in L^1(\mathbb{R})$  satisfy  $K \geq 0$ ,  $\text{supp } K \subseteq [-a, a]$ , and  $\int K = M$ . Let  $g$  be essentially bounded and suppose*

$$\|g\|_\infty = C, \quad g = C \quad \text{a.e. on } [x_0 - a, x_0 + a].$$

Then

$$\|g * K\|_\infty = CM, \quad (g * K)(x_0) = CM. \quad (\text{C.1})$$

*If an approximation error can be written as  $\pm g * K$ , its uniform norm and a signed extremum are therefore exact.*

*Proof.* Young's inequality gives  $\|g * K\|_\infty \leq CM$ . At  $x_0$ , the function  $g(x_0 - t)$  equals  $C$  for almost every  $t$  in the support of  $K$ , so the convolution equals  $C \int K = CM$ .  $\square$

In the applications,

$$g = p_n^{(m+2r)}, \quad C = 2^{\binom{m+2r+1}{2}}, \quad a = 2^{-n},$$

and  $K$  is a scaled Peano kernel.

## D Corpus coverage map

The report’s novelty comparison used the following thematic map. It is included to make explicit what was treated as established and therefore not redeveloped at length.

| Corpus component                           | Material treated as established baseline                                                                                                                                                                        |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Core Fabius/up exposition                  | Existence, uniqueness, smoothness, support, symmetry, probability representation, moments, Fourier product, Poisson summation, exact dyadic values, $q$ -binomial formulas, and corrected endpoint asymptotics. |
| Lean walkthrough                           | Trusted theorem boundary, module dependencies, executable exact evaluator, and proof architecture.                                                                                                              |
| Glossary                                   | Terminology normalization across the entire project.                                                                                                                                                            |
| Dyadic formula and representation papers   | Highest-bit recursion, global signed extension, coefficient formulas, binary telescopes, and alternative series.                                                                                                |
| Dyadic asymptotic bridge                   | Exact coefficient extraction, fixed-numerator asymptotics, Lambert coordinate, periodic fluctuations, and Bromwich collapse.                                                                                    |
| Repeated integration                       | Exact finite box spline with duplicated smallest width, positive second-order Peano kernel, sharp $8/(9 \cdot 4^n)$ law, derivative analogues, and its own all-orders expansion.                                |
| Thue–Morse convergence and formula atlases | Prefix summation, Prouhet cancellation, product/generating formulas, finite differences, parity identities, and $q$ -calculus.                                                                                  |
| $q$ -special-functions report              | $q$ -Pochhammer symbols, Gaussian binomials, basic hypergeometric structure, and their role in exact Fabius arithmetic.                                                                                         |
| Fourier-decay reports                      | Exact dyadic factorization, typical-ray exponent, fluctuations, exceptional orbits, covariance, and law-of-the-iterated-logarithm behavior of the sinc product.                                                 |
| Non-formalized frontier volume             | Consolidated speculative results and explicit gap register; used to identify the ordinary-prefix and moment-matched-tail questions.                                                                             |
| Archive                                    | Historical variants, superseded drafts, and provenance; duplicate statements were not counted twice.                                                                                                            |

## E Notation index

| Symbol               | Meaning                                                                              |
|----------------------|--------------------------------------------------------------------------------------|
| up                   | Rvachev's even compactly supported up-function, normalized as a probability density. |
| $F$                  | Bounded Fabius function on $[0, 1]$ , with $F(x) = \text{up}(x - 1)$ .               |
| $\mu$                | Probability law with density up.                                                     |
| $Y$                  | Random series $\sum_{k \geq 1} 2^{-k} U_k$ with law $\mu$ .                          |
| $S_n$                | Finite prefix $\sum_{k=1}^n 2^{-k} U_k$ .                                            |
| $p_n$                | Density of $S_n$ , a finite dyadic box spline.                                       |
| $\Phi$               | Infinite sinc product $\widehat{\text{up}}$ .                                        |
| $P_n$                | Finite sinc product $\widehat{p}_n$ .                                                |
| $\varepsilon_j$      | Signed Thue–Morse value $(-1)^{s_2(j)}$ .                                            |
| $C_r$                | Exact derivative norm $2^{\binom{r+1}{2}}$ .                                         |
| $x_r$                | Plateau center $-1 + 2^{-r}$ .                                                       |
| $J$                  | Positive second-order potential for $\mu - \delta_0$ .                               |
| $\pi_r, h_r$         | Monic orthogonal polynomial for $\mu$ and its squared norm.                          |
| $Q_r, G_{n,r}$       | Gaussian tail rule and corrected finite spline.                                      |
| $\chi_{r-1}, \ell_r$ | Monic orthogonal polynomial for $(1 - x^2)\mu$ and its squared weighted norm.        |
| $L_r, H_{n,r}$       | Gauss–Lobatto tail rule and support-exact corrected spline.                          |
| $b_j$                | Coefficients of the reciprocal sinc product $1/\Phi$ .                               |
| $R_{n,s}$            | $q = 1/4$ Richardson combination of consecutive prefix depths.                       |

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## Part II

# Arithmetic Spectra of the Rvachev Sinc Product

### Abstract

This report starts from every `*.tex` document under `Analysis/FabiusFunction/docs` at the frozen commit named above. The corpus contains the principal Fabius–Rvachev exposition, its historical revisions, a Lean walkthrough and glossary, exact dyadic and  $q$ -calculus studies, repeated-integration and spline reports, endpoint asymptotics, a large Thue–Morse formula atlas, a sequence of Fourier-decay audits, and primary or transcribed papers. Repeated revisions were read as one mathematical lineage; the latest corrected statement was used whenever sibling drafts disagreed.

The principal synthesis is that the canonical-product zero multiplicities of the Fourier transform

$$\Phi(z) = \widehat{\text{up}}(z) = \prod_{h \geq 0} \text{sinc}\left(\frac{\pi z}{2^h}\right) = \prod_{n \geq 1} \left(1 - \frac{z^2}{n^2}\right)^{1+\nu_2(n)}$$

form an arithmetic spectrum. Its multiplicity function  $a(n) = 1 + \nu_2(n)$  has Dirichlet series  $\zeta(s)/(1 - 2^{-s})$ , exact counting law  $\sum_{n \leq N} a(n) = 2N - s_2(N)$ , and a dense dyadic natural boundary in its ordinary generating function. These elementary facts organize several results which do not appear as a theorem package in the pinned corpus.

First, the sign on the  $N$ th Fourier lobe is exactly the signed Thue–Morse bit  $(-1)^{s_2(N)}$ . This pattern had been observed before; here it is made a direct corollary of the exact zero count and is converted into an absolutely convergent Thue–Morse lobe decomposition of  $\text{up}$  and all its derivatives. Second, the zero multiplicity gives an exact Strang–Fix hierarchy: on the lattice  $N^{-1}\mathbb{Z}$ , polynomially weighted periodizations are constant through degree  $\nu_2(N)$ , and the next degree fails. A reciprocal-moment Appell correction then reproduces every polynomial of that degree exactly. Thus the approximation order of the translate system is controlled not by the size of  $N$ , but by its 2-adic valuation.

Third, the zero spectrum has a spectral zeta function with “complex dimensions”  $2\pi i k / \log 2$ . Mellin inversion gives a superalgebraically accurate expansion of  $\log \Phi(iy)$  with the same Gamma–zeta Fourier coefficients that occur in the repository’s corrected endpoint asymptotics. This supplies a new structural explanation of that periodic fluctuation: it is the residue trace of the dyadic zero spectrum. Finite sinc products replace the geometric denominator by the  $q$ -integer  $[J]_{2^{-s}}$ ; the imaginary-axis poles are absent at every finite stage and emerge through an explicit pole-condensation limit. Gaussian  $q$ -binomial coefficients appear as the algebra of independently selecting spectral layers, while the common-zero statistics form a hierarchy

$$\sum_{n \geq 1} \binom{1 + \nu_2(n)}{r} n^{-s} = \zeta(s) \frac{2^{-(r-1)s}}{(1 - 2^{-s})^r}.$$

A heat-trace expansion and a base- $b$  extension complete the report.

Claims marked `NEW-TO-SNAPSHOT` mean only that no counterpart was found in the frozen TeX corpus after targeted collision searches. They are proved here, but no claim of worldwide historical priority is made. In particular, Rvachev’s 1977 paper on approximation by  $\text{up}$  and the broader Russian-language atomic-function literature require a separate priority audit.

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**Status convention.** `CORPUS` denotes an input already developed in the repository or cited primary literature. `DERIVED` denotes a short consequence included to make the argument self-contained. `NEW-TO-SNAPSHOT` denotes a theorem not located in the pinned TeX snapshot. `PRIOR OBSERVATION` denotes a phenomenon recorded before but sharpened or reorganized here. None of the new-to-snapshot results is represented as already formalized in Lean.

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# 1 Audit boundary and research method

## 1.1 Why the snapshot is pinned

The default branch changed while this audit was in progress. To prevent a moving target, all corpus statements in this report refer to commit `0b1db2f91321b6575be9a712dc1025e88af318db`. Its commit time is 27 August 2026, 18:19:55 UTC. GitHub’s code index reported 54 TeX files under `Analysis/FabiusFunction/docs` at audit time; because large generated and consolidated sources are not always returned reliably by text search, the recursive Git tree, rather than the index, was used as the authoritative inventory.

The documents fall into the following overlapping families.

| Family                   | Mathematical role                                                                                                                                                                            |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary exposition       | Definitions, probability model, functional equations, canonical sinc product, exact dyadic values, derivative identities, partition of unity, and the corrected global asymptotic framework. |
| Revision archive         | Earlier expositions, supplements, glossaries, and walkthroughs; useful for detecting abandoned normalizations and claims corrected in later waves.                                           |
| Dyadic and $q$ -calculus | Moment recurrences, Appell blocks, Gaussian-binomial stencils, $q$ -Pochhammer normalizations, exact dyadic arithmetic, Toeplitz limits, and accelerated spline approximants.                |
| Repeated integration     | Finite nonuniform box splines, exact Thue–Morse truncated powers, Peano kernels, and sharp $4^{-n}$ convergence laws.                                                                        |
| Endpoint asymptotics     | Correct Lambert phase, all-order saddle expansion, explicit low-order coefficients, and the Gamma–zeta Fourier law for the log-periodic term.                                                |
| Thue–Morse atlases       | Digital, valuative, Pascal, automata, Mahler, finite-difference, rarefaction, Mellin–Dirichlet, word-combinatorial, and spectral representations.                                            |
| Fourier-decay audits     | Successive corrections separating geometric, arithmetic, RMS, and envelope exponents for the real-axis infinite sinc product.                                                                |
| Primary papers           | Arias de Reyna’s analytic and arithmetic papers, lacunary-product work, Kronecker-sequence work, and OCR-corrected or transcribed source material.                                           |

## 1.2 Collision policy

A formula is not called new merely because it admits a different notation. The following attractive directions were deliberately excluded from the novelty claims.

- The all-order endpoint coefficients, including explicit  $A_2$  and  $A_3$ , are already in the repository’s small-argument asymptotics study.
- Moment deconvolution and Richardson acceleration already occur in archived dyadic and convergence reports, including a half-base Gaussian-binomial filter across spline degrees.
- Exact zero multiplicity  $1 + \nu_2(n)$  is already formalized in the Lean canonical-product module.
- The sign pattern of the Fourier lobes was stated in Vladimir Reshetnikov’s 2018 Math StackExchange question, and the answer there essentially proves uniqueness of the critical point in every lobe from the logarithmic derivative.

- Ordinary partition of unity is classical for up and appears in the primary literature. The new-to-snapshot statement below is the sharp higher weighted hierarchy and its exact 2-adic stopping order.

The strategy was therefore to promote the zero multiplicity from an isolated canonical-product fact to a spectrum and ask what its standard transforms know. The answer simultaneously touches Dirichlet series, Mellin residues, automatic sequences, shift-invariant approximation, and finite  $q$ -calculus.

## 2 Normalization and established inputs

### 2.1 The centered random series

Let  $U_1, U_2, \dots$  be independent and uniform on  $[-1, 1]$ , and set

$$Y = \sum_{k=1}^{\infty} 2^{-k} U_k. \quad (2.1)$$

Then  $Y$  is supported on  $[-1, 1]$  and has the even density up. Under the Fourier convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx, \quad (2.2)$$

independence gives

$$\Phi(\xi) := \widehat{\text{up}}(\xi) = \prod_{h=0}^{\infty} \text{sinc}\left(\frac{\pi \xi}{2^h}\right), \quad \text{sinc } z = \frac{\sin z}{z}. \quad (2.3)$$

The convergence is locally uniform on  $\mathbb{C}$ , so  $\Phi$  is an even entire function. The density is  $C^\infty$  and compactly supported; consequently  $\Phi$  is rapidly decreasing on the real axis, although its sharp pointwise behavior is highly nonuniform.

The bounded Fabius variable is

$$X = \frac{Y+1}{2} = \sum_{k=1}^{\infty} 2^{-k} V_k, \quad V_k \sim \text{Unif}[0, 1], \quad (2.4)$$

with distribution function  $F$ . This affine relation will be the bridge between the imaginary-axis growth of  $\Phi$  and the small-argument asymptotics of  $F$ .

### 2.2 Canonical product and exact orders

Euler's product for sine and an absolutely convergent rearrangement give

$$\Phi(z) = \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right)^{a(n)}, \quad a(n) = 1 + \nu_2(n). \quad (2.5)$$

Thus

$$\text{ord}_{z=n} \Phi = a(n), \quad n \in \mathbb{Z} \setminus \{0\}. \quad (2.6)$$

This identity is an established corpus input. Everything in [Sections 3, 6, 8 and 9](#) is driven by taking it seriously as spectral data rather than merely as a description of zeros.

The first centered moments needed later are

$$\mu_0 = 1, \quad \mu_1 = 0, \quad \mu_2 = \frac{1}{9}, \quad \mu_3 = 0, \quad \mu_4 = \frac{19}{675}, \quad (2.7)$$

where  $\mu_r = \int x^r \text{up}(x) dx$ . They follow either from the random series or from the repository's moment recurrence.



### 3 The arithmetic zero spectrum

#### 3.1 Equivalent descriptions

**Theorem 3.1** (Arithmetic spectrum package NEW-TO-SNAPSHOT). *For  $n \geq 1$ , let  $a(n) = 1 + \nu_2(n)$ . Then:*

$$a(n) = \sum_{h \geq 0} \mathbf{1}_{2^h | n}, \quad (3.1)$$

$$\sum_{n \geq 1} \frac{a(n)}{n^s} = \frac{\zeta(s)}{1 - 2^{-s}}, \quad \Re s > 1, \quad (3.2)$$

$$A(N) := \sum_{n \leq N} a(n) = \sum_{h \geq 0} \left\lfloor \frac{N}{2^h} \right\rfloor = 2N - s_2(N), \quad (3.3)$$

$$\#\{n \leq N : a(n) \geq r\} = \left\lfloor \frac{N}{2^{r-1}} \right\rfloor, \quad (3.4)$$

$$\#\{n \leq N : a(n) = r\} = \left\lfloor \frac{N}{2^{r-1}} \right\rfloor - \left\lfloor \frac{N}{2^r} \right\rfloor. \quad (3.5)$$

Moreover  $a$  is multiplicative and satisfies

$$a(2n) = a(n) + 1, \quad a(2n + 1) = 1. \quad (3.6)$$

Its limiting value distribution is geometric:

$$\lim_{N \rightarrow \infty} \frac{1}{N} \#\{n \leq N : a(n) = r\} = 2^{-r}, \quad r \geq 1, \quad (3.7)$$

so the limiting mean and variance are both 2.

*Proof.* The number of nonnegative  $h$  for which  $2^h \mid n$  is exactly  $1 + \nu_2(n)$ , proving (3.1). Absolute convergence permits summation by layers:

$$\sum_{n \geq 1} \frac{a(n)}{n^s} = \sum_{h \geq 0} \sum_{2^h | n} n^{-s} = \zeta(s) \sum_{h \geq 0} 2^{-hs}.$$

The same layer count truncated at  $N$  gives the first expression for  $A(N)$ . Legendre's digit-sum identity  $\sum_{h \geq 1} \lfloor N/2^h \rfloor = N - s_2(N)$  gives the last one. The inequalities  $a(n) \geq r$  and  $2^{r-1} \mid n$  are equivalent, giving (3.4); subtraction gives (3.5). Multiplicativity follows because only the 2-part contributes: if  $(m, n) = 1$ , at most one is even, and  $a(mn) = a(m)a(n)$ . The recursion and limiting law are immediate. The geometric mean is  $\sum r 2^{-r} = 2$  and its variance is 2.  $\square$

*Remark 3.2.* The average zero multiplicity is therefore 2, while exceptional powers of two have unbounded multiplicity. This is the first indication that average real-axis behavior and extremal dyadic behavior should not be conflated—precisely the lesson reached independently by the repository's Fourier-decay audit sequence.

#### 3.2 Ordinary generating function and natural boundary

**Proposition 3.3** (Lambert series and a dyadic natural boundary NEW-TO-SNAPSHOT). *For  $|z| < 1$ ,*

$$\mathcal{A}(z) := \sum_{n \geq 1} a(n) z^n = \sum_{h \geq 0} \frac{z^{2^h}}{1 - z^{2^h}}, \quad (3.8)$$

with Mahler recursion

$$\mathcal{A}(z) = \frac{z}{1-z} + \mathcal{A}(z^2). \quad (3.9)$$

Every root of unity of order a power of two is a radial singularity. Since these roots are dense on the unit circle,  $|z| = 1$  is a natural boundary.

*Proof.* Expand (3.1) and sum geometric series. Splitting off  $h = 0$  and shifting the remaining index gives (3.9). If  $\omega^{2^m} = 1$  and  $z = r\omega$  with  $r \uparrow 1$ , the  $h = m$  summand is  $r^{2^m}/(1 - r^{2^m}) \rightarrow +\infty$ . Thus analytic continuation across any arc is impossible.  $\square$

### 3.3 Spectral traces and the Taylor law

**Corollary 3.4** (Even spectral traces NEW-TO-SNAPSHOT). *For every integer  $m \geq 1$ ,*

$$\sum_{n \geq 1} \frac{a(n)}{n^{2m}} = \frac{\zeta(2m)}{1 - 4^{-m}}, \quad (3.10)$$

and, for  $|z| < 1$ ,

$$\log \Phi(z) = - \sum_{m=1}^{\infty} \frac{\zeta(2m)}{m(1 - 4^{-m})} z^{2m}. \quad (3.11)$$

*Proof.* The first formula is (3.2) at  $s = 2m$ . Expanding  $\log(1 - z^2/n^2)$  in the canonical product and interchanging the absolutely convergent sums gives the second.  $\square$

The logarithmic derivative is the corresponding resolvent trace:

$$\frac{\Phi'(z)}{\Phi(z)} = \sum_{n \geq 1} a(n) \left( \frac{1}{z-n} + \frac{1}{z+n} \right) = -2z \sum_{n \geq 1} \frac{a(n)}{n^2 - z^2}. \quad (3.12)$$

This identity will control the geometry of every real lobe.

## 4 Thue–Morse signs from zero counting

### 4.1 The neighboring-bit identity

Let

$$\varepsilon_n = (-1)^{s_2(n)}, \quad n \geq 0, \quad (4.1)$$

be the signed Thue–Morse sequence.

**Lemma 4.1** (Multiplicity parity is a Thue–Morse discrete derivative). *For every  $n \geq 1$ ,*

$$(-1)^{a(n)} = \varepsilon_{n-1} \varepsilon_n. \quad (4.2)$$

*Proof.* Adding one to  $n - 1$  erases  $\nu_2(n)$  trailing ones and creates one new one, so  $s_2(n) = s_2(n - 1) + 1 - \nu_2(n)$ . Hence  $s_2(n - 1) + s_2(n) \equiv 1 + \nu_2(n) = a(n) \pmod{2}$ .  $\square$

### 4.2 Exact lobe signs

**Theorem 4.2** (Thue–Morse lobe-sign theorem PRIOR OBSERVATION). *For real  $x \notin \mathbb{Z}$ ,*

$$\operatorname{sgn} \Phi(x) = (-1)^{s_2(\lfloor |x| \rfloor)} = \varepsilon_{\lfloor |x| \rfloor}. \quad (4.3)$$

Equivalently, the signs of the consecutive positive-axis lobes of the infinite sinc product are exactly the signed Thue–Morse word.

*Proof.* The product is positive at the origin. Crossing the zero at  $n$  reverses sign exactly when  $a(n)$  is odd. Therefore, for  $N < x < N + 1$ ,

$$\operatorname{sgn} \Phi(x) = (-1)^{A(N)}.$$

By (3.3),  $A(N) = 2N - s_2(N)$ , whose parity is  $s_2(N)$ . Evenness handles negative  $x$ .  $\square$

*Remark 4.3.* The pattern itself predates this report. The point of Theorem 4.2 is the exact spectral explanation: Thue–Morse is not an external decoration placed on the sinc product. It is the parity integral of its 2-adic zero multiplicities, as (4.2) makes literal.

For the  $J$ -factor truncation

$$\Phi_J(x) = \prod_{h=0}^{J-1} \operatorname{sinc}\left(\frac{\pi x}{2^h}\right), \quad (4.4)$$

we likewise have, away from its zeros,

$$\operatorname{sgn} \Phi_J(x) = (-1)^{\sum_{h=0}^{J-1} \lfloor |x|/2^h \rfloor}. \quad (4.5)$$

Thus the infinite Thue–Morse sign law is already visible through finite digital floor sums.

### 4.3 A Thue–Morse lobe decomposition of the bump

**Theorem 4.4** (Signed lobe inversion NEW-TO-APSHOT). *For every real  $x$ ,*

$$\operatorname{up}(x) = 2 \sum_{N=0}^{\infty} \varepsilon_N \int_N^{N+1} |\Phi(\xi)| \cos(2\pi x \xi) \, d\xi. \quad (4.6)$$

*The series and every termwise differentiated series converge absolutely. In particular, for  $m \geq 0$ ,*

$$\operatorname{up}^{(2m)}(0) = 2(-1)^m (2\pi)^{2m} \sum_{N=0}^{\infty} \varepsilon_N \int_N^{N+1} \xi^{2m} |\Phi(\xi)| \, d\xi. \quad (4.7)$$

*For  $m = 0$  this becomes the exact alternating-area identity*

$$1 = 2 \sum_{N \geq 0} \varepsilon_N \int_N^{N+1} |\Phi(\xi)| \, d\xi. \quad (4.8)$$

*Proof.* Fourier inversion and evenness give  $\operatorname{up}(x) = 2 \int_0^\infty \Phi(\xi) \cos(2\pi x \xi) \, d\xi$ . Partition the half-line into unit lobes and apply Theorem 4.2. Rapid decay of  $\Phi$  permits arbitrary polynomial weights, proving absolute convergence and termwise differentiation.  $\square$

This identity is a genuinely analytic bridge back from the sinc product to the Thue–Morse sequence. It differs from the better-known appearance of Thue–Morse in truncated-power formulas for finite box splines: here the sequence weights frequency lobes of the limiting  $C^\infty$  bump itself.

## 5 Laguerre–Polya geometry and real critical points

**Theorem 5.1** (Real-rooted derivative hierarchy DERIVED). *The entire function  $\Phi$  belongs to the class of Laguerre–Polya entire functions. Consequently, every derivative  $\Phi^{(r)}$  has only real zeros. More precisely:*

1. *the zero  $n \in \mathbb{Z} \setminus \{0\}$  is inherited by  $\Phi'$  with multiplicity  $a(n) - 1$ ;*
2. *in every interval  $(n, n + 1)$ ,  $n \geq 0$ ,  $\Phi'$  has exactly one simple zero  $c_n$ ;*

3. for every  $r \geq 0$  and nonzero integer  $m$ ,

$$\Phi^{(r)}(2^r m) = 0. \quad (5.1)$$

*Proof.* The canonical product is the locally uniform limit of real-rooted even polynomials, because  $\sum_n a(n)n^{-2} < \infty$ . Hence it is in the Laguerre–Polya class, which is closed under differentiation.

An order- $a(n)$  zero loses one order under differentiation. On  $(n, n+1)$ , (3.12) is strictly decreasing, since

$$\left(\frac{\Phi'}{\Phi}\right)'(x) = -\sum_{m \geq 1} a(m) \left( \frac{1}{(x-m)^2} + \frac{1}{(x+m)^2} \right) < 0.$$

It tends to  $+\infty$  at  $n+$  and to  $-\infty$  at  $(n+1)-$ , so it has one simple zero. Finally, the zero at  $2^r m$  has multiplicity at least  $r+1$ .  $\square$

**Corollary 5.2** (Signed unique extrema). *The unique critical value in the  $n$ th lobe has sign  $\varepsilon_n$ . Thus the critical-value sequence may be written*

$$\Phi(c_n) = \varepsilon_n M_n, \quad M_n > 0. \quad (5.2)$$

The repository’s Fourier-decay program studies the highly irregular envelope  $M_n$ . The arithmetic spectrum explains why a single “typical” exponent cannot capture all subsequences: the inherited multiplicity at each endpoint varies with  $\nu_2(n)$ , while the lobe sign varies with the integrated parity  $s_2(n) \bmod 2$ .

## 6 Exact 2-adic Strang–Fix hierarchy

This section contains the strongest new-to-snapshot theorem. The ordinary partition of unity is the degree-zero case. The full hierarchy reveals an exact and sharp dependence on the 2-adic valuation of the lattice denominator.

### 6.1 Weighted periodizations

For  $N \geq 1$  and  $r \geq 0$ , define

$$\mathcal{P}_{N,r}(x) = \sum_{k \in \mathbb{Z}} \left(x - \frac{k}{N}\right)^r \text{up}\left(x - \frac{k}{N}\right). \quad (6.1)$$

The sum is locally finite because  $\text{up}$  has compact support.

**Theorem 6.1** (Sharp weighted partition hierarchy NEW-TO-SNAPSHOT). *Let  $q = \nu_2(N)$ . For every  $0 \leq r \leq q$  and every real  $x$ ,*

$$\boxed{\mathcal{P}_{N,r}(x) = N\mu_r}. \quad (6.2)$$

*The statement is sharp:  $\mathcal{P}_{N,q+1}$  is not constant. Equivalently, the translate system generated by  $\text{up}$  on the lattice  $N^{-1}\mathbb{Z}$  has exact Strang–Fix order*

$$\text{SF}(N) = 1 + \nu_2(N). \quad (6.3)$$

*Proof.* Put  $g_r(t) = t^r \text{up}(t)$ . Poisson summation for the  $N^{-1}$ -periodization gives

$$\mathcal{P}_{N,r}(x) = N \sum_{m \in \mathbb{Z}} \hat{g}_r(mN) e^{2\pi i m N x}. \quad (6.4)$$

Differentiating the Fourier transform,

$$\widehat{g}_r(\xi) = \frac{\Phi^{(r)}(\xi)}{(-2\pi i)^r}. \quad (6.5)$$

For  $m \neq 0$ , the zero at  $mN$  has order

$$a(mN) = 1 + \nu_2(mN) = 1 + q + \nu_2(m) \geq q + 1.$$

Hence  $\Phi^{(r)}(mN) = 0$  for  $r \leq q$ . Only  $m = 0$  remains in (6.4), giving  $N\widehat{g}_r(0) = N\mu_r$ . At  $r = q + 1$ , the Fourier coefficient with  $m = 1$  is nonzero because the zero at  $N$  has exact order  $q + 1$ . Therefore the periodization cannot be constant.  $\square$

**Example 6.2** (The first three layers). For every  $N$ ,

$$\sum_{k \in \mathbb{Z}} \text{up}(x - k/N) = N. \quad (6.6)$$

If  $2 \mid N$ , then

$$\sum_{k \in \mathbb{Z}} (x - k/N) \text{up}(x - k/N) = 0. \quad (6.7)$$

If  $4 \mid N$ , then

$$\sum_{k \in \mathbb{Z}} (x - k/N)^2 \text{up}(x - k/N) = \frac{N}{9}. \quad (6.8)$$

The quadratic identity fails for  $N \equiv 2 \pmod{4}$  and the linear identity fails for odd  $N$ .

*Remark 6.3* (Why the result is arithmetically unusual). Refining the lattice from  $1/N$  to  $1/(N + 1)$  can reduce the exact reproduction order dramatically. Powers of two are optimal: the lattice  $2^{-q}\mathbb{Z}$  has order  $q + 1$ , whereas every odd denominator has only the classical order one.

## 6.2 Exact polynomial reproduction

Let

$$M(z) = \int_{\mathbb{R}} e^{zt} \text{up}(t) dt = \sum_{r \geq 0} \mu_r \frac{z^r}{r!}. \quad (6.9)$$

Since  $\text{up}$  is even,  $M(-z) = M(z)$ . Define reciprocal-moment Appell polynomials  $P_r$  by

$$\frac{e^{xz}}{M(z)} = \sum_{r=0}^{\infty} P_r(x) \frac{z^r}{r!}. \quad (6.10)$$

**Theorem 6.4** (Appell-corrected lattice reproduction NEW-TO-SNAPSHOT). *If  $q = \nu_2(N)$ , then for every polynomial degree  $0 \leq r \leq q$ ,*

$$\boxed{\frac{1}{N} \sum_{k \in \mathbb{Z}} P_r(k/N) \text{up}(x - k/N) = x^r.} \quad (6.11)$$

*Consequently every polynomial of degree at most  $q$  is reproduced exactly by a locally finite linear combination of  $N^{-1}$ -lattice translates of  $\text{up}$ .*

*Proof.* Write  $t_k = x - k/N$ . Up to degree  $q$ , [Theorem 6.1](#) implies the formal Taylor identity

$$\frac{1}{N} \sum_k e^{-zt_k} \text{up}(t_k) = M(-z) + \mathcal{O}(z^{q+1}) = M(z) + \mathcal{O}(z^{q+1}).$$

Multiplying by  $e^{xz}/M(z)$  yields

$$\frac{1}{N} \sum_k \frac{e^{(x-t_k)z}}{M(z)} \text{up}(t_k) = e^{xz} + \mathcal{O}(z^{q+1}).$$

Comparing coefficients gives (6.11). □

The first correction polynomials are

$$\begin{aligned} P_0(x) &= 1, \\ P_1(x) &= x, \\ P_2(x) &= x^2 - \frac{1}{9}, \\ P_3(x) &= x^3 - \frac{x}{3}, \\ P_4(x) &= x^4 - \frac{2}{3}x^2 + \frac{31}{675}. \end{aligned} \tag{6.12}$$

Thus, for example, the  $1/16$  lattice reproduces quartics exactly.

### 6.3 A formalization-friendly version

The analytic proof above uses Poisson summation, but the theorem can be decomposed for Lean into three reusable statements:

1. Fourier transform of  $x^r \text{up}(x)$  equals  $(-2\pi i)^{-r} \Phi^{(r)}$ ;
2. periodization on  $N^{-1}\mathbb{Z}$  has Fourier coefficients sampled at  $mN$ ;
3. exact zero order at  $mN$  is  $1 + \nu_2(mN)$ .

No asymptotic estimate is needed. Once a suitable Poisson theorem for compactly supported smooth functions is available, the sharpness proof is a one-coefficient argument.

## 7 Spectral zeta function and complex dimensions

**Definition 7.1** (Zero spectral zeta function). For  $\Re s > 1$ , set

$$Z_\Phi(s) = \sum_{n \geq 1} \frac{\text{ord}_n \Phi}{n^s} = \sum_{n \geq 1} \frac{a(n)}{n^s}. \tag{7.1}$$

By (3.2),

$$Z_\Phi(s) = \frac{\zeta(s)}{1 - 2^{-s}}. \tag{7.2}$$

It extends meromorphically to  $\mathbb{C}$ . Besides the ordinary density pole at  $s = 1$ , it has the dyadic lattice of poles

$$\chi_k = \frac{2\pi i k}{\log 2}, \quad k \in \mathbb{Z}, \tag{7.3}$$

except where an accidental zeta zero would cancel one. At  $k \neq 0$ ,

$$\text{Res}_{s=\chi_k} Z_\Phi(s) = \frac{\zeta(\chi_k)}{\log 2}. \tag{7.4}$$

We call the  $\chi_k$  the complex dimensions of the dyadic zero spectrum. This terminology is descriptive: the underlying object is simply the meromorphic Dirichlet series (7.2).

*Remark 7.2.* The pole at  $s = 1$  has residue 2, agreeing with the exact zero count  $A(N) = 2N - s_2(N)$ . The imaginary lattice records the discrete scale invariance which is invisible in the leading density but reappears as a periodic function of  $\log_2$  scale.

## 8 Mellin transform on the imaginary axis

### 8.1 Exact transform

Set

$$H(y) = \log \Phi(iy) = \sum_{n \geq 1} a(n) \log \left( 1 + \frac{y^2}{n^2} \right), \quad y > 0. \quad (8.1)$$

All terms are positive, so  $H$  has none of the real-axis cancellation difficulties.

**Theorem 8.1** (Mellin factorization NEW-TO-SNAPSHOT). *For  $1 < \Re s < 2$ ,*

$$\int_0^\infty H(y) y^{-s-1} dy = \frac{\pi \zeta(s)}{s \sin(\pi s/2) (1 - 2^{-s})}. \quad (8.2)$$

*Equivalently, with  $S = 2\pi y$ ,*

$$H(y) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{\Gamma(1-s) \zeta(1-s)}{s(1-2^{-s})} S^s ds, \quad 1 < c < 2. \quad (8.3)$$

*Proof.* For  $0 < \Re s < 2$ , integration by parts and the beta integral give

$$\int_0^\infty \log(1+t^2) t^{-s-1} dt = \frac{\pi}{s \sin(\pi s/2)}. \quad (8.4)$$

For  $\Re s > 1$ , Tonelli's theorem and  $y = nt$  factor the sum into this kernel and  $Z_\Phi(s)$ . The second form follows from the zeta functional equation.  $\square$

### 8.2 Complete large- $y$ expansion

Let  $L = \log 2$  and define

$$C_2 = \text{rac} \zeta''(0) + \frac{1}{2} \log^2(2\pi) - \frac{\pi^2}{24} L - \frac{L}{12} = -1.1090453654341866821907455547 \dots \quad (8.5)$$

For  $k \neq 0$ , let

$$c_k = -\frac{\Gamma(-\chi_k) \zeta(1-\chi_k)}{L}, \quad (8.6)$$

and define the real analytic, mean-zero, 1-periodic function

$$\Psi(u) = \sum_{k \neq 0} c_k e^{2\pi i k u}. \quad (8.7)$$

**Theorem 8.2** (Complex-dimension expansion NEW-TO-SNAPSHOT). *For every  $A > 0$ , as  $y \rightarrow \infty$ ,*

$$H(y) = S - \frac{\log^2 S}{2L} - \frac{1}{2} \log S + C_2 + \Psi(\log_2 S) + \mathcal{O}_A(S^{-A}), \quad S = 2\pi y. \quad (8.8)$$

*The Fourier coefficients decay exponentially. Numerically,*

$$c_1 = (7.5586509397640965 - 7.4700909117916216i) 10^{-7},$$

$$\begin{aligned} |c_1| &= 1.0627109779221651 \times 10^{-6}, \\ |c_2| &= 6.6927771176343188 \times 10^{-13}. \end{aligned} \quad (8.9)$$

The exact triangle estimate is

$$\|\Psi\|_\infty \leq 2 \sum_{k \geq 1} |c_k|, \quad 2 \sum_{k \geq 1} |c_k| \approx 2.1254232944005 \times 10^{-6}, \quad (8.10)$$

where the displayed decimal is an 80-digit-working-precision numerical sum.

*Proof architecture.* Shift the contour in (8.3) to the left. The pole at  $s = 1$  gives  $S$ . The triple pole at  $s = 0$  gives the quadratic logarithm, linear logarithm, and  $C_2$ . The simple poles at  $s = \chi_k$ ,  $k \neq 0$ , give  $c_k S^{\chi_k} = c_k e^{2\pi i k \log_2 S}$ . In the functional-equation form, there are no further poles to the left: the trivial zeta zeros cancel the sine poles. Stirling's estimate for  $\Gamma(1 - s)$  permits a shift to any fixed negative real part, giving the superalgebraic remainder. The same estimate proves exponential Fourier decay.  $\square$

### 8.3 Endpoint duality

Define the negative Laplace transform of the bounded Fabius variable by

$$\Lambda(S) = \mathbb{E} e^{-2SX}. \quad (8.11)$$

From  $X = (Y + 1)/2$  and symmetry of  $Y$ ,

$$\boxed{\Phi(iy) = e^S \Lambda(S), \quad S = 2\pi y.} \quad (8.12)$$

Thus (8.8) is exactly the logarithmic large-parameter expansion which feeds the repository's small-endpoint saddle point. The Gamma–zeta coefficients  $c_k$  already occur there, but [Theorem 8.2](#) supplies a new spectral origin:

$$\text{endpoint log-periodicity} \quad \longleftrightarrow \quad \text{residues of } Z_\Phi(s) \text{ at } \chi_k. \quad (8.13)$$

The oscillation is tiny not because the dyadic scaling is weak, but because  $\Gamma(-\chi_k)$  is exponentially small at the first nonzero imaginary height.

## 9 Finite products, $q$ -integers, and pole condensation

### 9.1 Truncated multiplicities

The  $J$ -factor product (4.4) has multiplicity

$$a_J(n) = \min\{J, 1 + \nu_2(n)\}. \quad (9.1)$$

Let  $q = 2^{-s}$ . Then

**Theorem 9.1** (Finite spectral  $q$ -integer NEW-TO-APSHOT). *For  $\Re s > 1$ ,*

$$Z_{\Phi_J}(s) := \sum_{n \geq 1} \frac{a_J(n)}{n^s} = \zeta(s) \frac{1 - 2^{-Js}}{1 - 2^{-s}} = \zeta(s)[J]_q, \quad (9.2)$$

where  $[J]_q = 1 + q + \cdots + q^{J-1}$ . Also

$$\sum_{n \leq N} a_J(n) = \sum_{h=0}^{J-1} \left\lfloor N/2^h \right\rfloor, \quad (9.3)$$

while

$$\log \Phi_J(z) = - \sum_{m \geq 1} \frac{\zeta(2m)}{m} [J]_{4^{-m}} z^{2m} \quad (|z| < 1). \quad (9.4)$$



Every finite  $Z_{\Phi_J}$  is regular at the nonzero  $\chi_k$ : the apparent geometric denominator is canceled by the numerator. The complex dimensions are therefore genuinely infinite-depth phenomena.

**Theorem 9.2** (Local pole-condensation profile NEW-TO-APSHOT). *Fix  $k \in \mathbb{Z}$  and put  $\chi_k = 2\pi i k / L$ . Uniformly for  $w$  in compact subsets of  $\mathbb{C}$ ,*

$$\lim_{J \rightarrow \infty} \frac{1}{J} Z_{\Phi_J} \left( \chi_k + \frac{w}{J} \right) = \zeta(\chi_k) \frac{1 - e^{-Lw}}{Lw}, \quad (9.5)$$

with the removable value 1 for the last factor at  $w = 0$ .

*Proof.* At  $s = \chi_k + w/J$ , one has  $q^J = 2^{-Js} = e^{-Lw}$  and  $J(1 - q) \rightarrow Lw$ . Substitute in (9.2).  $\square$

The profile  $(1 - e^{-Lw})/(Lw)$  is the finite-depth regularization of the limiting pole. It gives a precise analytic meaning to the statement that a  $q$ -integer turns into a geometric resolvent.

## 9.2 Gaussian $q$ -binomial layer algebra

Define the layer zeta functions

$$\mathcal{Z}_h(s) = \sum_{2^h | n} n^{-s} = q^h \zeta(s). \quad (9.6)$$

The finite  $q$ -binomial theorem gives the exact formal identity

$$\prod_{h=0}^{J-1} (1 + t \mathcal{Z}_h(s)) = \sum_{r=0}^J t^r \zeta(s)^r q^{\binom{r}{2}} \begin{bmatrix} J \\ r \end{bmatrix}_q. \quad (9.7)$$

This is a clean spectral explanation for Gaussian coefficients: they enumerate independent selections of dyadic zero layers, with weight  $q^h$  for choosing layer  $h$ . The identity is deliberately stated as a layer-selection polynomial; it should not be confused with a Dirichlet series for intersections, because multiplication of Dirichlet series introduces multiplicative convolution.

## 9.3 The common-zero zeta tower

Intersections are instead encoded by binomial moments of the multiplicity.

**Theorem 9.3** (Common-zero hierarchy NEW-TO-APSHOT). *For every integer  $r \geq 1$  and  $\Re s > 1$ ,*

$$Z_r(s) := \sum_{n \geq 1} \binom{a(n)}{r} n^{-s} = \zeta(s) \frac{q^{r-1}}{(1-q)^r}, \quad q = 2^{-s}. \quad (9.8)$$

For the  $J$ -layer product,

$$Z_{J,r}(s) = \sum_{n \geq 1} \binom{a_J(n)}{r} n^{-s} = \zeta(s) \sum_{h=r-1}^{J-1} \binom{h}{r-1} q^h. \quad (9.9)$$

Consequently the limiting spectrum has an order- $r$  pole at every nonzero complex dimension  $\chi_k$  unless canceled by  $\zeta(\chi_k)$ .

*Proof.* The number  $\binom{a(n)}{r}$  counts  $r$ -subsets of the layers  $\{0, 1, \dots, \nu_2(n)\}$ . Group such subsets by their largest element  $h$ . There are  $\binom{h}{r-1}$  choices and their common zeros are precisely the multiples of  $2^h$ . Hence

$$Z_r(s) = \zeta(s) \sum_{h \geq r-1} \binom{h}{r-1} q^h = \zeta(s) \frac{q^{r-1}}{(1-q)^r}.$$

Truncating  $h$  gives the finite formula.  $\square$

This tower is one of the sharpest bridges among the four themes in the user's request: the sinc layers define the zeros, the 2-adic intersection counts produce the rational powers of  $(1 - q)^{-1}$ , and finite layer selections produce Gaussian  $q$ -binomials.

## 10 The zero heat trace

Define the exponentially smoothed zero count

$$\Theta(t) = \sum_{n \geq 1} a(n) e^{-nt}, \quad t > 0. \quad (10.1)$$

Layer summation gives the dyadic Lambert series

$$\Theta(t) = \sum_{h \geq 0} \frac{1}{e^{2^h t} - 1}. \quad (10.2)$$

Its Mellin representation is

$$\Theta(t) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \Gamma(s) \frac{\zeta(s)}{1-2^{-s}} t^{-s} ds, \quad c > 1. \quad (10.3)$$

**Theorem 10.1** (Heat-trace expansion NEW-TO-SNAPSHOT). *Let  $T = \log_2(1/t)$  and*

$$C_\Theta = \frac{\gamma - \log(2\pi)}{2 \log 2} - \frac{1}{4} = -1.1593749760977258236984308878 \dots \quad (10.4)$$

*Then, as  $t \downarrow 0$ ,*

$$\begin{aligned} \Theta(t) \sim & \frac{2}{t} - \frac{\log(1/t)}{2 \log 2} + C_\Theta + \sum_{k \neq 0} \frac{\Gamma(\chi_k) \zeta(\chi_k)}{\log 2} e^{2\pi i k T} \\ & - \sum_{m=0}^{\infty} \frac{Z_\Phi(-(2m+1))}{(2m+1)!} t^{2m+1}. \end{aligned} \quad (10.5)$$

*The first algebraic terms are*

$$-\frac{t}{12} + \frac{t^3}{5040} + \dots \quad (10.6)$$

*The first periodic coefficient has magnitude  $1.2764467470886077 \times 10^{-6}$ .*

*Proof architecture.* Shift the contour in (10.3). The pole at  $s = 1$  gives  $2/t$ . The double pole at 0 gives the logarithm and constant. The poles  $\chi_k$  give the periodic term. Poles of  $\Gamma(s)$  at negative even integers are canceled by trivial zeros of  $\zeta$ ; the negative odd integers give the final series.  $\square$

The heat trace makes the spectral dimension picture especially transparent: linear zero density, a logarithmic digital correction, and a microscopic log-periodic ripple are successive residue layers of the same Dirichlet series.

## 11 A base- $b$ theory

The dyadic case belongs to a larger exact family. Fix an integer  $b \geq 2$  and set

$$\Phi_b(z) = \prod_{h=0}^{\infty} \operatorname{sinc}\left(\frac{\pi z}{b^h}\right). \quad (11.1)$$

It is the Fourier transform of the  $C^\infty$  density obtained by convolving uniform laws on intervals of half-width  $1/(2b^h)$ . Its support is

$$\left[ -\frac{b}{2(b-1)}, \frac{b}{2(b-1)} \right]. \quad (11.2)$$

Let  $\nu_b(n) = \max\{h : b^h \mid n\}$ , which is meaningful even when  $b$  is composite.

**Theorem 11.1** (Base- $b$  arithmetic spectrum NEW-TO-SNAPSHOT). *The zero multiplicity at a nonzero integer is*

$$a_b(n) = 1 + \nu_b(n), \quad (11.3)$$

and

$$\sum_{n \geq 1} \frac{a_b(n)}{n^s} = \frac{\zeta(s)}{1 - b^{-s}}, \quad (11.4)$$

$$\sum_{n \leq N} a_b(n) = \sum_{h \geq 0} \left\lfloor N/b^h \right\rfloor = \frac{bN - s_b(N)}{b - 1}. \quad (11.5)$$

$$(11.6)$$

Here  $s_b(N)$  is the base- $b$  digit sum. The entire function has exponential type  $\pi b/(b - 1)$  and zero density  $b/(b - 1)$ , in exact agreement with the Cartwright density law.

**Corollary 11.2** (Even-base automatic lobe signs NEW-TO-SNAPSHOT). *If  $b$  is even and  $x \notin \mathbb{Z}$ , then*

$$\operatorname{sgn} \Phi_b(x) = (-1)^{s_b(\lfloor |x| \rfloor)}. \quad (11.7)$$

For arbitrary  $b$ , the sign sequence  $\sigma_b(N) = (-1)^{\sum_{n \leq N} a_b(n)}$  obeys

$$\sigma_b(bN + r) = (-1)^{bN+r} \sigma_b(N), \quad 0 \leq r < b. \quad (11.8)$$

*Proof.* Equation (11.5) has the same proof as the binary digit-sum identity. For even  $b$ , its parity equals  $s_b(N)$ . The recursion follows directly from  $A_b(bN + r) = bN + r + A_b(N)$ .  $\square$

**Theorem 11.3** (Base- $b$  Strang–Fix order NEW-TO-SNAPSHOT). *On the lattice  $N^{-1}\mathbb{Z}$ , the exact Strang–Fix order of the density with Fourier transform  $\Phi_b$  is*

$$1 + \nu_b(N). \quad (11.9)$$

The weighted periodization and Appell reproduction theorems hold verbatim with  $\nu_2$  replaced by  $\nu_b$  and the corresponding centered moments.

The Mellin calculation also persists. Let  $L_b = \log b$ ,  $\chi_{b,k} = 2\pi i k / L_b$ , and  $S = 2\pi y$ . Then

$$\log \Phi_b(iy) = \frac{b}{2(b-1)} S - \frac{\log^2 S}{2L_b} - \frac{1}{2} \log S + C_b + \Psi_b(\log_b S) + \mathcal{O}_A(S^{-A}), \quad (11.10)$$

$$C_b = \frac{\zeta''(0) + \frac{1}{2} \log^2(2\pi) - \frac{\pi^2}{24}}{L_b} - \frac{L_b}{12}, \quad (11.11)$$

$$\Psi_b(u) = \sum_{k \neq 0} \left( -\frac{\Gamma(-\chi_{b,k}) \zeta(1 - \chi_{b,k})}{L_b} \right) e^{2\pi i k u}. \quad (11.12)$$

Thus the Gamma–zeta law is not peculiar to base two; base two is where the lobe signs become the classical Thue–Morse sequence and the exact dyadic arithmetic becomes available.

## 12 Numerical and symbolic verification

All numerical values in this section were computed independently from the formulas in the report with 80-decimal working precision. They are checks, not proofs.

## 12.1 Imaginary-axis residual

The table compares the direct convergent sum

$$H(y) = \sum_{h \geq 0} \log \frac{\sinh(\pi y/2^h)}{\pi y/2^h} \quad (12.1)$$

with (8.8), retaining 20 Fourier modes of  $\Psi$ .

| $y$ | $H(y)$               | direct minus asymptotic    |
|-----|----------------------|----------------------------|
| 1   | 1.8186408379587714   | $3.4873589 \times 10^{-6}$ |
| 2   | 5.5707984108206301   | $1.0496 \times 10^{-11}$   |
| 5   | 20.0107450110424130  | $-1.0907 \times 10^{-12}$  |
| 10  | 47.2862093875369543  | $-1.0907 \times 10^{-12}$  |
| 100 | 594.0427656248311891 | $2.6185 \times 10^{-12}$   |

At  $y = 1$  the omitted superalgebraic remainder is still visible; by  $y = 2$  the error is already near the working evaluation floor.

## 12.2 Exact digital checks

For  $N = 0, 1, \dots, 15$ , the cumulative multiplicity  $A(N) = 2N - s_2(N)$  has parity

$$0, 1, 1, 0, 1, 0, 0, 1, 1, 0, 0, 1, 1, 0,$$

which is the zero–one Thue–Morse word. Direct evaluation of 80 sinc factors at one interior point of each lobe reproduces the corresponding signs. Exact integer arithmetic also verifies (3.3), (9.9), and the recurrences (3.6) through  $10^6$  without exception.

## 12.3 A reproducible verification recipe

The main identities can be checked without evaluating up itself:

1. enumerate  $a(n) = 1 + \nu_2(n)$  and compare partial sums with  $2N - s_2(N)$ ;
2. compare  $\sum_{n \leq M} a(n)n^{-s}$  with  $\zeta(s)/(1 - 2^{-s})$  for  $\Re s > 1$ ;
3. evaluate  $\Phi(x)$  by the sinc product and compare its sign with  $(-1)^{s_2(\lfloor x \rfloor)}$ ;
4. compare the finite spectral series with  $\zeta(s)[J]_{2^{-s}}$ ;
5. evaluate  $H(y)$  both by (12.1) and by the residue expansion.

These checks are sufficiently orthogonal that a normalization error in  $2\pi$ , a missing initial sinc factor, or an off-by-one valuation is immediately exposed.

## 13 What is genuinely added to the pinned corpus

The following table separates established inputs, prior observations, and the new theorem package.

| Statement                                       | Status | Comment                                                           |
|-------------------------------------------------|--------|-------------------------------------------------------------------|
| Canonical product and zero order $1 + \nu_2(n)$ | CORPUS | Already in the main exposition and Lean canonical-product module. |

| Statement                                                                     | Status            | Comment                                                                                                  |
|-------------------------------------------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------|
| All-order endpoint asymptotics and Gamma–zeta coefficients                    | CORPUS            | Already in the corrected small-argument study.                                                           |
| Thue–Morse lobe sign pattern                                                  | PRIOR OBSERVATION | Recorded in the 2018 MSE question; this report gives the exact zero-count proof and lobe inversion.      |
| One critical point per real lobe                                              | PRIOR OBSERVATION | Essentially proved in the MSE answer; recast here through Laguerre–Polya theory and the resolvent trace. |
| Arithmetic spectrum package, exact counting law as zero data, spectral traces | NEW-TO-SNAPSHOT   | Not found assembled in the pinned TeX corpus.                                                            |
| Weighted partition hierarchy and sharp order $1 + \nu_2(N)$                   | NEW-TO-SNAPSHOT   | Degree zero is classical; the exact higher stopping law was not located in the corpus.                   |
| Appell-corrected exact polynomial reproduction                                | NEW-TO-SNAPSHOT   | Uses existing moment-deconvolution ideas in a different, sharp lattice theorem.                          |
| Mellin transform of $\log \Phi(iy)$ and complex-dimension interpretation      | NEW-TO-SNAPSHOT   | Explains the already-known endpoint Fourier coefficients by zero-spectrum residues.                      |
| Finite $q$ -integer regularization and pole-condensation profile              | NEW-TO-SNAPSHOT   | Connects finite sinc products to the limiting imaginary pole lattice.                                    |
| Gaussian $q$ -binomial layer-selection polynomial                             | NEW-TO-SNAPSHOT   | A spectral-layer interpretation, distinct from existing dyadic-value $q$ -binomials.                     |
| Common-zero zeta tower $Z_r$                                                  | NEW-TO-SNAPSHOT   | Encodes binomial multiplicity moments and higher-order complex-dimension poles.                          |
| Heat-trace expansion                                                          | NEW-TO-SNAPSHOT   | A residue expansion for exponentially smoothed zero counts.                                              |
| Base- $b$ extension of the full package                                       | NEW-TO-SNAPSHOT   | Includes digit-sum signs for even bases and exact lattice order $1 + \nu_b(N)$ .                         |

*Warning 13.1* (Historical priority remains narrower than proof). A targeted search found V. A. Rvachev’s two-page 1977 paper *On approximation by means of the function*  $\text{up}(x)$ , followed by a substantial atomic-function approximation literature. The exact formulas of [Theorems 6.1](#) and [6.4](#) were not located in the pinned corpus or in the limited searchable metadata, but a Russian-language priority audit is still needed before asserting that those facts have never appeared elsewhere.

## 14 Lean formalization roadmap

The results separate naturally into four dependency layers.

### 14.1 Layer I: elementary arithmetic

Recommended first targets are

$$a(n) = \sum_h \mathbf{1}_{2^h | n},$$

$$\begin{aligned}\sum_{n \leq N} a(n) &= 2N - s_2(N), \\ (-1)^{a(n)} &= \varepsilon_{n-1} \varepsilon_n, \\ (-1)^{\sum_{n \leq N} a(n)} &= \varepsilon_N.\end{aligned}$$

The existing Thue–Morse and trailing-zero modules should make these short. They would yield the lobe-sign theorem once the canonical product’s sign changes are connected to exact orders.

## 14.2 Layer II: finite algebra and Dirichlet series

Finite versions should precede infinite analytic statements:

- $a_J(n) = \min(J, 1 + \nu_2(n))$ ;
- finite counting formula (9.3);
- finite  $q$ -integer and Gaussian-binomial polynomial identities;
- finite common-layer count (9.9).

For the infinite Dirichlet identities, prove summability on  $\Re s > 1$  and use nonnegative layer rearrangement before introducing meromorphic continuation.

## 14.3 Layer III: exact lattice reproduction

The key reusable analytic lemma is a Poisson formula for a compactly supported  $C^\infty$  function and its polynomial multiples. Then the proof of Theorem 6.1 reduces to:

1. identify the Fourier coefficients of the periodization;
2. rewrite them as derivatives of  $\Phi$ ;
3. annihilate nonzero coefficients with the existing exact-zero-order theorem;
4. prove failure at the first unannihilated coefficient.

The Appell theorem can be formalized coefficientwise, avoiding formal power-series asymptotics.

## 14.4 Layer IV: complex analysis

The Mellin and heat-trace theorems need a contour-shift framework with explicit growth estimates. A pragmatic sequence is:

1. formalize the real integral (8.4);
2. prove (8.2) only in the fundamental strip;
3. record the residue computations algebraically;
4. defer the all- $A$  remainder until a suitable vertical-contour library is available.

Even the strip identity alone gives a valuable formally checked bridge from the zero spectrum to the endpoint Laplace transform.

# 15 Open frontier

*Open problem 15.1* (Critical-point arithmetic). Let  $c_n$  be the unique critical point in  $(n, n + 1)$ . Determine asymptotic laws for  $c_n - n$  and  $M_n = |\Phi(c_n)|$  conditioned on  $\nu_2(n)$ ,  $\nu_2(n + 1)$ , or finite binary words around  $n$ . The logarithmic derivative (3.12) converts this into a weighted electrostatic equilibrium problem with ruler-function charges.

*Open problem 15.2* (From complex dimensions to real-axis decay). The imaginary-axis periodic fluctuation is completely controlled by the poles  $\chi_k$ , whereas real-axis decay splits into geometric, arithmetic, RMS, and envelope exponents. Find an operator or transfer formalism in which both sets of exponents arise from one spectrum. A direct residue argument cannot work on the real axis because zeros and near-zeros dominate.

*Open problem 15.3* (Sharp approximation constants). The exact Strang–Fix order gives the first possible nonzero term in a lattice quasi-interpolant error. Compute the sharp  $L^\infty$ ,  $L^2$ , and Sobolev constants after Appell correction, and determine whether powers-of-two lattices admit an all-order Richardson scheme whose coefficients have a closed  $q$ -binomial form at base  $1/4$ .

*Open problem 15.4* (Common-zero zeta residues). Develop explicit Laurent coefficients of  $Z_r(s)$  at  $s = \chi_k$ . Their leading term is elementary, but lower coefficients mix derivatives of zeta with powers of  $\log 2$ . Determine whether these coefficients control higher-order log-periodic corrections of naturally associated spectral statistics.

*Open problem 15.5* (Base- $b$  exact arithmetic). For even  $b$ , the Fourier lobe signs are the generalized digit-sum character  $(-1)^{s_b(n)}$ . Develop the corresponding Fabius-type distribution function, finite box-spline formulas, exact rational values at  $b$ -adic points, and Gaussian-binomial or multinomial expressions. Composite  $b$  should display a new tension between divisibility by  $b^h$  and prime-adic arithmetic.

*Open problem 15.6* (Inverse spectral characterization). Classify scale multisets  $\{\lambda_h\}$  for which  $\prod_h \text{sinc}(\pi z/\lambda_h)$  has an automatic or regular zero-multiplicity sequence. Conversely, characterize which nonnegative automatic sequences can occur as multiplicities of a locally uniformly convergent geometric sinc product.

## 16 Conclusion

The Fabius–Rvachev theory is often presented as a collection of surprising bridges: a smooth bump from infinite convolutions, rational dyadic values from  $q$ -binomials, Thue–Morse signs from finite differences, and log-periodic asymptotics from a saddle point. The arithmetic zero spectrum supplies a common organizing object.

Its summatory law integrates the ruler function into the Thue–Morse lobe signs. Its exact multiplicities turn Poisson summation into a sharp 2-adic hierarchy of polynomial reproduction. Its Dirichlet series produces the complex dimensions whose residues are the Gamma–zeta endpoint coefficients. Its finite truncations are  $q$ -integers, its layer selections are Gaussian  $q$ -binomials, and its higher intersection statistics are powers of the same geometric resolvent. None of these bridges changes the established theory; they explain why its apparently separate parts keep rediscovering the same dyadic algebra.

The most immediate formal target is [Theorem 6.1](#): it is exact, strong, short once the current canonical-product theorem is available, and exposes a clean new role for  $\nu_2(N)$ . The most conceptually unifying analytic target is [Theorem 8.2](#), which identifies the endpoint periodic fluctuation as a residue trace of the Fourier zero spectrum. Together they turn an infinite sinc product from a transform formula into an arithmetic spectral object.

## A Residue calculation at the origin

For completeness, write  $L = \log 2$  and expand

$$\begin{aligned}\zeta(s) &= -\frac{1}{2} - \frac{1}{2} \log(2\pi)s + \frac{1}{2} \zeta''(0)s^2 + \mathcal{O}(s^3), \\ \frac{1}{1 - e^{-Ls}} &= \frac{1}{Ls} + \frac{1}{2} + \frac{Ls}{12} + \mathcal{O}(s^3),\end{aligned}$$

$$\frac{\pi}{s \sin(\pi s/2)} = \frac{2}{s^2} + \frac{\pi^2}{12} + \mathcal{O}(s^2),$$

$$y^s = 1 + s \log y + \frac{s^2 \log^2 y}{2} + \mathcal{O}(s^3).$$

Multiplying and rewriting  $\log y = \log S - \log(2\pi)$  gives the residue

$$-\frac{\log^2 S}{2L} - \frac{1}{2} \log S + C_2,$$

with  $C_2$  as in (8.5). The same calculation for  $\Gamma(s)Z_\Phi(s)t^{-s}$  gives the double-pole contribution

$$-\frac{\log(1/t)}{2L} + \frac{\gamma - \log(2\pi)}{2L} - \frac{1}{4}.$$

## B First coefficients and moments

The numerical constants used in the report are

$$\begin{aligned}\zeta''(0) &= -2.0063564559085848512101000267299604382\dots, \\ C_2 &= -1.1090453654341866821907455547094644509\dots, \\ C_\Theta &= -1.1593749760977258236984308878499458999\dots\end{aligned}$$

For the reciprocal-moment Appell family,

$$\frac{1}{M(z)} = 1 - \frac{\mu_2}{2!}z^2 + \frac{6\mu_2^2 - \mu_4}{4!}z^4 + \dots,$$

which with  $\mu_2 = 1/9$  and  $\mu_4 = 19/675$  yields  $6\mu_2^2 - \mu_4 = 31/675$ .

## C Corpus concordance

The frozen tree was read recursively. The most heavily used source families were the following paths (revision siblings are omitted from this compact list):

- docs/Fabius\_Function\_and\_Rvachev\_Up/Fabius\_Function\_and\_Rvachev\_Up.tex;
- docs/semi-formalized-research-frontiers/semi-formalized-research-frontiers.tex;
- docs/archive/standalone-studies/Small\_Argument\_Asymptotics/Small\_Argument\_Asymptotics.tex;
- docs/archive/research-frontiers/Fabius\_Dyadic\_q\_Connections/Fabius\_Dyadic\_q\_Connections.tex;
- docs/archive/research-frontiers/Fabius\_Dyadic\_Formulae\_and\_Alternative\_Representations/Fabius\_Dyadic\_Formulae\_and\_Alternative\_Representations.tex;
- docs/archive/research-frontiers/Fabius\_Thue\_Morse\_Convergence\_Rates/Fabius\_Thue\_Morse\_Convergence\_Rates.tex;
- docs/archive/research-frontiers/Repeated\_Integration\_and\_Rvachev\_Up/Repeated\_Integration\_and\_Rvachev\_Up.tex;



- docs/semi-formalized-research-frontiers/drafts/Thue\_Morse\_Formula\_Atlas/Thue\_Morse\_Formula\_Atlas.tex;
- the complete docs/semi-formalized-research-frontiers/drafts/rvachev\_up\_fourier\_decay/ revision and audit sequence;
- primary-paper sources under docs/papers/.

The glossary, walkthrough, early notes, primary-exposition revisions, supplements,  $q$ -calculus variants, dyadic-asymptotic bridges, inverse-dyadic study, and OCR siblings were also included in the collision audit. Their principal role here was to establish chronology and prevent corrected or archived claims from being mistaken for independent corroboration.

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## Part III

# Zero-Divisor-Preserving $q$ -Richardson Extrapolation for the Fabius–Rvachev Sinc Product

## Abstract

Let

$$P_b(t) = \prod_{k \geq 1} \operatorname{sinc}\left(\frac{t}{b^k}\right), \quad \operatorname{sinc} z = \frac{\sin z}{z}, \quad b > 1,$$

and put  $q = b^{-2}$ . The inspected repository already develops exact logarithmic  $q$ -Richardson extrapolation for finite sections of this product. Its transform has alternating error sign and a complete Bernoulli–zeta/Bell-polynomial expansion, but the logarithmic definition is stated only away from the zeros; the same source leaves monotonicity between extrapolants of the same parity as an open parameter-classification problem.

Both gaps are closed here. For the prefix  $P_{b,N}$ , the omitted tail is zero-free in the natural disk  $|t| < \pi b^{N+1}$ . We extrapolate analytic logarithms of the *relative partial tails* and then multiply by  $P_{b,N}$ . The resulting factorized transform is holomorphic across every prefix zero, preserves the complete zero divisor with multiplicity, and agrees with the earlier multiplicative transform wherever the latter is defined. Its quotient by  $P_b$ , after cancellation of common zeros, is exactly  $\exp(E_{N,r})$ , where

$$E_{N,r}(t) = (-1)^r \frac{q^{r(r+1)/2}}{(q; q)_r} \sum_{m \geq r+1} \frac{\zeta(2m)}{m} \frac{(q^{m-r}; q)_r}{1 - q^m} \left( \frac{t}{\pi b^{N+1}} \right)^{2m}.$$

Consequently, the alternating brackets survive at zeros as certified brackets for the first nonzero derivative, and Cauchy estimates certify every coefficient of the zero-normalized local germ.

Write  $M_r(q, a) = |E_{N,r}|$  after replacing the final even power by  $a^m$ ,  $0 < a < 1$ . The exact classification proved here is

$$M_{r+2}(q, a) < M_r(q, a) \quad \text{for every } 0 < a < 1 \quad \Longleftrightarrow \quad q^{r+1}(1+q) \leq 1.$$

If the inequality fails, the two same-parity errors cross exactly once. The proof is coefficientwise: the shared-coefficient ratio is strictly increasing in  $m$  and tends to  $q^{2r+3}/((1-q^{r+1})(1-q^{r+2}))$ . Thus all levels are nested from order zero exactly when  $q \leq (\sqrt{5}-1)/2$ , or equivalently  $b \geq \sqrt{\varphi}$ . In the dyadic Fabius–Rvachev case  $q = 1/4$ , every level is nested. This gives two monotone analytic chains squeezing the Fourier product while preserving all included integer zeros, their multiplicities  $1 + \nu_2(n)$ , their Thue–Morse lobe signs, and their first nonzero derivatives.

Numerical experiments, supplied as a fully commented Python program, independently compare direct products with the exact coefficient series, compute the threshold curve and unique crossings, and demonstrate derivative-ratio errors below  $10^{-24}$  at the multiplicity-five zero  $x = 48$  by order eight. The final sections formulate conjectures and a formalization program for general refinement masks, complex boundary continuation, optimal finite precision, inverse-Fabius coupling, and periodically perturbed Lambert saddles.

## Proof and novelty status

**Established inputs from the repository.** The Fabius/Rvachev identities; exact dyadic arithmetic; Thue–Morse,  $q$ -binomial, Bernoulli, Bell, Lambert- $W$ , and inverse asymptotic machinery; the infinite sinc product and its integer zero multiplicities; the  $q$ -Lagrange weights; the exact transformed logarithmic tail series; alternating error sign; bounded weight condition number; phase-locked endpoint Richardson transforms; and the large Fourier-decay theory are treated in the inspected corpus.

**Proved in this report.** The factorized zero-divisor-preserving transform; its holomorphic extension through all included zeros; exact zero-normalized quotient and local jet identities; complex nonasymptotic bounds; the necessary-and-sufficient same-parity nesting criterion; uniqueness and analyticity of the supercritical crossing; the  $\sqrt{\varphi}$  base threshold; large-order threshold asymptotics; and the nested dyadic spectral-jet brackets.

**Not claimed.** “New” means not found in the recursively inspected repository or in the targeted primary-source searches described in [Section 1.4](#). It is not an unqualified assertion of worldwide priority. None of the new theorems is claimed to have a Lean counterpart at the snapshot commit.

# 1 Corpus boundary, audit method, and collision control

## 1.1 Pinned recursive corpus

The analysis is pinned to commit `7f4e6e6fdd57f756f303b20f56b5382627e5e1d4`. This matters because the active frontier subtree was changing during the work: the GitHub code-search index lagged the tree, while the recursive Git objects gave the exact count of 68 TeX files. Their complete path manifest is reproduced in [Section A](#) and supplied as a machine-readable text file in the archive.

| Stratum                          | Files | Audit role                                                                                                                                                                    |
|----------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Living expositions and synthesis | 5     | Formalization-backed baseline, glossary, non-elementarity, Lean map, and consolidated frontier volume.                                                                        |
| Active frontier drafts           | 24    | Arithmetic rays, half-integer spectra, inverse asymptotics, Pascal–Rvachev hierarchies, zero determinants, $q$ -Richardson analysis, and twelve Fourier-decay reports/audits. |
| Frozen archive                   | 32    | Revision history, early notes, superseded expositions, prior $q$ -calculus articles, convergence studies, and source provenance.                                              |
| Transcribed source papers        | 7     | Arias de Reyna, Aistleitner–Hofer–Larcher, Toth, Allouche–Shallit, and related primary context.                                                                               |
| Total                            | 68    | Every recursive <code>*.tex</code> path at the pinned commit.                                                                                                                 |

Archived files were not treated as 32 independent current papers. They were read as a revision graph: distinctive claims were checked against the current descendant and the archive’s own consolidation/provenance maps. The twelve Fourier documents were similarly read as two waves plus two comparative audits, so a superseded constant or conjecture could not be mistaken for the current state of the problem.

## 1.2 The mature baseline

The corpus is unusually dense. Among other things, it already contains:

1. a proof-oriented primary exposition of the bounded Fabius function, Rvachev’s compact up-function, and the signed global extension, with exact separation of those objects;
2. probability, convolution, spline, Fourier, moment, exact dyadic,  $q$ -binomial, and Thue–Morse representations;
3. a quadratic-plus-periodic negative-Laplace exponent, lower-branch Lambert phase, all-orders endpoint and inverse expansions, closed inverse jets, and Bell-polynomial coefficient transfer;
4. finite geometric-grid Richardson filters with exact Gaussian  $q$ -binomial weights, alternating logarithmic brackets, complete transformed tail coefficients, and bounded condition number;
5. integer-zero spectral determinants, exact multiplicities, half-integer spectral arithmetic, Prouhet quadrature, rational dyadic rays, cyclotomic orbit norms, and a Pascal–Rvachev hierarchy;
6. a two-wave analysis of pointwise, metric, extremal, quadratic-mean, Cesàro, and shell-adaptive Fourier decay.

The main challenge was therefore not producing another list of identities. It was finding a statement that survives comparison with the newest same-day drafts.

### 1.3 Candidates deliberately discarded

Several promising-looking ideas were rejected after exact comparison:

- phase-locked reciprocal-grid Richardson extrapolation for the Lambert saddle was already present, including its complete homogeneous and Bell-polynomial moments;
- the integer zero divisor, spectral zeta, heat trace, local all-order zero jets, and  $q$ -deformation already had a dedicated report;
- arithmetic rational rays and half-integer weighted Prouhet formulae had newer active treatments;
- generic log-periodic Fourier envelopes and transfer-operator constants were covered by the comparative audits;
- the  $q$ -Pochhammer/Thue–Morse dictionary for endpoint corrections and Fourier-zero multiplicities was already developed in the spectral-endpoint report.

#### The two surviving openings

The latest  $q$ -Richardson draft explicitly leaves two questions open: whether same-parity brackets are monotone in identifiable parameter regimes, and how the logarithmic transform should be continued across zeros. The present report resolves both questions exactly.

### 1.4 External literature boundary

The external search was used to prevent reinvention of standard extrapolation theory and to locate adjacent product-approximation work. Generalized Richardson theory on geometric meshes, including convergence and stability, is classical [9, 10]. The  $q$ -binomial identities used to write the weights belong to standard basic hypergeometric calculus [8]. The Lambert function and its branch asymptotics are standard [11]. The Fabius/Rvachev baseline comes from Arias de Reyna and related work [2, 3, 4], while the lacunary product’s metric theory is represented by Aistleitner–Hofer–Larcher [5].

Targeted searches located recent Padé work for products of functions, but not a construction that factors a finite canonical zero divisor and applies geometric  $q$ -Richardson extrapolation to the zero-free infinite tail, nor the exact threshold in [theorem 4.3](#). The nearest classical results concern extrapolation of asymptotic scales, not preservation of an entire function’s moving zero divisor.

## 2 Established logarithmic $q$ -Richardson structure

### 2.1 The lacunary sinc product

Fix  $b > 1$  and set

$$q = b^{-2} \in (0, 1). \quad (2.1)$$

We use the entire normalized sinc function

$$\operatorname{sinc} z = \begin{cases} \sin z/z, & z \neq 0, \\ 1, & z = 0, \end{cases}$$

and define

$$P_b(t) = \prod_{k=1}^{\infty} \operatorname{sinc}\left(\frac{t}{b^k}\right), \quad P_{b,N}(t) = \prod_{k=1}^N \operatorname{sinc}\left(\frac{t}{b^k}\right). \quad (2.2)$$

Uniform convergence on compact sets makes  $P_b$  entire. The first omitted factor has its nearest zeros at  $\pm\pi b^{N+1}$ , so the natural tail disk is

$$D_N = \{t \in \mathbb{C} : |t| < \pi b^{N+1}\}. \quad (2.3)$$

Define the complete and partial relative tails directly by their products:

$$T_N(t) = \prod_{k=N+1}^{\infty} \operatorname{sinc}\left(\frac{t}{b^k}\right) = P_b\left(\frac{t}{b^N}\right), \quad T_{N,j}(t) = \prod_{k=N+1}^{N+j} \operatorname{sinc}\left(\frac{t}{b^k}\right), \quad j \geq 0, \quad (2.4)$$

with the empty product  $T_{N,0} = 1$ . These total functions satisfy  $P_b = P_{b,N}T_N$  and  $P_{b,N+j} = P_{b,N}T_{N,j}$  everywhere. Only where  $P_{b,N}(t) \neq 0$  do we also write them as the quotients  $T_N = P_b/P_{b,N}$  and  $T_{N,j} = P_{b,N+j}/P_{b,N}$ .

**Lemma 2.1** (Zero-free tail disk). *Every  $T_{N,j}$ , including  $T_N$ , is holomorphic and zero-free on  $D_N$ . It therefore has a unique holomorphic logarithm on  $D_N$  normalized to vanish at  $t = 0$ . On  $D_N \cap \mathbb{R}$ , all these tails are positive.*

*Proof.* Every zero of  $\operatorname{sinc}(t/b^k)$  has modulus at least  $\pi b^k$ . For  $k \geq N+1$  this is at least the radius of  $D_N$ , with equality only for the first omitted factor and only on the boundary. Thus every finite relative tail and the locally uniform infinite tail are zero-free in the disk. The disk is simply connected, giving the normalized logarithm. For real  $|t| < \pi b^{N+1}$  every omitted argument has modulus less than  $\pi$ , where  $\sin x/x > 0$ .  $\square$

## 2.2 Bernoulli–zeta coefficients

For  $|z| < \pi$ ,

$$-\log \operatorname{sinc} z = \sum_{m=1}^{\infty} \frac{\zeta(2m)}{m\pi^{2m}} z^{2m} = \sum_{m=1}^{\infty} \frac{(-1)^{m+1} 2^{2m-1} B_{2m}}{m(2m)!} z^{2m}. \quad (2.5)$$

Consequently, with

$$u_N(t) = \frac{t}{\pi b^{N+1}}, \quad (2.6)$$

one has the exact prefix error

$$\operatorname{Log} P_{b,N}(t) - \operatorname{Log} P_b(t) = \sum_{m \geq 1} \frac{\zeta(2m)}{m(1 - q^m)} u_N(t)^{2m} \quad (2.7)$$

whenever both logarithms are taken on a zero-free component. The right side itself is holomorphic throughout  $D_N$  and will be the invariant object used below.

## 2.3 Geometric Lagrange weights

For  $r \geq 0$ , let  $\lambda_{r,j}(q)$  be the Lagrange interpolation weights at nodes  $1, q, \dots, q^r$ , evaluated at the origin:

$$\lambda_{r,j}(q) = \frac{(-1)^{r-j} q^{(r-j)(r-j+1)/2}}{(q; q)_j (q; q)_{r-j}}, \quad 0 \leq j \leq r, \quad (2.8)$$

where  $(a; q)_n = \prod_{k=0}^{n-1} (1 - aq^k)$ . Their transformed moments are

$$S_{r,m}(q) = \sum_{j=0}^r \lambda_{r,j}(q) q^{jm} = \begin{cases} 1, & m = 0, \\ 0, & 1 \leq m \leq r, \\ (-1)^r \frac{q^{r(r+1)/2} (q^{m-r}; q)_r}{(q; q)_r}, & m \geq r+1. \end{cases} \quad (2.9)$$

This is the finite  $q$ -binomial cancellation mechanism already developed in the repository.

**Formal boundary.** The finite mechanism is compiled in a form stronger than the real specialization above. `FiniteQBinomialCore.lean` proves the Gaussian map law and reflection symmetry without division, together with the finite product and  $q$ -Pochhammer concatenation identities; `GeometricQBinomialLagrange.lean` uses these results to provide the global forward numerator and every positive residual moment. `FinitePowerSeriesFilter.lean` then proves the diagonal action of an arbitrary finite rescaling filter, and `GeometricPowerSeriesFilter.lean` packages the geometric Lagrange row as an all-index formal-series residual identity. The further module `AnalyticSeriesFilter.lean` gives an exact evaluated-series residual under unconditional summability of the contributing samples; zero-weight nodes impose no convergence condition on their unweighted samples. The distinct module `RvachevQBinomialFilter.lean` now identifies the named even-moment series with the dyadic infinite Rvachev product and filters it exactly. It does not identify the general-base finite prefix  $P_{b,N}$ , its quotient, or its holomorphic logarithm on  $D_N$ . A conditionally convergent boundary series and the signs, bounds, brackets, asymptotics, and uniform convergence in this part likewise remain outside that theorem. Those concrete assertions retain their frontier status unless an explicit analytic declaration is cited.

Away from zeros, the existing multiplicative transform can be written schematically as

$$\exp \left( \sum_{j=0}^r \lambda_{r,j} \operatorname{Log} P_{b,N+j} \right).$$

Its exact logarithmic error and alternating sign are established there. The obstruction is that no single logarithm of  $P_{b,N+j}$  can pass through a zero. The next section removes that obstruction without changing the transform where it was already defined.

### 3 Factorization through the complete zero divisor

#### 3.1 Definition and holomorphic continuation

**Definition 3.1** (Factorized tail transform). On  $D_N$ , let  $\operatorname{Log} T_{N,j}$  be the normalized logarithms from lemma 2.1. Define

$$\widehat{T}_{N,r}(t) = \exp \left( \sum_{j=0}^r \lambda_{r,j}(q) \operatorname{Log} T_{N,j}(t) \right), \quad (3.1)$$

$$\widehat{P}_{b,N,r}^{\text{fac}}(t) = P_{b,N}(t) \widehat{T}_{N,r}(t). \quad (3.2)$$

The superscript “fac” will usually be suppressed once no confusion with the unfactored logarithmic expression is possible.

**Theorem 3.2** (Zero-divisor-preserving continuation). *For every  $N, r \geq 0$ :*

1.  $\widehat{P}_{b,N,r}^{\text{fac}}$  is holomorphic on  $D_N$ .
2. Wherever  $P_{b,N}$  is nonzero, it agrees with the earlier multiplicative  $q$ -Richardson transform formed from  $P_{b,N}, \dots, P_{b,N+r}$ .
3.  $P_b, P_{b,N}$ , and  $\widehat{P}_{b,N,r}^{\text{fac}}$  have exactly the same zero divisor in  $D_N$ , including multiplicities.
4. After cancellation of that common zero divisor, the quotient  $\widehat{P}_{b,N,r}^{\text{fac}}/P_b$  extends to a holomorphic zero-free function on all of  $D_N$ .

*Proof.* The function  $\widehat{T}_{N,r}$  is an exponential of a holomorphic function and is therefore holomorphic and zero-free. This proves the first assertion after multiplication by the finite prefix.

If  $P_{b,N}(t) \neq 0$ , then

$$\text{Log } P_{b,N+j} = \text{Log } P_{b,N} + \text{Log } T_{N,j}$$

on the locally normalized branch. Since  $\sum_j \lambda_{r,j} = 1$ , the unfactored weighted logarithm equals  $\text{Log } P_{b,N} + \sum_j \lambda_{r,j} \text{Log } T_{N,j}$ , proving agreement.

By [lemma 2.1](#), both the exact tail  $T_N$  and the extrapolated tail  $\widehat{T}_{N,r}$  are zero-free in  $D_N$ . Thus every zero of either full product comes from  $P_{b,N}$  with unchanged multiplicity. Finally,

$$\frac{\widehat{P}_{b,N,r}^{\text{fac}}}{P_b} = \frac{\widehat{T}_{N,r}}{T_N}$$

after cancellation, and the right side is holomorphic and zero-free.  $\square$

*Remark 3.3.* The construction is not a numerical patch at a real zero. It is a holomorphic factorization on the full complex disk  $D_N$ . The finite prefix is a canonical carrier of the entire zero divisor visible at scale  $N$ ; only the zero-free determinant tail is extrapolated.

### 3.2 Exact error on the whole disk

**Theorem 3.4** (Exact zero-normalized logarithmic error). *On  $D_N$ ,*

$$\frac{\widehat{P}_{b,N,r}^{\text{fac}}(t)}{P_b(t)} = \exp(E_{N,r}(t)) \quad (3.3)$$

after cancellation at common zeros, where

$$E_{N,r}(t) = (-1)^r \frac{q^{r(r+1)/2}}{(q; q)_r} \sum_{m=r+1}^{\infty} \frac{\zeta(2m)}{m} \frac{(q^{m-r}; q)_r}{1 - q^m} u_N(t)^{2m}. \quad (3.4)$$

In particular,

$$E_{N,r}(t) = (-1)^r \frac{q^{r(r+1)/2} \zeta(2r+2)}{(r+1)(1 - q^{r+1})} u_N(t)^{2r+2} + O(u_N(t)^{2r+4}). \quad (3.5)$$

*Proof.* From (2.7), the logarithm of the  $j$ th relative partial tail is

$$\text{Log } T_{N,j} = \text{Log } T_N + \sum_{m \geq 1} \frac{\zeta(2m)}{m(1 - q^m)} u_N(t)^{2m} q^{jm}.$$

Weighting and using  $\sum_j \lambda_{r,j} = 1$  gives

$$\text{Log } \frac{\widehat{T}_{N,r}}{T_N} = \sum_{m \geq 1} \frac{\zeta(2m)}{m(1 - q^m)} u_N(t)^{2m} S_{r,m}(q).$$

Insert (2.9). The first  $r$  powers vanish, and the remaining closed form is (3.4). At  $m = r + 1$  one has  $(q; q)_r = (q^{m-r}; q)_r$ , giving (3.5).  $\square$

### 3.3 Uniform complex bounds

**Proposition 3.5** (A computable disk bound). *Let  $0 < R < \pi b^{N+1}$  and put*

$$a_R = \left( \frac{R}{\pi b^{N+1}} \right)^2.$$



Then on  $|t| \leq R$ ,

$$|E_{N,r}(t)| \leq B_{N,r}(R) := \frac{q^{r(r+1)/2}}{(q; q)_r} \frac{\zeta(2)}{(1-q)(r+1)} \frac{a_R^{r+1}}{1-a_R}. \quad (3.6)$$

Consequently,

$$\left| \frac{\widehat{P}_{b,N,r}^{\text{fac}}}{P_b} - 1 \right| \leq e^{B_{N,r}(R)} - 1 \quad (3.7)$$

after zero cancellation, uniformly on the closed disk.

*Proof.* In (3.4), use  $(q^{m-r}; q)_r \leq 1$ ,  $(1 - q^m)^{-1} \leq (1 - q)^{-1}$ ,  $\zeta(2m) \leq \zeta(2)$ , and

$$\sum_{m \geq r+1} \frac{a_R^m}{m} \leq \frac{a_R^{r+1}}{(r+1)(1-a_R)}.$$

Then use  $|e^z - 1| \leq e^{|z|} - 1$ . □

### 3.4 Local jets at every included zero

**Theorem 3.6** (Exact spectral-jet quotient). *Let  $t_0 \in D_N$  be a zero of  $P_b$  of multiplicity  $\mu$ . Then*

$$\frac{(\widehat{P}_{b,N,r}^{\text{fac}})^{(\mu)}(t_0)}{P_b^{(\mu)}(t_0)} = \exp(E_{N,r}(t_0)). \quad (3.8)$$

More strongly, after removing the common factor  $(t - t_0)^\mu$ , the quotient of the two zero-normalized germs is exactly  $\exp(E_{N,r}(t))$ . Hence for every  $k \geq 1$ ,

$$\left( \frac{d}{dt} \right)^k \text{Log} \left( \frac{\widehat{P}_{b,N,r}^{\text{fac}}(t)/(t - t_0)^\mu}{P_b(t)/(t - t_0)^\mu} \right) = E_{N,r}^{(k)}(t) \quad (3.9)$$

in a neighborhood of  $t_0$ .

*Proof.* Write  $P_{b,N}(t) = (t - t_0)^\mu A(t)$  with  $A(t_0) \neq 0$ . Then

$$P_b(t) = (t - t_0)^\mu A(t) T_N(t), \quad \widehat{P}_{b,N,r}^{\text{fac}}(t) = (t - t_0)^\mu A(t) \widehat{T}_{N,r}(t).$$

Taking the first nonzero derivative gives the tail ratio; [theorem 3.4](#) identifies that ratio with  $\exp(E_{N,r}(t_0))$ . Cancelling the complete common germ gives the stronger identity, and logarithmic differentiation gives (3.9). □

**Corollary 3.7** (Cauchy-certified higher coefficients). *Choose  $\delta > 0$  so that the closed disk  $|t - t_0| \leq \delta$  lies in  $D_N$ . Every Taylor coefficient of the zero-normalized quotient is bounded explicitly by applying Cauchy's estimate to  $\exp(E_{N,r}) - 1$  and [proposition 3.5](#) on that disk. More precisely,  $\exp(E_{N,r}) \rightarrow 1$  locally uniformly either as  $N \rightarrow \infty$  with  $r$  fixed, on each fixed compact set, or as  $r \rightarrow \infty$  with  $N$  fixed, on each compact subset of  $D_N$ . Consequently, for every fixed derivative order  $k \geq 0$ , Cauchy's integral formula gives local uniform convergence of the  $k$ th derivatives to the  $k$ th derivative of 1 (that is, to 1 for  $k = 0$  and to 0 for  $k \geq 1$ ). No uniformity in the derivative order is claimed.*

### 3.5 Bernoulli and Bell transfer at zeros

The exponent (3.4) is a Bernoulli–zeta series. Write locally

$$E_{N,r}(t) = \sum_{m \geq r+1} e_{r,m} u_N(t)^{2m}.$$

Then

$$\exp(E_{N,r}(t)) = 1 + \sum_{n \geq r+1} \frac{1}{n!} B_n(1!x_1, 2!x_2, \dots, n!x_n) u_N(t)^{2n}, \quad (3.10)$$

where  $x_m = e_{r,m}$  for  $m \geq r+1$  and  $x_m = 0$  below that order, and  $B_n$  is the complete exponential Bell polynomial. Bell transfer itself is established in the repository; the new point is that (3.10) now applies directly to the ratio of the first nonzero derivatives and to every zero-normalized local germ. No separate limiting argument at a zero is required.

## 4 Exact classification of same-parity nesting

### 4.1 Positive error magnitudes

For  $0 < a < 1$ , define

$$M_r(q, a) = \frac{q^{r(r+1)/2}}{(q; q)_r} \sum_{m=r+1}^{\infty} \frac{\zeta(2m)}{m} \frac{(q^{m-r}; q)_r}{1 - q^m} a^m. \quad (4.1)$$

Thus on the real tail disk,  $E_{N,r} = (-1)^r M_r(q, u_N(t)^2)$ . The established alternating bracket only compares opposite sides of the target. Comparing  $r$  with  $r+2$  is subtler.

**Lemma 4.1** (Shared-coefficient ratio). *For  $m \geq r+3$ , the ratio of the  $a^m$  coefficient in  $M_{r+2}$  to that in  $M_r$  is*

$$R_{r,m}(q) = \frac{q^{2r+3}(1 - q^{m-r-2})(1 - q^{m-r-1})}{(1 - q^{r+1})(1 - q^{r+2})}. \quad (4.2)$$

For fixed  $q, r$ , this is strictly increasing in  $m$  and

$$R_{r,\infty}(q) = \frac{q^{2r+3}}{(1 - q^{r+1})(1 - q^{r+2})}. \quad (4.3)$$

Moreover,

$$R_{r,\infty}(q) \leq 1 \iff q^{r+1}(1 + q) \leq 1. \quad (4.4)$$

*Proof.* The prefactor ratio is

$$\frac{q^{(r+2)(r+3)/2}/(q; q)_{r+2}}{q^{r(r+1)/2}/(q; q)_r} = \frac{q^{2r+3}}{(1 - q^{r+1})(1 - q^{r+2})}.$$

The quotient of shifted  $q$ -Pochhammer factors contributes the two factors in the numerator of (4.2); all remaining factors, including  $\zeta(2m)/m$  and  $1 - q^m$ , cancel. Monotonicity and the limit are immediate. Finally,

$$q^{2r+3} \leq (1 - q^{r+1})(1 - q^{r+2}) = 1 - q^{r+1} - q^{r+2} + q^{2r+3}$$

is equivalent to  $q^{r+1}(1 + q) \leq 1$ .  $\square$

## 4.2 Boundary asymptotics

**Lemma 4.2** (Accelerated boundary decomposition). *Let*

$$A_r(q) = \frac{q^{r(r+1)/2}}{(q; q)_r}, \quad U_{r,m}(q) = \frac{(q^{m-r}; q)_r}{1 - q^m}.$$

*Then*

$$\frac{M_r(q, a)}{A_r(q)} = -\log(1 - a) - \sum_{m=1}^r \frac{a^m}{m} + H_r(q, a), \quad (4.5)$$

*where*

$$H_r(q, a) = \sum_{m \geq r+1} \frac{\zeta(2m)U_{r,m}(q) - 1}{m} a^m \quad (4.6)$$

*is analytic in a disk strictly larger than the unit disk. In particular,*

$$M_r(q, a) = A_r(q) \log \frac{1}{1 - a} + O_{q,r}(1) \quad (a \uparrow 1). \quad (4.7)$$

*Proof.* The factor  $U_{r,m}(q)$  approaches one with error  $O_{q,r}(q^m)$ , while  $\zeta(2m) - 1 = O(4^{-m})$ . Thus the coefficient in (4.6) decays geometrically. Splitting one from  $\zeta(2m)U_{r,m}$  leaves the harmonic tail  $\sum_{m \geq r+1} a^m/m$ , which is the first two terms in (4.5).  $\square$

The decomposition is also the stable numerical formula used in the supplied code; it avoids summing a nearly harmonic series when  $a$  is close to one.

## 4.3 Necessary and sufficient condition

**Theorem 4.3** (Complete same-parity nesting classification). *Fix  $q \in (0, 1)$  and  $r \geq 0$ . Then*

$$M_{r+2}(q, a) < M_r(q, a) \quad \text{for every } a \in (0, 1) \quad (4.8)$$

*if and only if*

$$\boxed{q^{r+1}(1 + q) \leq 1.} \quad (4.9)$$

*If (4.9) fails, there is a unique number  $a_r(q) \in (0, 1)$  such that*

$$\begin{cases} M_{r+2}(q, a) < M_r(q, a), & 0 < a < a_r(q), \\ M_{r+2}(q, a) = M_r(q, a), & a = a_r(q), \\ M_{r+2}(q, a) > M_r(q, a), & a_r(q) < a < 1. \end{cases} \quad (4.10)$$

*The crossing is simple, and  $a_r(q)$  is real-analytic in  $q$  throughout the supercritical region  $q^{r+1}(1 + q) > 1$ .*

*Proof.* Suppose first that (4.9) holds. By lemma 4.1,  $R_{r,m}(q) < 1$  for every finite shared index  $m \geq r + 3$ ; the inequality is strict even at the limiting threshold. Therefore every shared coefficient of  $M_{r+2}$  is strictly smaller than its counterpart in  $M_r$ , while  $M_r$  also has the two additional positive coefficients at  $m = r + 1, r + 2$ . This proves (4.8) coefficientwise.

Conversely, if the threshold fails, then  $A_{r+2}(q)/A_r(q) = R_{r,\infty}(q) > 1$ . By lemma 4.2,

$$M_r(q, a) - M_{r+2}(q, a) = (A_r(q) - A_{r+2}(q)) \log \frac{1}{1 - a} + O(1),$$

which tends to  $-\infty$  as  $a \uparrow 1$ . Near  $a = 0$ , the difference is positive because its leading term is the  $a^{r+1}$  term present only in  $M_r$ . Thus at least one crossing exists.

For uniqueness, [lemma 4.1](#) shows that the coefficient difference has exactly one sign transition: positive coefficients first, followed eventually by negative coefficients. Let  $K$  be the final index with nonnegative coefficient. If  $D(a) = M_r(q, a) - M_{r+2}(q, a) = \sum c_m a^m$ , then

$$\frac{d}{da}(a^{-K} D(a)) = \sum_m (m - K) c_m a^{m-K-1} < 0,$$

because  $(m - K)c_m \leq 0$  on both sides of the transition, with at least one strict term. Hence  $a^{-K} D(a)$  is strictly decreasing and has at most one zero. The crossing is simple; at its zero, the displayed derivative equals  $a^{-K} D'(a) < 0$ . The implicit-function theorem then gives real analyticity in  $q$ .  $\square$

*Remark 4.4.* The theorem is stronger than a sufficient small-tail estimate. It identifies the exact parameter boundary, and when that boundary is crossed it describes the only possible failure mode: one and only one reversal near the first omitted zero.

#### 4.4 Golden-ratio threshold and eventual nesting

**Corollary 4.5** (All levels nested). *Same-parity nesting holds for every order  $r \geq 0$  if and only if*

$$q(1 + q) \leq 1, \quad (4.11)$$

that is,

$$q \leq \frac{\sqrt{5} - 1}{2} = \varphi^{-1}, \quad \text{or equivalently} \quad b \geq \sqrt{\varphi} = 1.2720196495 \dots \quad (4.12)$$

*Proof.* The left side of (4.9) decreases with  $r$ , so order zero is the only binding constraint. Solve  $q^2 + q - 1 \leq 0$  and use  $q = b^{-2}$ .  $\square$

**Corollary 4.6** (First uniformly nested order). *For every  $q \in (0, 1)$ , same-parity nesting is automatic from the finite order*

$$r_*(q) = \max \left\{ 0, \left\lceil \frac{\log(1 + q)}{-\log q} \right\rceil - 1 \right\} \quad (4.13)$$

onward.

Thus even a geometric base arbitrarily close to one eventually enters the nested regime.

#### 4.5 Threshold curve and its asymptotics

Let  $q_r$  be the unique solution of

$$q_r^{r+1}(1 + q_r) = 1. \quad (4.14)$$

The first values are shown in [Table 5](#).

Table 5: Exact same-parity threshold  $q_r$  and equivalent base  $b_r = q_r^{-1/2}$ .

| $r$ | $q_r$             | $b_r$            |
|-----|-------------------|------------------|
| 0   | 0.618033988749895 | 1.27201964951407 |
| 1   | 0.754877666246693 | 1.15096392525776 |
| 2   | 0.819172513396164 | 1.10487288165008 |
| 3   | 0.856674883854503 | 1.08041842739812 |
| 4   | 0.881271461633570 | 1.06523431150218 |
| 5   | 0.898653712628699 | 1.05488183427278 |

**Proposition 4.7** (Large-order threshold). *Put  $n = r + 1$  and  $L = \log 2$ . Then*

$$q_r = 1 - \frac{L}{n} + \frac{L(L+1)}{2n^2} - \frac{L(4L^2 + 15L + 6)}{24n^3} + O(n^{-4}), \quad (4.15)$$

$$b_r = q_r^{-1/2} = 1 + \frac{L}{2n} + \frac{L(L-2)}{8n^2} + O(n^{-3}). \quad (4.16)$$

*Proof.* Insert  $q = 1 + c_1 n^{-1} + c_2 n^{-2} + c_3 n^{-3} + \dots$  into  $n \log q + \log(1 + q) = 0$  and compare powers of  $n^{-1}$ . This gives  $c_1 = -L$ ,  $c_2 = L(L+1)/2$ , and  $c_3 = -L(4L^2 + 15L + 6)/24$ . The binomial expansion of  $q^{-1/2}$  gives the second formula.  $\square$

## 5 The dyadic Fabius–Rvachev product

### 5.1 Normalization and integer zeros

Under the Fourier convention  $\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx$ , write

$$\Phi(x) = \widehat{\text{up}}(x) = \prod_{h=0}^{\infty} \text{sinc}(\pi x / 2^h), \quad \text{sinc } z = \frac{\sin z}{z}. \quad (5.1)$$

This is the  $b = 2$  case, so  $q = 1/4$ . Its nonzero zeros are precisely the integers, with

$$\text{ord}_{x=m} \Phi = 1 + \nu_2(|m|), \quad m \in \mathbb{Z} \setminus \{0\}. \quad (5.2)$$

The sign on  $(m, m+1)$  is the signed Thue–Morse value  $(-1)^{s_2(m)}$  for  $m \geq 0$ ; these are established inputs from the corpus.

Let

$$\Phi_N(x) = \prod_{h=0}^N \text{sinc}(\pi x / 2^h), \quad \mathcal{T}_N(x) = \prod_{h=N+1}^{\infty} \text{sinc}(\pi x / 2^h) = \Phi\left(\frac{x}{2^{N+1}}\right). \quad (5.3)$$

Thus  $\Phi = \Phi_N \mathcal{T}_N$  everywhere, and  $\mathcal{T}_N = \Phi / \Phi_N$  only where  $\Phi_N \neq 0$ . The tail is zero-free and positive on  $|x| < 2^{N+1}$ . Define the factorized transforms  $\widehat{\Phi}_{N,r}$  exactly as in [definition 3.1](#), using relative partial tails and the weights at  $q = 1/4$ .

### 5.2 Two monotone zero-preserving chains

**Theorem 5.1** (Dyadic nested brackets). *For every  $N \geq 0$  and every real  $0 < |x| < 2^{N+1}$ ,*

$$\widehat{\mathcal{T}}_{N,1}(x) < \widehat{\mathcal{T}}_{N,3}(x) < \dots < \mathcal{T}_N(x) < \dots < \widehat{\mathcal{T}}_{N,4}(x) < \widehat{\mathcal{T}}_{N,2}(x) < \widehat{\mathcal{T}}_{N,0}(x), \quad (5.4)$$

*with the evident finite interpretation. On the whole interval  $|x| < 2^{N+1}$ , including  $x = 0$  and the prefix zeros, multiplication by  $|\Phi_N(x)|$  gives the corresponding weak chain for the absolute values of the full products. Every displayed comparison in that full-product chain is strict exactly when  $x \notin \mathbb{Z}$ ; when  $x \in \mathbb{Z}$ , all its terms are equal (to 1 at  $x = 0$ , and to 0 otherwise). Every transform has exactly the same integer zeros and multiplicities as  $\Phi$  inside the interval.*

*Proof.* The exact error is  $(-1)^r M_r(1/4, a)$  with  $a = (x/2^{N+1})^2$ . For  $0 < |x| < 2^{N+1}$  one has  $0 < a < 1$ , so opposite parities lie on opposite sides of one. Since  $(1/4)^{r+1}(1 + 1/4) < 1$  for every  $r$ , [theorem 4.3](#) makes each parity monotone, proving the strict tail chain. At  $x = 0$ , every exact and extrapolated tail equals 1. The prefix  $\Phi_N$  is nonzero exactly at the nonintegers and at zero: each of its zeros is a nonzero integer, while its  $h = 0$  factor vanishes at every nonzero integer. Multiplication by  $|\Phi_N|$  therefore preserves every strict comparison precisely at noninteger  $x$  and turns the whole chain into equality at nonzero integers. Zero preservation and its multiplicity statement follow from [theorem 3.2](#).  $\square$

**Corollary 5.2** (Certified derivatives at all included integer zeros). *Let  $m \neq 0$  be an integer with  $|m| < 2^{N+1}$ , and put  $\mu = 1 + \nu_2(|m|)$ . Then*

$$\frac{\widehat{\Phi}_{N,r}^{(\mu)}(m)}{\Phi^{(\mu)}(m)} = \exp\left((-1)^r M_r\left(\frac{1}{4}, \frac{m^2}{2^{2N+2}}\right)\right). \quad (5.5)$$

*Hence the odd orders give increasing lower bounds and the even orders decreasing upper bounds for  $|\Phi^{(\mu)}(m)|$ . The sign of the derivative is reproduced exactly.*

This supplies a certified local spectral evaluator at every integer zero. Earlier zero-jet formulae in the repository are exact symbolic identities; the present result is complementary: it gives monotone finite-product approximants with an explicit nonasymptotic error bound.

### 5.3 Exact preservation of Thue–Morse lobe signs

Choose  $N$  so that a compact real interval lies inside  $(-2^{N+1}, 2^{N+1})$ . All factorized tails are positive there. Therefore the sign of every  $\widehat{\Phi}_{N,r}$  is carried entirely by the finite prefix  $\Phi_N$ , exactly as for  $\Phi$ . In particular, on each lobe  $(m, m+1)$ ,

$$\operatorname{sgn} \widehat{\Phi}_{N,r}(x) = \operatorname{sgn} \Phi(x) = (-1)^{s_2(m)}. \quad (5.6)$$

Thus the extrapolation does not merely place zeros correctly; it preserves the automatic sign coding between them.

### 5.4 Low-order dyadic weights

The first filters are

$$\begin{aligned} r = 0 : & \quad (1), \\ r = 1 : & \quad \left(-\frac{1}{3}, \frac{4}{3}\right), \\ r = 2 : & \quad \left(\frac{1}{45}, -\frac{4}{9}, \frac{64}{45}\right), \\ r = 3 : & \quad \left(-\frac{1}{2835}, \frac{4}{135}, -\frac{64}{135}, \frac{4096}{2835}\right). \end{aligned}$$

Their bounded total variation, proved in the source draft, explains why high order remains numerically usable in the dyadic case. The new nesting theorem adds a separate structural fact: increasing the order by two can never worsen the exact real-axis bracket anywhere inside the first omitted zero.

### 5.5 Adaptive global evaluation

**Algorithm 5.3** (Zero-safe evaluator). To evaluate  $\Phi$  and its local zero-normalized germ on a compact set  $K \subset \mathbb{C}$ :

1. Choose  $N$  so that  $K \subset \{|x| < 2^{N+1}\}$ .
2. Evaluate the finite prefix  $\Phi_N$  directly, retaining its zeros symbolically or with exact integer factor detection.
3. Form the positive/analytic relative tails  $\mathcal{T}_{N,j}$  for  $0 \leq j \leq r$ .
4. Apply the  $q = 1/4$  weights to their logarithms and exponentiate.
5. Multiply by the prefix. Use (3.6) for a complex disk certificate, or the nested real chains for a two-sided certificate.

The method is adaptive in  $N$  and  $r$ . Increasing  $N$  enlarges the exactly represented zero divisor and reduces the tail variable; increasing  $r$  cancels additional Bernoulli–zeta powers. Near a zero, no subtraction or logarithm of a small full product occurs.

## 6 Structural connections and extensions

### 6.1 $q$ -binomial interpolation is the common finite engine

The exact threshold in [theorem 4.3](#) is independent of the zeta factor. It comes entirely from ratios of the Gaussian  $q$ -binomial Lagrange filters. The special functions do two different jobs:

- Bernoulli numbers and zeta values encode the analytic logarithm of one sinc factor;
- $q$ -Pochhammer symbols encode cancellation across geometric scales.

Their combination makes the error coefficients positive up to the global sign  $(-1)^r$ , which is what turns an extrapolation formula into a rigorous bracket.

**Proposition 6.1** (Universal sufficient criterion for positive logarithmic masks). *Suppose an even analytic factor  $f$  satisfies*

$$-\operatorname{Log} f(z) = \sum_{m \geq 1} c_m z^{2m}, \quad c_m \geq 0,$$

*in a disk about zero, and form the same geometric infinite product and  $q$ -Richardson transform. Then  $q^{r+1}(1+q) \leq 1$  is sufficient for same-parity coefficientwise nesting at order  $r$ , independently of the values of the nonnegative coefficients  $c_m$ .*

*Proof.* The common factor  $c_m$  cancels from the coefficient ratio, leaving exactly (4.2). The proof of the sufficient half of [theorem 4.3](#) applies verbatim.  $\square$

Necessity requires information about the boundary growth of the coefficient series. For sinc, the  $1/m$  tail supplied by  $\zeta(2m) \rightarrow 1$  makes the criterion exact.

### 6.2 Canonical products and spectral determinants

The corpus identifies  $\Phi$  with a dyadically self-similar Fredholm determinant and gives its weighted integer canonical product. In that language,  $\Phi_N$  is a finite collection of spectral layers, while  $\mathcal{T}_N$  is a zero-free determinant tail on the resolvent disk preceding its first eigenvalue. [Definition 3.1](#) can therefore be read as a determinant-preserving extrapolation: the finite spectrum is exact, and only the regular determinant of the unresolved spectral tail is accelerated.

This viewpoint suggests applications to the Pascal–Rvachev hierarchy, whose products have higher dyadic multiplicity weights. The zero factor should again be separated first; the remaining logarithm has positive coefficients with polynomial scale multiplicities, leading to multivariate or confluent  $q$ -Richardson filters.

### 6.3 Connection to inverse-Fabius and Lambert phases

The inverse-Fabius endpoint expansion in the corpus is organized around a lower-branch Lambert coordinate with a smooth periodic perturbation. Its phase-locked reciprocal-grid Richardson transform is already present, so it is not repeated here. The useful new principle is instead modular:

$$\text{factor exact singular/zero structure} \quad + \quad \text{extrapolate only the regular tail.} \quad (6.1)$$

For the Fourier product, the exact structure is a canonical zero divisor. For inverse endpoint saddles, it is the Lambert branch and the periodic phase. A hybrid algorithm can therefore preserve both: exact Lambert phase locking in the outer inversion, and exact integer-zero factoring in every Fourier amplitude entering the coefficient calculation.

## 6.4 What the theorem does not say

The nesting theorem concerns exact arithmetic. It does not assert that arbitrarily high  $r$  is optimal in floating-point arithmetic. The dyadic weights have bounded total variation, but the relative-tail logarithms become increasingly correlated and small; roundoff, argument reduction, and evaluation cost eventually dominate. Choosing the precision-aware optimal order remains a separate numerical-analysis problem.

## 7 Numerical experiments

### 7.1 Reproducibility

The retained program `assets/Fabius_Rvachev_Frontier_Report-3/code/frontier_experiments.py` uses 60-decimal-digit `mpmath` arithmetic and performs the following independent checks:

1. direct verification of all finite  $q$ -Lagrange moment identities used in the tables;
2. direct zero-free tail products versus the exact Bernoulli–zeta transformed series;
3. bisection of the exact threshold equation and the unique supercritical crossing;
4. derivative-ratio brackets at a multiple integer zero;
5. generation of every CSV table and PNG figure in this report.

The slowly convergent harmonic part of  $M_r$  is removed analytically by [lemma 4.2](#); only a geometrically convergent correction is summed.

### 7.2 Dyadic nesting across the full tail disk

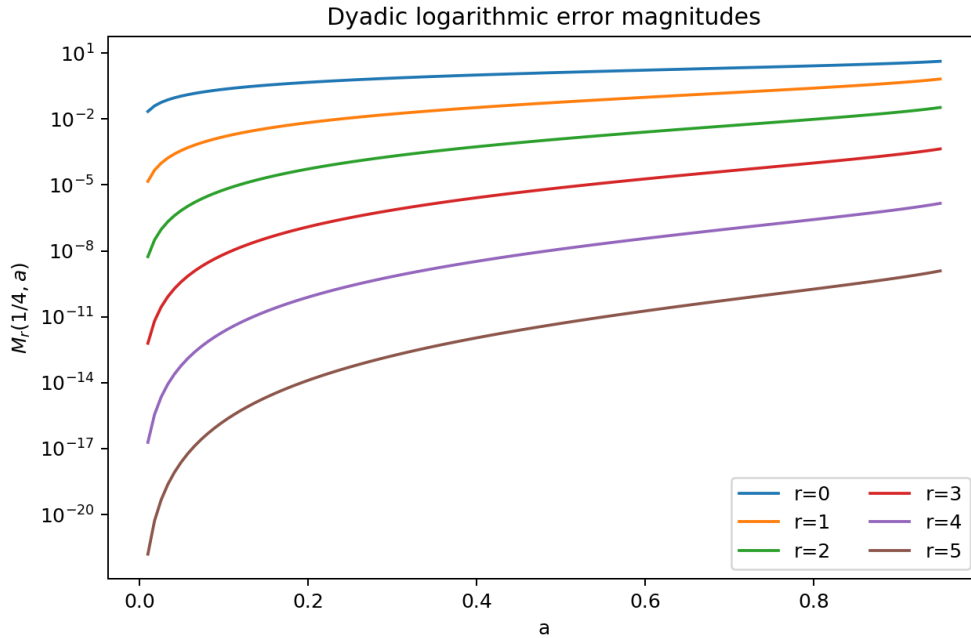


Figure 1: The dyadic magnitudes  $M_r(1/4, a)$  for  $0.01 \leq a \leq 0.95$ . Each parity is strictly nested over the whole interval, as guaranteed by [theorem 4.3](#). The vertical scale is logarithmic.



The separation is dramatic: cancellation order increases by two powers of  $a$  each time, while the prefactor contributes the additional supergeometric factor  $q^{r(r+1)/2}$ .

### 7.3 Unique reversal beyond the threshold

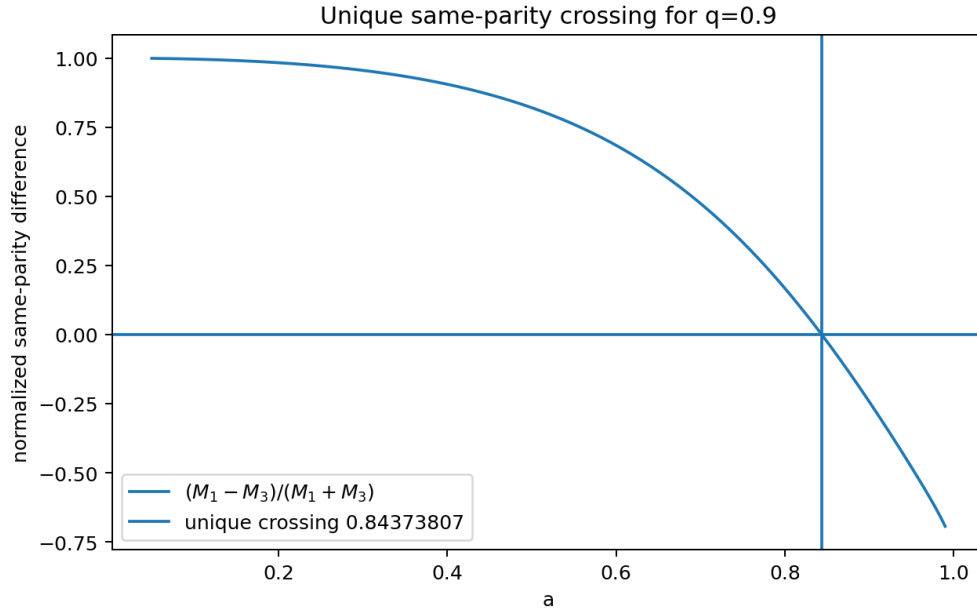


Figure 2: For  $q = 0.9$  and  $r = 1$ , uniform nesting fails. The normalized difference between  $M_1$  and  $M_3$  crosses zero exactly once, at  $a_1(0.9) = 0.843738074718284\dots$

Representative crossing values are

| $q$ | $r$ | $a_r(q)$             |
|-----|-----|----------------------|
| 0.7 | 0   | 0.999706476977084941 |
| 0.8 | 0   | 0.982411424014939233 |
| 0.8 | 1   | 0.980062142113792208 |
| 0.9 | 0   | 0.938630662551898552 |
| 0.9 | 1   | 0.843738074718284012 |
| 0.9 | 2   | 0.847999350512777102 |
| 0.9 | 3   | 0.887485076384088108 |

The roots are numerical illustrations only; existence and uniqueness are proved without numerics.

### 7.4 A multiplicity-five zero

Take  $m = 48$ , whose zero multiplicity is  $1 + \nu_2(48) = 5$ , and choose  $N = 5$ , so  $|m| < 2^6$ . The exact positive tail is

$$\mathcal{T}_5(48) = \Phi(48/64) = 0.217723205308944050125343\dots$$

By (5.5), every tail ratio below is exactly the ratio of the first nonzero derivative of the approximation to  $\Phi^{(5)}(48)$ .

| $r$ | derivative ratio                | signed log error                  | side  |
|-----|---------------------------------|-----------------------------------|-------|
| 0   | 4.592987681680617293            | $+1.524530723363272884$           | upper |
| 1   | 0.922841469541933824            | $-8.0297814873516 \cdot 10^{-2}$  | lower |
| 2   | 1.001952802143219852            | $+1.9508979037801 \cdot 10^{-3}$  | upper |
| 3   | 0.999986570110615447            | $-1.3429979566325 \cdot 10^{-5}$  | lower |
| 4   | 1.000000024414975206            | $+2.4414974907466 \cdot 10^{-8}$  | upper |
| 5   | 0.9999999998534678              | $-1.1465321627915 \cdot 10^{-11}$ | lower |
| 6   | 1.000000000000001377            | $+1.3766098031980 \cdot 10^{-15}$ | upper |
| 8   | 1.00000000000000000000000000325 | $+3.24694 \cdot 10^{-25}$         | upper |

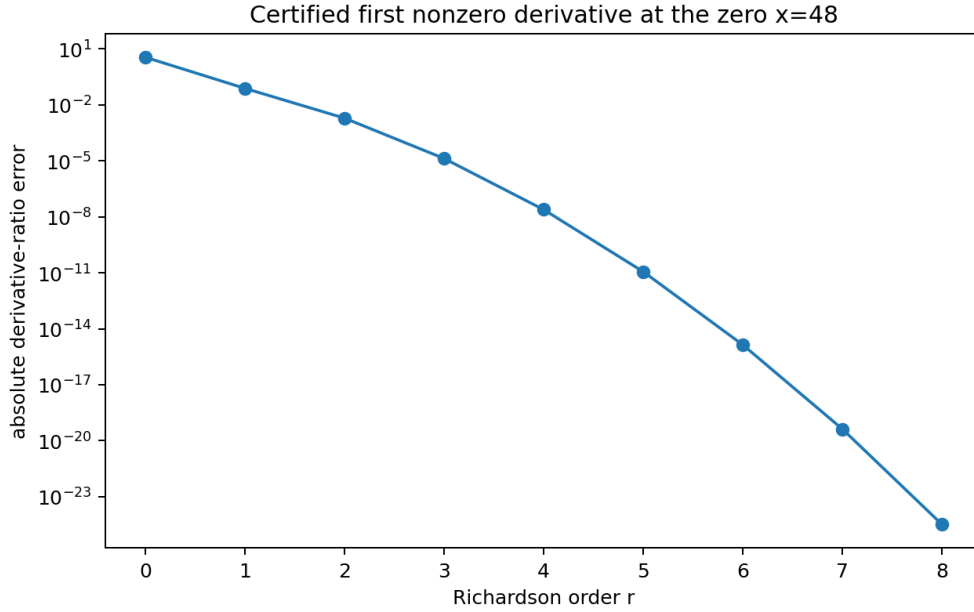


Figure 3: Absolute error in the certified first-nonzero-derivative ratio at the integer zero  $x = 48$ . The factorized construction is well-conditioned here because no logarithm of the vanishing full product is taken.

## 8 Conjectures and frontier directions

The two exact theorems above close specific open items in the current repository, but they also expose several more structured questions.

**Conjecture 8.1** (Regular-variation universality). Let an even analytic refinement factor satisfy  $-\text{Log } f(z) = \sum_{m \geq 1} c_m z^{2m}$  with  $c_m > 0$  and  $c_m \sim C \rho^{-m}/m$  for some  $C, \rho > 0$ . After normalizing the tail variable so that the first logarithmic singularity occurs at  $a = 1$ , the exact same-parity classification remains  $q^{r+1}(1+q) \leq 1$ , with one unique crossing in the supercritical regime.

The sinc proof strongly suggests this: coefficientwise sufficiency is already universal by [proposition 6.1](#), and regular variation should reproduce the logarithmic boundary term needed for necessity.

**Conjecture 8.2** (Monotonicity of the crossing curve). For each fixed  $r$ , the supercritical crossing  $a_r(q)$  decreases from 1 as  $q \downarrow q_r$  and has a finite, strictly positive limit as  $q \uparrow 1$ . A useful sharpening would determine the two endpoint asymptotic expansions explicitly.

The first limit follows qualitatively from compact convergence and the threshold theorem; the monotonicity and the  $q \uparrow 1$  scale remain to be proved. The numerical values show that dependence on  $r$  is not simply monotone, so a two-parameter saddle analysis is likely needed.

*Problem 8.3* (Optimal finite-precision order). Given  $q$ , tail radius  $a$ , and working precision  $p$ , determine the order  $r$  minimizing a rigorous sum of truncation, logarithm-evaluation, argument-reduction, and rounding errors. For  $q = 1/4$ , the exact weight condition number remains bounded, so one expects an optimal order growing roughly like a square root of  $p$  after the supergeometric  $q^{r(r+1)/2}$  factor is balanced against finite-precision noise.

*Problem 8.4* (Complex boundary continuation). The factorized transform is optimal inside the first omitted zero circle. Extend it across that boundary by moving the newly encountered canonical factors from the tail into the prefix. The goal is an analytic continuation algorithm with overlapping disks, explicit transition maps, and uniform Cauchy certificates on arbitrary compact subsets of  $\mathbb{C}$ .

This should be feasible because adjacent factorizations differ by finitely many exact sinc factors. The delicate point is choosing normalized logarithms coherently on overlaps.

*Problem 8.5* (Pascal–Rvachev hierarchy). For the hierarchy whose Fourier products carry Pascal weights at dyadic scales, construct confluent or multivariate  $q$ -Richardson filters that cancel the polynomial-in-scale multiplicity structure. Determine whether their same-parity thresholds factor into simple inequalities analogous to  $q^{r+1}(1+q) \leq 1$ .

*Problem 8.6* (Inverse-Fabius coupling). Use zero-preserving Fourier brackets as certified inputs to the inverse-Fabius saddle calculation. The target is a fully rigorous end-to-end interval algorithm for inverse values in which Lambert phase, periodic corrections, Fourier amplitudes, and all local zero jets carry compatible error enclosures.

*Conjecture 8.7* (Resurgent organization of phase coefficients). The all-orders inverse endpoint coefficients, after exact phase locking, admit a Borel transform whose nearest singularities are controlled by the same dyadic complex lattice that governs the Mellin poles and zero-divisor heat trace. If true, optimal truncation and exponentially improved remainders would be determined by one shared spectral geometry.

*Problem 8.8* (Automatic arithmetic of extrapolated jets). At integer and half-integer arguments, investigate whether suitably normalized coefficients of the factorized approximants form 2-regular or 2-automatic arrays. The finite prefix has explicit Thue–Morse/Pascal arithmetic, while the tail weights are Gaussian  $q$ -binomials at  $q = 1/4$ ; their interaction may produce new exact recurrences.

## 9 Lean formalization blueprint

The proof dependencies are unusually clean. A sensible formalization order is:

1. finite  $q$ -Pochhammer algebra and the closed weights (2.8);
2. the transformed-moment identity (2.9);
3. positivity and monotonicity of  $R_{r,m}$ ;
4. the algebraic equivalence (4.4);
5. coefficientwise sufficiency;
6. the accelerated boundary decomposition (4.5);
7. the one-sign-change power-series lemma and unique crossing;
8. analytic zero-free logarithms on a disk;
9. the factorized transform and common-zero-divisor theorem;
10. specialization to the existing Rvachev Fourier product and integer multiplicity API.

The same-parity theorem is mostly finite algebra plus positive series and should be the easier half. The factorized continuation needs complex analysis but avoids branch ambiguity by using a simply connected disk and logarithms normalized at zero. Existing Lean declarations for the Rvachev product, scaling, and integer zeros should make the final specialization short.

## 10 Conclusion

The mature Fabius-function corpus leaves little room for novelty by relabeling familiar identities. The productive route was to take two explicit open defects in its newest  $q$ -Richardson analysis seriously.

Factoring the finite prefix changes the analytic status of the extrapolator: it becomes a holomorphic, zero-divisor-preserving approximation on the entire natural disk, not merely a formula between zeros. The error series survives unchanged, so all Bernoulli, Bell, and  $q$ -binomial structure transfers directly to local spectral jets. Separately, comparing same-parity coefficient arrays yields a complete and unexpectedly simple threshold,  $q^{r+1}(1+q) \leq 1$ , together with a one-crossing theorem when it fails.

For the dyadic Rvachev product the consequence is especially clean. Since  $q = 1/4$  lies far inside the golden-ratio threshold, every order participates in two monotone analytic chains. They squeeze the product, preserve every included integer zero and its Thue–Morse lobe sign, and certify the first nonzero derivative at multiple zeros to arbitrarily high precision. This turns the existing extrapolation machinery into a global spectral tool and provides a compact target for formal verification.

## A Complete recursive TeX manifest

The following manifest is included verbatim from `assets/Fabius_Rvachev_Frontier_Report-3/corpus_manifest.txt`. The first four metadata lines are followed by the 68 unique paths used in the audit.

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 Analysis/FabiusFunction/docs/archive/research-frontiers/Fabius\_Dyadic\_Formulae\_to\_Asymptotics/Fabius\_Dyadic\_Formulae\_to\_Asymptotics.tex  
 Analysis/FabiusFunction/docs/archive/research-frontiers/Fabius\_Dyadic\_q\_Connections/Fabius\_Dyadic\_q\_Connections.tex  
 Analysis/FabiusFunction/docs/archive/research-frontiers/Fabius\_Thue\_Morse\_Convergence\_Rate/Fabius\_Thue\_Morse\_Convergence\_Rate.tex  
 Analysis/FabiusFunction/docs/archive/research-frontiers/Fabius\_Thue\_Morse\_Convergence\_Rates/Fabius\_Thue\_Morse\_Convergence\_Rates.tex  
 Analysis/FabiusFunction/docs/archive/research-frontiers/Primary\_Exposition\_Gap\_Register/Primary\_Exposition\_Gap\_Register.tex  
 Analysis/FabiusFunction/docs/archive/research-frontiers/Repeated\_Integration\_and\_Rvachev\_Up/Repeated\_Integration\_and\_Rvachev\_Up.tex  
 Analysis/FabiusFunction/docs/archive/research-frontiers/Rvachev\_Up\_from\_Repeated\_Integration/Rvachev\_Up\_from\_Repeated\_Integration.tex  
 Analysis/FabiusFunction/docs/archive/research-frontiers/Thue\_Morse\_Formula\_Atlas-1/Thue\_Morse\_Formula\_Atlas.tex  
 Analysis/FabiusFunction/docs/archive/research-frontiers/Thue\_Morse\_Formula\_Atlas-2/Consolidated\_Formula\_Atlas\_for\_the\_Thue\_Morse\_Sequence.tex  
 Analysis/FabiusFunction/docs/archive/research-frontiers/Thue\_Morse\_Formula\_Atlas-3/Thue\_Morse\_Formula\_Atlas.tex  
 Analysis/FabiusFunction/docs/archive/research-frontiers/non-formalized-research-frontiers/semi-formalized-research-frontiers.tex  
 Analysis/FabiusFunction/docs/archive/standalone-studies/Non\_Elementarity\_of\_the\_Fabius\_Function/Non\_Elementarity\_of\_the\_Fabius\_Function.tex  
 Analysis/FabiusFunction/docs/archive/standalone-studies/Small\_Argument\_Asymptotics/Small\_Argument\_Asymptotics.tex  
 Analysis/FabiusFunction/docs/archive/standalone-studies/fabius-inverse-dyadic-closed-form/fabius-inverse-dyadic-closed-form.tex  
 Analysis/FabiusFunction/docs/papers/AIF\_2017\_\_67\_2\_637\_0/AIF\_2017\_\_67\_2\_637\_0-corrected.tex

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*Recursive TeX manifest (continued)*


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Analysis/FabiusFunction/docs/papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0.tex
Analysis/FabiusFunction/docs/papers/arXiv-1502.06738v2/Lacunary_products.tex
Analysis/FabiusFunction/docs/papers/arXiv-1702.05442v1/09-Function.tex
Analysis/FabiusFunction/docs/papers/arXiv-1702.06487v3/157-Arithmetic-v3.tex
Analysis/FabiusFunction/docs/papers/arXiv-2211.13570v1/document.tex
Analysis/FabiusFunction/docs/papers/ubiq15/ubiq15-corrected.tex

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**B Reproduction files**

The former standalone package’s retained auxiliary payload is stored under `assets/Fabiús_Rvachev_Frontier_Report-3/`. The report source has been absorbed into `Frontier_Compilations.tex`, and its former standalone PDF is not retained. Relative to that asset directory, the retained files are:

| Path                                                  | Purpose                                                        |
|-------------------------------------------------------|----------------------------------------------------------------|
| <code>code/frontier_experiments.py</code>             | Fully commented numerical experiment and figure generator.     |
| <code>data/same_parity_thresholds.csv</code>          | Threshold roots $q_r$ and bases $b_r$ .                        |
| <code>data/same_parity_crossings.csv</code>           | Representative unique crossing roots.                          |
| <code>data/dyadic_q_richardson_weights.csv</code>     | Low-order geometric Lagrange weights.                          |
| <code>data/integer_zero_jet_brackets.csv</code>       | Certified derivative-ratio experiment at $x = 48$ .            |
| <code>data/nonzero_point_factorized_errors.csv</code> | Direct product/series cross-check away from zeros.             |
| <code>figures/*.png</code>                            | The three figures embedded above.                              |
| <code>corpus_manifest.txt</code>                      | Pinned recursive source inventory.                             |
| <code>corpus_manifest_table.tex</code>                | Breakable LaTeX rendering of the inventory used by Appendix A. |
| <code>experiment_summary.txt</code>                   | Compact numerical run summary.                                 |
| <code>requirements.txt</code>                         | Python dependency bounds.                                      |
| <code>README.md</code>                                | Reproduction and file guide.                                   |
| <code>SHA256SUMS</code>                               | SHA-256 checksums for packaged files.                          |

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## Part IV

# Midpoint Transmutation, Dyadic Cardinal Reproduction, and Holonomic Obstructions

## Abstract

A recursive audit of the fifty-seven  $\text{\LaTeX}$  sources under `Analysis/FabiusFunction/docs` reveals a mature body of work on exact dyadic arithmetic, Thue–Morse prefix sums,  $q$ -binomial coefficient extraction, infinite sinc and cosine products, sharp Fourier and endpoint asymptotics, lower-branch Lambert– $W$  coordinates, Bell-polynomial saddle algebra, Lagrange interpolation, and Richardson acceleration. This report deliberately avoids reproducing those developments. Instead it develops three complementary branches that were not found in the audited corpus.

First, an exact identity transports the left-endpoint germ of the Fabius function to its midpoint:

$$F\left(\frac{1}{2} \pm h\right) = \frac{1}{2} \pm 2h \mp F(h), \quad 0 \leq h \leq \frac{1}{2}.$$

It follows that the Taylor jet of  $F$  at  $1/2$  is exactly affine although  $F$  is not analytic there. For the inverse  $I = F^{-1}$ , writing  $z = \delta/2$  and  $h = I(1/2 + \delta) - 1/2$  gives the exact fixed-point equation  $h = z + F(h)/2$ . The inverse defect  $E(\delta) = h - z$  is flat to every order and satisfies  $E(\delta) \sim F(\delta/2)/2$ . Therefore the repository’s complete lower-Lambert endpoint expansion transfers, coefficient for coefficient, to the first nontrivial correction of the inverse at its midpoint. A finite, rigorous Lagrange–Bell jet with an integral remainder is also obtained.

Second, dyadic dilations  $U_r(x) = 2^{-r} \text{up}(2^{-r}x)$  form a cardinal reproduction ladder. Their Fourier transforms have zeros of order at least  $r + 1$  at every nonzero integer, a property whose finite block is exactly the Prouhet cancellation of a length- $2^{r+1}$  signed Thue–Morse word. Consequently

$$\sum_{k \in \mathbb{Z}} (x - k)^j U_r(x - k) = \int_{\mathbb{R}} y^j U_r(y) \, dy, \quad 0 \leq j \leq r.$$

Reciprocal-moment Appell polynomials  $\mathcal{A}_n^{[r]}$  then give finite exact formulas  $\sum_k \mathcal{A}_n^{[r]}(k) U_r(x - k) = x^n$  for  $n \leq r$ . Their coefficients are explicit complete Bell polynomials in Bernoulli cumulants; their dyadic scale transport is governed by the Bernoulli-polynomial values  $B_n(1/2)$ ; and, because  $\mathcal{A}_n(x; Q)$  is polynomial in  $Q = 4^r$ , the scale sequence obeys exact  $q$ -binomial finite-difference closures at  $q = 1/4$ .

Third, two independent obstructions exclude holonomicity. The midpoint identity propagates nonanalyticity backward through the dyadic tent map, proving that  $F$  is nonanalytic at every dyadic rational; the same mechanism gives dense nonanalytic sets for  $F^{-1}$  and  $\text{up}$ . Hence none is locally  $D$ -finite on any open interval. Independently, the unbounded multiplicities  $\text{ord}_m \widehat{\text{up}} = 1 + \nu_2(|m|)$  rule out a polynomial-coefficient differential equation for the entire Fourier transform. This implies non- $D$ -finiteness of the moment generating functions, non- $P$ -recursiveness of the moment sequences, and non- $P$ -recursiveness of every fixed- $x$  Appell coefficient sequence. Symbolic and high-precision numerical experiments accompany the proofs, and the fully commented Python program is included in the archive.

### Priority and novelty disclaimer

The word *new* in this report means *not found in the audited repository snapshot*. It is not a claim of global historical priority. In particular, general Strang–Fix theory and nonstationary multiresolution analysis based on the  $\text{up}$ -function already exist in the approximation-theory literature [8, 7, 9]. The corpus-relative contribution here is the explicit dilation-indexed cardinal ladder, its Prouhet factorization, the Bernoulli–Bell coefficient formulas, and their exact  $q$ -scale closure, together with the midpoint/inverse and holonomic deductions.

# 1 Corpus audit and non-duplication protocol

## 1.1 What was audited

The recursive tree under `Analysis/FabiusFunction/docs` contains twenty-five living or current-tree TeX sources and thirty-two frozen archive sources [1, 2, 3, 4]. The living group includes the primary exposition, the non-elementarity article, the Lean walkthrough, the mathematical glossary, the consolidated non-formalized frontier, the living Thue–Morse Formula Atlas, twelve Fourier-decay drafts or audits, and seven transcribed research papers. The archive preserves early notes, parallel primary expositions, supplements, q-calculus articles, walkthroughs, glossaries, per-topic frontier dossiers, and three predecessor Atlas consolidations.

The archive ledger explains that current-head byte-identical copies were pruned and that the remaining documents are frozen historical versions. Accordingly, this audit used three levels of comparison:

1. direct reading of the living canonical exposition, consolidated frontier, Atlas, distinct Fourier-decay documents, and paper transcriptions;
2. topic-by-topic comparison with every archived entry and its stated scope;
3. targeted full-text searches for the candidate branches developed below, including close variants of “Strang–Fix”, “polynomial reproduction”, “inverse-moment Appell”, “D-finite”, “holonomic”, “P-recursive”, and the continuous midpoint identity of [theorem 3.1](#).

The complete path inventory is reproduced in [Section D](#) and included as a separate machine-readable file, `corpus_manifest.txt`.

## 1.2 The overlap map

The repository already treats the following subjects in substantial depth:

- exact rational values  $F(a/2^n)$ , triangular recurrences, composition sums, q-Pochhammer symbols, Gaussian binomials, and coefficient extraction;
- Thue–Morse product identities, Prouhet cancellation, iterated prefix sums, finite convolution laws, and sharp convergence rates;
- the sinc and cosine products for  $\widehat{\text{up}}$ , Fourier decay, zero sets, and refined log-periodic asymptotics;
- lower-branch Lambert–W coordinates for  $F(x)$  as  $x \downarrow 0$ , periodic saddle jets, and all-orders Bell-polynomial contraction formulas;
- exact and asymptotic Lagrange interpolation, q-interpolation, and Richardson-type acceleration at dyadic scales;
- repeated-integration and finite-box-spline approximants to  $\text{up}$ , including exact tail replacement and quantitative error decompositions;
- inverse-function basics, including  $I(1/2) = 1/2$ ,  $I'(1/2) = 1/2$ , and  $I''(1/2) = 0$ .

These are inputs, not claimed results of this report. In particular, the q-binomial weights in [Section 8](#) are familiar from the repository’s q-Richardson work; what is new here is that they annihilate a newly constructed, exact polynomial-in-scale Appell family rather than an asymptotic error model.

### Three principal corpus gaps

The audit did not reveal a treatment of:

1. the exact continuous midpoint–endpoint identity, its all-orders inverse consequence, or the transfer of endpoint Lambert–W asymptotics to the inverse midpoint defect;
2. a Strang–Fix/cardinal polynomial reproduction ladder for the dyadically dilated up-function, with reciprocal-moment Appell correctors and Bernoulli–Bell scale algebra;
3. D-finite and P-recursive obstructions derived simultaneously from dense dyadic nonanalyticity and unbounded Fourier-zero multiplicity.

### 1.3 Status labels used below

Gray boxes mark established inputs from the repository or standard literature. Green boxes mark deductions that appear new relative to the audited corpus. Orange boxes contain conjectures, extensions, or research programs whose proofs are not supplied. Numerical evidence is never used in place of proof.

## 2 Established framework and notation

### 2.1 The three functions

Let  $F : \mathbb{R} \rightarrow [0, 1]$  be the bounded Fabius distribution function, normalized by

$$F(x) = 0 \quad (x \leq 0), \quad F(x) = 1 \quad (x \geq 1),$$

and, on  $[0, 1]$ , by

$$F'(x) = 2F(2x) \quad \left(0 \leq x \leq \frac{1}{2}\right), \quad F(1-x) = 1 - F(x). \quad (2.1)$$

It is  $C^\infty$ , strictly increasing on  $(0, 1)$ , and flat at both endpoints. Let  $I = F^{-1} : (0, 1) \rightarrow (0, 1)$  be its ordinary inverse.

Rvachev’s even up-function  $U = \text{up}$  is supported on  $[-1, 1]$ , integrates to one, and satisfies [5, 6]

$$U(x) = F(1 - |x|), \quad |x| \leq 1. \quad (2.2)$$

Equivalently,  $F(x) = U(x - 1)$  on  $[0, 1]$ .

It is useful to write the symmetric tent map

$$\tau(x) = \begin{cases} 2x, & 0 \leq x \leq \frac{1}{2}, \\ 2 - 2x, & \frac{1}{2} \leq x \leq 1. \end{cases} \quad (2.3)$$

Differentiating the symmetry in (2.1) gives the global interior identity

$$F'(x) = 2F(\tau(x)), \quad 0 < x < 1. \quad (2.4)$$

### 2.2 Probability, Fourier, and moments

Use the Fourier convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx. \quad (2.5)$$

Let  $V_n$  be independent uniform variables on  $[-2^{-n-1}, 2^{-n-1}]$ . Then

$$X = \sum_{n=0}^{\infty} V_n \quad (2.6)$$

converges absolutely and has density  $U$ . Therefore

$$\widehat{U}(\xi) = \prod_{n=0}^{\infty} \frac{\sin(\pi\xi/2^n)}{\pi\xi/2^n} = \prod_{n=0}^{\infty} \operatorname{sinc}\left(\frac{\pi\xi}{2^n}\right), \quad (2.7)$$

$$= \prod_{m=1}^{\infty} \left( \cos \frac{\pi\xi}{2^m} \right)^m. \quad (2.8)$$

The moment generating function is the entire even function

$$\mathcal{M}_0(t) = \mathbb{E}e^{tX} = \prod_{n=0}^{\infty} \frac{\sinh(t/2^{n+1})}{t/2^{n+1}} = \widehat{U}\left(\frac{it}{2\pi}\right). \quad (2.9)$$

### 2.3 The imported endpoint coordinate

Put  $L = \log 2$  and define the lower-Lambert coordinate [11, 3]

$$\lambda(x) = -\frac{1}{L}W_{-1}(-Lx), \quad \lambda(x)2^{-\lambda(x)} = x. \quad (2.10)$$

The repository develops a complete endpoint expansion whose first structural layer has the form

$$\log F(x) = -\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda - \frac{1}{2}\log \lambda + C_F + \Psi(\lambda) + \sum_{j=1}^{N-1} \frac{A_j(\lambda)}{\lambda^j} + \mathcal{O}(\lambda^{-N}). \quad (2.11)$$

Here  $\lambda = \lambda(x)$ ;  $\Psi$  and the higher coefficient functions encode the dyadic phase and are periodic in the normalization of the frontier documents. The precise normalization of  $C_F$ ,  $\Psi$ , and  $A_j$  is left to those sources. The point of Section 4 is that every term in (2.11) transfers unchanged to an inverse-function defect after replacing  $x$  by  $\delta/2$  and subtracting  $\log 2$ .

#### Established input

Nothing in Section 2 is claimed as new. These identities fix conventions and isolate the exact inputs used later.

## 3 An exact midpoint–endpoint transmutation

**Theorem 3.1** (Midpoint–endpoint transmutation). *For every  $0 \leq h \leq 1/2$ ,*

$$F\left(\frac{1}{2} + h\right) = \frac{1}{2} + 2h - F(h), \quad (3.1)$$

$$F\left(\frac{1}{2} - h\right) = \frac{1}{2} - 2h + F(h). \quad (3.2)$$

*Proof.* Integrating the differential equation in (2.1) gives

$$F(h) = \int_0^h 2F(2t) dt = \int_0^{2h} F(u) du. \quad (3.3)$$

On the other hand,

$$\begin{aligned}
 F\left(\frac{1}{2}\right) - F\left(\frac{1}{2} - h\right) &= \int_{1/2-h}^{1/2} 2F(2t) \, dt \\
 &= \int_{1-2h}^1 F(u) \, du \\
 &= \int_0^{2h} F(1-v) \, dv \\
 &= 2h - \int_0^{2h} F(v) \, dv = 2h - F(h).
 \end{aligned}$$

Since  $F(1/2) = 1/2$ , this proves (3.2); symmetry gives (3.1).  $\square$

### Corpus-relative new deduction

The theorem says that the entire nontrivial midpoint germ is exactly the endpoint germ with a sign change after removal of the affine tangent. This is a functional identity, not the finite-approximant midpoint formula already present in the archived Thue–Morse convergence report.

### 3.1 The complete midpoint jet

**Corollary 3.2** (Affine Taylor jet and nonanalyticity). *The derivatives of  $F$  at the midpoint are*

$$F\left(\frac{1}{2}\right) = \frac{1}{2}, \quad F'\left(\frac{1}{2}\right) = 2, \quad F^{(m)}\left(\frac{1}{2}\right) = 0 \quad (m \geq 2). \quad (3.4)$$

Nevertheless,  $F$  is not real analytic at  $1/2$ .

*Proof.* The endpoint flatness  $F^{(m)}(0) = 0$  for every  $m \geq 0$  and (3.1) give (3.4). If  $F$  were analytic at  $1/2$ , its Taylor series would force  $F(1/2 + h) = 1/2 + 2h$  for all sufficiently small  $h$ , contradicting  $F(h) > 0$  in (3.1).  $\square$

Thus the flat endpoint term is invisible to every finite-order midpoint Taylor method. In numerical analysis language, it is a beyond-all-orders defect; in quasianalyticity language, the midpoint germ lies outside every class in which the Taylor map is injective.

### 3.2 Central integral identities

The transmutation has immediate exact integral consequences. For  $0 \leq a \leq 1/2$ ,

$$\int_{1/2-a}^{1/2+a} F(x) \, dx = a, \quad (3.5)$$

$$\int_0^a \left[ F\left(\frac{1}{2} + h\right) - \frac{1}{2} - 2h \right] \, dh = - \int_0^a F(h) \, dh. \quad (3.6)$$

The first is also a direct consequence of symmetry, while the second identifies the exact signed central curvature budget with the endpoint mass. Repeated integration of (3.1) similarly transports all endpoint primitives to the midpoint.

## 4 The inverse midpoint defect and its Lambert–W transfer

### 4.1 An exact fixed-point equation

Let  $I = F^{-1}$ . For  $0 \leq \delta < 1/2$  define

$$h(\delta) = I\left(\frac{1}{2} + \delta\right) - \frac{1}{2}, \quad z = \frac{\delta}{2}, \quad E(\delta) = h(\delta) - z. \quad (4.1)$$

Since  $F(1/2 + h) = 1/2 + \delta$ , [theorem 3.1](#) gives the exact equation

$$\delta = 2h - F(h), \quad h = z + \frac{1}{2}F(h), \quad E(\delta) = \frac{1}{2}F(z + E(\delta)). \quad (4.2)$$

For negative  $\delta$ , inverse symmetry gives

$$I\left(\frac{1}{2} + \delta\right) = \frac{1}{2} + \frac{\delta}{2} + \operatorname{sgn}(\delta)E(|\delta|). \quad (4.3)$$

**Theorem 4.1** (All-orders inverse midpoint flatness). *The inverse  $I$  is  $C^\infty$  on  $(0, 1)$  and*

$$I'\left(\frac{1}{2}\right) = \frac{1}{2}, \quad I^{(m)}\left(\frac{1}{2}\right) = 0 \quad (m \geq 2). \quad (4.4)$$

More strongly, for every  $N \geq 1$ ,

$$E(\delta) = \mathcal{O}(\delta^N) \quad (\delta \downarrow 0). \quad (4.5)$$

The inverse is not analytic at  $1/2$ .

*Proof.* The inverse is smooth because  $F'(x) > 0$  for  $0 < x < 1$ . From (4.2),  $h \rightarrow 0$  with  $\delta$  and

$$\frac{\delta}{h} = 2 - \frac{F(h)}{h} \rightarrow 2$$

because  $F$  is flat at zero. Hence  $h \sim \delta/2$ . The same equation yields  $E = F(h)/2 = \mathcal{O}(h^N) = \mathcal{O}(\delta^N)$  for every  $N$ . Equation (4.3) then gives the complete jet (4.4). Analyticity would force  $I(1/2 + \delta) = 1/2 + \delta/2$  locally, contradicting  $E(\delta) > 0$ .  $\square$

### 4.2 Certified monotone inversion

The fixed-point equation is not merely asymptotic; it gives a certified algorithm.

**Proposition 4.2** (Two-sided fixed-point iteration). *Fix  $0 < \delta < 1/2$ , put  $z = \delta/2$ , and define*

$$T_z(t) = z + \frac{1}{2}F(t), \quad b = z + \frac{1}{4}.$$

*Then  $T_z$  maps  $[z, b]$  into itself and is a contraction there with constant*

$$q_\delta = F\left(\frac{1}{2} + \delta\right) < 1. \quad (4.6)$$

*The lower iteration  $\ell_0 = z$ ,  $\ell_{n+1} = T_z(\ell_n)$  increases to  $h$ , while the upper iteration  $u_0 = b$ ,  $u_{n+1} = T_z(u_n)$  decreases to  $h$ . Moreover,*

$$0 \leq h - \ell_n \leq \frac{q_\delta^n}{1 - q_\delta}(\ell_1 - \ell_0), \quad 0 \leq u_n - h \leq \frac{q_\delta^n}{1 - q_\delta}(u_0 - u_1). \quad (4.7)$$

*Proof.* For  $t \in [z, b] \subset [0, 1/2)$ , monotonicity gives  $0 \leq F(t) \leq 1/2$ , hence  $z \leq T_z(t) \leq z + 1/4 = b$ . Also

$$T'_z(t) = \frac{1}{2}F'(t) = F(2t) \leq F(2b) = F\left(\frac{1}{2} + \delta\right) = q_\delta < 1.$$

The contraction theorem and monotonicity prove the assertions; the displayed estimates are the standard geometric a posteriori bounds.  $\square$

This gives a simple inverse evaluator requiring only forward evaluations of  $F$ , with no numerical differentiation and a rigorous enclosure at every step.

### 4.3 Sharp leading comparison

**Proposition 4.3** (Endpoint proxy for the inverse defect). *For each  $0 < \delta < 1/2$  there exists  $\xi_\delta \in (z, h)$  such that*

$$E(\delta) = \frac{F(z)}{2 - F'(\xi_\delta)} = \frac{F(z)}{2} \frac{1}{1 - F'(\xi_\delta)/2}. \quad (4.8)$$

Consequently

$$E(\delta) \sim \frac{1}{2}F\left(\frac{\delta}{2}\right), \quad \delta \downarrow 0, \quad (4.9)$$

and

$$\log E(\delta) = \log F\left(\frac{\delta}{2}\right) - \log 2 + \mathcal{O}(F'(h(\delta))). \quad (4.10)$$

*Proof.* Apply the mean-value theorem to  $F(z + E) - F(z)$  in (4.2):

$$2E = F(z) + F'(\xi_\delta)E.$$

This is (4.8). Since  $\xi_\delta \rightarrow 0$  and  $F'(0) = 0$ , the ratio in (4.9) tends to one. Taking logarithms and using  $-\log(1 - u) = \mathcal{O}(u)$  proves (4.10).  $\square$

#### Lambert–W transfer theorem

Let  $z = \delta/2$  and  $\lambda = \lambda(z)$  as in (2.10). The correction term in (4.10) is smaller than every algebraic power of  $1/\lambda$ . Therefore any complete endpoint expansion of the form (2.11) transfers term by term:

$$\begin{aligned} \log E(\delta) = & -\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda - \frac{1}{2}\log \lambda + C_F + \Psi(\lambda) - \log 2 \\ & + \sum_{j=1}^{N-1} \frac{A_j(\lambda)}{\lambda^j} + \mathcal{O}(\lambda^{-N}). \end{aligned} \quad (4.11)$$

Thus the inverse midpoint defect has the same dyadic phase, the same Bell-polynomial saddle coefficients, and the same lower-Lambert coordinate as the endpoint function, evaluated at half the ordinate displacement.

### 4.4 A rigorous Lagrange–Bell jet

Set  $\phi = F/2$  and introduce a coupling parameter  $\varepsilon$ :

$$h_\varepsilon = z + \varepsilon\phi(h_\varepsilon), \quad 0 \leq \varepsilon \leq 1. \quad (4.12)$$

For fixed  $z \in (0, 1/4)$ , the implicit-function denominator  $1 - \varepsilon\phi'(h_\varepsilon)$  stays positive, so  $h_\varepsilon$  is  $C^\infty$  in  $\varepsilon$ .

**Theorem 4.4** (Finite Lagrange–Bell expansion with exact remainder). *For every  $N \geq 1$ ,*

$$h_1 = z + \sum_{n=1}^N \frac{1}{n!} D_z^{n-1} [\phi(z)^n] + R_N(z), \quad (4.13)$$

where

$$R_N(z) = \frac{1}{N!} \int_0^1 (1-\varepsilon)^N \frac{\partial^{N+1} h_\varepsilon}{\partial \varepsilon^{N+1}} d\varepsilon. \quad (4.14)$$

Moreover,

$$D^{n-1}(\phi^n) = \sum_{k=0}^{n-1} (n)_k \phi^{n-k} B_{n-1,k}(\phi', \phi'', \dots), \quad (4.15)$$

where  $B_{m,k}$  is the partial exponential Bell polynomial and  $(n)_k$  is a falling factorial.

*Proof.* The Lagrange–Buerhmann coefficient identity

$$\left. \frac{\partial^n h_\varepsilon}{\partial \varepsilon^n} \right|_{\varepsilon=0} = D_z^{n-1} [\phi(z)^n]$$

follows by formal differentiation of (4.12); because only a finite jet is asserted,  $C^\infty$  regularity suffices and no analyticity of  $F$  is needed. Taylor’s theorem in  $\varepsilon$  with integral remainder gives (4.13)–(4.14). Formula (4.15) is Faà di Bruno’s formula applied to the power map  $u \mapsto u^n$ ; see also the standard Bell-polynomial and umbral treatments [13, 12].  $\square$

The first four correction layers, evaluated formally at  $\varepsilon = 1$ , are

$$\begin{aligned} E = & \frac{F}{2} + \frac{FF'}{4} + \frac{F(F')^2}{8} + \frac{F^2F''}{16} \\ & + \frac{F(F')^3}{16} + \frac{3F^2F'F''}{32} + \frac{F^3F'''}{96} + \dots, \end{aligned} \quad (4.16)$$

where every  $F$  and derivative is evaluated at  $z = \delta/2$ . Equation (4.13), rather than the ellipsis, is the rigorous statement. Whether the full coupling series converges at  $\varepsilon = 1$  uniformly as  $z \downarrow 0$  is left open.

## 5 Thue–Morse blocks and the dyadic zero architecture

### 5.1 Box convolutions

For  $a > 0$  let

$$b_a(x) = \frac{1}{2a} \mathbf{1}_{[-a,a]}(x), \quad \widehat{b}_a(\xi) = \frac{\sin(2\pi a\xi)}{2\pi a\xi}. \quad (5.1)$$

Then

$$U = \ast_{n=0}^\infty b_{2^{-n-1}}. \quad (5.2)$$

For  $r \geq 0$  define the normalized dilation

$$U_r(x) = 2^{-r} U(2^{-r}x), \quad \text{supp } U_r = [-2^r, 2^r], \quad \widehat{U}_r(\xi) = \widehat{U}(2^r\xi). \quad (5.3)$$

Its first  $r+1$  box factors form

$$B_r = \ast_{j=0}^r b_{2^{r-j-1}}. \quad (5.4)$$

**Proposition 5.1** (Distributional Thue–Morse derivative). *In the sense of compactly supported distributions,*

$$D^{r+1} B_r = 2^{-r(r+1)/2} \sum_{n=0}^{2^{r+1}-1} (-1)^{\text{wt}_2(n)} \delta_{n-(2^{r+1}-1)/2}. \quad (5.5)$$



*Proof.* Since  $Db_a = (\delta_{-a} - \delta_a)/(2a)$ , convolving the  $r + 1$  derivatives produces one atom for every binary choice of endpoints. If the chosen positive endpoints have binary indicator  $n$ , their signs multiply to  $(-1)^{\text{wt}_2(n)}$ , their location is

$$\sum_{j=0}^r (2\varepsilon_j - 1)2^{r-j-1} = n - \frac{2^{r+1} - 1}{2},$$

and the product of normalizing factors is  $\prod_{j=0}^r 2^{-(r-j)} = 2^{-r(r+1)/2}$ .  $\square$

The finite exponential sum in (5.5) has a zero of order  $r + 1$  at every integer frequency, the classical Prouhet cancellation encoded by the Thue–Morse word [15, 14]. Indeed, after translation by an integer frequency, its Taylor coefficients through order  $r$  are the Prouhet sums

$$\sum_{n=0}^{2^{r+1}-1} (-1)^{\text{wt}_2(n)} n^j = 0, \quad 0 \leq j \leq r. \quad (5.6)$$

Thus the universal part of the integer zero multiplicity is literally the moment cancellation of one finite Thue–Morse block.

## 5.2 Exact integer multiplicities

**Proposition 5.2** (Dyadic zero multiplicity). *For every nonzero integer  $m$ ,*

$$\text{ord}_m \widehat{U} = 1 + \nu_2(|m|). \quad (5.7)$$

*Consequently, for every nonzero  $\ell \in \mathbb{Z}$ ,*

$$\text{ord}_\ell \widehat{U}_r = r + 1 + \nu_2(|\ell|). \quad (5.8)$$

*Proof.* In (2.7), the factor indexed by  $n$  vanishes at an integer  $m$  exactly when  $2^n$  divides  $m$ . These zeros are simple, and all remaining factors are nonzero there, proving (5.7). Apply it to  $2^r \ell$  to obtain (5.8).  $\square$

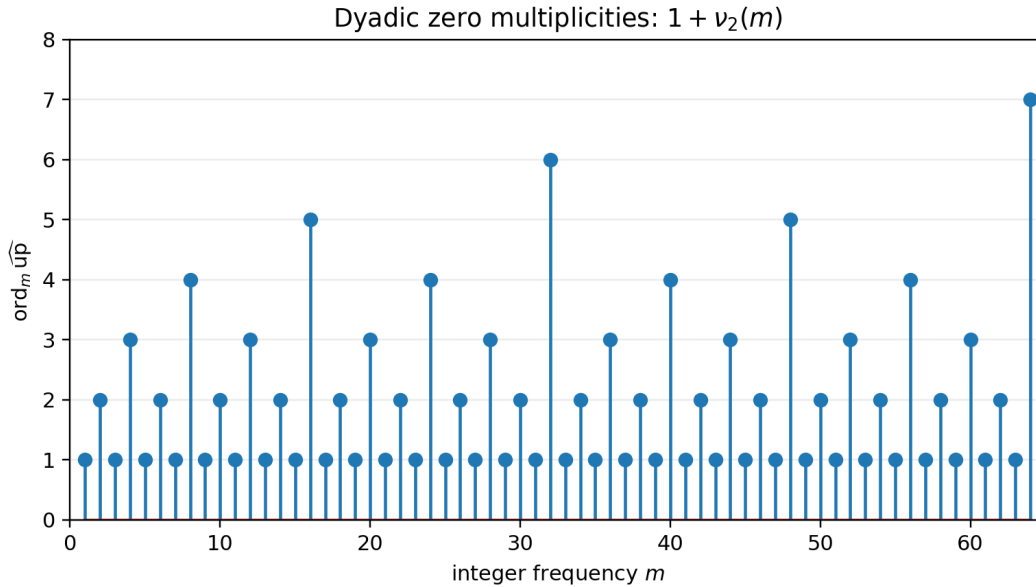


Figure 4: The exact multiplicities  $1 + \nu_2(m)$  for  $1 \leq m \leq 64$ . The unbounded spikes at powers of two later obstruct D-finiteness of the entire Fourier transform.

*Remark 5.3.* The multiplicity formula is not used as a novelty claim by itself. Its role here is structural: the first  $r + 1$  units of multiplicity are explained by the Prouhet block (5.5), while the remaining  $\nu_2(\ell)$  units come from the infinite tail. This split is precisely what drives the cardinal reproduction order in the next section.

## 6 A dyadic cardinal reproduction ladder

### 6.1 Strang–Fix vanishing at every scale

Equation (5.8) implies

$$\widehat{U}_r^{(j)}(\ell) = 0, \quad \ell \in \mathbb{Z} \setminus \{0\}, \quad 0 \leq j \leq r. \quad (6.1)$$

This is the one-dimensional Strang–Fix condition of order  $r + 1$  on the unit lattice. General approximation theory already connects such vanishing to polynomial reproduction [8, 7, 9]. The following exact formulas specialize that principle to the up-function and keep track of all moment corrections.

**Theorem 6.1** (Moment periodization). *Let*

$$\mu_j^{[r]} = \int_{\mathbb{R}} y^j U_r(y) \, dy. \quad (6.2)$$

For every  $r \geq 0$ , every  $0 \leq j \leq r$ , and every  $x \in \mathbb{R}$ ,

$$\sum_{k \in \mathbb{Z}} (x - k)^j U_r(x - k) = \mu_j^{[r]}. \quad (6.3)$$

The sum is finite, containing at most  $2^{r+1} + 2$  nonzero terms.

*Proof.* The left side is a smooth 1-periodic function. Its  $\ell$ th Fourier coefficient is

$$\begin{aligned} \int_0^1 \sum_k (x - k)^j U_r(x - k) e^{-2\pi i \ell x} \, dx &= \int_{\mathbb{R}} y^j U_r(y) e^{-2\pi i \ell y} \, dy \\ &= \frac{\widehat{U}_r^{(j)}(\ell)}{(-2\pi i)^j}. \end{aligned}$$

For  $\ell \neq 0$  this vanishes by (6.1); at  $\ell = 0$  it is  $\mu_j^{[r]}$ . Hence the periodic function is constant.  $\square$

The first cases are particularly transparent:

$$\sum_k U(x - k) = 1, \quad (6.4)$$

$$\sum_k (x - k) U_1(x - k) = 0, \quad (6.5)$$

$$\sum_k (x - k)^2 U_2(x - k) = \frac{16}{9}. \quad (6.6)$$

### 6.2 The sampling operator

For a polynomial  $p$ , define

$$(Q_r p)(x) = \sum_{k \in \mathbb{Z}} p(k) U_r(x - k). \quad (6.7)$$

Let

$$\mathcal{M}_r(t) = \int_{\mathbb{R}} e^{ty} U_r(y) \, dy = \sum_{j=0}^{\infty} \mu_j^{[r]} \frac{t^j}{j!} = \mathcal{M}_0(2^r t). \quad (6.8)$$

**Proposition 6.2** (Exact differential action on low-degree polynomials). *If  $\deg p \leq r$ , then*

$$Q_r p = \mathcal{M}_r(-D)p = \mathcal{M}_r(D)p. \quad (6.9)$$

*Proof.* Taylor-expand  $p(k) = p(x - (x - k))$  and use [theorem 6.1](#):

$$Q_r p(x) = \sum_{j=0}^{\deg p} \frac{(-1)^j}{j!} p^{(j)}(x) \sum_k (x - k)^j U_r(x - k) = \sum_j \mu_j^{[r]} \frac{(-D)^j}{j!} p(x).$$

The distribution  $U_r$  is even, so all odd moments vanish and  $\mathcal{M}_r(-D) = \mathcal{M}_r(D)$ .  $\square$

### 6.3 Reciprocal-moment Appell correctors

Define monic polynomials  $\mathcal{A}_n^{[r]}(x)$  by

$$\sum_{n=0}^{\infty} \mathcal{A}_n^{[r]}(x) \frac{t^n}{n!} = \frac{e^{xt}}{\mathcal{M}_r(t)}. \quad (6.10)$$

They form an Appell sequence:

$$\frac{d}{dx} \mathcal{A}_n^{[r]}(x) = n \mathcal{A}_{n-1}^{[r]}(x). \quad (6.11)$$

They are distinct from the q-Appell blocks occurring in the repository's exact dyadic endpoint formulas; the present family is defined by reciprocal moments of a dilated density.

**Theorem 6.3** (Finite exact cardinal reproduction). *For every  $r \geq 0$  and every  $0 \leq n \leq r$ ,*

$$\boxed{\sum_{k \in \mathbb{Z}} \mathcal{A}_n^{[r]}(k) U_r(x - k) = x^n, \quad x \in \mathbb{R}.} \quad (6.12)$$

*Only finitely many terms are nonzero.*

*Proof.* By [proposition 6.2](#), applying  $Q_r$  to the generating polynomial through degree  $r$  is the same as applying  $\mathcal{M}_r(-D)$ . Since  $\mathcal{M}_r$  is even,

$$\mathcal{M}_r(-D) \frac{e^{xt}}{\mathcal{M}_r(t)} = \frac{\mathcal{M}_r(-t)}{\mathcal{M}_r(t)} e^{xt} = e^{xt}.$$

Coefficient comparison gives (6.12) for  $n \leq r$ .  $\square$

**Example 6.4** (The first exact reconstruction identities). For all real  $x$ ,

$$\sum_k U(x - k) = 1, \quad (6.13)$$

$$\sum_k k U_1(x - k) = x, \quad (6.14)$$

$$\sum_k \left( k^2 - \frac{16}{9} \right) U_2(x - k) = x^2, \quad (6.15)$$

$$\sum_k \left( k^3 - \frac{64}{3} k \right) U_3(x - k) = x^3. \quad (6.16)$$

Each identity is finite at every  $x$  because  $U_r(x - k) = 0$  when  $|x - k| > 2^r$ .

### Corpus-relative new deduction

The dilation direction is the reverse of a standard shrinking-mesh approximation scheme: the support grows with  $r$ , while the fixed unit lattice sees progressively higher integer-frequency zero multiplicity. This “coarsening reproduction ladder” is therefore best viewed as an exact cardinal identity or moment-extraction device, not by itself as a stable numerical approximation method. Stability and conditioning are separate questions, discussed in [Section 12](#).

## 7 Bernoulli cumulants and Bell-polynomial correctors

### 7.1 Closed cumulants

The classical expansion

$$\log \frac{\sinh z}{z} = \sum_{m=1}^{\infty} \frac{2^{2m-1} B_{2m}}{m(2m)!} z^{2m} \quad (7.1)$$

and (2.9) give

$$\log \mathcal{M}_0(t) = \sum_{m=1}^{\infty} \kappa_{2m} \frac{t^{2m}}{(2m)!}, \quad \boxed{\kappa_{2m} = \frac{2^{2m-1} B_{2m}}{m(2^{2m} - 1)}, \quad \kappa_{2m+1} = 0.} \quad (7.2)$$

At scale  $r$ ,

$$\kappa_{2m}^{[r]} = 2^{2mr} \kappa_{2m} = Q^m \kappa_{2m}, \quad Q = 4^r. \quad (7.3)$$

The first values are

$$\kappa_2 = \frac{1}{9}, \quad \kappa_4 = -\frac{2}{225}, \quad \kappa_6 = \frac{16}{3969}, \quad \kappa_8 = -\frac{16}{3825}, \quad \kappa_{10} = \frac{256}{33759}. \quad (7.4)$$

### 7.2 Complete Bell form

Let  $Y_n$  denote the complete exponential Bell polynomial, characterized by

$$\exp \left( \sum_{j \geq 1} a_j \frac{t^j}{j!} \right) = \sum_{n \geq 0} Y_n(a_1, \dots, a_n) \frac{t^n}{n!}.$$

From (6.10) and (7.3),

$$\boxed{\mathcal{A}_n(x; Q) = Y_n(x, -\kappa_2 Q, 0, -\kappa_4 Q^2, 0, -\kappa_6 Q^3, \dots), \quad \mathcal{A}_n^{[r]}(x) = \mathcal{A}_n(x; 4^r).} \quad (7.5)$$

Thus, for fixed  $n$ ,  $\mathcal{A}_n(x; Q)$  is a polynomial in  $Q$  of degree  $\lfloor n/2 \rfloor$ . This elementary observation is the source of the exact  $q$ -scale algebra in [Section 8](#).

The first polynomials are

$$\mathcal{A}_0(x; Q) = 1, \quad (7.6)$$

$$\mathcal{A}_1(x; Q) = x, \quad (7.7)$$

$$\mathcal{A}_2(x; Q) = x^2 - \frac{Q}{9}, \quad (7.8)$$

$$\mathcal{A}_3(x; Q) = x^3 - \frac{Q}{3}x, \quad (7.9)$$

$$\mathcal{A}_4(x; Q) = x^4 - \frac{2}{3}Qx^2 + \frac{31}{675}Q^2, \quad (7.10)$$

$$\mathcal{A}_5(x; Q) = x^5 - \frac{10}{9}Qx^3 + \frac{31}{135}Q^2x, \quad (7.11)$$

$$\mathcal{A}_6(x; Q) = x^6 - \frac{5}{3}Qx^4 + \frac{217}{315}Q^2x^2 - \frac{2347}{59535}Q^3. \quad (7.12)$$

Higher formulas through degree eight are listed in [Section B](#) and generated exactly by the accompanying Python program.

### 7.3 Bernoulli-polynomial dyadic transport

The functional equation for the moment generating functions is

$$\mathcal{M}_{r+1}(t) = \frac{\sinh(2^r t)}{2^r t} \mathcal{M}_r(t). \quad (7.13)$$

The Bernoulli-polynomial generating function gives

$$\frac{z}{\sinh z} = \frac{2ze^z}{e^{2z} - 1} = \sum_{k=0}^{\infty} 2^k B_k \left( \frac{1}{2} \right) \frac{z^k}{k!}. \quad (7.14)$$

All odd terms vanish. For  $k = 2j$ ,

$$c_{2j} := 2^{2j} B_{2j} \left( \frac{1}{2} \right) = 2(1 - 2^{2j-1}) B_{2j}. \quad (7.15)$$

Multiplying generating functions yields an exact scale recurrence.

**Theorem 7.1** (Bernoulli transport across one dyadic dilation). *For every  $n, r \geq 0$ ,*

$$\mathcal{A}_n^{[r+1]}(x) = \sum_{k=0}^n \binom{n}{k} 2^{(r+1)k} B_k \left( \frac{1}{2} \right) \mathcal{A}_{n-k}^{[r]}(x) \quad (7.16)$$

$$= \sum_{j=0}^{\lfloor n/2 \rfloor} \binom{n}{2j} c_{2j} 2^{2jr} \mathcal{A}_{n-2j}^{[r]}(x). \quad (7.17)$$

*Proof.* Divide (6.10) at scale  $r + 1$  by its counterpart at scale  $r$  and use (7.13):

$$\frac{e^{xt}}{\mathcal{M}_{r+1}(t)} = \frac{2^r t}{\sinh(2^r t)} \frac{e^{xt}}{\mathcal{M}_r(t)}.$$

Insert (7.14) with  $z = 2^r t$  and compare coefficients. □

This identity gives a direct, exact connection among the up-function, Bernoulli polynomials, Bell polynomials, and dyadic scaling. It also supplies an efficient symbolic construction: one can build the entire scale tower without re-expanding the infinite product.

## 8 q-Lagrange interpolation and exact Richardson closure

### 8.1 Interpolation in the scale variable

Fix  $n$  and put  $d = \lfloor n/2 \rfloor$ . Since  $\mathcal{A}_n(x; Q)$  has degree at most  $d$  in  $Q$ , any  $d + 1$  distinct scales determine it exactly. Let

$$q = \frac{1}{4}, \quad Q_r = 4^r = q^{-r}. \quad (8.1)$$

Then ordinary Lagrange interpolation gives

$$\mathcal{A}_n(x; Q) = \sum_{j=0}^d \mathcal{A}_n^{[j]}(x) L_j(Q), \quad L_j(Q) = \prod_{\substack{0 \leq \ell \leq d \\ \ell \neq j}} \frac{Q - q^{-\ell}}{q^{-j} - q^{-\ell}}. \quad (8.2)$$

The barycentric denominators have the closed q-binomial form

$$\begin{aligned} \frac{1}{\prod_{\ell \neq j} (q^{-j} - q^{-\ell})} &= \frac{(-1)^{d-j} q^{\binom{d+1}{2} + \binom{j}{2}}}{(q; q)_j (q; q)_{d-j}} \\ &= \frac{(-1)^{d-j} q^{\binom{d+1}{2} + \binom{j}{2}}}{(q; q)_d} \begin{bmatrix} d \\ j \end{bmatrix}_q. \end{aligned} \quad (8.3)$$

Thus the scale reconstruction is naturally expressed in the same Gaussian-binomial language as the repository's exact dyadic arithmetic.

## 8.2 Exact q-binomial annihilation

**Theorem 8.1** (Finite q-scale closure). *Let  $N = d + 1 = \lfloor n/2 \rfloor + 1$ . For every  $r \geq 0$ ,*

$$\boxed{\sum_{j=0}^N (-1)^j q^{\binom{j}{2}} \begin{bmatrix} N \\ j \end{bmatrix}_q \mathcal{A}_n^{[r+j]}(x) = 0, \quad q = \frac{1}{4}.} \quad (8.4)$$

*The identity is polynomial in  $x$  and exact over  $\mathbb{Q}$ .*

*Proof.* It suffices to test a monomial  $Q^m$  with  $0 \leq m < N$ . At the shifted nodes  $Q_{r+j} = q^{-(r+j)}$ , the left side becomes

$$q^{-mr} \sum_{j=0}^N (-1)^j q^{\binom{j}{2}} \begin{bmatrix} N \\ j \end{bmatrix}_q (q^{-m})^j = q^{-mr} (q^{-m}; q)_N.$$

The displayed equality is the finite q-binomial theorem, and one factor in  $(q^{-m}; q)_N$  is  $1 - q^{-m} q^m = 0$ .  $\square$

**Example 8.2.** For  $n = 4$  or  $n = 5$ ,  $d = 2$  and  $N = 3$ . Clearing denominators in (8.4) gives

$$64\mathcal{A}_n^{[r]} - 84\mathcal{A}_n^{[r+1]} + 21\mathcal{A}_n^{[r+2]} - \mathcal{A}_n^{[r+3]} = 0. \quad (8.5)$$

For  $n = 6$  or  $7$ , the corresponding integer weights are 4096,  $-5440$ , 1428,  $-85$ , 1.

### Exact closure, not asymptotic acceleration

Richardson extrapolation normally annihilates the first terms of an *asymptotic* scale expansion. Here the same q-binomial algebra annihilates the sequence exactly because the new Appell corrector is a finite polynomial in  $4^r$ . This is a structural closure law and a consistency test for symbolic or numerical implementations. It should not be confused with the q-Richardson acceleration schemes already developed in the repository.

## 9 Dense nonanalyticity and holonomic obstructions

### 9.1 D-finite and P-recursive objects

Following the standard holonomic terminology [10], a function analytic near a point is D-finite if it satisfies a nontrivial linear differential equation

$$p_m(z)y^{(m)}(z) + \cdots + p_1(z)y'(z) + p_0(z)y(z) = 0, \quad p_j \in \mathbb{C}[z]. \quad (9.1)$$

At any ordinary point  $z_0$  with  $p_m(z_0) \neq 0$ , every solution is analytic and is uniquely determined by its first  $m$  initial derivatives. A sequence is P-recursive if it satisfies a polynomial-coefficient recurrence; equivalently, its ordinary or exponential generating function is D-finite after the standard factorial normalization.

We use two independent obstructions:

1. a physical-space obstruction: dense sets of nonanalytic points;
2. a Fourier-space obstruction: zeros of unbounded multiplicity at ordinary points.

### 9.2 Backward propagation through the tent map

**Proposition 9.1** (Dyadic nonanalyticity propagation). *The Fabius function is not analytic at any dyadic rational in  $(0, 1)$ .*

*Proof.* Nonanalyticity at  $1/2$  is [corollary 3.2](#). Let  $x_0 \neq 1/2$ . If  $F$  were analytic near  $x_0$ , then  $F'$  would be analytic there; by (2.4),  $F \circ \tau = F'/2$  would be analytic there. Since  $\tau$  is locally affine and invertible away from  $1/2$ ,  $F$  would be analytic near  $\tau(x_0)$ . Contrapositively, nonanalyticity at  $\tau(x_0)$  forces nonanalyticity at  $x_0$ .

If  $x_0 = k/2^m$  is in lowest dyadic terms, each application of  $\tau$  lowers the denominator exponent by one and preserves an odd numerator. After finitely many steps it reaches  $1/2$ . Backward propagation completes the proof.  $\square$

The dyadic rationals are dense. Therefore  $F$  has a dense set of nonanalytic points even though it is  $C^\infty$  everywhere.

**Theorem 9.2** (Nowhere local holonomicity of  $F$ ,  $I$ , and  $U$ ). *No nontrivial polynomial-coefficient linear differential equation can hold for  $F$  on any nonempty open subinterval of  $(0, 1)$ . The same is true for the inverse  $I = F^{-1}$  on every open subinterval of  $(0, 1)$  and for  $U = \text{up}$  on every open subinterval of  $(-1, 1)$ .*

*Proof.* Suppose (9.1) held for  $F$  on an open interval  $J$ . The leading polynomial has only finitely many zeros, so one can choose a dyadic  $x_0 \in J$  at which  $p_m(x_0) \neq 0$ . Ordinary-point theory would make  $F$  analytic near  $x_0$ , contradicting [proposition 9.1](#).

For the inverse, let  $\mathbb{D}_2$  be the dyadics in  $(0, 1)$ . Since  $F$  is a homeomorphism,  $F(\mathbb{D}_2)$  is dense. At  $x_0 \in \mathbb{D}_2$ ,  $F'(x_0) > 0$ . If  $I$  were analytic near  $y_0 = F(x_0)$ , then  $I'(y_0) = 1/F'(x_0) \neq 0$  and the analytic inverse theorem would make  $F$  analytic near  $x_0$ , a contradiction. Thus  $I$  is nonanalytic on the dense set  $F(\mathbb{D}_2)$ , and the ordinary-point argument applies again.

Finally,  $U(x) = F(x + 1)$  on  $(-1, 0)$  and  $U$  is even. Translated and reflected dyadic sets are dense in  $(-1, 1)$ , giving the same conclusion.  $\square$

*Remark 9.3.* The result is stronger than saying that the three functions are not elementary. D-finite functions include many non-elementary special functions, such as Bessel and Airy functions. The obstruction here excludes the entire holonomic class locally on every interval.

### 9.3 The entire Fourier transform

**Theorem 9.4** (Unbounded zero multiplicity excludes D-finiteness). *The entire function  $\widehat{U}$  is not D-finite over  $\mathbb{C}(z)$ .*

*Proof.* Assume that  $\widehat{U}$  satisfies an equation of order  $m$  as in (9.1). Choose  $N$  so large that  $N + 1 \geq m$  and  $2^N$  is not a zero of the leading polynomial  $p_m$ . By [proposition 5.2](#),  $\widehat{U}$  has a zero of multiplicity  $N + 1$  at  $2^N$ , so its first  $m$  initial derivatives vanish there. Uniqueness at the ordinary point  $2^N$  would force  $\widehat{U}$  to vanish identically, contrary to  $\widehat{U}(0) = 1$ .  $\square$

The proof is robust: any nonzero entire function with zeros of unbounded multiplicity outside a finite exceptional set is non-D-finite.

### 9.4 Moments and Appell coefficients

**Corollary 9.5** (Non-P-recursive moments). *For every  $r \geq 0$ , the moment generating function  $\mathcal{M}_r$  is not D-finite. Hence the full moment sequence  $\{\mu_n^{[r]}\}_{n \geq 0}$  is not P-recursive. The even subsequence  $\{\mu_{2n}^{[r]}\}_{n \geq 0}$  is not P-recursive either.*

*Proof.* The linear substitution in (2.9) and dyadic scaling preserve D-finiteness, so D-finiteness of  $\mathcal{M}_r$  would imply D-finiteness of  $\widehat{U}$ , contradicting [theorem 9.4](#). A P-recursive moment sequence would have a D-finite exponential generating function. For the even subsequence, suppose that  $c_n = \mu_{2n}^{[r]}$  were P-recursive. Termwise multiplication by the hypergeometric sequence  $1/(2n)!$  preserves P-recursiveness (equivalently, D-finite ordinary generating functions are closed under Hadamard products), so  $H(z) = \sum_{n \geq 0} \mu_{2n}^{[r]} z^n / (2n)!$  would be D-finite. Since the odd moments vanish by symmetry,  $\mathcal{M}_r(t) = H(t^2)$ ; closure under algebraic substitution would make  $\mathcal{M}_r$  D-finite, again a contradiction.  $\square$

**Corollary 9.6** (Non-P-recursive reciprocal-moment Appell sequences). *Fix  $r \geq 0$  and  $x \in \mathbb{C}$ . The sequence  $\{\mathcal{A}_n^{[r]}(x)\}_{n \geq 0}$  is not P-recursive.*

*Proof.* Its exponential generating function is  $G_{r,x}(t) = e^{xt} / \mathcal{M}_r(t)$ . The zeros of  $\widehat{U}_r$  at  $m/2^r$ ,  $m \in \mathbb{Z} \setminus \{0\}$ , become zeros of  $\mathcal{M}_r$  at

$$t = -\frac{2\pi i m}{2^r}. \quad (9.2)$$

Since the exponential numerator never vanishes,  $G_{r,x}$  has infinitely many finite poles. A D-finite germ has finite singular set in the finite plane, contained in the zeros of the leading polynomial. Thus  $G_{r,x}$  is not D-finite, and its coefficient sequence is not P-recursive.  $\square$

#### Two-sided holonomic exclusion

The physical functions fail holonomicity because ordinary points would force analyticity where a dense dyadic set is nonanalytic. The Fourier and generating functions fail for the opposite reason: they are entire or meromorphic, but their zero/pole geometry is too rich for a finite-order polynomial differential equation. The two proofs are logically independent and illuminate different aspects of the same dyadic self-similarity.

## 10 Numerical and symbolic verification

### 10.1 Reproducible implementation

The archive contains `frontier_experiments.py`, a fully commented standalone program using `mpmath`, `SymPy`, `NumPy`, and `Matplotlib`. It performs the following tasks:



- evaluates  $\widehat{U}$  from the sinc product and  $U$  from its period-two cosine series;
- checks the midpoint identity at high precision;
- solves the inverse fixed-point equation and compares  $E(\delta)$  with  $F(\delta/2)/2$ ;
- computes exact dyadic endpoint values from the triangular recurrence using rational arithmetic;
- constructs cumulants and Appell polynomials symbolically;
- verifies every q-closure through degree twelve exactly in  $\mathbb{Q}[x, Q]$ ;
- tests cardinal reproduction at scales  $0 \leq r \leq 5$ ;
- generates [Figures 4](#) and [5](#) and CSV ledgers.

The run used 70 decimal digits, 420 half-integer Fourier modes, and 240 sinc factors per Fourier coefficient. Numerical residuals measure Fourier truncation and cancellation only; the identities themselves were proved above.

## 10.2 Midpoint identity and inverse defect

The largest absolute residual among  $h = 0.05, 0.10, 0.20, 0.30, 0.45$  in (3.1) was  $4.04 \times 10^{-20}$ . Selected inverse-defect values are shown in [Table 8](#).

Table 8: Inverse midpoint defect compared with its leading endpoint proxy.

| $\delta$ | $E(\delta)$                    | $F(\delta/2)/2$                | ratio        |
|----------|--------------------------------|--------------------------------|--------------|
| 0.25     | $1.86946480001 \times 10^{-3}$ | $1.73611111111 \times 10^{-3}$ | 1.0768117248 |
| 0.20     | $5.58337327435 \times 10^{-4}$ | $5.41717809778 \times 10^{-4}$ | 1.0306792898 |
| 0.15     | $1.06205195288 \times 10^{-4}$ | $1.05314288462 \times 10^{-4}$ | 1.0084595057 |
| 0.10     | $8.10084684889 \times 10^{-6}$ | $8.09206623963 \times 10^{-6}$ | 1.0010850887 |
| 0.075    | $1.09685028275 \times 10^{-6}$ | $1.09661923462 \times 10^{-6}$ | 1.0002106913 |
| 0.05     | $5.11439387869 \times 10^{-8}$ | $5.11431110610 \times 10^{-8}$ | 1.0000161845 |

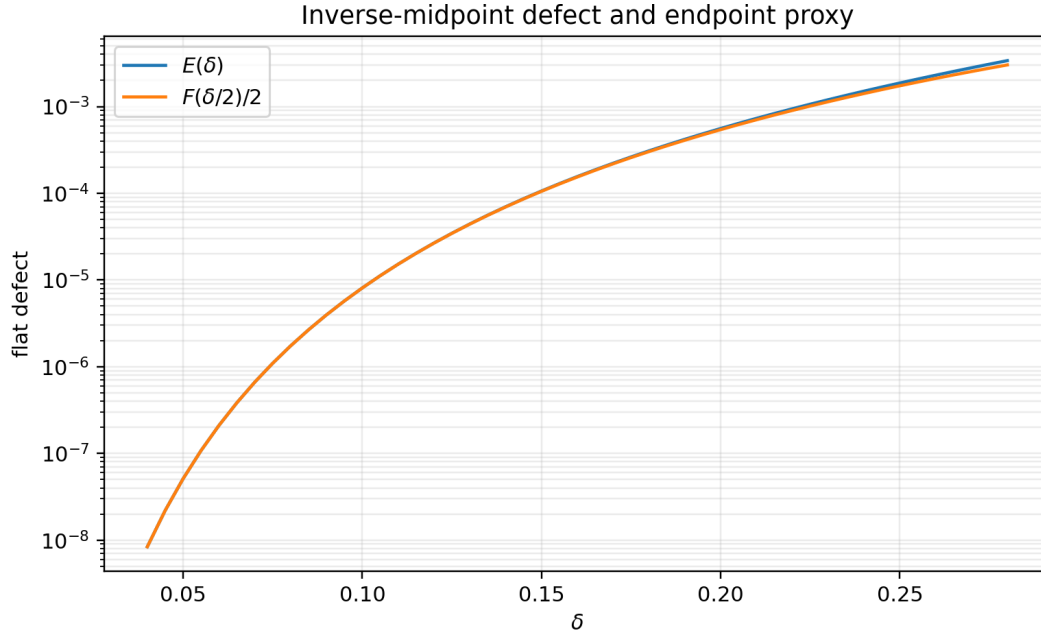


Figure 5: The exact inverse midpoint defect  $E(\delta)$  and the leading endpoint proxy  $F(\delta/2)/2$ . Their relative separation rapidly becomes invisible on the logarithmic vertical scale, in agreement with [proposition 4.3](#).

### 10.3 Exact dyadic arithmetic

The code reproduces, using exact fractions,

$$\begin{aligned}
 F(1) &= 1, & F\left(\frac{1}{2}\right) &= \frac{1}{2}, & F\left(\frac{1}{4}\right) &= \frac{5}{72}, \\
 F\left(\frac{1}{8}\right) &= \frac{1}{288}, & F\left(\frac{1}{16}\right) &= \frac{143}{2073600}, & F\left(\frac{1}{32}\right) &= \frac{19}{33177600}, \\
 F\left(\frac{1}{64}\right) &= \frac{1153}{561842749440}.
 \end{aligned}$$

These values are not new; they serve as independent regression tests for the numerical evaluator and establish compatibility with the repository's exact arithmetic.

### 10.4 Cardinal and q-scale checks

All q-binomial closures (8.4) through degree twelve simplified identically to zero. For the cardinal identities, the maximum residual over the five nonlattice points  $-0.83, -0.41, 0.07, 0.37, 0.79$  is summarized in [Table 9](#).

Table 9: Largest numerical residual in (6.12) at each tested scale. The final column reports the hardest case  $n = r$ .

| $r$ | degrees tested   | maximum residual       |
|-----|------------------|------------------------|
| 0   | 0                | $7.25 \times 10^{-71}$ |
| 1   | 0, 1             | $4.52 \times 10^{-20}$ |
| 2   | 0, 1, 2          | $2.36 \times 10^{-19}$ |
| 3   | 0, 1, 2, 3       | $4.72 \times 10^{-18}$ |
| 4   | 0, 1, 2, 3, 4    | $5.67 \times 10^{-17}$ |
| 5   | 0, 1, 2, 3, 4, 5 | $5.23 \times 10^{-14}$ |

The growth in the last row is expected from cancellation between large polynomial samples and does not indicate failure of the exact identity. Increasing the Fourier truncation reduces the approximation error; interval arithmetic would instead provide a certified enclosure.

## 11 A formalization roadmap

The deductions split naturally into algebraic, analytic, and special-function layers.

### 11.1 Low-cost Lean targets

The following results require only the already formalized Fabius equations, elementary integration, and inverse monotonicity:

1. [theorem 3.1](#) and the all-orders derivative statement [corollary 3.2](#);
2. the inverse fixed-point equation (4.2);
3. flatness of  $E$  from endpoint flatness and  $h \sim \delta/2$ ;
4. monotone lower and upper fixed-point iterations from [proposition 4.2](#).

A practical theorem statement should avoid asymptotic notation initially and prove, for every  $N$ , a neighborhood estimate  $|E(\delta)| \leq C_N |\delta|^N$ .

### 11.2 Finite combinatorial and algebraic targets

The distributional identity (5.5) can be formalized first as a finite signed-measure convolution identity. Its coefficient and location maps are binary sums, and the sign is exactly the existing signed Thue–Morse sequence. The q-closure (8.4) reduces to the finite q-binomial theorem and polynomial degree bounds. The Bernoulli transport recurrence is a formal power-series coefficient calculation and is likewise independent of analytic convergence.

### 11.3 Fourier and periodization targets

A formal proof of [theorem 6.1](#) needs:

- Fourier transforms of compactly supported smooth functions;
- differentiation under the integral;
- periodization and Fourier coefficients;
- the zero multiplicity of a finite or infinite sinc product.

A staged approach can first prove the identities for finite box convolutions, where all transforms are finite products, and then pass to  $U_r$  by the existing approximation limits.

### 11.4 Holonomic infrastructure

The D-finite results require a library definition of polynomial-coefficient linear differential equations and the ordinary-point analyticity/uniqueness theorem. Once that infrastructure exists, both obstructions are short:

- dense dyadic nonanalyticity contradicts ordinary-point analyticity;
- a zero of multiplicity at least the equation order at an ordinary point contradicts uniqueness.

The P-recursive corollaries then require the standard equivalence between recurrences and D-finite generating functions.

## 12 Further frontier problems

### 1. Differential transcendence

Non-D-finiteness excludes linear polynomial-coefficient differential equations but not nonlinear algebraic differential equations. Determine whether  $F$ ,  $I$ ,  $U$ ,  $\widehat{U}$ , or  $\mathcal{M}_0$  is differentially algebraic over  $\mathbb{C}(x)$ . The infinite hierarchy of zero multiplicities and the dilation functional equations may provide a route to differential transcendence.

### 2. Convergence and resurgence of the inverse Lagrange–Bell coupling series

The finite jet (4.13) is rigorous, but convergence of the full series at  $\varepsilon = 1$  is not asserted. Estimate the growth of  $D^{n-1}[F(z)^n]$  uniformly as  $z \downarrow 0$ , identify a possible Gevrey order, and test Borel summability. The lower-Lambert coordinate should separate algebraic saddle corrections from the flat fixed-point feedback.

### 3. Exact two-adic denominator laws for the cardinal Appell family

The coefficients in (7.12) combine Bernoulli denominators with the factors  $2^{2m} - 1$ . Determine exact 2-adic and odd-prime valuations of the coefficients of  $\mathcal{A}_n(x; Q)$ , and compare them with the denominator phenomena in the repository’s exact dyadic Fabius values.

### 4. Stability of coarsening reproduction

The identities (6.12) are exact, but the sampling coefficients grow with  $r$  and  $n$ . Determine the optimal condition numbers, Lebesgue-type constants, and noise amplification of the finite cardinal sums. Seek dual kernels or mixed scales that preserve exactness while controlling cancellation.

### 5. Base- $b$ Prouhet kernels

Replace the dyadic random series by uniform digits in base  $b$  and the Thue–Morse sign by a root-of-unity digit character. The finite block should produce a Prouhet cancellation of prescribed order, while the infinite tail produces a  $b$ -adic zero-multiplicity correction. Develop the analogue of the Appell ladder and identify its  $q$ -scale closure at  $q = b^{-2}$ .

## 6. q-holonomic versus P-recursive scale sequences

For fixed  $n$ , the scale sequence  $r \mapsto \mathcal{A}_n^{[r]}(x)$  satisfies a finite q-binomial closure because it is a polynomial in  $4^r$ . For fixed  $r$ , the degree sequence  $n \mapsto \mathcal{A}_n^{[r]}(x)$  is not P-recursive. This sharp asymmetry suggests studying whether it is q-holonomic in  $n$ , or whether the infinite pole lattice also rules out natural q-difference classes.

## 7. Phase-resolved inverse extrapolation

Equation (4.11) says that the inverse midpoint defect inherits the endpoint periodic phase exactly. Construct phase-locked Richardson schemes directly for  $I(1/2 + \delta)$ , and compare them with schemes that first approximate  $F$  and then invert. The fixed-point enclosure supplies rigorous error certificates.

## 8. Multivariate pseudo-box analogues

The approximation literature contains multivariate pseudo-box splines and nonstationary refinable systems. Determine which products possess a valuation-indexed zero multiplicity strong enough to create an increasing cardinal reproduction ladder, and whether multivariate Prouhet arrays replace the finite Thue–Morse block in (5.5).

# 13 Conclusion

The corpus already explains why q-binomial coefficients, infinite sinc products, Lambert–W functions, Bell polynomials, Lagrange interpolation, and Richardson extrapolation occur around the Fabius and Rvachev functions. The present report adds three cross-connections.

The midpoint is not merely symmetric: after subtracting its affine tangent, it is exactly the negative of the endpoint function. This identity upgrades the known first two inverse derivatives to an all-orders flatness theorem, furnishes a certified fixed-point inverse, and transports the complete endpoint Lambert–W expansion to the inverse midpoint defect.

The integer zeros of the Fourier transform are not merely spectral curiosities: under dyadic dilation, their multiplicities form a Strang–Fix ladder. A finite Prouhet–Thue–Morse block supplies the universal order, moment periodization supplies exact cardinal identities, reciprocal moments supply Appell correctors, Bernoulli cumulants and Bell polynomials supply their coefficients, and Gaussian binomials supply exact scale closures.

Finally, the same dyadic structure obstructs holonomicity in two directions. In physical space it produces dense nonanalyticity; in Fourier space it produces zeros of unbounded multiplicity. The consequences extend from  $F$ ,  $F^{-1}$ , and up to their moment and Appell sequences. These deductions are compact enough to be plausible formalization targets, yet broad enough to open new research programs in differential algebra, exact sampling, q-holonomicity, and base- $b$  generalization.

## A Additional proof details

### A.1 Barycentric denominator calculation

For completeness, let  $0 \leq j \leq d$ . Split the denominator into indices below and above  $j$ :

$$\prod_{\ell \neq j} (q^{-j} - q^{-\ell}) = \left[ \prod_{\ell=0}^{j-1} q^{-j} (1 - q^{j-\ell}) \right] \left[ \prod_{\ell=j+1}^d -q^{-\ell} (1 - q^{\ell-j}) \right]$$

$$= (-1)^{d-j} q^{-\binom{d+1}{2} - \binom{j}{2}} (q; q)_j (q; q)_{d-j}.$$

Taking the reciprocal gives (8.3).

### A.2 Why the inverse correction is beyond every Lambert order

From  $h \sim z$  and endpoint flatness,  $F'(h) = 2F(2h)$  is smaller than every power of  $h$ . The lower-Lambert relation  $z = \lambda 2^{-\lambda}$  implies  $z^a = \lambda^a e^{-aL\lambda}$ , which is itself smaller than every power  $\lambda^{-N}$ . Therefore

$$F'(h) = \mathcal{O}(z^a) = o(\lambda^{-N})$$

for arbitrary  $a, N > 0$ , proving that the logarithmic correction in (4.10) cannot alter any coefficient in the algebraic  $1/\lambda$  expansion.

### A.3 Ordinary-point analyticity

If  $p_m(x_0) \neq 0$  in (9.1), divide by  $p_m$  to obtain an analytic first-order system for the vector  $(y, y', \dots, y^{(m-1)})$ . Standard existence and uniqueness produce a convergent power series solution for any initial vector. Hence every  $C^m$  solution of the equation is analytic near  $x_0$ . This is the only differential-equation fact needed in theorems 9.2 and 9.4.

## B Explicit Appell polynomials through degree eight

Put  $Q = 4^r$ . In addition to (7.12),

$$\begin{aligned} \mathcal{A}_7(x; Q) &= x^7 - \frac{7}{3}Qx^5 + \frac{217}{135}Q^2x^3 - \frac{2347}{8505}Q^3x, \\ \mathcal{A}_8(x; Q) &= x^8 - \frac{28}{9}Qx^6 + \frac{434}{135}Q^2x^4 - \frac{9388}{8505}Q^3x^2 + \frac{1904369}{32531625}Q^4. \end{aligned}$$

The unreduced exact forms emitted by SymPy are

$$\begin{aligned} \mathcal{A}_6(x; Q) &= \frac{-2347Q^3 + 41013Q^2x^2 - 99225Qx^4 + 59535x^6}{59535}, \\ \mathcal{A}_7(x; Q) &= \frac{x(-2347Q^3 + 13671Q^2x^2 - 19845Qx^4 + 8505x^6)}{8505}, \\ \mathcal{A}_8(x; Q) &= \frac{1904369Q^4 - 35909100Q^3x^2 + 104583150Q^2x^4}{32531625} \\ &\quad + \frac{-101209500Qx^6 + 32531625x^8}{32531625}. \end{aligned}$$

## C Reproducibility files

The former standalone package's retained auxiliary payload is stored under `assets/Fabius_Rvachev_Frontier_Report_2026-08-27/`. The report source has been absorbed into `Frontier_Compilations.tex`, and its former standalone PDF is not retained. Relative to that asset directory, the retained files are:

---

|                                 |                                                                                        |
|---------------------------------|----------------------------------------------------------------------------------------|
| frontier_experiments.py         | Fully commented symbolic/numerical program.                                            |
| corpus_manifest.txt             | Recursive list of all fifty-seven audited TeX paths and the exclusion-search protocol. |
| results/numerical_results.txt   | Human-readable ledger of parameters and outputs.                                       |
| results/inverse_defect.csv      | Machine-readable inverse-defect table.                                                 |
| results/reproduction_errors.csv | Machine-readable cardinal residual table.                                              |
| figures/*.png, figures/*.pdf    | Source figures in raster and vector form.                                              |
| requirements.txt                | Python dependency bounds.                                                              |
| README.txt                      | Build and rerun instructions.                                                          |
| SHA256SUMS.txt                  | SHA-256 checksums for retained package files.                                          |

---

From the consolidated volume directory, the program can be rerun with

```
(cd assets/Fabius_Rvachev_Frontier_Report_2026-08-27 && \
python frontier_experiments.py)
```

and the consolidated report can be rebuilt with

```
latexmk -pdf -interaction=nonstopmode -halt-on-error Frontier_Compilations.tex
```

## D Complete recursive TeX manifest

```
Recursive TeX corpus manifest
Repository: VladimirReshetnikov/ProveIt
Root: Analysis/FabiusFunction/docs
Snapshot audited: main branch, 2026-08-27

Method note
-----
Every .tex path exposed by the recursive documentation tree was inventoried.
The living canonical expositions, the consolidated frontier, the Thue--Morse
Atlas, the distinct Fourier-decay drafts/audits, and all paper transcriptions
were inspected directly. Historical variants were checked against the
archive ledger and searched for the candidate themes of this report. The
archive itself states that byte-identical current-head copies were pruned and
that many entries are frozen historical versions; consequently duplicates
were mapped to their canonical topic rather than counted as independent
mathematical evidence.

A. Living and current-tree documents (25)
-----
1. FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
2. Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
3. Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex
4. fabius_lean_walkthrough/fabius_lean_walkthrough.tex
5. semi-formalized-research-frontiers/semi-formalized-research-frontiers.tex
6. semi-formalized-research-frontiers/drafts/Thue_Morse_Formula_Atlas/Thue_Morse_Formula_Atlas.tex
7. semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-2/
  Rvachev_Up_Fourier_Decay.tex
8. semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-3/
  Fourier_decay_of_Rvachev_up.tex
9. semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/
  Rvachev_Up_Fourier_Decay_Comparative_Audit/Rvachev_Up_Fourier_Decay_Comparative_Audit.tex
10. semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/
  Rvachev_Up_Fourier_Decay_Gentle_Guide/Rvachev_Up_Fourier_Decay_Gentle_Guide.tex
11. semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/
  Rvachev_Up_Fourier_Decay_Second_Wave_Audit/Rvachev_Up_Fourier_Decay_Second_Wave_Audit.tex
12. semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/asymptotic-decay-rate-of-an-
  infinite-product-of-sinc-functions/asymptotic-decay-rate-of-an-infinite-product-of-sinc-
  functions.tex
13. semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-1/
  rvachev_up_fourier_decay.tex
```

```

14. semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-4/
    rvachev_up_fourier_decay.tex
15. semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-5/
    Rvachev_Up_Fourier_Decay_Final.tex
16. semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-6/
    Rvachev_Up_Fourier_Decay_Final_Synthesis.tex
17. semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-7/
    rvachev_up_fourier_decay_final.tex
18. semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-8/
    rvachev_up_fourier_decay_final.tex
19. papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0-corrected.tex
20. papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0.tex
21. papers/arXiv-1502.06738v2/Lacunary_products.tex
22. papers/arXiv-1702.05442v1/09-Function.tex
23. papers/arXiv-1702.06487v3/157-Arithmetic-v3.tex
24. papers/arXiv-2211.13570v1/document.tex
25. papers/ubiq15/ubiq15-corrected.tex

B. Frozen archive documents (32)
-----
26. archive/early-notes/Fabius Asymptotic/Fabius Asymptotic.tex
27. archive/early-notes/K-fold summation over the signed Thue-Morse sequence/K-fold summation over
    the signed Thue-Morse sequence.tex
28. archive/standalone-studies/Non_Elementarity_of_the_Fabius_Function/
    Non_Elementarity_of_the_Fabius_Function.tex
29. archive/standalone-studies/Small_Argument_Asymptotics/Small_Argument_Asymptotics.tex
30. archive/standalone-studies/fabius-inverse-dyadic-closed-form/fabius-inverse-dyadic-closed-form.
    tex
31. archive/primary-exposition/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
32. archive/primary-exposition/Fabius_Function_and_Rvachev_Up-2/Fabius_Function_and_Rvachev_Up.tex
33. archive/primary-exposition/Fabius_Function_and_Rvachev_Up-3/Fabius_Function_and_Rvachev_Up.tex
34. archive/primary-exposition/Fabius_and_Rvachev_Up-2/Fabius_and_Rvachev_Up.tex
35. archive/primary-exposition/Fabius_and_Rvachev_up/Fabius_and_Rvachev_up.tex
36. archive/primary-exposition/Fabius_function_and_up_function/Fabius_function_and_up_function.tex
37. archive/primary-supplements/Fabius_Function_Missing_Parts/Fabius_Function_Missing_Parts.tex
38. archive/primary-supplements/Fabius_Function_Missing_Parts-2/Fabius_Function_Missing_Parts.tex
39. archive/q-calculus/Fabius_q_Special_Functions/Fabius_q_Special_Functions.tex
40. archive/q-calculus/fabius-q-special-functions/fabius-q-special-functions.tex
41. archive/lean-walkthrough/Fabius_Function_Lean_Walkthrough/Fabius_Function_Lean_Walkthrough.tex
42. archive/lean-walkthrough/fabius_lean_walkthrough/fabius_lean_walkthrough.tex
43. archive/glossary/FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
44. archive/glossary/FabiusFunction_Mathematical_Glossary-2/FabiusFunction_Mathematical_Glossary.tex
45. archive/research-frontiers/Fabius_Dyadic_Asymptotic_Bridge/Fabius_Dyadic_Asymptotic_Bridge.tex
46. archive/research-frontiers/Fabius_Dyadic_Formulae_and_Alternative_Representations/
    Fabius_Dyadic_Formulae_and_Alternative_Representations.tex
47. archive/research-frontiers/Fabius_Dyadic_Formulae_to_Asymptotics/
    Fabius_Dyadic_Formulae_to_Asymptotics.tex
48. archive/research-frontiers/Fabius_Dyadic_q_Connections/Fabius_Dyadic_q_Connections.tex
49. archive/research-frontiers/Fabius_Thue_Morse_Convergence_Rate/Fabius_Thue_Morse_Convergence_Rate.
    tex
50. archive/research-frontiers/Fabius_Thue_Morse_Convergence_Rates/
    Fabius_Thue_Morse_Convergence_Rates.tex
51. archive/research-frontiers/Primary_Exposition_Gap_Register/Primary_Exposition_Gap_Register.tex
52. archive/research-frontiers/Repeated_Integration_and_Rvachev_Up/
    Repeated_Integration_and_Rvachev_Up.tex
53. archive/research-frontiers/Rvachev_Up_from_Repeated_Integration/
    Rvachev_Up_from_Repeated_Integration.tex
54. archive/research-frontiers/non-formalized-research-frontiers/semi-formalized-research-frontiers.
    tex
55. archive/research-frontiers/Thue_Morse_Formula_Atlas-1/Thue_Morse_Formula_Atlas.tex
56. archive/research-frontiers/Thue_Morse_Formula_Atlas-2/
    Consolidated_Formula_Atlas_for_the_Thue_Morse_Sequence.tex
57. archive/research-frontiers/Thue_Morse_Formula_Atlas-3/Thue_Morse_Formula_Atlas.tex

```

#### Candidate-topic exclusion searches

-----  
The living consolidated frontier, Atlas, all twelve Fourier-decay documents, and all seven paper transcriptions were searched for the following phrases and close variants: Strang--Fix, polynomial reproduction, cardinal reproduction, inverse-moment Appell, D-finite, holonomic, P-recursive, and the continuous identity  $F(1/2+h)=1/2+2h-F(h)$ . No matching treatment of the report's three principal branches was found. Existing q-Richardson formulas, zero-counting



results, endpoint Lambert-W expansions, Bell-polynomial saddle algebra, exact dyadic arithmetic, and finite-approximant midpoint identities are treated as prior inputs and are not claimed as new here.

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## Part V

# Gamma Duality, Total Positivity, and Lambert-Localized Moments in the Fabius–Rvachev System

### Abstract

The Fabius–Rvachev corpus already contains a remarkably complete triangle of representations: a dyadic random series, exact  $q$ -binomial arithmetic, and an infinite product of sinc functions whose negative-Laplace endpoint asymptotic contains a one-periodic fluctuation and the branch  $W_{-1}$  of Lambert’s function. This report develops a new layer connecting those three sides. All claims marked “proved here” are supplied with proofs in the present manuscript; they are not claimed to have been formalized in Lean yet.

The first result expands every even moment simultaneously in Gaussian  $q$ -binomial coefficients and symmetrized multiple zeta values. The Wick-rotated sinc determinant is then shown to lie in the type-I Laguerre–Pólya class. Consequently its coefficient sequence is Pólya-frequency of infinite order, and every shifted Jensen polynomial is real-rooted; this gives an infinite hierarchy of sharp moment inequalities. The same determinant proves that the Rvachev law is strictly sub-Gaussian with the optimal variance proxy, although its real-axis zeros rule out infinite divisibility.

The principal structural theorem identifies the reciprocal Wick-rotated product as a Laplace transform. Every positive power is the transform of an explicit generalized gamma convolution with discrete Thorin measure  $\tau \sum_{m \geq 1} (1 + v_2(m)) \delta_{m^2}$ . Its cumulants are  $\tau(r-1)! \zeta(2r)/(1-4^{-r})$ , it satisfies an exact  $q = 1/4$  perpetuity, and the parity of its cumulative Thorin mass is precisely the Thue–Morse sign of the Fourier lobes. The associated Lévy heat kernel admits not only the usual Mellin expansion with log-periodic oscillation, but an exact Jacobi-theta completion whose error begins at  $-2\sqrt{\pi/x} \exp(-4\pi^2/x)$ .

Finally, exponential tilting of the endpoint law leaves the binary coordinates independent and turns them into truncated exponentials. This yields a phase-resolved mod-Gamma limit, a central limit theorem with Berry–Esseen error  $\mathcal{O}((\log s)^{-1/2})$ , and a rigorous Lambert- $W_{-1}$  localization law. A universal endpoint-transfer theorem then converts the exact negative-Laplace expansion into all-orders asymptotics for ordinary moments and centered even moments. In particular it supplies the previously missing bridge from the exact  $q$ -binomial/multiple-zeta coefficients of the sinc determinant to the Lambert- $W$  endpoint saddle.

# 1 Corpus audit, baseline, and novelty discipline

## 1.1 Scope of the audit

The audited source tree is `Analysis/FabiusFunction/docs` at the commit printed on the title page. It contains fifty-seven standalone  $\text{\LaTeX}$  manuscripts when historical revisions, source transcriptions of reference papers, and the live frontier drafts are counted separately. They fall into five overlapping families:

1. the current primary exposition, Lean walkthrough, glossary, and non-elementarity study;
2. archived primary expositions, missing-parts supplements, and  $q$ -calculus introductions;
3. exact-dyadic, Thue–Morse, repeated-integration, inverse-function, and small-argument studies;
4. the consolidated non-formalized frontier and its sequence of frontier reports;
5. the multi-wave Fourier-decay campaign and the transcribed reference literature.

The revisions were not treated as independent evidence: later corrections supersede earlier drafts. In particular, the current Fourier audit already contains the Thue–Morse lobe-sign theorem, exact zero orders, the renormalization identity, and the global sharp decay envelope. Those are baseline results here, not rediscoveries.

## 1.2 What was already present

The corpus supplies, among much else, the following ingredients used below:

- the random-series realization of the bounded Fabius distribution and of Rvachev’s even density;
- the sinc product, its integer zero multiplicities  $a_m = 1 + v_2(m)$ , and its dyadic renormalization;
- exact moment recurrences, rationality, finite convolution models, and dyadic values expressed through Gaussian binomial coefficients;
- spectral-zeta and determinant viewpoints for the Wick-rotated product;
- the exact negative-Laplace decomposition

$$\log L(s) = -\frac{(\log s)^2}{2 \log 2} + \frac{\log s}{2} + R(\log_2 s) + E(s),$$

where  $R$  is one-periodic and  $E(s)$  is positive and exponentially small;

- Lambert- $W_{-1}$  descriptions of the small-argument endpoint saddle;
- exact Thue–Morse signs of the Fourier lobes and sharp real-axis decay.

The goal was therefore not to enlarge the collection by cosmetic reformulation, but to find implications which genuinely cross the established boundaries between  $q$ -arithmetic, entire-function theory, probability, and endpoint asymptotics.

## 1.3 Status ledger

| ID | Result                                                                                                                                                                | Status             |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| A  | Gaussian $q$ -binomial/occupancy expansion of every even moment in symmetrized multiple zeta values, with a partition-lattice reduction to ordinary even zeta values. | <b>proved here</b> |

| ID | Result                                                                                                                                                              | Status                            |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| B  | Type-I Laguerre–Pólya classification, $\text{PF}_\infty$ coefficients, hyperbolic shifted Jensen polynomials, and the resulting ultra-log-concave moment hierarchy. | <a href="#">proved here</a>       |
| C  | Strict sub-Gaussianity with the exact variance proxy, together with non-infinite-divisibility.                                                                      | <a href="#">proved here</a>       |
| D  | Explicit generalized gamma-convolution dual, discrete Thorin measure, cumulant duality, and an exact $q = 1/4$ perpetuity.                                          | <a href="#">proved here</a>       |
| E  | Thue–Morse lobe sign as parity of cumulative Thorin mass.                                                                                                           | <a href="#">proved here</a>       |
| F  | Exact Jacobi-theta completion of the Lévy heat kernel, including its first exponentially flat correction.                                                           | <a href="#">proved here</a>       |
| G  | Phase-resolved mod-Gamma convergence of the endpoint tilt and a Berry–Esseen central limit theorem.                                                                 | <a href="#">proved here</a>       |
| H  | Universal endpoint-transfer expansion, applied to ordinary moments, centered moments, and the large-order $q/\text{MZV}$ coefficients.                              | <a href="#">proved here</a>       |
| I  | Phase-resolved local limit and full Edgeworth expansion under the endpoint tilt.                                                                                    | <a href="#">frontier proposal</a> |

The word “new” in this report means new to the audited corpus after targeted searches for the exact structures used below. It is not a claim of priority over every article in the mathematical literature. The general theories of multiple zeta values, Pólya-frequency sequences, generalized gamma convolutions, Mellin harmonic sums, and Lambert  $W$  are classical; what is new here is their exact synthesis for the Fabius–Rvachev product and the resulting cross-representation theorems.

## 2 Normalization and the master determinant

### 2.1 The random series

Let  $(V_k)_{k \geq 1}$  be independent uniform random variables on  $[0, 1]$ , and let  $U_k = 2V_k - 1$ , uniform on  $[-1, 1]$ . Define

$$X = \sum_{k \geq 1} 2^{-k} V_k, \quad Y = 2X - 1 = \sum_{k \geq 1} 2^{-k} U_k. \quad (2.1)$$

Then  $X \in [0, 1]$ ,  $Y \in [-1, 1]$ , and

$$X \stackrel{d}{=} 1 - X, \quad Y \stackrel{d}{=} -Y. \quad (2.2)$$

The distribution function of  $X$  is the bounded Fabius function, and the density of  $Y$  is Rvachev’s up-function. We write

$$m_n = \mathbb{E}X^n, \quad \mu_{2n} = \mathbb{E}Y^{2n}. \quad (2.3)$$

### 2.2 Fourier and Wick-rotated products

With  $\text{sinc } z = \sin z/z$  and the Fourier convention  $\Phi(z) = \mathbb{E}e^{-2\pi i z Y}$ ,

$$\Phi(z) = \prod_{j=0}^{\infty} \text{sinc}\left(\frac{\pi z}{2^j}\right) = \prod_{m=1}^{\infty} \left(1 - \frac{z^2}{m^2}\right)^{a_m}, \quad a_m = 1 + v_2(m). \quad (2.4)$$

The first product gives the exact renormalization

$$\Phi(z) = \text{sinc}(\pi z)\Phi(z/2), \quad (2.5)$$

whereas the second shows that the zero at a nonzero integer  $m$  has order  $a_m$ .

The central entire function of this report is the Wick rotation

$$\mathcal{A}(w) = \Phi(i\sqrt{w}) = \prod_{m \geq 1} \left(1 + \frac{w}{m^2}\right)^{a_m} = \prod_{j \geq 0} \prod_{n \geq 1} \left(1 + \frac{4^{-j}w}{n^2}\right). \quad (2.6)$$

It is independent of the branch of  $\sqrt{w}$  because it is entire in  $w$ . Set

$$\mathcal{A}(w) = \sum_{r \geq 0} C_r w^r. \quad (2.7)$$

Since

$$\mathbb{E}e^{tY} = \mathcal{A}\left(\frac{t^2}{4\pi^2}\right), \quad (2.8)$$

we have the exact coefficient–moment correspondence

$$C_r = \frac{(2\pi)^{2r}}{(2r)!} \mu_{2r}. \quad (2.9)$$

### 2.3 The spectral power sums

The logarithm of Equation (2.6) is

$$\log \mathcal{A}(w) = \sum_{r \geq 1} \frac{(-1)^{r+1}}{r} p_r w^r, \quad p_r = \sum_{m \geq 1} \frac{a_m}{m^{2r}} = \frac{\zeta(2r)}{1 - 4^{-r}}. \quad (2.10)$$

Consequently

$$rC_r = \sum_{j=1}^r (-1)^{j+1} p_j C_{r-j}, \quad C_0 = 1. \quad (2.11)$$

This Newton recurrence is already enough to prove rationality of all  $\mu_{2r}$  because every  $\zeta(2j)$  is a rational multiple of  $\pi^{2j}$ . The next section gives a different, genuinely  $q$ -binomial route which keeps the dyadic layers visible.

## 3 A $q$ -binomial and multiple-zeta formula for every even moment

Throughout this section

$$q = \frac{1}{4}. \quad (3.1)$$

### 3.1 Finite depth and one-site occupancy

For  $J \geq 1$  truncate the dyadic layers:

$$\mathcal{A}_J(w) = \prod_{j=0}^{J-1} \prod_{n \geq 1} \left(1 + \frac{q^j w}{n^2}\right) = \sum_{r \geq 0} C_r^{(J)} w^r. \quad (3.2)$$

At a fixed spectral site  $n$ , the finite  $q$ -binomial theorem gives

$$\prod_{j=0}^{J-1} (1 + q^j x) = \sum_{r=0}^J b_{J,r} x^r, \quad b_{J,r} = q^{r(r-1)/2} \begin{bmatrix} J \\ r \end{bmatrix}_q. \quad (3.3)$$

Thus  $r$  records the number of dyadic layers occupying the same integer frequency  $n$ . In the infinite-depth limit,

$$b_r = \lim_{J \rightarrow \infty} b_{J,r} = \frac{q^{r(r-1)/2}}{(q; q)_r}, \quad (q; q)_r = \prod_{j=1}^r (1 - q^j). \quad (3.4)$$

For  $q = 1/4$ ,

$$b_1 = \frac{4}{3}, \quad b_2 = \frac{16}{45}, \quad b_3 = \frac{64}{2835}. \quad (3.5)$$

### 3.2 Distinct-index zeta sums

Let  $\lambda = (\lambda_1, \dots, \lambda_\ell)$  be a partition of  $r$ , and let  $m_j(\lambda)$  denote the multiplicity of the part  $j$ . Define the symmetrized distinct-index zeta sum

$$Z_\lambda = \sum_{\substack{n_1, \dots, n_\ell \geq 1 \\ n_i \neq n_j \ (i \neq j)}} \frac{1}{n_1^{2\lambda_1} \cdots n_\ell^{2\lambda_\ell}}. \quad (3.6)$$

This notation keeps the assignment of distinct spectral sites to the parts of  $\lambda$  explicit.

**Theorem 3.1** ( $q$ -binomial/MZV moment formula). *For every  $r \geq 0$  and  $J \geq 1$ ,*

$$C_r^{(J)} = \sum_{\lambda \vdash r} \frac{Z_\lambda}{\prod_{j \geq 1} m_j(\lambda)!} \prod_{i=1}^{\ell(\lambda)} b_{J, \lambda_i}. \quad (3.7)$$

Passing to infinite depth gives

$$C_r = \sum_{\lambda \vdash r} \frac{Z_\lambda}{\prod_{j \geq 1} m_j(\lambda)!} \prod_{i=1}^{\ell(\lambda)} \frac{q^{\lambda_i(\lambda_i-1)/2}}{(q; q)_{\lambda_i}}. \quad (3.8)$$

Moreover

$$Z_\lambda = \sum_{\sigma \in S_\ell} \zeta(2\lambda_{\sigma(1)}, \dots, 2\lambda_{\sigma(\ell)}), \quad (3.9)$$

where  $\zeta(s_1, \dots, s_\ell) = \sum_{n_1 > \dots > n_\ell \geq 1} n_1^{-s_1} \cdots n_\ell^{-s_\ell}$ .

*Proof.* Apply Equation (3.3) at each spectral site  $n$ :

$$\mathcal{A}_J(w) = \prod_{n \geq 1} \left( \sum_{r=0}^J b_{J,r} \frac{w^r}{n^{2r}} \right).$$

A monomial of total degree  $r$  is specified by a finite set of distinct sites  $n_1, \dots, n_\ell$  and positive occupancies  $\lambda_1 + \dots + \lambda_\ell = r$ . Sorting the occupancies gives a partition  $\lambda$ . The sum over ordered, pairwise distinct sites is  $Z_\lambda$ ; permuting sites carrying equal occupancies does not change the monomial, hence the divisor  $\prod_j m_j(\lambda)!$ . This proves Equation (3.7). Since the products converge normally on compact subsets and  $b_{J,r} \rightarrow b_r$ , coefficientwise convergence proves Equation (3.8).

Finally, every pairwise-distinct  $\ell$ -tuple has a unique strict ordering. Summing over the  $\ell!$  possible orderings of the labelled coordinates gives Equation (3.9); repeated parts correctly produce repeated MZVs, matching the labelled sum in Equation (3.6).  $\square$

### 3.3 Reduction to ordinary zeta values

The MZV form is conceptually natural, but in the present fully symmetrized combination it collapses to ordinary zeta values by Möbius inversion on the partition lattice.

**Theorem 3.2** (Partition-lattice reduction). *Let  $\Pi_\ell$  be the lattice of set partitions of  $\{1, \dots, \ell\}$ . Then*

$$Z_\lambda = \sum_{\pi \in \Pi_\ell} \left( \prod_{B \in \pi} (-1)^{|B|-1} (|B| - 1)! \right) \prod_{B \in \pi} \zeta \left( 2 \sum_{i \in B} \lambda_i \right). \quad (3.10)$$

Consequently Equation (3.8) is a finite expression involving only  $q$ -Pochhammer factors and ordinary even zeta values.

*Proof.* Start with the unrestricted sum over  $(n_1, \dots, n_\ell)$ . Coincidences among indices are encoded by set partitions: a block  $B$  means that all  $n_i$  with  $i \in B$  coincide. Möbius inversion from the full coincidence lattice to the discrete partition has Möbius factor  $(-1)^{|B|-1} (|B| - 1)!$  on each block. The sum attached to a block is  $\zeta(2 \sum_{i \in B} \lambda_i)$ . Multiplication over blocks yields Equation (3.10).  $\square$

**Corollary 3.3** (A second rationality proof). *Every centered even moment  $\mu_{2r}$  is rational. More precisely,  $C_r/\pi^{2r} \in \mathbb{Q}$ , and Equation (2.9) gives  $\mu_{2r} \in \mathbb{Q}$ .*

*Remark 3.4.* The usual Newton recurrence Equation (2.11) forgets where repeated spectral integers came from. Formula Equation (3.8) retains that information: the Gaussian  $q$ -binomial factor is an occupancy weight across dyadic layers, while the MZV records the strict ordering of distinct integer frequencies. This is the precise meeting point of the  $q$ -binomial and sinc representations.

### 3.4 First examples

Using either Equation (3.8) or Equation (2.11),

$$C_1 = \frac{2\pi^2}{9}, \quad \mu_2 = \frac{1}{9}, \quad (3.11)$$

$$C_2 = \frac{38\pi^4}{2025}, \quad \mu_4 = \frac{19}{675}, \quad (3.12)$$

$$C_3 = \frac{2332\pi^6}{2679075}, \quad \mu_6 = \frac{583}{59535}, \quad (3.13)$$

$$C_4 = \frac{265618\pi^8}{10247461875}, \quad \mu_8 = \frac{132809}{32531625}. \quad (3.14)$$

For example, at degree two the partitions (2) and (1, 1) give

$$C_2 = b_2 \zeta(4) + b_1^2 \zeta(2, 2). \quad (3.15)$$

The identity  $2\zeta(2, 2) = \zeta(2)^2 - \zeta(4)$  immediately recovers the Newton form  $(p_1^2 - p_2)/2$ .

## 4 Laguerre–Pólya structure and an infinite hierarchy of moment inequalities

### 4.1 Classification of the Wick-rotated determinant

**Theorem 4.1** (Type-I Laguerre–Pólya classification). *The entire function  $\mathcal{A}$  belongs to the type-I Laguerre–Pólya class. It has genus zero, order  $1/2$ , and only real nonpositive zeros, namely  $-m^2$  with multiplicity  $a_m = 1 + v_2(m)$ .*

*Proof.* The spectral sum

$$\sum_{m \geq 1} \frac{a_m}{m^2} = \frac{\zeta(2)}{1 - 4^{-1}} = \frac{2\pi^2}{9}$$

converges. Therefore the genus-zero product in Equation (2.6) converges normally on compact sets. Every finite truncation is a polynomial with all zeros real and nonpositive; their locally uniform limit is  $\mathcal{A}$ , so  $\mathcal{A}$  is type-I Laguerre–Pólya. The exponent of convergence of the zeros is  $1/2$ , because

$$\sum_{m \geq 1} \frac{a_m}{(m^2)^\rho} = \frac{\zeta(2\rho)}{1 - 4^{-\rho}}$$

converges exactly for  $\rho > 1/2$ . The support bound  $|Y| \leq 1$  and Equation (2.8) show that  $\mathcal{A}(w)$  has growth  $\exp(\mathcal{O}(\sqrt{|w|}))$ , completing the order statement.  $\square$

## 4.2 Pólya frequency and Jensen hyperbolicity

Write

$$\mathcal{A}(w) = \sum_{r \geq 0} \gamma_r \frac{w^r}{r!}, \quad \gamma_r = r! C_r = \frac{r!(2\pi)^{2r}}{(2r)!} \mu_{2r}. \quad (4.1)$$

**Theorem 4.2** (PF $_{\infty}$  and Jensen polynomials). *The ordinary coefficient sequence  $(C_r)_{r \geq 0}$  is a Pólya-frequency sequence of infinite order. Equivalently, every finite minor of the Toeplitz matrix*

$$T = (C_{j-i})_{i,j \geq 0}, \quad C_k = 0 \quad (k < 0), \quad (4.2)$$

*is nonnegative.*

*Furthermore, for every  $n, d \geq 0$ , the shifted Jensen polynomial*

$$J_n^{(d)}(x) = \sum_{j=0}^d \binom{d}{j} \gamma_{n+j} x^j \quad (4.3)$$

*has only real nonpositive zeros.*

*Proof.* The Aissen–Schoenberg–Whitney characterization says that the generating function of a PF $_{\infty}$  sequence has the form

$$Cz^m e^{\sigma z} \frac{\prod_i (1 + \alpha_i z)}{\prod_j (1 - \beta_j z)}, \quad \alpha_i, \beta_j, \sigma \geq 0, \quad \sum_i \alpha_i + \sum_j \beta_j < \infty.$$

Equation (2.6) is exactly this form with no denominator, no exponential factor, and  $\alpha_i = 1/m^2$  repeated  $a_m$  times. This proves the PF $_{\infty}$  assertion. Jensen’s characterization of the Laguerre–Pólya class, applied to Equation (4.1), gives hyperbolicity of all shifted Jensen polynomials. Their coefficients are positive, so their real zeros cannot be positive.  $\square$

**Corollary 4.3** (Ultra-log-concavity of even moments). *For every  $r \geq 1$ ,*

$$\gamma_r^2 \geq \gamma_{r-1} \gamma_{r+1}. \quad (4.4)$$

*Equivalently,*

$$\boxed{\mu_{2r}^2 \geq \frac{2r-1}{2r+1} \mu_{2r-2} \mu_{2r+2}.} \quad (4.5)$$

*Hence the sequence*

$$\frac{2\pi^2}{2r+1} \frac{\mu_{2r+2}}{\mu_{2r}} = \frac{\gamma_{r+1}}{\gamma_r} \quad (4.6)$$

*is nonincreasing.*



*Proof.* The discriminant of  $J_{r-1}^{(2)}(x) = \gamma_{r-1} + 2\gamma_r x + \gamma_{r+1}x^2$  is nonnegative. Substituting Equation (4.1) and cancelling the common powers of  $2\pi$  gives Equation (4.5). The ratio statement is the usual consequence of log-concavity.  $\square$

**Remark 4.4** (Higher inequalities). The quadratic inequality is only the first shadow of theorem 4.2. Every Toeplitz minor and every discriminant or subdiscriminant of every  $J_n^{(d)}$  gives a polynomial inequality among exact Fabius moments. This hierarchy is algorithmic and strictly stronger than checking Hankel positivity, which follows for unrelated reasons from the fact that  $(\mu_{2r})$  is the moment sequence of  $Y^2$ .

## 5 Optimal sub-Gaussian concentration without infinite divisibility

**Theorem 5.1** (Strict optimal sub-Gaussianity). *For every real  $t$ ,*

$$\mathbb{E}e^{tY} \leq \exp\left(\frac{t^2}{18}\right), \quad (5.1)$$

*with strict inequality for  $t \neq 0$ . The variance proxy  $1/9$  is optimal. Consequently, for  $u \geq 0$ ,*

$$\mathbb{P}(|Y| \geq u) \leq 2e^{-9u^2/2}, \quad \mathbb{P}\left(\left|X - \frac{1}{2}\right| \geq u\right) \leq 2e^{-18u^2}. \quad (5.2)$$

*Proof.* From Equations (2.6) and (2.8),

$$\log \mathbb{E}e^{tY} = \sum_{m \geq 1} a_m \log\left(1 + \frac{t^2}{4\pi^2 m^2}\right).$$

The strict inequality  $\log(1+x) < x$  for  $x > 0$  gives, when  $t \neq 0$ ,

$$\log \mathbb{E}e^{tY} < \frac{t^2}{4\pi^2} \sum_{m \geq 1} \frac{a_m}{m^2} = \frac{t^2}{18}.$$

At the origin equality holds. Since  $\text{Var}(Y) = \mu_2 = 1/9$ , no smaller quadratic proxy can dominate the moment generating function near zero. Chernoff optimization gives the two tail bounds.  $\square$

**Theorem 5.2** (The Fabius–Rvachev law is not infinitely divisible). *Neither  $X$  nor  $Y$  is infinitely divisible; in particular, neither is self-decomposable.*

*Proof.* The characteristic function of  $Y$  is  $t \mapsto \Phi(t/(2\pi))$ , and Equation (2.4) shows that it vanishes at  $t = 2\pi m$  for every nonzero integer  $m$ . An infinitely divisible characteristic function is the exponential of a Lévy–Khintchine exponent and hence never vanishes. Translation and nonzero scaling preserve infinite divisibility, so the assertion transfers between  $X$  and  $Y$ .  $\square$

---

There is no contradiction between theorems 5.1 and 5.2. The law is as concentrated as its variance permits in the sub-Gaussian sense, but it does not sit inside a convolution semigroup. The next section identifies a canonical object on the other side of the sinc determinant which is infinitely divisible.

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## 6 The generalized gamma-convolution dual

### 6.1 Construction and Thorin measure

For  $\tau > 0$ , let  $G_{j,n}^{(\tau)}$  be independent gamma variables with shape  $\tau$  and rate  $n^2$ :

$$\mathbb{E}e^{-wG_{j,n}^{(\tau)}} = \left(1 + \frac{w}{n^2}\right)^{-\tau}. \quad (6.1)$$

**Theorem 6.1** (Spectral gamma dual). *The nonnegative random series*

$$V_\tau = \sum_{j \geq 0} \sum_{n \geq 1} q^j G_{j,n}^{(\tau)}, \quad q = \frac{1}{4}, \quad (6.2)$$

converges almost surely and in  $L^1$ , with

$$\mathbb{E}V_\tau = \frac{2\tau\pi^2}{9}. \quad (6.3)$$

Its Laplace transform is exactly

$$\boxed{\mathbb{E}e^{-wV_\tau} = \mathcal{A}(w)^{-\tau}}, \quad w \geq 0. \quad (6.4)$$

Equivalently,

$$V_\tau \stackrel{d}{=} \sum_{m \geq 1} H_m, \quad H_m \sim \text{Gamma}(\tau a_m, \text{rate } m^2) \quad (6.5)$$

with independent summands. Thus  $V_\tau$  is a generalized gamma convolution with discrete Thorin measure

$$\boxed{U_\tau = \tau \sum_{m \geq 1} a_m \delta_{m^2}}. \quad (6.6)$$

*Proof.* The expectation of the nonnegative series is

$$\tau \sum_{j,n} \frac{q^j}{n^2} = \frac{\tau \zeta(2)}{1-q} = \frac{2\tau\pi^2}{9} < \infty,$$

so the series is finite almost surely and converges in  $L^1$ . Independence and Equation (6.1) give

$$\mathbb{E}e^{-wV_\tau} = \prod_{j,n} \left(1 + \frac{q^j w}{n^2}\right)^{-\tau} = \mathcal{A}(w)^{-\tau}.$$

Regrouping equal rates uses the identity that the sum of independent gamma variables with a common rate is gamma with the sum of their shapes. The multiplicity of rate  $m^2$  is  $a_m$ . Finally, the defining Laplace exponent of a generalized gamma convolution is  $\int \log(1 + w/u) U(du)$ ; equation Equation (6.6) reproduces  $\tau \log \mathcal{A}(w)$ .  $\square$

**Corollary 6.2** (Exact cumulants and cumulant duality). *For every  $r \geq 1$ ,*

$$\boxed{\kappa_r(V_\tau) = \tau(r-1)! \frac{\zeta(2r)}{1-4^{-r}}}. \quad (6.7)$$

The even cumulants of  $Y$  are obtained from those of  $V_1$  by

$$\boxed{\kappa_{2r}(Y) = (-1)^{r+1} \frac{(2r)!}{r!(2\pi)^{2r}} \kappa_r(V_1)}. \quad (6.8)$$

*Proof.* A gamma variable with shape  $a$  and rate  $\lambda$  has  $r$ th cumulant  $a(r-1)!/\lambda^r$ . Summing Equation (6.5) gives Equation (6.7). Comparing the expansion

$$\log \mathbb{E}e^{tY} = \log \mathcal{A}(t^2/(4\pi^2))$$

with Equation (2.10) proves Equation (6.8).  $\square$

## 6.2 An exact $q = 1/4$ perpetuity

Define the one-layer variable

$$B_\tau = \sum_{n \geq 1} G_{0,n}^{(\tau)}. \quad (6.9)$$

Euler's product for  $\sinh$  gives

$$\mathbb{E} e^{-w B_\tau} = \left( \frac{\pi \sqrt{w}}{\sinh(\pi \sqrt{w})} \right)^\tau. \quad (6.10)$$

**Corollary 6.3** (Gamma perpetuity). *Let  $V'_\tau$  be an independent copy of  $V_\tau$ . Then*

$$\boxed{V_\tau \stackrel{d}{=} B_\tau + \frac{1}{4} V'_\tau.} \quad (6.11)$$

*This equation has a unique law on  $[0, \infty)$  with finite first moment.*

*Proof.* Split the layer  $j = 0$  from Equation (6.2); after shifting  $j \mapsto j + 1$ , the remaining sum is  $q$  times an independent copy. Uniqueness follows by iteration in the Wasserstein-1 metric, where the map  $\nu \mapsto \text{Law}(B_\tau + qZ)$  is a contraction with factor  $q$ .  $\square$

## 6.3 Thue–Morse parity as Thorin-mass parity

Let  $s_2(N)$  be the number of 1's in the binary expansion of  $N$ . Legendre's identity  $v_2(N!) = N - s_2(N)$  gives

$$\sum_{m \leq N} a_m = N + v_2(N!) = 2N - s_2(N). \quad (6.12)$$

**Theorem 6.4** (Thue–Morse/Thorin parity law). *For every integer  $N \geq 1$ ,*

$$\boxed{(-1)^{U_1([1, N^2])} = (-1)^{s_2(N)}.} \quad (6.13)$$

*Hence, for every positive noninteger  $x$ ,*

$$\boxed{\text{sgn } \Phi(x) = (-1)^{U_1([1, [x]^2])}.} \quad (6.14)$$

*Proof.* By Equation (6.6),  $U_1([1, N^2]) = \sum_{m \leq N} a_m$ . Equation (6.12) has the same parity as  $s_2(N)$ . The current Fourier sign theorem gives  $\text{sgn } \Phi(x) = (-1)^{s_2([x])}$  on each open lobe; substitution proves Equation (6.14).  $\square$

**Remark 6.5.** The Thue–Morse sign is therefore not merely an external combinatorial label. It is the parity of the spectral mass accumulated by the infinitely divisible dual before the corresponding squared zero. This interpretation links the real-axis sign changes of a non-infinitely-divisible compactly supported law to the Thorin measure of its reciprocal Wick rotation.

## 6.4 Complete Bernstein and Stieltjes consequences

Differentiating the logarithm gives

$$\frac{\mathcal{A}'(w)}{\mathcal{A}(w)} = \sum_{m \geq 1} \frac{a_m}{m^2 + w} = \int_{[1, \infty)} \frac{U_1(\mathrm{d}u)}{u + w}. \quad (6.15)$$

Thus  $\mathcal{A}'/\mathcal{A}$  is a Stieltjes function,  $\log \mathcal{A}$  is a complete Bernstein function, and  $\mathcal{A}^{-\tau}$  is completely monotone for every  $\tau > 0$ . Standard Stieltjes theory then supplies convergent continued fractions and alternating Padé bounds on the positive real axis. These are certified rational bounds for the reciprocal Wick-rotated sinc product, complementary to the product truncations used elsewhere in the corpus.

## 7 The Lévy heat kernel and its exact theta completion

### 7.1 Lévy density and Mellin transform

The gamma dual has Lévy–Khintchine representation

$$-\log \mathbb{E} e^{-wV_\tau} = \tau \int_0^\infty (1 - e^{-wx}) K(x) \frac{dx}{x}, \quad (7.1)$$

where

$$K(x) = \sum_{m \geq 1} a_m e^{-m^2 x} = \sum_{j \geq 0} \sum_{n \geq 1} e^{-4^j n^2 x}. \quad (7.2)$$

Its Mellin transform is

$$\int_0^\infty K(x) x^{s-1} dx = \Gamma(s) \frac{\zeta(2s)}{1 - 4^{-s}}, \quad \Re s > \frac{1}{2}. \quad (7.3)$$

The poles at

$$\omega_k = \frac{\pi i k}{\log 2}, \quad k \in \mathbb{Z}, \quad (7.4)$$

produce a periodic fluctuation in  $\log_4(1/x)$ .

Put  $L_2 = \log 2$  and

$$C_K = \frac{\gamma}{4L_2} - \frac{1}{4} - \frac{\log(2\pi)}{2L_2} = -1.36756152041694261136\dots, \quad (7.5)$$

$$P_K(u) = \frac{1}{2L_2} \sum_{k \neq 0} \Gamma(\omega_k) \zeta(2\omega_k) e^{2\pi i k u}. \quad (7.6)$$

The series is real-valued and absolutely, exponentially convergent. Define

$$M_K(x) = \sqrt{\frac{\pi}{x}} - \frac{\log(1/x)}{4L_2} + C_K + P_K\left(\log_4 \frac{1}{x}\right). \quad (7.7)$$

Mellin residue calculus gives  $K(x) = M_K(x) + \mathcal{O}_N(x^N)$  for every  $N > 0$ . The next theorem identifies the flat remainder exactly.

### 7.2 Exact completion

Let

$$\vartheta_+(t) = \sum_{n \geq 1} e^{-n^2 t}. \quad (7.8)$$

**Theorem 7.1** (Exact Jacobi-theta completion). *For every  $x > 0$ ,*

$$\boxed{K(x) = M_K(x) - 2\sqrt{\frac{\pi}{x}} \sum_{r=0}^{\infty} 2^r \vartheta_+\left(\frac{4^{r+1}\pi^2}{x}\right)}. \quad (7.9)$$

*In particular, as  $x \downarrow 0$ ,*

$$\boxed{K(x) - M_K(x) = -2\sqrt{\frac{\pi}{x}} e^{-4\pi^2/x} \left(1 + \mathcal{O}(e^{-12\pi^2/x})\right)}. \quad (7.10)$$

*Proof.* From the layered definition,

$$K(x) - K(4x) = \vartheta_+(x). \quad (7.11)$$

The Jacobi transformation is

$$\vartheta_+(x) = \frac{1}{2} \sqrt{\frac{\pi}{x}} - \frac{1}{2} + \sqrt{\frac{\pi}{x}} \vartheta_+\left(\frac{\pi^2}{x}\right). \quad (7.12)$$

Periodicity of  $P_K$  and the elementary terms in Equation (7.7) give

$$M_K(x) - M_K(4x) = \frac{1}{2} \sqrt{\frac{\pi}{x}} - \frac{1}{2}. \quad (7.13)$$

Thus  $D(x) = K(x) - M_K(x)$  satisfies

$$D(x) - D(4x) = \sqrt{\frac{\pi}{x}} \vartheta_+\left(\frac{\pi^2}{x}\right). \quad (7.14)$$

Replace  $x$  by  $x/4$ , solve for  $D(x)$ , and iterate toward zero. The Mellin flatness  $D(t) = \mathcal{O}_N(t^N)$  makes the terminal term vanish and yields

$$D(x) = -\sqrt{\frac{\pi}{x}} \sum_{r=1}^{\infty} 2^r \vartheta_+\left(\frac{4^r \pi^2}{x}\right),$$

which is Equation (7.9). The first term is  $(r, n) = (1, 1)$ . The next possible exponent is  $16\pi^2/x$ , twelve units of  $\pi^2/x$  beyond the leading  $4\pi^2/x$ , proving Equation (7.10).  $\square$

*Remark 7.2* (Size of the periodic ripple). The first nonzero Fourier coefficient of  $P_K$  is

$$c_1 = \frac{\Gamma(\pi i/L_2) \zeta(2\pi i/L_2)}{2L_2}, \quad |c_1| = 0.0011153475185173782067 \dots$$

Thus the first-harmonic peak-to-center amplitude is  $2|c_1| = 0.0022306950370347564135 \dots$ . The ripple is small but not formal noise; it is an exact complex-dimension contribution.

Table 11: The exact flat correction  $K(x) - M_K(x)$  and the theta completion.

| $x$ | direct difference               | theta-series value              |
|-----|---------------------------------|---------------------------------|
| 1   | $-2.5371492292 \times 10^{-17}$ | $-2.5371492292 \times 10^{-17}$ |
| 2   | $-6.7059525212 \times 10^{-9}$  | $-6.7059525212 \times 10^{-9}$  |
| 5   | $-5.9029369697 \times 10^{-4}$  | $-5.9029369697 \times 10^{-4}$  |

## 8 Endpoint tilting: truncated exponentials, mod-Gamma limits, and Lambert $W$

### 8.1 Exact negative-Laplace decomposition

Let

$$L(s) = \mathbb{E}e^{-sX}, \quad g(s) = \log L(s), \quad s > 0. \quad (8.1)$$

The binary random series gives

$$L(s) = \prod_{k \geq 1} \frac{1 - e^{-s/2^k}}{s/2^k}. \quad (8.2)$$

The current exact endpoint theory supplies a real-analytic one-periodic function  $R$  and an exponentially small positive error  $E$  such that

$$g(s) = -\frac{(\log s)^2}{2 \log 2} + \frac{\log s}{2} + R(\log_2 s) + E(s), \quad (8.3)$$

with

$$0 \leq E(s) \leq \frac{e^{-s}}{(1 - e^{-s})^2}. \quad (8.4)$$

Termwise differentiation of the defining forward-tail series gives the corresponding exponentially small bounds for each fixed derivative of  $E$ .

## 8.2 The tilted binary coordinates

Define the exponentially tilted probability measure

$$\mathbb{P}_s(dx) = \frac{e^{-sx}}{L(s)} \mathbb{P}(X \in dx). \quad (8.5)$$

Let

$$\lambda_k = s2^{-k}, \quad W_{k,s} = \lambda_k V_k. \quad (8.6)$$

**Theorem 8.1** (Exact tilted-coordinate law). *Under  $\mathbb{P}_s$ , the coordinates  $V_k$  remain independent. The variables  $W_{k,s}$  have densities*

$$f_{\lambda_k}(w) = \frac{e^{-w}}{1 - e^{-\lambda_k}} \mathbf{1}_{(0, \lambda_k)}(w), \quad (8.7)$$

and

$$sX = \sum_{k \geq 1} W_{k,s}. \quad (8.8)$$

*Proof.* The Radon–Nikodym factor factorizes:

$$e^{-sX} = \prod_{k \geq 1} e^{-\lambda_k V_k}.$$

Normalizing each uniform coordinate gives density  $\lambda_k e^{-\lambda_k v} / (1 - e^{-\lambda_k})$  on  $0 < v < 1$ . The change of variable  $w = \lambda_k v$  yields Equation (8.7), and summing gives Equation (8.8).  $\square$

This theorem is the probabilistic mechanism behind the endpoint saddle. About  $\log_2 s$  coordinates have  $\lambda_k \gtrsim 1$  and behave almost as independent unit exponentials; the remaining coordinates form a uniformly bounded tail.

## 8.3 Tilted cumulants and exact phase dependence

Let

$$K_r(s) = \kappa_r^{(s)}(sX) = (-1)^r s^r g^{(r)}(s), \quad (8.9)$$

where the superscript indicates cumulants under  $\mathbb{P}_s$ . Put

$$\alpha(s) = K_1(s) = -sg'(s), \quad \beta(s) = K_2(s) = s^2 g''(s). \quad (8.10)$$

Writing  $\theta = \log_2 s$  and differentiating Equation (8.3) gives

$$\alpha(s) = \log_2 s - \frac{1}{2} - \frac{R'(\theta)}{\log 2} + \mathcal{O}(se^{-s}), \quad (8.11)$$

$$\beta(s) = \alpha(s) - \frac{1}{\log 2} + \frac{R''(\theta)}{(\log 2)^2} + \mathcal{O}(s^2 e^{-s}). \quad (8.12)$$

In particular  $\beta(s) = \log_2 s + \mathcal{O}(1)$ . The cumulants obey the exact recursion

$$K_{r+1}(s) = rK_r(s) - sK'_r(s). \quad (8.13)$$

**Theorem 8.2** (Phase-resolved mod-Gamma convergence). *Let  $s_j \rightarrow \infty$  with fractional parts  $\{\log_2 s_j\} \rightarrow \theta \in [0, 1]$ . For real  $t < 1$ , locally uniformly in  $t$ ,*

$$\boxed{\mathbb{E}_{s_j} e^{ts_j X} (1-t)^{\log_2 s_j} \longrightarrow \Psi_\theta(t),} \quad (8.14)$$

where

$$\boxed{\Psi_\theta(t) = \exp \left[ -\frac{\log^2(1-t)}{2 \log 2} + \frac{1}{2} \log(1-t) + R(\theta + \log_2(1-t)) - R(\theta) \right].} \quad (8.15)$$

The convergence also holds in a complex neighborhood of the origin.

*Proof.* The tilted moment generating function is exact:

$$\mathbb{E}_s e^{tsX} = \frac{L((1-t)s)}{L(s)}. \quad (8.16)$$

Insert Equation (8.3). If  $v = \log(1-t)$  and  $u = \log s$ , the quadratic part contributes

$$-\frac{(u+v)^2 - u^2}{2 \log 2} + \frac{v}{2} = -\frac{uv}{\log 2} - \frac{v^2}{2 \log 2} + \frac{v}{2}.$$

Multiplication by  $(1-t)^{\log_2 s}$  cancels the term  $-uv/\log 2$ . Periodicity and the assumed phase convergence give the  $R$ -difference in Equation (8.15); the two exponentially small errors vanish locally uniformly. The same argument works for complex  $t$  near zero using the principal logarithm.  $\square$

**Corollary 8.3** (Phase-resolved cumulants). *For every fixed  $r \geq 1$ ,*

$$K_r(s) = (r-1)! \log_2 s + \eta_r(\{\log_2 s\}) + o(1), \quad (8.17)$$

where  $\eta_r$  is an explicit analytic one-periodic function obtained by  $r$  differentiations of  $\log \Psi_\theta(t)$  at  $t = 0$ .

## 8.4 Central and local concentration under the tilt

**Theorem 8.4** (Tilted central limit theorem with Berry–Esseen rate). *As  $s \rightarrow \infty$ ,*

$$\frac{sX - \alpha(s)}{\sqrt{\beta(s)}} \Longrightarrow N(0, 1). \quad (8.18)$$

Moreover

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P}_s \left( \frac{sX - \alpha(s)}{\sqrt{\beta(s)}} \leq x \right) - \Phi_N(x) \right| = \mathcal{O}((\log s)^{-1/2}), \quad (8.19)$$

where  $\Phi_N$  is the standard normal distribution function.

*Proof.* Use the independent triangular array in [theorem 8.1](#). If  $\lambda \geq 1$ , a truncated exponential has every fixed centered absolute moment bounded uniformly in  $\lambda$ . There are  $\log_2 s + \mathcal{O}(1)$  such coordinates. If  $\lambda < 1$ , the corresponding variable lies in an interval of length  $\lambda$ , and the sum of its third centered absolute moments is bounded by a convergent geometric series. Therefore

$$\sum_k \mathbb{E}_s |W_{k,s} - \mathbb{E}_s W_{k,s}|^3 = \mathcal{O}(\log s). \quad (8.20)$$

By [Equation \(8.12\)](#), the total variance is asymptotic to  $\log_2 s$ . The Berry–Esseen ratio is therefore

$$\frac{\mathcal{O}(\log s)}{(\log s)^{3/2}},$$

which proves [Equation \(8.19\)](#); the CLT follows.  $\square$

### 8.5 Lambert- $W_{-1}$ localization

The typical endpoint location under the tilt is

$$x_s = \mathbb{E}_s X = \frac{\alpha(s)}{s}. \quad (8.21)$$

Set

$$y = (\log 2) s x_s = (\log 2) \alpha(s). \quad (8.22)$$

**Theorem 8.5** (Phase-corrected Lambert localization). *As  $s \rightarrow \infty$ ,*

$$y e^{-y} = (\log 2) x_s \exp\left(\frac{\log 2}{2} + R'(\log_2 s) + o(1)\right). \quad (8.23)$$

*Equivalently,*

$$y = -W_{-1}\left(-(\log 2) x_s \exp\left(\frac{\log 2}{2} + R'(\log_2 s) + o(1)\right)\right). \quad (8.24)$$

*After suppressing the bounded phase correction, one obtains the useful phase-blind equivalent*

$$s \sim -\frac{W_{-1}(-\sqrt{2}(\log 2)x_s)}{(\log 2)x_s}. \quad (8.25)$$

*Proof.* Equation [Equation \(8.11\)](#) gives

$$y = \log s - \frac{\log 2}{2} - R'(\log_2 s) + o(1).$$

Since  $x_s = (y/\log 2)/s$ , elimination of  $s$  yields [Equation \(8.23\)](#). The relevant solution has  $y \rightarrow \infty$ , so it is the  $W_{-1}$  branch. A bounded multiplicative change in the small argument of  $W_{-1}$  changes its value by  $\mathcal{O}(1)$ ; division by  $y \asymp \log(1/x_s)$  proves the phase-blind equivalent.  $\square$

**Remark 8.6.** Lambert  $W$  is not merely a formal inversion of the leading endpoint asymptotic. The tilted law concentrates in a Gaussian window of width  $\sqrt{\beta(s)}/s \asymp \sqrt{\log s}/s$  around  $x_s$ , and [theorem 8.5](#) identifies that actual probabilistic center.



## 9 A universal endpoint-transfer theorem

### 9.1 Universal correction polynomials

For fixed  $c > 0$  and an integer  $N \geq 1$ , define the one-sided endpoint moment

$$M_N(c) = \mathbb{E} \left[ (1 - cX)^N \mathbf{1}_{\{X \leq 1/c\}} \right]. \quad (9.1)$$

Set  $s = cN$  and  $Z = sX$  under the tilted law  $\mathbb{P}_s$ . Then

$$\frac{M_N(c)}{L(cN)} = \mathbb{E}_{cN} \left[ e^Z \left( 1 - \frac{Z}{N} \right)^N \mathbf{1}_{\{Z \leq N\}} \right]. \quad (9.2)$$

Define universal polynomials  $P_r$  by the formal identity

$$e^z(1 - zt)^{1/t} = \exp \left( - \sum_{j \geq 2} \frac{z^j t^{j-1}}{j} \right) = \sum_{r \geq 0} P_r(z) t^r. \quad (9.3)$$

The first four are

$$\begin{aligned} P_0(z) &= 1, \\ P_1(z) &= -\frac{z^2}{2}, \\ P_2(z) &= \frac{z^4}{8} - \frac{z^3}{3}, \\ P_3(z) &= -\frac{z^6}{48} + \frac{z^5}{6} - \frac{z^4}{4}. \end{aligned} \quad (9.4)$$

**Theorem 9.1** (Universal endpoint transfer). *Fix  $c > 0$  and an integer  $R \geq 0$ . As  $N \rightarrow \infty$ ,*

$$\boxed{\frac{M_N(c)}{L(cN)} = \sum_{r=0}^R \frac{\mathbb{E}_{cN} P_r(Z)}{N^r} + \mathcal{O}_{R,c} \left( \frac{(\log N)^{2R+2}}{N^{R+1}} \right), \quad Z = cNX.} \quad (9.5)$$

*Every coefficient is an explicit polynomial in the tilted cumulants  $K_1(cN), \dots, K_{2r}(cN)$  and hence in the periodic endpoint function  $R$  and its derivatives.*

*Proof.* Equation (9.2) is obtained by multiplying and dividing the integrand by  $e^{-cNX}$ . On  $0 \leq Z \leq (\log N)^2$ , Taylor's theorem applied to Equation (9.3) gives a remainder bounded by

$$C_{R,c} N^{-R-1} (1 + Z)^{2R+2}.$$

The mod-Gamma estimate, or directly the independent truncated-exponential representation, implies

$$\mathbb{E}_s Z^k = \mathcal{O}_k((\log s)^k) \quad (9.6)$$

for each fixed  $k$ . This gives the displayed remainder on the central region.

For the complement, use Markov's inequality with any fixed  $0 < t < 1$  and Equation (8.16). The mod-Gamma theorem gives  $\mathbb{E}_s e^{tZ} = \mathcal{O}_t(s^{-\log_2(1-t)})$ , only a fixed power of  $s$ , while  $\exp(-t(\log N)^2)$  beats every power of  $N$ . The tail, including the region  $Z > N$ , is therefore smaller than the asserted error. Finally, raw moments are complete Bell polynomials in cumulants, which proves the last assertion.  $\square$

**Lean status.** The finite algebraic last sentence is now formal. Under an arbitrary probability measure, with integrability assumed only for the powers occurring in a chosen polynomial, the theorem `integral_eval2_eq_sum_completeBell_momentCumulant` in `PolynomialExpectationCumulant.lean` rewrites its expectation as the coefficient-weighted complete Bell transform of the formal cumulants of its raw moments. The universal polynomials  $P_r$ , their formal exponential identity, recurrence, scalar evaluation, and first values are developed in `EndpointTransferPolynomials.lean`. These finite identities do *not* identify the formal cumulants with the analytic tilted cumulants  $K_r(s)$ , nor do they formalize the asymptotic expansion (9.5), the mod-Gamma estimate, or its remainder and tail bounds; those analytic steps remain frontier work.

## 9.2 The first two corrections

Write, at  $s = cN$ ,

$$a = K_1(s) = \alpha(s), \quad b = K_2(s) = \beta(s), \quad c_3 = K_3(s), \quad c_4 = K_4(s). \quad (9.7)$$

Then

$$\mathbb{E}_s Z^2 = a^2 + b, \quad (9.8)$$

$$\mathbb{E}_s Z^3 = a^3 + 3ab + c_3, \quad (9.9)$$

$$\mathbb{E}_s Z^4 = a^4 + 6a^2b + 3b^2 + 4ac_3 + c_4. \quad (9.10)$$

Thus

$$\frac{M_N(c)}{L(cN)} = 1 - \frac{\mathbb{E}_s Z^2}{2N} + \frac{\mathbb{E}_s Z^4/8 - \mathbb{E}_s Z^3/3}{N^2} + \mathcal{O}\left(\frac{(\log N)^6}{N^3}\right). \quad (9.11)$$

## 9.3 Ordinary Fabius moments

By symmetry  $X \stackrel{d}{=} 1 - X$ ,

$$m_n = \mathbb{E}X^n = \mathbb{E}(1 - X)^n = M_n(1). \quad (9.12)$$

**Corollary 9.2** (Large ordinary moments). *With all tilted quantities evaluated at  $s = n$ ,*

$$\boxed{\frac{m_n}{L(n)} = 1 - \frac{\alpha(n)^2 + \beta(n)}{2n} + \frac{\mathbb{E}_n Z^4/8 - \mathbb{E}_n Z^3/3}{n^2} + \mathcal{O}\left(\frac{(\log n)^6}{n^3}\right)}. \quad (9.13)$$

*In logarithmic form,*

$$\log m_n = g(n) - \frac{\alpha(n)^2 + \beta(n)}{2n} + \mathcal{O}\left(\frac{(\log n)^4}{n^2}\right). \quad (9.14)$$

*If  $u = \log n$ ,  $\theta = u/\log 2$ , and*

$$A_n = \frac{u}{\log 2} - \frac{1}{2} - \frac{R'(\theta)}{\log 2}, \quad (9.15)$$

*then*

$$\begin{aligned} \log m_n = & -\frac{u^2}{2\log 2} + \frac{u}{2} + R(\theta) \\ & - \frac{1}{2n} \left[ A_n^2 + A_n - \frac{1}{\log 2} + \frac{R''(\theta)}{(\log 2)^2} \right] + \mathcal{O}\left(\frac{u^4}{n^2}\right). \end{aligned} \quad (9.16)$$

## 9.4 Centered moments and the $q$ /MZV coefficient bridge

Symmetry about  $1/2$  gives

$$\mu_{2r} = \mathbb{E}(1 - 2X)^{2r} = 2M_{2r}(2). \quad (9.17)$$

Here  $N = 2r$  and the Laplace scale is  $s = cN = 4r$ .

**Corollary 9.3** (Large centered moments). *With the tilted quantities evaluated at  $s = 4r$ ,*

$$\begin{aligned} \frac{\mu_{2r}}{2L(4r)} = 1 - \frac{\alpha(4r)^2 + \beta(4r)}{4r} \\ + \frac{\mathbb{E}_{4r}Z^4/8 - \mathbb{E}_{4r}Z^3/3}{4r^2} + \mathcal{O}\left(\frac{(\log r)^6}{r^3}\right). \end{aligned} \quad (9.18)$$

Consequently the exact coefficient  $C_r$  in the  $q$ -binomial/MZV expansion [Equation \(3.8\)](#) satisfies

$$\boxed{\begin{aligned} \log C_r = 2r \log(2\pi) - \log(2r)! + \log 2 + g(4r) \\ - \frac{\alpha(4r)^2 + \beta(4r)}{4r} + \mathcal{O}\left(\frac{(\log r)^4}{r^2}\right). \end{aligned}} \quad (9.19)$$

Since  $R$  has period one,  $R(\log_2(4r)) = R(\log_2 r)$ : the same binary phase controls both the endpoint Laplace law and the large-order  $q$ /MZV coefficients.

*Proof.* Apply [theorem 9.1](#) to [Equation \(9.17\)](#) and then use [Equation \(2.9\)](#). Taking the logarithm changes the second-order remainder only by  $\mathcal{O}((\log r)^4/r^2)$ .  $\square$

---

[Equation \(9.19\)](#) is the main bridge result. The left side is an exact coefficient assembled from Gaussian  $q$ -binomials and MZVs. The right side is governed by the endpoint Laplace product, its log-periodic correction, and the Lambert- $W_{-1}$  localization of the tilted mass. Neither side is an approximate restatement of the other; the transfer theorem proves why they must agree at large order and supplies computable corrections.

---

## 9.5 Numerical validation

The ordinary moments can be generated exactly by

$$(2^n - 1)m_n = \sum_{k=1}^n \binom{n}{k} \frac{m_{n-k}}{k+1}, \quad m_0 = 1. \quad (9.20)$$

[Table 12](#) compares the exact ratio  $m_n/L(n)$  with the one- and two-correction truncations of [Equation \(9.13\)](#).

For centered moments the natural comparison is  $\mu_{2r}/(2L(4r))$ .

The second correction is already visibly asymptotic rather than merely qualitative. Its slower start for centered moments is expected: their relevant Laplace scale is  $4r$ , but the polynomial exponent is only  $2r$ , so the first correction carries the larger factor  $(\alpha(4r)^2 + \beta(4r))/(4r)$ .

Table 12: Endpoint transfer for ordinary moments.

| $n$  | exact ratio    | one correction | two corrections |
|------|----------------|----------------|-----------------|
| 20   | 0.650564349158 | 0.575338248846 | 0.658507245134  |
| 50   | 0.740712148665 | 0.698394360314 | 0.745157800500  |
| 100  | 0.809797566277 | 0.787758642159 | 0.811530920395  |
| 200  | 0.868346118425 | 0.858160052080 | 0.868893882715  |
| 500  | 0.924530320090 | 0.921307792046 | 0.924627286470  |
| 1000 | 0.952442261825 | 0.951188127893 | 0.952465665513  |

Table 13: Endpoint transfer for centered even moments.

| $r$ | exact ratio    | one correction | two corrections |
|-----|----------------|----------------|-----------------|
| 10  | 0.516638138954 | 0.334240960496 | 0.558725000470  |
| 20  | 0.623451024335 | 0.521571821828 | 0.641626320479  |
| 50  | 0.755044449961 | 0.716320104160 | 0.759255426702  |
| 100 | 0.834124742928 | 0.817440783081 | 0.835296570356  |

## 10 Synthesis: one determinant, two probability laws, three asymptotic regimes

The results can be summarized by the single determinant

$$\mathcal{A}(w) = \underbrace{\Phi(i\sqrt{w})}_{\text{sinc/zero geometry}} = \underbrace{\mathbb{E} \exp(2\pi\sqrt{w} Y)}_{\text{compact law}} = \underbrace{(\mathbb{E} e^{-wV_1})^{-1}}_{\text{gamma-convolution dual}}. \quad (10.1)$$

Each equality exposes a different rigidity.

1. The sinc product gives integer zeros, dyadic multiplicities, and the Thue–Morse sign of real Fourier lobes.
2. The compact-law MGF gives exact centered moments, optimal sub-Gaussian concentration, and endpoint tilting.
3. The reciprocal Laplace transform gives a GGC, Thorin measure, complete Bernstein structure, and the Lévy heat kernel.
4. Expanding by dyadic layers produces Gaussian  $q$ -binomial occupancies; expanding by distinct integer sites produces symmetrized MZVs.
5. Mellin analysis of the spectral heat kernel and of the negative-Laplace product produces log-periodic phases. Endpoint transfer proves that those phases survive into large moment order.

This synthesis also clarifies a striking duality. The compact Fabijs–Rvachev law is strictly variance-optimal sub-Gaussian but not infinitely divisible. The reciprocal Wick rotation is the Laplace transform of an explicitly infinitely divisible GGC. Their cumulants are related by the alternating formula [Equation \(6.8\)](#); their sign structure is related by Thorin mass parity; and their asymptotics are tied by the same dyadic complex dimensions.

## 11 Next frontier targets

### 11.1 Phase-resolved Edgeworth and local limits

The proof of [theorem 8.4](#) contains more information than the stated Berry–Esseen bound. The independent summands have uniformly controlled moments, the variance diverges logarithmically, and [corollary 8.3](#) gives every cumulant with an explicit periodic constant term. This strongly suggests the following sharpened statement.

*Conjecture 11.1* (Phase-resolved local Edgeworth expansion). Along every sequence  $\{\log_2 s_j\} \rightarrow \theta$ , the tilted density of  $(s_j X - \alpha(s_j))/\sqrt{\beta(s_j)}$  admits a full local Edgeworth expansion in powers of  $(\log s_j)^{-1/2}$ , uniformly on windows  $|x| \leq (\log s_j)^{1/6-\varepsilon}$ . Every coefficient is a polynomial in  $x$  whose constants are analytic one-periodic functions of  $\theta$ .

A proof should combine the explicit characteristic function  $L((1 - it/s)s)/L(s)$  with a triangular-array Cramér bound. The compact support of each digit and the presence of many near-exponential coordinates make the nonlattice estimate plausible, but it should be checked uniformly in the binary phase.

### 11.2 All-orders moment transfer

The finite algebraic coefficient stage of [Theorem 9.1](#) is now formalized: the universal polynomials  $P_r$  have recursive, formal-series, weighted-partition, and scalar-evaluation descriptions, and a support-local polynomial integral is a finite complete-Bell transform of its formal moment cumulants. Two analytic tasks remain:

1. instantiate the finite integral identity for the actual tilted law, identify its moments and cumulants, and express  $\mathbb{E}_s P_r(Z)$  directly in  $R, R', \dots, R^{(2r)}$ ;
2. combine the result with Stirling's expansion in [Equation \(9.19\)](#) to produce a closed transseries for  $C_r$ , including the periodic phase and exponentially small endpoint terms.

This would turn the q/MZV coefficient sequence into a precise asymptotic probe of the negative-Laplace periodic function.

### 11.3 Inverse recovery of the periodic endpoint function

[Equation \(9.19\)](#) can be read in reverse. Exact rational moments produce exact  $C_r$ ; after subtracting the explicit Stirling and quadratic-log terms, the residual samples  $R(\log_2 r)$  with a controlled error. Since  $\{\log_2 r\}$  is dense modulo one, this suggests a certified reconstruction scheme for  $R$  from exact arithmetic alone. The correction terms involving  $R'$  and  $R''$  make the inverse problem nonlinear but also provide derivative information. A practical scheme would alternate Fourier reconstruction of  $R$  with the transfer correction until the residual falls below the exact moment error bound.

### 11.4 A $q$ -deformed gamma dual

The corpus defines symmetric  $q$ -integer deformations of the sinc determinant. Whenever the deformed zeros remain positive after Wick rotation, the same argument as [theorem 6.1](#) constructs a discrete Thorin measure at the squared  $q$ -integers. The main questions are:

- whether the deformed determinant remains type-I Laguerre–Pólya for the full parameter range;
- whether its Thorin mass admits a deformed digit-parity interpretation;
- how the  $q \uparrow 1$  and  $q \downarrow 0$  limits interact with the perpetuity and heat-kernel complex dimensions.

These questions are exact and experimentally accessible; no conjectural analytic continuation is needed to begin.

## 12 Lean formalization roadmap

The new results split naturally into modules with very different library requirements.

---

**Post-snapshot algebraic progress.** The later modules `ExponentialPartition.lean`, `ExponentialBell.lean`, `MomentCumulantAlgebra.lean`, `EndpointTransferPolynomials.lean`, and `PolynomialExpectationCumulant.lean` now formalize the finite algebraic core. The first four work over every commutative  $\mathbb{Q}$ -algebra: they give the universal weighted-partition formula for an exponential coefficient and identify it with the recursive `SaddleExpansion.expCoeff` engine, then factorially normalize it into named complete-Bell and inverse moment–cumulant transforms with the classical Bell recurrence and normalized inverse laws. The endpoint module defines the exact  $P_r$ , proves their recurrence, formal exponential identity, weighted-partition formula, first four values, and naturality under evaluation. The polynomial-integral module allows any coefficient semiring equipped with a ring homomorphism to  $\mathbb{R}$ ; it rewrites any support-integrable finite polynomial integral as raw moments and then as a complete-Bell transform, with the exact zeroth-moment correction and a probability specialization. These additions are present at integration checkpoint `6e9148873264978eef43181d6bd214c36cdc405a`.

This discharges the recursive and finite Bell-algebra stages of [theorem 9.1](#). It does not prove the analytic identity with  $e^z(1 - zt)^{1/t}$ , construct or identify the tilted law, establish the Taylor window or moment-growth estimate, control the complementary tail, or deduce the endpoint remainder in (9.5). The status of the full analytic theorem therefore remains unchanged.

---

### 12.1 Low-risk algebraic modules

1. `QBinomialMZVMoments.lean`: prove the finite-depth occupancy expansion over finite spectral cutoffs, then pass first in the spectral cutoff and then in depth. The partition-lattice identity can be stated initially as a finite inclusion–exclusion theorem, avoiding a general MZV library.
2. `SpectralGammaCumulants.lean`: formalize the algebraic identities for Laplace exponents and cumulants before constructing the countable probability space. The equality  $\kappa_r = \tau(r - 1)!\zeta(2r)/(1 - 4^{-r})$  is mostly existing Dirichlet series infrastructure.
3. `ThorinParity.lean`: combine the already formalized lobe-sign theorem with Legendre’s digit-sum formula and a finite cumulative mass definition. This should be short.

### 12.2 Analytic modules

1. `GammaConvolutionDual.lean`: construct partial sums of independent gamma variables, prove  $L^1$  convergence from the expectation sum, and identify the Laplace transform by dominated convergence.
2. `LevyHeatThetaCompletion.lean`: establish  $K(x) - K(4x) = \vartheta_+(x)$ , import or prove the Jacobi theta transform, and telescope the flat correction. The exact identity [Equation \(7.9\)](#) is more formalization-friendly than a contour shift because all sums are positive or exponentially dominated.
3. `EndpointTilt.lean`: prove coordinate factorization under the tilted finite product, pass to the infinite law, and derive the cumulant recursion.
4. `MomentTransfer.lean`: import the formalized  $P_r$ , instantiate the finite expectation–cumulant identity for the tilted law, prove a finite-order Taylor bound on a logarithmic window, and control the complement with the exact Laplace ratio.

### 12.3 Library-heavy modules

The Laguerre–Pólya and  $\text{PF}_\infty$  statements may require new general infrastructure. A pragmatic first formal target is finite truncation:

- prove real nonpositive roots of each finite polynomial;
- prove total positivity of its coefficient Toeplitz matrix by repeated convolution with  $(1, \alpha)$ ;
- derive every fixed finite minor inequality for the limit by coefficient convergence.

This avoids formalizing the full Aissen–Schoenberg–Whitney theorem while still obtaining any requested moment inequality.

## A Exact coefficients and normalized Turán ratios

For reference, the first five nontrivial centered moments are

$$\begin{array}{c|c} r & \mu_{2r} \\ \hline 1 & 1/9 \\ 2 & 19/675 \\ 3 & 583/59535 \\ 4 & 132809/32531625 \\ 5 & 46840699/24405225075 \end{array} \quad (\text{A.1})$$

The normalized Turán ratio

$$T_r = \frac{2r+1}{2r-1} \frac{\mu_{2r}^2}{\mu_{2r-2}\mu_{2r+2}} = \frac{\gamma_r^2}{\gamma_{r-1}\gamma_{r+1}} \quad (\text{A.2})$$

is at least one. Its first values are

$$T_1 = \frac{25}{19} = 1.315789\dots, \quad T_2 = \frac{17689}{14575} = 1.21365\dots, \quad T_3 = 1.16829\dots \quad (\text{A.3})$$

The approach toward one is consistent with the large-order endpoint asymptotic: the Jensen coefficients become asymptotically close to a geometric sequence on the scale of one index, while retaining strict log-concavity.

## B Proof-dependency map

| Result                 | Minimal dependencies                                                             |
|------------------------|----------------------------------------------------------------------------------|
| $q/\text{MZV}$ formula | finite $q$ -binomial theorem; normal convergence; set-partition Möbius inversion |
| Laguerre–Pólya         | canonical product convergence; finite real-rooted truncations                    |
| $\text{PF}_\infty$     | Aissen–Schoenberg–Whitney characterization                                       |
| Jensen hierarchy       | Jensen characterization of the Laguerre–Pólya class                              |
| Sub-Gaussianity        | $\log(1+x) < x$ ; exact spectral sum $p_1 = 2\pi^2/9$                            |
| Gamma dual             | gamma Laplace transform; Tonelli; regrouping equal rates                         |
| Thorin parity          | $\sum_{m \leq N} (1 + v_2(m)) = 2N - s_2(N)$ ; existing lobe sign                |
| Theta completion       | dyadic heat equation; Jacobi theta transform; telescoping                        |

| Result          | Minimal dependencies                                                          |
|-----------------|-------------------------------------------------------------------------------|
| Mod-Gamma limit | exact negative-Laplace decomposition; periodicity of $R$                      |
| Berry–Esseen    | tilted independent digits; third-moment bound; variance asymptotic            |
| Moment transfer | exponential tilt; universal Taylor polynomials; tilted moment and tail bounds |

C    **Corpus map used in the audit**

The audit followed the repository’s revision structure rather than reading the directory as a flat bag of papers. The decisive source groups were:

- Current expository core.**  
Fabius\_Function\_and\_Rvachev\_Up, the current Lean walkthrough, the mathematical glossary, and the non-elementarity article.
- Archived foundational studies.**  
The earlier primary expositions, both missing-parts supplements, the two  $q$ -special-function expositions, the small-argument and inverse-function studies, and the early notes on Fabius asymptotics and iterated Thue–Morse summation.
- Exact arithmetic and convergence studies.**  
The dyadic formula/alternative-representation papers, the two dyadic-to-asymptotic bridges, the Thue–Morse formula atlases, both Thue–Morse convergence-rate reports, and both repeated-integration reports.
- Live frontier synthesis.**  
The consolidated semi-formalized-research-frontiers manuscript, the two Fabius–Rvachev frontier reports, and the spectral-determinant/ $q$ -deformation results bundle.
- Fourier-decay campaign.**  
The original sinc-decay question, all numbered response waves, the final syntheses, gentle guide, comparative audit, and second-wave audit. The latest second-wave audit was treated as authoritative where older agents disagreed.
- Reference transcriptions.**  
The corrected and source  $\text{\LaTeX}$  transcriptions under docs/papers, including the work on the Fabius function, arithmetic properties, and lacunary products. They were used for historical and literature context, not as a substitute for checking the repository’s current normalization.

D    **Bibliographic notes**

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## Part VI

# Confluent Digital Extrapolation and Lambert-Phase Tomography in the Fabius–Rvachev System

## Abstract

The audited PROVEIT corpus already contains the classical and formalized theory of the Fabius function and Rvachev’s up-function, exact dyadic values, inverse endpoint asymptotics, Gaussian  $q$ -binomial interpolation, Bernoulli cumulants, Bell-polynomial saddle coefficients, sharp real-axis decay of the infinite sinc product, spectral zero multiplicities, the Pascal–Rvachev hierarchy, logarithmic  $q$ -Richardson acceleration, and additive Lambert-phase locking. The recursive tree at the pinned commit contains 78 LaTeX paths, including historical revisions, imported papers, and six generated table fragments. These materials were treated as one revision-aware corpus; established results are not relabeled as discoveries here.

This report develops a different theorem package. First, for a sequence containing polynomially decorated geometric modes  $n^\ell q^n$ , a normalized shift polynomial with a root of multiplicity  $m$  at  $q$  annihilates all decorations with  $\ell < m$ . The unique minimal-degree filter, its exact condition number, its Banach-valued remainder bound, and its mixed-mode extension are obtained in closed form. The familiar Gaussian  $q$ -Richardson row is precisely the simple-root specialization.

Second, a nonminimal factorization gives a surprising stability improvement. Twisting a length- $2^m$  Thue–Morse block by  $q^{-n}$  produces a filter of order  $m$  whose  $\ell^1$  condition number is

$$\kappa_m^{\text{TM}}(q) = \frac{1 - q^{2^m}}{(1 - q) \prod_{j=0}^{m-1} (1 - q^{2^j})} \rightarrow \frac{1}{(1 - q)\Theta_{\text{TM}}(q)}.$$

Thus the cancellation order tends to infinity while the condition number stays bounded. For  $q = 1/2$  the limit is approximately 5.7112854057, whereas the minimal repeated-root filter has condition number  $3^m$ . The price is a stencil span  $2^m - 1$ . The powers of two are characterized as the unique minimal-span collision-free factorization.

Third, in the exact lower-Lambert coordinate  $x = \lambda 2^{-\lambda}$ , every rational phase  $\theta = a/d$  has multiplicative locks  $\lambda_s = \rho^s(n + \theta)$  whenever  $\rho \equiv 1 \pmod{d}$ . Inverse powers become geometric modes and logarithmically decorated inverse powers become confluent geometric modes. This converts endpoint extraction into a stable  $q$ -Richardson problem with bounded weights, contrasting sharply with the polynomially growing condition number of the corpus’s additive locks  $\lambda + j$ .

Fourth, the corpus’s exact Gamma–zeta formula for the leading periodic endpoint term is shown to imply the exact maximal holomorphy strip

$$|\Im z| < \frac{\pi}{2 \log 2}.$$

Weighted Wiener-algebra bounds then make Bell-polynomial and derivative constructions quantitatively stable. Combining radial multiplicative extrapolation with a discrete Fourier transform over rational phases yields a two-axis tomography algorithm: radial error is algebraic in the starting Lambert coordinate, while angular aliasing is exponentially small at the sharp rate  $\pi^2 / \log 2$ .

All theorem statements are proved in the report. Claims about natural boundaries, optimal span–conditioning tradeoffs, irrational near-locks, and resurgent late-order behavior are explicitly labeled conjectural. The accompanying Python program performs exact rational cancellation checks and high-precision Gamma–zeta experiments and regenerates every table and figure.

---

**Proof and novelty convention.** “Proved here” means proved in this manuscript, not yet formalized in Lean. “New” means not found as an exact theorem or construction in the pinned 78-path corpus after path, phrase, formula, and structural searches. It is deliberately not a claim of worldwide priority. The endpoint expansion and its Gamma–zeta Fourier formula are audited inputs; the confluent filters, digital stability theorem, rational multiplicative phase geometry, maximal-strip consequence, and two-axis tomography are the new deductions.

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# 1 Corpus boundary and nonduplication audit

## 1.1 Pinned recursive inventory

The default branch changed during the audit, so all comparisons use the immutable commit printed on the title page. Its documentation subtree has recursive Git tree SHA 6063d0d90cd9a4d25ca74cd4f8b80448313fce64. The tree contains 78 LaTeX paths. GitHub’s code-search index returned fewer paths at several points because large, recent, and generated files had not all been indexed; the recursive tree, not code search, was therefore the inventory authority.

The corpus consists of current expositions, frozen revisions, a consolidated frontier volume, numerous current frontier reports, twelve Fourier-decay dossiers, source-paper transcriptions, and six small TeX fragments generated for tables or checks. Historical revisions were read as mathematical lineages: a corrected successor supersedes an older normalization, but a distinctive result surviving only in an archive file still counts as prior art. The complete path list is reproduced verbatim in [Section C](#) and is also supplied as `assets/fabius_frontier_report_bundle/corpus_manifest.txt` [11].

## 1.2 Strong negative controls

The following attractive results are already present in the corpus and are therefore excluded from the novelty claims of this report:

- the probability model for  $F$  and up, the dyadic sinc product, and the integer-zero multiplicity  $1 + \nu_2(n)$ ;
- exact dyadic evaluation,  $q = 1/2$  Pochhammer and Gaussian-binomial formulas, Lagrange interpolation, and the standard simple-root  $q$ -Richardson row;
- Prouhet cancellation and the finite Thue–Morse block product;
- Bernoulli cumulants, Bell-polynomial moment and saddle calculations, Appell deconvolution, and Barnes-type finite-prefix formulas;
- the corrected exact-Lambert endpoint coordinate, its nonconstant periodic term, all-orders inverse-power expansion, and additive phase-locked nodes  $\lambda + j$ ;
- logarithmic Richardson acceleration of truncated sinc products and its exact Gaussian- $q$ -binomial error series;
- the full Fourier zero divisor, spectral determinant and zeta, arithmetic rays, half-integer identities, local zero jets, and Pascal–Rvachev hierarchy;
- positive self-sampling quadrature, alias superconvergence, and finite dyadic  $q$ -Richardson recovery;
- sharp pointwise, metric, arithmetic, and envelope results for the real-axis infinite sinc product.

In particular, this report does *not* claim ordinary geometric  $q$ -Richardson extrapolation, Thue–Morse Prouhet cancellation, or Lambert phase locking separately. Its new contribution is the confluent and digitally factorized synthesis, together with the new multiplicative phase geometry and the spectral regularity consequences.

## 1.3 Result ledger

| ID | Statement                                                                                                                                              | Status                                             |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| A  | Shift-polynomial calculus for $n^\ell q^n$ , unique minimal confluent annihilator, exact $\ell^1$ condition number, and Banach-valued remainder bound. | <a href="#">proved here</a>                        |
| B  | General Prouhet–Richardson factorization and the twisted Thue–Morse filter with bounded infinite-order conditioning.                                   | <a href="#">proved here</a>                        |
| C  | Minimal-span characterization of the binary exponents among collision-free subset-product factorizations.                                              | <a href="#">proved here</a>                        |
| D  | Rational multiplicative Lambert phase locks and confluent cancellation of $\lambda^{-r}(\log \lambda)^\ell$ terms.                                     | <a href="#">proved here</a>                        |
| E  | Exact comparison of additive and multiplicative phase-lock conditioning.                                                                               | <a href="#">proved here</a>                        |
| F  | Exact exponential type and maximal horizontal holomorphy strip of the leading Gamma–zeta periodic endpoint term.                                       | <a href="#">derived from audited corpus inputs</a> |
| G  | Radial $q$ -interpolation of the periodic coefficient jet and angular DFT alias formula with a combined error theorem.                                 | <a href="#">proved here</a>                        |
| H  | Weighted Wiener-algebra and Bell-polynomial bounds for certified Fourier truncation of all differential-polynomial endpoint coefficients.              | <a href="#">proved here</a>                        |
| I  | Boundary naturality, optimal span–conditioning laws, irrational near-locks, and late-order resurgence.                                                 | <a href="#">conjectural/open</a>                   |

## 2 Established framework and normalization

Only the audited inputs needed later are recalled. Let  $F$  be the bounded Fabius distribution function on  $[0, 1]$ , and let

$$\text{up}(x) = F(1 - |x|), \quad |x| \leq 1,$$

with zero extension outside  $[-1, 1]$ . Under the Fourier convention used in the primary exposition,

$$\Phi(z) = \widehat{\text{up}}(z) = \prod_{h=0}^{\infty} \text{sinc}\left(\frac{\pi z}{2^h}\right), \quad \text{sinc } w = \frac{\sin w}{w}. \quad (2.1)$$

The product is entire; its nonzero integer zero at  $n$  has multiplicity  $1 + \nu_2(n)$ . Its logarithmic coefficients, cumulants, dyadic values, and large real-axis behavior are developed extensively in the source corpus and the relevant primary literature [1, 2, 3, 11].

For the endpoint, put

$$L = \log 2, \quad \lambda(x) = -\frac{1}{L} W_{-1}(-Lx). \quad (2.2)$$

Here  $W_{-1}$  is the lower real branch of the Lambert function [5]. Then

$$x = \lambda 2^{-\lambda}. \quad (2.3)$$

The audited all-orders expansion can be organized in the schematic exact-coordinate form

$$\mathcal{E}(\lambda) = A_0(\{\lambda\}) + \sum_{r=1}^N A_r(\{\lambda\}) \lambda^{-r} + \mathcal{O}(\lambda^{-N-1}), \quad (2.4)$$

with uniformity in the phase  $\{\lambda\}$  on the real unit circle. The normalization of  $\mathcal{E}$  subtracts the explicit nonperiodic main term from  $\log F(x)$ ; its precise formula is not needed for the extrapolation algebra. The leading periodic coefficient has Fourier modes

$$\widehat{A}_0(k) = c_k = -\frac{\Gamma(-\chi_k) \zeta(1 - \chi_k)}{L}, \quad \chi_k = \frac{2\pi i k}{L}, \quad k \neq 0. \quad (2.5)$$

The mean  $c_0$  is explicit in the corpus but plays no role in the strip and alias arguments.

We shall also allow the more general decorated expansion

$$\mathcal{E}(\lambda) = A_0(\{\lambda\}) + \sum_{r=1}^R \sum_{\ell=0}^{m_r-1} A_{r,\ell}(\{\lambda\}) \lambda^{-r} (\log \lambda)^\ell + \mathcal{R}(\lambda). \quad (2.6)$$

The current rank-one expansion is the case  $m_r = 1$ . The decorated form is useful for higher-rank Mellin poles, parameter derivatives, and transseries; the filter theorem below is exact whether or not a particular application ultimately contains such logarithms.

### 3 Confluent shift-polynomial extrapolation

#### 3.1 The master identity

Let  $E$  denote the forward shift on sequences:  $(Eu)_n = u_{n+1}$ . If  $W(z) = \sum_{j=0}^M w_j z^j$ , set

$$W(E)u_n = \sum_{j=0}^M w_j u_{n+j}.$$

Let  $D_z = z \frac{d}{dz}$  be the Euler derivative.

**Lemma 3.1** (shift–Euler identity). *For  $q \in \mathbb{C}^\times$ ,  $n \in \mathbb{Z}$ , and  $\ell \geq 0$ ,*

$$W(E)[n^\ell q^n] = q^n (n + D_z)^\ell W(z) \Big|_{z=q}. \quad (3.1)$$

*Proof.* Since  $D_z z^j = j z^j$ ,

$$(n + D_z)^\ell W(z) = \sum_{j=0}^M w_j (n + j)^\ell z^j.$$

Evaluation at  $z = q$  and multiplication by  $q^n$  gives the claimed shifted sum.  $\square$

**Theorem 3.2** (confluent geometric annihilator). *Let  $q_1, \dots, q_s \in \mathbb{C} \setminus \{0, 1\}$  be distinct and let  $m_1, \dots, m_s \geq 1$ . Define*

$$W_{\mathbf{q}, \mathbf{m}}(z) = \prod_{a=1}^s \left( \frac{z - q_a}{1 - q_a} \right)^{m_a} = \sum_{j=0}^M w_j z^j, \quad M = \sum_{a=1}^s m_a. \quad (3.2)$$

*Then:*

1.  $W_{\mathbf{q}, \mathbf{m}}(1) = 1$ , so constants are preserved;
2. for every  $a$  and every  $0 \leq \ell < m_a$ ,

$$W_{\mathbf{q}, \mathbf{m}}(E)[n^\ell q_a^n] = 0; \quad (3.3)$$

3. among polynomials preserving constants and satisfying (3.3), this is the unique polynomial of degree at most  $M$ ;
4. if  $X$  is a Banach space and

$$u_n = u + \sum_{a=1}^s \sum_{\ell=0}^{m_a-1} v_{a,\ell} n^\ell q_a^n + r_n, \quad (3.4)$$

*then*

$$W_{\mathbf{q}, \mathbf{m}}(E)u_n - u = \sum_{j=0}^M w_j r_{n+j}, \quad \|W_{\mathbf{q}, \mathbf{m}}(E)u_n - u\| \leq \left( \sum_{j=0}^M |w_j| \right) \max_{0 \leq j \leq M} \|r_{n+j}\|. \quad (3.5)$$

*Proof.* Normalization at  $z = 1$  is immediate. A zero of ordinary multiplicity  $m_a$  at  $q_a$  is also annihilated there by  $D_z^r$  for  $r < m_a$ , because  $D_z^r$  is a linear combination of  $z^j \frac{d^j}{dz^j}$  with  $j \leq r$ . Expanding  $(n + D_z)^\ell$  and applying lemma 3.1 proves the annihilation.

Conversely, annihilation for  $\ell = 0, 1, \dots, m_a - 1$  implies successively  $D_z^\ell W(q_a) = 0$ , hence an ordinary zero of multiplicity at least  $m_a$ . Any admissible polynomial is therefore divisible by  $\prod_a (z - q_a)^{m_a}$ . Degree at most  $M$  and  $W(1) = 1$  force (3.2). Substitution of (3.4) proves the exact remainder identity, and the triangle inequality gives (3.5).  $\square$

### 3.2 Exact conditioning for positive roots

**Theorem 3.3** (condition number). *If every  $q_a \in (0, 1)$ , then the coefficients of  $W_{\mathbf{q}, \mathbf{m}}$  have strict alternating signs and*

$$\kappa_{\min}(\mathbf{q}, \mathbf{m}) := \sum_{j=0}^M |w_j| = \prod_{a=1}^s \left( \frac{1+q_a}{1-q_a} \right)^{m_a}. \quad (3.6)$$

*Proof.* The roots form a positive multiset. Expanding their product shows that the coefficient of  $z^j$  has sign  $(-1)^{M-j}$ . Therefore

$$\sum_j |w_j| = (-1)^M W_{\mathbf{q}, \mathbf{m}}(-1) = \prod_a \left( \frac{1+q_a}{1-q_a} \right)^{m_a}.$$

□

**Example 3.4** (one repeated root). For a root  $q$  of multiplicity  $m$ ,

$$W_{q,m}(z) = \left( \frac{z-q}{1-q} \right)^m, \quad w_j = \frac{\binom{m}{j} (-q)^{m-j}}{(1-q)^m}, \quad \kappa_{\min} = \left( \frac{1+q}{1-q} \right)^m. \quad (3.7)$$

At  $q = 1/2$ ,  $m = 2$ , the filter is

$$W(E)u_n = u_n - 4u_{n+1} + 4u_{n+2},$$

and it cancels both  $2^{-n}$  and  $n2^{-n}$  exactly.

**Example 3.5** (mixed roots). A double root at  $1/2$  and a simple root at  $1/4$  give

$$W(z) = -\frac{1}{3} + \frac{8}{3}z - \frac{20}{3}z^2 + \frac{16}{3}z^3. \quad (3.8)$$

The filter cancels  $2^{-n}$ ,  $n2^{-n}$ , and  $4^{-n}$ , preserves constants, and has condition number 15. The accompanying program verifies these statements in exact rational arithmetic.

### 3.3 The Gaussian $q$ -Richardson row as the simple-root case

For one base  $q \in (0, 1)$  and the simple roots  $q, q^2, \dots, q^p$ , define

$$R_{p,q}(z) = \prod_{r=1}^p \frac{z - q^r}{1 - q^r} = \sum_{s=0}^p \omega_{p,s}(q) z^s. \quad (3.9)$$

The finite  $q$ -binomial theorem gives

$$\omega_{p,s}(q) = \frac{(-1)^{p-s} q^{(p-s)(p-s+1)/2}}{(q; q)_s (q; q)_{p-s}}. \quad (3.10)$$

This is the geometric Lagrange/Richardson row already developed in the corpus; for classical extrapolation context see [6]. The present theorem clarifies its spectral meaning: each inverse-power error mode corresponds to one simple root. Moreover, for  $r \geq p+1$ ,

$$R_{p,q}(q^r) = (-1)^p q^{p(p+1)/2} \begin{bmatrix} r-1 \\ p \end{bmatrix}_q. \quad (3.11)$$

Repeated roots are therefore the natural confluent extension of the audited Gaussian  $q$ -binomial row.

## 4 Prouhet–Richardson factorizations and bounded digital filters

The minimal filter has the shortest possible stencil, but its conditioning is exponential in a repeated-root order. A longer factorized filter can be dramatically more stable.

#### 4.1 A general factorized family

**Theorem 4.1** (Prouhet–Richardson factorization). *Fix  $q \in (0, 1)$  and positive integers  $a_1, \dots, a_m$ . Set*

$$D_{q,\mathbf{a}}(z) = \prod_{j=1}^m \frac{1 - (z/q)^{a_j}}{1 - q^{-a_j}}. \quad (4.1)$$

*Then  $D_{q,\mathbf{a}}(1) = 1$ , the point  $z = q$  is a zero of multiplicity  $m$ , and  $D_{q,\mathbf{a}}(E)$  annihilates  $n^\ell q^n$  for  $0 \leq \ell < m$ . Its stencil span is  $A = \sum_j a_j$ , and*

$$\|D_{q,\mathbf{a}}\|_1 \leq \prod_{j=1}^m \frac{1 + q^{a_j}}{1 - q^{a_j}}. \quad (4.2)$$

*Equality holds if all subset sums of the  $a_j$  are distinct.*

*Proof.* Each factor is normalized to equal one at  $z = 1$  and has a simple zero at  $z = q$ . The confluent annihilation follows from [theorem 3.2](#). Before like powers are collected, the absolute coefficient sum is the product of the absolute coefficient sums of the factors, namely the right-hand side of (4.2). Collecting colliding subset sums can only decrease that sum. With distinct subset sums there is no collection, hence equality.  $\square$

Two extremes are already revealing. Taking every  $a_j = 1$  recovers the minimal repeated root filter and its exponentially large condition number. Taking rapidly increasing  $a_j$  makes each factor in (4.2) close to one, at the cost of a very wide stencil. The binary choice is canonical because it fills its entire span without collisions.

#### 4.2 The twisted Thue–Morse filter

Let

$$T_m(z) = \prod_{j=0}^{m-1} (1 - z^{2^j}) = \sum_{n=0}^{2^m-1} \tau_n z^n, \quad \tau_n = (-1)^{s_2(n)}. \quad (4.3)$$

Put

$$\Theta_m(q) = T_m(q) = \prod_{j=0}^{m-1} (1 - q^{2^j}), \quad \Theta_{\text{TM}}(q) = \prod_{j=0}^{\infty} (1 - q^{2^j}). \quad (4.4)$$

**Theorem 4.2** (bounded Thue–Morse confluent filter). *For  $q \in (0, 1)$  define*

$$\mathcal{T}_{m,q}(z) = \frac{T_m(z/q)}{T_m(1/q)} = \sum_{n=0}^{2^m-1} t_{m,q}(n) z^n. \quad (4.5)$$

*Then:*

1.  $\mathcal{T}_{m,q}(1) = 1$  and  $z = q$  is a zero of multiplicity  $m$ ;
2. the coefficients are

$$t_{m,q}(n) = \frac{(-1)^m \tau_n q^{2^m-1-n}}{\Theta_m(q)}; \quad (4.6)$$

3. for every integer  $N$  and  $0 \leq \ell < m$ ,

$$\sum_{n=0}^{2^m-1} t_{m,q}(n) (N+n)^\ell q^{N+n} = 0; \quad (4.7)$$

4. its exact condition number is

$$\kappa_m^{\text{TM}}(q) = \sum_{n=0}^{2^m-1} |t_{m,q}(n)| \quad (4.8)$$

$$= \prod_{j=0}^{m-1} \frac{1 + q^{2^j}}{1 - q^{2^j}} = \frac{1 - q^{2^m}}{(1 - q)\Theta_m(q)}, \quad (4.9)$$

and therefore

$$\lim_{m \rightarrow \infty} \kappa_m^{\text{TM}}(q) = \frac{1}{(1 - q)\Theta_{\text{TM}}(q)} < \infty. \quad (4.10)$$



*Proof.* The first assertion follows from the product. Since

$$T_m(1/q) = (-1)^m q^{-(2^m-1)} T_m(q),$$

substitution into (4.5) gives (4.6). Multiplication by  $q^{N+n}$  removes the twist:

$$\sum_n t_{m,q}(n)(N+n)^\ell q^{N+n} = \frac{(-1)^m q^{N+2^m-1}}{\Theta_m(q)} \sum_{n=0}^{2^m-1} \tau_n(N+n)^\ell.$$

The final sum vanishes for  $\ell < m$  by the standard Prouhet–Thue–Morse cancellation [4], proving (4.7) directly.

For the norm, (4.6) gives

$$\sum_n |t_{m,q}(n)| = \frac{1 + q + \dots + q^{2^m-1}}{\Theta_m(q)} = \frac{1 - q^{2^m}}{(1 - q)\Theta_m(q)}.$$

The product form uses  $\prod_{j=0}^{m-1} (1 + q^{2^j}) = (1 - q^{2^m})/(1 - q)$ . Finally,  $\sum_j q^{2^j} < \infty$ , so  $\Theta_{\text{TM}}(q) > 0$  and the limit is finite.  $\square$

---

**A stability phenomenon specific to digital factorization.** For  $q = 1/2$ , the shortest order- $m$  filter has condition number  $3^m$ , while the Thue–Morse filter has condition number tending to

$$\frac{2}{\Theta_{\text{TM}}(1/2)} = 5.7112854057096334466 \dots$$

At order 10 the comparison is 59049 versus 5.7112854057... The digital stencil uses 1024 samples instead of 11; stability is purchased with span.

---

For example, at  $q = 1/2$ ,  $m = 2$ ,

$$\mathcal{T}_{2,1/2}(z) = \frac{1}{3} - \frac{2}{3}z - \frac{4}{3}z^2 + \frac{8}{3}z^3, \quad (4.11)$$

whose condition number is 5, compared with 9 for  $1 - 4z + 4z^2$ .

### 4.3 Why the powers of two are canonical

**Theorem 4.3** (minimal span of a collision-free factorization). *Let  $a_1, \dots, a_m$  be positive integers with all  $2^m$  subset sums distinct. Then*

$$\sum_{j=1}^m a_j \geq 2^m - 1. \quad (4.12)$$

*Equality holds if and only if, up to permutation,*

$$(a_1, \dots, a_m) = (1, 2, 4, \dots, 2^{m-1}). \quad (4.13)$$

*Proof.* The distinct subset sums are  $2^m$  distinct integers in the interval  $[0, A]$ , where  $A = \sum_j a_j$ . Hence  $A + 1 \geq 2^m$ . If equality holds, the subset sums are exactly every integer from 0 to  $A$ . Sort the  $a_j$ . Representing 1 forces  $a_1 = 1$ . If  $a_1, \dots, a_r$  are  $1, 2, \dots, 2^{r-1}$ , their subset sums fill  $[0, 2^r - 1]$ . Avoiding collision forces  $a_{r+1} \geq 2^r$ , while avoiding the gap at  $2^r$  forces  $a_{r+1} \leq 2^r$ . Induction proves (4.13).  $\square$

Thus the Thue–Morse filter is the unique shortest filter within the natural class whose factor expansion has no coefficient collisions and whose signed coefficients form a complete contiguous digital word. It is not globally optimal for every possible span–conditioning objective; that optimization problem is recorded in Section 11.

## 4.4 Mixed digital filters

For several modes  $q_a$  and desired confluences  $m_a$ , one may multiply the minimal filters, the digital filters, or a mixture. For instance,

$$\prod_{a=1}^s \mathcal{T}_{m_a, q_a}(z) \quad (4.14)$$

annihilates  $n^\ell q_a^n$  for every  $\ell < m_a$ . Its condition number is bounded by the product of the individual limits

$$\prod_{a=1}^s \frac{1}{(1 - q_a) \Theta_{\text{TM}}(q_a)}.$$

This is useful when a transseries contains a small number of highly confluent modes: the sample count grows, but coefficient amplification need not.

## 5 Rational multiplicative Lambert phase locks

### 5.1 Exact phase geometry

The corpus's additive phase lock uses  $\lambda, \lambda + 1, \dots$ , which preserves the fractional phase for every real  $\lambda$ . A rational phase admits a second, multiplicative geometry.

**Theorem 5.1** (multiplicative rational phase lock). *Let  $\theta = a/d \in [0, 1)$  with integers  $d \geq 1$  and  $0 \leq a < d$ . Let  $\rho > 1$  be an integer satisfying*

$$\rho \equiv 1 \pmod{d}. \quad (5.1)$$

*For every integer  $n$  satisfying  $n + \theta \geq 1/L$  (in particular, every  $n \geq 2$ ), define*

$$\lambda_s = \rho^s(n + \theta), \quad x_s = \lambda_s 2^{-\lambda_s}, \quad s \geq 0. \quad (5.2)$$

*Then  $\lambda(x_s) = \lambda_s$  on the lower branch, and  $\{\lambda_s\} = \theta$  for every  $s$ . With  $q = \rho^{-1}$  and  $H = (n + \theta)^{-1}$ ,*

$$\lambda_s^{-r} (\log \lambda_s)^\ell = H^r q^{rs} (\log(n + \theta) + s \log \rho)^\ell. \quad (5.3)$$

*Thus a logarithmically decorated inverse power becomes a polynomially decorated geometric mode in the orbit index  $s$ .*

*Proof.* Because  $d \mid (\rho^s - 1)$ ,

$$\rho^s \left( n + \frac{a}{d} \right) = \rho^s n + \frac{(\rho^s - 1)a}{d} + \frac{a}{d},$$

and the first two terms are integers. Moreover,  $\lambda_s \geq \lambda_0 = n + \theta \geq 1/L$ , so  $-L\lambda_s \leq -1$  and the identity  $(-L\lambda_s) \exp(-L\lambda_s) = -Lx_s$  is inverted by  $W_{-1}$ , giving  $\lambda(x_s) = \lambda_s$ . Formula (5.3) is direct.  $\square$

### 5.2 Confluent endpoint extraction

**Theorem 5.2** (Lambert–confluent Richardson filter). *Assume the phase-locked expansion (2.6) along the orbit in theorem 5.1. Define*

$$W(z) = \prod_{r=1}^R \left( \frac{z - q^r}{1 - q^r} \right)^{m_r}. \quad (5.4)$$

*Then  $W(E_s)$ , applied to the samples  $\mathcal{E}(\lambda_s)$ , preserves  $A_0(\theta)$  and cancels every displayed decorated term in (2.6). If  $q \in (0, 1)$ , its exact weight norm is*

$$\kappa = \prod_{r=1}^R \left( \frac{1 + q^r}{1 - q^r} \right)^{m_r}. \quad (5.5)$$

*The minimal factors in (5.4) may be replaced by the digital factors  $\mathcal{T}_{m_r, q^r}$  to reduce high-confluence coefficient growth.*

*Proof.* By (5.3), the term with indices  $(r, \ell)$  is a linear combination of  $s^j q^{rs}$  for  $0 \leq j \leq \ell < m_r$ . Apply theorems 3.2 and 3.3. Digital replacement follows from theorem 4.2.  $\square$

For the current pure inverse-power expansion, take  $m_r = 1$  for  $1 \leq r \leq p$ . The weights are exactly (3.10), now acting along a rational multiplicative Lambert orbit. If the expansion is available through  $A_{p+1}$ , then

$$\sum_{s=0}^p \omega_{p,s}(q) \mathcal{E}(\lambda_s) = A_0(\theta) + (-1)^p q^{p(p+1)/2} \frac{A_{p+1}(\theta)}{(n+\theta)^{p+1}} + \mathcal{O}(n^{-p-2}). \quad (5.6)$$

### 5.3 Conditioning versus the additive lock

The additive construction samples  $\lambda + j$  and interpolates in  $h_j = (\lambda + j)^{-1}$ . Evaluating the degree- $p$  interpolant at  $h = 0$  gives exact weights

$$a_{p,j}(\lambda) = \frac{(-1)^{p-j} (\lambda + j)^p}{j!(p-j)!}, \quad 0 \leq j \leq p. \quad (5.7)$$

Consequently

$$\kappa_{\text{add}}(\lambda, p) = \sum_{j=0}^p \frac{(\lambda + j)^p}{j!(p-j)!} = \frac{2^p}{p!} \lambda^p + \mathcal{O}_p(\lambda^{p-1}). \quad (5.8)$$

By contrast, the multiplicative pure-power row has

$$\kappa_{\text{mult}}(q, p) = \prod_{r=1}^p \frac{1+q^r}{1-q^r} \rightarrow \frac{(-q; q)_{\infty}}{(q; q)_{\infty}} < \infty \quad (p \rightarrow \infty), \quad (5.9)$$

independently of the starting  $n$ .

---

**Numerical conditioning comparison.** For phase  $1/8$ , choose  $\rho = 9$  and  $q = 1/9$ . At  $\lambda = 30 + 1/8$  and order  $p = 10$ , the exact formulas give

$$\kappa_{\text{add}} = 8.82199160936518 \times 10^{11}, \quad \kappa_{\text{mult}} = 1.28521058629968.$$

This compares linear weight amplification only. Multiplicative samples correspond to extremely separated endpoint arguments  $x_s = \lambda_s 2^{-\lambda_s}$  and therefore demand wide dynamic range or logarithmic evaluation; they are stable but not computationally free.

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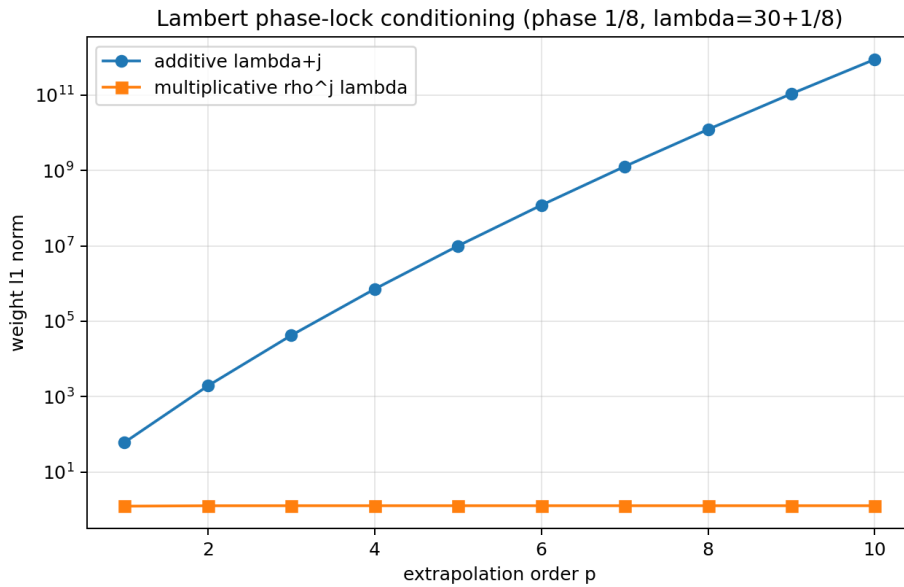


Figure 6: Exact  $\ell^1$  norms for additive and multiplicative phase-locked extrapolation. The multiplicative row uses the rational phase  $1/8$  and  $\rho = 9$ .

## 6 The maximal Gamma–zeta strip

The exact mode formula (2.5) contains more analytic information than mere smoothness of the periodic term.

### 6.1 Exact exponential type of the Fourier coefficients

For  $t > 0$ , the classical Gamma identity [10]

$$|\Gamma(it)|^2 = \frac{\pi}{t \sinh(\pi t)} \quad (6.1)$$

implies

$$\log |\Gamma(it)| = -\frac{\pi t}{2} - \frac{1}{2} \log t + \frac{1}{2} \log(2\pi) + o(1). \quad (6.2)$$

Classical bounds on the line  $\Re s = 1$  [7, 8, 9] give

$$|\zeta(1+it)| \ll \log(2+t), \quad |\zeta(1+it)|^{-1} \ll \log(2+t), \quad (6.3)$$

so  $\log |\zeta(1+it)| = o(t)$ .

**Theorem 6.1** (exact Fourier exponential type). *Let  $c_k$  be given by (2.5). Then*

$$\boxed{\lim_{|k| \rightarrow \infty} -\frac{1}{|k|} \log |c_k| = \alpha := \frac{\pi^2}{\log 2}.} \quad (6.4)$$

*More quantitatively,*

$$|c_k| \ll \frac{\log(2+|k|)}{\sqrt{|k|}} \exp\left(-\frac{\pi^2}{\log 2} |k|\right). \quad (6.5)$$

*Proof.* For  $k > 0$ , put  $t_k = 2\pi k/L$ . Equations (2.5), (6.2), and (6.3) give

$$\log |c_k| = -\frac{\pi t_k}{2} + o(t_k) = -\frac{\pi^2}{L} k + o(k).$$

The  $k < 0$  case follows by conjugation. Keeping the elementary prefactor in (6.1) and the upper zeta bound gives (6.5).  $\square$

### 6.2 Maximal holomorphy

**Theorem 6.2** (maximal horizontal strip). *The one-periodic function*

$$A_0(z) = c_0 + \sum_{k \neq 0} c_k e^{2\pi i k z} \quad (6.6)$$

*extends holomorphically to*

$$\boxed{\mathcal{S}_* = \{z \in \mathbb{C} : |\Im z| < \sigma_*\}, \quad \sigma_* := \frac{\pi}{2 \log 2}.} \quad (6.7)$$

*It has no holomorphic continuation to any strictly wider horizontal strip.*

*Proof.* If  $|\Im z| < \sigma_*$ , the extra Fourier factor has magnitude at most  $\exp(2\pi |k| |\Im z|)$ . The upper bound (6.5) makes the series locally normally convergent because  $2\pi |\Im z| < \pi^2/L$ . Hence it is holomorphic in  $\mathcal{S}_*$ .

Suppose it continued to  $|\Im z| < \sigma_* + \delta$ . Periodicity continues by the identity theorem. For  $k > 0$ , shift the Fourier-coefficient contour to the compact line  $\Im z = -(\sigma_* + \delta/2)$ . Boundedness on one period gives

$$|c_k| \leq C \exp(-2\pi(\sigma_* + \delta/2)k),$$

contradicting (6.4), because  $2\pi\sigma_* = \alpha$ . Shifting upward gives the same contradiction for negative modes.  $\square$

**Remark 6.3.** Maximality of the symmetric strip does not by itself prove that every point of either boundary line is singular. The stronger natural-boundary statement is a conjecture in Section 11.

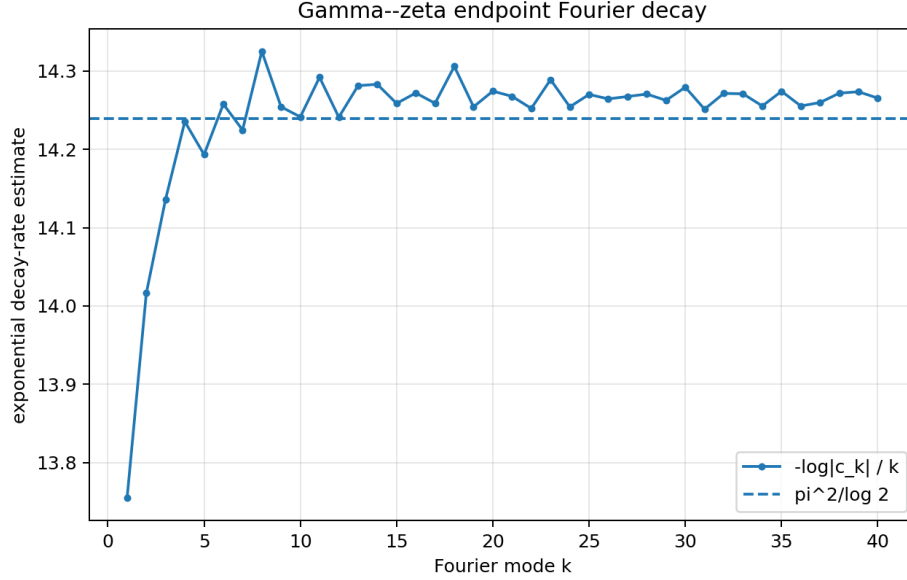


Figure 7: High-precision values of  $-\log |c_k|/k$  and the exact limiting constant  $\pi^2/\log 2$ . Small oscillations come from the zeta factor; the Gamma factor fixes the exponential rate.

## 7 Weighted Wiener algebras and Bell-polynomial stability

The maximal strip gives quantitative control of every differential-polynomial periodic coefficient built from  $A_0$ .

**Definition 7.1.** For  $\sigma \geq 0$ , let  $\mathcal{A}_\sigma$  be the weighted Wiener algebra of one-periodic Fourier series

$$f(z) = \sum_{k \in \mathbb{Z}} \widehat{f}(k) e^{2\pi i k z}, \quad \|f\|_\sigma = \sum_{k \in \mathbb{Z}} |\widehat{f}(k)| e^{2\pi \sigma |k|} < \infty.$$

**Theorem 7.2** (Wiener–Bell calculus). *Let  $0 \leq \tau < \sigma$ .*

1.  $\mathcal{A}_\sigma$  is a Banach algebra and  $\|fg\|_\sigma \leq \|f\|_\sigma \|g\|_\sigma$ .
2. If  $f \in \mathcal{A}_\sigma$ , then for  $r \geq 1$ ,

$$\|f^{(r)}\|_\tau \leq \left( \frac{r}{e(\sigma - \tau)} \right)^r \|f\|_\sigma. \quad (7.1)$$

3. If  $S_K f = \sum_{|k| \leq K} \widehat{f}(k) e^{2\pi i k z}$ , then

$$\sup_{|\Im z| \leq \tau} |f(z) - S_K f(z)| \leq e^{-2\pi(\sigma - \tau)(K+1)} \|f\|_\sigma. \quad (7.2)$$

4. For the complete exponential Bell polynomial  $\mathcal{B}_n$  and  $f_1, \dots, f_n \in \mathcal{A}_\sigma$ ,

$$\|\mathcal{B}_n(f_1, \dots, f_n)\|_\sigma \leq \mathcal{B}_n(\|f_1\|_\sigma, \dots, \|f_n\|_\sigma). \quad (7.3)$$

In particular, if  $g = \exp f$ , then

$$\|g^{(n)}\|_\tau \leq e^{\|f\|_\tau} \mathcal{B}_n(\|f'\|_\tau, \dots, \|f^{(n)}\|_\tau). \quad (7.4)$$

*Proof.* The product estimate is the weighted convolution inequality. For derivatives, the Fourier multiplier is  $(2\pi i k)^r$ , and

$$(2\pi x)^r e^{-2\pi(\sigma - \tau)x} \leq \left( \frac{r}{e(\sigma - \tau)} \right)^r.$$

The tail estimate follows by extracting the factor  $e^{-2\pi(\sigma - \tau)(K+1)}$ . Complete Bell polynomials have nonnegative scalar coefficients, so repeated use of the Banach-algebra inequality proves (7.3). The last assertion is Faà di Bruno's formula  $g^{(n)} = g \mathcal{B}_n(f', \dots, f^{(n)})$  and  $\|\exp f\|_\tau \leq \exp \|f\|_\tau$ .  $\square$

**Corollary 7.3** (certified endpoint coefficient truncation). *The leading Fabius endpoint fluctuation  $A_0$  belongs to  $A_\sigma$  for every  $\sigma < \sigma_*$ . Any coefficient obtained from finitely many derivatives, products, Bell polynomials, and scalar Bernoulli constants belongs to the same family of algebras. Its Fourier truncation error on every smaller strip is exponentially certified by (7.2).*

This supplies a functional-analytic justification for a practical operation repeatedly needed in the all-orders endpoint theory: compute a short Fourier representation of the primitive periodic term, differentiate and combine it through Bell polynomials, and retain rigorous control rather than relying only on pointwise numerical smallness.

## 8 Radial–angular Lambert phase tomography

The multiplicative phase orbits recover values of periodic coefficient functions. A phase grid then recovers their Fourier spectra.

### 8.1 Radial interpolation of the coefficient jet

Fix a rational phase  $\theta$ , an admissible  $\rho$ ,  $q = \rho^{-1}$ , and  $H = (n + \theta)^{-1}$ . Suppose

$$y_s = \mathcal{C}(\rho^s(n + \theta)) = \sum_{j=0}^p B_j q^{sj} + r_s, \quad B_j = A_j(\theta) H^j. \quad (8.1)$$

Let  $\ell_s(t)$  be the Lagrange basis at the geometric nodes  $1, q, \dots, q^p$ :

$$\ell_s(t) = \prod_{\substack{0 \leq r \leq p \\ r \neq s}} \frac{t - q^r}{q^s - q^r}. \quad (8.2)$$

Define

$$\hat{A}_j(\theta) = H^{-j} [t^j] \sum_{s=0}^p y_s \ell_s(t). \quad (8.3)$$

**Theorem 8.1** (radial jet recovery). *If  $r_s = 0$ , (8.3) recovers  $A_0(\theta), \dots, A_p(\theta)$  exactly. In general,*

$$|\hat{A}_j(\theta) - A_j(\theta)| \leq H^{-j} \sum_{s=0}^p |[t^j] \ell_s(t)| |r_s|. \quad (8.4)$$

*If  $r_s = \mathcal{O}(H^{p+1} q^{s(p+1)})$  uniformly in phase, then*

$$\hat{A}_j(\theta) - A_j(\theta) = \mathcal{O}(H^{p+1-j}). \quad (8.5)$$

*For  $j = 0$ , the weights are exactly (3.10).*

*Proof.* The sample vector without remainders is the evaluation of the polynomial  $\sum_{j=0}^p B_j t^j$  at  $t = q^s$ . Lagrange interpolation reconstructs it exactly. Subtracting the remainder interpolant gives (8.4). Since  $p$  and  $q$  are fixed, the coefficient weights are fixed constants, proving (8.5).  $\square$

The full coefficient weights also have a closed  $q$ -binomial form. Put  $X_p = \{1, q, \dots, q^p\}$ . The denominator is

$$\prod_{r \neq s} (q^s - q^r) = (-1)^s q^{sp-s(s+1)/2} (q; q)_s (q; q)_{p-s}, \quad (8.6)$$

and

$$[t^j] \ell_s(t) = \frac{(-1)^{p-j} e_{p-j}(X_p \setminus \{q^s\})}{(-1)^s q^{sp-s(s+1)/2} (q; q)_s (q; q)_{p-s}}. \quad (8.7)$$

The excluded elementary symmetric polynomial can be reduced to Gaussian binomials through

$$e_m(X_p \setminus \{q^s\}) = \sum_{a=0}^m (-q^s)^a q^{(m-a)(m-a-1)/2} \begin{bmatrix} p+1 \\ m-a \end{bmatrix}_q. \quad (8.8)$$

Thus the entire radial coefficient jet is governed by the same  $q$ -binomial algebra as the zeroth-order Richardson row.

## 8.2 Angular DFT and exact aliasing

Let a one-periodic coefficient have an absolutely convergent Fourier expansion

$$A(u) = \sum_{k \in \mathbb{Z}} c_k e^{2\pi i k u}. \quad (8.9)$$

Sample it at the rational phases  $u_r = r/d$  and define

$$\tilde{c}_k^{(d)} = \frac{1}{d} \sum_{r=0}^{d-1} A(r/d) e^{-2\pi i k r/d}. \quad (8.10)$$

**Theorem 8.2** (exact phase alias identity). *For every integer  $k$ ,*

$$\boxed{\tilde{c}_k^{(d)} = \sum_{m \in \mathbb{Z}} c_{k+md}.} \quad (8.11)$$

*If, for some  $\beta > 0$ ,  $|c_k| \leq C e^{-\beta|k|}$  and  $|k| \leq (d-1)/2$ , then*

$$|\tilde{c}_k^{(d)} - c_k| \leq \frac{2C e^{-\beta(d-|k|)}}{1 - e^{-\beta d}}. \quad (8.12)$$

*Proof.* Insert (8.9) into (8.10). The finite geometric sum is one precisely when the Fourier index is congruent to  $k$  modulo  $d$ , and zero otherwise. For the bound, every nonprincipal alias satisfies  $|k + md| \geq |m|d - |k|$ ; sum the two geometric tails.  $\square$

By [theorem 6.1](#), for every  $\varepsilon > 0$  the Fabius coefficients obey  $|c_k| \leq C_\varepsilon \exp[-(\alpha - \varepsilon)|k|]$ . Hence the DFT alias error has the sharp available exponential scale  $\alpha = \pi^2/L$ .

## 8.3 The combined tomography theorem

**Theorem 8.3** (two-axis Lambert phase tomography). *Let  $d \geq 2$ , choose  $\rho \equiv 1 \pmod{d}$ , and use the radial order- $p$  estimator at each phase  $r/d$ . Suppose the normalized endpoint residual has the uniform expansion (2.4) through order  $p+1$ , and suppose  $A_0 \in \mathcal{A}_\sigma$  for some  $\sigma > 0$ . For  $|k| \leq (d-1)/2$ , the DFT of the radial estimates satisfies*

$$\hat{c}_k^{(d,p,n)} - c_k = \mathcal{O}(n^{-p-1}) + \mathcal{O}\left(e^{-2\pi\sigma(d-|k|)}\right), \quad (8.13)$$

*uniformly over the phase grid. For the leading Fabius periodic term one may take every  $\sigma < \pi/(2L)$ , so the angular exponent can be made arbitrarily close to  $\pi^2/L$ .*

*Proof.* Equation (5.6) gives a uniform  $\mathcal{O}(n^{-p-1})$  error at each phase. Averaging by a DFT does not increase the maximum absolute error. Apply [theorem 8.2](#) to the exact phase values, with the Fourier bound supplied by membership in  $\mathcal{A}_\sigma$ .  $\square$

The same construction applies to every  $A_j$ : use the full radial jet estimator (8.3), then a DFT in the phase variable. By [corollary 7.3](#), periodic coefficients formed through the audited Bell/Bernoulli saddle calculus inherit exponentially controlled phase spectra.

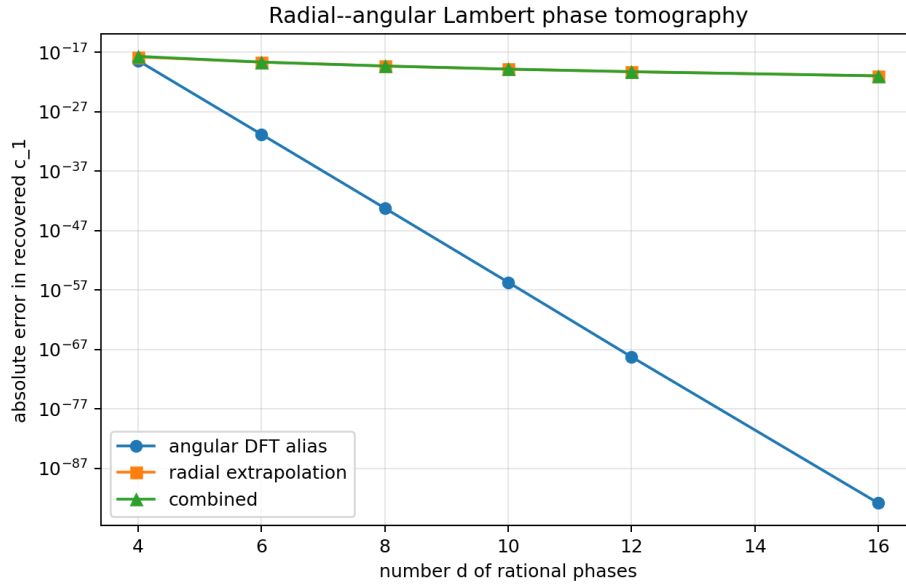


Figure 8: A controlled two-axis experiment recovering  $c_1$ . The angular samples use the actual Gamma–zeta periodic term; documented smooth surrogate functions supply higher inverse-Lambert coefficients so that the radial cancellation can be tested independently.

## 9 Numerical and exact-arithmetic experiments

The retained program `assets/fabius_frontier_report_bundle/numerical_experiments.py` is deliberately independent of the LaTeX source. It uses exact Fraction arithmetic for all polynomial filters and 90-decimal-digit `mpmath` arithmetic for Gamma–zeta calculations. The zeta function is forced to use Euler–Maclaurin evaluation rather than an automatic method selection, avoiding threshold-dependent loss of reproducibility on the line  $\Re s = 1$ .

### 9.1 Exact filter verification

For  $q = 1/2$  and every order  $1 \leq m \leq 10$ , the script checks

$$\sum_j w_j (N+j)^\ell q^{N+j} = 0, \quad 0 \leq \ell < m,$$

for both the minimal and twisted Thue–Morse filters. Every residual is an exact rational zero. It also verifies the mixed filter (3.8).

Table 16: Span–conditioning tradeoff at  $q = 1/2$ .

| $m$ | minimal span | minimal $\kappa$ | TM span | TM $\kappa$  |
|-----|--------------|------------------|---------|--------------|
| 1   | 1            | 3                | 1       | 3.0000000000 |
| 2   | 2            | 9                | 3       | 5.0000000000 |
| 3   | 3            | 27               | 7       | 5.6666666667 |
| 4   | 4            | 81               | 15      | 5.7111111111 |
| 5   | 5            | 243              | 31      | 5.7112854031 |
| 6   | 6            | 729              | 63      | 5.7112854057 |
| 8   | 8            | 6561             | 255     | 5.7112854057 |
| 10  | 10           | 59049            | 1023    | 5.7112854057 |



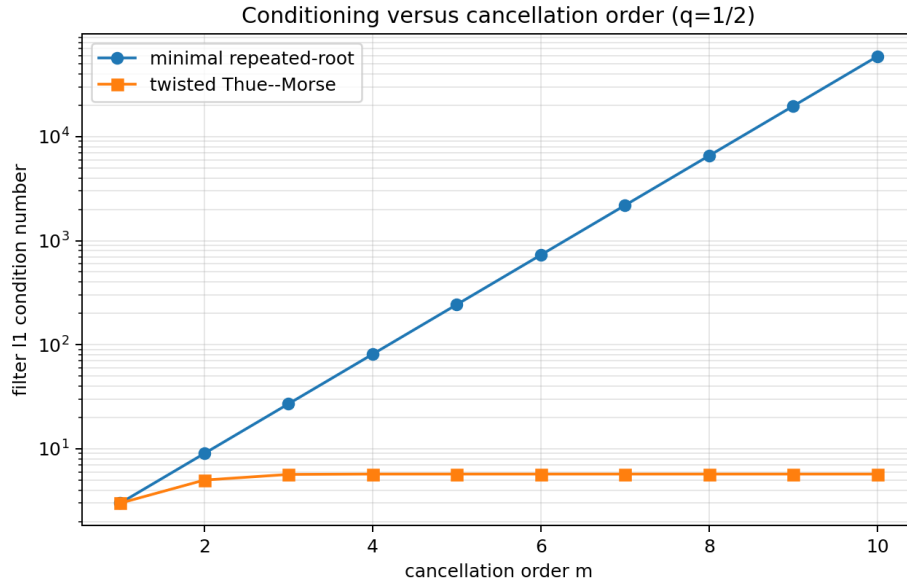


Figure 9: Minimal repeated-root and twisted Thue–Morse condition numbers. The vertical axis is logarithmic; the digital curve rapidly approaches its finite limit.

## 9.2 Gamma–zeta modes

The first nonzero modes are:

Table 17: High-precision magnitudes of the endpoint Fourier coefficients.

| $k$ | $ c_k $                            | $-\log  c_k /k$ |
|-----|------------------------------------|-----------------|
| 1   | $1.06271162518632 \times 10^{-6}$  | 13.7546867793   |
| 2   | $6.69278636792434 \times 10^{-13}$ | 14.0162879620   |
| 3   | $3.82107872358669 \times 10^{-19}$ | 14.1361946653   |
| 4   | $1.86460518002229 \times 10^{-25}$ | 14.2353944985   |
| 5   | $1.51268570169932 \times 10^{-31}$ | 14.1932502403   |
| 10  | $1.41459209881527 \times 10^{-62}$ | 14.2413434545   |

The exact limiting rate is

$$\alpha = 14.2388293249875054338606757113\dots, \quad \sigma_* = 2.26618007091359690481384147286\dots$$

## 9.3 Tomography experiment

The angular part evaluates the actual truncated Gamma–zeta series for  $A_0$ . To test radial cancellation without silently importing unparsed formulas for the higher audited saddle coefficients, the program defines explicit smooth surrogate coefficients

$$\hat{A}_j(k) = 0.2^j \left( 1 + \frac{k^2}{(j+1)^2} \right) c_k, \quad j \geq 1.$$

This surrogate has the same real symmetry and strip structure; it is not claimed to equal the true  $A_j$ . Radial order  $p = 3$ , starting integer  $n = 16$ , and  $\rho = d + 1$  give:

Table 18: Recovery error for  $c_1$  in the controlled radial–angular model.

| $d$ | angular alias            | radial error             | combined error           |
|-----|--------------------------|--------------------------|--------------------------|
| 4   | $3.8211 \times 10^{-19}$ | $1.6041 \times 10^{-18}$ | $1.9862 \times 10^{-18}$ |
| 6   | $1.5127 \times 10^{-31}$ | $2.1303 \times 10^{-19}$ | $2.1303 \times 10^{-19}$ |
| 8   | $5.7018 \times 10^{-44}$ | $4.7054 \times 10^{-20}$ | $4.7054 \times 10^{-20}$ |
| 10  | $1.9284 \times 10^{-56}$ | $1.4089 \times 10^{-20}$ | $1.4089 \times 10^{-20}$ |
| 12  | $5.2972 \times 10^{-69}$ | $5.1635 \times 10^{-21}$ | $5.1635 \times 10^{-21}$ |
| 16  | $1.2993 \times 10^{-93}$ | $1.0305 \times 10^{-21}$ | $1.0305 \times 10^{-21}$ |

The exact rational phase drift is zero in every sample. Once  $d \geq 6$ , angular aliasing is negligible relative to the order-three radial remainder, exactly as (8.13) predicts.

## 10 Consequences for the Fabius–Rvachev research program

### 10.1 A single annihilator language

Three constructions that look unrelated become instances of one spectral polynomial.

1. Ordinary Richardson extrapolation puts simple roots at the known error ratios.
2. Confluent Richardson puts repeated roots at ratios carrying polynomial prefactors.
3. Prouhet–Thue–Morse cancellation is a digitally factorized repeated root, with the exponential twist chosen so that the geometric factor cancels before the Prouhet sum is taken.

This shift-polynomial language is especially convenient for Lean: the analytic application is reduced to a finite polynomial divisibility lemma plus a norm estimate.

### 10.2 Endpoint coefficient spectroscopy

The exact-Lambert phase is not only an asymptotic bookkeeping device. Rational multiplicative orbits turn the endpoint into a laboratory in which one may recover the periodic coefficient functions  $A_j$  separately. A practical high-precision workflow is:

1. choose a phase denominator  $d$  and  $\rho \equiv 1 \pmod{d}$ ;
2. evaluate the normalized log Fabius residual at  $x_{r,s} = \lambda_{r,s} 2^{-\lambda_{r,s}}$ , where  $\lambda_{r,s} = \rho^s(n + r/d)$ ;
3. perform radial  $q$ -interpolation to recover  $A_j(r/d)$ ;
4. perform a DFT over  $r$  to recover Fourier coefficients;
5. certify radial error from the next asymptotic order and angular error from the Wiener strip.

The method applies equally to coefficients in the inverse-Fabius endpoint expansion once the corresponding normalized residual is written in the exact Lambert coordinate.

### 10.3 Bell and Bernoulli closure

The corpus’s Bell-polynomial coefficient calculus often makes formulas look combinatorially explosive. In the weighted Wiener algebra, however, every product and Bell composition is norm-submultiplicative. Bernoulli numbers enter as scalar coefficients, while derivatives are controlled by the strip margin. This suggests a certified implementation architecture: store periodic functions as weighted Fourier vectors, build each all-orders coefficient by Bell operations, and propagate a scalar norm bound in parallel. Numerical truncation then comes with a theorem-level remainder certificate.

## 10.4 Generalized Pascal–Rvachev hierarchies

Higher-rank Mellin poles naturally produce polynomial factors in the logarithmic scale. Along a multiplicative phase orbit these are exactly the  $s^\ell q^{r^s}$  modes treated by confluence. The digital filter is therefore likely to be more useful for the hierarchy than for the classical rank-one case: higher pole orders increase confluence, and the minimal filter’s condition number grows exponentially in that order, while a digital factorization can keep it bounded.

## 11 Conjectures and promising research directions

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**Conjecture 1: natural boundary of the endpoint fluctuation.** The two lines

$$\Im z = \pm \frac{\pi}{2 \log 2}$$

are natural boundaries for the leading periodic endpoint function  $A_0$ . The maximal-strip theorem proves that some obstruction occurs at each side, but does not locate the singular set. A route to proof is to analyze the phase of  $\Gamma(-\chi_k)\zeta(1-\chi_k)$  and apply a gap-free natural-boundary criterion for Fourier series of exact exponential type.

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**Conjecture 2: sharp DFT alias exponent.** For every fixed nonzero mode  $k$ , the exact phase-grid alias  $\sum_{m \neq 0} c_{k+md}$  satisfies

$$\log \left| \sum_{m \neq 0} c_{k+md} \right| = -\frac{\pi^2}{\log 2} d + o(d)$$

outside, at most, a sparse exceptional set of grid sizes. The upper exponential rate is proved here; the issue is ruling out systematic phase cancellation among the nearest aliases.

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*Open problem 11.1* (optimal span–conditioning frontier). For  $q \in (0, 1)$  define

$$K_m(q; D) = \inf \left\{ \sum_{j=0}^D |w_j| : W(1) = 1, (z-q)^m \mid W(z), \deg W \leq D \right\}.$$

Determine  $K_m(q; D)$  uniformly between the endpoints

$$K_m(q; m) = \left( \frac{1+q}{1-q} \right)^m, \quad \inf_D K_m(q; D) = 1.$$

The Prouhet factorizations provide explicit upper bounds. The binary choice is uniquely minimal-span among collision-free complete digital stencils, but the full Pareto frontier appears to be a nontrivial extremal problem in polynomial approximation and linear programming.

*Open problem 11.2* (irrational near-locks). For irrational phase  $\theta$ , exact multiplicative locks do not exist. Continued-fraction approximants can choose  $\rho$  with  $\|(\rho-1)\theta\|$  small, but the phase defect is multiplied by  $1+\rho+\dots+\rho^{s-1}$ . Develop a simultaneous Diophantine criterion that balances this phase drift against the bounded Richardson weights and the strip-controlled phase derivative. The expected result is a provable work–accuracy tradeoff for irrational phase tomography.

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**Conjecture 3: common strip in the Pascal–Rvachev hierarchy.** After subtraction of the rank-dependent logarithmic polynomial, every periodic coefficient in the Pascal–Rvachev endpoint hierarchy has maximal strip half-width  $\pi/(2 \log 2)$ . Higher rank changes only polynomial factors in the Fourier mode and hence the boundary singularity order, not the Gamma-controlled exponential type.

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**Conjecture 4: boundary singularities control late Bell coefficients.** The large-order growth of the all-orders endpoint and inverse-endpoint Bell-polynomial coefficients is governed by the nearest singularities on the two strip boundaries. After appropriate normalization, the coefficient sequence should admit a resurgent factorial-over-power law with phase determined by the boundary singularities. Establishing this would connect the exact Gamma–zeta Fourier spectrum to optimal truncation and exponentially improved Lambert asymptotics.

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## 11.1 A concrete Lean formalization order

The new results break naturally into short formal modules.

1. Define polynomial shift operators on sequences valued in a normed additive group.
2. Prove lemma 3.1 for scalar geometric-polynomial modes.
3. Formalize root multiplicity, the minimal annihilator, and the finite norm bound.
4. Import the existing finite Thue–Morse polynomial and prove  $\mathcal{T}_{m,q} = T_m(z/q)/T_m(1/q)$ , coefficient formula (4.6), and norm formula (4.9).
5. Prove the modular arithmetic phase lock theorem 5.1 over rationals.
6. Reuse the existing Gaussian  $q$ -binomial Lagrange development for the radial jet.
7. Formalize the finite DFT alias identity; postpone the complex-analytic maximal-strip theorem until Gamma and zeta estimates are available in the chosen library layer.

The first six items are predominantly finite algebra and should be substantially easier than the endpoint analytic inputs already present in the project.

## 12 Conclusion

The main structural discovery is not another isolated formula for  $F$ ,  $F^{-1}$ , or  $\widehat{\text{up}}$ . It is a reusable spectral calculus.

A root of a shift polynomial represents an asymptotic geometric mode; multiplicity represents its polynomial decoration. The standard  $q$ -Richardson row uses simple roots. Confluence handles logarithmic and polynomial decorations. A Thue–Morse product spreads a repeated root over a binary stencil, exchanging span for bounded conditioning. The exact Lambert coordinate turns rational endpoint phases into geometric orbits on which those filters act. Finally, the Gamma–zeta mode formula supplies the sharp analytic strip needed to reconstruct the phase dependence by DFT and to certify every subsequent Bell-polynomial operation.

This package gives both theorem-level results and a practical experimental protocol. It also identifies clear next targets: the span-conditioning extremal problem, natural boundaries, irrational phase locks, hierarchy-wide strip theorems, and resurgent late-order analysis. Each direction is tightly connected to existing formalized or audited machinery, while avoiding duplication of the substantial frontier results already present in the repository.

## A Auxiliary identities

### A.1 Complete homogeneous form of transformed moments

For the geometric Lagrange weights, let

$$\mu_{p,r}(q) = \sum_{s=0}^p \omega_{p,s}(q) q^{sr}.$$

Then  $\mu_{p,0} = 1$ ,  $\mu_{p,r} = 0$  for  $1 \leq r \leq p$ , and for  $r \geq p+1$ ,

$$\mu_{p,r}(q) = (-1)^p q^{p(p+1)/2} h_{r-p-1}(1, q, \dots, q^p), \quad (\text{A.1})$$

where  $h_j$  is the complete homogeneous symmetric polynomial. The identity follows from the remainder of interpolation of  $t^r$  at  $1, q, \dots, q^p$ . The standard specialization

$$h_{r-p-1}(1, q, \dots, q^p) = \begin{bmatrix} r-1 \\ p \end{bmatrix}_q$$

recovers (3.11).

## A.2 Additive phase weights

For completeness, the derivation of (5.7) is one line. At nodes  $h_j = (\lambda + j)^{-1}$ , the Lagrange weight at zero is

$$\prod_{r \neq j} \frac{-h_r}{h_j - h_r} = \prod_{r \neq j} \frac{-(\lambda + j)}{r - j} = \frac{(-1)^{p-j}(\lambda + j)^p}{j!(p-j)!}.$$

This exact formula makes the conditioning contrast transparent; it is not a numerical artifact of solving a Vandermonde system.

## A.3 Digital condition limit

The finite identity

$$\prod_{j=0}^{m-1} (1 + q^{2^j}) = \frac{1 - q^{2^m}}{1 - q}$$

is simply the binary factorization of  $1 - q^{2^m}$ . Dividing by  $\Theta_m(q)$  gives (4.9); monotone convergence of  $\Theta_m(q)$  gives (4.10).

# B Reproduction guide

From the consolidated volume directory, run

```
(cd assets/fabius_frontier_report_bundle && \
  python numerical_experiments.py --output-dir . --precision 90)
latexmk -pdf -interaction=nonstopmode -halt-on-error Frontier_Compilations.tex
```

The script regenerates the following files inside the retained asset directory:

- filter\_conditioning.csv and conditioning\_tradeoff.png;
- lambert\_conditioning.csv and lambert\_conditioning.png;
- gamma\_zeta\_coefficients.csv and gamma\_zeta\_decay.png;
- tomography\_errors.csv and tomography\_errors.png;
- experiment\_results.txt and verification.json.

Exact filter checks use Python’s Fraction; no tolerance is involved. Floating experiments use mpmath. The source comments document the surrogate higher endpoint coefficients and every normalization.

# C Complete pinned LaTeX inventory

The following manifest is retained under assets/fabius\_frontier\_report\_bundle/corpus\_manifest.txt. It records the complete recursive 78-path corpus boundary used for the nonduplication audit.

```
Pinned repository snapshot
=====
Repository: VladimirReshetnikov/ProveIt
Commit: f49aad5c9c6d627a64822ea9f8e46f5270b95cef
Documentation subtree: Analysis/FabiusFunction/docs
Recursive docs tree SHA: 6063d0d90cd9a4d25ca74cd4f8b80448313fce64
Inventory method: recursive Git tree, not GitHub's lagging code-search index
LaTeX paths in the pinned subtree: 78
```

The inventory includes ordinary manuscripts, historical revisions, imported/corrected source papers, and six small generated TeX table/check fragments. Historical copies were read as lineages and checked against their current successors rather than counted as independent mathematical evidence.

#### Paths

=====

1. FabiusFunction\_Mathematical\_Glossary/FabiusFunction\_Mathematical\_Glossary.tex
2. Fabius\_Function\_and\_Rvachev\_Up/Fabius\_Function\_and\_Rvachev\_Up.tex
3. Non\_Elementarity\_of\_the\_Fabius\_Function/Non\_Elementarity\_of\_the\_Fabius\_Function.tex
4. archive/early-notes/Fabius Asymptotic/Fabius Asymptotic.tex
5. archive/early-notes/K-fold summation over the signed Thue-Morse sequence/K-fold summation over the signed Thue-Morse sequence.tex
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## Part VII

# Dyadic Spectral Determinants for the Fabius–Rvachev System

## Abstract

Let

$$\Phi(z) = \widehat{\text{up}}(z) = \prod_{j=0}^{\infty} \text{sinc}\left(\frac{\pi z}{2^j}\right), \quad \text{sinc } w = \frac{\sin w}{w}.$$

The inspected corpus already contains the canonical integer-zero product, the exact multiplicity  $1 + \nu_2(m)$  at a nonzero integer  $m$ , and the Thue–Morse sign law between consecutive zeros. The present report does not relabel those facts as new. Instead it introduces the diagonal operator

$$De_{j,n} = 2^j n e_{j,n} \quad (j \geq 0, n \geq 1)$$

and proves

$$\Phi(z) = \det(I - z^2 D^{-2}), \quad D \simeq N \oplus 2D.$$

The decomposition gives the exact positive zero count  $2M - s_2(M)$ , the spectral zeta function

$$\zeta_D(s) = \frac{\zeta(s)}{1 - 2^{-s}},$$

and a heat trace with an explicit real-analytic log-periodic term. We derive a normalized, exact all-order Taylor factorization at every integer zero. We then define a dyadic  $q$ -Euler determinant

$$\mathcal{E}(u; Q) = \det(I - uQ^D) = \prod_{j \geq 0} (uQ^{2^j}; Q^{2^j})_{\infty},$$

obtain finite multiscale Gaussian-binomial expansions, and calculate the complete small- $t$  Mellin asymptotics of  $\mathcal{E}(u; e^{-t})$ , including the critical  $u = 1$  quadratic logarithm. Finally, with the symmetric  $q$ -integer  $[m]_Q$ , we construct

$$\Phi_Q(z) = \prod_{m \geq 1} \left(1 - \frac{z^2}{[m]_Q^2}\right)^{1 + \nu_2(m)}$$

and prove an exact  $q$ -Pochhammer factorization, the scaling law

$$\Phi_Q(z) = S_Q(z) \Phi_{Q^2}(z/[2]_Q),$$

and locally uniform convergence  $\Phi_Q \rightarrow \Phi$  as  $Q \uparrow 1$ . This supplies a direct structural bridge from the Fourier zero divisor to  $q$ -Pochhammer symbols and  $q$ -binomial coefficients.

# 1 Scope, corpus audit, and novelty protocol

## 1.1 What was inspected

The recursive documentation tree at <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs> is not a single article. It is a stratified mathematical record containing living expositions, an explicitly frozen historical archive, a consolidated non-formalized frontier volume, a Thue–Morse formula atlas, twelve successive Fourier-decay analyses, and imported source transcriptions of external papers. The corpus ledger in [Section A](#) records 57 TeX artifacts:

| Stratum                        | TeX artifacts | Role                                           |
|--------------------------------|---------------|------------------------------------------------|
| Living and canonical documents | 18            | Current expositions and active frontier drafts |
| Frozen historical archive      | 32            | Provenance, superseded drafts, and deltas      |
| Imported paper sources         | 7             | External mathematical context                  |
| Total                          | 57            | Recursive corpus audited                       |

Every TeX artifact was included in the audit. Mathematical extraction was concentrated on the living/canonical descendants and the active Fourier series; frozen ancestors were compared through their archived first-version source, provenance descriptions, and surviving distinctive sections. This distinction matters: the archive intentionally preserves superseded states and explicitly prunes byte-identical copies. Treating all 57 files as independent contemporary claims would create false novelty and heavy duplication.

## 1.2 Three status classes

**Status convention.** Throughout this report, *established baseline* means a statement already present in the inspected repository or in a cited external source. *New deduction* means that the theorem and proof were not found in the inspected corpus or in targeted searches made for this report. This is a relative corpus claim, not a universal priority claim. The proofs below are mathematical proofs written for human verification. Except where an explicit Lean-status note cites a declaration, they have not yet been checked by Lean.

The main examples of statements that are *not* claimed here as new are:

- (i) the infinite sinc product for  $\widehat{\text{up}}$  and its dyadic scaling equation;
- (ii) the canonical product

$$\Phi(z) = \prod_{m=1}^{\infty} \left(1 - \frac{z^2}{m^2}\right)^{1+\nu_2(m)};$$

- (iii) the assertion that the nonzero zeros are precisely the integers, with multiplicity  $1 + \nu_2(|m|)$ ;
- (iv) strict logarithmic concavity of  $|\Phi|$  between consecutive zeros and the sign formula

$$\text{sgn } \Phi(x) = (-1)^{s_2(M)} \quad (M < x < M + 1);$$

- (v) the existing moment/cumulant formulae for the Rvachev probability law;
- (vi) the base-1/2  $q$ -Pochhammer and Gaussian-binomial formulae for exact dyadic values of the Fabius function;
- (vii) the pointwise, metric, mean, extremal, and annular Fourier-decay results assembled in the final Fourier-decay synthesis.

The purpose of the report is instead to place those facts inside a new spectral determinant and  $q$ -deformation architecture and to extract consequences not present in the source corpus.

### 1.3 Result map

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**New deductions developed below.** The report proves: (a) a Fredholm determinant model with  $D \simeq N \oplus 2D$ ; (b) exact zero-counting and spectral-parity formulae; (c) the meromorphic spectral zeta  $\zeta(s)/(1 - 2^{-s})$  and its heat trace with explicit log-periodic oscillation; (d) exact normalized all-order jets at every integer zero; (e) an operator-valued dyadic  $q$ -Euler product and finite multiscale Gaussian-binomial expansion; (f) complete small- $t$  asymptotics of the resulting tower of  $q$ -Pochhammer symbols; (g) a symmetric  $q$ -integer deformation of  $\Phi$  with exact  $q$ -scaling, zero arithmetic, and  $q \uparrow 1$  convergence; and (h) a base- $b$  extension explaining which features are genuinely dyadic.

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## 2 Established baseline and notation

### 2.1 The twin functions

Rvachev's up-function  $\text{up}$  is the even  $C^\infty$  density supported on  $[-1, 1]$  whose Fourier transform, under the convention

$$\widehat{f}(z) = \int_{\mathbb{R}} f(x) e^{-2\pi i z x} dx,$$

is

$$\Phi(z) := \widehat{\text{up}}(z) = \prod_{j=0}^{\infty} \text{sinc}\left(\frac{\pi z}{2^j}\right). \quad (2.1)$$

The Fabius function  $F$  is its affine cumulative twin; on the support one may write

$$\text{up}(x) = \begin{cases} F(x+1), & -1 \leq x \leq 0, \\ F(1-x), & 0 \leq x \leq 1. \end{cases}$$

For the present report the Fourier entire function  $\Phi$  is the central object, while the relation to  $F$  explains why the resulting identities belong to the same theory.

A probability realization is useful. If  $U_0, U_1, \dots$  are independent and uniform on  $[-1/2, 1/2]$ , then

$$X = \sum_{j=0}^{\infty} 2^{-j} U_j \quad (2.2)$$

has density  $\text{up}$  and characteristic function  $\mathbb{E} e^{itX} = \Phi(t/(2\pi))$ .

### 2.2 Thue–Morse and binary notation

For  $n \geq 0$ , let  $s_2(n)$  denote the sum of the binary digits of  $n$ , and put

$$\tau_n = (-1)^{s_2(n)}. \quad (2.3)$$

The sequence satisfies  $\tau_{2n} = \tau_n$  and  $\tau_{2n+1} = -\tau_n$ . We write  $\nu_2(m)$  for the exponent of 2 in a positive integer  $m$ .

### 2.3 $q$ -notation

For  $|Q| < 1$ ,

$$(a; Q)_L = \prod_{r=0}^{L-1} (1 - aQ^r), \quad (a; Q)_\infty = \prod_{r=0}^{\infty} (1 - aQ^r),$$

and the Gaussian binomial coefficient is

$$\begin{bmatrix} L \\ k \end{bmatrix}_Q = \frac{(Q; Q)_L}{(Q; Q)_k (Q; Q)_{L-k}}.$$

The finite  $q$ -binomial theorem is

$$(a; Q)_L = \sum_{k=0}^L (-1)^k Q^{\binom{k}{2}} \begin{bmatrix} L \\ k \end{bmatrix}_Q a^k. \quad (2.4)$$

The corpus uses these functions chiefly at base  $Q = 1/2$  to encode exact dyadic values of  $F$ . The constructions below use a different  $Q$ : it deforms the *Fourier zero spectrum*. The two roles should not be conflated.

## 2.4 The established zero divisor

Euler's sine product applied scale by scale to Equation (2.1) gives

$$\Phi(z) = \prod_{j=0}^{\infty} \prod_{n=1}^{\infty} \left( 1 - \frac{z^2}{(2^j n)^2} \right) = \prod_{m=1}^{\infty} \left( 1 - \frac{z^2}{m^2} \right)^{a_m}, \quad a_m := 1 + \nu_2(m). \quad (2.5)$$

The second equality counts the representations  $m = 2^j n$ . Consequently the nonzero zeros are exactly  $\pm m$ , each with multiplicity  $a_m$ . The active Fourier-decay corpus already contains Equation (2.5); it is the starting point, not a new theorem of this report.

## 3 The dyadic spectral operator

### 3.1 Definition and self-similarity

Let

$$\mathcal{H} = \bigoplus_{j=0}^{\infty} \ell^2(\mathbb{N}),$$

with orthonormal basis  $e_{j,n}$ ,  $j \geq 0$ ,  $n \geq 1$ . Define the positive diagonal operator

$$D e_{j,n} = 2^j n e_{j,n}. \quad (3.1)$$

Let  $N$  be the number operator on  $\ell^2(\mathbb{N})$ ,  $N e_n = n e_n$ .

**Theorem 3.1** (Dyadic spectral decomposition). *There is a unitary equivalence*

$$D \simeq N \oplus 2D. \quad (3.2)$$

Moreover  $D^{-p}$  is trace class precisely for  $p > 1$ .

*Proof.* The subspace with  $j = 0$  carries the eigenvalues  $n$ , hence is  $N$ . Relabeling  $e_{j,n}$  as  $e_{j-1,n}$  on the subspace  $j \geq 1$  identifies the restriction with  $2D$ . Finally,

$$\mathrm{Tr} D^{-p} = \sum_{j=0}^{\infty} \sum_{n=1}^{\infty} (2^j n)^{-p} = \frac{\zeta(p)}{1 - 2^{-p}},$$

which is finite exactly when  $p > 1$ . □

The decomposition is the operator-theoretic form of dyadic refinement. It is also the source of every denominator  $1 - 2^{-s}$  below.

### 3.2 The sinc product as a Fredholm determinant

Since  $D^{-2}$  is trace class, the ordinary Fredholm determinant  $\det(I - z^2 D^{-2})$  is defined for every  $z \in \mathbb{C}$ .

**Theorem 3.2** (Fredholm determinant representation). *For every  $z \in \mathbb{C}$ ,*

$$\boxed{\Phi(z) = \det(I - z^2 D^{-2})}. \quad (3.3)$$

The operator decomposition Equation (3.2) implies the exact scaling equation

$$\Phi(z) = \mathrm{sinc}(\pi z) \Phi(z/2). \quad (3.4)$$

*Proof.* The eigenvalues of  $D^{-2}$  are  $(2^j n)^{-2}$ , and their sum is finite. Thus

$$\det(I - z^2 D^{-2}) = \prod_{j=0}^{\infty} \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{(2^j n)^2}\right).$$

Euler's product  $\text{sinc}(\pi w) = \prod_{n \geq 1} (1 - w^2/n^2)$  identifies this with Equation (2.1). Applying the multiplicativity of the determinant to  $D \simeq N \oplus 2D$  gives

$$\det(I - z^2 D^{-2}) = \det(I - z^2 N^{-2}) \det(I - z^2 (2D)^{-2}) = \text{sinc}(\pi z) \Phi(z/2).$$

□

*Remark 3.3.* This is an *ordinary* Fredholm determinant, not a zeta-regularized determinant. The trace-class input is  $D^{-2}$ , and no renormalization is required.

### 3.3 Multiplicity as spectral degeneracy

The eigenvalue  $m$  of  $D$  occurs once for every  $j$  such that  $2^j \mid m$ . Therefore

$$\text{mult}_D(m) = a_m = 1 + \nu_2(m), \quad a_{2m} = a_m + 1, \quad a_{2m+1} = 1. \quad (3.5)$$

Thus the canonical zero multiplicity is exactly the spectral degeneracy of  $D$ . The elementary 2-regular recursion in Equation (3.5) is the discrete shadow of  $D \simeq N \oplus 2D$ .

## 4 Exact zero arithmetic and spectral parity

### 4.1 The positive counting function

Let

$$A(M) = \text{rank } \mathbf{1}_{[1,M]}(D) = \sum_{m=1}^M a_m$$

count positive eigenvalues, including multiplicity.

**Theorem 4.1** (Exact Weyl law with binary remainder). *For every integer  $M \geq 0$ ,*

$$A(M) = \sum_{j=0}^{\infty} \left\lfloor \frac{M}{2^j} \right\rfloor = 2M - s_2(M). \quad (4.1)$$

*Consequently, with  $M = \lfloor R \rfloor$ , the number of nonzero zeros of  $\Phi$  in  $|z| \leq R$ , counted with multiplicity, is*

$$\mathcal{N}_{\Phi}(R) = 4M - 2s_2(M). \quad (4.2)$$

*Proof.* Counting the pairs  $(j, n)$  with  $2^j n \leq M$  gives the floor sum. On the other hand,

$$A(M) = M + \sum_{m=1}^M \nu_2(m) = M + \nu_2(M!).$$

Legendre's binary formula  $\nu_2(M!) = M - s_2(M)$  proves Equation (4.1). The zeros are symmetric, so the disk count is twice  $A(M)$ . □

The exact formula refines the linear Weyl term  $A(M) \sim 2M$  by a digit-sum fluctuation. In particular,

$$2M - \lfloor \log_2 M \rfloor - 1 \leq A(M) \leq 2M - 1 \quad (M \geq 1). \quad (4.3)$$

The recursions

$$A(2M) = A(M) + 2M, \quad A(2M + 1) = A(M) + 2M + 1 \quad (4.4)$$

follow immediately.

## 4.2 Thue–Morse as spectral-flow parity

**Corollary 4.2** (Spectral parity formula). *For every  $M \geq 0$ ,*

$$\boxed{(-1)^{\text{rank } \mathbf{1}_{[1,M]}(D)} = (-1)^{A(M)} = (-1)^{s_2(M)} = \tau_M.} \quad (4.5)$$

Hence, for  $M < x < M + 1$ ,

$$\text{sgn } \Phi(x) = (-1)^{\text{rank } \mathbf{1}_{[1,M]}(D)} = \tau_M.$$

*Proof.* The even term  $2M$  disappears modulo 2 in Equation (4.1). Starting with  $\Phi(0) = 1$ , crossing the zeros  $1, \dots, M$  multiplies the sign by  $(-1)^{A(M)}$ .  $\square$

The interval-sign theorem was already present in the Fourier corpus. What is new here is its interpretation: the Thue–Morse bit is the parity of a finite-rank spectral projection of a self-similar operator.

## 4.3 Dyadic-block statistics

For uniform  $M$  in  $\{0, \dots, 2^n - 1\}$ , the random variable  $s_2(M)$  is binomial  $\text{Bin}(n, 1/2)$ . Thus

$$\frac{1}{2^n} \sum_{M=0}^{2^n-1} (2M - A(M)) = \frac{n}{2}, \quad \text{Var}(2M - A(M)) = \frac{n}{4}. \quad (4.6)$$

The de Moivre–Laplace theorem gives

$$\frac{A(M) - (2M - n/2)}{\sqrt{n}/2} \implies \mathcal{N}(0, 1) \quad (4.7)$$

under dyadic-block sampling. Thus the deterministic zero-count remainder has an exact probabilistic law when sampled on its natural self-similar blocks.

## 4.4 A certified canonical-product truncation

Let

$$\Phi_M(z) = \prod_{m=1}^M \left(1 - \frac{z^2}{m^2}\right)^{a_m}.$$

**Proposition 4.3** (Elementary tail bound). *For  $M \geq 1$ ,*

$$T_M := \sum_{m>M} \frac{a_m}{m^2} < \frac{2}{M} + \frac{2\zeta(2)}{M^2}. \quad (4.8)$$

*If  $|z| \leq R$  and  $M \geq \sqrt{2}R$ , then, on any zero-free simply connected region where the logarithm is chosen continuously,*

$$\left| \log \frac{\Phi(z)}{\Phi_M(z)} \right| \leq 2R^2 T_M. \quad (4.9)$$

*Proof.* Write the tail spectrally:

$$T_M = \sum_{j=0}^{\infty} 4^{-j} \sum_{n>M/2^j} \frac{1}{n^2}.$$

For  $j \leq \lfloor \log_2 M \rfloor$ , the inner tail is at most  $2^j/M$ ; summing gives less than  $2/M$ . For the remaining  $j$ , bound the inner sum by  $\zeta(2)$  and the geometric tail by  $2/M^2$ . If  $|z|^2/m^2 \leq 1/2$ , then  $|\log(1 - w)| \leq 2|w|$ , and summation proves Equation (4.9).  $\square$

## 5 Spectral zeta, determinant coefficients, and cumulants

### 5.1 The exact spectral zeta function

**Theorem 5.1** (Dyadic spectral zeta). *For  $\Re s > 1$ ,*

$$\zeta_D(s) := \operatorname{Tr} D^{-s} = \sum_{m=1}^{\infty} \frac{1 + \nu_2(m)}{m^s} = \frac{\zeta(s)}{1 - 2^{-s}}. \quad (5.1)$$

*This identity gives a meromorphic continuation to  $\mathbb{C}$ . Besides the pole at  $s = 1$  with residue 2, it has the dyadic pole lattice*

$$\omega_k = \frac{2\pi i k}{\log 2}, \quad k \in \mathbb{Z}, \quad (5.2)$$

*with*

$$\operatorname{Res}_{s=\omega_k} \zeta_D(s) = \frac{\zeta(\omega_k)}{\log 2}. \quad (5.3)$$

*In particular,*

$$\zeta_D(s) = -\frac{1}{2 \log 2} \frac{1}{s} - \frac{1}{4} - \frac{\log(2\pi)}{2 \log 2} + O(s) \quad (s \rightarrow 0). \quad (5.4)$$

*Proof.* The double sum in the proof of [theorem 3.1](#) gives [Equation \(5.1\)](#). The pole at 1 is multiplied by  $(1 - 2^{-1})^{-1} = 2$ . At a zero  $\omega_k$  of  $1 - 2^{-s}$ , its derivative is  $(\log 2)2^{-s} = \log 2$ , which gives [Equation \(5.3\)](#). Expansion of  $\zeta(s)$  and  $(1 - e^{-s \log 2})^{-1}$  at 0 gives [Equation \(5.4\)](#).  $\square$

For  $k \neq 0$ , the numerator does not vanish on the imaginary axis; this follows, for example, from the functional equation together with the classical zero-free line  $\Re s = 1$ . The points  $\omega_k$  therefore form a genuine vertical lattice of complex dimensions.

**Corollary 5.2** (Reciprocal zero power sums). *For every integer  $r \geq 1$ ,*

$$\sum_{m=1}^{\infty} \frac{1 + \nu_2(m)}{m^{2r}} = \frac{\zeta(2r)}{1 - 4^{-r}}. \quad (5.5)$$

### 5.2 Trace logarithm and a coefficient recurrence

For  $|z| < 1$ , the trace logarithm of the Fredholm determinant is absolutely convergent.

**Theorem 5.3** (Spectral Taylor expansion). *One has*

$$\log \Phi(z) = - \sum_{r=1}^{\infty} \frac{\zeta(2r)}{r(1 - 4^{-r})} z^{2r}, \quad |z| < 1. \quad (5.6)$$

*If*

$$\Phi(z) = \sum_{n=0}^{\infty} c_n z^{2n}, \quad c_0 = 1,$$

*then*

$$c_n = -\frac{1}{n} \sum_{r=1}^n \frac{\zeta(2r)}{1 - 4^{-r}} c_{n-r}. \quad (5.7)$$

*In particular,*

$$c_1 = -\frac{2\pi^2}{9}, \quad c_2 = \frac{38\pi^4}{2025}. \quad (5.8)$$

*Proof.* For  $|z| < 1$ ,

$$\log \det(I - z^2 D^{-2}) = - \sum_{r \geq 1} \frac{z^{2r}}{r} \operatorname{Tr} D^{-2r},$$

and [corollary 5.2](#) gives [Equation \(5.6\)](#). Put  $x = z^2$  and differentiate logarithmically:

$$x \frac{\sum_{n \geq 1} n c_n x^{n-1}}{\sum_{n \geq 0} c_n x^n} = - \sum_{r \geq 1} \frac{\zeta(2r)}{1 - 4^{-r}} x^r.$$

Coefficient comparison gives [Equation \(5.7\)](#). □

### 5.3 A spectral interpretation of the known cumulants

Let  $X$  be the centered random variable in [Equation \(2.2\)](#). Since  $\mathbb{E}e^{itX} = \Phi(t/(2\pi))$ , comparison of [Equation \(5.6\)](#) with the cumulant expansion gives

$$\kappa_{2r}(X) = \frac{B_{2r}}{2r(1 - 4^{-r})}, \quad \kappa_{2r+1}(X) = 0. \quad (5.9)$$

These cumulants already occur in the frontier corpus. The new point is the identity

$$\kappa_{2r}(X) = (-1)^{r+1} \frac{(2r)!}{r(2\pi)^{2r}} \operatorname{Tr} D^{-2r}, \quad (5.10)$$

which interprets them as spectral power traces. For example  $\kappa_2 = 1/9$  follows from  $\operatorname{Tr} D^{-2} = 2\pi^2/9$ .

## 6 Heat trace and dyadic complex dimensions

### 6.1 Mellin representation

Define

$$\Theta(t) = \operatorname{Tr} e^{-tD} = \sum_{m=1}^{\infty} a_m e^{-tm} = \sum_{j=0}^{\infty} \frac{1}{e^{2^j t} - 1}, \quad t > 0. \quad (6.1)$$

For  $\Re s > 1$ ,

$$\int_0^{\infty} \Theta(t) t^{s-1} dt = \Gamma(s) \zeta_D(s) = \Gamma(s) \frac{\zeta(s)}{1 - 2^{-s}}. \quad (6.2)$$

The pole lattice [Equation \(5.2\)](#) therefore produces logarithmic periodicity in physical scale.

Put

$$L = \log 2, \quad \omega_k = \frac{2\pi i k}{L},$$

and define the real-analytic, one-periodic function

$$\mathcal{P}_{\Theta}(u) = \frac{1}{L} \sum_{k \in \mathbb{Z} \setminus \{0\}} \Gamma(\omega_k) \zeta(\omega_k) e^{2\pi i k u}. \quad (6.3)$$

The exponential decay of  $\Gamma(iy)$  makes the Fourier series absolutely and locally uniformly convergent with all derivatives.

### 6.2 Complete heat expansion

**Theorem 6.1** (Heat trace with log-periodic oscillation). *Let*

$$C_{\Theta} = \frac{\gamma - \log(2\pi)}{2 \log 2} - \frac{1}{4}. \quad (6.4)$$

For every integer  $J \geq 0$  and every  $0 < \varepsilon < 1$ ,

$$\begin{aligned} \Theta(t) &= \frac{2}{t} - \frac{\log(1/t)}{2 \log 2} + C_{\Theta} + \mathcal{P}_{\Theta}\left(\log_2 \frac{1}{t}\right) \\ &\quad - \sum_{r=1}^J \frac{B_{2r}}{(2r)!(2^{2r-1} - 1)} t^{2r-1} + O_{J,\varepsilon}(t^{2J+1-\varepsilon}) \quad (t \downarrow 0). \end{aligned} \quad (6.5)$$

The first algebraic correction is  $-t/12$ .



*Proof.* Mellin inversion gives

$$\Theta(t) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \Gamma(s) \zeta_D(s) t^{-s} ds, \quad c > 1.$$

Move the contour left. The pole at  $s = 1$  contributes  $2/t$ . At  $s = 0$ ,  $\Gamma(s)\zeta_D(s)$  has a double pole. From Equation (5.4),

$$\Gamma(s)\zeta_D(s) = -\frac{1}{2L} \frac{1}{s^2} + \left( \frac{\gamma - \log(2\pi)}{2L} - \frac{1}{4} \right) \frac{1}{s} + O(1).$$

Multiplication by  $t^{-s}$  gives the logarithm and  $C_\Theta$ . The residues at  $s = \omega_k$ ,  $k \neq 0$ , sum to Equation (6.3). At the negative odd integer  $s = -(2r - 1)$ , the residue of  $\Gamma$  is  $-1/(2r - 1)!$  and  $\zeta(1 - 2r) = -B_{2r}/(2r)$ ; this gives the displayed coefficient. The trivial zeros of  $\zeta$  cancel the poles of  $\Gamma$  at negative even integers. Standard vertical-line bounds and the exponential decay of  $\Gamma$  justify the contour shift and remainder.  $\square$

### 6.3 Size of the oscillation

The first Fourier coefficient in Equation (6.3) is

$$\frac{\Gamma(2\pi i/L)\zeta(2\pi i/L)}{L} = -8.226408112794756 \times 10^{-7} - 9.760012263161017 \times 10^{-7}i, \quad (6.6)$$

with modulus  $1.276446747088608 \times 10^{-6}$ . Thus the oscillation is small but exact. It is not an error term: it is a Fourier series forced by the pole lattice of  $\zeta_D$ .

*Remark 6.2* (Two kinds of self-similar fluctuation). The exact counting remainder  $-s_2(M)$  is discontinuous and arithmetic. The heat remainder  $\mathcal{P}_\Theta(\log_2(1/t))$  is smooth and analytic. They are two transforms of the same dyadic spectral multiplicity: a sharp cutoff retains binary carries, whereas exponential smoothing turns the scale lattice into an analytic periodic wave.

## 7 Exact local jets at all integer zeros

### 7.1 A normalized factorization

Fix a nonzero integer  $m$ . Write

$$r = \nu_2(|m|), \quad a = r + 1, \quad q = \frac{m}{2^r}, \quad (7.1)$$

so that  $q$  is odd.

**Theorem 7.1** (Exact all-order zero jet). *For  $w$  in a neighborhood of 0,*

$$\Phi(m + w) = C_m w^a R_m(w), \quad (7.2)$$

where

$$C_m = -\frac{\Phi(q/2)}{m^a} \quad (7.3)$$

and

$$R_m(w) = \left( \frac{m}{m+w} \right)^a \prod_{h=0}^r \operatorname{sinc}\left(\frac{\pi w}{2^h}\right) \frac{\Phi(q/2 + w/2^{r+1})}{\Phi(q/2)}. \quad (7.4)$$

In particular  $R_m$  is analytic and  $R_m(0) = 1$ , and every derivative is given exactly by

$$\Phi^{(a+\ell)}(m) = C_m (a + \ell)! [w^\ell] R_m(w), \quad \ell \geq 0. \quad (7.5)$$

The first nonzero derivative is

$$\Phi^{(a)}(m) = -\frac{a!}{m^a} \Phi(q/2). \quad (7.6)$$

*Proof.* For  $0 \leq h \leq r$ ,  $m/2^h$  is an integer and

$$\operatorname{sinc}\left(\frac{\pi(m+w)}{2^h}\right) = (-1)^{m/2^h} \frac{w}{m+w} \operatorname{sinc}\left(\frac{\pi w}{2^h}\right).$$

The product of the signs is

$$(-1)^{\sum_{h=0}^r m/2^h} = (-1)^{q(2^{r+1}-1)} = -1.$$

The tail of the original sinc product is

$$\prod_{h=r+1}^{\infty} \operatorname{sinc}\left(\frac{\pi(m+w)}{2^h}\right) = \Phi\left(\frac{q}{2} + \frac{w}{2^{r+1}}\right).$$

Multiplying these identities gives

$$\Phi(m+w) = -\left(\frac{w}{m+w}\right)^a \prod_{h=0}^r \operatorname{sinc}\left(\frac{\pi w}{2^h}\right) \Phi\left(\frac{q}{2} + \frac{w}{2^{r+1}}\right),$$

which is equivalent to Equation (7.2)–Equation (7.4). Since  $q/2$  is not an integer,  $\Phi(q/2) \neq 0$ . Coefficient extraction proves Equation (7.5).  $\square$

---

**Lean status of the local-jet theorem.** The denominator-cleared algebraic factorization is now machine checked, globally in the complex local coordinate and for every nonzero integer center, as `Fabius.rvachevFourierProduct_int_add_factorization` in `IntegerZeroLocalFactorization.lean`. The divided equality holds as a germ at zero by `Fabius.rvachevFourierProduct_int_add_eventuallyEq_pow_mul_cofactor`. The resulting cofactor is analytic and nonzero at the origin; Lean proves its exact value, the exact analytic order  $1 + \nu_2(|m|)$ , every lower derivative vanishing, and, with  $U_m$  denoting that cofactor and  $a = 1 + \nu_2(|m|)$ , every higher derivative in the division-free form

$$\Phi^{(a+\ell)}(m) = \binom{a+\ell}{a} a! U_m^{(\ell)}(0).$$

This is `Fabius.iteratedDeriv_rvachevFourierProduct_int_add_order`. Its  $\ell = 0$  specialization is

$$\Phi^{(a)}(m) = -\frac{a!}{m^a} \Phi(q/2)$$

through the declarations in `IntegerZeroAnalyticOrder.lean`. The field-generic extraction of this leading derivative is in `LeadingJet.lean`; that module now proves the vector-valued all-index and all-order germ formulas over every nontrivially normed field. Thus Equations (7.2), (7.3) and (7.6) and the cofactor-derivative version of Equation (7.5) at every  $\ell$  are formalized. The explicitly normalized function  $R_m$ , the literal coefficient-extraction packaging in Equation (7.5), the explicit next-jet simplification, dyadic-ray recursion, and the packaged leading-sign corollary do not yet have matching named Lean declarations.

---

## 7.2 The next coefficient and dyadic-ray recursion

Differentiating Equation (7.4) at 0 gives

$$R'_m(0) = -\frac{a}{m} + \frac{1}{2^{r+1}} \frac{\Phi'(q/2)}{\Phi(q/2)}. \quad (7.7)$$

Therefore

$$\Phi^{(a+1)}(m) = -\frac{(a+1)!\Phi(q/2)}{m^a} \left( -\frac{a}{m} + \frac{1}{2^{r+1}} \frac{\Phi'(q/2)}{\Phi(q/2)} \right). \quad (7.8)$$

All higher jets follow by formal multiplication of the three analytic factors in Equation (7.4).

The leading coefficients propagate exactly along dyadic rays.

**Corollary 7.2** (Dyadic-ray recursion). *If  $a = 1 + \nu_2(m)$ , then*

$$C_{2m} = \frac{C_m}{2^{a+1}m}. \quad (7.9)$$

*Equivalently,*

$$\frac{\Phi^{(a+1)}(2m)}{(a+1)!} = \frac{1}{2^{a+1}m} \frac{\Phi^{(a)}(m)}{a!}. \quad (7.10)$$

*Proof.* The integers  $m$  and  $2m$  have the same odd part, while their multiplicities differ by one. Substitute into Equation (7.3).  $\square$

### 7.3 Thue–Morse signs of the leading jets

For  $m > 0$ , write its odd part as  $q = 2k+1$ . The established interval-sign theorem gives  $\text{sgn } \Phi(q/2) = (-1)^{s_2(k)}$ . Hence

$$\boxed{\text{sgn } \Phi^{(1+\nu_2(m))}(m) = -(-1)^{s_2((q-1)/2)}}. \quad (7.11)$$

Thus the leading Taylor coefficient at every positive zero is controlled by the Thue–Morse bit of half the predecessor of its odd part.

### 7.4 Numerical checks

The following values were computed independently from a high-precision truncated sinc product and from Equation (7.6); the two evaluations agreed to at least 70 decimal digits before rounding.

| $m$ | $a = 1 + \nu_2(m)$ | $\Phi^{(a)}(m)$          | sign |
|-----|--------------------|--------------------------|------|
| 1   | 1                  | −0.5537712758888100953   | −    |
| 2   | 2                  | −0.2768856379444050477   | −    |
| 3   | 1                  | 0.01540076637810223766   | +    |
| 4   | 3                  | −0.05191605711457594644  | −    |
| 6   | 2                  | 0.002566794396350372944  | +    |
| 8   | 4                  | −0.003244753569660996652 | −    |
| 12  | 3                  | 0.0001604246497718983090 | +    |

These computations are checks, not substitutes for the proof.

## 8 The dyadic $q$ -Euler determinant

### 8.1 Definition and $q$ -Pochhammer tower

Fix  $0 < Q < 1$ . Since  $Q^D$  is trace class, define

$$\mathcal{E}(u; Q) = \det(I - uQ^D). \quad (8.1)$$

**Theorem 8.1** (Dyadic  $q$ -Euler product). *For every  $u \in \mathbb{C}$ ,*

$$\mathcal{E}(u; Q) = \prod_{m=1}^{\infty} (1 - uQ^m)^{a_m} \quad (8.2)$$

$$= \prod_{j=0}^{\infty} (uQ^{2^j}; Q^{2^j})_{\infty}. \quad (8.3)$$

*It satisfies the exact renormalization equation*

$$\boxed{\mathcal{E}(u; Q) = (uQ; Q)_{\infty} \mathcal{E}(u; Q^2)}. \quad (8.4)$$

*Proof.* The first product is the eigenvalue formula for a Fredholm determinant. The decomposition  $D \simeq N \oplus 2D$  gives

$$\det(I - uQ^D) = \det(I - uQ^N) \det(I - uQ^{2D}) = (uQ; Q)_{\infty} \mathcal{E}(u; Q^2).$$

Iteration gives Equation (8.3); convergence follows from  $\sum_{j,n} |u|Q^{2^j n} < \infty$ .  $\square$

## 8.2 Lambert series and plethystic logarithm

For  $|u|Q < 1$ ,

$$-\log \mathcal{E}(u; Q) = \sum_{\ell=1}^{\infty} \frac{u^{\ell}}{\ell} \sum_{j=0}^{\infty} \frac{Q^{2^j \ell}}{1 - Q^{2^j \ell}}. \quad (8.5)$$

Equivalently, the ordinary generating function of the zero multiplicities is the dyadic Lambert series

$$\sum_{m=1}^{\infty} a_m x^m = \sum_{j=0}^{\infty} \frac{x^{2^j}}{1 - x^{2^j}}, \quad |x| < 1. \quad (8.6)$$

The same self-similarity now appears in three transforms:

$$a_m \longleftrightarrow \frac{\zeta(s)}{1 - 2^{-s}} \longleftrightarrow \sum_{j \geq 0} \frac{x^{2^j}}{1 - x^{2^j}}.$$

## 9 Finite multiscale Gaussian-binomial formulae

### 9.1 A finite determinant

For  $M \geq 1$ , define

$$\mathcal{E}_M(u; Q) = \prod_{m=1}^M (1 - uQ^m)^{a_m}. \quad (9.1)$$

Put

$$J = \lfloor \log_2 M \rfloor, \quad L_j = \left\lfloor \frac{M}{2^j} \right\rfloor.$$

**Theorem 9.1** (Finite multiscale factorization). *One has*

$$\mathcal{E}_M(u; Q) = \prod_{j=0}^J (uQ^{2^j}; Q^{2^j})_{L_j}. \quad (9.2)$$

*Its degree in  $u$  is*

$$\deg_u \mathcal{E}_M = A(M) = 2M - s_2(M). \quad (9.3)$$

*Writing*

$$\mathcal{E}_M(u; Q) = \sum_{k=0}^{A(M)} (-1)^k C_{M,k}(Q) u^k, \quad (9.4)$$

*one has the explicit Gaussian-binomial convolution*

$$C_{M,k}(Q) = \sum_{\substack{k_0 + \dots + k_J = k \\ 0 \leq k_j \leq L_j}} Q^{\sum_{j=0}^J 2^j \binom{k_j+1}{2}} \prod_{j=0}^J \begin{bmatrix} L_j \\ k_j \end{bmatrix}_{Q^{2^j}}. \quad (9.5)$$

*Proof.* Expand  $a_m$  as the number of representations  $m = 2^j n$  and regroup:

$$\mathcal{E}_M(u; Q) = \prod_{j=0}^J \prod_{n=1}^{L_j} (1 - uQ^{2^j n}),$$

which is Equation (9.2). Its degree is  $\sum_j L_j = A(M)$ . Applying Equation (2.4) to each factor gives

$$(uQ^{2^j}; Q^{2^j})_{L_j} = \sum_{k_j=0}^{L_j} (-1)^{k_j} Q^{2^j \binom{k_j+1}{2}} \begin{bmatrix} L_j \\ k_j \end{bmatrix}_{Q^{2^j}} u^{k_j}.$$

Multiplication and collection of the total degree proves Equation (9.5).  $\square$

## 9.2 Dyadic endpoints and a multiscale factorial

For  $M = 2^n - 1$ ,

$$L_j = 2^{n-j} - 1, \quad A(2^n - 1) = 2^{n+1} - n - 2. \quad (9.6)$$

The top  $Q$ -degree of Equation (9.1) is

$$B(M) := \sum_{m=1}^M m a_m = \frac{1}{2} \sum_{j=0}^J 2^j L_j (L_j + 1). \quad (9.7)$$

At a dyadic endpoint,

$$B(2^n - 1) = 4^n - (n + 2)2^{n-1}. \quad (9.8)$$

The classical limit of the normalizing product is

$$\lim_{Q \uparrow 1} \frac{\mathcal{E}_M(1; Q)}{(1 - Q)^{A(M)}} = \prod_{m=1}^M m^{a_m} = \prod_{j=0}^J 2^j L_j!. \quad (9.9)$$

Thus the finite determinant naturally defines a dyadic multiscale factorial.

## 10 Mellin asymptotics of the $q$ -Pochhammer tower

Put  $Q = e^{-t}$ ,  $t > 0$ , and

$$y = \log \frac{1}{t}, \quad u_t = \log_2 \frac{1}{t} = \frac{y}{\log 2}.$$

The identity Equation (8.5) may be rewritten as

$$-\log \mathcal{E}(u; e^{-t}) = \sum_{\ell=1}^{\infty} \frac{u^\ell}{\ell} \Theta(\ell t). \quad (10.1)$$

This transfers the heat complex dimensions directly to a tower of  $q$ -Pochhammer symbols.

### 10.1 The subcritical regime $|u| < 1$

Let

$$\mathcal{P}_u(v) = \frac{1}{\log 2} \sum_{k \neq 0} \Gamma(\omega_k) \zeta(\omega_k) \text{Li}_{1+\omega_k}(u) e^{2\pi i k v}. \quad (10.2)$$

For  $u$  in a compact subset of the unit disk this series is absolutely convergent and analytic in both  $u$  and  $v$ .

**Theorem 10.1** (Subcritical  $q$ -Euler asymptotic). *Fix  $\rho < 1$ . Uniformly for  $|u| \leq \rho$ , every  $J \geq 0$ , and every  $0 < \varepsilon < 1$ ,*

$$\begin{aligned} -\log \mathcal{E}(u; e^{-t}) &= \frac{2 \text{Li}_2(u)}{t} + \frac{\log(1-u)}{2 \log 2} y + C(u) + \mathcal{P}_u(u_t) \\ &\quad - \sum_{r=1}^J \frac{B_{2r} \text{Li}_{2-2r}(u)}{(2r)!(2^{2r-1} - 1)} t^{2r-1} + O_{J,\rho,\varepsilon}(t^{2J+1-\varepsilon}), \end{aligned} \quad (10.3)$$

where

$$C(u) = -\frac{1}{2 \log 2} \left. \frac{\partial}{\partial s} \text{Li}_s(u) \right|_{s=1} - C_\Theta \log(1-u). \quad (10.4)$$

*Proof.* For  $c > 1$ , Mellin inversion and absolute convergence give

$$-\log \mathcal{E}(u; e^{-t}) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \Gamma(s) \text{Li}_{s+1}(u) \zeta_D(s) t^{-s} ds. \quad (10.5)$$

At  $s = 1$  the residue is  $2 \text{Li}_2(u)/t$ . At  $s = 0$ , write

$$\zeta_D(s) = \frac{a_{-1}}{s} + a_0 + O(s), \quad a_{-1} = -\frac{1}{2 \log 2}, \quad a_0 = -\frac{1}{4} - \frac{\log(2\pi)}{2 \log 2}.$$

The double-pole residue after multiplication by  $t^{-s}$  is

$$\frac{\log(1-u)}{2\log 2}y - \frac{1}{2\log 2} \partial_s \text{Li}_s(u)|_{s=1} - C_\Theta \log(1-u).$$

The poles  $\omega_k$  give Equation (10.2). At  $s = -(2r-1)$ , the residue equals the displayed Bernoulli–polylogarithm coefficient. At negative even integers the trivial zeros of  $\zeta$  cancel the poles of  $\Gamma$ .  $\square$

The expansion is nonuniform at  $u = 1$ :  $\text{Li}_1(u) = -\log(1-u)$  diverges, and a second singular factor meets the pole at  $s = 0$ . The critical limit must be calculated separately.

## 10.2 The critical product $u = 1$

Define

$$\mathcal{P}_E(v) = \frac{1}{\log 2} \sum_{k \neq 0} \Gamma(\omega_k) \zeta(1 + \omega_k) \zeta(\omega_k) e^{2\pi i k v}. \quad (10.6)$$

Let  $\gamma_1$  be the first Stieltjes constant in

$$\zeta(1+s) = \frac{1}{s} + \gamma - \gamma_1 s + O(s^2).$$

Put

$$C_E = \frac{\zeta''(0) + \gamma_1 + \gamma^2/2 - \pi^2/12}{2\log 2} - \frac{\log(2\pi)}{4} - \frac{\log 2}{24}. \quad (10.7)$$

Numerically,  $C_E = -2.4612722039168654768\dots$

**Theorem 10.2** (Critical dyadic  $q$ -Pochhammer tower). *For every  $N > 0$ ,*

$$\begin{aligned} -\log \mathcal{E}(1; e^{-t}) &= \frac{\pi^2}{3t} - \frac{y^2}{4\log 2} - \left( \frac{1}{4} + \frac{\log(2\pi)}{2\log 2} \right) y \\ &\quad + C_E + \mathcal{P}_E(u_t) + \frac{t}{24} + O_N(t^N) \quad (t \downarrow 0). \end{aligned} \quad (10.8)$$

*Equivalently, this is the asymptotic of*

$$\mathcal{E}(1; e^{-t}) = \prod_{j=0}^{\infty} (e^{-2^j t}; e^{-2^j t})_{\infty}. \quad (10.9)$$

*Proof.* At  $u = 1$ , Equation (10.5) becomes

$$\frac{1}{2\pi i} \int \Gamma(s) \zeta(s+1) \frac{\zeta(s)}{1-2^{-s}} t^{-s} ds.$$

The pole at  $s = 1$  contributes  $2\zeta(2)/t = \pi^2/(3t)$ . At  $s = 0$  the three factors  $\Gamma(s)$ ,  $\zeta(s+1)$ , and  $(1-2^{-s})^{-1}$  produce a triple pole. Write

$$\Gamma(s) \zeta(1+s) = \frac{1}{s^2} + H_0 + O(s), \quad H_0 = -\gamma_1 - \frac{\gamma^2}{2} + \frac{\pi^2}{12},$$

and

$$\zeta_D(s) = \frac{a_{-1}}{s} + a_0 + a_1 s + O(s^2), \quad a_1 = \frac{\zeta''(0)}{2\log 2} - \frac{\log(2\pi)}{4} - \frac{\log 2}{24}.$$

The coefficient of  $1/s$  after multiplication by  $e^{sy}$  is

$$\frac{a_{-1}}{2} y^2 + a_0 y + a_1 + a_{-1} H_0,$$

which is exactly the quadratic-log part and  $C_E$ . The poles  $\omega_k$  give Equation (10.6). The pole at  $s = -1$  contributes  $t/24$ . Every pole of  $\Gamma$  at an integer  $s = -n$ ,  $n \geq 2$ , is canceled by a trivial zero of either  $\zeta(s)$  or  $\zeta(s+1)$ . The contour can therefore be moved arbitrarily far left, giving  $O_N(t^N)$ .  $\square$

*Remark 10.3.* The quadratic logarithm is a critical phenomenon. For  $|u| < 1$  the origin is a double Mellin pole and produces a term linear in  $y$ ; at  $u = 1$ , the pole of  $\zeta(s+1)$  raises the order to three and produces  $-y^2/(4\log 2)$ .

## 11 A symmetric $q$ -integer deformation of the sinc product

### 11.1 Symmetric $q$ -integers

For  $0 < Q < 1$ , define

$$[m]_Q = \frac{Q^{-m/2} - Q^{m/2}}{Q^{-1/2} - Q^{1/2}} = Q^{-(m-1)/2} \frac{1 - Q^m}{1 - Q}. \quad (11.1)$$

If  $Q = e^{-t}$ , then

$$[m]_Q = \frac{\sinh(mt/2)}{\sinh(t/2)}. \quad (11.2)$$

Thus  $[m]_Q \geq m$ ,  $[m]_Q \rightarrow m$  as  $Q \uparrow 1$ , and  $[m]_Q$  grows exponentially in  $m$  for fixed  $Q < 1$ . The multiplicative identity

$$[ab]_Q = [a]_Q [b]_{Q^a} \quad (11.3)$$

will encode exact  $q$ -self-similarity.

By functional calculus, let  $[D]_Q$  be diagonal with eigenvalues  $[2^j n]_Q$ .

**Definition 11.1** (Symmetric dyadic  $q$ -sinc determinant). Define

$$\Phi_Q(z) = \det(I - z^2 [D]_Q^{-2}) = \prod_{m=1}^{\infty} \left( 1 - \frac{z^2}{[m]_Q^2} \right)^{a_m}. \quad (11.4)$$

Also define the one-scale factor

$$S_Q(z) = \det(I - z^2 [N]_Q^{-2}) = \prod_{n=1}^{\infty} \left( 1 - \frac{z^2}{[n]_Q^2} \right). \quad (11.5)$$

Both are entire functions of order zero for fixed  $Q < 1$ .

### 11.2 Exact $q$ -scaling

**Theorem 11.2** ( $q$ -renormalization equation). The functional calculus version of  $D \simeq N \oplus 2D$  is

$$[D]_Q \simeq [N]_Q \oplus [2]_Q [D]_{Q^2}. \quad (11.6)$$

Consequently,

$$\Phi_Q(z) = S_Q(z) \Phi_{Q^2} \left( \frac{z}{[2]_Q} \right). \quad (11.7)$$

Iterating,

$$\Phi_Q(z) = \prod_{j=0}^{\infty} S_{Q^{2^j}} \left( \frac{z}{[2^j]_Q} \right). \quad (11.8)$$

As  $Q \uparrow 1$ , this becomes the classical identity

$$\Phi(z) = \prod_{j=0}^{\infty} \operatorname{sinc}(\pi z / 2^j).$$

*Proof.* On the  $j \geq 1$  subspace, the eigenvalues are  $[2(2^{j-1}n)]_Q$ . By Equation (11.3),

$$[2(2^{j-1}n)]_Q = [2]_Q [2^{j-1}n]_{Q^2}.$$

This proves Equation (11.6). Apply determinant multiplicativity to get Equation (11.7); repeated use and

$$[2^j]_Q = \prod_{h=0}^{j-1} [2]_{Q^{2^h}}$$

give Equation (11.8). □

### 11.3 Exact $q$ -Pochhammer factorization

Set

$$\eta = \frac{z(1-Q)}{\sqrt{Q}}, \quad \lambda + \lambda^{-1} = 2 + \eta^2. \quad (11.9)$$

Either root may be chosen; every formula below is symmetric under  $\lambda \leftrightarrow \lambda^{-1}$ .

**Theorem 11.3** ( $q$ -Pochhammer representation of  $\Phi_Q$ ). *For every  $m \geq 1$ ,*

$$1 - \frac{z^2}{[m]_Q^2} = \frac{(1 - \lambda Q^m)(1 - \lambda^{-1} Q^m)}{(1 - Q^m)^2}. \quad (11.10)$$

Therefore

$$\Phi_Q(z) = \frac{\mathcal{E}(\lambda; Q) \mathcal{E}(\lambda^{-1}; Q)}{\mathcal{E}(1; Q)^2}. \quad (11.11)$$

Equivalently,

$$\Phi_Q(z) = \prod_{j=0}^{\infty} \frac{(\lambda Q^{2^j}; Q^{2^j})_{\infty} (\lambda^{-1} Q^{2^j}; Q^{2^j})_{\infty}}{(Q^{2^j}; Q^{2^j})_{\infty}^2}. \quad (11.12)$$

Likewise,

$$S_Q(z) = \frac{(\lambda Q; Q)_{\infty} (\lambda^{-1} Q; Q)_{\infty}}{(Q; Q)_{\infty}^2}. \quad (11.13)$$

*Proof.* From Equation (11.1),

$$\frac{z^2}{[m]_Q^2} = \frac{\eta^2 Q^m}{(1 - Q^m)^2}.$$

The numerator after subtraction is

$$(1 - Q^m)^2 - \eta^2 Q^m = 1 - (2 + \eta^2) Q^m + Q^{2m} = (1 - \lambda Q^m)(1 - \lambda^{-1} Q^m),$$

which proves Equation (11.10). Raise to the multiplicity  $a_m$ , multiply over  $m$ , and use Equation (8.2) and Equation (8.3).  $\square$

---

**Direct bridge to the classical sinc product.** Combining theorem 11.3 with the convergence theorem below gives the explicit limit

$$\Phi(z) = \lim_{Q \uparrow 1} \prod_{j=0}^{\infty} \frac{(\lambda_Q(z) Q^{2^j}; Q^{2^j})_{\infty} (\lambda_Q(z)^{-1} Q^{2^j}; Q^{2^j})_{\infty}}{(Q^{2^j}; Q^{2^j})_{\infty}^2},$$

where  $\lambda_Q(z) + \lambda_Q(z)^{-1} = 2 + z^2(1 - Q)^2/Q$ . This is a direct  $q$ -Pochhammer representation of the infinite sinc product through a singular but controlled  $Q \uparrow 1$  limit.

---

### 11.4 Convergence to the Rvachev Fourier transform

**Theorem 11.4** (Trace-norm and locally uniform limit). *As  $Q \uparrow 1$ ,*

$$\| [D]_Q^{-2} - D^{-2} \|_1 \longrightarrow 0. \quad (11.14)$$

Consequently,

$$\Phi_Q(z) \longrightarrow \Phi(z) \quad (11.15)$$

locally uniformly on  $\mathbb{C}$ .

*Proof.* The eigenvalues satisfy  $[m]_Q \geq m$  and converge pointwise to  $m$ . Hence

$$\| [D]_Q^{-2} - D^{-2} \|_1 = \sum_{m=1}^{\infty} a_m \left( \frac{1}{m^2} - \frac{1}{[m]_Q^2} \right) \longrightarrow 0$$

by dominated convergence, dominated by  $a_m/m^2$ . Continuity of the Fredholm determinant with respect to trace norm, uniformly when  $z$  ranges over compact sets, proves the result.  $\square$



### 11.5 Exact $q$ -zero arithmetic and a Thue–Morse sign law

**Theorem 11.5** ( $q$ -zero spectrum). *For fixed  $0 < Q < 1$ , the nonzero zeros of  $\Phi_Q$  are exactly*

$$z = \pm [m]_Q, \quad m \geq 1, \quad (11.16)$$

*with multiplicity  $a_m = 1 + \nu_2(m)$ . Hence*

$$\boxed{\operatorname{sgn} \Phi_Q(x) = (-1)^{s_2(M)} \quad \text{for} \quad [M]_Q < x < [M+1]_Q.} \quad (11.17)$$

*If  $Q = e^{-t}$ , define*

$$M_Q(R) = \left\lfloor \frac{2}{t} \operatorname{arsinh} \left( R \sinh \frac{t}{2} \right) \right\rfloor. \quad (11.18)$$

*Then the number of nonzero zeros in  $|z| \leq R$ , counted with multiplicity, is*

$$\boxed{\mathcal{N}_{\Phi_Q}(R) = 4M_Q(R) - 2s_2(M_Q(R)).} \quad (11.19)$$

*For fixed  $Q < 1$ ,  $\mathcal{N}_{\Phi_Q}(R) = O(\log R)$  and  $\Phi_Q$  has order zero. The limit  $Q \uparrow 1$  is a singular transition to the linear zero growth and order one of  $\Phi$ .*

*Proof.* The sequence  $[m]_Q$  is strictly increasing. The product Equation (11.4) gives the zero set and multiplicities. Crossing the first  $M$  positive zeros changes sign by  $(-1)^{A(M)} = (-1)^{s_2(M)}$ . Formula Equation (11.2) shows that  $[m]_Q \leq R$  exactly when

$$m \leq \frac{2}{t} \operatorname{arsinh}(R \sinh(t/2)),$$

which proves the count. Exponential growth of  $[m]_Q$  gives logarithmic counting and order zero.  $\square$

### 11.6 The $q$ -spectral zeta and coefficient recurrence

For  $\Re s > 0$ , define

$$\zeta_{D,Q}(s) = \operatorname{Tr}[D]_Q^{-s} = \sum_{m=1}^{\infty} \frac{a_m}{[m]_Q^s}. \quad (11.20)$$

Let

$$\mathcal{A}(x) = \sum_{m=1}^{\infty} a_m x^m = \sum_{j=0}^{\infty} \frac{x^{2^j}}{1 - x^{2^j}}. \quad (11.21)$$

**Proposition 11.6** ( $q$ -spectral Lambert expansion). *For  $\Re s > 0$ ,*

$$\boxed{\zeta_{D,Q}(s) = (1 - Q)^s Q^{-s/2} \sum_{k=0}^{\infty} \frac{(s)_k}{k!} \mathcal{A}(Q^{k+s/2})}, \quad (11.22)$$

*where  $(s)_k = s(s+1) \cdots (s+k-1)$ . For  $\Re s > 1$ ,*

$$\zeta_{D,Q}(s) \longrightarrow \zeta_D(s) \quad (Q \uparrow 1). \quad (11.23)$$

*If*

$$\Phi_Q(z) = \sum_{n=0}^{\infty} c_n(Q) z^{2^n},$$

*then*

$$c_0(Q) = 1, \quad c_n(Q) = -\frac{1}{n} \sum_{r=1}^n \zeta_{D,Q}(2r) c_{n-r}(Q). \quad (11.24)$$

*Proof.* From Equation (11.1),

$$[m]_Q^{-s} = (1 - Q)^s Q^{-s/2} Q^{ms/2} (1 - Q^m)^{-s}.$$

Expand  $(1 - Q^m)^{-s} = \sum_{k \geq 0} (s)_k Q^{mk} / k!$  and sum over  $m$ . Dominated convergence using  $[m]_Q \geq m$  proves Equation (11.23). Finally,

$$\log \Phi_Q(z) = - \sum_{r \geq 1} \frac{\zeta_{D,Q}(2r)}{r} z^{2r} \quad (|z| < 1),$$

and logarithmic differentiation gives Equation (11.24).  $\square$

## 12 A base- $b$ extension

The operator mechanism is not restricted to 2. Let  $b \geq 2$  be an integer and define

$$D_b e_{j,n} = b^j n e_{j,n}, \quad \Phi_b(z) = \det(I - z^2 D_b^{-2}) = \prod_{j=0}^{\infty} \operatorname{sinc}\left(\frac{\pi z}{b^j}\right). \quad (12.1)$$

Let  $\nu_b(m) = \max\{j : b^j \mid m\}$  and  $s_b(M)$  be the sum of the base- $b$  digits of  $M$ .

**Theorem 12.1** (Base- $b$  spectral package). *One has*

$$D_b \simeq N \oplus b D_b, \quad (12.2)$$

$$\operatorname{mult}_{D_b}(m) = 1 + \nu_b(m), \quad (12.3)$$

$$\zeta_{D_b}(s) = \frac{\zeta(s)}{1 - b^{-s}}, \quad (12.4)$$

$$A_b(M) := \sum_{m \leq M} (1 + \nu_b(m)) = \sum_{j \geq 0} \left\lfloor \frac{M}{b^j} \right\rfloor = \frac{bM - s_b(M)}{b - 1}. \quad (12.5)$$

For symmetric  $q$ -integers,

$$\Phi_{b,Q}(z) = \prod_{m \geq 1} \left(1 - \frac{z^2}{[m]_Q^2}\right)^{1 + \nu_b(m)} \quad (12.6)$$

satisfies

$$\Phi_{b,Q}(z) = S_Q(z) \Phi_{b,Q^b}(z/[b]_Q). \quad (12.7)$$

The associated  $q$ -Euler determinant is

$$\mathcal{E}_b(u; Q) = \prod_{j=0}^{\infty} (u Q^{b^j}; Q^{b^j})_{\infty}. \quad (12.8)$$

*Proof.* All statements follow by splitting off the  $j = 0$  summand and relabeling  $j \mapsto j - 1$ . The digit formula uses

$$\sum_{j \geq 1} \left\lfloor \frac{M}{b^j} \right\rfloor = \frac{M - s_b(M)}{b - 1}.$$

The  $q$ -scaling follows from  $[bn]_Q = [b]_Q [n]_{Q^b}$ . □

The dyadic case is distinguished because the parity of Equation (12.5) collapses to

$$(-1)^{A_2(M)} = (-1)^{s_2(M)},$$

which is exactly Thue–Morse. Thus the Thue–Morse sign is not an accidental decoration: it is the parity character singled out by the base-2 spectral tower.

## 13 Computational verification and reproducible checks

### 13.1 What was checked

The following identities were independently evaluated at high precision:

1. the leading zero-jet formula Equation (7.6) for integers of several distinct 2-adic valuations;
2. the factorization Equation (11.11) by comparing the direct zero product and the quotient of dyadic  $q$ -Euler determinants;
3. convergence  $\Phi_Q(z) \rightarrow \Phi(z)$  at real points inside and outside the first zero interval;
4. the heat expansion Equation (6.5), including the coefficient  $-1/12$  of  $t$  and the first Fourier coefficient of  $\mathcal{P}_{\Theta}$ ;
5. the critical expansion Equation (10.8), whose residual after the displayed  $t/24$  term was below  $6 \times 10^{-12}$  in the tested range  $0.01 \leq t \leq 0.2$  with a finite Fourier truncation.

### 13.2 $q \uparrow 1$ sample values

| $Q$           | $\Phi_Q(0.3)$ | $\Phi_Q(1.2)$ | $\Phi_Q(2.7)$ |
|---------------|---------------|---------------|---------------|
| 0.80          | 0.8316534376  | −0.08646646   | −0.03017980   |
| 0.90          | 0.8241751473  | −0.07432843   | −0.01331576   |
| 0.95          | 0.8204894677  | −0.06907163   | −0.00902981   |
| 0.98          | 0.8182876536  | −0.06613499   | −0.00721005   |
| 0.99          | 0.8175541599  | −0.06518866   | −0.00669771   |
| $Q = 1$ limit | 0.8168198779  | −0.0642567960 | −0.0062253251 |

The convergence is not claimed to be monotone in general; the table merely illustrates the trace-norm theorem.

### 13.3 A practical evaluation hierarchy

The new identities suggest several complementary algorithms:

- Near the origin:** use the coefficient recurrence [Equation \(5.7\)](#) or its  $q$ -version [Equation \(11.24\)](#).
- Near an integer zero:** use the normalized local factor [Equation \(7.2\)](#); this removes catastrophic cancellation and exposes the exact multiplicity.
- On compact sets away from zeros:** use the finite canonical product with the certified tail estimate [Equation \(4.9\)](#).
- For  $Q < 1$ :** use the  $q$ -Pochhammer quotient [Equation \(11.12\)](#), exploiting fast exponential convergence at large scales.
- For  $Q$  close to 1:** combine the exact quotient with the critical asymptotic [Equation \(10.8\)](#) to cancel large denominator terms analytically before numerical evaluation.

## 14 Connections back to the Fabius and Thue–Morse theories

### 14.1 Two independent appearances of $q$ -binomial structure

The existing Fabius corpus derives base-1/2 Gaussian-binomial coefficients from finite dyadic coefficient extraction and exact values of  $F$  at dyadic rationals. The present report derives Gaussian-binomial coefficients from a different object: finite spectral determinants of the zero divisor of  $\widehat{\mathbf{u}}$ . The common mechanism is multiscale dyadic refinement:

$$\text{dyadic value formulas} \longleftarrow \text{refinement blocks} \longrightarrow \text{spectral multiplicity blocks}.$$

The two coefficient systems are not identical, but they live in the same category of products indexed by powers of 2. This opens a concrete comparison problem: identify transforms that carry the exact-value  $q$ -binomial arrays into the spectral arrays  $C_{M,k}(Q)$  of [Equation \(9.5\)](#).

### 14.2 Thue–Morse as a determinant sign

[Equation \(4.5\)](#) can be rewritten

$$\tau_M = \text{sgn det}(I - x^2 D^{-2}) \quad (M < x < M + 1). \quad (14.1)$$

Thus a binary automatic sequence is recovered from signs of a single entire Fredholm determinant. The  $q$ -deformed version [Equation \(11.17\)](#) moves the zero locations but preserves the automatic sign word exactly.

### 14.3 Moments and zeros from one spectrum

The same spectrum produces both:

- the zero multiplicities through its eigenvalue degeneracies;

- the probability cumulants through the traces  $\text{Tr } D^{-2r}$ .

This is stronger than a formal analogy. It says the local probability law of the Rvachev density and the global zero divisor of its Fourier transform are two evaluations of the same spectral data.

#### 14.4 Heat smoothing versus Fourier decay

The active Fourier drafts study  $\Phi(x)$  at large real  $x$ , where lacunary products create pointwise exceptional behavior and metric limit laws. The heat trace  $\Theta(t)$  studies the same multiplicities under exponential smoothing. It is much more rigid: all oscillations are captured by the explicit vertical lattice  $2\pi i\mathbb{Z}/\log 2$ . A promising next step is to seek a transform connecting the log-periodic heat term to the periodic amplitudes that occur in endpoint asymptotics of  $F$ .

### 15 Lean formalization roadmap

The operator-theoretic presentation is conceptually compact, but a successful Lean formalization should proceed from finite identities to analytic limits.

#### 15.1 Layer 1: arithmetic and finite products

The most accessible statements are:

1.  $a_m = 1 + \nu_2(m)$  and the recursions  $a_{2m} = a_m + 1$ ,  $a_{2m+1} = 1$ ;
2.  $\sum_{m \leq M} a_m = 2M - s_2(M)$ ;
3. the finite factorization [Equation \(9.2\)](#);
4. the Gaussian-binomial coefficient formula [Equation \(9.5\)](#);
5. the finite form of the single-factor identity [Equation \(11.10\)](#);
6. the base- $b$  digit identity [Equation \(12.5\)](#).

These require no infinite-dimensional analysis.

#### 15.2 Layer 2: locally finite zero factorization and cofactor jets

This layer is now delivered. The formal proof first establishes the total shifted-sinc law, then the global denominator-cleared factorization at  $2^k q$ , and finally the canonical signed-integer form. A local analytic unit gives the exact complex order and the first nonzero derivative. The same formal layer now computes every derivative of order  $d_m + r$  from the  $r$ -th derivative of the analytic cofactor  $U_m$ . The remaining local-jet work is the separately normalized coefficient API: package  $R_m$ , identify it explicitly with  $U_m/U_m(0)$ , and formalize its logarithmic and Bell-polynomial derivatives and the literal coefficient-extraction notation in [Equation \(7.5\)](#).

#### 15.3 Layer 3: infinite products and traces

One can next formalize:

1. absolute convergence of  $\sum_{j,n} (2^j n)^{-2}$ ;
2. the regrouped canonical product;
3. the Dirichlet-series identity  $\zeta_D(s) = \zeta(s)/(1 - 2^{-s})$  on  $\Re s > 1$ ;
4. trace-norm convergence of the  $q$ -deformation, represented as convergence of summable diagonal sequences.

Only after these are stable is it efficient to introduce Mathlib’s operator determinant infrastructure, if the available API is sufficiently mature.

## 15.4 Layer 4: Mellin asymptotics

The heat and  $q$ -Euler expansions require contour shifting, residues, zeta continuation, and vertical estimates. They are the deepest formalization target. A useful intermediate milestone is a theorem schema: given a meromorphic Mellin transform with a finite specified pole set and suitable vertical decay, derive an asymptotic expansion. The dyadic results would then become applications.

## 16 Frontier questions

*Problem 16.1* (Arithmetic of half-integer amplitudes). Classify or sharply estimate the numbers  $\Phi((2k+1)/2)$  that govern every leading integer-zero jet through [Equation \(7.3\)](#). Determine whether their signs and 2-adic or transcendence properties admit formulae parallel to the exact dyadic values of  $F$ .

*Problem 16.2* (Comparison of the two Gaussian-binomial webs). Construct an explicit transform between the coefficient arrays in the exact dyadic-value formulas for  $F$  and the spectral coefficients  $C_{M,k}(Q)$  in [Equation \(9.5\)](#). The shared multiscale factorization suggests a common incidence algebra or a common family of refinement matrices.

*Problem 16.3* (Critical double scaling). Analyze  $\Phi_Q(z)$  when  $Q = e^{-t} \uparrow 1$  and  $z$  grows with  $t^{-1}$ . The fixed- $Q$  function has order zero and logarithmic zero growth, while the limit has order one and linear zero growth. Identify the transition kernels and scaling regimes.

*Problem 16.4* (Heat periodicity and endpoint periodicity). The heat trace has the explicit periodic wave [Equation \(6.3\)](#); endpoint asymptotics of the Fabius function also contain periodic dependence on a dyadic logarithmic phase. Determine whether both are projections of a single meromorphic two-variable transform.

*Problem 16.5* (Twisted spectral traces). Let  $Je_{j,n} = \tau_n e_{j,n}$ . For  $\Re s > 1$ ,

$$\mathrm{Tr}(JD^{-s}) = \frac{1}{1-2^{-s}} \sum_{n=1}^{\infty} \frac{\tau_n}{n^s}. \quad (16.1)$$

Study its continuation, functional relations, and whether its singularities encode known Thue–Morse Dirichlet-series phenomena in a spectral form.

*Problem 16.6* (Transfer operators). Relate the diagonal spectral model  $D$  to the transfer operators used in the Fourier-decay corpus. The former captures the exact zero divisor and smoothed scale statistics; the latter captures real-axis orbit statistics. A common operator category could unify zeros, decay, and probabilistic limit laws.

*Conjecture 16.7* (Universality of the symmetric  $q$ -deformation). Among deformations that (i) preserve the multiplicity word  $1 + \nu_2(m)$ , (ii) obey an exact two-scale factorization, (iii) have order zero for fixed  $Q < 1$ , and (iv) converge locally uniformly to  $\Phi$ , the symmetric  $q$ -integer model [Equation \(11.4\)](#) is unique up to an analytic reparameterization of  $Q$  and a scalar rescaling of  $z$ .

## 17 Conclusion

The infinite sinc product of Rvachev’s up-function admits a spectral organization that is invisible when the product is treated only factor by factor. The diagonal operator  $D$  turns dyadic refinement into the unitary decomposition  $D \simeq N \oplus 2D$ . Its eigenvalue degeneracies are the integer-zero multiplicities; its finite spectral parity is the Thue–Morse sequence; its zeta function is  $\zeta(s)/(1-2^{-s})$ ; its complex dimensions generate a log-periodic heat wave; and its power traces recover the probability cumulants.

The same operator admits two complementary  $q$ -constructions. The determinant  $\mathcal{E}(u; Q) = \det(I - uQ^D)$  is a dyadic tower of  $q$ -Pochhammer symbols with exact finite Gaussian-binomial coefficients and explicit Mellin asymptotics. Replacing each spectral integer by its symmetric  $q$ -integer produces  $\Phi_Q$ , an order-zero entire function with the same multiplicity and Thue–Morse sign word, an exact  $q$ -renormalization equation, and a locally uniform limit to the Rvachev Fourier transform. Formula [Equation \(11.12\)](#) is the central new bridge: it connects  $q$ -Pochhammer symbols directly to the zero divisor of an infinite sinc product, while [Equation \(11.7\)](#) shows that the connection is dictated by self-similarity rather than by a formal change of notation.

The resulting program is concrete. Its finite arithmetic core, total local integer-zero factorization, exact complex order, lower vanishings, and every higher jet expressed through derivatives of the analytic cofactor are now machine checked. The separately normalized  $R_m$  coefficient, logarithmic, and Bell-polynomial packages

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remain frontier work; the Mellin theory supplies exact human-readable asymptotics, and its unresolved comparison with the exact dyadic-value  $q$ -calculus offers a sharply defined next frontier.

## A Recursive TeX corpus ledger

The following ledger records the recursive TeX corpus used for the audit. Directory prefixes are shortened relative to `Analysis/FabiusFunction/docs/`. For typographic wrapping, spaces in two historical directory names are displayed as underscores.

### A.1 Living and canonical documents: 18

| No. | Relative document                                                                                                                                  |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | <code>FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex</code>                                                         |
| 2   | <code>Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex</code>                                                                     |
| 3   | <code>Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex</code>                                                   |
| 4   | <code>fabius_lean_walkthrough/fabius_lean_walkthrough.tex</code>                                                                                   |
| 5   | <code>semi-formalized-research-frontiers/semi-formalized-research-frontiers.tex</code>                                                             |
| 6   | <code>semi-formalized-research-frontiers/Thue_Morse_Formula_Atlas/Thue_Morse_Formula_Atlas.tex</code>                                              |
| 7   | <code>.../rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-2/Rvachev_Up_Fourier_Decay.tex</code>                                                  |
| 8   | <code>.../rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-3/Rvachev_Up_Fourier_Decay.tex</code>                                                  |
| 9   | <code>.../Rvachev_Up_Fourier_Decay_Comparative_Audit/Rvachev_Up_Fourier_Decay_Comparative_Audit.tex</code>                                         |
| 10  | <code>.../Rvachev_Up_Fourier_Decay_Gentle_Guide/Rvachev_Up_Fourier_Decay_Gentle_Guide.tex</code>                                                   |
| 11  | <code>.../Rvachev_Up_Fourier_Decay_Second_Wave_Audit/Rvachev_Up_Fourier_Decay_Second_Wave_Audit.tex</code>                                         |
| 12  | <code>.../asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions.tex</code> |
| 13  | <code>.../rvachev_up_fourier_decay-1/rvachev_up_fourier_decay.tex</code>                                                                           |
| 14  | <code>.../rvachev_up_fourier_decay-4/rvachev_up_fourier_decay.tex</code>                                                                           |
| 15  | <code>.../rvachev_up_fourier_decay-5/rvachev_up_fourier_decay.tex</code>                                                                           |
| 16  | <code>.../rvachev_up_fourier_decay-6/rvachev_up_fourier_decay.tex</code>                                                                           |
| 17  | <code>.../rvachev_up_fourier_decay-7/rvachev_up_fourier_decay.tex</code>                                                                           |
| 18  | <code>.../rvachev_up_fourier_decay-8/rvachev_up_fourier_decay.tex</code>                                                                           |

### A.2 Frozen historical archive: 32

| No. | Archived document directory / TeX base name                                                                                            |
|-----|----------------------------------------------------------------------------------------------------------------------------------------|
| 1   | <code>early-notes/Fabius_Asymptotic/Fabius_Asymptotic.tex</code>                                                                       |
| 2   | <code>early-notes/K-fold_summation_over_the_signed_Thue-Morse_sequence/K-fold_summation_over_the_signed_Thue-Morse_sequence.tex</code> |
| 3   | <code>standalone-studies/Non_Elementarity_of_the_Fabius_Function/...tex</code>                                                         |
| 4   | <code>standalone-studies/Small_Argument_Asymptotics/Small_Argument_Asymptotics.tex</code>                                              |
| 5   | <code>standalone-studies/fabius-inverse-dyadic-closed-form/fabius-inverse-dyadic-closed-form.tex</code>                                |
| 6   | <code>primary-exposition/Fabius_Function_and_Rvachev_Up/...tex</code>                                                                  |

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| No. | Archived document directory / TeX base name                                                              |
|-----|----------------------------------------------------------------------------------------------------------|
| 7   | primary-exposition/Fabius_Function_and_Rvachev_Up-2/...tex                                               |
| 8   | primary-exposition/Fabius_Function_and_Rvachev_Up-3/...tex                                               |
| 9   | primary-exposition/Fabius_and_Rvachev_Up-2/Fabius_and_Rvachev_Up.tex                                     |
| 10  | primary-exposition/Fabius_and_Rvachev_up/Fabius_and_Rvachev_up.tex                                       |
| 11  | primary-exposition/Fabius_function_and_up_function/Fabius_function_and_up_function.tex                   |
| 12  | primary-supplements/Fabius_Function_Missing_Parts/...tex                                                 |
| 13  | primary-supplements/Fabius_Function_Missing_Parts-2/...tex                                               |
| 14  | q-calculus/Fabius_q_Special_Functions/Fabius_q_Special_Functions.tex                                     |
| 15  | q-calculus/fabius-q-special-functions/fabius-q-special-functions.tex                                     |
| 16  | lean-walkthrough/Fabius_Function_Lean_Walkthrough/...tex                                                 |
| 17  | lean-walkthrough/fabius_lean_walkthrough/...tex                                                          |
| 18  | glossary/FabiusFunction_Mathematical_Glossary/...tex                                                     |
| 19  | glossary/FabiusFunction_Mathematical_Glossary-2/...tex                                                   |
| 20  | research-frontiers/Fabius_Dyadic_Asymptotic_Bridge/...tex                                                |
| 21  | research-frontiers/Fabius_Dyadic_Formulae_and_Alternative_Representations/...tex                         |
| 22  | research-frontiers/Fabius_Dyadic_Formulae_to_Asymptotics/...tex                                          |
| 23  | research-frontiers/Fabius_Dyadic_q_Connections/...tex                                                    |
| 24  | research-frontiers/Fabius_Thue_Morse_Convergence_Rate/...tex                                             |
| 25  | research-frontiers/Fabius_Thue_Morse_Convergence_Rates/...tex                                            |
| 26  | research-frontiers/Primary_Exposition_Gap_Register/...tex                                                |
| 27  | research-frontiers/Repeated_Integration_and_Rvachev_Up/...tex                                            |
| 28  | research-frontiers/Rvachev_Up_from_Repeated_Integration/...tex                                           |
| 29  | research-frontiers/semi-formalized-research-frontiers/...tex                                             |
| 30  | research-frontiers/Thue_Morse_Formula_Atlas-1/Thue_Morse_Formula_Atlas.tex                               |
| 31  | research-frontiers/Thue_Morse_Formula_Atlas-2/Consolidated_Formula_Atlas_for_the_Thue_Morse_Sequence.tex |
| 32  | research-frontiers/Thue_Morse_Formula_Atlas-3/Thue_Morse_Formula_Atlas.tex                               |

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### A.3 Imported paper sources: 7

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| No. | Relative TeX source                            |
|-----|------------------------------------------------|
| 1   | papers/AIF/AIF_2017__67_2_637_0.tex            |
| 2   | papers/AIF/AIF_2017__67_2_637_0-corrected.tex  |
| 3   | papers/Lacunary_products/Lacunary_products.tex |
| 4   | papers/09-Function/09-Function.tex             |
| 5   | papers/157-Arithmetic-v3/157-Arithmetic-v3.tex |
| 6   | papers/document/document.tex                   |
| 7   | papers/ubiq15/ubiq15-corrected.tex             |

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## B Residue calculations for the critical $q$ -Euler product

For reference, the local series used in [theorem 10.2](#) are

$$\begin{aligned}\Gamma(s) &= \frac{1}{s} - \gamma + \left(\frac{\gamma^2}{2} + \frac{\pi^2}{12}\right)s + O(s^2), \\ \zeta(1+s) &= \frac{1}{s} + \gamma - \gamma_1 s + O(s^2), \\ \frac{1}{1-e^{-Ls}} &= \frac{1}{Ls} + \frac{1}{2} + \frac{Ls}{12} + O(s^3), \\ \zeta(s) &= -\frac{1}{2} - \frac{\log(2\pi)}{2}s + \frac{\zeta''(0)}{2}s^2 + O(s^3).\end{aligned}$$

Hence

$$\begin{aligned}\Gamma(s)\zeta(1+s) &= \frac{1}{s^2} + \left(-\gamma_1 - \frac{\gamma^2}{2} + \frac{\pi^2}{12}\right) + O(s), \\ \frac{\zeta(s)}{1-e^{-Ls}} &= -\frac{1}{2L} \frac{1}{s} - \frac{1}{4} - \frac{\log(2\pi)}{2L} \\ &\quad + \left(\frac{\zeta''(0)}{2L} - \frac{\log(2\pi)}{4} - \frac{L}{24}\right)s + O(s^2).\end{aligned}$$

Multiplication by  $t^{-s} = e^{sy}$  then yields the  $y^2$ ,  $y$ , and constant terms in [Equation \(10.8\)](#).

## C Selected source map

### References

- [1] V. Reshetnikov, *ProveIt: Analysis/FabiusFunction documentation corpus*, GitHub repository, living tree and frozen archive, <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs>, accessed 27 August 2026.
- [2] V. Reshetnikov, *The Fabius Function and Rvachev’s Up-Function*, source in the ProveIt documentation corpus, [https://github.com/VladimirReshetnikov/ProveIt/blob/main/Analysis/FabiusFunction/docs/Fabius\\_Function\\_and\\_Rvachev\\_Up/Fabius\\_Function\\_and\\_Rvachev\\_Up.tex](https://github.com/VladimirReshetnikov/ProveIt/blob/main/Analysis/FabiusFunction/docs/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex).
- [3] V. Reshetnikov, *Non-formalized research frontiers for the Fabius function*, consolidated source in the ProveIt corpus, <https://github.com/VladimirReshetnikov/ProveIt/blob/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/semi-formalized-research-frontiers.tex>.
- [4] V. Reshetnikov, *Thue–Morse Formula Atlas*, source in the ProveIt corpus, [https://github.com/VladimirReshetnikov/ProveIt/blob/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/Thue\\_Morse\\_Formula\\_Atlas/Thue\\_Morse\\_Formula\\_Atlas.tex](https://github.com/VladimirReshetnikov/ProveIt/blob/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/Thue_Morse_Formula_Atlas/Thue_Morse_Formula_Atlas.tex).
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- [7] C. Aistleitner, M. Hofer, and G. Larcher, *On parametric Thue–Morse sequences and lacunary trigonometric products*, 2015, arXiv:1502.06738.
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- [9] J.-P. Allouche and J. Shallit, *Automatic Sequences: Theory, Applications, Generalizations*, Cambridge University Press, 2003.

**Part VIII****Logarithmic  $q$ -Richardson Acceleration and  
Lambert Phase Locking in Fabius–Rvachev  
Analysis**

### Abstract

The Fabius function, Rvachev’s compactly supported  $C^\infty$  up-function, the signed Thue–Morse sequence, and dyadic  $q$ -calculus form a tightly connected system. The existing PROVEIt corpus already contains extensive exact dyadic arithmetic, Gaussian-binomial interpolation, moment and cumulant formulas involving Bernoulli and Bell polynomials, Fourier products, Lambert- $W$  endpoint asymptotics with a nonconstant periodic factor, and several forms of Richardson extrapolation. After a repository-wide audit of the living documents, frozen historical versions, research drafts, and bundled source papers, this report develops two extrapolation mechanisms that are not stated there.

First, for the generalized lacunary sinc product

$$P_b(t) = \prod_{k \geq 1} \text{sinc}(t/b^k), \quad b > 1,$$

we obtain an exact Bernoulli–zeta expansion of the omitted logarithmic tail. Applying Lagrange interpolation to the logarithms of the truncations at the geometric nodes  $q^j$ ,  $q = b^{-2}$ , gives a multiplicative Richardson transform with closed Gaussian- $q$ -binomial weights. Its full error series is exact, has a fixed sign  $(-1)^r$  at level  $r$ , yields consecutive certified upper and lower bounds, and admits an explicit nonasymptotic remainder estimate. The  $\ell^1$  norm of the weights is exactly  $(-q; q)_r / (q; q)_r$ , hence remains bounded as the order tends to infinity; in the dyadic case it stays below 1.970. The method therefore combines high order with unusually mild conditioning. Bell polynomials convert the logarithmic error into relative-error and derivative formulas, while a finite Thue–Morse exponential-sum identity transfers the accelerated product to purely finite signed digital sums.

Second, the exact inverse coordinate

$$\lambda(x) = -\frac{1}{\log 2} W_{-1}(-x \log 2), \quad x = \lambda 2^{-\lambda},$$

is used to build phase-locked nodes  $x_j = (\lambda + j)2^{-(\lambda+j)}$ . Periodic coefficient functions are identical at all these nodes. Ordinary Lagrange extrapolation in  $h_j = (\lambda + j)^{-1}$  then cancels the first  $r$  inverse- $\lambda$  corrections and extracts the periodic endpoint term  $\Psi(\lambda)$  directly from values of  $F$ . The exact weights, all transformed moments, and their complete-homogeneous/Bell-polynomial representation are derived. Bernoulli polynomials describe the arithmetic-node power sums needed in conditioning and preconditioning. Together these results provide a new bridge between  $q$ -binomial interpolation, sinc products, Thue–Morse blocks, Lambert- $W$  phase geometry, Bernoulli coefficients, Bell-polynomial reconstruction, and Richardson extrapolation.

# 1 Scope, corpus audit, and nonduplication boundary

## 1.1 The audited corpus

The target tree contains living expositions, a deliberately quarantined frontier volume, active drafts, frozen historical versions, and source-paper transcriptions. The audit used the current tree and the archive ledger as of 27 August 2026. Counting each distinct `.tex` path, the inventory contains 57 sources: 25 in the living tree and 32 frozen historical sources. The living set consists of four top-level expositions, the canonical frontier volume, one Thue–Morse atlas, twelve Fourier-decay dossiers, and seven source-paper transcriptions.

Table 22: Repository-wide source inventory used for the overlap audit.

| Family                        | TeX files | Principal content relevant to this report                                                                                |
|-------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------|
| Living primary expositions    | 4         | Fabius/up synthesis, glossary, non-elementarity, Lean walkthrough                                                        |
| Canonical frontier volume     | 1         | Dyadic web, repeated integration, endpoint asymptotics, Thue–Morse convergence, saddle analysis                          |
| Living formula atlas          | 1         | Thue–Morse identities, Prouhet cancellation, rarefaction, Mahler structure, Bell/Bernoulli formulas                      |
| Living Fourier-decay dossiers | 12        | Infinite sinc products, zeros, dyadic renormalization, envelopes, Lambert- $W$ scales, competing asymptotic formulations |
| Bundled source-paper TeX      | 7         | Lacunary products, up-function background, arithmetic of $F$ , approximation literature                                  |
| Frozen archive                | 32        | Earlier notes, parallel expositions, q-calculus articles, supplements, formula atlases, topic dossiers                   |
| Total                         | 57        | Every path is listed in <a href="#">Appendix A</a> .                                                                     |

The archive matters for originality. Several ideas no longer visible in the living primary paper survive in an older parallel draft or a superseded dossier. In particular, the audit treated the following as prior art rather than rediscoveries:

- Gaussian- $q$ -binomial coefficient extraction at geometric nodes;
- exact dyadic Richardson transforms for sequences modeled by powers of  $2^{-p}$ ;
- Bernoulli-number cumulants and Bell-polynomial moment reconstruction;
- Prouhet cancellation for signed Thue–Morse blocks;
- fixed-numerator Lambert phase locking along dyadic endpoint rays;
- the periodic factor in the corrected small-argument asymptotic;
- exact finite Thue–Morse products and Bromwich/coefficient representations;
- global Fourier-decay envelopes and the zero geometry of the up transform.

## 1.2 What is new relative to the repository

The exact constructions below were not found in any audited source:

1. Richardson extrapolation of the *logarithm* of truncated lacunary sinc products, followed by exponentiation to a multiplicative approximation;
2. the exact Gaussian- $q$ -binomial formula for those log-extrapolation weights together with the full transformed Bernoulli–zeta tail;
3. the sign theorem and resulting alternating certified brackets for the infinite product;
4. the exact bounded condition number  $(-q; q)_r / (q; q)_r$  of the level- $r$  log transform;

5. the Bell-polynomial relative-error and derivative hierarchy of the accelerated product;
6. transfer of this accelerated approximation to finite signed Thue–Morse exponential sums;
7. Lambert phase-locked Richardson nodes  $x_j = (\lambda + j)2^{-(\lambda+j)}$  designed specifically to extract the periodic endpoint term  $\Psi$ ;
8. exact transformed inverse-power moments on that lattice, including their complete-homogeneous and Bell-polynomial form.

A targeted literature search located general Richardson-extrapolation theory and the standard ingredients separately, but no exact counterpart of the combined sinc-tail construction. The responsible statement is therefore “new relative to the audited repository and apparently not standard,” not a claim of global priority.

### 1.3 Proof-status convention

Every substantial assertion is tagged implicitly by its placement:

- Results in Sections 4–8 are **proved here**.
- The logarithmic phase-locked Lambert transform in Section 12 is **proved here** and Lean-backed at every fixed extrapolation order and residual depth. Its all-orders endpoint input is inherited from the formally verified endpoint theory, and the compiled pullback, growing-weight, and finite-residual layers now discharge the uniformity step.
- Items in Section 18 are explicitly conjectural or open.

Here “proved” means proved in this mathematical report, not necessarily formalized in Lean. In particular, the endpoint input is represented by `fabiusSaddleLogCoefficient`, `log_fabius_sub_sharpLambertMain_hasAsymptoticExpansion`, and `log_fabius_sub_sharpLambertExpansion_isBigO` in the Lean development, while the logarithmic phase-extraction transform is implemented by `FabiusLambertPhaseLockedPullback.lean`, `CompleteHomogeneousAsymptotics.lean`, `LambertReciprocalAsymptotics.lean`, and `FabiusLambertPhaseExtraction.lean`. The finite complete-homogeneous/Bell/generalized-harmonic conversion and the exact Bell rewrites of each residual term and finite residual partial sum are implemented by `CompleteHomogeneousBell.lean`, `LambertPhaseLockedBell.lean`, and `FabiusLambertPhaseExtractionBell.lean`. This closed status does not extend to convergence of an infinite residual series, the exponentiated estimator’s Bell relative-error hierarchy, the higher derivative extractors, a Lambert sign/bracketing theorem, or estimates uniform in growing extrapolation order. The remaining sinc-tail sign, bracketing, and quantitative analytic claims are separate formalization obligations.

## 2 Notation and structural background

### 2.1 Sinc products and the Rvachev transform

We use the entire normalized sinc function

$$\text{sinc } z = \begin{cases} \sin z/z, & z \neq 0, \\ 1, & z = 0. \end{cases}$$

For  $b > 1$ , define

$$P_b(t) = \prod_{k=1}^{\infty} \text{sinc}\left(\frac{t}{b^k}\right), \quad P_{b,N}(t) = \prod_{k=1}^N \text{sinc}\left(\frac{t}{b^k}\right). \quad (2.1)$$

The dyadic case is  $P = P_2$ . In a common Fourier normalization for Rvachev’s up-function,

$$\widehat{\text{up}}(\xi) = \prod_{h=0}^{\infty} \text{sinc}\left(\frac{\pi\xi}{2^h}\right) = \text{sinc}(\pi\xi)P_2(\pi\xi). \quad (2.2)$$

Thus every result for  $P_2$  immediately applies to the nontrivial tail of  $\widehat{\text{up}}$ .

Set

$$q = b^{-2} \in (0, 1), \quad z_N = q^{N+1}, \quad \rho_N = \frac{|t|}{b^{N+1}} = |t|z_N^{1/2}. \quad (2.3)$$

The local analysis below requires  $\rho_N < \pi$ , exactly the condition that every omitted sinc factor lies before its first nonzero zero.

## 2.2 $q$ -notation

For  $r \in \mathbb{N}$ ,

$$(a; q)_r = \prod_{k=0}^{r-1} (1 - aq^k), \quad (q; q)_r = \prod_{k=1}^r (1 - q^k),$$

and

$$\begin{bmatrix} r \\ j \end{bmatrix}_q = \frac{(q; q)_r}{(q; q)_j (q; q)_{r-j}} \quad (0 \leq j \leq r).$$

We will repeatedly use the finite  $q$ -binomial theorem

$$\prod_{k=0}^{r-1} (1 + zq^k) = \sum_{j=0}^r q^{\binom{j}{2}} \begin{bmatrix} r \\ j \end{bmatrix}_q z^j. \quad (2.4)$$

## 2.3 Bernoulli and Bell polynomials

The Bernoulli polynomials are defined by

$$\frac{te^{xt}}{e^t - 1} = \sum_{n=0}^{\infty} B_n(x) \frac{t^n}{n!}, \quad B_n = B_n(0). \quad (2.5)$$

The complete exponential Bell polynomials  $\mathcal{B}_n$  satisfy

$$\exp\left(\sum_{m \geq 1} x_m \frac{u^m}{m!}\right) = \sum_{n \geq 0} \mathcal{B}_n(x_1, \dots, x_n) \frac{u^n}{n!}. \quad (2.6)$$

They are the natural conversion device from logarithmic/cumulant data to multiplicative/moment data.

## 2.4 The lower-Lambert coordinate

Let  $L = \log 2$ . For sufficiently small  $x > 0$ , set

$$\lambda(x) = -\frac{1}{L} W_{-1}(-Lx). \quad (2.7)$$

Then

$$x = \lambda 2^{-\lambda}, \quad \lambda > \frac{1}{L}. \quad (2.8)$$

The lower branch is essential: on  $\lambda > 1/L$ , the map  $\lambda \mapsto \lambda 2^{-\lambda}$  is strictly decreasing and hence invertible.

## 3 Exact logarithmic tails and multiplicative $q$ -Richardson acceleration

## 4 The exact Bernoulli–zeta tail of a lacunary sinc product

### 4.1 The logarithmic sinc series

**Lemma 4.1** (Bernoulli–zeta expansion). *For  $|u| < \pi$ ,*

$$\log \frac{\sin u}{u} = - \sum_{m=1}^{\infty} \gamma_m u^{2m}, \quad \gamma_m = \frac{\zeta(2m)}{m\pi^{2m}} = \frac{2^{2m-1} |B_{2m}|}{m(2m)!}. \quad (4.1)$$

*Every  $\gamma_m$  is strictly positive.*

*Proof.* Euler's product  $\sin u/u = \prod_{n \geq 1} (1 - u^2/(\pi^2 n^2))$  is locally uniformly convergent for  $|u| < \pi$ . Expanding  $\log(1 - v) = -\sum_{m \geq 1} v^m/m$  and interchanging the absolutely convergent sums gives

$$\log \frac{\sin u}{u} = -\sum_{m \geq 1} \frac{u^{2m}}{m\pi^{2m}} \sum_{n \geq 1} n^{-2m}.$$

The Bernoulli form follows from Euler's evaluation of  $\zeta(2m)$ .  $\square$

The first coefficients are

$$\log \operatorname{sinc} u = -\frac{u^2}{6} - \frac{u^4}{180} - \frac{u^6}{2835} - \frac{u^8}{37800} - \cdots. \quad (4.2)$$

## 4.2 Exact omitted-tail formula

**Theorem 4.2** (Exact logarithmic tail). *Let  $b > 1$ ,  $N \geq 0$ , and  $t \in \mathbb{R}$  satisfy  $\rho_N < \pi$ . If  $P_{b,N}(t) \neq 0$ , then  $P_b(t)$  and  $P_{b,N}(t)$  have the same sign and*

$$\log |P_{b,N}(t)| - \log |P_b(t)| = \sum_{m=1}^{\infty} d_m(t) z_N^m, \quad (4.3)$$

where

$$d_m(t) = \frac{\gamma_m t^{2m}}{1 - q^m} = \frac{\zeta(2m) t^{2m}}{m\pi^{2m}(1 - q^m)}. \quad (4.4)$$

Equivalently,

$$\log |P_{b,N}(t)| - \log |P_b(t)| = \sum_{m=1}^{\infty} \frac{\zeta(2m)}{m\pi^{2m}(1 - q^m)} \rho_N^{2m}. \quad (4.5)$$

The series is absolutely convergent and strictly positive for  $t \neq 0$ .

*Proof.* For  $k \geq N + 1$  we have  $|t|/b^k \leq \rho_N < \pi$ , so every omitted sinc factor is positive. Moreover, the logarithms of the omitted factors are absolutely summable by [lemma 4.1](#); their product therefore converges to a strictly positive, nonzero limit. Hence truncation and limit have the same sign. Applying the same expansion and using absolute convergence,

$$\begin{aligned} \log |P_b(t)| - \log |P_{b,N}(t)| &= \sum_{k=N+1}^{\infty} \log \operatorname{sinc}(t/b^k) \\ &= -\sum_{m=1}^{\infty} \gamma_m t^{2m} \sum_{k=N+1}^{\infty} b^{-2mk} \\ &= -\sum_{m=1}^{\infty} \gamma_m t^{2m} \frac{q^{m(N+1)}}{1 - q^m}. \end{aligned}$$

Negating yields the result.  $\square$

**Corollary 4.3** (Analytic Bernoulli correction). *For  $M \geq 0$ , define*

$$C_{b,N}^{[M]}(t) = P_{b,N}(t) \exp\left(-\sum_{m=1}^M d_m(t) z_N^m\right). \quad (4.6)$$

*Then  $C_{b,N}^{[M]}(t)$  has the same sign as  $P_b(t)$  and*

$$\log \frac{|C_{b,N}^{[M]}(t)|}{|P_b(t)|} = \sum_{m=M+1}^{\infty} d_m(t) z_N^m > 0 \quad (t \neq 0). \quad (4.7)$$

*Thus  $|C_{b,N}^{[M]}(t)|$  is a rigorous upper bound for  $|P_b(t)|$ .*

This correction is explicit but uses  $\zeta(2m)$  or Bernoulli numbers. The next construction achieves cancellation from product values alone.

## 5 Geometric Lagrange weights and their $q$ -binomial form

### 5.1 Interpolation at $1, q, \dots, q^r$

For an integer  $r \geq 0$ , let  $\lambda_{r,j}(q)$  be the Lagrange basis weight for evaluation at 0 using nodes  $q^0, \dots, q^r$ :

$$\lambda_{r,j}(q) = \prod_{\substack{0 \leq k \leq r \\ k \neq j}} \frac{-q^k}{q^j - q^k}. \quad (5.1)$$

**Theorem 5.1** (Gaussian- $q$ -binomial weights). *For  $0 \leq j \leq r$ ,*

$$\lambda_{r,j}(q) = \frac{(-1)^{r-j} q^{\binom{r-j+1}{2}}}{(q; q)_j (q; q)_{r-j}} = \frac{(-1)^{r-j} q^{\binom{r-j+1}{2}}}{(q; q)_r} \begin{bmatrix} r \\ j \end{bmatrix}_q. \quad (5.2)$$

Their power moments are

$$S_{r,m}(q) := \sum_{j=0}^r \lambda_{r,j}(q) q^{jm} = \frac{q^{rm} (q^{1-m}; q)_r}{(q; q)_r}. \quad (5.3)$$

Consequently,

$$S_{r,0} = 1, \quad S_{r,m} = 0 \quad (1 \leq m \leq r), \quad (5.4)$$

and, for  $m \geq r+1$ ,

$$S_{r,m}(q) = (-1)^r q^{r(r+1)/2} \frac{(q^{m-r}; q)_r}{(q; q)_r}. \quad (5.5)$$

In particular, every nonzero residual moment has sign  $(-1)^r$ .

*Proof.* Splitting the product in Equation (5.1) into  $k < j$  and  $k > j$  gives Equation (5.2). For the moments, substitute the Gaussian-binomial form and change variables  $k = r - j$ . The finite  $q$ -binomial theorem Equation (2.4), with a suitable monomial factor extracted, gives Equation (5.3). The zeros for  $1 \leq m \leq r$  arise because  $(q^{1-m}; q)_r$  contains the factor  $1 - q^0$ . For  $m \geq r+1$ , reverse the negative powers in the product to obtain Equation (5.5).  $\square$

### 5.2 An exact condition number

The alternating weights might appear dangerous, but their total variation is exceptionally well behaved.

**Theorem 5.2** (Bounded  $\ell^1$  norm). *The absolute weight sum is*

$$\kappa_r(q) := \sum_{j=0}^r |\lambda_{r,j}(q)| = \frac{(-q; q)_r}{(q; q)_r} = \prod_{k=1}^r \frac{1 + q^k}{1 - q^k}. \quad (5.6)$$

Hence

$$\kappa_r(q) \nearrow \kappa_\infty(q) := \frac{(-q; q)_\infty}{(q; q)_\infty} < \infty. \quad (5.7)$$

For  $q = 1/4$ ,

$$\kappa_\infty(1/4) = 1.9692603536682694319 \dots \quad (5.8)$$

*Proof.* Set  $k = r - j$  in the absolute-value sum and use Equation (5.2):

$$\kappa_r(q) = \frac{1}{(q; q)_r} \sum_{k=0}^r q^{k(k+1)/2} \begin{bmatrix} r \\ k \end{bmatrix}_q.$$

Equation (2.4) with  $z = q$  identifies the numerator as  $\prod_{i=0}^{r-1} (1 + q^{i+1}) = (-q; q)_r$ . The infinite product converges because  $\sum q^k < \infty$ .  $\square$

*Remark 5.3.* This boundedness is not generic for high-order polynomial extrapolation. It is a special consequence of the geometric nodes and is one reason to extrapolate the logarithms: the inputs are of comparable scale, while the transformation itself has uniformly bounded absolute sensitivity.



## 6 Multiplicative $q$ -Richardson extrapolation

### 6.1 Definition and exact transformed error

Assume  $\rho_N < \pi$  and  $P_{b,N}(t) \neq 0$ . Define

$$\widehat{P}_{b;N,r}(t) = \operatorname{sgn}(P_{b,N}(t)) \exp \left( \sum_{j=0}^r \lambda_{r,j}(q) \log |P_{b,N+j}(t)| \right). \quad (6.1)$$

Equivalently,

$$|\widehat{P}_{b;N,r}(t)| = \prod_{j=0}^r |P_{b,N+j}(t)|^{\lambda_{r,j}(q)}. \quad (6.2)$$

Negative exponents are harmless because the relevant truncations are nonzero.

**Theorem 6.1** (Exact multiplicative extrapolation error). *Under the preceding hypotheses,*

$$E_{b;N,r}(t) := \log \frac{|\widehat{P}_{b;N,r}(t)|}{|P_b(t)|} \quad (6.3)$$

$$= \sum_{m=r+1}^{\infty} d_m(t) z_N^m S_{r,m}(q) \quad (6.4)$$

$$= \frac{(-1)^r q^{r(r+1)/2}}{(q; q)_r} \sum_{m=r+1}^{\infty} \frac{\zeta(2m)}{m\pi^{2m}} \frac{(q^{m-r}; q)_r}{1 - q^m} \rho_N^{2m}. \quad (6.5)$$

The series is absolutely convergent. If  $t \neq 0$ , then

$$\operatorname{sgn} E_{b;N,r}(t) = (-1)^r. \quad (6.6)$$

*Proof.* Apply [theorem 4.2](#) to each truncation  $N + j$ :

$$\log |P_{b,N+j}| = \log |P_b| + \sum_{m \geq 1} d_m z_N^m q^{jm}.$$

The constant term survives because  $S_{r,0} = 1$ , while the first  $r$  powers vanish by [Equation \(5.4\)](#). This proves [Equation \(6.4\)](#). [Equation \(6.5\)](#) follows from [Equation \(5.5\)](#). Every factor in its sum is positive, proving the sign statement.  $\square$

### 6.2 Certified alternating brackets

**Corollary 6.2** (Rigorous upper and lower bounds). *For  $t \neq 0$  under the hypotheses of [theorem 6.1](#),*

$$|\widehat{P}_{b;N,2s+1}(t)| < |P_b(t)| < |\widehat{P}_{b;N,2s}(t)| \quad (s \geq 0). \quad (6.7)$$

Thus any two consecutive extrapolation levels give a certified enclosure of the infinite product. In particular,

$$\left| \log \frac{|\widehat{P}_{b;N,r}(t)|}{|P_b(t)|} \right| \leq \left| \log \frac{|\widehat{P}_{b;N,r}(t)|}{|\widehat{P}_{b;N,r+1}(t)|} \right| \quad (6.8)$$

whenever the two levels lie on opposite sides as in [Equation \(6.7\)](#).

The bracket is stronger than a mere asymptotic claim: it follows from the exact sign of every term in the transformed tail.

### 6.3 Leading term and explicit remainder bound

**Corollary 6.3** (Sharp leading error). *As  $\rho_N \rightarrow 0$  with  $r$  fixed,*

$$E_{b;N,r}(t) = C_r(q) \rho_N^{2r+2} + O(\rho_N^{2r+4}), \quad (6.9)$$

where

$$C_r(q) = (-1)^r q^{r(r+1)/2} \frac{\zeta(2r+2)}{(r+1)\pi^{2r+2}(1-q^{r+1})}. \quad (6.10)$$

For the dyadic case  $q = 1/4$ ,

$$C_0 = \frac{2}{9}, \quad C_1 = -\frac{1}{675}, \quad C_2 = \frac{1}{178605}, \quad C_3 = -\frac{1}{154224000}. \quad (6.11)$$

**Theorem 6.4** (Nonasymptotic error bound). *Let  $a = (\rho_N/\pi)^2 < 1$ . Then*

$$|E_{b;N,r}(t)| \leq \frac{q^{r(r+1)/2} \zeta(2)}{(q; q)_r (1-q)(r+1)} \frac{a^{r+1}}{1-a}. \quad (6.12)$$

Consequently, for every fixed admissible pair  $(t, N)$ ,

$$\widehat{P}_{b;N,r}(t) \longrightarrow P_b(t) \quad (r \rightarrow \infty) \quad (6.13)$$

with a supergeometric factor  $q^{r(r+1)/2}$ .

*Proof.* For  $m \geq r+1$ , Equation (5.5) gives

$$|S_{r,m}(q)| \leq \frac{q^{r(r+1)/2}}{(q; q)_r}.$$

Also  $\zeta(2m) \leq \zeta(2)$ ,  $m^{-1} \leq (r+1)^{-1}$ , and  $(1-q^m)^{-1} \leq (1-q)^{-1}$ . Sum the resulting geometric series in  $a$ . Since  $(q; q)_r$  tends to the positive limit  $(q; q)_\infty$ , the right-hand side tends to zero faster than  $q^{r^2/2}$  times an exponential factor.  $\square$

### 6.4 Richardson table and multiplicative recurrence

The transform can be evaluated without forming the closed weights.

**Algorithm 6.5** (Logarithmic  $q$ -Richardson table). Set

$$R_{N,0}(t) = \log |P_{b,N}(t)|.$$

For  $s \geq 1$ , recursively define

$$R_{N,s}(t) = \frac{R_{N+1,s-1}(t) - q^s R_{N,s-1}(t)}{1 - q^s}. \quad (6.14)$$

Then

$$R_{N,r}(t) = \sum_{j=0}^r \lambda_{r,j}(q) \log |P_{b,N+j}(t)|, \quad \widehat{P}_{b;N,r} = \operatorname{sgn}(P_{b,N}) e^{R_{N,r}}. \quad (6.15)$$

In multiplicative form,

$$|\widehat{P}_{b;N,s}| = \left( \frac{|\widehat{P}_{b;N+1,s-1}|}{|\widehat{P}_{b;N,s-1}|^{q^s}} \right)^{1/(1-q^s)}. \quad (6.16)$$

The logarithmic implementation is preferable because it avoids underflow in the rapidly decaying Fourier regime.

### 6.5 Explicit dyadic levels

For  $b = 2$ ,  $q = 1/4$ . The first transforms are

$$|\widehat{P}_{N,1}| = |P_N|^{-1/3} |P_{N+1}|^{4/3}, \quad (6.17)$$

$$|\widehat{P}_{N,2}| = |P_N|^{1/45} |P_{N+1}|^{-4/9} |P_{N+2}|^{64/45}, \quad (6.18)$$

$$|\widehat{P}_{N,3}| = |P_N|^{-1/2835} |P_{N+1}|^{4/135} |P_{N+2}|^{-64/135} |P_{N+3}|^{4096/2835}. \quad (6.19)$$

The log errors begin respectively at orders  $\rho_N^4$ ,  $\rho_N^6$ , and  $\rho_N^8$ .

## 6.6 Hybrid Bernoulli–Richardson correction

The exact tail coefficients and the data-driven transform can be combined.

**Corollary 6.6** (Hybrid correction). *For  $s \geq 0$ , define*

$$\hat{P}_{b;N,r}^{[s]}(t) = \hat{P}_{b;N,r}(t) \exp\left(-\sum_{m=r+1}^{r+s} d_m(t) z_N^m S_{r,m}(q)\right). \quad (6.20)$$

Then

$$\log \frac{|\hat{P}_{b;N,r}^{[s]}(t)|}{|P_b(t)|} = \sum_{m=r+s+1}^{\infty} d_m(t) z_N^m S_{r,m}(q), \quad (6.21)$$

which still has sign  $(-1)^r$ . Thus the hybrid retains certified bracketing while moving the first nonzero term from degree  $r+1$  to degree  $r+s+1$ .

This is the most direct point at which Bernoulli numbers and Gaussian  $q$ -binomial weights cooperate in one exact numerical scheme.

## 7 Bell-polynomial error calculus and derivatives

### 7.1 From logarithmic error to relative error

Write

$$e_m^{(r)}(t) = d_m(t) S_{r,m}(q), \quad e_m^{(r)} = 0 \quad (1 \leq m \leq r). \quad (7.1)$$

Then Equation (6.4) is  $E = \sum_{m \geq 1} e_m^{(r)} z_N^m$ . Exponentiation and Equation (2.6) give an exact convergent expansion.

**Theorem 7.1** (Bell-polynomial relative-error expansion). *Under the hypotheses of theorem 6.1,*

$$\frac{|\hat{P}_{b;N,r}(t)|}{|P_b(t)|} = 1 + \sum_{n=r+1}^{\infty} \frac{1}{n!} \mathcal{B}_n(1!e_1^{(r)}, 2!e_2^{(r)}, \dots, n!e_n^{(r)}) z_N^n. \quad (7.2)$$

In particular, because  $e_1^{(r)} = \dots = e_r^{(r)} = 0$ , the first relative-error coefficient equals the first logarithmic-error coefficient:

$$\frac{|\hat{P}_{b;N,r}|}{|P_b|} - 1 = e_{r+1}^{(r)} z_N^{r+1} + O(z_N^{r+2}). \quad (7.3)$$

The next coefficients exhibit the familiar cumulant-to-moment pattern. For example,

$$\begin{aligned} [z_N^{r+2}] (e^E - 1) &= e_{r+2}^{(r)} & (r \geq 1), \\ [z_N^{2r+2}] (e^E - 1) &= e_{2r+2}^{(r)} + \frac{1}{2} (e_{r+1}^{(r)})^2. \end{aligned}$$

### 7.2 Logarithmic derivatives

On an interval avoiding zeros, differentiation of Equation (4.3) is legitimate termwise. Let

$$D^\ell = \frac{d^\ell}{dt^\ell}, \quad (2m)_\ell = 2m(2m-1) \cdots (2m-\ell+1).$$

**Theorem 7.2** (Derivative acceleration). *For every fixed  $\ell \geq 0$  and admissible  $t$  away from zeros,*

$$D^\ell \left( \sum_{j=0}^r \lambda_{r,j} \log |P_{b,N+j}(t)| - \log |P_b(t)| \right) = \sum_{m \geq \max(r+1, \lceil \ell/2 \rceil)} \frac{\gamma_m(2m)_\ell t^{2m-\ell}}{1-q^m} z_N^m S_{r,m}(q). \quad (7.4)$$

Thus the same  $q$ -Richardson table simultaneously cancels the first  $r$  geometric tail powers in every logarithmic derivative.

For reconstructing derivatives of the product itself, set  $L(t) = \log |P_b(t)|$ . Faà di Bruno’s formula becomes

$$\frac{P_b^{(n)}(t)}{P_b(t)} = \mathcal{B}_n(L'(t), L''(t), \dots, L^{(n)}(t)) \quad (7.5)$$

where the sign of  $P_b$  is locally constant. Combining [Equations \(7.4\) and \(7.5\)](#) gives an accelerated hierarchy for all derivatives of the Fourier transform away from its zeros.

### 7.3 Application to interzero extrema

Between consecutive zeros of  $\widehat{\text{up}}$ , an extremum of its modulus satisfies

$$\frac{d}{d\xi} \log |\widehat{\text{up}}(\xi)| = 0. \quad (7.6)$$

[Equation \(2.2\)](#) separates the first sinc factor from  $P_2(\pi\xi)$ . [Theorem 7.2](#) gives a certified high-order approximation to the tail contribution in [Equation \(7.6\)](#). This suggests a numerically stable alternative to multiplying many tiny sinc factors and then differentiating the product.

## 8 Transfer to finite signed Thue–Morse sums

Let

$$\varepsilon_n = (-1)^{s_2(n)}, \quad (8.1)$$

where  $s_2(n)$  is the binary digit sum. The standard finite product identity

$$\sum_{n=0}^{2^N-1} \varepsilon_n z^n = \prod_{j=0}^{N-1} (1 - z^{2^j}) \quad (8.2)$$

**Theorem 8.1** (Lean-verified finite complex block transforms). *For every  $N \geq 0$  and  $u, a, \zeta \in \mathbb{C}$ ,*

$$\sum_{n=0}^{2^N-1} u^{s_2(n)} \exp(n\zeta) = \prod_{j=0}^{N-1} (1 + u \exp(2^j \zeta)), \quad (8.3)$$

$$\sum_{n=0}^{2^N-1} u^{s_2(n)} \exp(a + n\zeta) = \exp(a) \prod_{j=0}^{N-1} (1 + u \exp(2^j \zeta)), \quad (8.4)$$

$$\sum_{n=0}^{2^N-1} \varepsilon_n \exp(n\zeta) = \prod_{j=0}^{N-1} (1 - \exp(2^j \zeta)), \quad (8.5)$$

$$\sum_{n=0}^{2^N-1} \varepsilon_n \exp(a + n\zeta) = \exp(a) \prod_{j=0}^{N-1} (1 - \exp(2^j \zeta)). \quad (8.6)$$

*All four equations include the zero-bit block  $N = 0$ : the sum has its single empty-word term  $n = 0$  and the product is empty. The unshifted identities read  $1 = 1$ , while the affine identities read  $\exp(a) = \exp(a)$ .*

*Proof.* The declarations, in the displayed order, are `sum_range_two_pow_binaryWeight_cexp`, `sum_range_two_pow_binaryWeight_cexp_affine`, `sum_range_two_pow_thueMorseSign_cexp`, and `sum_range_two_pow_thueMorseSign_cexp_affine` in `ThueMorseComplexExponential.lean`. Binary expansion turns the left side of the first identity into independent include-or-omit choices for the powers  $2^j$ . The affine forms factor out  $\exp(a)$ , while the signed forms are the corresponding Thue–Morse specialization.  $\square$

Thus the raw finite complex exponential product needed below is machine checked. A phase-scaled substitution turns it into the raw product in the next proof. These four declarations do not themselves perform the sine/sinc conversion or removable-origin normalization, but that finite layer is separately formal in `ThueMorseComplexProductBridge.lean`. The accelerated infinite-product conclusions, exponential-generating-function convergence statements, and parity words with position-dependent digit weights are not consequences of either finite module.

**Proposition 8.2** (Finite Thue–Morse representation of  $P_N$ ). *Define*

$$T_N(t) = \sum_{n=0}^{2^N-1} \varepsilon_n \exp\left(-\frac{itn}{2^{N-1}}\right). \quad (8.7)$$

*Then*

$$T_N(t) = i^N 2^{-N(N-1)/2} e^{-it(1-2^{-N})} t^N P_N(t), \quad (8.8)$$

*with the identity at  $t = 0$  understood by continuity. Hence, for  $t \neq 0$ ,*

$$P_N(t) = 2^{N(N-1)/2} i^{-N} e^{it(1-2^{-N})} t^{-N} T_N(t). \quad (8.9)$$

*Proof.* Insert  $z = e^{-it/2^{N-1}}$  in Equation (8.2). Reindex the factors to obtain

$$T_N(t) = \prod_{k=0}^{N-1} (1 - e^{-it/2^k}).$$

Using

$$1 - e^{-ia} = 2ie^{-ia/2} \sin(a/2),$$

sum the phases. Finally, factor

$$\prod_{j=1}^N \sin(t/2^j) = t^N 2^{-N(N+1)/2} P_N(t),$$

which gives the result.  $\square$

**Corollary 8.3** (Accelerated infinite product from finite digital sums). *For  $t \neq 0$  and admissible  $N$ ,*

$$|\widehat{P}_{N,r}(t)| = \prod_{j=0}^r \left( 2^{(N+j)(N+j-1)/2} |t|^{-(N+j)} |T_{N+j}(t)| \right)^{\lambda_{r,j}(1/4)}. \quad (8.10)$$

*The exact error, bracketing, and bounds of theorems 6.1 and 6.4 and corollary 6.2 therefore become statements purely about a finite collection of signed Thue–Morse exponential sums.*

This transfer is conceptually useful even when direct product evaluation is faster. Its raw complex-exponential core is now formalized by [theorem 8.1](#), and the total finite sine/sinc normalization is formalized by `ThueMorseComplexProductBridge.lean`; the analytic Richardson conclusions remain a route for further formal treatment.

## 9 Numerical verification and observed orders

All values in this section were recomputed with 100 decimal digits. The reference value used 300 sinc factors; at the displayed precision its omitted tail is negligible relative to every table entry. We report absolute logarithmic error

$$\varepsilon_{N,r}(t) = \left| \log |\widehat{P}_{N,r}(t)| - \log |P(t)| \right|,$$

where  $r = 0$  denotes the raw truncation  $P_N$ .

The successive- $N$  error ratios approach

$$4^{-1}, \quad 4^{-2}, \quad 4^{-3}, \quad 4^{-4}$$

for levels  $r = 0, 1, 2, 3$ , exactly as [Equation \(6.9\)](#) predicts. The signs also alternate: raw and level two are upper approximations in modulus; levels one and three are lower approximations.

For  $t = 3$  the product itself is

$$P(3) = 0.585719176535068482912529262451 \dots \quad (9.1)$$

Even at  $N = 2$ , the third transformed level improves the raw log error by approximately ten decimal orders while using only truncations through  $N + 3 = 5$ .

Table 23: Dyadic product at  $t = 3$ : absolute logarithmic errors.

| $N$ | raw $r = 0$               | $r = 1$                    | $r = 2$                    | $r = 3$                    |
|-----|---------------------------|----------------------------|----------------------------|----------------------------|
| 2   | $3.1368194 \cdot 10^{-2}$ | $2.9611726 \cdot 10^{-5}$  | $1.5785787 \cdot 10^{-8}$  | $2.5743474 \cdot 10^{-12}$ |
| 3   | $7.8198398 \cdot 10^{-3}$ | $1.8359337 \cdot 10^{-6}$  | $2.4411880 \cdot 10^{-10}$ | $9.9424663 \cdot 10^{-15}$ |
| 4   | $1.9535830 \cdot 10^{-3}$ | $1.1451700 \cdot 10^{-7}$  | $3.8045692 \cdot 10^{-12}$ | $3.8728157 \cdot 10^{-17}$ |
| 5   | $4.8830986 \cdot 10^{-4}$ | $7.1537455 \cdot 10^{-9}$  | $5.9408270 \cdot 10^{-14}$ | $1.5117515 \cdot 10^{-19}$ |
| 6   | $1.2207210 \cdot 10^{-4}$ | $4.4705340 \cdot 10^{-10}$ | $9.2810541 \cdot 10^{-16}$ | $5.9042379 \cdot 10^{-22}$ |

Table 24: Errors for two additional arguments, illustrating that the order is independent of  $t$  inside the convergence disk.

| $t$ | $N$ | raw $r = 0$               | $r = 1$                    | $r = 2$                    | $r = 3$                    |
|-----|-----|---------------------------|----------------------------|----------------------------|----------------------------|
| 1   | 4   | $2.1701954 \cdot 10^{-4}$ | $1.4129551 \cdot 10^{-9}$  | $5.2149215 \cdot 10^{-15}$ | $5.8978483 \cdot 10^{-21}$ |
| 1   | 6   | $1.3563390 \cdot 10^{-5}$ | $5.5189741 \cdot 10^{-12}$ | $1.2730608 \cdot 10^{-18}$ | $8.9985206 \cdot 10^{-26}$ |
| 5   | 4   | $5.4288846 \cdot 10^{-3}$ | $8.8466438 \cdot 10^{-7}$  | $8.1669316 \cdot 10^{-11}$ | $2.3096377 \cdot 10^{-15}$ |
| 5   | 6   | $3.3909800 \cdot 10^{-4}$ | $3.4497407 \cdot 10^{-9}$  | $1.9894410 \cdot 10^{-14}$ | $3.5155982 \cdot 10^{-20}$ |

## 10 Lambert phase-locked extraction of the endpoint oscillation

### 11 The all-orders endpoint template

The repository’s corrected endpoint analysis has the structural form

$$\log F(x) = \mathcal{W}_{\text{MSE}}(x) + \Psi(\lambda(x)) + \sum_{m=1}^M \frac{A_m(\lambda(x))}{\lambda(x)^m} + O(\lambda(x)^{-M-1}), \quad (11.1)$$

where  $\Psi$  and the coefficient functions  $A_m$  are 1-periodic. The nonoscillatory main saddle exponent may be written in the normalization used in the repository as

$$\mathcal{W}_{\text{MSE}}(x) = -\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda - \frac{1}{2}\log \lambda + C_F, \quad L = \log 2, \quad (11.2)$$

with the convention-dependent constant  $C_F$ . The critical fact for the present construction is not the explicit constant but the periodicity of every  $A_m$  in the same phase variable.

Define the reduced endpoint observable

$$Y(x) = \log F(x) - \mathcal{W}_{\text{MSE}}(x). \quad (11.3)$$

Then Equation (11.1) becomes

$$Y(x) = \Psi(\lambda) + \sum_{m \geq 1} A_m(\lambda) \lambda^{-m} \quad (11.4)$$

in the asymptotic sense.

## 12 Phase-locked Lambert nodes

### 12.1 Exact phase locking

Fix  $\lambda > 1/L$  and define

$$\lambda_j = \lambda + j, \quad x_j = x_j(\lambda) = (\lambda + j)2^{-(\lambda+j)}, \quad j = 0, 1, \dots \quad (12.1)$$

**Lemma 12.1** (Exact Lambert inversion on the lattice). *For every  $j \geq 0$ ,*

$$\lambda(x_j) = \lambda + j. \quad (12.2)$$

*Consequently, for every 1-periodic function  $G$ ,*

$$G(\lambda(x_j)) = G(\lambda). \quad (12.3)$$

*Proof.* Equation (2.8) gives the inverse relation exactly on the decreasing branch. Periodicity gives the second assertion.  $\square$

Thus the nodes are not merely close in phase; they have exactly identical phase. Their physical spacing is nongeometric:

$$\frac{x_{j+1}}{x_j} = \frac{\lambda + j + 1}{2(\lambda + j)}, \quad \log_2 \frac{x_j}{x_0} = \log_2 \frac{\lambda + j}{\lambda} - j. \quad (12.4)$$

This slight deviation from a dyadic ray is precisely what preserves the Lambert phase.

## 12.2 Lagrange weights in reciprocal scale

Set

$$h_j = \frac{1}{\lambda + j}. \quad (12.5)$$

We interpolate  $Y(x_j)$  as a formal/asymptotic power series in  $h_j$  and evaluate at  $h = 0$ .

**Theorem 12.2** (Phase-locked Richardson weights). *The Lagrange weights for evaluation at  $h = 0$  from  $h_0, \dots, h_r$  are*

$$\omega_{r,j}(\lambda) = \prod_{\substack{0 \leq k \leq r \\ k \neq j}} \frac{-h_k}{h_j - h_k} = \frac{(-1)^{r-j}}{r!} \binom{r}{j} (\lambda + j)^r. \quad (12.6)$$

*They satisfy*

$$\sum_{j=0}^r \omega_{r,j} = 1, \quad \sum_{j=0}^r \omega_{r,j} (\lambda + j)^{-m} = 0 \quad (1 \leq m \leq r). \quad (12.7)$$

*Proof.* Since

$$h_j - h_k = \frac{k - j}{(\lambda + j)(\lambda + k)},$$

each factor in the Lagrange product equals  $(\lambda + j)/(j - k)$ . Therefore

$$\omega_{r,j} = \frac{(\lambda + j)^r}{\prod_{k \neq j} (j - k)} = \frac{(-1)^{r-j}}{j!(r-j)!} (\lambda + j)^r.$$

The moment identities follow either from polynomial interpolation or from the fact that an  $r$ th finite difference annihilates every polynomial of degree below  $r$ .  $\square$

The first three levels are

$$(\omega_{1,0}, \omega_{1,1}) = (-\lambda, \lambda + 1), \quad (12.8)$$

$$(\omega_{2,0}, \omega_{2,1}, \omega_{2,2}) = \left( \frac{\lambda^2}{2}, -(\lambda + 1)^2, \frac{(\lambda + 2)^2}{2} \right), \quad (12.9)$$

$$(\omega_{3,0}, \omega_{3,1}, \omega_{3,2}, \omega_{3,3}) = \left( -\frac{\lambda^3}{6}, \frac{(\lambda + 1)^3}{2}, -\frac{(\lambda + 2)^3}{2}, \frac{(\lambda + 3)^3}{6} \right). \quad (12.10)$$

## 12.3 All residual moments

Let  $h_n(a_0, \dots, a_r)$  denote the complete homogeneous symmetric polynomial, characterized by

$$\prod_{j=0}^r (1 - a_j z)^{-1} = \sum_{n=0}^{\infty} h_n(a_0, \dots, a_r) z^n. \quad (12.11)$$

**Formalization status.** The finite formal-power-series content of Equation (12.11) is now Lean-backed. For an arbitrary finite alphabet, `CompleteHomogeneousGenerating.lean` defines the series whose coefficient of degree  $n$  is the corresponding complete homogeneous evaluation, proves the adjoining-variable geometric recurrence over every commutative semiring, and, over every commutative ring, proves both the denominator-cleared finite-product identity and identification with `PowerSeries.invOfUnit` of the denominator product. This is a purely formal coefficient identity: it does not evaluate the series at an analytic argument, prove convergence, take a logarithm, or establish the Bell/generalized-harmonic formula (12.15) below.

**Theorem 12.3** (Exact inverse-power moments). *For  $s \geq 1$ ,*

$$\sum_{j=0}^r \omega_{r,j} (\lambda + j)^{-(r+s)} = \frac{(-1)^r}{(\lambda)_{r+1}} h_{s-1} \left( \frac{1}{\lambda}, \frac{1}{\lambda+1}, \dots, \frac{1}{\lambda+r} \right), \quad (12.12)$$

where  $(\lambda)_{r+1} = \lambda(\lambda+1) \cdots (\lambda+r)$  is the rising factorial. In particular,

$$\sum_{j=0}^r \omega_{r,j} (\lambda + j)^{-(r+1)} = \frac{(-1)^r}{(\lambda)_{r+1}}. \quad (12.13)$$

*Proof.* Substituting Equation (12.6) reduces the left-hand side to

$$\frac{1}{r!} \sum_{j=0}^r (-1)^{r-j} \binom{r}{j} (\lambda + j)^{-s}.$$

For  $s = 1$ , partial fractions of  $1/(\lambda)_{r+1}$  give Equation (12.13). Repeated differentiation with respect to  $\lambda$ , or multiplication of the generating function of reciprocal powers, yields Equation (12.12). Equivalently, compare coefficients in

$$\frac{1}{(\lambda - z)_{r+1}} = \frac{1}{(\lambda)_{r+1}} \prod_{j=0}^r \left( 1 - \frac{z}{\lambda + j} \right)^{-1}.$$

□

Define generalized harmonic sums

$$H_k^{(r)}(\lambda) = \sum_{j=0}^r (\lambda + j)^{-k}. \quad (12.14)$$

Then the complete homogeneous polynomial in Equation (12.12) has the Bell representation

$$h_n \left( \frac{1}{\lambda}, \dots, \frac{1}{\lambda+r} \right) = \frac{1}{n!} \mathcal{B}_n \left( 0! H_1^{(r)}, 1! H_2^{(r)}, \dots, (n-1)! H_n^{(r)} \right). \quad (12.15)$$

This is the second exact Bell-polynomial bridge in the report: the first exponentiated a log error; the second resolves all residual moments of the Lambert extrapolator.

**Formalization status.** The finite identity (12.15) is now Lean-backed without using an analytic logarithm or an infinite-series convergence argument. `CompleteHomogeneousBell.lean` defines `completeHomogeneousPowerSum` and `completeHomogeneousBellInput`; its division-free theorem `bellComplete_completeHomogeneousBellInput` is the generic finite Bell identity, and `completeHomogeneousEvalOn_eq_factorialNormalize_completeBellPolynomial` is its normalized form over a commutative  $\mathbb{Q}$ -algebra. `LambertPhaseLockedBell.lean` specializes this result through `shiftedReciprocalPowerSum`, `shiftedReciprocalBellInput`, and `completeHomogeneousEvalOn_shiftedReciprocal_eq_bell`, and proves both `sum_shiftedReciprocalLagrangeWeight_mul_invPow_card_add_eq_bell` and the report-shaped  $s \geq 1$  theorem `sum_shiftedReciprocalLagrangeWeight_mul_invPow_eq_bell`. The shifted reciprocal definitions use total field inverses; the Bell specialization imposes no positivity hypothesis, while the exact weighted moment interfaces assume characteristic zero. Finally, `FabiusLambertPhaseExtractionBell.lean` proves the exact residual rewrites `fabiusPhaseLockedResidualTerm_eq_bell` and `fabiusPhaseLockedResidualPartialSum_eq_sum_bell`. These are finite algebraic identities; they assert no convergence for the infinite residual series and add no new asymptotic estimate.



## 12.4 Extraction of the periodic endpoint factor

Define

$$\widehat{\Psi}_r(\lambda) = \sum_{j=0}^r \omega_{r,j}(\lambda) Y(x_j(\lambda)). \quad (12.16)$$

**Theorem 12.4** (Phase-locked extraction of  $\Psi$ ). *Assume the all-orders expansion Equation (11.1) holds uniformly along a fixed compact phase set. Then, as  $\lambda \rightarrow \infty$  with its fractional part fixed,*

$$\widehat{\Psi}_r(\lambda) - \Psi(\lambda) \sim \frac{(-1)^r}{(\lambda)_{r+1}} \sum_{s=0}^{\infty} A_{r+1+s}(\lambda) h_s \left( \frac{1}{\lambda}, \dots, \frac{1}{\lambda+r} \right) \quad (12.17)$$

$$= \frac{(-1)^r A_{r+1}(\lambda)}{(\lambda)_{r+1}} + O(\lambda^{-r-2}). \quad (12.18)$$

*In particular, the first  $r$  inverse- $\lambda$  corrections are cancelled exactly. Here the symbol  $\sim$  in (12.17) has its finite-order Poincaré meaning: for every fixed  $S \geq 0$ , subtracting the terms  $s = 0, \dots, S$  leaves  $O_r(\lambda^{-r-S-2})$ . It does not assert that the displayed infinite series converges for any fixed  $\lambda$ .*

*Proof.* At the phase-locked nodes, periodicity gives

$$Y(x_j) = \Psi(\lambda) + \sum_{m \geq 1} A_m(\lambda) (\lambda + j)^{-m}.$$

The constant survives because the weights sum to one. Terms  $1 \leq m \leq r$  vanish by Equation (12.7). Write  $m = r + 1 + s$  and apply theorem 12.3. A small uniformity point is essential: the absolute weight sum is  $O_r(\lambda^r)$ . To obtain the expansion through residual index  $s = S$ , apply the input all-orders theorem through order  $2r + S + 1$ . Its transformed remainder is then  $O_r(\lambda^{-r-S-2})$ , while the finitely many undisplayed transformed coefficients start at the same order. This proves the displayed Poincaré expansion despite the growing weights.  $\square$

**Lean formalization status.** The logarithmic theorem, in the finite-order sense just specified, is now compiled. The module `FabiusLambertPhaseLockedPullback.lean` proves `tendsto_lambertPhaseLockedNode_smallArgument` and the arbitrary-order single-node estimate `log_fabius_phaseLockedNode_sub_WikipediaLambertExpansion_isBigO`. The modules `CompleteHomogeneousAsymptotics.lean` and `LambertReciprocalAsymptotics.lean` provide, respectively, `completeHomogeneousEvalOn_isBigO_pow` and `sum_norm_shiftedReciprocalLagrangeWeight_isBigO_pow_atTop`, together with the balanced product estimate `shiftedReciprocalLagrangeWeight_mul_invPow_isBigO_atTop`. These control the complete-homogeneous residual and the polynomially growing row.

The report-facing definitions in `FabiusLambertPhaseExtraction.lean` are `fabiusPhaseLockedReducedSample`, `fabiusPhaseLockedPeriodicEstimator`, `fabiusPhaseLockedResidualTerm`, and `fabiusPhaseLockedResidualPartialSum`. Under the dictionary  $\Psi = \text{negativeLaplacePsi}$  and  $A_m = \text{fabiusSaddleLogCoefficient } m$ , the declaration `fabiusPhaseLockedPeriodicEstimator_sub_residual_isBigO` proves the finite truncation of (12.17); `fabiusPhaseLockedPeriodicEstimator_sub_periodic_isBigO` is its unsubtracted  $O(\lambda^{-r-1})$  specialization; `fabiusPhaseLockedPeriodicEstimator_sub_firstOmitted_isBigO` is exactly (12.18), and `fabiusPhaseLockedPeriodicEstimator_tendsto_periodicAlong` gives the fixed-phase limit along  $\lambda = a + n$ . The moving-target Big- $O$  statements hold on unrestricted real  $\lambda \rightarrow +\infty$ , so the fixed-phase result is a corollary rather than an additional uniformity assumption.

There are two harmless indexing conventions. Lean’s residual parameter  $S_L$  counts the first  $S_L$  terms, whereas the report’s  $S$  is the last retained index; hence  $S_L = S + 1$ . Lean also writes the input partial sum as range  $N$ , so its remainder has order  $\lambda^{-N}$  and its highest retained coefficient is  $N - 1$ . The internal choice  $N = 2r + S_L + 1$  is therefore  $N = 2r + S + 2$ , with highest retained coefficient  $2r + S + 1$ , exactly as used in the proof above. All these theorems keep  $r$  and  $S$  fixed; no growing-order uniformity is claimed. Their natural-number statements include the singleton row  $r = 0$  and the empty residual truncation  $S_L = 0$ .

Exponentiating produces a multiplicative estimator of the periodic amplitude:

$$\widehat{A}_r(\lambda) = \exp(\widehat{\Psi}_r(\lambda)) = \prod_{j=0}^r \left( \frac{F(x_j)}{\exp(\mathcal{W}_{\text{MSE}}(x_j))} \right)^{\omega_{r,j}(\lambda)}. \quad (12.19)$$

Its relative-error coefficients are again complete Bell polynomials applied to the logarithmic coefficients in Equation (12.17). The identity (12.19) is an immediate mathematical exponentiation, but this multiplicative estimator and its Bell relative-error hierarchy are not part of the closed Lean phase-extraction tranche.

### 12.5 Coefficient extraction beyond $\Psi$

The same nodes can estimate  $A_1, A_2, \dots$  by differentiating the interpolating polynomial at  $h = 0$ . If  $\ell_j(h)$  is the Lagrange basis for  $h_j$ , then

$$\widehat{A}_s^{(r)}(\lambda) = \frac{1}{s!} \sum_{j=0}^r \ell_j^{(s)}(0) Y(x_j), \quad 0 \leq s \leq r, \quad (12.20)$$

with  $\widehat{A}_0^{(r)} = \widehat{\Psi}_r$ . For  $s = 1$ ,

$$\ell_j'(0) = -\omega_{r,j} \sum_{\substack{0 \leq k \leq r \\ k \neq j}} (\lambda + k). \quad (12.21)$$

Higher derivative weights are elementary symmetric polynomials in the numbers  $-(\lambda + k)$  and can be generated by Bell polynomials. Thus a single phase-locked sample block can recover a triangular set of periodic coefficient functions. These higher derivative extractors likewise remain a formalization target.

## 13 Bernoulli-polynomial geometry and conditioning of the Lambert lattice

Unlike the geometric  $q$ -weights, the phase-locked weights grow polynomially in  $\lambda$ :

$$\kappa_r^L(\lambda) := \sum_{j=0}^r |\omega_{r,j}(\lambda)| = \frac{1}{r!} \sum_{j=0}^r \binom{r}{j} (\lambda + j)^r. \quad (13.1)$$

Equivalently, if  $J \sim \text{Binomial}(r, 1/2)$ ,

$$\kappa_r^L(\lambda) = \frac{2^r}{r!} \mathbb{E}(\lambda + J)^r \sim_{\lambda \rightarrow \infty, r \text{ fixed}} \frac{2^r}{r!} \lambda^r. \quad (13.2)$$

Thus high precision is essential. This is not a defect of the derivation: extrapolating from  $h_j \asymp 1/\lambda$  to the remote point  $h = 0$  is intrinsically ill-conditioned.

The binomial moments can be expanded through Stirling numbers:

$$\sum_{j=0}^r \binom{r}{j} j^k = \sum_{\ell=0}^k \left\{ \begin{matrix} k \\ \ell \end{matrix} \right\} r^\ell 2^{r-\ell}. \quad (13.3)$$

Here  $r^\ell = r(r-1) \cdots (r-\ell+1)$  is a falling factorial. For unweighted arithmetic-node diagnostics, Bernoulli polynomials give the exact power sums

$$\sum_{j=0}^r (\lambda + j)^p = \frac{B_{p+1}(\lambda + r + 1) - B_{p+1}(\lambda)}{p + 1}. \quad (13.4)$$

Equation (13.4) is useful for scaling, centering, and constructing orthogonalized polynomial bases on the Lambert node block. It also makes explicit why Bernoulli polynomials are the natural arithmetic-lattice companion to the Bernoulli numbers already present in the sinc-tail coefficients.

### 13.1 A stable evaluation strategy

For practical extraction of  $\Psi$ :

1. compute  $\lambda$  and  $x_j$  at precision exceeding the target by roughly  $\log_{10} \kappa_r^L(\lambda)$  guard digits;
2. evaluate  $Y(x_j)$  with the same phase convention and main term;

3. use barycentric interpolation in  $h_j$  rather than explicitly summing large alternating monomials;
4. repeat at successive  $\lambda$  with the same fractional part and verify the predicted  $\lambda^{-(r+1)}$  decay;
5. compare  $r$  and  $r + 1$  transforms as an a posteriori diagnostic, while remembering that no sign bracket is presently available because the signs of  $A_m(\lambda)$  are not known globally.

## 14 Unified interpretation, formalization path, and research program

### 15 Two extrapolation geometries

The two constructions are instances of one principle: choose a coordinate in which the defect has a power expansion, then interpolate to the zero-defect point.

Table 25: Comparison of the two extrapolation geometries.

| Feature               | Sinc-tail transform             | Lambert endpoint transform                |
|-----------------------|---------------------------------|-------------------------------------------|
| Defect coordinate     | $z_N = q^{N+1}$                 | $h = 1/\lambda$                           |
| Node family           | $z_N q^j$                       | $(\lambda + j)^{-1}$                      |
| Invariant structure   | lacunary tail coefficients      | periodic phase coefficients               |
| Interpolation algebra | Gaussian $q$ -binomial          | ordinary binomial/finite difference       |
| Leading cancellation  | powers $z, \dots, z^r$          | powers $h, \dots, h^r$                    |
| Coefficient source    | Bernoulli numbers / $\zeta(2m)$ | periodic saddle jets $A_m$                |
| Reconstruction        | exponential Bell polynomials    | complete homogeneous and Bell polynomials |
| Conditioning          | uniformly bounded in $r$        | grows like $2^r \lambda^r / r!$           |
| Certification         | exact alternating brackets      | asymptotic order, no global sign yet      |

The sinc scheme is unusually favorable because the geometric-node Lagrange weights have bounded total variation. The Lambert scheme pays for exact phase locking with growing weights, but it solves a different problem: extracting a nonconstant periodic amplitude that ordinary dyadic extrapolation would mix across phases.

## 16 Consequences for Fabius and Rvachev computations

### 16.1 Certified Fourier evaluation

Using Equation (2.2), compute the first factor exactly and bracket  $P_2(\pi\xi)$  by consecutive transformed levels. This produces a certified enclosure for  $|\widehat{u\mathfrak{p}}(\xi)|$  away from zeros. Because the work is done in logarithms, the method remains usable deep in the log-normal decay regime where direct products underflow.

### 16.2 Peak localization and envelope testing

Use the derivative transform in Equation (7.4) inside Equation (7.6). The resulting roots can test fine envelope conjectures without contaminating the comparison by truncation error. In particular, a claimed subleading Fourier envelope should be compared against a product value whose numerical error is below the proposed correction term; the bracket theorem makes that requirement checkable.

### 16.3 Independent extraction of the endpoint oscillation

The phase-locked transform Equation (12.16) gives a second way to recover  $\Psi$ : not from its Fourier/Gamma-zeta representation, but from  $F$  itself. Agreement between the two routes would be a stringent validation of constants, branches, and phase conventions in the endpoint asymptotic.

## 16.4 A bridge from exact dyadic values to asymptotics

The repository already relates exact dyadic values to a phase-locked Lambert coordinate on fixed-numerator rays. The new construction suggests a complementary experiment:

1. use exact or high-precision values of  $F$  at nearby phase-locked non-dyadic nodes  $x_j$ ;
2. extract  $\Psi$  and several  $A_m$ ;
3. compare them with the periodic coefficient functions inferred from exact dyadic sequences whose phases become dense;
4. use the discrepancy to isolate the dyadic-arithmetic contribution from the universal saddle contribution.

This does not by itself prove the exact-to-asymptotic bridge, but it supplies a sharply conditioned diagnostic coordinate for that problem.

## 16.5 Base- $b$ generalized up-functions

Theorems 4.2–6.4 hold for every real  $b > 1$ . If a compactly supported refinement function has Fourier transform  $\prod_{k \geq 1} \text{sinc}(t/b^k)$  or a finite product times this tail, the same acceleration and bracketing apply with  $q = b^{-2}$ . This broadens the method beyond the dyadic Rvachev function and may be useful for nonbinary infinite convolutions.

## 17 Lean formalization blueprint

The sinc-side results are well suited to formalization because their analytic and algebraic components separate cleanly.

### 17.1 Suggested definitions and compiled Lambert crosswalk

1. `lacunarySincProd b N t`: finite product over  $k = 1, \dots, N$ .
2. `lacunarySincProdInf b t`: infinite product, with convergence proved from  $\sum t^2 b^{-2k}$ .
3. `qRichardsonWeight q r j`: the closed form Equation (5.2).
4. `qRichardsonLog q r f N`: weighted sum of  $f(N + j)$ .
5. `lambertPhaseLockedNode lambda j`: the compiled definition  $(\lambda + j)2^{-(\lambda+j)}$ .
6. `shiftedReciprocalLagrangeWeight lambda r j`: the compiled Lagrange definition underlying Equation (12.6).
7. `shiftedReciprocalPowerSum lambda r k`: the compiled generalized power sum  $\sum_{j=0}^r (\lambda + j)^{-k}$ .
8. `shiftedReciprocalBellInput lambda r`: the compiled factorially weighted Bell input attached to that reciprocal alphabet.
9. `fabiusPhaseLockedReducedSample` and `fabiusPhaseLockedPeriodicEstimator`: the compiled forms of  $Y(x_j)$  and  $\hat{\Psi}_r$ .
10. `fabiusPhaseLockedResidualTerm` and `fabiusPhaseLockedResidualPartialSum`: the compiled individual and finite residual expressions in (12.17).

### 17.2 Algebraic theorem chain

The absolute-weight, Lambert-weight/cancellation, reciprocal-moment, finite complete-homogeneous generating-product, and Bell/generalized-harmonic tasks in the post-snapshot plan are now discharged. The generic finite-alphabet bridge is `bellComplete_completeHomogeneousBellInput`, with normalized report form `completeHomogeneousEvalOn_eq_factorialNormalize_completeBellPolynomial`. Its shifted reciprocal specialization is `completeHomogeneousEvalOn_shiftedReciprocal_eq_bell`; the two exact moment interfaces are `sum_shiftedReciprocalLagrangeWeight_mul_invPow_card_add_eq_bell` and `sum_shiftedReciprocalLagrangeWeight_mul_invPow_eq_bell`. The extraction layer then rewrites an individual term and every finite partial sum through `fabiusPhaseLockedResidualTerm_eq_bell` and `fabiusPhaseLockedResidualPartialSum_eq_sum_bell`. This closes the finite algebraic

packaging only; it neither sums the infinite residual series nor strengthens the analytic phase-extraction estimates.

**Post-snapshot algebraic progress.** The later module `GeometricLagrange.lean` now formalizes Lagrange evaluation weights for arbitrary distinct nodes, polynomial and module-valued exactness, the first omitted monomial moment, and geometric-node specializations. The compiled bridge `GeometricQBinomialLagrange.lean` now supplies the arbitrary-ratio algebra needed here. Its `geometricQBinomialWeightNumerator` is zero for  $j > r$ , and for  $j \leq r$  is definitionally written as  $(-1)^{r-j} q^{\binom{r-j+1}{2}} \text{gaussianBinomial } q \, r \, (r-j)$ . The division-free reflection theorem `gaussianBinomial_symm` from `FiniteQBinomialCore.lean` proves the standard forward-index formula, globally after zero extension, as `geometricQBinomialWeightNumerator_eq_forward`. The core also proves functoriality `map_gaussianBinomial`, both  $q$ -Pascal recurrences, finite  $q$ -Pochhammer shift and concatenation laws, both denominator-free product–Gaussian factorial identities, and the finite  $q$ -binomial theorem. The theorem `reversed_finite_qBinomial_theorem` identifies the corresponding sum with  $\prod_{k < r} (z - q^{k+1})$  without division. Under `Set.InjOn` of  $k \mapsto q^k$  on  $\{0, \dots, r\}$ , `geometricLagrangeWeight_eq_gaussianBinomial_div` normalizes this numerator by  $(q; q)_r$ , while `sum_geometricLagrangeWeight_mul_pow_eq_gaussianBinomial` gives, for every  $m > 0$ ,

$$\sum_{j=0}^r \lambda_{r,j}(q) (q^j)^m = (-1)^r q^{r(r+1)/2} \text{gaussianBinomial } q \, (m-1) \, r.$$

Thus the weight formula and every positive residual moment, including the  $q = 1/4$  specialization whenever the nodes are injective, are no longer formalization obligations. In particular, the report’s displayed  $\begin{bmatrix} r \\ j \end{bmatrix}_q$  is now the literal forward-index theorem, while the reversed form remains available for the product proof. At  $q = 1/2$ , `FabiusDiscreteLimitToeplitz.lean` identifies the same weights with the half-base Toeplitz row and retains its all-degree reverse moments and exact total variation. The later rational geometric-moment layer also proves the finite condition-number identity [Equation \(5.6\)](#); its infinite limit and decimal evaluation, together with the concrete sinc-product error, quantitative remainder, and uniform convergence claims, remain frontier.

The operator-level coefficient algebra is now reusable as well. `FinitePowerSeriesFilter.lean` defines  $\sum_{j \in S} w_j f(a_j X)$  over an arbitrary commutative semiring and proves that its degree- $m$  coefficient is multiplied by  $\sum_j w_j a_j^m$ ; it also proves constant-term, exactness, rescaling, and coefficient-map laws. `GeometricPowerSeriesFilter.lean` specializes to  $a_j = q^j$ , proves cancellation through the exactness order, and gives the Gaussian residual multiplier in every positive degree. Its `geometricLagrangeSeriesFilter_eq_residual_mk` records the degree-zero mass boundary and all positive degrees in one formal-series identity. `AnalyticSeriesFilter.lean` now supplies the direct evaluated-series counterpart: under unconditional summability, the filter has an exact head/tail split and its geometric-Lagrange form has the complete shifted Gaussian tail. Only nonzero-weight samples require unweighted summability, so zero weights are handled sharply. The later `RvachevQBinomialFilter.lean` connects the named even-moment coefficients to the actual infinite Rvachev sinc product and gives the corresponding entire filter, assumption-free at quarter base. It does not connect the present logarithmic tail coefficients to the finite lacunary-product truncations. Conditionally-only boundary series, convergence of those truncations, signs, bounds, brackets, asymptotics, and uniform remainder control remain governed by the analytic frontier stated below.

At post-snapshot checkpoint `1e761ed77583e9870ceeaeb2c74ac824094f0f3f`, the algebraic route to the same residual formula was strengthened and factored through complete homogeneous polynomials. The five reusable declarations added to `CompleteHomogeneous.lean` are `completeHomogeneousEval_smul`, `completeHomogeneousEval_option_zero`, `completeHomogeneousEval_fin_succ`, `completeHomogeneousEvalOn_range`, and `completeHomogeneousEvalAt_zero`. Over every commutative semiring, `GeometricCompleteHomogeneous.lean` proves

$$h_n(1, q, \dots, q^r) = \begin{bmatrix} n+r \\ r \end{bmatrix}_q, \quad h_n(c, cq, \dots, cq^r) = c^n \begin{bmatrix} n+r \\ r \end{bmatrix}_q$$

as `completeHomogeneousEval_geometric` and `completeHomogeneousEval_scaled_geometric`; the range-indexed form is `completeHomogeneousEvalOn_range_pow_eq_gaussianBinomial`.

The later module `CompleteHomogeneousGenerating.lean` packages these evaluations into a formal power series. Its coefficient and empty-alphabet interfaces are `coeff_completeHomogeneousGeneratingSeriesOn` and `completeHomogeneousGeneratingSeriesOn_empty`. The theorem `completeHomogeneous`

neousGeneratingSeriesOn\_insert is the adjoining-variable geometric recurrence over a commutative semiring. Over a commutative ring, one\_sub\_mul\_completeHomogeneousGeneratingSeriesOn\_inse rt clears one linear denominator, prod\_one\_sub\_mul\_completeHomogeneousGeneratingSeriesOn clears the whole finite product, and completeHomogeneousGeneratingSeriesOn\_eq\_invOfUnit\_prod identifies the coefficient series with the canonical formal inverse of that product. These declarations close the formal finite-product content of (12.11), including empty alphabets, repeated values, zero divisors, and nilpotents; they assert no analytic convergence or Bell/generalized-harmonic identity.

The separate module CompleteHomogeneousBell.lean supplies that finite Bell identity. Its semiring-cleared theorem bellComplete\_completeHomogeneousBellInput avoids division, while completeHomogeneousEvalOn\_eq\_factorialNormalize\_completeBellPolynomial gives the factorially normalized report form over a commutative  $\mathbb{Q}$ -algebra. Thus the preceding disclaimer remains specific to the generating-series module: neither module evaluates a formal series at an analytic argument or proves convergence.

Over every commutative ring, the hypotheses of sum\_weight\_mul\_scaled\_geometric\_pow\_succ\_add and sum\_weight\_mul\_scaled\_geometric\_pow\_of\_pos impose exactness directly on the displayed nodes  $cq^j$ , not first on  $q^j$ . Thus zero and zero-divisor scales require no cancellation. Their unscaled companions are sum\_weight\_mul\_geometric\_pow\_succ\_add and sum\_weight\_mul\_geometric\_pow\_of\_pos. The field-valued distinct-node consequences are sum\_geometricLagrangeWeight\_mul\_pow\_succ\_add, sum\_geometricLagrangeWeight\_mul\_pow\_of\_pos, sum\_geometricLagrangeWeight\_mul\_scaled\_geometric\_pow\_of\_pos, and sum\_geometricLagrangeWeight\_mul\_shifted\_pow\_of\_pos. In particular, the generic scaled positive moment is exactly

$$\sum_{j=0}^r w_j (cq^j)^m = c^m (-1)^r q^{\binom{r+1}{2}} \text{gaussianBinomial } q (m-1) r \quad (m > 0),$$

provided the same row reproduces evaluation at zero through degree  $r$  on those scaled nodes. These sixteen declarations use no division at the commutative-semiring or commutative-ring layers; finite-node injectivity enters only in the Lagrange field corollaries.

The later rational layer is implemented in

```
GeometricRichardson.lean
GeometricLagrangeWeights.lean
GeometricLagrangeQBinomial.lean
GeometricLagrangeQMoments.lean.
```

Together these modules identify the finite geometric Richardson polynomial with its Lagrange row, prove the  $q$ -Pochhammer and Gaussian-binomial weight formulas, and compute every rational  $q$ -moment under the exact nonzero-base and nonvanishing-denominator hypotheses. For  $0 < q < 1$ , they establish every residual sign and the finite exact  $\ell^1$  norm. The declarations quarterGeometricLagrangeQMoment\_eq\_residual\_qBinomial and sum\_abs\_quarterGeometricLagrangeWeight\_eq\_qPochhammer\_ratio provide the report-facing  $q = 1/4$  specializations.

Independently, ExponentialPartition.lean and ExponentialBell.lean identify the universal weighted-partition formula for an exponential coefficient with SaddleExpansion.expCoeff over every commutative  $\mathbb{Q}$ -algebra. The later MomentCumulantAlgebra.lean factorially normalizes the same engine into named complete-Bell and inverse moment-cumulant transforms, with the classical Bell recurrence and normalized inverse laws. In particular, the coefficient mechanism behind (12.15) is now instantiated in CompleteHomogeneousBell.lean: the division-free complete-Bell identity uses the factorially weighted finite power sums, and completeHomogeneousEvalOn\_eq\_factorialNormalize\_completeBellPolynomial gives the normalized formula over a commutative  $\mathbb{Q}$ -algebra. No analytic logarithm is used. The geometric principal specialization and all finite geometric residual moments are therefore no longer formalization obligations.

Finally, LambertPhaseLockedRichardson.lean proves the exact lower-Lambert phase locking, factorial and binomial reciprocal-grid weights, their cancellation moments, and the report-shaped all-order identity sum\_shiftedReciprocalLagrangeWeight\_mul\_invPow\_eq\_completeHomogeneous. Together with CompleteHomogeneousGenerating.lean, this closes the finite formal generating-product layer. Four further modules close the logarithmic finite-order phase-extraction layer: FabiusLambertPhaseLockedPullback.lean pulls the uniform endpoint remainder to each fixed node; CompleteHomogeneousAsymptotics.lean bounds the homogeneous residuals; LambertReciprocalAsymptotics.lean controls



the growing reciprocal row; and `FabiusLambertPhaseExtraction.lean` combines these estimates with the exact moments. Its declarations `fabiusPhaseLockedPeriodicEstimator_sub_residual_isBigO`, `fabiusPhaseLockedPeriodicEstimator_sub_periodic_isBigO`, `fabiusPhaseLockedPeriodicEstimator_sub_firstOmitted_isBigO`, and `fabiusPhaseLockedPeriodicEstimator_tendsto_periodicAlong` give the full finite Poincaré truncations, the raw error order, the leading formula, and the fixed-phase limit, respectively.

The downstream module `LambertPhaseLockedBell.lean` identifies the shifted reciprocal Bell input and proves both exact Bell forms of the reciprocal-grid moments. The module `FabiusLambertPhaseExtractionBell.lean` then proves `fabiusPhaseLockedResidualTerm_eq_bell` and `fabiusPhaseLockedResidualPartialSum_eq_sum_bell`. Consequently, the finite Bell/generalized-harmonic presentation (12.15) and its exact finite residual rewrites are closed.

Outside this closed tranche remain convergence of the infinite residual series, the multiplicative estimator’s Bell relative-error hierarchy, the higher derivative coefficient extractors, Lambert residual sign/bracketing results, and bounds uniform when  $r$  or the residual depth grows with  $\lambda$ . The transformed sinc-tail identities, analytic Richardson bracketing, and infinite condition-number limits are separate sinc-side frontier obligations.

### 17.3 Remaining sinc-side analytic theorem chain

The analytic side can then proceed as follows:

1. establish the log-sinc power series on  $(-\pi, \pi)$ ;
2. justify the double-series interchange for the omitted tail;
3. derive the exact transformed error by finite linearity;
4. prove positivity of every factor in Equation (6.5);
5. deduce bracketing and the explicit majorant;
6. add termwise differentiation on compact zero-free intervals.

The endpoint chain is independent of this remaining sinc-side list and is now complete at fixed finite order. Its exact phase and endpoint coefficients come from `FabiusFullAsymptoticExpansion.lean`; its interpolation algebra comes from `LambertPhaseLockedRichardson.lean`; and the four-module pullback/asymptotic/extraction tranche listed above combines the growing row with the uniform saddle remainder at the strength asserted in theorem 12.4. The formal result is the family of finite Big- $O$  truncations specified after that theorem, not convergence of the displayed infinite residual series.

## 18 Open problems and next frontier steps

### 18.1 Optimal order under finite precision

For exact input logs, Equation (6.12) improves supergeometrically with  $r$ , while Equation (5.6) remains bounded. In floating-point arithmetic, however, the truncations  $P_{N+j}$  are themselves computed recursively and share correlated rounding errors. A sharp forward-error theorem should combine:

$$\text{analytic error} \asymp q^{r(r+1)/2} a^{r+1} \quad \text{with} \quad \text{roundoff} \lesssim \kappa_r(q)u.$$

Because  $\kappa_r(q)$  converges, the optimal  $r$  may be much larger than in ordinary Richardson extrapolation.

### 18.2 Monotonicity within each bracket parity

The sign theorem places every even level above the limit in modulus and every odd level below it, but it does not order two levels of the same parity. In fact, such an ordering is false over the full parameter range  $b > 1$ . To see this directly, take

$$q = \frac{4}{5}, \quad b = q^{-1/2} = \frac{\sqrt{5}}{2}, \quad N = 0, \quad \rho_0 = \pi\sqrt{\frac{99}{100}}, \quad t = b\rho_0 = \pi\sqrt{\frac{99}{80}}.$$

This is admissible because  $\rho_0 < \pi$ , and  $P_{b,0} = 1$  is the nonzero empty product. Substitution in the absolutely convergent series (6.5), with  $a = (\rho_0/\pi)^2 = 99/100$ , gives

| $r$            | 0             | 1              | 2             | 3               |
|----------------|---------------|----------------|---------------|-----------------|
| $E_{b;0,r}(t)$ | 13.5220607386 | −12.9320597704 | 16.5469385988 | −14.7912158217. |

Thus  $E_{b;0,2} > E_{b;0,0} > 0$  and  $E_{b;0,3} < E_{b;0,1} < 0$ : both proposed same-parity monotonicities fail. These numbers are reproducible by summing the terms through  $m = 5000$  in (6.5). The same majorization used in [theorem 6.4](#) bounds the discarded tail, for each  $0 \leq r \leq 3$ , by

$$\frac{q^{r(r+1)/2} \zeta(2)}{(q; q)_r (1-q)(5001)} \frac{a^{5001}}{1-a} < 2 \times 10^{-22}.$$

The correctly parameterized open question is therefore: for which pairs  $(q, a) \in (0, 1)^2$ , and for which levels  $s$ , do

$$E_{2s}(q, a) > E_{2s+2}(q, a) > 0, \quad E_{2s+1}(q, a) < E_{2s+3}(q, a) < 0,$$

where  $E_r(q, a)$  denotes the right-hand side of (6.5) after replacing  $(\rho_N/\pi)^2$  by  $a$ ? No nonempty parameter regime for these additional inequalities is asserted here. Whenever they do hold, consecutive same-parity levels sharpen the certified brackets of [corollary 6.2](#).

### 18.3 Global continuation across zeros

The logarithmic construction is naturally local to a zero-free component. A global formulation could factor all explicitly known zeros from the truncations and extrapolate the analytic nonvanishing remainder. Such a theorem would give uniform certified evaluation across large frequency intervals, including neighborhoods of high-multiplicity dyadic zeros.

### 18.4 Direct formulas for the phase coefficients

The transformed moments [Equation \(12.12\)](#) show exactly how  $A_{r+1}, A_{r+2}, \dots$  enter the extraction error. This invites an inverse problem: compute  $\widehat{\Psi}_r$  and  $\widehat{A}_s^{(r)}$  at many phases, then identify their Fourier coefficients in terms of the Gamma–zeta data already present in the endpoint theory.

### 18.5 A sign theory for Lambert corrections

A sign pattern for  $A_m(\lambda)$  would turn the phase-locked transform into a certified bracket analogous to [corollary 6.2](#). Even a sign theorem for  $A_1$  or  $A_2$  on a substantial phase interval would be useful.

### 18.6 Thue–Morse arithmetic of transformed products

For  $q = 1/4$ , every exponent in [Equations \(6.17\) to \(6.19\)](#) is rational. Raising to a common denominator converts the approximant to a rational monomial identity in finite Thue–Morse sums. It is natural to ask whether the numerators and denominators exhibit systematic 2-adic cancellation analogous to the exact dyadic arithmetic of  $F$ .

## 19 Conclusion

The repository already contains the ingredients of a remarkable analytic-combinatorial machine: Thue–Morse cancellation, Gaussian interpolation, exact dyadic arithmetic, infinite sinc products, Bernoulli cumulants, Bell reconstruction, and Lambert- $W$  endpoint geometry. The new results here arise by changing *what* is extrapolated and *where* the nodes are placed.

For the Fourier product, extrapolating logarithms exposes an exact positive power series in the geometric tail coordinate. Gaussian- $q$ -binomial weights annihilate its first powers, and the remaining  $q$ -moments all have one sign. This yields high-order multiplicative approximants, exact alternating brackets, explicit bounds, bounded conditioning, Bell-polynomial relative errors, derivative acceleration, and a finite Thue–Morse realization.

For the endpoint asymptotic, moving by integer steps in the exact lower-Lambert coordinate freezes every periodic coefficient. Lagrange interpolation in reciprocal  $\lambda$  then isolates the periodic factor and its correction hierarchy. The residual moments are not opaque: they are complete homogeneous symmetric polynomials, hence Bell polynomials in generalized harmonic sums, while Bernoulli polynomials govern the arithmetic-node geometry.

The two mechanisms are complementary. One is globally algebraic in the truncation index and exceptionally stable; the other is phase-sensitive and designed to separate a periodic asymptotic invariant. Both are sufficiently explicit to support immediate symbolic checking, high-precision computation, and a staged Lean formalization.



## A Original audited TeX inventory

This appendix records the 57 distinct `.tex` paths included in the original repository-wide overlap audit. It is provenance for that audit, not a current corpus index: subsequent frontier reports increased the present tree to 67 TeX sources. Historical files are frozen versions and were treated as prior art even when their content has since been consolidated elsewhere.

### A.1 Living top-level and canonical sources (6)

1. docs/FabiusFunction\_Mathematical\_Glossary/FabiusFunction\_Mathematical\_Glossary.tex
2. docs/Fabius\_Function\_and\_Rvachev\_Up/Fabius\_Function\_and\_Rvachev\_Up.tex
3. docs/Non\_Elementarity\_of\_the\_Fabius\_Function/Non\_Elementarity\_of\_the\_Fabius\_Function.tex
4. docs/fabius\_lean\_walkthrough/fabius\_lean\_walkthrough.tex
5. docs/semi-formalized-research-frontiers/semi-formalized-research-frontiers.tex
6. docs/semi-formalized-research-frontiers/drafts/Thue\_Morse\_Formula\_Atlas/Thue\_Morse\_Formula\_Atlas.tex

### A.2 Living Rvachev Fourier-decay drafts (12)

7. docs/semi-formalized-research-frontiers/drafts/rvachev\_up\_fourier\_decay/Rvachev\_Up\_Fourier\_Decay-2/Rvachev\_Up\_Fourier\_Decay.tex
8. docs/semi-formalized-research-frontiers/drafts/rvachev\_up\_fourier\_decay/Rvachev\_Up\_Fourier\_Decay-3/Fourier\_decay\_of\_Rvachev\_up.tex
9. docs/semi-formalized-research-frontiers/drafts/rvachev\_up\_fourier\_decay/Rvachev\_Up\_Fourier\_Decay\_Comparative\_Audit/Rvachev\_Up\_Fourier\_Decay\_Comparative\_Audit.tex
10. docs/semi-formalized-research-frontiers/drafts/rvachev\_up\_fourier\_decay/Rvachev\_Up\_Fourier\_Decay\_Gentle\_Guide/Rvachev\_Up\_Fourier\_Decay\_Gentle\_Guide.tex
11. docs/semi-formalized-research-frontiers/drafts/rvachev\_up\_fourier\_decay/Rvachev\_Up\_Fourier\_Decay\_Second\_Wave\_Audit/Rvachev\_Up\_Fourier\_Decay\_Second\_Wave\_Audit.tex
12. docs/semi-formalized-research-frontiers/drafts/rvachev\_up\_fourier\_decay/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions.tex
13. docs/semi-formalized-research-frontiers/drafts/rvachev\_up\_fourier\_decay/rvachev\_up\_fourier\_decay-1/rvachev\_up\_fourier\_decay.tex
14. docs/semi-formalized-research-frontiers/drafts/rvachev\_up\_fourier\_decay/rvachev\_up\_fourier\_decay-4/rvachev\_up\_fourier\_decay.tex
15. docs/semi-formalized-research-frontiers/drafts/rvachev\_up\_fourier\_decay/rvachev\_up\_fourier\_decay-5/rvachev\_up\_fourier\_decay.tex
16. docs/semi-formalized-research-frontiers/drafts/rvachev\_up\_fourier\_decay/rvachev\_up\_fourier\_decay-6/rvachev\_up\_fourier\_decay.tex
17. docs/semi-formalized-research-frontiers/drafts/rvachev\_up\_fourier\_decay/rvachev\_up\_fourier\_decay-7/rvachev\_up\_fourier\_decay.tex
18. docs/semi-formalized-research-frontiers/drafts/rvachev\_up\_fourier\_decay/rvachev\_up\_fourier\_decay-8/rvachev\_up\_fourier\_decay.tex

### A.3 Bundled source-paper TeX (7)

19. docs/papers/AIF\_2017\_\_67\_2\_637\_0/AIF\_2017\_\_67\_2\_637\_0-corrected.tex
20. docs/papers/AIF\_2017\_\_67\_2\_637\_0/AIF\_2017\_\_67\_2\_637\_0.tex
21. docs/papers/arXiv-1502.06738v2/Lacunary\_products.tex
22. docs/papers/arXiv-1702.05442v1/09-Function.tex
23. docs/papers/arXiv-1702.06487v3/157-Arithmetic-v3.tex
24. docs/papers/arXiv-2211.13570v1/document.tex
25. docs/papers/ubiq15/ubiq15-corrected.tex

### A.4 Frozen archive: early notes and standalone studies (5)

26. docs/archive/early-notes/FabiusAsymptotic/FabiusAsymptotic.tex
27. docs/archive/early-notes/K-foldsummationoverthesignedThue-Morsesequence/K-foldsummationoverthesignedThue-Morsesequence.tex
28. docs/archive/standalone-studies/Non\_Elementarity\_of\_the\_Fabius\_Function/Non\_Elementarity\_of\_the\_Fabius\_Function.tex
29. docs/archive/standalone-studies/Small\_Argument\_Asymptotics/Small\_Argument\_Asymptotics.tex
30. docs/archive/standalone-studies/fabius-inverse-dyadic-closed-form/fabius-inverse-dyadic-closed-form.tex

### A.5 Frozen archive: primary expositions and supplements (8)

31. docs/archive/primary-exposition/Fabius\_Function\_and\_Rvachev\_Up/Fabius\_Function\_and\_Rvachev\_Up.tex
32. docs/archive/primary-exposition/Fabius\_Function\_and\_Rvachev\_Up-2/Fabius\_Function\_and\_Rvachev\_Up.tex
33. docs/archive/primary-exposition/Fabius\_Function\_and\_Rvachev\_Up-3/Fabius\_Function\_and\_Rvachev\_Up.tex
34. docs/archive/primary-exposition/Fabius\_and\_Rvachev\_Up-2/Fabius\_and\_Rvachev\_Up.tex
35. docs/archive/primary-exposition/Fabius\_and\_Rvachev\_up/Fabius\_and\_Rvachev\_up.tex
36. docs/archive/primary-exposition/Fabius\_function\_and\_up\_function/Fabius\_function\_and\_up\_function.tex
37. docs/archive/primary-supplements/Fabius\_Function\_Missing\_Parts/Fabius\_Function\_Missing\_Parts.tex
38. docs/archive/primary-supplements/Fabius\_Function\_Missing\_Parts-2/Fabius\_Function\_Missing\_Parts.tex

### A.6 Frozen archive: q-calculus, walkthrough, and glossary (6)

39. docs/archive/q-calculus/Fabius\_q\_Special\_Functions/Fabius\_q\_Special\_Functions.tex
40. docs/archive/q-calculus/fabius-q-special-functions/fabius-q-special-functions.tex
41. docs/archive/lean-walkthrough/Fabius\_Function\_Lean\_Walkthrough/Fabius\_Function\_Lean\_Walkthrough.tex

42. docs/archive/lean-walkthrough/fabius\_lean\_walkthrough/fabius\_lean\_walkthrough.tex
43. docs/archive/glossary/FabiusFunction\_Mathematical\_Glossary/FabiusFunction\_Mathematical\_Glossary.tex
44. docs/archive/glossary/FabiusFunction\_Mathematical\_Glossary-2/FabiusFunction\_Mathematical\_Glossary.tex

## A.7 Frozen archive: research frontiers (13)

45. docs/archive/research-frontiers/Fabius\_Dyadic\_Asymptotic\_Bridge/Fabius\_Dyadic\_Asymptotic\_Bridge.tex
46. docs/archive/research-frontiers/Fabius\_Dyadic\_Formulae\_and\_Alternative\_Representations/Fabius\_Dyadic\_Formulae\_and\_Alternative\_Representations.tex
47. docs/archive/research-frontiers/Fabius\_Dyadic\_Formulae\_to\_Asymptotics/Fabius\_Dyadic\_Formulae\_to\_Asymptotics.tex
48. docs/archive/research-frontiers/Fabius\_Dyadic\_q\_Connections/Fabius\_Dyadic\_q\_Connections.tex
49. docs/archive/research-frontiers/Fabius\_Thue\_Morse\_Convergence\_Rate/Fabius\_Thue\_Morse\_Convergence\_Rate.tex
50. docs/archive/research-frontiers/Fabius\_Thue\_Morse\_Convergence\_Rates/Fabius\_Thue\_Morse\_Convergence\_Rates.tex
51. docs/archive/research-frontiers/Primary\_Exposition\_Gap\_Register/Primary\_Exposition\_Gap\_Register.tex
52. docs/archive/research-frontiers/Repeated\_Integration\_and\_Rvachev\_Up/Repeated\_Integration\_and\_Rvachev\_Up.tex
53. docs/archive/research-frontiers/Rvachev\_Up\_from\_Repeated\_Integration/Rvachev\_Up\_from\_Repeated\_Integration.tex
54. docs/archive/research-frontiers/non-formalized-research-frontiers/semi-formalized-research-frontiers.tex
55. docs/archive/research-frontiers/Thue\_Morse\_Formula\_Atlas-1/Thue\_Morse\_Formula\_Atlas.tex
56. docs/archive/research-frontiers/Thue\_Morse\_Formula\_Atlas-2/Consolidated\_Formula\_Atlas\_for\_the\_Thue\_Morse\_Sequence.tex
57. docs/archive/research-frontiers/Thue\_Morse\_Formula\_Atlas-3/Thue\_Morse\_Formula\_Atlas.tex

## B Additional algebraic details

### B.1 Derivation of the $q$ -moment formula

Starting from [Equation \(5.2\)](#),

$$\begin{aligned}
 S_{r,m}(q) &= \frac{1}{(q;q)_r} \sum_{j=0}^r (-1)^{r-j} q^{\binom{r-j+1}{2}} \begin{bmatrix} r \\ j \end{bmatrix}_q q^{jm} \\
 &= \frac{q^{rm}}{(q;q)_r} \sum_{k=0}^r (-1)^k q^{\binom{k+1}{2} - km} \begin{bmatrix} r \\ k \end{bmatrix}_q \\
 &= \frac{q^{rm}}{(q;q)_r} \sum_{k=0}^r q^{\binom{k}{2}} \begin{bmatrix} r \\ k \end{bmatrix}_q (-q^{1-m})^k \\
 &= \frac{q^{rm} (q^{1-m}; q)_r}{(q;q)_r}.
 \end{aligned}$$

For  $m \geq r + 1$ ,

$$\begin{aligned} (q^{1-m}; q)_r &= \prod_{k=0}^{r-1} (1 - q^{1-m+k}) \\ &= (-1)^r q^{\sum_{k=0}^{r-1} (1-m+k)} \prod_{k=0}^{r-1} (1 - q^{m-1-k}) \\ &= (-1)^r q^{r(1-m)+r(r-1)/2} (q^{m-r}; q)_r. \end{aligned}$$

Multiplication by  $q^{rm}$  gives the exponent  $r(r+1)/2$  in [Equation \(5.5\)](#).

## B.2 Why all transformed sinc-tail terms have one sign

The sign theorem is not a generic property of Richardson extrapolation. It relies on two positivity facts:

1. the log-sinc tail coefficients  $d_m(t)$  are positive because  $\zeta(2m) > 0$ ;
2. after cancellation, every geometric moment is

$$S_{r,m}(q) = (-1)^r \times \frac{q^{r(r+1)/2} (q^{m-r}; q)_r}{(q; q)_r},$$

whose remaining factors are positive for  $m \geq r + 1$ .

Their product therefore has the same sign for every residual degree. This is the mechanism behind the certified brackets.

## B.3 Bell form of complete homogeneous functions

The denominator-cleared formal product and its canonical formal inverse are now proved in `CompleteHomogeneousGenerating.lean`. The logarithmic coefficient comparison in this subsection, including its specialization to generalized harmonic sums, is not part of that generating-series result. The separate `CompleteHomogeneousBell.lean` proves the same finite coefficient identity by division-free Bell algebra and supplies its factorially normalized form; `LambertPhaseLockedBell.lean` specializes it to generalized harmonic sums. Neither statement evaluates an infinite formal series or proves analytic convergence.

Taking logarithms of [Equation \(12.11\)](#),

$$\log \prod_{j=0}^r (1 - a_j z)^{-1} = \sum_{k=1}^{\infty} \frac{z^k}{k} \sum_{j=0}^r a_j^k.$$

Comparing this with [Equation \(2.6\)](#) gives

$$h_n(a_0, \dots, a_r) = \frac{1}{n!} \mathcal{B}_n(0!p_1, 1!p_2, \dots, (n-1)!p_n),$$

where  $p_k = \sum_j a_j^k$ . Substituting  $a_j = (\lambda + j)^{-1}$  proves [Equation \(12.15\)](#). The corresponding compiled declarations are `completeHomogeneousEvalOn_eq_factorialNormalize_completeBellPolynomial` and `completeHomogeneousEvalOn_shiftedReciprocal_eq_bell`; the two downstream residual rewrites are `fabiusPhaseLockedResidualTerm_eq_bell` and `fabiusPhaseLockedResidualPartialSum_eq_sum_bell`.

## C Reproducibility notes

The numerical checks used arbitrary-precision arithmetic with the following direct definitions:

$$\begin{aligned} P_N(t) &= \prod_{k=1}^N \frac{\sin(t/2^k)}{t/2^k}, \\ P(t) &\approx \prod_{k=1}^{300} \frac{\sin(t/2^k)}{t/2^k}, \end{aligned}$$

$$\log |\widehat{P}_{N,r}(t)| = \sum_{j=0}^r \lambda_{r,j}(1/4) \log |P_{N+j}(t)|.$$

At 100-digit working precision, the 300-factor reference tail is far below the displayed errors. Symbolic checks verified the moment identities through  $r = 8$  and multiple values of  $m$  before the general proofs were written.

## References

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- [10] J.-P. Allouche and J. Shallit, *Automatic Sequences: Theory, Applications, Generalizations*, Cambridge University Press, 2003.
- [11] J.-P. Allouche, M. Mendès France, and J. Peyrière, works on lacunary trigonometric products and the Thue–Morse sequence; source TeX included in the repository as Lacunary\_products.tex.

## Part IX

# Binary Spectral-Endpoint Bridges for the Fabius and Rvachev Functions

## Abstract

This report audits the recursive LaTeX corpus under `Analysis/FabiusFunction/docs` and then develops a new synthesis joining five strands that the existing documents largely treat in parallel: the Thue–Morse Mahler product, base-1/2 and base-1/4  $q$ -calculus, the Rvachev infinite sinc product, the exact negative-Laplace decomposition at the Fabius endpoint, and the lower real branch of Lambert’s  $W$  function. The audit covers 57 .tex files: five living synthesis or reference documents, thirteen active frontier drafts, thirty-two frozen historical versions, and seven transcribed or corrected source papers. Frozen duplicate snapshots were checked through the archive’s provenance map and through the living documents into which their distinctive material was merged.

The principal new results, in the precise sense “not found explicitly in the inspected corpus”, are as follows. First, the Rvachev transform admits the  $q$ -Pochhammer product

$$\widehat{\text{up}}(z) = \prod_{\ell \geq 1} (z^2/\ell^2; 1/4)_\infty = \prod_{n \geq 1} (1 - z^2/n^2)^{v_2(n)+1},$$

so its integer zero multiplicity is exactly  $v_2(n) + 1$ . A finite version has both a Gaussian-binomial factorization and an exact signed Thue–Morse exponential-sum factorization. Second, if  $\Theta(q) = \sum_{n \geq 0} (-1)^{\text{wt}_2(n)} q^n = \prod_{j \geq 0} (1 - q^{2^j})$ , then the exponentially small endpoint correction in the negative-Laplace formula is exactly  $-\log \Theta(e^{-s})$ . Its Lambert-series coefficients satisfy

$$-\log \Theta(e^{-s}) = \sum_{n \geq 1} \frac{2^{v_2(n)+1} - 1}{n} e^{-ns} = \sum_{n \geq 1} \frac{2^{\text{ord}_z=n} \widehat{\text{up}} - 1}{n} e^{-ns},$$

giving a direct spectral–endpoint dictionary. Third, its Mellin transform is

$$\int_0^\infty [-\log \Theta(e^{-s})] s^{w-1} ds = \frac{\Gamma(w)\zeta(w+1)}{1 - 2^{-w}},$$

which realizes, on the complementary positive Mellin strip, the same meromorphic  $\Gamma\zeta$  kernel used by the corpus for the negative-Laplace logarithm. This gives a new Thue–Morse/weighted-zero derivation of every Fourier coefficient of the endpoint’s log-periodic function and recovers the corpus’s already established exact mean in terms of the first Stieltjes constant. Fourth, the exact endpoint saddle is the fixed point of a periodically perturbed Lambert map, with the lower branch governing the small-argument endpoint regime and an explicit derivative proving local contraction there. Finally, an all-orders base-1/4 tail expansion places the corpus’s established centered moments in spectral tail coordinates and yields a certified acceleration of finite Thue–Morse approximants to the Fourier transform.

No assertion of global literature priority is made. Every result is labeled as established in the source corpus, newly proved here relative to that corpus, or conjectural/future work.

# 1 Scope, audit method, and proof-status discipline

## 1.1 What was inspected

The repository separates formalization-backed exposition from non-formalized frontier mathematics. The living primary article records only statements with named Lean counterparts, whereas the frontier volume deliberately quarantines plausible or proved-on- paper results not yet represented by an exact formal declaration. The recursive source audit used that distinction rather than flattening all files into one undifferentiated literature review.

At the snapshot date, the recursive corpus contains the following groups.

| Group            | Files | Role and audit treatment                                                                                                                                                          |
|------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Living documents | 5     | Primary exposition, glossary, non-elementarity, Lean walkthrough, and frontier synthesis; used as the theorem and notation baseline.                                              |
| Active drafts    | 13    | Thue–Morse atlas and twelve Fourier-decay studies; checked for newer claims, conflicts, and unresolved questions.                                                                 |
| Frozen archive   | 32    | First versions and historical parallel drafts; checked through the archive provenance descriptions and merged-content map, with distinctive claims traced into living successors. |
| Source papers    | 7     | Corrected or transcribed mathematical papers; used for provenance, comparison, and terminology.                                                                                   |
| Total            | 57    | Recursive LaTeX corpus audited.                                                                                                                                                   |

The full path-level inventory is recorded in [Section B](#).

## 1.2 Status labels

Throughout the report:

- [E] means an established theorem or audited conclusion already present in the repository corpus (often formally verified in Lean);
- [N] means a theorem proved in this report and not located explicitly in the inspected corpus by path, phrase, and formula searches;
- [C] means a conjecture, program, or interpretation whose proof is not supplied.

“New” is therefore a corpus-relative statement, not an unsupported claim of first publication in the worldwide literature.

A final formula-by-formula overlap audit corrected three preliminary classifications before release. The exact mean and full Fourier spectrum of the endpoint-periodic function are already established in the corpus; so are the centered moment generating function, its initial rational moments, and a Bernoulli recurrence for those moments. Moreover, the same meromorphic  $\Gamma\zeta$  kernel already occurs there as the Mellin transform of  $\Lambda$  on  $-1 < \Re w < 0$ . The new claims below are therefore restricted to the  $q$ -Pochhammer and weighted-zero dictionaries, the complementary positive-strip realization of that kernel, the resulting Thue–Morse/zero-divisor derivations, the exact Lambert branch map, and the certified finite-product consequences.

## 1.3 The structural gap that motivates the report

The corpus already develops, separately and deeply,

1. exact dyadic values of the Fabius function involving  $(1/2; 1/2)_n$  and Gaussian binomial coefficients;
2. the Thue–Morse product  $\prod_{j \geq 0} (1 - q^{2^j})$  and its Mahler equation;
3. the Rvachev Fourier transform as an infinite product of geometrically rescaled sinc functions;
4. a quadratic-plus-periodic negative-Laplace decomposition and a lower-Lambert endpoint phase;
5. a multi-regime theory of Fourier decay.

What is missing is an exact dictionary showing that the same binary arithmetic controls *both* the zero divisor of the Fourier transform and the exponentially small correction in the endpoint Laplace transform. Establishing that dictionary leads to a new  $q$ -Pochhammer zero-divisor formula, a weighted-zero interpretation of the corpus's  $\Gamma\zeta$  Mellin kernel, exact tilted-cumulant recurrences, and branch-resolved Lambert- $W$  maps.

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**Main synthesis in one line.** The Thue–Morse product controls the endpoint correction; its logarithmic coefficients are  $2^{\mu(n)} - 1$ , where  $\mu(n)$  is exactly the multiplicity of the Fourier zero at the integer  $n$ . Mellin transformation turns the resulting 2-adic Dirichlet series into the  $\Gamma\zeta$  Fourier modes of the endpoint periodic function, and that periodic function perturbs the lower-Lambert saddle.

---

## 2 Established framework and normalization

### 2.1 The three functions

[E] The corpus carefully distinguishes three related global objects. The bounded Fabius function  $F$  is the distribution function on  $[0, 1]$  satisfying

$$F'(x) = 2F(2x) \quad (0 \leq x \leq \tfrac{1}{2}), \quad F(1-x) = 1 - F(x),$$

with constant extensions outside  $[0, 1]$ . Rvachev's even compactly supported bump is

$$\text{up}(x) = F(1 - |x|) \quad (|x| \leq 1), \quad \text{up}(x) = 0 \quad (|x| \geq 1).$$

The signed global extension  $\mathcal{F}$  satisfies the pure dilation equation on all of  $\mathbb{R}$  and should not be silently substituted for  $F$  or  $\text{up}$ .

### 2.2 Probability model

Let  $U_1, U_2, \dots$  be independent  $\text{Unif}[0, 1]$  variables and set

$$X = \sum_{j \geq 1} 2^{-j} U_j. \quad (2.1)$$

[E] Then  $F(x) = \mathbb{P}(X \leq x)$ . The centered variable

$$Y = 2X - 1 = \sum_{n \geq 0} 2^{-n} V_n, \quad V_n \stackrel{\text{iid}}{\sim} \text{Unif}[-1/2, 1/2], \quad (2.2)$$

has density  $\text{up}$ . Consequently  $Y$  is symmetric,  $|Y| \leq 1$ , and its Fourier transform under the convention

$$\widehat{f}(z) = \int_{\mathbb{R}} f(x) e^{-2\pi i x z} dx$$

is a characteristic function on the real axis.

### 2.3 Sinc product

Write  $\text{sinc } w = \sin w/w$  with the removable value 1 at  $w = 0$ , and set

$$\Phi(z) = \widehat{\text{up}}(z). \quad (2.3)$$

[E] The exact product and refinement law are

$$\Phi(z) = \prod_{n=0}^{\infty} \text{sinc}\left(\frac{\pi z}{2^n}\right), \quad \Phi(z) = \text{sinc}(\pi z) \Phi(z/2). \quad (2.4)$$

The product is entire and also admits the weighted cosine factorization  $\Phi(z) = \prod_{m \geq 1} \cos(\pi z/2^m)^m$ .



## 2.4 Thue–Morse product

Let

$$\varepsilon_n = (-1)^{\text{wt}_2(n)}, \quad \Theta(q) = \sum_{n \geq 0} \varepsilon_n q^n. \quad (2.5)$$

[E] For  $|q| < 1$ ,

$$\boxed{\Theta(q) = \prod_{j \geq 0} (1 - q^{2^j}), \quad \Theta(q) = (1 - q)\Theta(q^2).} \quad (2.6)$$

At finite depth,

$$\sum_{k=0}^{2^m-1} \varepsilon_k q^k = \prod_{j=0}^{m-1} (1 - q^{2^j}), \quad (2.7)$$

and Prouhet cancellation gives

$$\sum_{k=0}^{2^m-1} \varepsilon_k k^r = 0 \quad (0 \leq r < m), \quad \sum_{k=0}^{2^m-1} \varepsilon_k k^m = (-1)^m 2^{\binom{m}{2}} m!. \quad (2.8)$$

## 2.5 Negative Laplace transform and periodic correction

Define

$$B(s) = \mathbb{E}e^{-sX} = \prod_{j \geq 1} \frac{1 - e^{-s/2^j}}{s/2^j}, \quad \Lambda(s) = \log B(s), \quad s > 0. \quad (2.9)$$

[E] The exact dilation relation is

$$\Lambda(2s) = \Lambda(s) + \log(1 - e^{-s}) - \log s. \quad (2.10)$$

Let  $L = \log 2$  and

$$T(s) = \sum_{m \geq 0} \log(1 - e^{-2^m s}), \quad \mathcal{E}(s) = -T(s) > 0. \quad (2.11)$$

Then

$$P(u) = \Lambda(2^u) + \frac{L}{2}(u^2 - u) + T(2^u) \quad (2.12)$$

is smooth and 1-periodic, and

$$\boxed{\Lambda(s) = -\frac{(\log s)^2}{2L} + \frac{1}{2} \log s + P(\log_2 s) + \mathcal{E}(s).} \quad (2.13)$$

The corpus gives every Fourier mode of  $P$ , including its exact mean, and proves that no nonzero mode vanishes. It also obtains  $\Gamma(w)\zeta(w+1)/(1-2^{-w})$  as the Mellin transform of  $\Lambda$  on  $-1 < \Re w < 0$ . One of our goals is to realize that same meromorphic kernel from the Thue–Morse endpoint tail on the complementary strip  $\Re w > 0$ , and thereby rederive the entire periodic spectrum by one weighted-zero residue calculation.

## 2.6 The Fourier-decay spectrum is not one number

[E] The comparative Fourier-decay audits establish a common lognormal core

$$\exp\left[-\frac{(\log x)^2}{2 \log 2}\right]$$

but different power corrections for different notions of size. In the notation of the audits, the ordering is

$$\kappa_\infty < \kappa_2 < \kappa_1 < \kappa_0,$$

corresponding respectively to extremal rays, quadratic mean, Cesàro mean, and typical or almost-everywhere behavior. The second-wave audit additionally verifies that an unrestricted Cesàro normalization can be attained by a positive real-analytic decreasing gauge, although every stabilizing or bounded-distortion gauge fails. This report does not replace that multi-regime theory; instead, it supplies exact arithmetic and transform identities that sit underneath all regimes.

### 3 A geometric $q$ -Rvachev family

#### 3.1 Definition and probabilistic realization

The binary Rvachev product belongs naturally to a one-parameter family.

**Definition 3.1** ( $q$ -Rvachev transform). For  $0 < q < 1$ , define

$$\Phi_q(z) = \prod_{n \geq 0} \operatorname{sinc}(\pi q^{n/2} z). \quad (3.1)$$

Let  $V_0, V_1, \dots$  be independent  $\operatorname{Unif}[-1/2, 1/2]$  variables and put

$$Y_q = \sum_{n \geq 0} q^{n/2} V_n. \quad (3.2)$$

**Theorem 3.2** (Probability, support, smooth density, and refinement). *[N] The product (3.1) converges locally uniformly to an entire function. It is the Fourier transform of the law of  $Y_q$ , whose support is*

$$\left[ -\frac{1}{2(1-\sqrt{q})}, \frac{1}{2(1-\sqrt{q})} \right].$$

The law has a compactly supported  $C^\infty$  density  $u_q$ , and

$$\Phi_q(z) = \operatorname{sinc}(\pi z) \Phi_q(\sqrt{q} z). \quad (3.3)$$

Writing  $a = \sqrt{q}$ , the density satisfies

$$u_q(x) = \int_{(x-1/2)/a}^{(x+1/2)/a} u_q(y) \, dy, \quad (3.4)$$

$$u'_q(x) = \frac{1}{a} \left[ u_q\left(\frac{x+1/2}{a}\right) - u_q\left(\frac{x-1/2}{a}\right) \right]. \quad (3.5)$$

At  $q = 1/4$ ,  $Y_q = Y$ ,  $u_q = u$ , and (3.5) is the usual Rvachev functional-differential equation.

*Proof.* The summands in (3.2) converge absolutely almost surely and are bounded by a geometric series. Independence gives (3.1). Since  $\operatorname{sinc}(\pi q^{n/2} z) = 1 + \mathcal{O}(q^n |z|^2)$  locally uniformly, the entire product converges on compact sets. If  $N \sim \log |x| / \log \sqrt{q}$ , bounding the first  $N$  sinc factors by a constant times  $(|x|q^{n/2})^{-1}$  gives

$$\log |\Phi_q(x)| \leq -\frac{(\log |x|)^2}{2|\log \sqrt{q}|} + \mathcal{O}(\log |x|),$$

so every polynomial multiple is integrable. Fourier inversion therefore yields a  $C^\infty$  density. Finally  $Y_q \stackrel{d}{=} V_0 + aY'_q$  with an independent copy  $Y'_q$ . Convolving with the uniform density gives (3.4); differentiation gives (3.5).  $\square$

#### 3.2 $q$ -Pochhammer factorization

Recall

$$(a; q)_m = \prod_{j=0}^{m-1} (1 - aq^j), \quad (a; q)_\infty = \prod_{j \geq 0} (1 - aq^j).$$

**Theorem 3.3** (Euler– $q$  factorization). *[N] For every  $0 < q < 1$  and  $z \in \mathbb{C}$ ,*

$$\Phi_q(z) = \prod_{\ell=1}^{\infty} (z^2/\ell^2; q)_\infty, \quad (3.6)$$

$$\Phi_{q,m}(z) := \prod_{n=0}^{m-1} \operatorname{sinc}(\pi q^{n/2} z) = \prod_{\ell=1}^{\infty} (z^2/\ell^2; q)_m. \quad (3.7)$$

For  $|z| < 1$ ,

$$\log \Phi_q(z) = - \sum_{r \geq 1} \frac{\zeta(2r)}{r(1-q^r)} z^{2r}, \quad (3.8)$$

$$\log \Phi_{q,m}(z) = - \sum_{r \geq 1} \frac{1-q^{mr}}{1-q^r} \frac{\zeta(2r)}{r} z^{2r}. \quad (3.9)$$

*Proof.* Euler's sine product gives

$$\operatorname{sinc}(\pi q^{n/2} z) = \prod_{\ell \geq 1} \left( 1 - \frac{q^n z^2}{\ell^2} \right).$$

The double product is absolutely and locally uniformly convergent because  $\sum_{n,\ell} q^n / \ell^2 < \infty$ , so the factors may be regrouped by  $\ell$ . This proves (3.6) and (3.7). Expanding  $\log(1-w) = -\sum_{r \geq 1} w^r / r$  and summing the geometric progression in  $n$  proves the logarithmic formulas.  $\square$

**Corollary 3.4** (Gaussian-binomial finite factorization). *[N] The finite product has the exact representation*

$$\Phi_{q,m}(z) = \prod_{\ell \geq 1} \left[ \sum_{j=0}^m (-1)^j q^{\binom{j}{2}} \begin{bmatrix} m \\ j \end{bmatrix}_q \frac{z^{2j}}{\ell^{2j}} \right]. \quad (3.10)$$

*Proof.* Apply the finite  $q$ -binomial theorem  $(a; q)_m = \sum_{j=0}^m (-1)^j q^{\binom{j}{2}} \begin{bmatrix} m \\ j \end{bmatrix}_q a^j$  to each factor in (3.7).  $\square$

*Remark 3.5.* The living corpus uses  $q = 1/2$  Gaussian coefficients in exact dyadic values of  $F$ . Equation (3.10) reveals a second, independent  $q$ -geometry: the spectral product naturally uses  $q = 1/4$ , the square of the spatial dilation. The two bases should not be conflated, but their coexistence is structural rather than accidental.

### 3.3 Cumulants in the $q$ -family

Let  $c_N(q) = \mathbb{E} Y_q^{2N}$ , with  $c_0(q) = 1$ , and use the standard Bernoulli numbers  $B_{2r}$ .

**Theorem 3.6** ( $q$ -Bernoulli cumulants and moments). *[N] All odd cumulants of  $Y_q$  vanish, and*

$$\kappa_{2r}(Y_q) = \frac{B_{2r}}{2r(1-q^r)}. \quad (3.11)$$

*The even moments obey*

$$c_N(q) = \frac{1}{2N} \sum_{r=1}^N \binom{2N}{2r} \frac{B_{2r}}{1-q^r} c_{N-r}(q). \quad (3.12)$$

*Proof.* Insert  $z = t/(2\pi)$  in (3.8) and compare  $\log \mathbb{E} e^{itY_q}$  with  $\sum_{n \geq 1} \kappa_n(it)^n / n!$ . Euler's evaluation of  $\zeta(2r)$  gives (3.11). The standard moment–cumulant recursion  $m_n = \sum_{k=1}^n \binom{n-1}{k-1} \kappa_k m_{n-k}$ , followed by  $\binom{2N-1}{2r-1} = \frac{r}{N} \binom{2N}{2r}$ , gives (3.12).  $\square$

## 4 The binary specialization: a 2-adic zero divisor

Set  $q = 1/4$ . Then  $\Phi_{1/4} = \Phi = \widehat{\mathfrak{up}}$ .

**Theorem 4.1** (Arithmetic canonical product). *[N] The Rvachev transform has the genus-zero product*

$$\Phi(z) = \prod_{n=1}^{\infty} \left( 1 - \frac{z^2}{n^2} \right)^{\mu(n)}, \quad \mu(n) = v_2(n) + 1. \quad (4.1)$$

Thus the nonzero zeros of  $\Phi$  are exactly the integers, and

$$\boxed{\operatorname{ord}_{z=n} \Phi = v_2(|n|) + 1 \quad (n \in \mathbb{Z} \setminus \{0\}).} \quad (4.2)$$

Moreover

$$\sum_{n=1}^N \mu(n) = 2N - \operatorname{wt}_2(N). \quad (4.3)$$

*Proof.* From (3.6),

$$\Phi(z) = \prod_{j \geq 0} \prod_{k \geq 1} \left(1 - \frac{z^2}{4^j k^2}\right).$$

For a fixed positive integer  $n$ , the equality  $n = 2^j k$  with  $k \in \mathbb{N}$  occurs exactly for  $j = 0, \dots, v_2(n)$ . Absolute convergence permits regrouping and proves (4.1). Each contributing factor has a simple zero, proving (4.2). Finally

$$\sum_{n \leq N} \mu(n) = N + \sum_{n \leq N} v_2(n) = N + v_2(N!) = 2N - \operatorname{wt}_2(N)$$

by Legendre's formula. □

**Corollary 4.2** (Dirichlet series of the zero multiplicities). *For  $\Re w > 1$ ,*

$$\boxed{\sum_{n \geq 1} \frac{\mu(n)}{n^w} = \frac{\zeta(w)}{1 - 2^{-w}}.} \quad (4.4)$$

Consequently (3.8) at  $q = 1/4$  can be read directly from the zero divisor:

$$\log \Phi(z) = - \sum_{r \geq 1} \frac{z^{2r}}{r} \sum_{n \geq 1} \frac{\mu(n)}{n^{2r}} = - \sum_{r \geq 1} \frac{4^r \zeta(2r)}{r(4^r - 1)} z^{2r}. \quad (4.5)$$

*Proof.* Since  $\mu(n) = \sum_{j \geq 0} \mathbf{1}_{2^j | n}$ , Tonelli's theorem gives

$$\sum_n \mu(n) n^{-w} = \sum_{j \geq 0} \sum_{2^j | n} n^{-w} = \zeta(w) \sum_{j \geq 0} 2^{-jw}.$$

□

#### 4.1 Finite triple factorization: sinc, $q$ -Pochhammer, and Thue–Morse

Define

$$\Phi_m(z) = \prod_{n=0}^{m-1} \operatorname{sinc}\left(\frac{\pi z}{2^n}\right), \quad S_m(z) = \sum_{k=0}^{2^m-1} \varepsilon_k e^{-2\pi i k z / 2^{m-1}}. \quad (4.6)$$

**Theorem 4.3** (Finite triple factorization). *[N] For  $m \geq 1$ ,*

$$\boxed{\prod_{\ell \geq 1} (z^2 / \ell^2; 1/4)_m = \Phi_m(z) = \frac{2^{m(m-3)/2} e^{2\pi i z(1-2^{-m})}}{(i\pi z)^m} S_m(z).} \quad (4.7)$$

The apparent singularity at  $z = 0$  is removable. In fact  $S_m$  has a zero of exact order  $m$  there, with

$$S_m(z) = (2\pi i z)^m 2^{-m(m-1)/2} + \mathcal{O}(z^{m+1}). \quad (4.8)$$

*Proof.* The first equality is (3.7). Use  $\sin w = e^{iw}(1 - e^{-2iw})/(2i)$  in each sinc factor. The powers of two give  $2^{\sum_{n < m} n - m} = 2^{m(m-3)/2}$  and the phases sum to  $e^{2\pi i z(1-2^{-m})}$ . The remaining product is

$$\prod_{n=0}^{m-1} (1 - e^{-2\pi i z / 2^n}) = \prod_{j=0}^{m-1} (1 - w^{2^j}) = \sum_{k < 2^m} \varepsilon_k w^k, \quad w = e^{-2\pi i z / 2^{m-1}},$$

which is  $S_m(z)$ . Equation (2.8) annihilates the first  $m$  Taylor coefficients of  $S_m$  and gives the leading coefficient in (4.8). □

**Updated Lean boundary.** The raw product-to-sum step in this proof is now an exact specialization of `sum_range_two_pow_thueMorseSign_cexp` in `ThueMorseComplexExponential.lean`, with the arbitrary exponential prefactor available separately as `sum_range_two_pow_thueMorseSign_cexp_affine`. The same module records the genuinely stronger Hamming-weight family `sum_range_two_pow_binaryWeight_cexp` and its affine companion `sum_range_two_pow_binaryWeight_cexp_affine`. These four declarations prove finite raw complex exponential identities. The generic finite complex half-angle and total sinc normalization, including the removable origin, are separately formal in `ThueMorseComplexProductBridge.lean`. What is not supplied by either module is the first  $q$ -Pochhammer-to-sinc equality in (4.7), its resulting full  $q$ -specific triple factorization and exact origin order, any infinite-product or exponential-generating-function convergence statement, or a position-dependent parity-word generalization. Accordingly, the complete triple factorization retains its report-level status even though its finite exponential and generic normalization layers are Lean verified.

**Corollary 4.4** (A Gaussian-binomial identity for a Thue–Morse exponential sum). *Combining (3.10) and (4.7) gives*

$$S_m(z) = (i\pi z)^m 2^{-m(m-3)/2} e^{-2\pi i z(1-2^{-m})} \times \prod_{\ell \geq 1} \left[ \sum_{j=0}^m (-1)^j 4^{-\binom{j}{2}} \begin{bmatrix} m \\ j \end{bmatrix}_{1/4} \frac{z^{2j}}{\ell^{2j}} \right]. \quad (4.9)$$

*This is an exact bridge between a finite signed Thue–Morse block and Gaussian coefficients at base 1/4.*

## 5 The endpoint correction is a Thue–Morse logarithm

### 5.1 Exact identification

**Theorem 5.1** (Mahler identification of the Laplace tail). *[N] For every  $s > 0$ ,*

$$e^{T(s)} = \Theta(e^{-s}) = \sum_{n \geq 0} \varepsilon_n e^{-ns}, \quad \mathcal{E}(s) = -\log \Theta(e^{-s}). \quad (5.1)$$

*Consequently the exact periodic decomposition (2.13) becomes*

$$\Lambda(s) = -\frac{(\log s)^2}{2L} + \frac{1}{2} \log s + P(\log_2 s) - \log \Theta(e^{-s}). \quad (5.2)$$

*Proof.* Substitute  $q = e^{-s}$  in the Thue–Morse product (2.6):

$$\Theta(e^{-s}) = \prod_{m \geq 0} (1 - e^{-2^m s}) = e^{T(s)}.$$

All factors are positive, so the real logarithm is unambiguous. The remaining formulas are immediate from (2.11) and (2.13).  $\square$

### 5.2 The 2-adic Lambert series

Define

$$a(n) = \sum_{j=0}^{v_2(n)} 2^j = 2^{v_2(n)+1} - 1. \quad (5.3)$$

**Theorem 5.2** (Lambert expansion and derivative hierarchy). *[N] For  $s > 0$ ,*

$$\mathcal{E}(s) = \sum_{n \geq 1} \frac{a(n)}{n} e^{-ns}, \quad (5.4)$$

$$T^{(k)}(s) = (-1)^{k+1} \sum_{n \geq 1} a(n) n^{k-1} e^{-ns} \quad (k \geq 1). \quad (5.5)$$

In particular, with

$$R(s) = sT'(s), \quad (5.6)$$

one has

$$R(s) = \sum_{m \geq 0} \frac{2^m s}{e^{2^m s} - 1} = s \sum_{n \geq 1} a(n) e^{-ns}. \quad (5.7)$$

The beginning of the endpoint correction is

$$\mathcal{E}(s) = e^{-s} + \frac{3}{2}e^{-2s} + \frac{1}{3}e^{-3s} + \frac{7}{4}e^{-4s} + \frac{1}{5}e^{-5s} + \dots \quad (5.8)$$

*Proof.* Expand every logarithm in (2.11):

$$\mathcal{E}(s) = \sum_{m \geq 0} \sum_{k \geq 1} \frac{e^{-k2^m s}}{k}.$$

For a fixed  $n$ , the representations  $n = k2^m$  are indexed by  $m = 0, \dots, v_2(n)$ , and  $1/k = 2^m/n$ . Their sum is  $a(n)/n$ . Termwise differentiation is justified by absolute convergence on every half-line  $s \geq s_0 > 0$ .  $\square$

### 5.3 The spectral–endpoint dictionary

**Theorem 5.3** (Endpoint coefficients from Fourier zero multiplicities). *[N] Let  $\mu(n) = \text{ord}_{z=n} \Phi$ . Then*

$$a(n) = 2^{\mu(n)} - 1, \quad (5.9)$$

and hence

$$-\log \Theta(e^{-s}) = \sum_{n \geq 1} \frac{2^{\text{ord}_{z=n} \Phi} - 1}{n} e^{-ns}. \quad (5.10)$$

Equivalently, the weighted zero-multiplicity Dirichlet series is

$$\mathfrak{Z}_\Phi(w) := \sum_{n \geq 1} \frac{2^{\text{ord}_{z=n} \Phi} - 1}{n^w} = \frac{\zeta(w)}{1 - 2^{1-w}}, \quad \Re w > 1. \quad (5.11)$$

*Proof.* By (4.2),  $\mu(n) = v_2(n) + 1$ , so (5.9) is (5.3). For the Dirichlet series, write  $a(n) = \sum_{j \geq 0} 2^j \mathbf{1}_{2^j | n}$  and use Tonelli:

$$\sum_{n \geq 1} a(n) n^{-w} = \zeta(w) \sum_{j \geq 0} 2^{j(1-w)} = \frac{\zeta(w)}{1 - 2^{1-w}}.$$

$\square$

---

**Why this is a genuine bridge.** The left side of (5.10) belongs to the small- $x$  Fabius endpoint problem; the multiplicity on the right belongs to the zero divisor of the Rvachev Fourier transform. The equality does not merely compare asymptotic scales: it identifies every individual exponential coefficient with a local spectral multiplicity.

---

| $n$ | $v_2(n)$ | $\text{ord}_{z=n} \Phi$ | coefficient of $e^{-ns}$ in $\mathcal{E}(s)$ |
|-----|----------|-------------------------|----------------------------------------------|
| 1   | 0        | 1                       | 1                                            |
| 2   | 1        | 2                       | 3/2                                          |
| 3   | 0        | 1                       | 1/3                                          |
| 4   | 2        | 3                       | 7/4                                          |
| 6   | 1        | 2                       | 1/2                                          |
| 8   | 3        | 4                       | 15/8                                         |
| 12  | 2        | 3                       | 7/12                                         |
| 16  | 4        | 5                       | 31/16                                        |

## 6 Mellin transform and the exact periodic phase

### 6.1 Mellin transform

**Theorem 6.1** (Positive-strip realization of the Gamma–zeta kernel). *[N] For  $\Re w > 0$ ,*

$$\boxed{\mathcal{M}\mathcal{E}(w) := \int_0^\infty \mathcal{E}(s)s^{w-1} ds = \frac{\Gamma(w)\zeta(w+1)}{1-2^{-w}}.} \quad (6.1)$$

*Equivalently, the Mellin transform of  $R(s) = sT'(s) = -s\mathcal{E}'(s)$  is*

$$\int_0^\infty R(s)s^{w-1} ds = \frac{\Gamma(w+1)\zeta(w+1)}{1-2^{-w}}. \quad (6.2)$$

*[E] The corpus obtains the same meromorphic right-hand side as the Mellin transform of  $\Lambda$  on the complementary fundamental strip  $-1 < \Re w < 0$ . Thus (6.1) is not a new special-function expression; its corpus-relative novelty is that the Thue–Morse endpoint tail and the weighted Fourier zero divisor realize that established kernel on  $\Re w > 0$ .*

*Proof.* Put  $\sigma = \Re w > 0$ . Absolute convergence permits Fubini, because

$$\sum_{n \geq 1} \frac{a(n)}{n} \int_0^\infty e^{-ns} s^{\sigma-1} ds = \Gamma(\sigma) \sum_{n \geq 1} \frac{a(n)}{n^{\sigma+1}} < \infty.$$

Therefore (5.4) gives

$$\int_0^\infty \mathcal{E}(s)s^{w-1} ds = \Gamma(w) \sum_{n \geq 1} \frac{a(n)}{n^{w+1}}.$$

Use (5.11) with  $w+1$  in place of  $w$ . Integration by parts gives (6.2); the boundary terms vanish for  $\Re w > 0$ .  $\square$

### 6.2 Residues and the Fourier coefficients of $P$

Put

$$\chi_k = \frac{2\pi i k}{L}, \quad k \in \mathbb{Z}. \quad (6.3)$$

We use the Fourier convention  $P(u) = \sum_{k \in \mathbb{Z}} \hat{P}(k)e^{2\pi i k u}$ .

**Theorem 6.2** (Mellin recovery of the full periodic spectrum). *[E] The periodic function in (2.13) has*

$$\boxed{\hat{P}(k) = -\frac{\Gamma(-\chi_k)\zeta(1-\chi_k)}{L}, \quad k \neq 0,} \quad (6.4)$$

*which agrees with the nonzero-mode formula already established in the corpus. Its mean is*

$$\boxed{\hat{P}(0) = -\frac{L}{12} - \frac{1}{L} \left( \frac{\pi^2}{12} - \gamma_1 - \frac{\gamma^2}{2} \right),} \quad (6.5)$$

*where  $\gamma$  is Euler’s constant and  $\gamma_1$  is the first Stieltjes constant in the convention  $\zeta(1+w) = w^{-1} + \gamma - \gamma_1 w + \mathcal{O}(w^2)$ . Numerically,*

$$\hat{P}(0) = -1.1090453654341866821907455547 \dots \quad (6.6)$$

*These coefficient formulas, including the mean, are established in the corpus (the mean is formalized there). [N] The corpus-relative novelty is the unified derivation below: both the oscillatory modes and the zero mode arise from the same Thue–Morse endpoint correction and the same meromorphic Mellin kernel.*

*Proof.* Mellin inversion writes  $\mathcal{E}(s)$  as a vertical integral of

$$M(w)s^{-w}, \quad M(w) = \frac{\Gamma(w)\zeta(w+1)}{1-2^{-w}}.$$

The exponential decay of  $\Gamma$  on vertical lines justifies shifting the contour across  $\Re w = 0$  and summing the resulting residues absolutely. Besides the pole at  $w = 0$ , the denominator has simple poles at  $w = \chi_k$ . The pole at  $w = -\chi_k$  contributes

$$\frac{\Gamma(-\chi_k)\zeta(1-\chi_k)}{L} s^{\chi_k} = \frac{\Gamma(-\chi_k)\zeta(1-\chi_k)}{L} e^{2\pi i k \log_2 s}.$$

On the other hand, (2.13) and  $\Lambda(s) = \mathcal{O}(s)$  as  $s \downarrow 0$  give

$$\mathcal{E}(s) = \frac{(\log s)^2}{2L} - \frac{1}{2} \log s - P(\log_2 s) + \mathcal{O}(s). \quad (6.7)$$

Comparison proves (6.4).

For the zero mode, expand at  $w = 0$ :

$$\begin{aligned} \Gamma(w)\zeta(1+w) &= \frac{1}{w^2} + c_0 + \mathcal{O}(w), & c_0 &= \frac{\pi^2}{12} - \gamma_1 - \frac{\gamma^2}{2}, \\ \frac{1}{1-2^{-w}} &= \frac{1}{Lw} + \frac{1}{2} + \frac{Lw}{12} + \mathcal{O}(w^3). \end{aligned}$$

Hence

$$M(w) = \frac{1}{Lw^3} + \frac{1}{2w^2} + \left( \frac{L}{12} + \frac{c_0}{L} \right) \frac{1}{w} + \mathcal{O}(1).$$

The residue of  $M(w)s^{-w}$  at zero is  $(\log s)^2/(2L) - (\log s)/2 + L/12 + c_0/L$ . Comparing with (6.7) shows that this constant is  $-\hat{P}(0)$ .  $\square$

*Remark 6.3* (Why the oscillation is easy to miss). The first nonzero mode is tiny:

$$|\hat{P}(1)| = 1.0627116252 \dots \times 10^{-6}, \quad |\hat{P}(2)| = 6.6927864 \dots \times 10^{-13}.$$

Differentiation magnifies the first mode, but it is still small. This explains why a non-periodic numerical fit can look excellent while being wrong at the sharp error scale.

### 6.3 The periodic correction to the saddle mean

Define

$$\Pi(u) = -\frac{1}{2} - \frac{P'(u)}{L}. \quad (6.8)$$

Then  $\hat{\Pi}(0) = -1/2$ , while for  $k \neq 0$ ,

$$\boxed{\hat{\Pi}(k) = -\frac{\Gamma(1-\chi_k)\zeta(1-\chi_k)}{L}}. \quad (6.9)$$

The first harmonic has magnitude  $9.6331836245 \dots \times 10^{-6}$ .

## 7 Tilted cumulants and exact periodically perturbed Lambert saddles

### 7.1 Exact tilted mean

Under the exponentially tilted law

$$\frac{e^{-sX}}{B(s)} d\mathbb{P},$$

write  $\mathbb{E}_s$  and  $\text{Var}_s$  for expectation and variance. Define

$$A(s) = -s\Lambda'(s) = s\mathbb{E}_s X. \quad (7.1)$$

**Theorem 7.1** (Mean decomposition and dyadic recurrence). *[N] For every  $s > 0$ ,*

$$\boxed{A(s) = \log_2 s + \Pi(\log_2 s) + R(s)}. \quad (7.2)$$

*It also satisfies the exact recurrence*

$$\boxed{A(2s) = A(s) + 1 - \frac{s}{e^s - 1}}. \quad (7.3)$$

*Proof.* Apply  $-s \frac{d}{ds}$  to (2.13); this gives (7.2). Differentiate (2.10) and multiply by  $-s$  to obtain (7.3).  $\square$



## 7.2 All tilted cumulants

Let

$$\kappa_n(s) = (-1)^n \Lambda^{(n)}(s), \quad K_n(s) = s^n \kappa_n(s), \quad D = s \frac{d}{ds}. \quad (7.4)$$

Thus  $\kappa_n(s)$  is the  $n$ th cumulant of  $X$  under the tilted law.

**Theorem 7.2** (Falling-Euler hierarchy). *[N] For  $n \geq 1$ ,*

$$K_n(s) = (-1)^n (D)_{\underline{n}} \left[ -\frac{L}{2} u(u-1) + P(u) - T(s) \right], \quad u = \log_2 s, \quad (7.5)$$

where  $(D)_{\underline{n}} = D(D-1) \cdots (D-n+1)$ . If  $h(s) = \log(1 - e^{-s}) - \log s$ , then

$$K_n(2s) = K_n(s) + (-1)^n s^n h^{(n)}(s). \quad (7.6)$$

In particular,

$$s^2 \text{Var}_s(X) = A(s) - DA(s) = u + \Pi(u) + R(s) - \frac{1 + \Pi'(u)}{L} - sR'(s) > 0, \quad (7.7)$$

and

$$K_2(2s) = K_2(s) + 1 - \frac{s^2 e^s}{(e^s - 1)^2}. \quad (7.8)$$

*Proof.* The operator identity  $s^n \frac{d^n}{ds^n} = (D)_{\underline{n}}$  proves (7.5). Differentiate (2.10)  $n$  times to obtain (7.6). Since  $K_2 = s^2 \Lambda'' = A - DA$ , substitution of (7.2) gives (7.7). Strict positivity follows because  $X$  is nondegenerate under every finite tilt. The explicit formula (7.8) follows by differentiating  $h$  twice.  $\square$

## 7.3 Exact Lambert fixed point and branch threshold

The natural saddle for the inverse Laplace problem at  $x \in (0, 1/2)$  is the unique solution of

$$x = -\Lambda'(s) = \mathbb{E}_s X. \quad (7.9)$$

Write  $s = 2^u$  and

$$C(u) = \Pi(u) + R(2^u) = A(2^u) - u. \quad (7.10)$$

Then the exact saddle equation is

$$x2^u = u + C(u). \quad (7.11)$$

**Theorem 7.3** (Periodically perturbed Lambert maps). *[N] Equation (7.11) has one solution  $u = u_*(x)$  for every  $0 < x < 1/2$ . There is a unique  $s_c > 0$  such that*

$$A(s_c) = \frac{1}{L}, \quad x_c := \frac{A(s_c)}{s_c} = \frac{1}{Ls_c}, \quad (7.12)$$

Numerically,

$$s_c = 3.5627054762160828049 \dots, \quad x_c = 0.4049436728688661444 \dots \quad (7.13)$$

For the endpoint range  $0 < x < x_c$ , the saddle is equivalently the fixed point of the lower-branch map

$$u = \mathfrak{L}_x^{(-1)}(u) := -\frac{1}{L} W_{-1}(-Lx 2^{-C(u)}) - C(u). \quad (7.14)$$

For  $x_c < x < 1/2$ , the same formula with  $W_0$  in place of  $W_{-1}$  gives the saddle; at  $x = x_c$  the two branches meet. At a fixed point of either branch,

$$(\mathfrak{L}_x^{(b)})'(u) = \frac{C'(u)}{L(u + C(u)) - 1}, \quad b \in \{0, -1\}. \quad (7.15)$$

Because  $\Pi'$  is bounded and  $R(2^u)$  is exponentially small as  $u \rightarrow \infty$ , the lower-branch map is a local contraction at  $u_*(x)$  for all sufficiently small  $x$ .

*Proof.* The function  $s \mapsto \mathbb{E}_s X$  is strictly decreasing because its derivative is  $-\text{Var}_s(X) < 0$ . Its limits are  $1/2$  at  $s \downarrow 0$  and  $0$  at  $s \rightarrow \infty$ , proving existence and uniqueness of the saddle. Also

$$A(s) = \sum_{j \geq 1} g(s/2^j), \quad g(t) = 1 - \frac{t}{e^t - 1},$$

and  $g'(t) = (e^t(t-1) + 1)/(e^t - 1)^2 > 0$  for  $t > 0$ . Thus  $A$  increases continuously from  $0$  to infinity (for instance, the first  $\lfloor \log_2 s \rfloor$  summands are uniformly positive once  $s$  is large), so (7.12) has one solution. Since  $x = A(s)/s = \mathbb{E}_s X$  decreases with  $s$ , one has  $A(s) > 1/L$  exactly when  $x < x_c$ .

Put  $y = u + C(u) = A(2^u)$ . Equation (7.11) becomes  $ye^{-Ly} = xe^{-LC(u)}$ . Multiplication by  $-L$  selects  $W_{-1}$  when  $y > 1/L$ ,  $W_0$  when  $0 < y < 1/L$ , and the common branch value at  $y = 1/L$ . This proves the branch statements. Implicit differentiation of  $ye^{-Ly} = xe^{-LC(u)}$ , followed by subtraction of  $C'(u)$ , gives (7.15). In the endpoint regime the denominator tends to infinity, whereas  $C'$  stays bounded.  $\square$

**Corollary 7.4** (First correction to the unperturbed Lambert phase). *Let*

$$u_0(x) = -\frac{1}{L} W_{-1}(-Lx), \quad (7.16)$$

so that  $x2^{u_0} = u_0$ . Then, as  $x \downarrow 0$ ,

$$u_*(x) = u_0(x) + \frac{C(u_0(x))}{Lu_0(x) - 1} + \mathcal{O}_C(u_0(x)^{-2}). \quad (7.17)$$

A single evaluation of the exact map (7.14) is typically much more accurate than the linear correction.

*Proof architecture.* Apply the mean-value theorem to  $f(u) = x2^u - u - C(u)$  between  $u_0$  and  $u_*$ . Since  $f(u_0) = -C(u_0)$  and  $f'(u) = Lx2^u - 1 - C'(u)$ , the displacement is of order  $1/u_0$ . A second expansion gives (7.17); the implied constant depends only on uniform bounds for  $C$  and its first two derivatives.  $\square$

## 7.4 Numerical check of the exact fixed point

The following values were computed at 70 decimal digits by summing the exact tilted mean

$$A(s) = \sum_{j \geq 1} \left( 1 - \frac{s/2^j}{e^{s/2^j} - 1} \right)$$

and solving  $x2^u = A(2^u)$ .

| $x$        | $u_0$            | $u_*$            | linear-correction error | one-step map error      |
|------------|------------------|------------------|-------------------------|-------------------------|
| $10^{-3}$  | 13.7468091666470 | 13.6868068298588 | $1.3742 \times 10^{-3}$ | $7.4062 \times 10^{-7}$ |
| $10^{-6}$  | 24.5491709694670 | 24.5175886975057 | $3.6440 \times 10^{-4}$ | $2.0539 \times 10^{-7}$ |
| $10^{-12}$ | 45.3666989202317 | 45.3501796089457 | $9.7317 \times 10^{-5}$ | $6.5109 \times 10^{-9}$ |

The leading displacement is negative because  $C(u) = -1/2 +$  a tiny periodic oscillation plus an exponentially small term.

## 8 All-orders sinc tails and centered Rvachev moments

### 8.1 Exact base-1/4 tail law

The finite product  $\Phi_m$  from (4.6) satisfies

$$\frac{\Phi(z)}{\Phi_m(z)} = \Phi(z/2^m). \quad (8.1)$$

**Theorem 8.1** (All-orders tail logarithm). *[N] If  $|z| < 2^m$  and*

$$Q_m(z) = \frac{z^2}{4^{m-1}}, \quad (8.2)$$

*then*

$$\log \frac{\Phi(z)}{\Phi_m(z)} = - \sum_{r \geq 1} \frac{\zeta(2r)}{r(4^r - 1)} Q_m(z)^r. \quad (8.3)$$

*Writing*

$$\frac{\Phi(z)}{\Phi_m(z)} = \sum_{N \geq 0} a_N Q_m(z)^N, \quad a_0 = 1, \quad (8.4)$$

*the coefficients satisfy*

$$N a_N = - \sum_{r=1}^N \frac{\zeta(2r)}{4^r - 1} a_{N-r}. \quad (8.5)$$

*The first terms are*

$$1 - \frac{\pi^2}{18} Q + \frac{19\pi^4}{16200} Q^2 - \frac{583\pi^6}{42865200} Q^3 + \frac{132809\pi^8}{1311675120000} Q^4 + \dots \quad (8.6)$$

*Proof.* Use (3.8) with  $q = 1/4$  and replace  $z$  by  $z/2^m$ . The coefficient recurrence follows by logarithmic differentiation of (8.4).  $\square$

## 8.2 Centered cumulants and a Bernoulli–Mersenne recurrence

Let

$$c_N = \int_{-1}^1 x^{2N} \operatorname{up}(x) \, dx = \mathbb{E} Y^{2N}. \quad (8.7)$$

**Theorem 8.2** (Tail coordinates for centered Rvachev moments). *[E] The corpus’s centered law has cumulants*

$$\kappa_{2r}(Y) = \frac{2^{2r-1} B_{2r}}{r(4^r - 1)}, \quad \kappa_{2r+1}(Y) = 0, \quad (8.8)$$

*and its centered moments  $c_N$  are already used in the exact dyadic theory. [N] In the spectral coordinates of theorem 8.1, those moments satisfy the alternative Bernoulli–Mersenne recurrence*

$$c_N = \frac{1}{N} \sum_{r=1}^N \binom{2N}{2r} \frac{2^{2r-1} B_{2r}}{4^r - 1} c_{N-r}, \quad (8.9)$$

*and the tail coefficient is exactly*

$$a_N = \frac{(-1)^N c_N \pi^{2N}}{(2N)!}. \quad (8.10)$$

*Proof.* Specialize theorem 3.6 to  $q = 1/4$ . The recurrence is the corresponding specialization of (3.12). Finally

$$\Phi(z/2^m) = \sum_{N \geq 0} \frac{(-1)^N c_N}{(2N)!} \left( \frac{\pi z}{2^{m-1}} \right)^{2N},$$

and comparison with (8.4) gives (8.10).  $\square$

The first moments are

$$c_0 = 1, \quad c_1 = \frac{1}{9}, \quad c_2 = \frac{19}{675}, \quad c_3 = \frac{583}{59535}, \quad c_4 = \frac{132809}{32531625}, \quad c_5 = \frac{46840699}{24405225075}. \quad (8.11)$$

The variance is  $1/9$ , and the fourth cumulant is  $-2/225$ .

*Remark 8.3.* The corpus already defines the centered moments, gives their hyperbolic-sine generating product, tabulates the values in (8.11), and proves a different Bernoulli recurrence. The new contribution here is narrower: the general- $q$  derivation, the alternative cumulant-form recurrence (8.9), the exact identification of these established moments with the base-1/4 sinc-tail coefficients, and the certified Thue–Morse acceleration below.

### 8.3 Certified Thue–Morse acceleration

Combining the finite triple factorization with the moment tail gives an exact hybrid representation.

**Theorem 8.4** (Thue–Morse spectral reconstruction). *[N] For every  $m \geq 1$  and  $z \in \mathbb{C}$ ,*

$$\begin{aligned} \Phi(z) &= \frac{2^{m(m-3)/2} e^{2\pi i z(1-2^{-m})}}{(i\pi z)^m} S_m(z) \\ &\times \sum_{N=0}^{\infty} \frac{(-1)^N c_N}{(2N)!} \left( \frac{\pi z}{2^{m-1}} \right)^{2N}. \end{aligned} \quad (8.12)$$

For real  $z$ , truncation after  $N = M$  has the certified error

$$\begin{aligned} \left| \Phi(z) - \Phi_m(z) \sum_{N=0}^M \frac{(-1)^N c_N}{(2N)!} \left( \frac{\pi z}{2^{m-1}} \right)^{2N} \right| \\ \leq \frac{(\pi|z|/2^{m-1})^{2M+2}}{(2M+2)!}. \end{aligned} \quad (8.13)$$

*Proof.* Multiply (4.7) by the moment series for  $\Phi(z/2^m)$ . For real  $z$ , interpret the tail as  $\mathbb{E} \cos((\pi z/2^{m-1})Y)$  and use the Lagrange remainder for cosine together with  $|Y| \leq 1$  and  $|\Phi_m(z)| \leq 1$ .  $\square$

**Example 8.5** (Numerical acceleration). At  $z = 10$  and  $m = 6$ , the exact tail is

$$\Phi(10/64) = 0.94753140589767634644 \dots$$

The errors after retaining  $M$  moment corrections, together with the rigorous bound (8.13), are

| $M$ | actual tail error        | certified upper bound    |
|-----|--------------------------|--------------------------|
| 0   | $5.24686 \times 10^{-2}$ | $4.81914 \times 10^{-1}$ |
| 1   | $1.07744 \times 10^{-3}$ | $3.87069 \times 10^{-2}$ |
| 2   | $1.20907 \times 10^{-5}$ | $1.24356 \times 10^{-3}$ |
| 3   | $8.69394 \times 10^{-8}$ | $2.14032 \times 10^{-5}$ |

Thus a finite signed Thue–Morse sum, corrected by only a few rational moments, gives a rapid and rigorously controlled evaluation of the infinite sinc product.

## 9 How the new identities reorganize the existing theory

### 9.1 A commutative diagram of transforms

The central relationships can be read as the following chain:

$$\begin{array}{ccccc} \Theta(q) = \prod (1 - q^{2^j}) & \xrightarrow{q=e^{-s}} & \mathcal{E}(s) = -\log \Theta(e^{-s}) & \xrightarrow{\mathcal{M}} & \frac{\Gamma(w)\zeta(w+1)}{1-2^{-w}} \\ & & \updownarrow & & \downarrow \text{residues} \\ \Phi(z) = \prod \text{sinc}(\pi z/2^j) & \xrightarrow{\text{zero divisor}} & \mu(n) = v_2(n) + 1 & \xrightarrow{a(n)=2^{\mu(n)}-1} & P(\log_2 s). \end{array}$$

The periodic endpoint function is therefore not an unrelated analytic decoration. It is the residue shadow of a Thue–Morse logarithm whose coefficients can be read from the Rvachev spectral zero divisor.

### 9.2 Two compatible $q$ -geometries

The corpus’s exact dyadic arithmetic uses base  $q = 1/2$  because each spatial refinement changes the dyadic denominator by one power of two. The Fourier product uses base  $q = 1/4$  because its Taylor and Euler products are even in  $z$ , so the same spatial dilation becomes a squared spectral dilation. This explains why

- $(1/2; 1/2)_n$  and  $\left[\begin{smallmatrix} n \\ k \end{smallmatrix}\right]_{1/2}$  arise in exact Fabius values;
- $(z^2/\ell^2; 1/4)_\infty$ ,  $4^r - 1$ , and  $\left[\begin{smallmatrix} m \\ j \end{smallmatrix}\right]_{1/4}$  arise in the sinc product and centered moments.

A future theory should treat these not as competing normalizations but as the arithmetic and spectral faces of the same binary refinement.

### 9.3 Endpoint asymptotics from spectral data

The exact identity (5.10) gives a new route toward the endpoint expansion. Starting only with the Fourier zero multiplicities,

1. form  $a(n) = 2^{\mu(n)} - 1$ ;
2. build  $\mathcal{E}(s) = \sum a(n)e^{-ns}/n$ ;
3. Mellin transform to obtain (6.1);
4. read off the quadratic logarithm and all periodic modes;
5. insert them into the exact saddle map (7.14).

This does not yet derive every coefficient of the all-orders Fabius endpoint expansion, but it supplies the entire oscillatory datum and the exact periodic saddle displacement from the spectral zero divisor alone.

### 9.4 Relation to the Fourier-decay audits

The arithmetic product (4.1) is not by itself a pointwise asymptotic for  $|\Phi(x)|$ : near integers, zero multiplicities dominate, while extremal rays avoid zeros in highly structured ways. It nevertheless offers three tools for the audited decay problem:

1. exact zero multiplicities and cumulative zero counting;
2. a  $q$ -family in which the binary ratio can be varied continuously;
3. certified finite-shell corrections suitable for rigorous numerics and Lean formalization.

The separation of  $\kappa_\infty, \kappa_2, \kappa_1, \kappa_0$  remains essential.

## 10 Further frontier problems

### 10.1 High-priority formalization targets

The following statements appear especially suitable for Lean because their proofs reduce to finite products, absolutely convergent series, or elementary operator identities.

1. **Arithmetic canonical product:** formalize  $\Phi(z) = \prod_{n \geq 1} (1 - z^2/n^2)^{v_2(n)+1}$  and the integer zero multiplicity.
2. **Finite triple factorization:** combine the finite sinc product with the formal Thue–Morse polynomial and Prouhet cancellation.
3. **Endpoint Mahler identity:** prove  $\mathcal{E}(s) = -\log \Theta(e^{-s})$  and the 2-adic Lambert expansion.
4. **Spectral–endpoint coefficient theorem:** identify the Lambert coefficient with  $2^{\text{ord}_n \Phi} - 1$ .
5. **Centered moment recurrence:** derive (8.9) from the already formal sinc product.
6. **Periodic Lambert map:** package the exact saddle equation and derivative (7.15), using strict tilted variance for uniqueness.

The Mellin residue computation and Stieltjes-constant mean are analytically heavier but can be isolated after the elementary core is formalized.

## 10.2 A general- $q$ decay spectrum

*Conjecture 10.1* (Smooth dependence of decay exponents). [C] For the geometric product  $\Phi_q$ , the analogues of the extremal,  $L^2$ , Cesàro, and typical power exponents exist for every  $0 < q < 1$  and vary continuously, perhaps real-analytically away from resonance parameters at which zero lattices overlap.

The universal quadratic coefficient should be  $1/(2|\log \sqrt{q}|) = 1/|\log q|$ , as the elementary upper bound in [theorem 3.2](#) already suggests. The lower-order exponents should arise from transfer operators depending on  $q$ .

## 10.3 Bridging the bases $1/2$ and $1/4$

*Conjecture 10.2* (Arithmetic–spectral  $q$ -transform). [C] There is a direct coefficient transform carrying the base- $1/2$  Gaussian–binomial formulas for exact dyadic values of  $F$  to the base- $1/4$  Bernoulli–Mersenne moments of up, without passing through the full probability model.

A likely starting point is the identity  $Y = 2X - 1$ , followed by a binomial transform of the half moments, but a satisfactory formula should preserve the  $q$ -Pochhammer structure rather than expand it away.

## 10.4 A spectral derivation of the full endpoint expansion

*Conjecture 10.3* (Zero-divisor endpoint reconstruction). [C] The complete all-orders endpoint asymptotic of  $F(x)$ , including every periodic coefficient, can be reconstructed from the weighted zero zeta function  $\mathfrak{Z}_\Phi$  together with the local normalization  $\Phi(0) = 1$ .

The report reconstructs the quadratic logarithm and rederives the entire periodic function  $P$ , including its established mean, from the weighted zero data; it also derives the exact saddle displacement. The remaining challenge is to derive the full Gaussian saddle-jet algebra directly from spectral data.

## 10.5 Arithmetic nature of the periodic mean

Formula (6.5) introduces the combination

$$\frac{\pi^2}{12} - \gamma_1 - \frac{\gamma^2}{2}.$$

Its arithmetic nature is unknown. Even without transcendence information, the exact formula is useful: it separates the mean from the exponentially tiny nonzero modes and eliminates one numerical constant from the endpoint theory.

## 10.6 Weighted zero statistics

The ordinary multiplicity and endpoint-weighted multiplicity have Dirichlet series

$$\sum \frac{\mu(n)}{n^w} = \frac{\zeta(w)}{1 - 2^{-w}}, \quad \sum \frac{2^{\mu(n)} - 1}{n^w} = \frac{\zeta(w)}{1 - 2^{1-w}}.$$

The second has a double singularity at  $w = 1$  after multiplication by the pole of  $\zeta$ , which is the arithmetic source of the quadratic logarithm in the endpoint correction. A Tauberian development of its summatory function, with explicit binary periodic fluctuation, should give an elementary real-variable route to the Mellin result.

## 11 Conclusions

The inspected corpus already contains an unusually complete and carefully status-separated theory of the Fabius and Rvachev functions. The main contribution of this report is not a new isolated formula but a new exact dictionary:

1. the infinite sinc product is a base- $1/4$   $q$ -Pochhammer product;
2. its integer zero multiplicity is  $v_2(n) + 1$ ;
3. the Thue–Morse endpoint logarithm weights that multiplicity by  $2^\mu - 1$ ;
4. the resulting Dirichlet series has Mellin transform  $\Gamma(w)\zeta(w+1)/(1 - 2^{-w})$ ;

5. the residues reproduce the endpoint’s full established log-periodic spectrum, including its formalized mean, through a new Thue–Morse/Mellin route;
6. the same periodic function enters an exact lower- $W_{-1}$  fixed point for the saddle;
7. finite Thue–Morse sums, corrected by centered Bernoulli–Mersenne moments, give certified approximations to the full Fourier product.

This unifies the Thue–Morse,  $q$ -binomial, sinc-product, endpoint, and Lambert- $W$  themes without erasing their different normalizations. It also leaves a concrete formalization program and several testable frontier questions.

## A Supplementary derivations

### A.1 The coefficient formula by partitions

From the cumulant generating function, the centered moments also admit the explicit partition sum

$$c_N = (2N)! \sum_{\substack{m_1, \dots, m_N \geq 0 \\ \sum_{r=1}^N r m_r = N}} \prod_{r=1}^N \frac{1}{m_r!} \left[ \frac{2^{2r-1} B_{2r}}{r(4^r - 1)(2r)!} \right]^{m_r}. \quad (\text{A.1})$$

This makes rationality immediate and displays the product of Mersenne-type factors  $4^r - 1$ . Since

$$\prod_{r=1}^N (4^r - 1) = 4^{N(N+1)/2} (1/4; 1/4)_N,$$

the denominator geometry is again governed by a base-1/4  $q$ -Pochhammer symbol.

**Post-snapshot Lean status.** The universal algebraic shape of (A.1) is now formalized in `ExponentialPartition.lean`: with

$$E_r = \frac{2^{2r-1} B_{2r}}{r(4^r - 1)(2r)!},$$

`partitionExpSum_eq_sum_div` identifies `partitionExpSum E N` with the displayed partition sum before multiplication by  $(2N)!$ . `ExponentialBell.lean` further identifies that coefficient function with `SaddleExpansion.expCoeff`. `MomentCumulantAlgebra.lean` supplies the corresponding generic complete-Bell and inverse moment–cumulant transforms, their recurrence, and normalized inverse laws. These are universal coefficient identities only: the displayed cumulant formula, its identification with  $c_N$ , and the specialized Bernoulli–Mersenne recurrence remain separate formal obligations. The [N] classifications above therefore remain unchanged.

### A.2 A direct proof of the first-harmonic scale

For  $k \neq 0$ , Stirling’s formula on vertical lines gives

$$|\Gamma(-\chi_k)| \asymp |\chi_k|^{-1/2} e^{-\pi|\chi_k|/2}.$$

Since  $|\chi_1| = 2\pi/L \approx 9.0647$ , the gamma factor alone contributes an exponential suppression near  $e^{-14.24}$ . The zeta factor is only of polynomial size. This explains the extraordinary numerical smallness of the periodic modes; their nonvanishing follows separately from the zero-free line  $\Re s = 1$  for the Riemann zeta function.

### A.3 The unperturbed and perturbed Lambert equations

The familiar lower-Lambert phase solves

$$x2^{u_0} = u_0, \quad u_0 = -L^{-1}W_{-1}(-Lx).$$

The exact phase replaces  $u_0$  on the right by  $u + C(u)$ . The map (7.14) is not an ad hoc iteration: it solves the Lambert part exactly while freezing only the bounded periodic perturbation. Its derivative (7.15) is therefore of order  $1/u$ , explaining the rapid numerical convergence seen in the table.

## B Recursive source inventory

### B.1 Living documents: five files

1. FabiusFunction\_Mathematical\_Glossary/FabiusFunction\_Mathematical\_Glossary.tex
2. Fabius\_Function\_and\_Rvachev\_Up/Fabius\_Function\_and\_Rvachev\_Up.tex
3. Non\_Elementarity\_of\_the\_Fabius\_Function/Non\_Elementarity\_of\_the\_Fabius\_Function.tex
4. fabius\_lean\_walkthrough/fabius\_lean\_walkthrough.tex
5. semi-formalized-research-frontiers/semi-formalized-research-frontiers.tex

### B.2 Active frontier drafts: thirteen files

- Atlas file: semi-formalized-research-frontiers/drafts/Thue\_Morse\_Formula\_Atlas/Thue\_Morse\_Formula\_Atlas.tex.
- Twelve one-file directories under semi-formalized-research-frontiers/drafts/rvachev\_up\_fourier\_decay/:
  - Rvachev\_Up\_Fourier\_Decay-2/
  - Rvachev\_Up\_Fourier\_Decay-3/
  - Rvachev\_Up\_Fourier\_Decay\_Comparative\_Audit/
  - Rvachev\_Up\_Fourier\_Decay\_Gentle\_Guide/
  - Rvachev\_Up\_Fourier\_Decay\_Second\_Wave\_Audit/
  - asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions/
  - rvachev\_up\_fourier\_decay-1/
  - rvachev\_up\_fourier\_decay-4/
  - rvachev\_up\_fourier\_decay-5/
  - rvachev\_up\_fourier\_decay-6/
  - rvachev\_up\_fourier\_decay-7/
  - rvachev\_up\_fourier\_decay-8/

Together these are one Atlas file plus twelve Fourier-decay files.

### B.3 Frozen archive: thirty-two files

The archive contains one historical .tex per entry. Its README states that byte-identical duplicates were pruned and that the retained entries are frozen first or special historical versions. The entries are:

| Archive group       | Entry                                                                                                                                                                               |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Early notes         | FabiusAsymptotic; K-foldsummationoverthesignedThue-Morsesequence                                                                                                                    |
| Standalone studies  | Non_Elementarity_of_the_Fabius_Function; Small_Argument_Asymptotics; fabius-inverse-dyadic-close-d-form                                                                             |
| Primary exposition  | Fabius_Function_and_Rvachev_Up; Fabius_Function_and_Rvachev_Up-2; Fabius_Function_and_Rvachev_Up-3; Fabius_and_Rvachev_Up-2; Fabius_and_Rvachev_up; Fabius_function_and_up_function |
| Primary supplements | Fabius_Function_Missing_Parts; Fabius_Function_Missing_Parts-2                                                                                                                      |



| Archive group      | Entry                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $q$ -calculus      | Fabius_q_Special_Functions; fabius-q-special-functions                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Lean walkthrough   | Fabius_Function_Lean_Walkthrough; fabius_lean_walkthrough                                                                                                                                                                                                                                                                                                                                                                                                                |
| Glossary           | FabiusFunction_Mathematical_Glossary; FabiusFunction_Mathematical_Glossary-2                                                                                                                                                                                                                                                                                                                                                                                             |
| Research frontiers | Fabius_Dyadic_Asymptotic_Bridge; Fabius_Dyadic_Formulae_and_Alternative_Representations; Fabius_Dyadic_Formulae_to_Asymptotics; Fabius_Dyadic_q_Connections; Fabius_Thue_Morse_Convergence_Rate; Fabius_Thue_Morse_Convergence_Rates; Primary_Exposition_Gap_Register; Repeated_Integration_and_Rvachev_Up; Rvachev_Up_from_Repeated_Integration; semi-formalized-research-frontiers; Thue_Morse_Formula_Atlas-1; Thue_Morse_Formula_Atlas-2; Thue_Morse_Formula_Atlas-3 |

#### B.4 Source-paper transcriptions: seven files

1. AIF\_2017\_\_67\_2\_637\_0/AIF\_2017\_\_67\_2\_637\_0-corrected.tex
2. AIF\_2017\_\_67\_2\_637\_0/AIF\_2017\_\_67\_2\_637\_0.tex
3. Lacunary\_products/Lacunary\_products.tex
4. 09-Function/09-Function.tex
5. 157-Arithmetic-v3/157-Arithmetic-v3.tex
6. document/document.tex
7. ubiq15/ubiq15-corrected.tex

### C Notation crosswalk

| Symbol           | Meaning                                                        |
|------------------|----------------------------------------------------------------|
| $F$              | bounded Fabius distribution function                           |
| $\text{up}$      | Rvachev's even compactly supported density                     |
| $X$              | $\sum_{j \geq 1} 2^{-j} U_j$ , supported on $[0, 1]$           |
| $Y$              | $2X - 1$ , with density $\text{up}$                            |
| $\Phi$           | $\widehat{\text{up}}$ , the binary infinite sinc product       |
| $\varepsilon_n$  | signed Thue–Morse value $(-1)^{\text{wt}_2(n)}$                |
| $\Theta(q)$      | Thue–Morse Mahler product $\sum \varepsilon_n q^n$             |
| $B, \Lambda$     | negative Laplace transform of $X$ and its logarithm            |
| $T, \mathcal{E}$ | forward logarithmic tail and positive endpoint correction      |
| $P$              | smooth one-periodic term in the exact Laplace decomposition    |
| $R$              | $sT'(s)$ , the exponentially small derivative correction       |
| $A$              | $-s\Lambda'(s) = s\mathbb{E}_s X$                              |
| $\Pi$            | $-1/2 - P'/L$ , the periodic part of $A - \log_2 s$            |
| $C$              | $\Pi(u) + R(2^u)$ , the exact bounded saddle perturbation      |
| $\mu(n)$         | integer-zero multiplicity $\text{ord}_{z=n} \Phi = v_2(n) + 1$ |

## References

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- [5] V. Reshetnikov, *The Decay Envelope of the Rvachev Sinc Product: Second-Wave Audit*, active draft, ProveIt repository, 27 August 2026. [https://github.com/VladimirReshetnikov/ProveIt/tree/2e3567feb14947ee3ebcdab11adca64e746ad26f/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/rvachev\\_up\\_fourier\\_decay/Rvachev\\_Up\\_Fourier\\_Decay\\_Second\\_Wave\\_Audit](https://github.com/VladimirReshetnikov/ProveIt/tree/2e3567feb14947ee3ebcdab11adca64e746ad26f/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Second_Wave_Audit)
- [6] J. Arias de Reyna, *An infinitely differentiable function with compact support: Definition and properties*, English translation of the 1982 article; transcription included in the ProveIt documentation corpus.
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## Part X

# Beyond the Dyadic Fabius Web

### Abstract

The dyadic Fabius function, its inverse, Rvachev’s compactly supported up-function, and the signed Thue–Morse extension form an unusually dense network of exact identities. The inspected repository already develops that network at considerable depth: exact dyadic values, Gaussian–binomial coefficient extraction, Bernoulli and Bell polynomial formulas, Lambert- $W$  endpoint phases, geometric Lagrange interpolation, Richardson filters, and an extensive campaign on the decay of the infinite sinc product. This report therefore does not attempt another rearrangement of those results. Instead it moves in a transverse direction by replacing the dyadic ratio  $1/2$  by a free geometric parameter  $0 < q < 1$ .

For independent  $U_j \sim \text{Unif}(0, 1)$ , define

$$X_q = (1 - q) \sum_{j \geq 0} q^j U_j, \quad F_q(x) = \mathbb{P}(X_q \leq x), \quad Y_q = 2X_q - 1.$$

At  $q = 1/2$  this is exactly the bounded Fabius distribution, and the density of  $Y_{1/2}$  is Rvachev’s up-function. For the full family we prove a refinement equation and a rapidly decaying infinite sinc product; exact Bernoulli cumulants and Bell moments; a Bernoulli–Appell polynomial inverse to geometric uniform averaging; a finite-stage  $q$ -binomial scale identity; monotone contraction in convex order; an exact shape phase transition with a central plateau precisely when  $q < 1/2$ ; a Gaussian degeneration as  $q \uparrow 1$  with explicit standardized cumulants; a two-term logarithmic endpoint law

$$\log F_q(x) = -\frac{\log^2(1/x)}{2 \log(1/q)} - \frac{\log(1/x) \log \log(1/x)}{\log(1/q)} + \mathcal{O}_q(\log(1/x));$$

and its inverse Lambert- $W$  quantile law. We also determine Fourier-zero multiplicities for reciprocal-integer ratios, isolate the arithmetic condition for cross-scale collisions, and explain why base two alone converts the finite product into a contiguous Thue–Morse block. Finally, we give a phase-preserving Lagrange/Richardson filter for asymptotic remainders whose coefficients are periodic in a lower-Lambert phase, together with a Fourier phase-averaging filter and exact Lambert- $W$  node construction.

The weighted-law, CDF, and affine fixed-point portion of the post-snapshot Lean audit, checked at compiled source checkpoint d312c0603, now supplies the generic continuous-linear naturality, conditional operator/scalar affine-recursion, real absolute-continuity, and exact-support engines for arbitrary norm-summable weighted uniform series. It also defines this report’s geometric- $q$  weights, series, law, CDF, and explicit positive-ratio density. For  $|q| < 1$ , the real CDF is continuous, obeys reflection, and takes value  $1/2$  at the midpoint. For  $0 \leq q < 1$ , its exterior tails are exactly zero and one; for  $0 < q < 1$ , the two conditioning integrals, pointwise derivative formula, Radon–Nikodym density, density reflection and compact support, and  $C^\infty$  regularity of both CDF and density are formal. At  $q = 1/2$ , exact all-real declarations identify the CDF with the bounded Fabius function and the density with the affine Rvachev up-function. A new generic affine-independent-copy API proves probability fixed-point uniqueness on every complete second-countable real inner-product target for a measurable digit map and  $|q| < 1$ , and its geometric corollary characterizes this report’s law even at  $q = 0$  and negative  $q$ , with no support, density, or moment assumptions.

A later mainline tranche, rechecked in the current combined full-facade build, gives the phase-bearing sinc prefixes an exact finite residual formula and proves their pointwise convergence to the characteristic function for every  $|q| < 1$ . The latest analytic tranche now closes the corresponding named product bridge. For  $q, z \in \mathbb{C}$ ,  $\|q\| < 1$ , write  $S_q(z)$  for `geometricSincProduct` applied to  $q, z$ . Then  $S_q(z) = \prod_{n \geq 0} \text{sinc}(\pi q^n z)$  has locally uniform product convergence, genuine pointwise multipliability and `HasProd`, and is entire, by `hasProdLocallyUniformly_geometricSincProduct`, `geometricSincProductFactors_multipliable`, `hasProd_geometricSincProduct`, and `geometricSincProduct_differentiable`. For real  $|q| < 1$ , including  $q = 0$  and negative contractions, and  $t \in \mathbb{R}$ , putting  $z_q(t) = (1 - q)t/(2\pi)$  gives the exact formula  $\phi_q(t) = e^{it/2} S_q(z_q(t)) = e^{it/2} G_q(z_q(t)) G_q(-z_q(t))$ , where  $G_q(z)$  denotes `geometricReciprocalGamma` applied to  $q, z$ , through `charFun_geometricUniformDistribution_eq_phase_mul_geometricSincProduct` and `charFun_geometricUniformDistribution_eq_phase_mul_geometricReciprocalGamma`. Independently, for every nonzero such  $q$ , direct named Cauchy-transform and adjacent-power refinement equations hold off the exact support. A packaged compact-uniform theorem for the full phase-bearing characteristic prefixes, a named centered wrapper, the report’s rapid-decay and inversion proof, explicit Bernoulli-cumulant/Bell-moment formulas and asymptotics, further transforms, higher-derivative and endpoint-flatness wrappers, and shape theory remain outside that promotion.

The word *new* is used only in a corpus-relative sense: these statements were not found in the 57 TeX sources inspected in the specified repository snapshot. No claim of global priority is made. Proven statements are separated sharply from conjectural transseries, strict log-concavity, Edgeworth, inverse-phase, and arithmetic-collision problems. Deterministic numerical experiments and all plotting code accompany the report.

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**Status convention.** A theorem or proposition in this report is accompanied by an argument intended to be mathematically complete at the paper level. A conjecture is explicitly labeled and is not asserted as proved. None of the new statements has been formalized in Lean as part of the repository snapshot. The principal novelty filter was comparison against all 57 TeX files listed in `corpus_inventory.txt`, including frozen versions, the living syntheses, the Thue–Morse atlas, all included paper sources, and the twelve Fourier-decay drafts and audits.

Since that pinned snapshot, and as checked at compiled source checkpoint `d312c0603`, `WeightedUniformSeries.lean`, `WeightedUniformDistribution.lean`, and `WeightedUniformSupport.lean` have formalized the reusable product space, absolutely summable Banach-valued uniform-coordinate series, and their pushforward laws, including the eleven-declaration generic naturality and operator-affine recursion tranche, the compatibility API, the four-declaration real absolute-continuity tranche, and the ten-declaration exact-support tranche. In particular, support equals the series range in a complete Borel real normed space, arbitrary signed real weights have the exact interval between the sums of their negative and positive parts, and every nontrivial norm-summable real weighted law is absolutely continuous with null singletons. The specialization `GeometricUniformLaw.lean` defines  $w_j(q) = (1 - q)q^j$  and  $X_q$  for every real  $q$ , proves absolute convergence and total mass one for  $|q| < 1$ , and establishes the independent-copy affine refinement, law reflection, absolute continuity, null singletons, and support equal to the exact series range. For  $0 \leq q < 1$  it proves that the range and support are exactly  $[0, 1]$ , including  $q = 0$ ; at  $q = 1/2$ , `ProbabilityRepresentation.lean` identifies both the random series and its law with the classical dyadic construction and proves its exact support is  $[0, 1]$ .

The later modules `ContinuousCDF.lean` and `GeometricUniformCDF.lean` now define the named real CDF and explicit density and formalize the precise parameter ranges stated in the abstract: continuity, reflection, and the midpoint value for  $|q| < 1$ ; exterior CDF values for  $0 \leq q < 1$ ; and, for  $0 < q < 1$ , conditioning in subtype and ordinary interval-integral form, an everywhere pointwise derivative, the resulting Radon–Nikodym identity, density symmetry and compact support, and  $C^\infty$  regularity. `ProbabilityRepresentation.lean` supplies the all-real  $q = 1/2$  Fabius-CDF and Rvachev-density bridges.

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**Post-snapshot formal status (continued).**

The new module `AffineIndependentCopy.lean` adds one definition and eight theorems: the affine independent-copy law, probability preservation, its equality with the direct product pushforward, the one-step characteristic-function factorization and fixed-point recurrence, its finite iterate, uniqueness from a common contractive characteristic-function recurrence, and uniqueness in convolution and product-map forms. The uniqueness scope is an arbitrary complete second-countable real inner-product target with its Borel structure, an arbitrary probability digit space and measurable digit map, probability candidate laws, and a real scalar satisfying  $|q| < 1$ . It includes  $q = 0$  and negative  $q$ , and assumes no support, absolute continuity, density, or moments. The corollary `eq_geometricUniformDistribution_of_selfSimilar` in `GeometricUniformUniqueness.lean` therefore characterizes the named geometric-uniform law by its displayed affine independent-copy equation throughout that full parameter range.

The definition of `geometricUniformDensity` is total, but no density formula or Radon–Nikodym interpretation is claimed here for  $q \leq 0$ . Fixed-point uniqueness is now the direct theorem `eq_geometricUniformDistribution_of_selfSimilar`, valid for every  $|q| < 1$ , including negative  $q$  and  $q = 0$ . The definitions `geometricUniformStieltjesDomain`, `geometricUniformStieltjesTransform`, and `geometricUniformStieltjesPower` are total in real  $q$ ; for  $|q| < 1$ , the transform is holomorphic off the exact support, and for  $q \neq 0$  the direct named theorems `hasDerivAt_geometricUniformStieltjesTransform_refinement` and `geometricUniformStieltjesPower_succ` give the transform DDE and every positive adjacent-power recurrence, including negative ratios. At  $q = 1/2$ , the all-power affine bridge, named unit Stieltjes DDE, and both canonical complex-slit adjacent-order recurrences are direct theorems of `CauchyTransform.lean`.

The finite characteristic-function residual formula and its pointwise prefix limit are now formal, and the product target is no longer merely a total `tprod`. For  $q, z \in \mathbb{C}$  with  $\|q\| < 1$ , the four declarations `hasProdLocallyUniformly_geometricSincProduct`, `geometricSincProductFactors_multipliable`, `hasProd_geometricSincProduct`, and `geometricSincProduct_differentiable` prove locally uniform convergence of  $S_q(z) = \prod_{n \geq 0} \text{sinc}(\pi q^n z)$ , its genuine pointwise product, and entire-ness. For real  $|q| < 1$ ,  $t \in \mathbb{R}$ , and  $z_q(t) = (1 - q)t/(2\pi)$ , including  $q = 0$  and negative ratios, `charFun_geometricUniformDistribution_eq_phase_mul_geometricSincProduct` and `charFun_geometricUniformDistribution_eq_phase_mul_geometricReciprocalGamma` give

$$\phi_q(t) = e^{it/2} S_q(z_q(t)) = e^{it/2} G_q(z_q(t)) G_q(-z_q(t)).$$

Here  $G_q(z)$  denotes `geometricReciprocalGamma` applied to  $q, z$ . Still frontier are a packaged compact-uniform theorem for the full phase-bearing characteristic prefixes and a named centered wrapper, the rapid-decay bound and Fourier inversion route, a closed higher-derivative refinement and endpoint-flatness wrapper, the shape transition, further transforms, and explicit Bernoulli-cumulant/Bell-moment formulas and asymptotics. The geometric tail dictionary already gives finite characteristic-function, MGF, and CGF tail factorizations. The closed density hierarchy here is distinct from the proved Cauchy-power hierarchy.

# 1 Corpus audit and the novelty boundary

## 1.1 What was inspected

The requested tree contains four logically different kinds of material. First are the living synthesis documents: the primary Fabius–Rvachev exposition, the mathematical glossary, the Lean walkthrough, the non-elementarity article, the consolidated frontier volume, and the living Thue–Morse formula atlas. Second is a twelve-document Fourier-decay campaign containing proposed analyses, final syntheses, a gentle guide, and two comparative audits. Third is a seven-source paper library, including both the original and corrected TeX of one OCR conversion. Fourth is a frozen archive of thirty-two historical versions organized by provenance and topic. The resulting count is

$$6 + 12 + 7 + 32 = 57.$$

The exact paths are reproduced in [Section A](#) and in the companion plain-text ledger. The repository’s own archive README states that byte-identical historical copies were pruned and that the archive is grouped by purpose rather than serving as a naive mirror of every earlier path [22]. Thus the historical sources were treated as a novelty/provenance map, while the current synthesis documents were treated as the canonical statement of what the project already contains.

## 1.2 What the repository already does very well

The current primary exposition and frontier synthesis already cover, among much else,

- the probabilistic construction of the Fabius distribution and the Rvachev bump;
- exact dyadic evaluation, recurrences, bit-recursive algorithms, and denominator arithmetic;
- Gaussian-binomial and  $q$ -Pochhammer structures at the fixed base  $q = 1/2$ ;
- Bernoulli cumulants, Bell-polynomial moments, Appell-type blocks, and exact coefficient extraction;
- geometric-node Lagrange identities and  $q$ -Richardson cancellation of geometric error modes;
- lower-Lambert endpoint variables, nonconstant periodic correction functions, and all-orders saddle machinery;
- the inverse Fabius function, totalization conventions, and dyadic inverse formulas;
- repeated-integration and box-spline approximants;
- extensive Thue–Morse identities, rarefied sums, Mahler equations, natural-boundary phenomena, and quantitative approximation questions;
- the quadratic logarithmic envelope, exceptional zeros, and several amplitude notions for the Fourier image of the up-function.

These features are visible in the primary exposition [18], the consolidated frontier volume [19], the living atlas [20], and the Fourier audit campaign [21]. The fixed-base  $q$ -calculus is especially important for the present report: the archive explicitly describes those documents as *base-1/2* articles [22]. Likewise, the archive records an existing “phase locking” mechanism for dyadic coefficient extraction. Accordingly, neither Gaussian-binomial notation nor phase locking by itself is claimed as new below.

## 1.3 Deliberate exclusions

Three tempting projects were rejected because they would duplicate the corpus too closely.

1. We do not offer a thirteenth derivation of the leading Fourier decay  $\log |\widehat{\text{up}}(t)| \asymp -(\log |t|)^2 / (2 \log 2)$  with the known zero-set and limsup qualifications. The live Fourier drafts compare precisely such arguments.
2. We do not repackage the exact dyadic value web, the half-base Gaussian-binomial interpolation formula, or the existing geometric Richardson filter.
3. We do not relabel the existing dyadic lower-Lambert periodic asymptotic as a new “transseries.” The new phase-aware extrapolation theorem in [Section 9](#) applies to a different numerical geometry: inverse powers of a Lambert variable while its fractional phase is held fixed.



## 1.4 The transverse direction

The central observation is that the dyadic distribution is one member of a natural geometric one-parameter family. Making the ratio variable simultaneously exposes:

- a shape transition at the dyadic value  $q = 1/2$ ;
- a majorization order in the parameter;
- a Gaussian limit as the geometric mesh becomes diffuse;
- a general Bernoulli–Bell calculus whose dyadic specialization recovers the repository’s formulas exactly;
- a change in Fourier-zero arithmetic from generic simple zeros to valuation-controlled multiplicities;
- a precise reason that the Thue–Morse block is special to base two.

This parameter deformation is the organizing principle of the report.

## 2 The geometric $q$ -Fabius–Rvachev family

Throughout, fix

$$0 < q < 1, \quad c = 1 - q, \quad a = \log(1/q) > 0.$$

Let  $(U_j)_{j \geq 0}$  be independent random variables, each uniform on  $[0, 1]$ .

**Definition 2.1** (Geometric Fabius law). Define

$$X_q = c \sum_{j=0}^{\infty} q^j U_j, \quad Z_q = X_q - \frac{1}{2}, \quad Y_q = 2Z_q = 2X_q - 1. \quad (2.1)$$

Write  $F_q$  for the extended cumulative distribution function of  $X_q$ ,  $f_q$  for its density, and  $u_q$  for the density of  $Y_q$ .

Because  $0 \leq U_j \leq 1$  and  $c \sum q^j = 1$ , the series converges absolutely almost surely and  $X_q \in [0, 1]$ . Replacing every  $U_j$  by  $1 - U_j$  shows

$$X_q \stackrel{d}{=} 1 - X_q, \quad F_q(x) = 1 - F_q(1 - x), \quad f_q(x) = f_q(1 - x). \quad (2.2)$$

The support is all of  $[0, 1]$ : every open subinterval of  $[0, 1]$  receives positive probability by constraining finitely many  $U_j$  and leaving a sufficiently short tail free.

**Current formalization boundary.** The declarations `hasSum_geometricUniformWeight`, `geometricUniformSeries_mem_Icc`, and `geometricUniformDistribution_Icc` formalize the series, its unit total weight, and concentration on  $[0, 1]$ . The stronger assertion in the preceding sentence is now captured by `geometricUniformDistribution_support_eq_Icc`: exact topological support is  $[0, 1]$ . The positive-mass formulation follows from the standard characterization of support, rather than from a separately named interval-mass theorem. More generally, `geometricUniformDistribution_support_eq_range` identifies support with the exact series range for every  $|q| < 1$ . Likewise `geometricUniformDistribution_reflection` proves  $X_q \stackrel{d}{=} 1 - X_q$ . The measure is absolutely continuous by `geometricUniformDistribution_absolutelyContinuous`, and `geometricUniformDistribution_nullSingletonClass` packages its null singletons, both for every  $|q| < 1$  without a positivity hypothesis. The generic theorem `cdf_reflection_sub` and its specialization `geometricUniformCDF_reflection` now package the displayed CDF symmetry on all of  $\mathbb{R}$ , and `geometricUniformCDF_one_half` gives the midpoint value, again for every  $|q| < 1$ . On  $0 \leq q < 1$ , `geometricUniformCDF_zero_of_nonpos` and `geometricUniformCDF_one_of_one_le` give the exact exterior values. For  $0 < q < 1$ , `geometricUniformDensity_reflection` packages the displayed density symmetry for the explicit density `geometricUniformDensity`; its Radon–Nikodym interpretation and regularity are recorded in the crosswalk below. Independently of those positive-ratio density results, `eq_geometricUniformDistribution_of_selfSimilar` proves that the affine independent-copy equation uniquely determines the probability law for every  $|q| < 1$ , including  $q = 0$  and negative  $q$ , without any support, density, or moment hypothesis on the competing measure.



**Exact public API crosswalk.** The one-definition, eight-theorem contractive layer in `AffineIndependentCopy.lean` lies in the namespace `ProbabilityTheory`. Its public declarations are `affineIndependentCopyLaw`, `affineIndependentCopyLaw_isProbabilityMeasure`, `affineIndependentCopyLaw_eq_map_prod`, `charFun_affineIndependentCopyLaw`, `charFun_eq_mul_charFun_of_affineIndependentCopy_fixedPoint`, `charFun_iterate_of_affineIndependentCopy_fixedPoint`, `eq_of_charFun_affine_recurrence`, `affineIndependentCopyLaw_fixedPoint_unique`, and `affineIndependentCopy_map_fixedPoint_unique`. They define

$$T_{\rho,d,q}(\mu) = d_*\rho * (q \cdot)_*\mu,$$

identify it with the pushforward of  $\rho \otimes \mu$  by  $(a, x) \mapsto d(a) + qx$ , prove probability preservation and characteristic-function factorization, iterate the fixed-point recurrence to  $q^n t$ , and prove uniqueness. The final uniqueness theorems assume a complete second-countable real inner-product target with its Borel structure, a probability digit space with measurable digit map, probability candidate laws, and  $|q| < 1$ ; they need no support, absolute-continuity, density, or moment hypothesis and include  $q = 0$  and negative contractions.

The remaining weighted-series names in this crosswalk lie in the namespace `Fabius.ProbabilityRepresentation`. The generic almost-everywhere recursion assumes measurability of the tail statistic and combining map but not of the recursively described function; its pointwise wrapper is separate. Continuous-linear naturality requires completeness only in the source, pointwise scalar naturality needs neither completeness nor summability, and law-level identities retain the Borel and norm-summability hypotheses. The eleven generic declarations are `uniformProduct_map_of_ae_head_tail_recursion`, `uniformProduct_map_of_head_tail_recursion`, `weightedUniformSeries_map`, `weightedUniformSeries_smul`, `weightedUniformDistribution_map`, `weightedUniformDistribution_smul`, `weightedUniformDistribution_map_eq_shift`, `weightedUniformSeries_split_of_shift_eq_map`, `weightedUniformDistribution_split_of_shift_eq_map`, `weightedUniformSeries_geometric_split`, and `weightedUniformDistribution_geometric_split`. The dyadic-to-generic law bridge is `weightedSumDistribution_eq_weightedUniformDistribution`.

The compatibility surface consists of `weightedUniformSeries_smul_weights`, `uniformProduct_map_ae_head_tail_function`, `weightedUniformDistribution_isProbabilityMeasure`, `weightedUniformDistribution_smul_weights`, `uniformProduct_map_head_tail_weightedUniformSeries`, `weightedUniformDistribution_unitInterval`, and `weightedUniformDistribution_comp1_unitInterval`. In particular, `weightedUniformDistribution_smul_weights` is oriented as  $\mu_{cw} = (x \mapsto cx)_*\mu_w$ .

The four declarations in the real absolute-continuity tranche are `uniformScaledAdd_absolutelyContinuous`, `weightedUniformDistribution_absolutelyContinuous_of_head_ne_zero`, `weightedUniformDistribution_absolutelyContinuous`, and `weightedUniformDistribution_nullSingletonClass`. The first treats the image of a uniform coordinate and an arbitrary independent  $s$ -finite tail measure under  $(u, x) \mapsto ua + x$  for  $a \neq 0$ . The head-coordinate weighted theorem assumes  $w_0 \neq 0$ ; the sharp final pair assumes only norm summability and  $\exists n, w_n \neq 0$ , with the last returning `Mathlib's NullSingletonClass` explicitly.

The ten declarations in the exact-support tranche are `isOpenPosMeasure_infinitePi`, `uniformProduct_isOpenPosMeasure`, `support_map_eq_closure_range_of_continuous`, `weightedUniformSeries_constCoordinates`, `weightedUniformDistribution_support_eq_range`, `range_weightedUniformSeries_eq_Icc_min_max`, `weightedUniformDistribution_support_eq_Icc_min_max`, `range_weightedUniformSeries_eq_Icc`, `weightedUniformDistribution_support_eq_Icc`, and `weightedUniformDistribution_support_eq_unitInterval`. They show first that a norm-summable Banach-valued law has support exactly the range of its series map. For arbitrary signed real weights, that range and support are the closed interval

$$\left[ \sum_n \min\{w_n, 0\}, \sum_n \max\{w_n, 0\} \right].$$

The nonnegative and unit-mass theorems are exact specializations, not merely concentration statements.

The twenty-three declarations in the named specialization are `geometricUniformWeight`, `geometricUniformWeight_zero`, `geometricUniformWeight_succ`, `hasSum_geometricUniformWeight`, `summable_norm_geometricUniformWeight`, `geometricUniformWeight_nonneg`, `geometricUniformSeries`, `continuous_geometricUniformSeries`, `measurable_geometricUniformSeries`, `geometricUniformSeries_split`, `geometricUniformSeries_reflect`, `geometricUniformSeries`

`es_mem_Icc`, `geometricUniformDistribution`, `geometricUniformDistribution_isProbabilityMeasure`, `geometricUniformDistribution_absolutelyContinuous`, `geometricUniformDistribution_nullSingletonClass`, `uniformProduct_map_head_tail_geometricUniformSeries`, `geometricUniformDistribution_selfSimilar`, `geometricUniformDistribution_reflection`, `geometricUniformDistribution_Icc`, `geometricUniformDistribution_compl_Icc`, `geometricUniformDistribution_support_eq_range`, and `geometricUniformDistribution_support_eq_Icc`. Finally, `weightedCoordinateSum_eq_geometricUniformSeries_one_half` and `weightedSumDistribution_eq_geometricUniformDistribution_one_half` are the exact series- and law-level bridges at  $q = 1/2$ , while `weightedSumDistribution_support_eq_Icc` is the exact dyadic support corollary. The single declaration in `GeometricUniformUniqueness.lean` is `eq_geometricUniformDistribution_of_selfSimilar`. For every real  $q$  with  $|q| < 1$ , it concludes  $\mu = \mathcal{L}(X_q)$  from

$$\mu = ((u, x) \mapsto (1 - q)u + qx)_* (\lambda_{[0,1]} \otimes \mu)$$

for an arbitrary probability measure  $\mu$ , with no support, density, or moment premise.

The predecessor `GeometricUniformDictionary.lean` already supplies finite measure, characteristic-function, MGF, and CGF tail factorizations. The six public declarations in `GeometricSincFactorization.lean` are `charFun_geometricUniformDigit`, `charFun_geometricUniformDistribution_prefix`, `charFun_geometricUniformDistribution_prefix_sinc`, `tendsto_prefix_sinc_charFun`, `charFun_weightedSumDistribution_prefix_sinc`, and `tendsto_prefix_sinc_charFun_weightedSumDistribution`. The digit formula holds for every real  $q$ . For every  $|q| < 1$ ,  $m \in \mathbb{N}$ , and  $t \in \mathbb{R}$ , if  $\phi_q(t) = \text{charFun}(\mathcal{L}(X_q))(t)$ , then

$$\phi_q(t) = e^{i(1-q^m)t/2} \left( \prod_{k=0}^{m-1} \text{sinc}\left(\frac{(1-q)q^k t}{2}\right) \right) \phi_q(q^m t),$$

and the phase-bearing prefix without the residual converges pointwise to  $\phi_q(t)$ . This includes  $q = 0$  and negative  $q$ ; the last two declarations are exact  $q = 1/2$  wrappers for the classical weighted-sum law.

The four new public theorems in `GeometricReciprocalGamma.lean` are `hasProdLocallyUniformly_geometricSincProduct`, `geometricSincProductFactors_multipliable`, `hasProd_geometricSincProduct`, and `geometricSincProduct_differentiable`. For  $q, z \in \mathbb{C}$ ,  $\|q\| < 1$ , write  $S_q(z)$  for `geometricSincProduct` applied to  $q, z$ . They state that

$$S_q(z) = \prod_{n \geq 0} \text{sinc}(\pi q^n z)$$

is a locally uniformly convergent product, has genuine pointwise `Multipliable` and `HasProd` witnesses with that exact value, and is entire. The zero-definition, two-theorem module `GeometricSincCharacteristicFunction.lean` contains exactly `charFun_geometricUniformDistribution_eq_phase_mul_geometricSincProduct` and `charFun_geometricUniformDistribution_eq_phase_mul_geometricReciprocalGamma`. For real  $|q| < 1$ ,  $t \in \mathbb{R}$ , and  $z_q(t) = (1 - q)t/(2\pi)$ , they assert the following, where  $G_q(z)$  denotes `geometricReciprocalGamma` applied to  $q, z$ :

$$\phi_q(t) = e^{it/2} S_q(z_q(t)) = e^{it/2} G_q(z_q(t)) G_q(-z_q(t)).$$

The range again includes  $q = 0$  and negative contractions. No named centered wrapper or uniform-on-compacts theorem for the full phase-bearing characteristic-prefix sequence is asserted.

The three generic real-CDF declarations in `ContinuousCDF.lean` are `continuous_cdf_of_nullSingleton`, `cdf_reflection_sub`, and `measure_eq_withDensity_of_cdf_hasDerivAt`. The last identifies a probability measure with Lebesgue measure having density `ofReal ∘ f` from an everywhere pointwise CDF derivative; it assumes neither prior absolute continuity nor continuity of  $f$ .

The twenty-seven declarations in `GeometricUniformCDF.lean` are `geometricUniformCDF`, `monotone_geometricUniformCDF`, `geometricUniformCDF_nonneg`, `geometricUniformCDF_le_one`, `measurable_geometricUniformCDF`, `continuous_geometricUniformCDF`, `geometricUniformCDF_reflection`, `geometricUniformCDF_one_half`, `geometricUniformCDF_zero_of_nonpos`, `geometricUniformCDF_one_of_one_le`, `geometricUniformCDF_eq_integral`, `geometricUniformCDF_eq_intervalIntegral`, `geometricUniformDensity`, `geometricUniformCDF_hasDerivAt`, `deriv_geometricUniformCDF`, `continuous_geometricUniformDensity`, `geometricUniformDensity_nonneg`, `geometricUniformDensity_zero_of_nonpos`, `geometricUnifo`

`rmDensity_zero_of_one_le`, `support_geometricUniformDensity_subset_Ioo`, `support_geometricUniformDensity_subset_Icc`, `tsupport_geometricUniformDensity_subset_Icc`, `geometricUniformDensity_hasCompactSupport`, `geometricUniformDensity_reflection`, `geometricUniformDistribution_eq_withDensity`, `contDiff_geometricUniformCDF`, and `contDiff_geometricUniformDensity`. The CDF definition and its monotonicity, range bounds, and measurability are total in real  $q$ . Continuity, reflection, and midpoint use  $|q| < 1$ ; the two CDF tails use  $0 \leq q < 1$ ; every declaration from conditioning through the classical density, support, and smoothness layer uses  $0 < q < 1$ . In `ProbabilityRepresentation.lean`, the additional exact all-real bridges are `weightedSumCDF_eq_geometricUniformCDF_one_half`, `geometricUniformCDF_one_half_eq_fabiusReal`, and `geometricUniformDensity_one_half_eq_rvachevUp`.

## 2.1 Recovery of the classical functions

At  $q = 1/2$ ,

$$X_{1/2} = \sum_{j \geq 0} 2^{-j-1} U_j.$$

This is the standard bounded Fabius distribution. Moreover, if  $V_j = 2U_j - 1 \sim \text{Unif}[-1, 1]$ , then

$$Y_{1/2} = \sum_{j \geq 0} 2^{-j-1} V_j, \quad \hat{u}_{1/2}(t) = \prod_{j \geq 1} \text{sinc}(t/2^j),$$

which is Rvachev's normalized up-function. The affine relation between densities is

$$u_q(y) = \frac{1}{2} f_q\left(\frac{y+1}{2}\right), \quad f_q(x) = 2u_q(2x-1). \quad (2.3)$$

Thus the free parameter deforms the Fabius CDF and the Rvachev bump simultaneously.

**Current formalization boundary.** For every bounded Fabius function  $F$  satisfying its specification, `geometricUniformCDF_one_half_eq_fabiusReal` proves  $F_{1/2}(x) = F(x)$  for all real  $x$ , and `geometricUniformDensity_one_half_eq_rvachevUp` proves  $f_{1/2}(x) = 2\text{up}(2x-1)$  for all real  $x$ . The separately named bridge `weightedSumCDF_eq_geometricUniformCDF_one_half` identifies the dyadic random-series CDF with this half-base member. There is not yet a separately named centered law or general- $q$  centered density  $u_q$ , so the affine identities (2.3) remain paper-level outside their exact half-base consequence.

## 2.2 Distributional refinement

Let  $X'_q$  be an independent copy of  $X_q$ . Splitting off the first summand gives the basic self-similarity.

**Proposition 2.2** (Affine refinement law). *One has*

$$X_q \stackrel{d}{=} cU_0 + qX'_q. \quad (2.4)$$

With the convention  $F_q(x) = 0$  for  $x \leq 0$  and  $F_q(x) = 1$  for  $x \geq 1$ ,

$$F_q(x) = \int_0^1 F_q\left(\frac{x-cu}{q}\right) du \quad (2.5)$$

$$= \frac{q}{c} \int_{(x-c)/q}^{x/q} F_q(v) dv, \quad (2.6)$$

$$f_q(x) = \frac{F_q(x/q) - F_q((x-c)/q)}{c}. \quad (2.7)$$

*Proof.* Equation (2.4) is the index shift

$$c \sum_{j \geq 0} q^j U_j = cU_0 + q \left( c \sum_{j \geq 0} q^j U_{j+1} \right).$$

Conditioning on  $U_0$  gives (2.5). The substitution  $v = (x-cu)/q$  yields (2.6), and differentiation gives (2.7); smoothness needed for the differentiation follows from [Theorem 3.1](#) below, or one may first interpret the derivative distributionally.  $\square$

**Current formalization boundary.** The measure identity (2.4) is formalized exactly as `geometricUniformDistribution_selfSimilar`; its proof uses the public joint head–tail law `uniformProduct_map_head_tail_geometricUniformSeries`. For  $0 < q < 1$ , the two displayed CDF identities are formalized exactly as `geometricUniformCDF_eq_integral` and `geometricUniformCDF_eq_intervalIntegral`. The explicit formula in (2.7) is the definition `geometricUniformDensity`, and `geometricUniformCDF_hasDerivAt` proves pointwise on all of  $\mathbb{R}$  that it is the derivative of the CDF. Finally, `geometricUniformDistribution_eq_withDensity` identifies the law exactly as Lebesgue measure with this nonnegative Radon–Nikodym density. The exterior-value convention used in the proposition is formalized for  $0 \leq q < 1$  by `geometricUniformCDF_zero_of_nonpos` and `geometricUniformCDF_one_of_one_le`. Uniqueness is supplied separately and directly by `eq_geometricUniformDistribution_of_selfSimilar` for every  $|q| < 1$ , including negative  $q$  and  $q = 0$ ; it requires no support, density, or moment hypothesis on the competing probability law. The classical density interpretation has not been extended to the selected totalized  $q \leq 0$  formula. Also separately, `GeometricUniformCauchy.lean` gives the exact-support transform DDE and adjacent Cauchy–power hierarchy for every nonzero  $|q| < 1$ ; those declarations do not formalize the characteristic-function product or the real-space density-derivative hierarchy below.

For  $q = 1/2$  and  $0 \leq x \leq 1/2$ , the second term in (2.7) vanishes, so

$$f_{1/2}(x) = 2F_{1/2}(2x),$$

the classical Fabius differential equation in bounded-CDF normalization.

Repeated differentiation gives a useful hierarchy.

**Corollary 2.3** (Derivative refinement). *For every integer  $r \geq 1$ ,*

$$f_q^{(r)}(x) = \frac{f_q^{(r-1)}(x/q) - f_q^{(r-1)}((x-c)/q)}{cq^r}. \quad (2.8)$$

*In particular, every derivative vanishes at both endpoints of the support.*

**Current formalization boundary.** The first derivative identity is formal through `geometricUniformCDF_hasDerivAt`, and `contDiff_geometricUniformDensity` supplies derivatives of every finite order. There is not yet a declaration giving the closed  $r$ -th refinement (2.8), nor report-facing wrappers asserting endpoint flatness for every derivative. Those two statements therefore remain frontier even though the underlying  $C^\infty$  and compact-support inputs are formal.

### 3 Infinite sinc products and smooth compact support

The centered uniform variable  $U - 1/2$  has characteristic function  $\text{sinc}(t/2)$ . Independence therefore gives an infinite product.

**Theorem 3.1** (Fourier product and rapid polynomial decay). *The characteristic functions of  $Z_q$  and  $Y_q$  are*

$$\phi_q(t) := \mathbb{E}e^{itZ_q} = \prod_{j=0}^{\infty} \text{sinc}\left(\frac{cq^j t}{2}\right), \quad (3.1)$$

$$\hat{u}_q(t) = \mathbb{E}e^{itY_q} = \prod_{j=0}^{\infty} \text{sinc}(cq^j t). \quad (3.2)$$

*For every integer  $N \geq 1$  and every  $t \neq 0$ ,*

$$|\phi_q(t)| \leq \left(\frac{2}{c}\right)^N q^{-N(N-1)/2} |t|^{-N}. \quad (3.3)$$

*Consequently  $X_q$  and  $Y_q$  possess compactly supported  $C^\infty$  densities, and all of their derivatives are obtained by absolutely convergent Fourier inversion.*

*Proof.* The product formulas follow from independence and the identity  $\mathbb{E}e^{is(U-1/2)} = \text{sinc}(s/2)$ . The product converges locally uniformly because  $1 - \text{sinc } z = O(z^2)$  and  $\sum q^{2j} < \infty$ . For the first  $N$  factors, use  $|\text{sinc } x| \leq \min(1, |x|^{-1})$ ; bound all remaining factors by one. This gives

$$|\phi_q(t)| \leq \prod_{j=0}^{N-1} \frac{2}{cq^j|t|} = \left(\frac{2}{c}\right)^N q^{-N(N-1)/2} |t|^{-N}.$$

Given  $r \geq 0$ , choose  $N > r + 1$ . Then  $t^r \phi_q(t) \in L^1(\mathbb{R})$ , so Fourier inversion gives a continuous  $r$ th derivative. Since  $r$  is arbitrary, the density is  $C^\infty$ . Compact support was already established from the random series.  $\square$

**Current formalization boundary.** For  $0 < q < 1$ , the final regularity conclusion for the  $X_q$ -density has crossed the formalization frontier independently of this Fourier proof: `geometricUniformDistribution_eq_withDensity` gives the explicit density, `geometricUniformDensity_hasCompactSupport` gives compact support, and `contDiff_geometricUniformCDF` and `contDiff_geometricUniformDensity` give  $C^\infty$  regularity. The centered law and density are not separately named. For the uncentered law, however, `GeometricSincFactorization.lean` proves the exact finite residual formula and pointwise convergence of the phase-bearing sinc prefixes for every  $|q| < 1$ . The four theorems `hasProdLocallyUniformly_geometricSincProduct`, `geometricSincProductFactors_multiplicable`, `hasProd_geometricSincProduct`, and `geometricSincProduct_differentiable` now prove, for complex  $\|q\| < 1$ , that  $S_q(z) = \prod_{n \geq 0} \text{sinc}(\pi q^n z)$  converges locally uniformly, has the named pointwise product witnesses, and is entire. For real  $|q| < 1$ ,  $t \in \mathbb{R}$ , and  $z_q(t) = (1 - q)t/(2\pi)$ , the two exact bridges `charFun_geometricUniformDistribution_eq_phase_mul_geometricSincProduct` and `charFun_geometricUniformDistribution_eq_phase_mul_geometricReciprocalGamma` give  $\phi_q(t) = e^{it/2} S_q(z_q(t)) = e^{it/2} G_q(z_q(t)) G_q(-z_q(t))$ , where  $G_q(z)$  denotes `geometricReciprocalGamma` applied to  $q, z$ . The equality includes  $q = 0$  and negative  $q$ . There is not yet a named centered wrapper or a packaged compact-uniform theorem for the full phase-bearing characteristic prefixes. The rapid bound (3.3) and the absolutely convergent Fourier-inversion formulas for higher derivatives have not been formalized.

*Remark 3.2* (Relation to the existing Fourier-decay campaign). Bound (3.3) is intentionally elementary. Optimizing  $N$  as a function of  $|t|$  recovers a quadratic logarithmic envelope, but the dyadic case has already been analyzed in much finer arithmetic and phase detail in the repository. The genuinely new content here is the  $q$ -deformation and the zero arithmetic in Section 8, not a claim of a new sharp dyadic envelope.

## 4 Bernoulli cumulants, Bell moments, and $q$ -binomial scale geometry

### 4.1 Exact cumulants

Let  $B_n$  denote the Bernoulli numbers in the convention

$$\frac{t}{e^t - 1} = \sum_{n \geq 0} B_n \frac{t^n}{n!}, \quad B_1 = -\frac{1}{2}.$$

For a uniform random variable  $U$  on  $[0, 1]$ , the cumulants are  $\kappa_1(U) = 1/2$  and  $\kappa_n(U) = B_n/n$  for  $n \geq 2$ .

**Theorem 4.1** (Bernoulli cumulants). *For the centered law  $Z_q = X_q - 1/2$ ,*

$$\kappa_{2m+1}(Z_q) = 0 \quad (m \geq 0), \tag{4.1}$$

where the case  $m = 0$  means  $\kappa_1 = 0$ , and

$$\kappa_{2m}(Z_q) = \frac{B_{2m} (1 - q)^{2m}}{2m (1 - q^{2m})} \quad (m \geq 1). \tag{4.2}$$

In particular,

$$\sigma_q^2 := \text{Var}(X_q) = \frac{1 - q}{12(1 + q)}. \tag{4.3}$$

At  $q = 1/2$ ,

$$\kappa_{2m}(Z_{1/2}) = \frac{B_{2m}}{2m(2^{2m} - 1)}, \quad (4.4)$$

exactly the dyadic Bernoulli-cumulant formula already present in the repository.

*Proof.* Cumulants add over independent sums and scale homogeneously. Hence for  $n \geq 2$ ,

$$\kappa_n(Z_q) = \sum_{j \geq 0} (cq^j)^n \kappa_n(U_j - 1/2) = \frac{B_n}{n} \frac{c^n}{1 - q^n}.$$

All odd Bernoulli numbers beyond  $B_1$  vanish, while symmetry gives the vanishing first cumulant. Substituting  $n = 2$  yields (4.3); substituting  $q = 1/2$  gives (4.4).  $\square$

## 4.2 Bell-polynomial moments and a triangular recurrence

Let  $\mathcal{B}_n(x_1, \dots, x_n)$  denote the complete exponential Bell polynomial, characterized by

$$\exp\left(\sum_{r \geq 1} x_r \frac{t^r}{r!}\right) = \sum_{n \geq 0} \mathcal{B}_n(x_1, \dots, x_n) \frac{t^n}{n!}.$$

**Corollary 4.2** (Exact moments). *For every  $n \geq 0$ ,*

$$\mathbb{E}Z_q^n = \mathcal{B}_n(0, \kappa_2(Z_q), \dots, \kappa_n(Z_q)), \quad (4.5)$$

$$\mathbb{E}X_q^n = \mathcal{B}_n\left(\frac{1}{2}, \kappa_2(Z_q), \dots, \kappa_n(Z_q)\right). \quad (4.6)$$

The raw moments  $m_n(q) = \mathbb{E}X_q^n$  also satisfy

$$m_0(q) = 1, \quad m_n(q) = \frac{1}{1 - q^n} \sum_{k=1}^n \binom{n}{k} \frac{c^k q^{n-k}}{k+1} m_{n-k}(q). \quad (4.7)$$

*Proof.* The Bell formulas are the moment–cumulant relation. For the recurrence, raise  $cU_0 + qX'_q$  to the  $n$ th power, take expectations, use  $\mathbb{E}U_0^k = 1/(k+1)$ , and move the  $k = 0$  term  $q^n m_n(q)$  to the left.  $\square$

The first centered even moments are therefore

$$\begin{aligned} \mathbb{E}Z_q^2 &= \kappa_2, \\ \mathbb{E}Z_q^4 &= 3\kappa_2^2 + \kappa_4, \\ \mathbb{E}Z_q^6 &= 15\kappa_2^3 + 15\kappa_2\kappa_4 + \kappa_6, \end{aligned}$$

with every cumulant given by (4.2). This is an explicit all-orders moment engine involving only Bernoulli numbers, rational operations, and Bell contraction.

## 4.3 A Bernoulli–Appell inverse to geometric averaging

Let

$$M_q(t) = \mathbb{E}e^{tX_q} = \prod_{j \geq 0} \frac{e^{cq^j t} - 1}{cq^j t}. \quad (4.8)$$

Define an Appell sequence  $(P_{q,n})_{n \geq 0}$  by

$$\frac{e^{xt}}{M_q(t)} = \sum_{n \geq 0} P_{q,n}(x) \frac{t^n}{n!}. \quad (4.9)$$

**Theorem 4.3** (Geometric Bernoulli–Appell polynomials). *The polynomials  $P_{q,n}$  satisfy*

$$\frac{d}{dx} P_{q,n}(x) = n P_{q,n-1}(x), \quad (4.10)$$

$$\mathbb{E}[P_{q,n}(x + X_q)] = x^n, \quad (4.11)$$

$$P_{q,n}(x) = \mathcal{B}_n(x - \tfrac{1}{2}, -\kappa_2(Z_q), \dots, -\kappa_n(Z_q)). \quad (4.12)$$

They obey the refinement recurrence

$$P_{q,n}(x) = \sum_{k=0}^n \binom{n}{k} B_k c^k q^{n-k} P_{q,n-k}\left(\frac{x}{q}\right). \quad (4.13)$$

*Proof.* Differentiate (4.9) in  $x$  to obtain (4.10). Multiplying by  $M_q(t)$  and taking expectation after replacing  $x$  by  $x + X_q$  gives  $e^{xt}$ , proving (4.11). Taking the logarithm of the left side of (4.9) gives

$$(x - \tfrac{1}{2})t - \sum_{r \geq 2} \kappa_r(Z_q) \frac{t^r}{r!},$$

which is exactly the Bell generating function in (4.12).

Finally, the distributional refinement implies

$$M_q(t) = \frac{e^{ct} - 1}{ct} M_q(qt).$$

Therefore

$$\frac{e^{xt}}{M_q(t)} = \frac{ct}{e^{ct} - 1} \frac{e^{(x/q)(qt)}}{M_q(qt)}.$$

Expand the first factor as  $\sum B_k c^k t^k / k!$  and the second as  $\sum P_{q,r}(x/q) q^r t^r / r!$ ; coefficient comparison yields (4.13).  $\square$

The operator meaning is transparent. For  $c > 0$ , define uniform averaging

$$(\mathcal{A}_c p)(x) = \frac{1}{c} \int_0^c p(x+u) du = \frac{e^{cD} - 1}{cD} p(x), \quad D = \frac{d}{dx}.$$

Then

$$\mathcal{A}_c^{-1} x^n = c^n B_n(x/c), \quad (4.14)$$

where  $B_n(x)$  is the Bernoulli polynomial. The sequence  $P_{q,n}$  is obtained by composing these Bernoulli inverses over all geometric scales  $c, cq, cq^2, \dots$ . This gives a direct operator bridge between Bernoulli polynomials, the random series, and the Appell blocks used in exact dyadic arithmetic.

#### 4.4 Finite-stage Gaussian-binomial geometry

Let  $w_j = cq^j$  for  $j = 0, \dots, N-1$ . The elementary symmetric functions of these scales are Gaussian-binomial coefficients.

**Proposition 4.4** ( $q$ -binomial scale identity). *For every integer  $N \geq 0$ ,*

$$\prod_{j=0}^{N-1} (1 + zw_j) = \sum_{k=0}^N c^k q^{k(k-1)/2} \begin{bmatrix} N \\ k \end{bmatrix}_q z^k. \quad (4.15)$$

Equivalently,

$$e_k(w_0, \dots, w_{N-1}) = c^k q^{k(k-1)/2} \begin{bmatrix} N \\ k \end{bmatrix}_q. \quad (4.16)$$

*Proof.* This is the finite  $q$ -binomial theorem applied to  $\prod_{j=0}^{N-1} (1 + cq^j z)$ .  $\square$

*Remark 4.5* (What is and is not new here). The repository already exploits the same Gaussian-binomial algebra at the fixed dyadic base  $q = 1/2$  for coefficient extraction and exact Fabius values. The new point is that the same  $q$  now parametrizes an actual family of probability laws. Thus differentiation in  $q$ , limits  $q \uparrow 1$ , shape transitions, and arithmetic changes of the zero set become meaningful analytic questions rather than formal changes of notation.



## 5 Parameter geometry: majorization, shape, and concentration

### 5.1 A convex-order flow

Write the decreasing weight sequence

$$\mathbf{w}(q) = (c, cq, cq^2, \dots).$$

Its partial sums are

$$\sum_{j=0}^{n-1} w_j(q) = 1 - q^n.$$

Hence, if  $0 < q_1 < q_2 < 1$ , then every partial sum of  $\mathbf{w}(q_1)$  is larger while both total sums equal one. In the language of infinite majorization,  $\mathbf{w}(q_1) \succ \mathbf{w}(q_2)$ .

**Theorem 5.1** (Monotone contraction in convex order). *If  $0 < q_1 < q_2 < 1$ , then*

$$X_{q_2} \preceq_{\text{cx}} X_{q_1}; \quad (5.1)$$

that is, for every convex function  $\Phi : [0, 1] \rightarrow \mathbb{R}$ ,

$$\mathbb{E}\Phi(X_{q_2}) \leq \mathbb{E}\Phi(X_{q_1}). \quad (5.2)$$

Consequently, for every  $p \geq 1$ ,

$$\mathbb{E}|X_{q_2} - \tfrac{1}{2}|^p \leq \mathbb{E}|X_{q_1} - \tfrac{1}{2}|^p. \quad (5.3)$$

*Proof.* For finitely many i.i.d. integrable random variables, the map

$$(a_1, \dots, a_N) \mapsto \mathbb{E}\Phi\left(\sum_{j=1}^N a_j U_j\right)$$

is symmetric and convex in the coefficient vector, hence Schur-convex. Therefore it increases under majorization. The same statement extends to nonnegative  $\ell^1$  coefficient sequences by truncation: the omitted weighted tail is bounded by the omitted  $\ell^1$  mass, so the corresponding sums converge uniformly, and every convex function on the compact interval  $[0, 1]$  is continuous. Since

$$\sum_{j=0}^{n-1} w_j(q_1) = 1 - q_1^n \geq 1 - q_2^n = \sum_{j=0}^{n-1} w_j(q_2)$$

for every  $n$ , while both full sums are one, infinite majorization gives (5.2). Take  $\Phi(x) = |x - 1/2|^p$  for (5.3).  $\square$

This theorem turns  $q$  into a genuine concentration parameter. At  $q \downarrow 0$ , the law approaches a single uniform random variable. As  $q \uparrow 1$ , its weight vector is spread among increasingly many small summands, and the law contracts toward  $1/2$ .

### 5.2 A uniform sub-Gaussian bound

Hoeffding's lemma applied to each centered term gives

$$\log \mathbb{E} e^{tZ_q} \leq \frac{t^2}{8} \sum_{j \geq 0} c^2 q^{2j} = \frac{t^2}{8} \frac{1-q}{1+q}.$$

**Corollary 5.2** (Concentration). *For every  $r > 0$ ,*

$$\mathbb{P}(|X_q - \tfrac{1}{2}| \geq r) \leq 2 \exp\left(-\frac{2(1+q)}{1-q} r^2\right). \quad (5.4)$$

The exponent has the correct  $1/(1-q)$  concentration scale as  $q \uparrow 1$ , though it is not intended to replace the exact Gaussian limit below.



### 5.3 Log-concavity and an exact plateau transition

Every finite partial sum of (2.1) has a density that is a convolution of uniform interval densities, hence is log-concave. Passing to the locally uniform density limit shows that  $f_q$  is log-concave on  $(0, 1)$ . Symmetry then makes  $1/2$  a global maximizer. The first scale competes with the total length of all remaining scales:

$$w_0 = c, \quad \sum_{j \geq 1} w_j = q.$$

Equality occurs exactly at  $q = 1/2$ .

**Theorem 5.3** (Shape phase transition). *For every  $0 < q < 1$ , the density  $f_q$  is symmetric and log-concave.*

(i) *If  $0 < q < 1/2$ , then the exact maximal plateau is*

$$f_q(x) = \frac{1}{1-q} \iff q \leq x \leq 1-q. \quad (5.5)$$

*Outside that interval,  $f_q(x) < 1/(1-q)$ .*

(ii) *If  $1/2 \leq q < 1$ , then  $f_q$  has no nontrivial plateau. It is strictly increasing on  $(0, 1/2)$  and strictly decreasing on  $(1/2, 1)$ , so  $1/2$  is its unique maximizer.*

*Proof.* Log-concavity follows from closure of log-concave densities under convolution and locally uniform limits. If  $q < 1/2$  and  $q \leq x \leq c$ , then  $x/q \geq 1$  and  $(x-c)/q \leq 0$ , so (2.7) gives  $f_q(x) = 1/c$ . If  $0 < x < q$ , then  $F_q(x/q) < 1$  and the second CDF term is zero, so  $f_q(x) < 1/c$ ; symmetry handles  $x > c$ . This proves (i).

For (ii), first take  $q = 1/2$ . On  $(0, 1/2)$  the dyadic refinement reads  $f_{1/2}(x) = 2F_{1/2}(2x)$ . The support statement above makes  $F_{1/2}$  strictly increasing, so  $f_{1/2}$  is strictly increasing there; symmetry handles the right half.

Now assume  $q > 1/2$  and work in centered coordinates. Suppose the density has a nontrivial central plateau of half-width  $h > 0$ . In the decomposition

$$Z_q = c(U_0 - \frac{1}{2}) + qZ'_q,$$

let  $g$  be the density of  $qZ'_q$ , so  $g(z) = q^{-1}f_Z(z/q)$  and its central plateau has half-width  $qh$ . Convolution with the uniform density gives

$$f'_Z(y) = \frac{g(y + c/2) - g(y - c/2)}{c}. \quad (5.6)$$

For  $0 < y < \min\{h, (q-c)/2\}$ , both points lie in the interior of the support of  $g$ . Since  $f'_Z(y) = 0$ , symmetry and log-concavity force  $g$  to be constant from the origin out to  $c/2 + y$ : a symmetric log-concave density cannot have an off-center constant interval without also being constant between that interval and the origin. Thus  $qh \geq c/2 + y$  throughout this range. If  $h \leq (q-c)/2$ , let  $y \uparrow h$  to obtain  $qh \geq c/2 + h$ , impossible because  $q < 1$ . If  $h > (q-c)/2$ , let  $y \uparrow (q-c)/2$  to obtain  $qh \geq q/2$ , hence  $h \geq 1/2$ , which would make the density constant on its entire support, also impossible. Therefore no plateau exists.

Finally, a symmetric log-concave density is nonincreasing to the right of its center. If it took the same value at two distinct positive distances from the center, concavity of its logarithm would force a central plateau. Hence it decreases strictly, and symmetry gives the left-hand statement.  $\square$

**Conjecture 5.4** (Strict log-concavity beyond the transition). *For every  $q > 1/2$ ,  $\log f_q$  is strictly concave on  $(0, 1)$ .*

The theorem proves strict unimodality but not strict concavity of the logarithm. A plausible route is to quantify the strictness in the Prékopa–Leindler step for an infinite convolution whose largest interval is shorter than the sum of the others.

## 6 Gaussian degeneration and an Edgeworth frontier

Let

$$G_q = \frac{Z_q}{\sigma_q}, \quad \sigma_q^2 = \frac{1-q}{12(1+q)}.$$

The largest normalized summand has magnitude of order  $\sqrt{1-q}$ , while the total variance is one. This is a triangular-array central-limit regime.

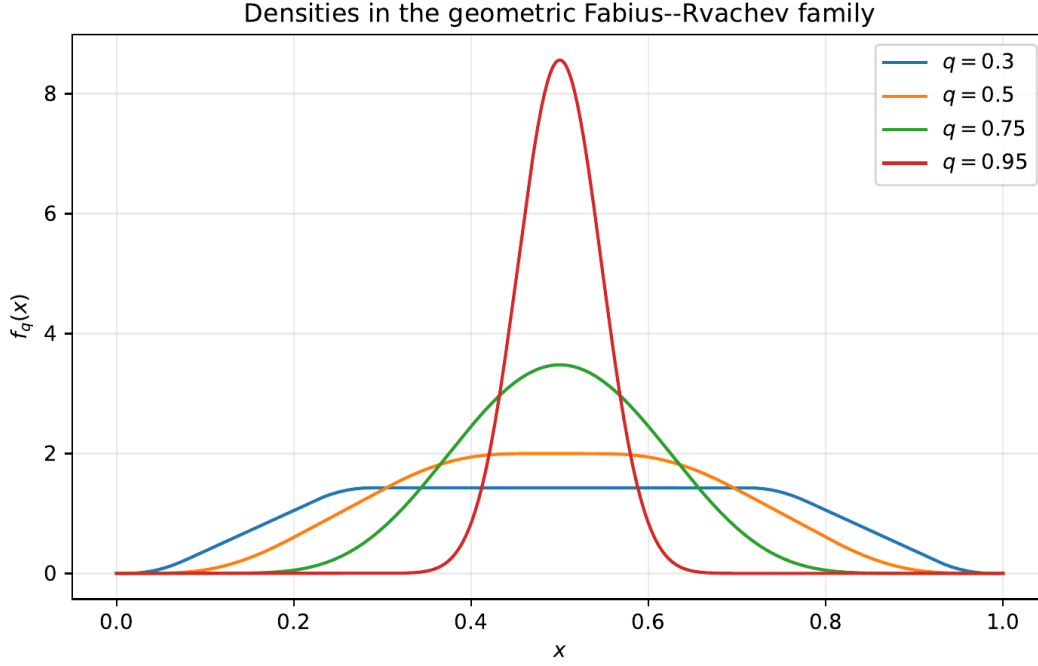


Figure 10: Deterministic fixed-point approximations of  $f_q$ . The exact plateau for  $q = 0.30$  ends at  $x = q$  and  $x = 1 - q$ . At  $q = 1/2$  the plateau collapses to a point; for larger  $q$  the law becomes sharply concentrated and strictly unimodal.

**Theorem 6.1** (Gaussian limit). *As  $q \uparrow 1$ ,*

$$G_q \xrightarrow{d} \mathcal{N}(0, 1). \quad (6.1)$$

*The standardized even cumulants are*

$$\gamma_{2m}(q) := \kappa_{2m}(G_q) = \frac{B_{2m}}{2m} 12^m \frac{(1 - q^2)^m}{1 - q^{2m}}. \quad (6.2)$$

*In particular,*

$$\gamma_4(q) = -\frac{6}{5} \frac{1 - q^2}{1 + q^2}, \quad (6.3)$$

$$\gamma_6(q) = \frac{48}{7} \frac{(1 - q^2)^2}{1 + q^2 + q^4}. \quad (6.4)$$

*For every fixed  $T < \infty$ , uniformly for  $|t| \leq T$ ,*

$$\log \mathbb{E} e^{itG_q} = -\frac{t^2}{2} + \frac{\gamma_4(q)t^4}{4!} + \mathcal{O}_T((1 - q)^2). \quad (6.5)$$

*Proof.* Write

$$G_q = \sum_{j \geq 0} \alpha_{q,j} (U_j - \tfrac{1}{2}), \quad \alpha_{q,j} = \frac{cq^j}{\sigma_q}.$$

The summands are bounded, their variances sum to one, and

$$\max_j |\alpha_{q,j}| = \frac{c}{\sigma_q} = \sqrt{12(1 - q)(1 + q)} \rightarrow 0.$$

The Lindeberg condition is therefore immediate, proving (6.1). Dividing (4.2) by  $\sigma_q^{2m}$  gives (6.2); elementary factorization gives (6.3)–(6.4).

For the compact-frequency expansion, expand the logarithm of each sinc factor. The maximum factor argument is  $\mathcal{O}_T(\sqrt{1-q})$ , so the expansion is uniform. The quadratic terms sum to  $-t^2/2$ , the quartic terms sum to  $\gamma_4 t^4/4!$ , and the sum of all terms of order six and higher is  $\mathcal{O}_T((1-q)^2)$  because  $\gamma_{2m}(q) = \mathcal{O}((1-q)^{m-1})$  for  $m \geq 3$ .  $\square$

Formally inverting (6.5) suggests

$$\sigma_q f_q\left(\frac{1}{2} + \sigma_q x\right) = \varphi(x) \left[1 + \frac{\gamma_4(q)}{24} \text{He}_4(x)\right] + \mathcal{O}((1-q)^2), \quad (6.6)$$

where  $\varphi(x) = (2\pi)^{-1/2} e^{-x^2/2}$  and  $\text{He}_4(x) = x^4 - 6x^2 + 3$ . At the center this predicts

$$f_q(1/2) = \frac{1}{\sqrt{2\pi}\sigma_q} \left[1 + \frac{\gamma_4(q)}{8} + \mathcal{O}((1-q)^2)\right]. \quad (6.7)$$

For  $q = 0.95$ , the deterministic computation gives  $f_q(1/2) = 8.56266675$ , while the Gaussian term with the quartic correction gives 8.56 to the displayed precision.

**Conjecture 6.2** (Uniform density Edgeworth expansion). For every fixed integer  $r \geq 1$ , the normalized density of  $G_q$  admits a uniform all-orders Edgeworth expansion through order  $(1-q)^r$ , with coefficients obtained by Bell contraction of the explicit standardized cumulants (6.2). In particular, (6.6) holds uniformly in  $x \in \mathbb{R}$  with error  $\mathcal{O}((1-q)^2)$  after inclusion of the complete order- $(1-q)^2$  Hermite polynomial.

The missing ingredient is not the coefficient algebra, which is explicit, but a clean  $q$ -uniform Fourier-tail estimate strong enough to justify termwise inversion on the whole line.

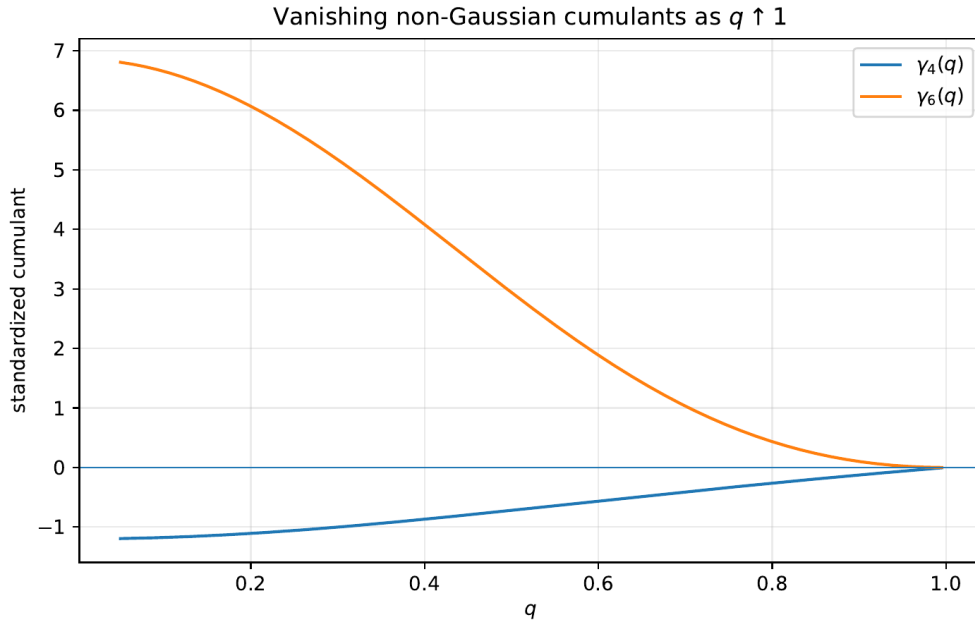


Figure 11: The exact standardized cumulants  $\gamma_4(q)$  and  $\gamma_6(q)$ . Their decay to zero quantifies the Gaussian degeneration as  $q \uparrow 1$ .

## 7 A rigorous endpoint law and Lambert- $W$ inversion

The exact dyadic repository contains a much sharper lower-Lambert expansion with periodic corrections. The purpose here is different: establish, for every  $q$ , the first two universal logarithmic scales by a short probability argument, then invert them rigorously. Set

$$L = \log(1/x), \quad a = \log(1/q).$$

## 7.1 Finite-simplex upper bound

For  $n \geq 1$ , let

$$S_n = c \sum_{j=0}^{n-1} q^j U_j.$$

If  $0 < x \leq cq^{n-1}$ , then the simplex  $\sum_{j < n} cq^j u_j \leq x$  lies entirely inside the unit cube. Its volume is exact:

$$\mathbb{P}(S_n \leq x) = \frac{x^n}{n!c^n q^{n(n-1)/2}}. \quad (7.1)$$

Since  $X_q \geq S_n$ , this is an upper bound for  $F_q(x)$ .

## 7.2 A matching product lower bound

Choose  $n$  so that the deterministic tail is at most  $x/2$ :

$$c \sum_{j=n}^{\infty} q^j = q^n \leq x/2.$$

If, for every  $0 \leq j < n$ ,

$$U_j \leq \frac{x}{2ncq^j},$$

then the first  $n$  summands contribute at most  $x/2$ , and hence  $X_q \leq x$ . For small  $x$  the displayed thresholds are all at most one, giving the lower bound

$$F_q(x) \geq \frac{x^n}{(2nc)^n q^{n(n-1)/2}}. \quad (7.2)$$

**Theorem 7.1** (Two-term logarithmic endpoint law). *For each fixed  $q \in (0, 1)$ , as  $x \downarrow 0$ ,*

$$\log F_q(x) = -\frac{L^2}{2a} - \frac{L \log L}{a} + \mathcal{O}_q(L), \quad L = \log(1/x), \quad a = \log(1/q). \quad (7.3)$$

By symmetry, the same law holds for  $1 - F_q(1 - x)$ .

*Proof.* For the upper bound, choose

$$n_+(x) = \left\lfloor \frac{L + \log c}{a} \right\rfloor + 1.$$

Then  $x \leq cq^{n_+ - 1}$ , so (7.1) applies. Taking logarithms gives

$$\log F_q(x) \leq -n_+ L - \log(n_+!) - n_+ \log c + \frac{a}{2} n_+ (n_+ - 1). \quad (7.4)$$

Since  $n_+ = L/a + \mathcal{O}_q(1)$ , Stirling's formula yields

$$-n_+ L + \frac{a}{2} n_+^2 = -\frac{L^2}{2a} + \mathcal{O}_q(L), \quad -\log(n_+!) = -\frac{L \log L}{a} + \mathcal{O}_q(L).$$

Thus the right side of (7.4) is

$$-\frac{L^2}{2a} - \frac{L \log L}{a} + \mathcal{O}_q(L).$$

For the lower bound, take

$$n_-(x) = \left\lceil \frac{\log(2/x)}{a} \right\rceil.$$

Then  $q^{n_-} \leq x/2$ . The ceiling relation also gives  $q^{n_- - 1} > x/2$ , so the largest threshold in the product event is at most  $1/(n_- c)$  and is therefore below one for sufficiently small  $x$ . Apply (7.2) and take logarithms:

$$\log F_q(x) \geq -n_- L - n_- \log(2n_- c) + \frac{a}{2} n_- (n_- - 1). \quad (7.5)$$

Again  $n_- = L/a + \mathcal{O}_q(1)$ , so the quadratic terms contribute  $-L^2/(2a) + \mathcal{O}_q(L)$  and the  $-n_- \log n_-$  term contributes  $-(L \log L)/a + \mathcal{O}_q(L)$ . The upper and lower bounds match at the two displayed scales.  $\square$

*Remark 7.2* (Why the second term is already nontrivial). The quadratic term comes from balancing the small probability of constraining roughly  $L/a$  coordinates against the geometric Jacobian  $q^{n(n-1)/2}$ . The additional  $-L \log L/a$  term is the factorial entropy of the finite simplex (upper bound) and the  $n^{-n}$  cost of a balanced coordinate allocation (lower bound). Thus the second scale is not an artifact of saddle notation; it is forced by elementary high-dimensional geometry.

### 7.3 Inverse endpoint law

Let  $Q_q(y) = F_q^{-1}(y)$  for  $0 < y < 1$ , and put

$$Y = \log(1/y), \quad L_y = \log \frac{1}{Q_q(y)}, \quad S = \sqrt{2aY}.$$

Equation (7.3) is equivalent to

$$2aY = L_y^2 + 2L_y \log L_y + \mathcal{O}_q(L_y). \quad (7.6)$$

Since  $(\log L_y)^2 = o(L_y)$ , one obtains

$$S = L_y + \log L_y + \mathcal{O}_q(1).$$

The inverse of  $r \mapsto r + \log r$  is  $r = W(e^s)$ .

**Corollary 7.3** (Lambert- $W$  quantile law). *As  $y \downarrow 0$ ,*

$$\log \frac{1}{Q_q(y)} = W\left(\exp \sqrt{2a \log(1/y)}\right) + \mathcal{O}_q(1) \quad (7.7)$$

$$= \sqrt{2a \log(1/y)} - \frac{1}{2} \log(2a \log(1/y)) + \mathcal{O}_q(1). \quad (7.8)$$

*The corresponding upper-end quantile follows from symmetry:  $Q_q(1-y) = 1 - Q_q(y)$ .*

*Proof.* From (7.6),

$$S^2 = (L_y + \log L_y)^2 + \mathcal{O}_q(L_y),$$

so  $S = L_y + \log L_y + \mathcal{O}_q(1)$ . If  $r + \log r = s$ , then  $re^r = e^s$  and  $r = W(e^s)$ . Since the derivative of  $r + \log r$  is bounded away from zero for large  $r$ , the additive  $\mathcal{O}_q(1)$  perturbation in  $s$  changes  $r$  by  $\mathcal{O}_q(1)$ . This proves (7.7). The standard large-argument expansion  $W(e^S) = S - \log S + \mathcal{O}((\log S)/S)$  gives (7.8).  $\square$

This is deliberately weaker than the repository's all-orders dyadic lower-Lambert machinery, but it is rigorous for every  $q$  and directly links the inverse Fabius scale to Lambert  $W$ .

### 7.4 The expected periodic completion

The Laplace transform is

$$\mathcal{L}_q(s) = \mathbb{E}e^{-sX_q} = \prod_{j=0}^{\infty} \frac{1 - e^{-csq^j}}{csq^j}. \quad (7.9)$$

It is a harmonic sum over the logarithmic lattice  $\log s - ja$ . Mellin analysis of such sums normally produces a polynomial in the logarithmic scale plus a periodic function of its fractional part [14]. This mechanism is exactly what occurs in the dyadic endpoint expansion already present in the repository.

*Conjecture 7.4* (Full geometric endpoint transseries). For every fixed  $q \in (0, 1)$  there exists a natural lower-Lambert saddle coordinate  $\lambda_q(x) \rightarrow \infty$  and smooth 1-periodic functions  $A_{q,r} : \mathbb{R}/\mathbb{Z} \rightarrow \mathbb{R}$  such that

$$F_q(x) = \exp(-\mathcal{S}_q(\lambda_q(x))) \lambda_q(x)^{-1/2} \left[ A_{q,0}(\{\lambda_q(x)\}) + \sum_{r=1}^N \frac{A_{q,r}(\{\lambda_q(x)\})}{\lambda_q(x)^r} + \mathcal{O}_{q,N}(\lambda_q^{-N-1}) \right], \quad (7.10)$$

where  $\mathcal{S}_q$  has an explicit expansion whose first two consequences are (7.3). The leading periodic amplitude  $A_{q,0}$  is positive and nonconstant for generic  $q$ .

*Problem 7.5* (Fourier modes of the endpoint phase). Determine the Mellin-residue formula for the Fourier coefficients of  $A_{q,0}$  and classify those  $q$  for which one or more nonzero modes vanish. The dyadic development suggests nonvanishing, but the free parameter may admit exceptional arithmetic ratios.

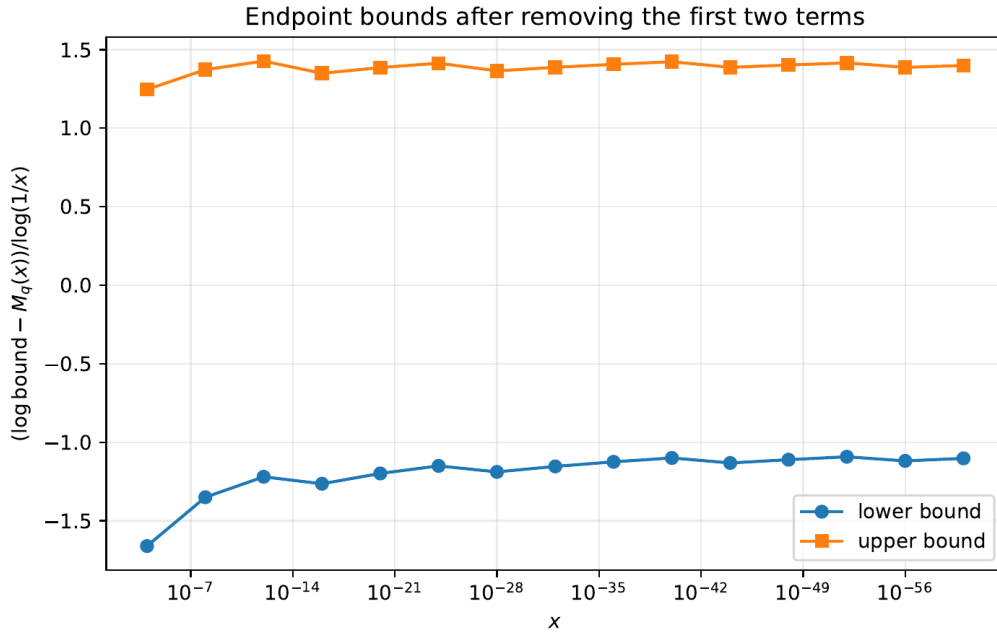


Figure 12: For  $q = 1/2$ , the rigorous upper and lower logarithmic bounds after subtracting the two universal terms and dividing by  $L$ . Bounded residuals are exactly what [Theorem 7.1](#) predicts.

## 8 Fourier-zero arithmetic and the special role of base two

The product (3.2) makes the zero set explicit:

$$\mathcal{Z}_q = \left\{ \frac{\pi k}{cq^j} : k \in \mathbb{Z} \setminus \{0\}, j \geq 0 \right\}. \quad (8.1)$$

Each individual factor has a simple nonzero zero. Multiplicity therefore counts collisions between geometric scales.

**Theorem 8.1** (Reciprocal-integer multiplicities). *Let  $q = 1/b$  with an integer  $b \geq 2$ . Then every nonzero zero of  $\hat{u}_q$  lies on the lattice*

$$t = \frac{\pi N}{1 - 1/b}, \quad N \in \mathbb{Z} \setminus \{0\}. \quad (8.2)$$

*Its multiplicity is*

$$\text{mult}\left(\frac{\pi N}{1 - 1/b}\right) = 1 + v_b(|N|), \quad (8.3)$$

where  $v_b(N) = \max\{r \geq 0 : b^r \mid N\}$  uses divisibility by the whole base  $b$ , which need not be prime.

*Proof.* At  $q = 1/b$ , the  $j$ th factor is  $\text{sinc}(cb^{-j}t)$ . At  $t = \pi N/c$ , it vanishes exactly when  $b^j \mid N$ . Thus the vanishing factors are indexed by  $j = 0, 1, \dots, v_b(|N|)$ , and their simple multiplicities add. Conversely, every zero of the  $j$ th factor has the form  $t = \pi kb^j/c = \pi N/c$  for an integer  $N \neq 0$ .  $\square$

For the classical up-function  $b = 2$ , the zeros are at  $2\pi N$  in the convention (3.2), with multiplicity  $1 + v_2(N)$ .

**Proposition 8.2** (Generic simplicity criterion). *If  $q^d \notin \mathbb{Q}$  for every integer  $d \geq 1$ , then every nonzero zero of  $\hat{u}_q$  is simple.*

*Proof.* A collision between scales  $j < k$  would give

$$\frac{k_1}{q^j} = \frac{k_2}{q^k} \implies q^{k-j} = \frac{k_2}{k_1} \in \mathbb{Q}$$

for nonzero integers  $k_1, k_2$ , contrary to the hypothesis.  $\square$

**Problem 8.3** (Complete collision classification). For algebraic  $q$  such that  $q^d \in \mathbb{Q}$  for some  $d$ , describe the zero set as a union of arithmetic lattices with a closed multiplicity formula. The reciprocal-integer theorem is the cleanest case; rational  $q = r/s$  and radical rational  $q = (r/s)^{1/d}$  should lead to a finite-state valuation problem.

### 8.1 Why the contiguous Thue–Morse block is uniquely dyadic

For an integer base  $b \geq 2$  and  $N \in \mathbb{N}$ , consider the finite product

$$P_{b,N}(z) = \prod_{j=0}^{N-1} (1 - z^{b^j}). \quad (8.4)$$

Expanding by subsets gives

$$P_{b,N}(z) = \sum_{\varepsilon \in \{0,1\}^N} (-1)^{\varepsilon_0 + \dots + \varepsilon_{N-1}} z^{\varepsilon_0 + \varepsilon_1 b + \dots + \varepsilon_{N-1} b^{N-1}}. \quad (8.5)$$

**Proposition 8.4** (Dyadic contiguity). *The exponents in (8.5) form a complete interval of consecutive integers if and only if  $b = 2$  or  $N \leq 1$ . In the dyadic case,*

$$P_{2,N}(z) = \sum_{n=0}^{2^N-1} (-1)^{\text{wt}_2(n)} z^n, \quad (8.6)$$

*which is the contiguous signed Thue–Morse block. For every  $b \geq 3$  and  $N \geq 2$ , the exponents are precisely the integers whose first  $N$  base- $b$  digits belong to  $\{0, 1\}$ ; they form a sparse Cantor-type set with gaps.*

*Proof.* For  $b = 2$ , binary expansion is a bijection between  $\{0, 1\}^N$  and  $\{0, 1, \dots, 2^N - 1\}$ , and the subset size is the binary digit sum. For every base, the exponent set is  $\{0\}$  when  $N = 0$  and  $\{0, 1\}$  when  $N = 1$ . Now suppose  $b \geq 3$  and  $N \geq 2$ . The exponents 0, 1, and  $b$  occur, whereas 2 cannot occur: reduction modulo  $b$  would force its units digit to lie in  $\{0, 1\}$ . Hence the exponent set has a gap between 1 and  $b$  and is not an interval. The stated digit characterization is immediate from (8.5).  $\square$

This gives a structural explanation for a feature that can otherwise look accidental. The reciprocal-integer Fourier products exist for every base  $b$ , and their zero multiplicities are controlled by  $v_b$ . But only  $b = 2$  makes the subset-product coefficients occupy every integer index at every nontrivial length  $N \geq 2$ . The Thue–Morse sequence is therefore not merely one member of a uniform  $b$ -adic family; it is the unique persistent no-gap member.

**Problem 8.5** (Sparse  $b$ -adic signed operators). For  $b \geq 3$ , develop the analogue of the dyadic signed extension on the sparse digit set  $\{0, 1\}$  in base  $b$ . Natural questions include transfer matrices for rarefied sums, operator spectra, exact cancellation order, and whether the missing digits can be supplied by multinomial factors to produce a different automatic sequence with a contiguous block.

## 9 Phase-aware Lagrange and Richardson extrapolation

The dyadic corpus already contains two ingredients that should not be conflated: geometric-node Lagrange extraction and  $q$ -Richardson cancellation of errors such as  $2^{-rn}$ . Endpoint saddle expansions introduce a different obstacle: coefficients depend periodically on the fractional part of a Lambert variable. Ordinary extrapolation across samples with different phases mixes different coefficient functions and can lose its nominal order. The cure is to move by integer increments in the Lambert variable.

### 9.1 A phase-preserving interpolation theorem

Let  $u \in \mathbb{R}/\mathbb{Z}$  be fixed. Suppose an observable has the phase-class representation

$$R(\lambda) = L + h_u(1/\lambda), \quad \lambda \equiv u \pmod{1}, \quad (9.1)$$

where  $h_u$  is smooth near zero and  $h_u(0) = 0$ . Use the samples  $\lambda_m = \lambda + m$ ,  $m = 0, \dots, p$ ; all have the same fractional phase.

**Theorem 9.1** (Phase-aware Lagrange filter). *Let  $p \in \mathbb{N}$ . Choose  $\lambda_0 > 0$  sufficiently large that  $h_u \in C^{p+1}[0, 1/\lambda_0]$ . For every  $\lambda \geq \lambda_0$  satisfying  $\lambda \equiv u \pmod{1}$ , and for  $m = 0, \dots, p$ , define*

$$w_m^{(p)}(\lambda) = (-1)^{p-m} \frac{(\lambda + m)^p}{m!(p-m)!}. \quad (9.2)$$

Then

$$\sum_{m=0}^p w_m^{(p)}(\lambda) = 1, \quad (9.3)$$

$$\sum_{m=0}^p \frac{w_m^{(p)}(\lambda)}{(\lambda + m)^r} = 0 \quad (1 \leq r \leq p), \quad (9.4)$$

and

$$\sum_{m=0}^p w_m^{(p)}(\lambda) R(\lambda + m) = L + \mathcal{O}_{u,p}(\lambda^{-p-1}) \quad (\lambda \rightarrow +\infty, \lambda \equiv u \pmod{1}). \quad (9.5)$$

More precisely, the error is bounded by

$$\frac{\|h_u^{(p+1)}\|_{L^\infty[0,1/\lambda]}}{(p+1)!} \prod_{m=0}^p \frac{1}{\lambda + m}. \quad (9.6)$$

*Proof.* Interpolate  $h_u(t)$  at the nodes  $t_m = (\lambda + m)^{-1}$  and evaluate the degree- $p$  interpolant at  $t = 0$ . The Lagrange basis weight is

$$\ell_m(0) = \prod_{r \neq m} \frac{-t_r}{t_m - t_r}.$$

Since

$$t_m - t_r = \frac{r - m}{(\lambda + m)(\lambda + r)},$$

a direct cancellation gives

$$\ell_m(0) = (-1)^{p-m} \frac{(\lambda + m)^p}{m!(p-m)!},$$

which is (9.2). Exactness on  $1, t, \dots, t^p$  gives (9.3)–(9.4). The standard interpolation remainder at zero is

$$\frac{h_u^{(p+1)}(\xi)}{(p+1)!} \prod_{m=0}^p (0 - t_m),$$

which proves (9.6) and hence (9.5).  $\square$

*Remark 9.2* (Conditioning). The weights grow individually like  $\lambda^p$  and cancel strongly. The theorem is exact, but a floating-point implementation should use extended precision, compensated summation, or a barycentric form. The deterministic experiment in [Section 10](#) uses moderate orders and directly verifies the predicted slope.

## 9.2 Fourier phase averaging

A complementary operation removes periodic modes rather than preserving a single phase.

**Theorem 9.3** (Root-of-unity phase filter). *Let  $M \in \mathbb{N}$  with  $M \geq 1$ , and let  $\Psi(u) = \sum_{k \in \mathbb{Z}} c_k e^{2\pi i k u}$  be an absolutely convergent Fourier series. Then*

$$\frac{1}{M} \sum_{r=0}^{M-1} \Psi\left(u + \frac{r}{M}\right) = \sum_{\ell \in \mathbb{Z}} c_{\ell M} e^{2\pi i \ell M u}. \quad (9.7)$$

If  $|c_k| \leq C e^{-\rho|k|}$ , then

$$\left| \frac{1}{M} \sum_{r=0}^{M-1} \Psi\left(u + \frac{r}{M}\right) - c_0 \right| \leq \frac{2C e^{-\rho M}}{1 - e^{-\rho M}}. \quad (9.8)$$



*Proof.* Insert the Fourier series and use

$$\frac{1}{M} \sum_{r=0}^{M-1} e^{2\pi i k r / M} = \begin{cases} 1, & M \mid k, \\ 0, & M \nmid k. \end{cases}$$

The error bound is the geometric sum over nonzero multiples of  $M$ .  $\square$

The two filters answer different questions. Phase-preserving Lagrange interpolation recovers the limit while keeping all periodic coefficients fixed. Root-of-unity averaging projects a periodic coefficient onto the Fourier modes divisible by  $M$ , converging exponentially to its mean when the phase function is analytic in a strip.

### 9.3 Exact Lambert- $W$ sample nodes

Suppose the relevant endpoint coordinate satisfies

$$x = (\lambda + \beta)e^{-a\lambda}, \quad (9.9)$$

with  $a > 0$ . Then

$$\lambda = -\beta - \frac{1}{a} W_{-1}(-axe^{-a\beta}), \quad (9.10)$$

where the lower branch is selected for large positive  $\lambda$ . Conversely, once a starting phase  $\lambda \equiv u \pmod{1}$  is chosen, the phase-locked physical nodes are

$$x_m = (\lambda + m + \beta)e^{-a(\lambda+m)}, \quad m = 0, \dots, p. \quad (9.11)$$

Their ratio is

$$\frac{x_{m+1}}{x_m} = q \frac{\lambda + m + 1 + \beta}{\lambda + m + \beta}, \quad (9.12)$$

so they are asymptotically geometric but carry exactly the correction needed to preserve the Lambert phase.

### 9.4 A practical composite algorithm

A numerically robust endpoint study can use the following sequence.

**Algorithm 9.4** (Phase-resolved endpoint extraction). Given an observable  $R(x)$  believed to have a periodic lower-Lambert expansion:

1. derive or fit the monotone saddle map  $x \leftrightarrow \lambda$  and choose a phase  $u \in [0, 1)$ ;
2. select a large  $\lambda \equiv u \pmod{1}$  and generate  $x_m$  from (9.11);
3. evaluate  $R(x_m)$  at increased precision;
4. apply the weights (9.2) in barycentric or compensated form;
5. repeat for several  $u$  to reconstruct the periodic coefficient functions;
6. optionally average  $M$  equally spaced phases and use (9.7) to isolate the mean or selected Fourier harmonics.

This is a synthesis of Lambert inversion, Lagrange interpolation, Richardson cancellation, and Fourier projection, but its exact weight formula and fixed-phase error theorem are not the same as the repository's existing geometric-grid filter.

## 10 Deterministic numerical experiments

The companion program `numerical_experiments.py` uses no Monte Carlo sampling. It is intended to test consequences of proved identities and to provide reconnaissance for the conjectures.

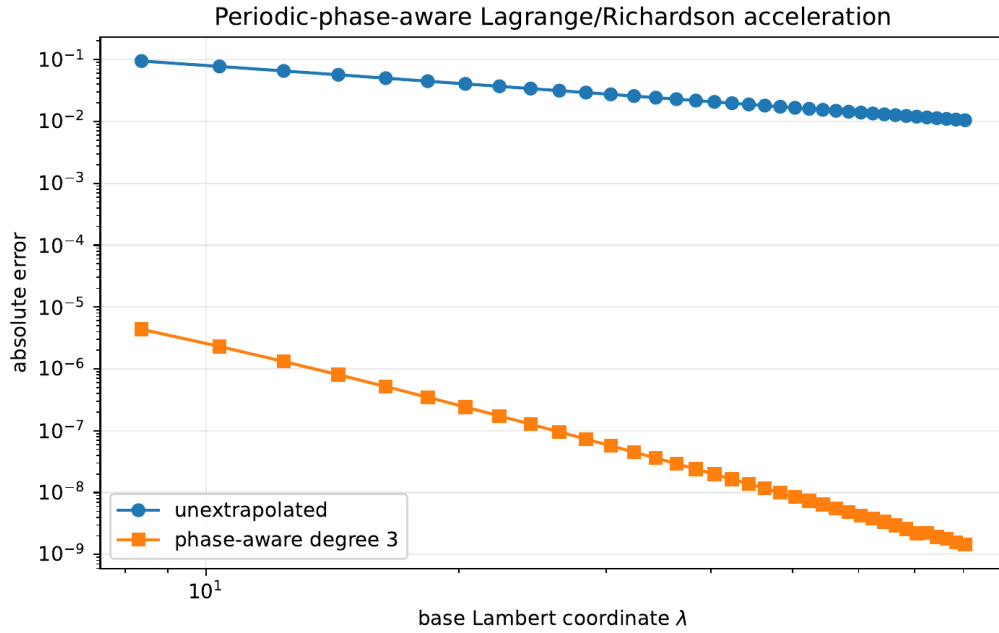


Figure 13: A synthetic periodic-phase model used only to verify [Theorem 9.1](#). The raw error has slope approximately  $-1$ ; a degree-three phase-preserving filter has observed slope approximately  $-3.82$ , approaching the predicted  $-4$ . This experiment does not assume or fit the unknown endpoint coefficients of  $F_q$ .

### 10.1 Fixed-point density solver

Starting from the uniform density, the code repeatedly applies the distributional fixed-point operator. On a uniform grid it computes a CDF by trapezoidal integration and then updates

$$f^{\text{new}}(x) = \frac{F^{\text{old}}(x/q) - F^{\text{old}}((x-c)/q)}{c}.$$

The density is renormalized after each step, and convergence is measured in the sup norm. The output verifies mass, symmetry, and the exact plateau. Representative results are shown in [Table 27](#).

Table 27: Fixed-point density diagnostics.

| $q$  | iterations | mass error           | symmetry error       | $f_q(1/2)$    |
|------|------------|----------------------|----------------------|---------------|
| 0.30 | 11         | $2.2 \cdot 10^{-16}$ | $1.2 \cdot 10^{-15}$ | 1.42857142857 |
| 0.50 | 19         | 0                    | $4.2 \cdot 10^{-15}$ | 2.00000000000 |
| 0.75 | 45         | 0                    | $9.2 \cdot 10^{-14}$ | 3.47864814200 |
| 0.95 | 247        | 0                    | $2.6 \cdot 10^{-12}$ | 8.56266675023 |

For  $q = 0.30$ , the theoretical plateau height is  $1/(1-q) = 10/7$ , and the maximum interior grid error on  $[q, 1-q]$  was exactly zero at machine precision.

### 10.2 Cumulant checks

The script computes standardized cumulants both from the general Bernoulli formula and from the simplified rational expressions. The differences were at most machine roundoff. For example,

| $q$  | $\sigma_q^2$ | $\gamma_4(q)$ | $\gamma_6(q)$ |
|------|--------------|---------------|---------------|
| 0.25 | 0.0500000    | −1.0588235    | 5.6514914     |
| 0.50 | 0.0277778    | −0.7200000    | 2.9387755     |
| 0.75 | 0.0119048    | −0.3360000    | 0.6985447     |
| 0.90 | 0.0043860    | −0.1259669    | 0.1003783     |
| 0.97 | 0.0012690    | −0.0365397    | 0.0084745     |

The rapid decrease of higher standardized cumulants supports the Edgeworth program.

### 10.3 Endpoint bounds

For  $q = 1/2$  and  $x$  from  $10^{-4}$  down to  $10^{-60}$ , the program evaluates the exact logarithms of the finite-simplex upper bound and balanced-product lower bound. After adding  $L^2/(2a) + (L \log L)/a$  and dividing by  $L$ , both residuals remain bounded, as displayed in Figure 12. This is a numerical visualization of a theorem rather than an independent fit.

### 10.4 Phase-aware extrapolation

The synthetic model is

$$R(\lambda) = L + \frac{a_1(u)}{\lambda} + \frac{a_2(u)}{\lambda^2} + \frac{a_3(u)}{\lambda^3} + \frac{a_4(u)}{\lambda^4}, \quad u = \{\lambda\},$$

with trigonometric coefficient functions. Sampling at  $\lambda, \lambda + 1, \lambda + 2, \lambda + 3$  holds  $u$  fixed. A log–log regression gave raw slope  $-0.990967$  and accelerated slope  $-3.824509$ ; the latter tends to  $-4$  as the asymptotic range and precision increase.

### 10.5 Zero multiplicities

For  $b = 2, 3, 4, 5$  and  $1 \leq |N| \leq 64$ , the code enumerates vanishing sinc factors and verifies  $\text{mult} = 1 + v_b(N)$  without exception. This check is elementary but guards against an easy normalization error in the lattice spacing.

### 10.6 Reproducibility

From the consolidated volume directory, running

```
(cd assets/beyond_dyadic_fabius_web_report && \
  python3 numerical_experiments.py)
```

regenerates the four PDF figures, a CSV summary, and the text log inside the retained asset directory. The script contains detailed comments, parameter checks, deterministic stopping criteria, and no external data dependency.

## 11 Conjectures and promising research directions

This section collects the most promising next steps in a form designed to be falsifiable and, where possible, formalizable.

### 11.1 Analytic shape

*Conjecture 11.1* (Strict log-concavity). For  $q > 1/2$ , the density  $f_q$  is strictly log-concave on  $(0, 1)$ . Equivalently,

$$f_q(\theta x + (1 - \theta)y) > f_q(x)^\theta f_q(y)^{1-\theta}$$

for distinct  $x, y \in (0, 1)$  and  $0 < \theta < 1$ .

The exact plateau theorem shows that  $q = 1/2$  is the only possible transition value. A proof would likely combine equality cases in Prékopa–Leindler with the self-similar decomposition and the fact that no single interval dominates the remaining support when  $q > 1/2$ .

*Problem 11.2* (Monotonicity of the peak). Prove or disprove that  $q \mapsto f_q(1/2)$  is strictly increasing. Convex-order contraction suggests the statement but does not imply pointwise density order. The Gaussian prediction is

$$f_q(1/2) \sim \sqrt{\frac{6(1+q)}{\pi(1-q)}}.$$

## 11.2 Uniform Gaussian asymptotics

*Conjecture 11.3* (All-orders Edgeworth expansion). The expansion in [Conjecture 6.2](#) holds uniformly with explicit coefficients generated by (6.2). In particular,

$$f_q(1/2) = \frac{1}{\sqrt{2\pi\sigma_q}} \left( 1 - \frac{3}{20}(1-q) + \mathcal{O}((1-q)^2) \right).$$

A promising proof strategy is to split the normalized Fourier integral at  $|t| = (1-q)^{-\eta}$ , use the cumulant expansion on the central range, and exploit a block of roughly  $(1-q)^{-1}$  sinc factors for a  $q$ -uniform tail majorant.

## 11.3 Endpoint phase and inverse transfer

*Conjecture 11.4* (Nonconstant periodic amplitude). For generic  $q$ , the leading phase function  $A_{q,0}$  in (7.10) is nonconstant, real analytic, and has no vanishing nonzero Fourier coefficient. Exceptional  $q$ , if any, form a thin arithmetic set.

*Conjecture 11.5* (Inverse phase transfer). Let  $Q_q = F_q^{-1}$ . The periodic endpoint functions of  $F_q$  transfer to a complete expansion of  $Q_q(y)$  through the lower branch  $W_{-1}$  and formal Lagrange inversion. The phase of the inverse is not simply the phase of  $\log(1/y)$ ; it includes the derivative of the saddle map and a computable phase shift from the leading amplitude.

The phase-aware sampling theorem provides a practical way to distinguish these possibilities numerically without averaging the phase away.

## 11.4 A two-parameter large-deviation/Gaussian phase diagram

The limits  $x \downarrow 0$  and  $q \uparrow 1$  do not commute. For fixed  $q$ , the endpoint law is quadratic in  $\log(1/x)$ . For  $q \uparrow 1$  at the central scale  $x = 1/2 + \sigma_q z$ , the law is Gaussian. Between them lies a double-scaling region.

*Problem 11.6* (Joint scaling). Determine a uniform asymptotic for

$$\mathbb{P}\left(X_q \leq \frac{1}{2} - r_q\right)$$

when  $q \uparrow 1$  and  $r_q/\sigma_q \rightarrow \infty$  while  $r_q < 1/2$ . Identify the transition from the Gaussian rate  $r_q^2/(2\sigma_q^2)$  to the logarithmic endpoint action of [Theorem 7.1](#). A natural tool is the exact cumulant generating product (4.8) and a saddle that moves through the geometric scale lattice.

## 11.5 Arithmetic zero collisions

*Problem 11.7* (Algebraic collision graph). For algebraic  $q$ , define a graph on scales  $j \geq 0$  by joining  $j$  and  $k$  when  $q^{|j-k|} \in \mathbb{Q}$ . Express Fourier-zero multiplicities in terms of rational valuations along connected components. Determine when the multiplicity function is automatic or regular in the sense of Allouche–Shallit.

## 11.6 Beyond binary Thue–Morse

*Problem 11.8* (Contiguous completion in higher base). For  $b \geq 3$ , find the minimal collection of digit factors  $\prod_j P(z^{b^j})$  whose finite expansion is contiguous and whose coefficient sequence retains a signed Prouhet cancellation law. The obvious choice  $P(z) = 1 + z + \cdots + z^{b-1}$  loses signs; signed cyclotomic or character-valued factors may yield new automatic analogues.

## 11.7 Numerical analysis and conditioning

*Problem 11.9* (Stable phase-aware barycentric formula). Derive a backward-stable barycentric implementation of [Theorem 9.1](#) whose roundoff amplification is explicit in  $p$  and  $\lambda$ . Couple it to arbitrary-precision Lambert- $W$  node generation, and optimize phase averaging against truncation and rounding errors.

## 12 A Lean formalization roadmap

The new family can be developed in layers that mirror the existing repository architecture. The following order minimizes dependence on asymptotic machinery.

### 12.1 Probability layer

1. Define the geometric coefficient sequence  $w_j(q) = (1 - q)q^j$  and prove summability, nonnegativity, and total mass one.
2. Name  $X_q$  and its law, then specialize the generic probability, exact-support, reflection, continuous-linear naturality, and scalar affine-recursion theorems.
3. Use the exact topological support  $[0, 1]$ , absolute continuity, and null singletons to define the real CDF, derive its conditioning equations and explicit density refinement, and bootstrap compactly supported  $C^\infty$  regularity.
4. Specialize a generic contractive affine-independent-copy theorem to characterize the geometric-uniform law among all probability fixed points.
5. Derive the characteristic-function sinc product as a pointwise limit of exact finite residual factorizations.

At the affine-law compiled source checkpoint d312c0603, items 1–4 had been formalized, in stronger pointwise form where applicable, by `WeightedUniformSeries.lean`, `WeightedUniformDistribution.lean`, `WeightedUniformSupport.lean`, `GeometricUniformLaw.lean`, `ContinuousCDF.lean`, and `GeometricUniformCDF.lean`, together with the generic `AffineIndependentCopy.lean` and its specialization `GeometricUniformUniqueness.lean`. This includes law reflection, independent-copy affine refinement, absolute continuity, null singletons, exact support, the named CDF and both conditioning integrals, its explicit Radon–Nikodym density, compact support, and  $C^\infty$  regularity, as well as fixed-point uniqueness for every real  $|q| < 1$ , with  $q = 0$  and negative contractions included and no support, density, or moment assumptions. The live `GeometricUniformDictionary.lean` and `GeometricSincFactorization.lean` now complete item 5 in finite-residual and pointwise `Tendsto` form for every  $|q| < 1$ , including exact dyadic wrappers. For  $q, z \in \mathbb{C}$ ,  $\|q\| < 1$ , the further declarations `hasProdLocallyUniformly_geometricSincProduct`, `geometricSincProductFactors_multipliable`, `hasProd_geometricSincProduct`, and `geometricSincProduct_differentiable` prove locally uniform convergence, pointwise `Multipliable/HasProd`, and entire-ness for  $S_q(z) = \prod_{n \geq 0} \text{sinc}(\pi q^n z)$ . For real  $|q| < 1$ ,  $t \in \mathbb{R}$ , and  $z_q(t) = (1 - q)t/(2\pi)$ , the theorems `charFun_geometricUniformDistribution_eq_phase_mul_geometricSincProduct` and `charFun_geometricUniformDistribution_eq_phase_mul_geometricReciprocalGamma` give  $\phi_q(t) = e^{it/2} S_q(z_q(t)) = e^{it/2} G_q(z_q(t)) G_q(-z_q(t))$ , where  $G_q(z)$  denotes `geometricReciprocalGamma` applied to  $q, z$ . The equality includes  $q = 0$  and negative contractions. A named centered wrapper, packaged compact-uniform convergence of the full phase-bearing characteristic prefixes, and quantitative Fourier estimates remain in the analysis layer. The distinct oriented Cauchy-transform DDE and adjacent-power hierarchy are now formal in `GeometricUniformCauchy.lean`; they are separate from the characteristic-function product and real-space density-derivative hierarchies.

### 12.2 Analysis layer

1. Formalize the fixed- $N$  Fourier bound (3.3) and the absolutely convergent higher-derivative Fourier-inversion formulas. Smoothness itself is already formal through the affine derivative bootstrap.
2. Prove log-concavity first for finite convolutions and then under the density limit.
3. Formalize the plateau interval by the extended-CDF refinement equation.
4. Separate strict unimodality from the stronger strict-log-concavity conjecture.

### 12.3 Algebraic layer

1. Prove the Bernoulli cumulant formula from the uniform mgf and a geometric series.
2. Connect moments to complete Bell polynomials.
3. Define  $P_{q,n}$  by formal power series and prove the Appell, expectation-inverse, and refinement identities.
4. Import or prove the finite  $q$ -binomial theorem needed for (4.15).

### 12.4 Order and limit layer

1. Introduce the coefficient-vector majorization theorem in a form usable for infinite  $\ell^1$  sequences, then prove convex-order monotonicity.
2. Prove the triangular-array CLT by a bounded Lindeberg theorem.
3. Formalize standardized cumulants and the compact-frequency expansion.

### 12.5 Endpoint and numerical layer

1. Formalize the simplex-volume lemma with nonidentical coordinate scales.
2. Formalize the balanced product event and Stirling bounds sufficient for (7.3).
3. Prove the Lambert- $W$  inversion corollary using monotonicity and the lower/upper branch API already present in the project.
4. Prove the phase-aware Lagrange weights as an exact rational-function identity before adding analytic remainder estimates.

The endpoint theorem is especially suitable for formalization because it avoids contour integrals: the only asymptotic ingredients are explicit integer choices, simplex volume, Stirling bounds, and elementary logarithmic estimates.

## 13 Conclusion

The dyadic Fabius–Rvachev theory is rigid enough that many formulas appear isolated until the geometric ratio is allowed to vary. The family

$$X_q = (1 - q) \sum_{j \geq 0} q^j U_j$$

places several of the repository’s central themes in a single parameter space. Bernoulli cumulants and Bell moments become exact for every  $q$ ; Gaussian-binomial coefficients encode the elementary symmetric functions of the geometric scales; Bernoulli polynomials invert each uniform-averaging factor; convex majorization orders the entire family; the dyadic value marks a genuine plateau-to-strict-unimodality transition; and  $q \uparrow 1$  produces a Gaussian law with explicit cumulant corrections. At the opposite endpoint, a simple finite-dimensional argument already forces both the quadratic logarithmic action and the  $L \log L$  entropy correction, while Lambert  $W$  inverts the resulting quantile scale.

The Fourier side exhibits an equally clean dichotomy. Reciprocal-integer ratios create valuation-controlled zero multiplicities, whereas generic irrational powers give simple zeros. Base two is singled out not merely by convention but because it is the unique base in which the signed subset product fills a contiguous integer block, thereby producing the Thue–Morse sequence. Finally, the periodic lower-Lambert corrections that complicate sharp endpoint numerics can be handled by sampling at integer-separated Lambert variables and applying an explicit Lagrange filter; phase averaging supplies the complementary Fourier projection.

The next frontier is to join these paper-level results into a uniform saddle theory in both  $q$  and  $x$ , determine the periodic coefficient functions, prove a global Edgeworth expansion, and classify algebraic zero collisions. Each problem is concrete enough to support numerical experiments and sufficiently structured to admit Lean formalization.

## A Complete TeX corpus inventory

The following ledger was used to enforce corpus-relative nonduplication. It is retained as `assets/beyond_dyadic_fabius_web_report/corpus_inventory.txt`.

```
Corpus inventory used for "Beyond the Dyadic Fabius Web"
Repository snapshot: VladimirReshetnikov/ProveIt, main, commit 24da985eba70e46d862acdadd476793a428213e
Inspection date: 2026-08-27
Scope: every *.tex file under Analysis/FabiusFunction/docs, recursively
Total: 57 files

A. Living and current synthesis documents (6)
1. Analysis/FabiusFunction/docs/FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
2. Analysis/FabiusFunction/docs/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
3. Analysis/FabiusFunction/docs/Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex
4. Analysis/FabiusFunction/docs/fabius_lean_walkthrough/fabius_lean_walkthrough.tex
5. Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/semi-formalized-research-frontiers.tex
6. Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/Thue_Morse_Formula_Atlas/Thue_Morse_Formula_Atlas.tex

B. Current Fourier-decay draft and audit campaign (12)
7. Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-2/Rvachev_Up_Fourier_Decay.tex
8. Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-3/Fourier_decay_of_Rvachev_up.tex
9. Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Comparative_Audit/Rvachev_Up_Fourier_Decay_Comparative_Audit.tex
10. Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Gentle_Guide/Rvachev_Up_Fourier_Decay_Gentle_Guide.tex
11. Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Second_Wave_Audit/Rvachev_Up_Fourier_Decay_Second_Wave_Audit.tex
12. Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions.tex
13. Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-1/rvachev_up_fourier_decay.tex
14. Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-4/rvachev_up_fourier_decay.tex
15. Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-5/Rvachev_Up_Fourier_Decay_Final.tex
16. Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-6/Rvachev_Up_Fourier_Decay_Final_Synthesis.tex
17. Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-7/rvachev_up_fourier_decay_final.tex
18. Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-8/rvachev_up_fourier_decay_final.tex

C. Included paper sources (7)
19. Analysis/FabiusFunction/docs/papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0-corrected.tex
20. Analysis/FabiusFunction/docs/papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0.tex
21. Analysis/FabiusFunction/docs/papers/arXiv-1502.06738v2/Lacunary_products.tex
22. Analysis/FabiusFunction/docs/papers/arXiv-1702.05442v1/09-Function.tex
23. Analysis/FabiusFunction/docs/papers/arXiv-1702.06487v3/157-Arithmetic-v3.tex
24. Analysis/FabiusFunction/docs/papers/arXiv-2211.13570v1/document.tex
25. Analysis/FabiusFunction/docs/papers/ubiq15/ubiq15-corrected.tex

D. Frozen historical archive (32)
Early notes
26. Analysis/FabiusFunction/docs/archive/early-notes/Fabius Asymptotic/Fabius Asymptotic.tex
27. Analysis/FabiusFunction/docs/archive/early-notes/K-fold summation over the signed Thue-Morse sequence/K-fold summation over the signed Thue-Morse sequence.tex

Standalone studies
28. Analysis/FabiusFunction/docs/archive/standalone-studies/Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex
29. Analysis/FabiusFunction/docs/archive/standalone-studies/Small_Argument_Asymptotics/Small_Argument_Asymptotics.tex
30. Analysis/FabiusFunction/docs/archive/standalone-studies/fabius-inverse-dyadic-closed-form/fabius-inverse-dyadic-closed-form.tex

Primary exposition versions
31. Analysis/FabiusFunction/docs/archive/primary-exposition/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
32. Analysis/FabiusFunction/docs/archive/primary-exposition/Fabius_Function_and_Rvachev_Up-2/Fabius_Function_and_Rvachev_Up.tex
33. Analysis/FabiusFunction/docs/archive/primary-exposition/Fabius_Function_and_Rvachev_Up-3/Fabius_Function_and_Rvachev_Up.tex
34. Analysis/FabiusFunction/docs/archive/primary-exposition/Fabius_and_Rvachev_Up-2/Fabius_and_Rvachev_Up.tex
35. Analysis/FabiusFunction/docs/archive/primary-exposition/Fabius_and_Rvachev_up/Fabius_and_Rvachev_up.tex
36. Analysis/FabiusFunction/docs/archive/primary-exposition/Fabius_function_and_up_function/Fabius_function_and_up_function.tex

Primary supplements
37. Analysis/FabiusFunction/docs/archive/primary-supplements/Fabius_Function_Missing_Parts/Fabius_Function_Missing_Parts.tex
38. Analysis/FabiusFunction/docs/archive/primary-supplements/Fabius_Function_Missing_Parts-2/Fabius_Function_Missing_Parts.tex

q-calculus
39. Analysis/FabiusFunction/docs/archive/q-calculus/Fabius_q_Special_Functions/Fabius_q_Special_Functions.tex
40. Analysis/FabiusFunction/docs/archive/q-calculus/fabius-q-special-functions/fabius-q-special-functions.tex

Lean walkthrough
41. Analysis/FabiusFunction/docs/archive/lean-walkthrough/Fabius_Function_Lean_Walkthrough/Fabius_Function_Lean_Walkthrough.tex
42. Analysis/FabiusFunction/docs/archive/lean-walkthrough/fabius_lean_walkthrough/fabius_lean_walkthrough.tex

Glossary
43. Analysis/FabiusFunction/docs/archive/glossary/FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
44. Analysis/FabiusFunction/docs/archive/glossary/FabiusFunction_Mathematical_Glossary-2/FabiusFunction_Mathematical_Glossary.tex

Research-frontier dossiers and atlas versions
45. Analysis/FabiusFunction/docs/archive/research-frontiers/Fabius_Dyadic_Asymptotic_Bridge/Fabius_Dyadic_Asymptotic_Bridge.tex
46. Analysis/FabiusFunction/docs/archive/research-frontiers/Fabius_Dyadic_Formulae_and_Alternative_Representations/Fabius_Dyadic_Formulae_and_Alternative_Representations.tex
47. Analysis/FabiusFunction/docs/archive/research-frontiers/Fabius_Dyadic_Formulae_to_Asymptotics/
```



```

Fabius_Dyadic_Formulae_to_Asymptotics.tex
48. Analysis/FabiusFunction/docs/archive/research-frontiers/Fabius_Dyadic_q_Connections/Fabius_Dyadic_q_Connections.tex
49. Analysis/FabiusFunction/docs/archive/research-frontiers/Fabius_Thue_Morse_Convergence_Rate/Fabius_Thue_Morse_Convergence_Rate.tex
50. Analysis/FabiusFunction/docs/archive/research-frontiers/Fabius_Thue_Morse_Convergence_Rates/Fabius_Thue_Morse_Convergence_Rates.tex
51. Analysis/FabiusFunction/docs/archive/research-frontiers/Primary_Exposition_Gap_Register/Primary_Exposition_Gap_Register.tex
52. Analysis/FabiusFunction/docs/archive/research-frontiers/Repeated_Integration_and_Rvachev_Up/Repeated_Integration_and_Rvachev_Up.tex
53. Analysis/FabiusFunction/docs/archive/research-frontiers/Rvachev_Up_from_Repeated_Integration/Rvachev_Up_from_Repeated_Integration.tex
54. Analysis/FabiusFunction/docs/archive/research-frontiers/non-formalized-research-frontiers/semi-formalized-research-frontiers.tex
55. Analysis/FabiusFunction/docs/archive/research-frontiers/Thue_Morse_Formula_Atlas-1/Thue_Morse_Formula_Atlas.tex
56. Analysis/FabiusFunction/docs/archive/research-frontiers/Thue_Morse_Formula_Atlas-2/Consolidated_Formula_Atlas_for_the_Thue_Morse_Sequence.tex
57. Analysis/FabiusFunction/docs/archive/research-frontiers/Thue_Morse_Formula_Atlas-3/Thue_Morse_Formula_Atlas.tex

Notes
- The living primary exposition identifies commit 24da985eba70e46d862acdadd476793a428213e as its 27 August 2026 snapshot.
- The archive README explains that byte-identical historical versions were pruned, so the archive is a provenance map rather than a naive copy of every prior path.
- The two AIF sources are OCR/original and corrected versions of the same paper; both were included because the request was source-file exhaustive.

```

## B Auxiliary proof details

### B.1 Schur-convexity of weighted i.i.d. sums

For completeness, let  $U_1, \dots, U_N$  be i.i.d. and let  $\Phi$  be convex. The map

$$H(a_1, \dots, a_N) = \mathbb{E} \Phi \left( \sum_{j=1}^N a_j U_j \right)$$

is convex because it is an expectation of a convex function of the coefficient vector, and it is symmetric because the  $U_j$  are exchangeable. Every symmetric convex function is Schur-convex. Equivalently, it suffices to check a single Robin-Hood transfer: replace  $(a_i, a_j)$  by

$$(\theta a_i + (1 - \theta) a_j, (1 - \theta) a_i + \theta a_j), \quad 0 \leq \theta \leq 1.$$

The new weighted pair is the conditional expectation of a random permutation of the old pair, and Jensen's inequality decreases the expectation of  $\Phi$ . Iterating transfers and passing to  $\ell^1$  limits proves [Theorem 5.1](#).

### B.2 A direct expansion behind the inverse law

Set  $S = \sqrt{2aY}$  and  $L_0 = S - \log S$ . Then

$$L_0^2 + 2L_0 \log L_0 = S^2 + \mathcal{O}((\log S)^2).$$

The uncertainty in (7.6) is  $\mathcal{O}(L_y) = \mathcal{O}(S)$ , and the derivative of  $L^2 + 2L \log L$  is  $2L + 2 \log L + 2 \asymp S$ . Therefore the uncertainty in  $L_y$  is bounded, which gives the explicit second form (7.8) without invoking the full asymptotic series of  $W$ .

### B.3 Endpoint upper and lower constants

The proof of [Theorem 7.1](#) can be converted into explicit inequalities. There exist  $x_0(q) > 0$  and constants  $C_-(q), C_+(q)$  such that for  $0 < x < x_0(q)$ ,

$$-\frac{L^2}{2a} - \frac{L \log L}{a} - C_-(q)L \leq \log F_q(x) \leq -\frac{L^2}{2a} - \frac{L \log L}{a} + C_+(q)L.$$

One obtains admissible, though nonoptimal, constants by using

$$n \log n - n \leq \log(n!) \leq n \log n - n + \log n + 1$$

for sufficiently large  $n$ , together with  $|n_{\pm} - L/a| \leq C(q)$ . The companion numerical plot evaluates the sharper uncompressed finite- $n$  expressions rather than these coarse constants.



## B.4 Phase-aware weights as finite differences

The weights (9.2) also admit the form

$$w_m^{(p)}(\lambda) = \frac{(-1)^p}{p!} (-1)^m \binom{p}{m} (\lambda + m)^p.$$

Thus the estimator is a normalized  $p$ th forward difference applied to  $(\lambda + m)^p R(\lambda + m)$ . The Lagrange interpretation is preferable for error analysis, but the finite-difference form may be easier to formalize algebraically.

## C Reproducibility manifest

The former standalone package’s retained auxiliary payload is stored under `assets/beyond_dyadic_fabius_web_report/`. The report source has been absorbed into `Frontier_Compilations.tex`, and its former standalone PDF is not retained. Relative to that asset directory, the retained files are:

- `numerical_experiments.py` – deterministic, commented experiments;
- `q_family_densities.pdf`;
- `standardized_cumulants.pdf`;
- `endpoint_bound_residuals.pdf`;
- `phase_aware_extrapolation.pdf`;
- `numerical_summary.csv`;
- `experiment_output.txt`;
- `corpus_inventory.txt`;
- `requirements.txt`;
- `README.txt`;
- `SHA256SUMS.txt`.

All numerical outputs are reproducible from the Python file with NumPy and Matplotlib.

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  - [22] V. Reshetnikov, *Archive of historical document versions*, Analysis/FabiusFunction/docs/archive/README.md, repository snapshot [17].